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Technical Report

3rd Generation Partnership Project; Technical Specification Group Radio Access Network; Study on solutions for Ambient IoT (Internet of Things) in NR

(Release 20)





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Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

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- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
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In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

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should not indicates a recommendation not to do something

may indicates permission to do something

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The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

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In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

Introduction

In recent years, IoT has attracted much attention in the wireless communication world. More 'things' are expected to be interconnected for improving productivity efficiency and increasing comforts of life. Further reduction of size, complexity, and power consumption of IoT devices can enable the deployment of tens or even hundreds of billions of IoT devices for various applications and provide added value across the entire value chain. It is impossible to power all the IoT devices by battery that needs to be replaced or recharged manually, which leads to high maintenance cost, serious environmental issues, and even safety hazards for some use cases, for example, wireless sensors in electrical power, and petroleum industries.

Most of the existing wireless communication devices are powered by batteries that need to be replaced or recharged manually. The automation and digitization of various industries opens numerous new markets requiring new IoT technologies of supporting batteryless devices with no energy storage capability or devices with energy storage that do not need to be replaced or recharged manually.

An example type of application is asset identification, which presently has to resort mainly to barcodes and RFID in most industries. The main advantage of these two technologies is the ultra-low complexity and small form factor of the tags. However, the limited reading range of a few meters usually requires handheld scanning which leads to labour intensive and time-consuming operations, or RFID portals/gates which leads to costly deployments. Moreover, the lack of interference management scheme results in severe interference between RFID readers and capacity problems, especially in case of dense deployment. It is hard to support a large-scale network with seamless coverage for RFID. Other examples are sensor applications, which have characteristics such as the operating conditions or cost requirements, etc., which mean they cannot be addressed by 3GPP's LPWA technologies. In general, the use cases for these sensors span indoor and outdoor applications. Applications are also shown to have requirements for positioning, with accuracy targets varying between indoor and outdoor scenarios.

In contrast, this study investigates solutions for Ambient IoT, a new IoT technology to open new markets within 3GPP systems, whose number of connections and/or device density can be orders of magnitude higher than existing 3GPP IoT technologies, and which can provide complexity and power consumption orders-of-magnitude lower than existing 3GPP LPWA technologies such as NB-IoT and LTE-MTC. TSG RAN has completed a Rel-18 RAN-level SI on Ambient IoT, producing TR 38.848 which provides a terminological and scoping framework for future discussions of Ambient IoT. This has defined representative use cases, deployment scenarios, connectivity topologies, Ambient IoT devices, design targets, and required functionalities; it also conducted a preliminary feasibility assessment.

The SI reported in this present TR is now to investigate solutions in detail at RAN-WG level for Ambient IoT in 3GPP.

1 Scope

1.1 Indoor scenarios

The overall objective of the SI is to study a harmonized air interface design with minimized differences (where necessary) for Ambient IoT to enable the following devices:

- ~1 μW peak power consumption, has energy storage, initial sampling frequency offset (SFO) up to 10^X ppm, neither R2D nor D2R amplification in the device. The device's D2R transmission is backscattered on a carrier wave provided externally.
- ≤ a few hundred μW peak power consumption has energy storage, initial sampling frequency offset (SFO) up to 10^X ppm, both R2D and/or D2R amplification in the device. The device's D2R transmission may be generated internally by the device, or be backscattered on a carrier wave provided externally.

Referring to the definitions in [2, TR 38.848], this is done in the context of:

- Deployment scenario 1 (indoor-to-indoor) with Topology 1, and indoor microcell basestation.
- Deployment scenario 2 (indoor-to-outdoor) with Topology 2 and indoor UE as intermediate node under network control, and outdoor macrocell basestation.

The spectrum considered is FR1 licensed spectrum in FDD, which can be in-band to NR, in guard-band to NR/LTE, or in standalone band(s). The traffic types considered are DO-DTT and DT, focusing on indoor inventory and indoor command representative use cases. The study also assesses whether the harmonized air interface can address the DO-A use case.

Study of the design of energy harvesting signal/waveform is out of scope in Rel-19.

1.2 Outdoor scenarios

The objectives of the SI are to:

- Study necessary and feasible changes to Ambient IoT specified in TS 38.291 [228] to support outdoor scenarios
- Identify D2R RSRP-like measurement and the involved signal(s)/channel(s) which are feasible for Device 2b/Device C for more accurate outdoor Device localization than based on Reader-ID

for the following devices:

- ≤ a few hundred μW peak power consumption, has energy storage, IF envelope detector receiver or ZIF receiver, initial sampling frequency offset (SFO) up to 10Y ppm, R2D and/or D2R amplification in the device. The device's D2R transmission is generated internally by the device.
- ≤ 1 mW to ≤ 10 mW peak power consumption, has energy storage, IF envelope detector receiver or ZIF receiver, initial sampling frequency offset (SFO) up to 10Y ppm, R2D and/or D2R amplification in the device. The device's D2R transmission is generated internally by the device.

Referring to the definitions in [2], this is done in the context of deployment scenario 4 (outdoor-to-outdoor) with Topology 1, with no external carrier wave. The spectrum considered is FR1 licensed spectrum in FDD in-band to NR and in standalone band(s) with R2D in DL spectrum and D2R in UL spectrum. The traffic types considered are DO-DTT, DT and DO-A, focusing on outdoor inventory, outdoor sensor and outdoor command representative use cases. The study assumes that the Ambient IoT BS readers are deployed on the same sites as existing outdoor NR macro BSs and assumes the same air interface for both the device types. The study also addresses positioning for outdoor scenarios.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
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[358] R1-2509274, "Study on other procedures for Ambient IoT outdoor for active device", NTT DOCOMO, INC., RAN1#123, Dallas, United States, November 2025.

[359] R1-2509324, "Discussion on other procedures for Ambient IoT", TCL, RAN1#123, Dallas, United States, November 2025.

[360] R1-2509345, "Discussion on other procedures for Ambient IoT", CEWiT, RAN1#123, Dallas, United States, November 2025.

[361] R1-2600071, "Other Air Interface Procedures", FUTUREWEI, RAN1#124, Gothenburg, Sweden, February 2026.

[362] R1-2600090, "Study on other procedures for A-IoT", Huawei, HiSilicon, RAN1#124, Gothenburg, Sweden, February 2026.

[363] R1-2600103, "Discussion on other procedure for Ambient IoT", Spreadtrum, UNISOC, RAN1#124, Gothenburg, Sweden, February 2026.

[364] R1-2600181, "Discussion on other procedures for Rel-20 A-IoT", OPPO, RAN1#124, Gothenburg, Sweden, February 2026.

[365] R1-2600212, "Other procedures of A-IoT", Ericsson, RAN1#124, Gothenburg, Sweden, February 2026.

[366] R1-2600329, "Study of general procedures of A-IoT enhancement for device 2b/C", CATT, RAN1#124, Gothenburg, Sweden, February 2026.

[367] R1-2600370, "Other procedures for AIoT", Nokia, RAN1#124, Gothenburg, Sweden, February 2026.

[368] R1-2600380, "Discussion on other procedures", CMCC, RAN1#124, Gothenburg, Sweden, February 2026.

[369] R1-2600419, "Discussion on other procedures for Device 2b/C", Xiaomi, RAN1#124, Gothenburg, Sweden, February 2026.

[370] R1-2600453, "Discussion on other procedures for active Ambient IoT device", ZTE Corporation, Sanechips, RAN1#124, Gothenburg, Sweden, February 2026.

[371] R1-2600493, "Discussion on Other Procedures for R20 AIoT", vivo, RAN1#124, Gothenburg, Sweden, February 2026.

[372] R1-2600547, "Other procedures for Device 2b/C", LG Electronics, RAN1#124, Gothenburg, Sweden, February 2026.

[373] R1-2600661, "Study on other procedures for Device 2b and C", NEC, RAN1#124, Gothenburg, Sweden, February 2026.

[374] R1-2600683, "Discussion on other procedures for Ambient IoT", China Telecom, RAN1#124, Gothenburg, Sweden, February 2026.

[375] R1-2600713, "Discussion on A-IoT other procedures", Panasonic, RAN1#124, Gothenburg, Sweden, February 2026.

[376] R1-2600745, "Study on other procedures of air interface for Device 2b/C", Samsung, RAN1#124, Gothenburg, Sweden, February 2026.

[377] R1-2600817, "On other procedures for device 2b/C", Apple, RAN1#124, Gothenburg, Sweden, February 2026.

[378] R1-2600967, "Discussion on other procedures", Sharp, RAN1#124, Gothenburg, Sweden, February 2026.

[379] R1-2600993, "Discussion on other procedures for A-IoT", ETRI, RAN1#124, Gothenburg, Sweden, February 2026.

[380] R1-2601022, "Discussion on other procedures for R20 Ambient IoT", Lenovo, RAN1#124, Gothenburg, Sweden, February 2026.

[381] R1-2601076, "Procedures for Active AIoT Devices", InterDigital, Inc., RAN1#124, Gothenburg, Sweden, February 2026.

[382] R1-2601124, "System information and DO-A aspects for Ambient IoT outdoor devices", Sony, RAN1#124, Gothenburg, Sweden, February 2026.

[383] R1-2601169, "Study on other procedures for Ambient IoT outdoor for active device", NTT DOCOMO, INC., RAN1#124, Gothenburg, Sweden, February 2026.

[385] R1-2601262, "Study of other procedures for Device 2b/C", Qualcomm Incorporated, RAN1#124, Gothenburg, Sweden, February 2026.

[386] R1-2601293, "Discussion on other procedures for Device 2b/C", Quectel, RAN1#124, Gothenburg, Sweden, February 2026.

[387] R1-2601301, "Discussion on procedures for A-IoT active device", ASUSTeK, RAN1#124, Gothenburg, Sweden, February 2026.

[388] R1-2601341, "Discussion on other procedures for Ambient IoT", TCL, RAN1#124, Gothenburg, Sweden, February 2026.

[389] R1-2601347, "Discussion on other procedures for Device 2b/C", KT Corp., RAN1#124, Gothenburg, Sweden, February 2026.

[390] R1-2601442, "Discussion on other aspects of Rel-20 AIoT", IIT Kanpur, RAN1#124, Gothenburg, Sweden, February 2026.

[391] R1-2601450, "Discussion on other procedures for R20 A-IoT", Fraunhofer HHI, Fraunhofer IIS, RAN1#124, Gothenburg, Sweden, February 2026.

[392] R1-2601731, "Ambient IoT evaluation results spreadsheet for TR38.769", Moderator (Huawei), RAN1#124, Gothenburg, Sweden, February 2026.

[393] R4-2602113, "Collection of formal simulation results for A-IoT", Moderator (LGE), RAN4#118, Gothenburg, Sweden, February 2026.

[394] R4-2523061, "Way Forward for [117][321] Study of A-IoT for outdoor", Moderator (LGE), RAN4#117, Dallas, USA, November 2025.

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

For the purposes of the study, this TR uses the following terms:

Device 1: ~1 μW peak power consumption, has energy storage, initial sampling frequency offset (SFO) up to 10^X ppm, neither R2D nor D2R amplification in the device. The device's D2R transmission is backscattered on a carrier wave provided externally.

Device 2a: ≤ a few hundred μW peak power consumption, has energy storage, initial sampling frequency offset (SFO) up to 10^X ppm, both R2D and/or D2R amplification in the device. The device's D2R transmission is backscattered on a carrier wave provided externally.

Device 2b: ≤ a few hundred μW peak power consumption, has energy storage, initial sampling frequency offset (SFO) up to 10^X ppm, both R2D and/or D2R amplification in the device. The device's D2R transmission is generated internally by the device.

Device C: ≤ 1 mW to ≤ 10 mW peak power consumption, has energy storage, IF envelope detector receiver or ZIF receiver, initial sampling frequency offset (SFO) up to 10^Y ppm, R2D and/or D2R amplification in the device. The device's D2R transmission is generated internally by the device.

D1T1: Deployment scenario 1 with connectivity topology 1, according to TR 38.848 [2].

D2T2: Deployment scenario 2 with connectivity topology 2, according to TR 38.848 [2].

D4T1: Deployment scenario 1 with connectivity topology 1, according to TR 38.848 [2].

Inventory: The service provided by the network to discover and acquire the identifier of A-IoT device(s).

Command: The service provided by the network to send the operation instruction to the A-IoT device (e.g. read, write, etc.).

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

1SB	Single sideband
2SB	Double sideband
A-IoT	Ambient IoT
A-IoT RAN	Ambient IoT Radio Access Network
AOA	Angle of arrival
BFSK	Binary frequency-shift keying
BPSK	Binary phase-shift keying
CAP	Clock-acquisition part (of R-TAS)
CFO	Carrier-frequency offset
CP	Cyclic prefix
CW	Carrier-wave
CW2D	Carrier-wave, or carrier-wave node, to device
D2R	Device to reader
DFT-s-OFDM	DFT-spread OFDM
DO-A	Device-originated autonomous
DO-DTT	Device-originated by device-terminated trigger
DT	Device-terminated
ED	Envelope detector
FAR	False alarm rate
FDR	False detection rate
FEC	Forward error-correction code
FR	Frequency Range
IF	Intermediate frequency
IoT	Internet of Things
LLS	Link-level simulation
LO	Local oscillator
LPWA	Low-power, wide-area
LTE-MTC	Long Term Evolution – Machine Type Communication
MCS	Modulation and coding scheme

MDR	Missed detection rate
MSK	Minimum-shift keying
NB-IoT	Narrowband IoT
OOK	On-off keying
PDRCH	Physical device-to-reader channel
PIE	Pulse interval encoding
PRDCH	Physical reader-to-device channel
R2D	Reader to device
RF	Radio frequency
RFID	Radio frequency identification
RMa	Rural macro (channel model)
R-TAS	R2D timing acquisition signal
SER	Sample error rate
SFO	Sampling-frequency offset
SFS	Small frequency shift
SIP	Start indicator part (of R-TAS)
UMa	Urban macro (channel model)
UMi	Urban micro (channel model)
ZIF	Zero IF

4 Evaluation methodology

4.1 Remaining details of RAN design targets

TR 38.848 [2] sets a number of RAN design targets. In [3], in particular three aspects of design targets beyond those in TR 38.848 are to be studied:

- Applicable maximum distance target value(s): The maximum distance targets are set separately for device 1, device 2a, device 2b, respectively. The distance target is compared to the evaluation results which assume the agreed mandatory evaluation assumptions.

  For device 1 in D1T1: 15 m

  For device 2a in D1T1: 25 m

  For device 2b in D1T1: 50 m

  For device 1 in D2T2: 10 m

  For device 2a in D2T2: 15 m

  For device 2b in D2T2: 40 m

- Refined definition of latency:
  - For the use case of "inventory-only":
    - The time interval between the time that the A-IoT paging is sent from reader to a A-IoT device and the time that the inventory report is successfully received at reader from the A-IoT device, i.e., for completing Step A and Step B (as per RAN2 agreements).
  - For the use case of "inventory and command":
    - The time interval between the time that the A-IoT paging is sent from reader and the time that
      - The command is successfully received at A-IoT device, i.e., for completing Step A, Step B, Step C1 (as per RAN2 agreements), if Step C2 is optional and not used (note: pending RAN2 decision on optionality of Step C2).
      - The response is successfully received at Reader, i.e., for completing Step A, Step B, Step C1, and Step C2 (as per RAN2 agreements).
  - See Clause 6.3.1 for descriptions of the Steps of these procedures.

Expected value of latency is calculated according a X% re-attempt probability to each attempt.

Note: The "successfully received" (i.e., decoding successfully)

Note: The latency is evaluated for a single A-IoT device.

Note: Time for energy harvesting is not included in the definition of latency.

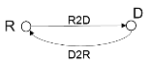
- 2D distribution of devices: A-IoT devices are dropped uniformly distributed over the horizontal area. See Table 4.2.2-2.

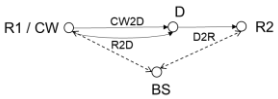
4.2 Evaluation scenarios and assumptions

4.2.1 Evaluation scenarios

The following scenarios are defined for the purpose of potential evaluation.

Table 4.2.1-1: Evaluation scenarios

Scenario	CW Inside/outside topology	Diagram of the scenario	Description of the scenario	Device 1/2a/2b/C	CW spectrum	D2R spectrum	R2D spectrum
D1T1-A1	CW inside topology		CW node inside topology 1 'CW' in CW2D and 'R2' in D2R are different 'CW' in CW2D and 'R1' in R2D are same 'R1' in R2D and 'R2' in D2R are different	Device 1, 2a	Case 1-1 (inside topology, DL) Case 1-2 (inside topology, UL)	Same as CW	At least DL
D1T1-A2			CW node inside topology 1 Same 'CW' and 'R' node for CW2D, D2R and R2D		Same as D1T1-A1	Same as CW	At least DL
D1T1-B	CW outside topology		CW node outside topology 1 'CW' in CW2D and 'R' in D2R are different 'CW' in CW2D and 'R' in R2D are different 'R' in R2D and 'R' in D2R are same		Case 1-4 (outside topology, UL)	Same as CW	At least DL
D1T1-C D4T1-C	No CW		No CW Node.	Device 2b Device C	N/A	UL	At least DL for D1T1.

							DL for D4T1.
D2T2-A1	CW inside topology		CW node inside topology 2 'CW' in CW2D and 'R2' in D2R are different 'CW' in CW2D and 'R1' in R2D are same 'R1' in R2D and 'R2' in D2R are different BS communicates with R1 and R2	Device 1, 2a	Case 2-2 (inside topology, UL)	Same as CW	UL
D2T2-A2			CW node inside topology 2 Same 'CW' and 'R' node for CW2D, D2R and R2D BS communicates with R		Same as D2T2-A1	Same as CW	UL
D2T2-B	CW outside topology		CW node outside topology 2 'CW' in CW2D and 'R' in D2R are different 'CW' in CW2D and 'R' in R2D are different 'R' in R2D and 'R' in D2R are same BS communicates with R		Case 2-3 (outside topology, DL) Case 2-4 (outside topology, UL)	Same as CW	UL
D2T2-C	No CW		No CW Node. BS communicates with R	Device 2b	N/A	At least UL	UL
Note: This table is for the case where D2R is in the same spectrum as CW2D.							

4.2.2 Evaluation assumptions

The following tables of coverage evaluation assumptions for link-level simulation are considered for indoor scenarios and outdoor scenarios, respectively. (M) indicates a value mandatory for evaluation, (O) indicates optional for evaluation. If there are any differences between devices, they are for evaluation purposes only.

Table 4.2.2-1: Coverage evaluation assumptions for link-level simulation for indoor scenarios

No.	Parameters		Assumptions
R2D/D2R common parameters			
[0a]	Carrier frequency		Refer to link budget template
[0b]	SCS		15 kHz as baseline
[0c]	Block structure		Blocks as agreed in 9.4.2.3, or other blocks reported by companies
[0d]	Channel model		**R2D**: For D2T2: BS pathloss model is reused for intermediate UE with antenna height = 1.5 m [0D]-Alt1: InF-DL NLOS, with TDL-A [0D]-Alt2: InH-Office LOS, with TDL-D For D1T1: InF-DH NLOS, with TDL-A **D2R:** For D2T2: BS pathloss model is reused for intermediate UE with antenna height = 1.5 m [0D]-Alt1: InF-DL NLOS, with TDL-A [0D]-Alt2: InH-Office LOS, with TDL-D For D1T1: InF-DH NLOS, with TDL-A
[0e]	Delay spread		An RMS delay spread of 30 ns (M) and [150] ns (O) is considered for TDL-A channel model. An RMS delay spread of 30 ns is considered for TDL-D channel model.
[0f]	Device velocity		3 km/h
[0g]	Number of Tx/Rx chains for Ambient IoT device		1
[0h1]	BS	Number of antenna elements	2 or 4
[0h2]		Number of TXRUs	2 or 4
[0j1]	Intermediate UE	Number of antenna elements	1 or 2
[0j2]		Number of TXRUs	1 or 2
[0m]	Reference data rate		1 kbps (M) 5 - 7 kbps (M) 48 - 60 kbps (O) 0.1 kbps for message size of 20 bits or 96 bits (O) Other data rates can be reported by companies Note 1: Companies to report the exact data rate. Note 2: The exact data rate is close to the values listed above. Note 3: The exact data rate is calculated by dividing the ~~total~~ message size (excluding CRC) by the total transmission time including applicable overheads(e.g., CRC, pre/mid/post-ambles if present). Note 4: The exact data rate may be related to coding scheme, repetition and etc. Note 5: All data rates considered are for evaluation purpose only
[0n]	Message size		{20 bits, 96 bits, 400 bits} are considered for message size. Note 1: companies to report the M value and chip length used for each message

		size Note 2: CRC is not included for the message size
[0p]	BLER target	1%, 10%
[0q]	Sampling frequency	Companies to report the sampling frequency (e.g., 1.92Msps or other feasible values if any) Initial SFO (Sampling Frequency Offset) (Fe): (M) Randomly select a value from the range of [0.1 ~ 1] *10^4 ppm for device 2, (M) Randomly select a value from the range of [0.1 ~ 1] * 10^5 ppm for device 1, (O) Randomly select a value from the range of [0.1 ~ 1] *10^5 ppm for device 2, Note: For random selection, the value is randomly selected per simulation drop, according to a uniform distribution Note: Above values are only for sampling purpose. Note: Above assumptions are only for LLS evaluation purpose only for R2D and D2R. The timing drift ΔT over a time T is modelled as ΔT = ±Fe * T. Note: SFO after clock calibration can be applied to Fe. CFO for device 2b: 100 ppm (M) 200 ppm (O) 1000 ppm (O, only as initial CFO) Drift rate of 0.1 or 1 ppm/s. Companies to report which value they used. Note: Above assumptions are for LLS evaluation purpose only
[0r]	Device 1/2a/2b	Options are as follows, Device 1, RF-ED Device 2a, RF-ED Device 2b, RF-ED/IF-ED/ZIF
R2D specific parameters		
[1a]	Transmission bandwidth	180 kHz as baseline. Other larger values are not precluded.
[1b]	ED bandwidth	The ED bandwidth is the bandwidth for calculating the noise/interference (if any) power: For evaluations, the value(s) of ED bandwidth is 20 MHz for RF-ED, [180] kHz for IF/ZIF receiver. Note: this does not imply that a A-IoT device supports sampling clock rate as large as RF ED bandwidth.
[1c]	BB LPF	[X]-order Butterworth/RC filter with cutoff frequency at half of R2D transmission bandwidth. Companies to report X = {3, 5}.
[1d]	Waveform	OOK waveform generated by OFDM modulator
[1e]	Modulation	OOK Companies to report, e.g., OOK-1, OOK-4 with M chips per OFDM symbol
[1f]	Line code	Companies to report, e.g., Manchester, PIE
[1g]	FEC	No FEC as baseline
[1h]	ADC bit width	1-bit for device 1 4-bit for device 2
[1j]	Detection/decoding method for Line code	Companies to report
D2R specific parameters		
[2a1]	Transmission bandwidth	**[2a1]-Alt1 (M):** DSB X kHz is considered for D2R transmission bandwidth. The value is for two sidebands, i.e., the total transmission bandwidth for DSB is X kHz **[2a1]-Alt2:** SSB X kHz is considered for D2R transmission bandwidth. The value is for one sideband, i.e., the total transmission bandwidth for SSB is X kHz. Not applicable for at least device 1 X = {[15 (M)], [180 (O)]}, other values are not precluded and reported by companies
[2a2]	[OOK/BPSK/BFSK chip rate]	Companies to report

[2a3]	Receiver bandwidth	D2R receiver bandwidth is the bandwidth used at the reader side to filter out the D2R signals for calculating noise and interference (if any) power. Assume the receiver matches the transmitter's modulation, i.e., to receiver uses SSB when transmitter uses SSB, receiver uses DSB when transmitter uses DSB. Companies to report the value, and further down-selection of the values and DSB/SSB is not precluded.
[2b]	Waveform (CW)	Companies to report waveform, e.g., unmodulated single tone, multi-tone(multiple unmodulated single tone)
[2d]	Modulation	Companies to report modulation, e.g., OOK, BPSK, BFSK
[2e]	Line code	Companies to report, e.g., Manchester encoding, FM0 encoding, Miller encoding, no line coding
[2g]	FEC	Companies to report, e.g., CC, No FEC
[2h]	ADC bit width	Companies to report, e.g., 11-bit
[2j]	D2R receiver	Companies to report, e.g., coherent receiver / non-coherent receiver
	Other assumptions	
[3a]	Other assumptions	To be reported by company
[3b]	Note: Companies to report required SINR/SNR/CINR/CNR according to BLER target.	

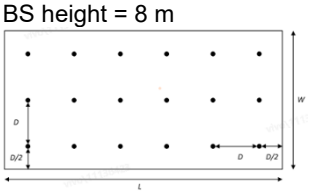
Table 4.2.2-1a: Coverage evaluation assumptions for link-level simulation for outdoor scenarios

No.	Parameters		Assumptions
	R2D/D2R common parameters		
[0a]	Carrier frequency		Refer to link budget template
[0b]	SCS		15 kHz as baseline
[0c]	Block structure		Blocks as agreed in Rel-19 with update reported by companies
[0d]	Channel model		TDL-C
[0e]	Delay spread		An RMS delay spread of 300 ns
[0f]	Device velocity		3 km/h Companies can also report 10 km/h
[0g]	Number of Tx/Rx chains for Ambient IoT device		1
[0h1]	BS	Number of antenna elements	2 or 4
[0h2]		Number of TXRUs	2 or 4
[0m]	Reference data rate		1 kbps (M) 5 - 7 kbps (M) 48 - 60 kbps (O) 0.1 kbps for message size of 20 bits or 96 bits (O) Other data rates can be reported by companies Note 1: Companies to report the exact data rate. Note 2: The exact data rate is close to the values listed above. Note 3: The exact data rate is calculated by dividing the total message size (excluding CRC) by the total transmission time including applicable overheads (e.g., CRC, pre/mid/post-ambles if present). Note 4: The exact data rate may be related to coding scheme, repetition and etc. Note 5: All data rates considered are for evaluation purpose only
[0n]	Message size		{20 bits, 96 bits, 400 bits} are considered for message size. Note 1: companies to report the M value and chip length used for each message size Note 2: CRC is not included for the message size
[0p]	BLER target		1%, 10%
[0q]	Sampling frequency		Companies to report the sampling frequency (e.g., 1.92Msps or other feasible values if any) Initial SFO (Sampling Frequency Offset) (Fe): For Device 2b: Randomly select a value from the range of 10^3~10^4 ppm For Device C: 50 ppm SFO after clock sync/calibration: For Device 2b: 10^3 (M) ppm, 10^2 (O) ppm For Device C: 10 ppm Note: For random selection, the value is randomly selected per simulation drop, according to a uniform distribution

		Note: Above values are only for sampling purpose. Note: Above assumptions are only for LLS evaluation purpose only for R2D and D2R. The timing drift ΔT over a time T is modelled as ΔT = ±Fe * T. Note: SFO after clock calibration can be applied to Fe. Initial CFO: For Device 2b: Randomly select a value from the range of 10^3~10^4 ppm For Device C: 50 ppm CFO after clock sync/calibration: For Device 2b: 100 (M) ppm, 200 (O) ppm, 50 (O) ppm For Device C: 10 ppm Drift rate of 0.1 or 1 ppm/s. Companies to report which value they used. Note: Above assumptions are for LLS evaluation purpose only
[0r]	Device 2b/C	ZIF/IF-ED
R2D specific parameters		
[1a]	Transmission bandwidth	180kHz Other values e.g. 360kHz, 540kHz or other multiples of 180kHz are not precluded and up to company report
[1b]	ED bandwidth	Up to company report with considering: R2D transmission bandwidth Device Rx LO uncertainty (e.g. CFO ppm assumption)
[1c]	BB LPF	[X]-order Butterworth/RC filter with cutoff frequency at half of ED bandwidth. Companies to report X = {3, 5}.
[1d]	Waveform	OOK waveform generated by OFDM modulator
[1e]	Modulation	OOK-4 Companies to report M chips per OFDM symbol
[1f]	Line code	Manchester
[1g]	FEC	Companies who report evaluations using FEC should also report evaluations without FEC
[1h]	ADC bit width	4-bit
[1j]	Detection/decoding method for Line code	Companies to report
[1k]	Repetition	Companies who report R2D evaluations using repetition should also report R2D evaluations without repetition
D2R specific parameters		
[2a1]	Transmission bandwidth	15kHz Other values are not precluded and up to company report Companies to report which of 2SB or 1SB modulation is assumed for D2R transmission bandwidth
[2a2]	Chip rate	Companies to report
[2a3]	Receiver bandwidth	D2R receiver bandwidth is the bandwidth used at the reader side to filter out the D2R signals for calculating noise and interference (if any) power. Assume the receiver matches the transmitter's modulation, i.e., to receiver uses 1SB when transmitter uses 1SB, receiver uses 2SB when transmitter uses 2SB. Companies to report the value, and further down-selection of the values and 2SB/1SB is not precluded.
[2d]	Modulation	Companies to report modulation, e.g., OOK, BPSK
[2e]	Line code	Companies to report, e.g., Manchester encoding, no line coding
[2g]	FEC	Companies to report, e.g., CC, No FEC
[2h]	ADC bit width	Companies to report, e.g., 11-bit
[2j]	D2R receiver	Companies to report, e.g., coherent receiver / non-coherent receiver
[2k]	Repetition	Companies to report based on update of Rel-19 if any
Other assumptions		
[3a]	Other assumptions	To be reported by company
[3b]	Note: Companies to report required SINR/SNR/CINR/CNR according to BLER target.	

The following layouts are used for evaluation purposes for indoor scenarios.

Table 4.2.2-2: Assumptions on layout for D1T1 and D2T2

Parameter	Assumptions for D1T1	Assumptions for D2T2	
Scenario	InF-DH	InH-office	InF-DL
Hall size	120x60 m	120 x50 m	300x150 m
Room height	10 m	3m	10 m
Sectorization	None		
BS deployment / Intermediate UE dropping	18 BSs on a square lattice with spacing D, located D/2 from the walls. L=120m x W=60m; D=20m BS height = 8 m 	L=120m x W=50m; Intermediate UE height = 1.5 m Option 1: Intermediate UEs are assumed to be deployed in a grid with D = e.g. 10 m / 20 m. Option 2: Intermediate UEs are assumed to be uniformly distributed over the horizontal area. Number of UE is either 1 or 2.	L=300m x W=150m; Intermediate UE height = 1.5 m Option 1: The intermediate UEs are assumed to be deployed in a grid with D = e.g. 10 m / 20 m. Option 2: Intermediate UEs are assumed to be uniformly distributed over the horizontal area. Number of UE is either 1 or 2.
Device distribution	Device Height= 1.5 m A-IoT devices drop uniformly distributed over the horizontal area	Device Height= 1.5 m A-IoT devices drop uniformly distributed over the horizontal area Option 1: The devices within the maximum distance, which is obtained by the coverage evaluation, from each intermediate UE are involved in the evaluations. Option 2: The devices within the applicable maximum distance target are involved in the evaluations.	Device Height= 1.5m A-IoT devices drop uniformly distributed over the horizontal area Option 1: The devices within the maximum distance, which is obtained by the coverage evaluation, from each intermediate UE are involved in the evaluations. Option 2: The devices within the applicable maximum distance target are involved in the evaluations.
Device mobility (horizontal plane only)	3 kph	3 kph	3 kph
NOTE: The CW distribution in D1T1-B or D2T2-B is reported by companies			

For evaluation purposes for outdoor scenarios, UMa and UMi both use a hexagonal grid with 19 site and 3 sectors per site.

4.3 Link budget

4.3.1 Receiver sensitivity

The study uses the following definitions for receiver sensitivity.

Budget-Alt1: Receiver sensitivity is derived by a predefined threshold and no link-level simulation is needed for link budget calculation

The results rely on the received sensitivity and maximum transmit power, and directly calculate the maximum distance / pathloss based on these values and other related parameters. The link-level simulation performances, such as required SINR, can be satisfied for such case and no link-level simulation is needed for link budget calculation.

Budget-Alt1 receiver sensitivity values are summarized in Annex A.

Budget-Alt2: Receiver sensitivity is derived by required SINR which is given by LLS results

The results rely on link-level simulation results, e.g., required SINR which corresponds to detail LLS assumptions (e.g., BW, coding, data rate). And based on the required SINR, the received sensitivity can be calculated and then the maximum distance / pathloss can be derived.

Note: For noise power, a noise figure value needs to be provided.

4.3.2 Link budget templates

Link budget is calculated according to the following Table 4.3.2-1 for indoor scenarios or Table 4.3.2-2 for outdoor scenarios. (M) denotes the value is mandatory to be evaluated. (O) denotes the value can be optionally evaluated.

Table 4.3.2-1: Link budget template for indoor scenarios

No.	Item	Reader-to-Device	Device-to-Reader
(0) System configuration			
[0A]	Scenarios	D1T1-A1/A2/B/C D2T2-A1/A2/B/C	D1T1-A1/A2/B/C D2T2-A1/A2/B/C
[0A1]	CW case	N/A	1-1/1-2/1-4/2-2/2-3/2-4
[0B]	Device 1/2a/2b	Device 1/2a/2b	Device 1/2a/2b
[0C]	Center frequency (MHz)	900MHz (M), 2GHz (O)	900MHz (M), 2GHz (O)
[0D]	Topology/Pathloss model	For D2T2: BS pathloss model is reused for intermediate UE with antenna height = 1.5 m [0D]-Alt1: InF-DL NLOS [0D]-Alt2: InH-Office LOS For D1T1: InF-DH NLOS	For D2T2: BS pathloss model is reused for intermediate UE with antenna height = 1.5 m [0D]-Alt1: InF-DL NLOS [0D]-Alt2: InH-Office LOS For D1T1: InF-DH NLOS
(1) Transmitter			
[1D]	Number of Tx antenna elements / TxRU/ Tx chains modelled in LLS	For BS: 2(M) or 4(O) antenna elements for 0.9 GHz For Intermediate UE: 1(M) or 2(O)	1
[1E]	Total Tx Power (dBm)	For BS in DL spectrum for indoor - [1E]-R2D-Alt1: 33dBm(M), - [1E]-R2D-Alt2: 38dBm(O), - [1E]-R2D-Alt3: 24dBm(M) - Companies to report if PSD constraints are imposed (company to report the condition for applying PSD constraints in Row [5A]) For UL spectrum for indoor, - [1E]-R2D-Alt4:23dBm (M) - [1E]-R2D-Alt5:26dBm(O)	For device 1/2a: (see note 1) - [1E]-D2R-Alt1: For scenarios ‘B’, the device Tx Power is calculated by CW received power which can be derived by at least CW2D distance (m) value and other related factors. - [1E]-D2R-Alt2: For scenarios ‘A1’ and ‘A2’, the device Tx Power is calculated by assuming CW2D pathloss = D2R pathloss. For device 2b: For scenarios ‘C’ - [1E]-D2R-Alt3: -20 dBm(M) - [1E]-D2R-Alt4: -10 dBm(O)
[1E1]	CW Tx power (dBm)	N/A	For scenario ‘A1’, ‘A2’ and ‘B’ - Report a value from the candidate values [1E]-R2D-Alt1 / [1E]-R2D-Alt2 / [1E]-R2D-Alt3 from [1E]-R2D if CW in DL spectrum

			- Report a value from the candidate values [1E]-R2D-Alt4 / [1E]-R2D-Alt5 from [1E]-R2D if CW in UL spectrum. Note: only applicable for device 1/2a
[1E2]	CW Tx antenna gain (dBi)	N/A	Company to report, the value equals: - UE Tx ant gain, or - BS Tx ant gain Note: Only applicable for device 1/2a
[1E3]	CW2D distance (m)	N/A	For scenarios 'B' D1T1-B: - 5m, - 10m, - 20m - CW2D distance is derived assuming CW node is located with the same position as 'R1' in 'A1' scenario. (See note 1) D2T2-B: - 5m, - 10m, For scenarios 'A1' and 'A2': Calculated (see note 1), (i.e., CW2D distance is calculated by assuming CW2D pathloss = D2R pathloss) Note 1: Only applicable for device 1/2a. Note 2: Companies to report which value(s) are evaluated.
[1E4]	CW2D pathloss (dB)	N/A	Calculated (see note1) Note 1: Only applicable for device 1/2a Note 2: For CW2D pathloss model, use the same pathloss model as used for R2D/D2R.
[1E5]	CW received power (dBm)	N/A	Calculated (see note1) Note: Only applicable for device 1/2a
[1F]	Transmission Bandwidth used for the evaluated channel (Hz)	180kHz(M), 360kHz(O), 1.08MHz(O)	Refer to LLS table [2a1]
[1G]	Tx antenna gain (dBi)	For BS for indoor, 6 dBi(M), 2dBi(M) For intermediate UE, 0 dBi	For A-IoT device, 0dBi
[1H]	Ambient IoT backscatter loss (dB) due to Modulation factor	N/A	OOK: 6 dB PSK: 0 dB FSK: Y dB It is applicable for device 1 and 2a. Companies to report and justify their assumptions for Y. Companies to report in row 3D if they assume any additional related loss.
[1J]	Ambient IoT on-object antenna penalty	N/A	0.9dB or 4.7dB
[1K]	Ambient IoT backscatter amplifier gain (dB)	N/A	10 dB (M) 15 dB (O) Note: Only for device 2a
[1N]	Cable, connector, combiner, body losses, etc. (dB)	For BS, X dB, X <=3 to be reported by companies with justification provided in row 5A	N/A

		For intermediate UE, 1 dB	
[1M]	EIRP (dBm)	Calculated (see Note 1)	Calculated (see Note 1)
(2) Receiver			
[2A]	Number of receive antenna elements / TxRU / chains modelled in LLS	Same as [1D]-D2R	Same as [1D]-R2D
[2B]	Bandwidth used for the evaluated channel (Hz)	Refer to LLS table [1b] ED bandwidth	Refer to LLS table [2a3]
[2C]	Receiver antenna gain (dBi)	same as [1G]-D2R	Same as [1G]-R2D
[2X]	Cable, connector, combiner, body losses, etc. (dB)	N/A	Same as [1N]-R2D
[2D]	Receiver Noise Figure (dB)	For RF-ED receiver 20dB, Device 2 For IF/ZIF receiver 15dB, Device 2 Other values can be optionally used and reported by companies	For BS as reader: 5dB For intermediate UE as reader: 7dB
[2E]	Thermal Noise power spectrum density (dBm/Hz)	-174	-174
[2F]	Noise Power (dBm)	Calculated (see Note 1)	Calculated (see Note 1)
[2G]	Required SNR/CNR	Reported by companies for Budget-Alt2	Reported by companies for Budget-Alt2
[2H]	Ambient IoT on-object antenna penalty	0.9 dB or 4.7 dB	Not applicable
[2J]	Budget-Alt1/ Budget-Alt2	Budget-Alt1/ Budget-Alt2 (see note1)	Budget-Alt2
[2K]	CW cancellation (dB)	N/A	Companies to report for scenario A2/A1/B for BS and intermediate UE. Notes: - Only applicable for device 1/2a - The value provided is for the unmodulated single-tone CW. The value for other cases (e.g. multi-tone CW) can be reported by companies.
[2K1]	Remaining CW interference (dBm)	N/A	Calculated (see Note 1) Note: only applicable for device 1/2a
[2K2]	Receiver sensitivity loss(dB)	N/A	Calculated (see Note 1) Note: only applicable for device 1/2a
[2L]	Receiver Sensitivity (dBm)	For Budget-Alt1 For device 1 (RF-ED), for example: {-30 dBm, -36 dBm, -40 dBm, etc} For device 2 (RF-ED), for example: {-40 dBm, -45 dBm, etc} For Budget-Alt2 Calculated (see note1)	Calculated (see Note 1) Note 2L: the receiver sensitivity includes the receiver sensitivity loss [2K2], i.e. after CW cancellation at least if 'A2' scenario is used
(3) System margins			
[3A]	Shadow fading margin (dB)	For D1T1: 4 dB For D2T2: 3dB for InH-LOS	For D1T1: 4 dB For D2T2: 3dB for InH-LOS

		7.2dB for InF-DL-NLOS	7.2dB for InF-DL-NLOS
[3B]	polarization mismatching loss (dB)	3 dB	3 dB
[3C]	BS selection/macro-diversity gain (dB)	0 dB Note: only applicable for D1T1	0 dB Note: only applicable for D1T1
[3D]	Other gains (dB) (if any please specify)	Reported by companies with justification	Reported by companies with justification
(4) MPL / distance			
[4A]	MPL (dB)	Calculated (see Note 1)	Calculated (see Note 1)
[4B]	Distance (m)	Calculated (see Note 1)	Calculated (see Note 1)
(5) Other			
[5A]	Other notes	Companies to report	Companies to report

Note 1: Calculated values are derived according to the following.

[1E3]

- For scenarios where CW2D distance is calculated by assuming CW2D pathloss = D2R pathloss, [1E3] is derived by assuming pathloss is [1E4] and use the pathloss formula as agreed.

[1E4]

- For scenarios where CW2D distance is calculated by assuming CW2D pathloss = D2R pathloss
  - [1E4] = 0.5* ([1E1] + [1E2] - [1N](R2D) + [2C] (R2D) - [2H](R2D) - 2*[3A] - 2*[3B] +[3D](R2D) + [1K] - [1H] + [1G] - [1J] + [2C] - [2X] - [2L] + [3C] + [3D])
  - [1K] is only for device 2a
- Otherwise
  - [1E4] is derived according to pathloss formula by assume distance is [1E3]

[1E5]

- [1E5] = [1E1] + [1E2] - [1N](R2D) - [1E4] + [2C] (R2D) - [2H](R2D) - [3A] - [3B] + [3D](R2D)

[1E]

- [1E] = [1E5]+ [1K] - [1H]
- [1K] is only for device 2a

[1M]:

- For R2D,
  - [1M] = [1E] + [1G] - [1N]
- For D2R
  - [1M] = [1E] + [1G] - [1J]

[2F]:

- [2F] = [2D] + [2E] +*lin2dB*([2B])

[2G]

- For the R2D LLS for ED, CINR/CNR is reported, where CINR/CNR is defined as the ratio of signal power spectral density in the transmission bandwidth to the noise and interference (if any) power spectral density in the device ED channel bandwidth.
- For R2D ZIF receiver, report the same metrics (i.e., CNR/CINR, signal transmission bandwidth, ED bandwidth) as agreed for RF-ED/IF receiver.
- For the D2R LLS, the SINR/SNR is reported and it is defined as the ratio of signal power to noise and interference (if any) power in the receiver bandwidth. Receiver bandwidth is the bandwidth used at the reader side to filter the D2R signals for calculating noise and interference (if any) power.
- On/off keying backscatter loss (including DC removal loss) is not taken into account in the LLS and is included in link budget table [1H].

[2J]

- For R2D link in the coverage evaluation, for device 1
  - Budget-Alt1 is used (note: receiver architecture is RF ED)
- For R2D link in the coverage evaluation for device 2,
  - Budget-Alt1 is used if receiver architecture is RF ED
  - Budget-Alt2 is used if receiver architecture is IF/ZIF ED

Note A: this does not preclude to have LLS for device 1 and 2 R2D link with RF-ED if needed.

Note B: For device 2 R2D link with RF-ED, *Budget-Alt1* is mandatory, *Budget-Alt2* is optional.

Note C: this does not imply all M values are achievable with the sensitivity given by *Budget-Alt1* for RF ED

Note D: For device 2 with an RF ED-based receiver on the R2D link, if the receiver sensitivity derived from *Budget-Alt2*, assuming a noise figure in [2D] from the table, exceeds the receiver sensitivity based on *Budget-Alt1*, then *Budget-Alt2* is applied.

[2K1]:

- [2K1] = [1E1] + [1E2] -[1N](R2D) + [2C] - [2X] - [2K]

[2K2]:

- $[2K2] = \text{lin2dB}\left(1 + \frac{\text{dB2lin}([2K1])}{\text{dB2lin}([2F])}\right)$

[2L]:

- For R2D and *Budget-Alt2*,
  - [2L] = [2G] - lin2dB([2B] / [1F]) + [2F]

Note E: The term ‘lin2dB([2B] / [1F])’ is applied due to scaling from CNR/CINR to SNR/SINR.

- For D2R,
  - [2L] = [2G] + [2F] + [2K2], device 1/2a
  - [2L] = [2G] + [2F], device 2b

[4A]

- For R2D
  - [4A] = [1M] + [2C] - [2H] - [2L] - [3A] - [3B] + [3C] + [3D]
- For D2R
  - [4A] = [1M] + [2C] - [2X] - [2L] - [3A] - [3B] + [3C] + [3D]

[4B]

- [4B] is derived by assuming pathloss is [4A] and using the pathloss formula as agreed.

Table 4.3.2-2: Link budget template for outdoor scenarios

No.	Item	Reader-to-Device	Device-to-Reader
(0) System configuration			
[0A]	Scenarios	D4T1-C	D4T1-C
[0B]	Device 2b/C	Device 2b/C	Device 2b/C
[0C]	Center frequency (MHz)	900MHz (M), 2GHz (O)	900MHz (M), 2GHz (O)
[0D]	Pathloss model	UMa NLOS defined in TR38.901 BS antenna height: 25 m Device antenna height: 1.5 m RMa NLOS defined in TR38.901 BS antenna height: 35 m Device antenna height: 1.5 m Avg. street width: 20 m Avg. building height: 5 m Note: companies can also report to use RMa LOS defined in TR38.901 UMi NLOS defined in TR38.901 BS antenna height: 10 m Device antenna height: 1.5 m	UMa NLOS defined in TR38.901 BS antenna height: 25 m Device antenna height: 1.5 m RMa NLOS defined in TR38.901 BS antenna height: 35 m Device antenna height: 1.5 m Avg. street width: 20 m Avg. building height: 5 m Note: companies can also report to use RMa LOS defined in TR38.901 UMi NLOS defined in TR38.901 BS antenna height: 10 m Device antenna height: 1.5 m
(1) Transmitter			
[1D]	Number of Tx antenna elements / TxRU/ Tx chains modelled in LLS	Refer to LLS table [0h1] and [0h2]	1
[1E]	Total Tx Power (dBm)	For standalone: 43 dBm For in-band: 38 or 33 dBm over R2D transmission BW Note: companies should report any PSD limitation they assume in their RAN1 evaluations	Device 2b: {-10, -20} dBm Device C: {-3, 0, 5} dBm
[1F]	Transmission Bandwidth used for the evaluated channel (Hz)	Refer to LLS table [1a]	Refer to LLS table [2a1]
[1G]	Tx antenna gain (dBi)	17dBi	For A-IoT device, 0dBi
[1J]	Ambient IoT on-object antenna penalty	N/A	0dB or 0.9dB or 4.7dB
[1N]	Cable, connector, combiner, body losses, etc. (dB)	3dB	N/A
[1M]	EIRP (dBm)	Calculated (see Note 1)	Calculated (see Note 1)
(2) Receiver			
[2A]	Number of receive antenna elements / TxRU / chains modelled in LLS	Same as [1D]-D2R	Same as [1D]-R2D
[2B]	Bandwidth used for the evaluated channel (Hz)	Refer to LLS table [1b] ED bandwidth	Refer to LLS table [2a3]
[2C]	Receiver antenna gain (dBi)	same as [1G]-D2R	Same as [1G]-R2D

[2X]	Cable, connector, combiner, body losses, etc. (dB)	N/A	Same as [1N]-R2D
[2D]	Receiver Noise Figure (dB)	Device 2b: 15 dB Device C: 13 dB	For BS as reader: 5dB
[2E]	Thermal Noise power spectrum density (dBm/Hz)	-174	-174
[2E1]	Receiver interference density (dBm/Hz)	-169.3 dBm/Hz Companies can also report other higher value(s) for R2D or -999 dBm/Hz along with their assumption	-165.7 dBm/Hz Companies can also report -999 dBm/Hz along with their assumption
[2F]	Effective Noise Power (dBm)	Calculated (see Note 1)	Calculated (see Note 1)
[2G]	Required SNR/CNR	Reported by companies for Budget-Alt2	Reported by companies for Budget-Alt2
[2H]	Ambient IoT on-object antenna penalty	0dB or 0.9 dB or 4.7 dB	Not applicable
[2J]	Budget-Alt1/ Budget-Alt2	Budget-Alt2	Budget-Alt2
[2L]	Receiver Sensitivity (dBm)	Calculated (see Note 1)	Calculated (see Note 1)
(3) System margins			
[3A]	Shadow fading margin (dB)	For UMa NLOS: 6dB For RMa NLOS: 8dB For UMi NLOS: 7.82dB	For UMa NLOS: 6dB For RMa NLOS: 8dB For UMi NLOS: 7.82dB
[3B]	polarization mismatching loss (dB)	3 dB	3 dB
[3C]	BS selection/macro-diversity gain (dB)	0 dB	0 dB
[3D]	Other gains (dB) (if any please specify)	Reported by companies with justification	Reported by companies with justification
[3E]	Penetration margin (dB)	{0, 20, 80} dB	{0, 20, 80} dB
(4) MPL / distance			
[4A]	MPL (dB)	Calculated (see Note 1)	Calculated (see Note 1)
[4B]	Distance (m)	Calculated (see Note 1)	Calculated (see Note 1)
(5) Other			
[5A]	Other notes	Companies to report	Companies to report

Note 1: Calculated values are derived according to the following.

[1M]:

- For R2D,
  - [1M] = [1E] + [1G] - [1N]
- For D2R
  - [1M] = [1E] + [1G] - [1J]

[2F]:

- [2F] = lin2dB(dB2lin([2D] + [2E]) + dB2lin([2E1])) +lin2dB([2B])

[2G]

- For the R2D LLS for ED, CINR/CNR is reported, where CINR/CNR is defined as the ratio of signal power spectral density in the transmission bandwidth to the noise and interference (if any) power spectral density in the device ED channel bandwidth.
- For R2D ZIF receiver, report the same metrics (i.e., CNR/CINR, signal transmission bandwidth, ED bandwidth) as agreed for RF-ED/IF receiver.
- For the D2R LLS, the SINR/SNR is reported and it is defined as the ratio of signal power to noise and interference (if any) power in the receiver bandwidth. Receiver bandwidth is the bandwidth used at the reader side to filter the D2R signals for calculating noise and interference (if any) power.

[2L]:

- For R2D,
  - [2L] = [2G] - lin2dB([2B] / [1F]) + [2F]

Note A: The term 'lin2dB([2B] / [1F])' is applied due to scaling from CNR/CINR to SNR/SINR.

- For D2R,
  - [2L] = [2G] + [2F]

[4A]

- For R2D
  - [4A] = [1M] + [2C] - [2H] - [2L] - [3A] - [3B] + [3C] + [3D] - [3E]
- For D2R
  - [4A] = [1M] + [2C] - [2X] - [2L] - [3A] - [3B] + [3C] + [3D] - [3E]

[4B]

- [4B] is derived by assuming pathloss is [4A] and using the pathloss formula as agreed.

4.4 R2D waveform generation

With reference to the R2D waveform described in Clause 6.1.1.1, for evaluation purposes the waveform for DFT-s-OFDM is generated as follows:

1. The time domain OOK signal is the *M* chips of one OFDM symbol.
2. A chip is represented (e.g. upsampled) by *L* samples
   - Companies to report *L*
3. An *N*'-points DFT is performed on the samples of one OFDM symbol to obtain the frequency domain signal.
   - Companies to report *N*', e.g. *N*'=128 or equal to *X*
4. Map the frequency domain signal obtained by N'-points DFT to the *X* subcarriers of $B_{tx,R2D}$.
   - Companies report how to map and report *X*
5. An *N*-points IDFT is performed to obtain the time domain signal.
   - Companies to report *N*, and how value was selected

Note: Companies report whether/how CP samples are added.

4.5 Energy harvesting/storage assumptions for outdoor scenarios

The following energy harvesting/storage assumptions on the energy source and energy storage capacity used to obtain the reported data rates were provided during the study.

Table 4.5-1: Energy harvesting/storage assumptions

Source	Energy harvesting/storage assumptions
Source [230]	Energy source: Solar power/light Energy storage capacity: ~300 mF capacitor for low data rates ~60 mF capacitor for middle data rates ~40 mF capacitor for high data rates High voltage 2.7 V, low voltage 1.1 V Reported data rate: Low data rate: ~ 0.1 kbps for D2R and ~ 1 kbps for R2D Middle data rate: ~ 1 kbps for D2R and ~ 1 kbps for R2D High data rate: ~ 5 kbps for D2R and ~ 5 kbps for R2D Other information: For D2R transmission, the consumed energy per hour is For 0.1 kbps data rate: ~50 mJ For 1 kbps data rate: ~6.25 mJ For 5-7 kbps data rate: ~1.6 mJ For R2D transmission, the consumed energy per hour is For 1 kbps data rate: 4 mJ For 5-7 kbps data rate: 0.8 mJ

Source [235]	Energy source: Energy sources with high energy conversion efficiency, e.g., solar, vibration, thermal, etc., can be considered. Energy storage capacity: >=1mJ for 1mW peak power consumption and >=10mJ for 10mW peak power consumption. Reported data rate: For D2R, data rate is 1kbps. Other information: When considering energy storage capacity, device at least can finish one D2R transmission. For 1000bits and 1kbps data rate, the D2R transmission time is 1000ms. For device C, the peak power consumption is ≤ 1 mW to ≤ 10 mW as agreed. Thus, for 10mW peak power consumption the energy is at least larger than 10mW*1000ms, i.e., 10mJ. Assuming the rated maximum voltage is 3V and the minimum available voltage is 2V, it can be derived that the capacitance of capacitor should be larger than 4mF. And for 1mW peak power consumption. The energy is at least larger than 1mW*1000ms, i.e., 1mJ corresponding to 400uF capacitance of capacitor.
Source [232]	Energy source: Solar or other energy source are used. RF energy harvesting is deprioritized Energy storage capacity: > 15 mJ for device 2b > 75 mJ for device C Reported data rate: ~ 4 kbps for R2D transmission ~ 0.1 kbps for D2R transmission Other information: The energy storage of > 15 mJ (device 2c) and > 75 mJ (device C) is considered to support an entire CBRA process for multiple devices. For a single PDRCH transmission, assuming a D2R data rate of 0.1 kbps and a TBS of 400 bits, the transmission takes approximately 4 seconds. This requires Device 2b (with a Tx power consumption of 800 uW) to have an energy storage capacity greater than 3.2 mJ and Device C (with a Tx power consumption of 2 mW) greater than 8 mJ to support a single D2R transmission.
Source [231]	Energy source: Solar/light energy harvesting Energy storage capacity: 180-300 mF for Device 2b with voltage = 3.3 V. 180-1700 mF for Device C with voltage = 5V Reported data rate: R2D data rate is 12-15 kbps with M=2 D2R data rate is 5-7 kbps Other information: Energy conversion efficiency of energy harvester is considered as 80-90%. Device 2b power consumption is considered as 480 μW to 770 μW. Device 2b power consumption is considered as 1 mW to 10 mW. Solar panel size for device 2b is 20x20 mm^2 and Device C is 30x25 mm^2.

Source [239]	Energy source: Solar energy Energy storage capacity: For device 2b, 10~100μF with voltage of 3V, i.e., 30~300uJ; For device C, 0.5~5mF with voltage of 3V, i.e., 1.5~15mJ; Reported data rate: 1 kbps or 5 – 7 kbps Other information: Device 2b and Device C harvest energy at similar rate.
Source [233]	For device 2b Energy source: RF signal/Solar Energy storage capacity: 10s ~ 100s uF < 1mF Reported data rate: 1.7/2.5/5kbps Other information: Device 2b, message size= 96 bits, tx duration = 2.28ms, device power consumption = 500uW, voltage = 1V, assumed only 50% of energy can be used. For device C Energy source: Solar Energy storage capacity: 1m ~ 10mF Reported data rate: 1.7/2.5/5kbps Other information: Device C, message size= 96 bits, tx duration = 2.28ms, device power consumption = 5mW, voltage = 1V, assumed only 50% of energy can be used.
Source [234]	Energy source: Solar, thermal, etc. Energy storage capacity: For device 2b with 500uW peak power consumption: 0.1kbps: ~5mJ/~20mF 1kbps: ~500uJ/~2mF 7kbps: ~71.4uJ~285uF For device C with 10mW peak power consumption: 0.1kbps: ~100mJ/~400mF 1kbps: ~10mJ/~40mF 7kbps: ~1.4mJ/~5.7mF Reported data rate: 0.1 kbps, 1kbps, 7kbps Other information: For the maximum message size, 1000 bits is assumed for single PRDCH/PDRCH transmission. For voltage, max voltage as 2V and min voltage as 1V are assumed. For power consumption of each device type, 500uW is assumed for device 2b and 10mW is assumed for device C. It should be noted that the reported capacity is for single PRDCH/PDRCH transmission. To sustain whole procedure of DT, DO-DTT, DO-A, larger capacity is required.

Source [236]	Energy sources: Solar, thermal, vibration etc. Energy storage capacity: For device 2b with 500uW peak power consumption: •Low rate: 0.1kbps: ~5000 μJ /~5000 μF •Middle rate: 1kbps: ~500 μJ /~500 μF •High rate: 5kbps: ~100 μJ /~100 μF For device C with 10mW peak power consumption: Low rate: 0.1kbps: ~100 mJ /~100 mF Middle rate: 1kbps: ~10mJ /~10 mF High rate: 5kbps: ~1 mJ /~1000 μF Reported data rate: 01.kbps, 1kbps, 5kbps Other information: It is observed that using RF energy harvesting is challenging as the energy source for Device 2b based on the following description. Typical leakage current for 500 μF capacitors is at least higher than 3 μA. With a supply voltage of 2 volts, the leakage current translates to ~ -22.2 dBm leakage power. Using RF harvesting is challenging in this case. For the maximum message size, 1000 bits is assumed for single PRDCH/PDRCH transmission. The rated voltage is 2V and min voltage is 1.4V (70% of the rated voltage). It should be noted that the reported capacity is for single PRDCH/PDRCH transmission. To sustain whole procedure of DT, DO-DTT, DO-A, larger capacity is required.
Source [238]	Energy source: Solar Energy storage capacity: ~100mF Reported data rate: 0.1kbps, 1kbps, 5~7kbps, 48~60kbps Other information: Assumptions of power consumption: 100uW and 1mW for R2D reception 200uW and 5mW for D2R transmission
Source [240]	Energy source: Solar Energy storage capacity: >10μJ for device 2b, 100μJ for device C Reported data rate: 5.3 kbps for DL 20 bits, 6.7 kbps for UL 20 bits, 7.2 kbps for DL 96 bits, 8.4 kbps for UL 96 bits; with an average of 40% charging time

Source [237]	Energy source: RF signal/ Solar Energy storage capacity: Device 2b: 10s μF to a few 100 μF Device C: a few mF Reported data rates: 1kbps, 7kbps, and 50 kbps Other information: For device 2b: with tx power of -10dBm we considered the power consumption of 400 μW, to transmit 400 bits with the data rate of 1 kbps, the needed storage capacity is around 140 μF. For device C: with tx power of 5dBm we considered the power consumption of 12.6 mW, to transmit 400 bits with the data rate of 1 kbps, the needed storage capacity is around 4.5 mF.

5 Ambient IoT device architectures

5.1 ~1 μW devices (Device 1)

The architecture of such a device is summarised in Figure 5.1-1, with the blocks described as follows.

- **Antenna** could be either shared or separate for RF energy harvester and receiver/transmitter.
- **Matching network** is to match impedance between antenna and other components (including RF energy harvester and receiver related blocks).
- **RF energy harvester** can include **rectifier** performing RF signal (AC) to DC conversion.
- **Energy storage** (e.g., capacitor) stores harvested energy from RF energy harvester.
- **Power management unit (PMU)** manages storing energy to energy storage from energy harvester and supplying power to active component blocks which needs power supply.
- **Digital BB logic** includes functional blocks like encoder, decoder, controller, etc.
- **Memory** can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing device ID, etc, and 2) registers for temporarily keeping any information required for its operation only while energy is available in energy storage.
- **Clock generator** provides required clock signal(s).
- **Reception related blocks**
  - **RF BPF** for improving selectivity.
    - Depending on implementation, it may not exist. RAN4 RF requirement (if any, e.g., ACS) and peak power consumption target also need to be considered.
  - **RF Envelope Detector** converts RF signal to baseband.
  - **BB LPF** can filter out harmonics and high frequency components to improve input signal quality to comparator.
    - Depending on implementation, it may not exist. Presence of BB LPF is assumed for the study.
  - **Comparator** determines high/low of input signal.
- **Transmission related blocks**

- **Backscatter modulator** switches impedance to modulate backscattered signal with transmitted signal from BB logic.

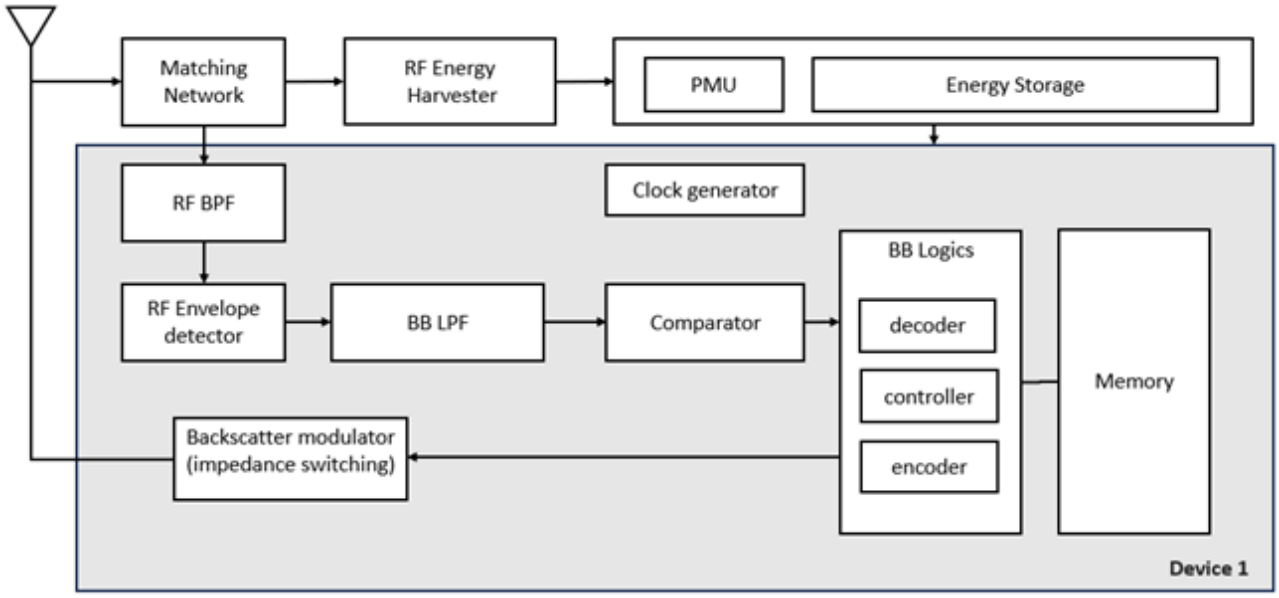


Figure 5.1-1: Architecture of device 1

5.2 ≤ a few hundred μW devices (Device 2)

5.2.1 External carrier wave (Device 2a)

The architecture of device 2a is summarised in Figure 5.2.1-1, with the blocks described as follows.

- **Antenna** could be either shared or separate for RF energy harvester (if present) and receiver/transmitter.
- **Matching network** is to match impedance between antenna and other components (including RF energy harvester (if present) and receiver related blocks).
- **Energy harvester**.
- **Energy storage** (e.g., capacitor) stores harvested energy from energy harvester.
- **Power management unit (PMU)** manages storing energy to energy storage from energy harvester and suppling power to active component blocks which needs power supply.
- **Digital BB logic** includes functional blocks like encoder, decoder, controller, etc.
- **Memory** can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing device ID, etc, and 2) registers for temporarily keeping any information required for its operation only while energy is available in energy storage.
- **Clock generator** provides required clock signal(s).
- **Reflection amplifier** can amplify reflected backscattered signal.
  - At least one of R2D/CW2D and D2R could be amplified by either reflection amplifier or LNA.
- **Reception related blocks**
  - **RF BPF** filter for improving selectivity.
    - Depending on implementation, it may not exist. RAN4 RF requirement (if any, e.g., ACS) and peak power consumption target also need to be considered.

- **LNA** for improving signal strength and sensitivity of receiver, if present.
  - At least one of R2D/CW2D and D2R could be amplified by either reflection amplifier or LNA.
- **RF envelope detector (RF-ED)** detects envelope from RF signal.
- **BB amplifier** amplifies BB signal to improve signal strength.
- **BB LPF** can filter out harmonics and high frequency components to improve input signal quality to comparator/ADC.
  - Depending on implementation, it may not exist.
- **Comparator** or **N-bit ADC**

- **Transmission related blocks**
  - **Backscatter modulator** switches impedance to modulate backscattered signal with tx signal from BB logics.
  - **Large Frequency shifter (**e.g., tens of MHz**)** for shifting backscattered signal from one frequency (e.g., FDD-DL frequency) to another frequency (e.g., FDD-UL frequency).

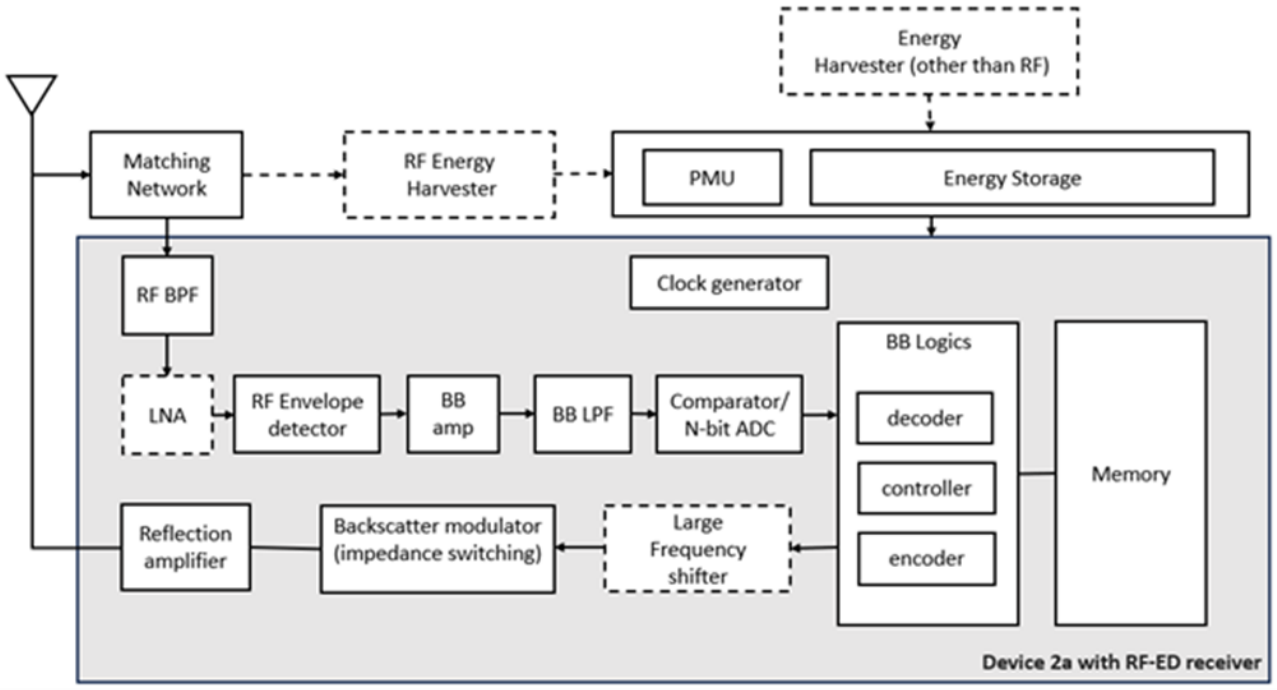


Figure 5.2.1-1: Architecture of device 2a

IF envelope detector receiver and ZIF receiver for device 2a are de-prioritized.

5.2.1.1 Reflection amplifier

For the reflection amplifier block, the following characteristics are considered for device 2a:

- Direction of amplification
  - Uni-directional reflection amplifier (baseline) can amplify backscattered signal in D2R which can improve D2R link budget.
  - Bi-directional amplifier can amplify both signal in R2D and backscatter signal in D2R at least when R2D and D2R are in the same spectrum.

  - Bi-directional amplifier has higher complexity, higher noise figure, and reduced isolation between tx and rx path.
- Amplification gain ranges from 10 to 20 dB.
- Power consumption of reflection amplifier is in the range of a tens of uW to 100s of uW.
- Reflection amplifier can operate in FDD frequency bands.
- Reflection amplifier bandwidth can support 10s of MHz.
- Note: reflection amplifier can get unstable when the input power exceeds a certain value, which may be frequency-dependent.

5.2.1.2 Large frequency shifter

For the large frequency shifter block, it is observed that:

- Large frequency shift can be used in shifting reflected signal in tens of MHz, e.g., from FDD DL to FDD UL frequency or vice versa.
- Large frequency shift consumes 10s of uW to 100s of uW.
- Large frequency shift is not feasible for device 1.
- Large frequency shift requires a clock for IF generation which is accurate enough to avoid large guard band and interference to adjacent channels/bands.
- Large frequency shift requires image suppression and may require harmonics suppression
- Note: details of image suppression and harmonics suppression are not discussed in RAN1
- Large frequency shift may allow the reader to avoid implementing in-band full duplex capability for scenarios e.g., D1T1-A2 and D2T2-A2.
- Large frequency shift may result in e.g., 5 kHz - 50 kHz of frequency uncertainty in target frequency for clock accuracy of e.g., 0.01% - 0.1% assuming the large frequency shift range is 50 MHz.

Large frequency shift may be technically feasible in terms of power consumption, and device complexity.

Remaining technical feasibility of large frequency shift in terms of image suppression, harmonics suppression may be relevant to device emission requirement, spectrum usage. Additional complexity and power consumption may be needed to achieve image suppression and harmonics suppression.

The necessity or effectiveness of large frequency shift may depend on spectrum for R2D/CW2D/D2R, full duplex capability of reader, whether device 1 and 2a are supported by a single reader, spectrum regulation/usage, frequency error, coverage, etc.

5.2.2 Internally-generated carrier wave (Device 2b)

5.2.2.1 RF envelope detector receiver

The architecture of device 2b with an RF envelope detector receiver is summarised in Figure 5.2.2.1-1, with the blocks described as follows.

- **Antenna** could be either shared or separate for RF energy harvester (if present) and receiver/transmitter.
- **Matching network** is to match impedance between antenna and other components (including RF energy harvester (if present) and receiver related blocks).
- **Energy harvester** for harvesting energy from e.g., RF signal, solar, vibration/movement, temperature difference, etc
- **Energy storage** (e.g., capacitor) stores harvested energy from energy harvester.

- **Power management unit (PMU)** manages storing energy to energy storage from energy harvester and suppling power to active component blocks which needs power supply.
- **Digital BB logic** includes functional blocks like encoder, decoder, controller, etc.
- **Memory** can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing device ID, etc, and 2) registers for temporarily keeping any information required for its operation only while energy is available in energy storage.
- **Clock generator** provides required clock signal(s).
- **Reception related blocks**
  - **RF BPF** filter for improving selectivity.
    - Depending on implementation, it may not exist. RAN4 RF requirement (if any, e.g., ACS) and peak power consumption target also need to be considered.
  - **LNA** for improving signal strength and sensitivity of receiver, if present
  - **RF envelope detector (RF-ED)** detects envelope from RF signal.
  - **BB amplifier** amplifies BB signal to improve signal strength.
  - **BB LPF** can filter out harmonics and high frequency components to improve input signal quality to comparator/ADC.
    - Depending on implementation, it may not exist.
  - Comparator or N-bit ADC
- **Transmission related blocks**
  - **Tx Modulator**: baseband bits are modulated according to modulation scheme. This block could be the part of BB logic.
  - **Digital to Analog Converter (DAC)** converts digital signal to analog signal.
  - **Low pass filter** for filtering out undesired signal
  - **Mixer** performs up converting baseband signal to RF range.
  - **Local oscillator (LO)** for carrier frequency generation
    - FLL(/PLL) can be used for frequency synthesis
      - Depending on implementation, FLL(/PLL) may not exist.
  - **Power amplifier (PA)** amplifies tx signal, if present
  - Details on transmitter related blocks depends on tx waveform/modulation.

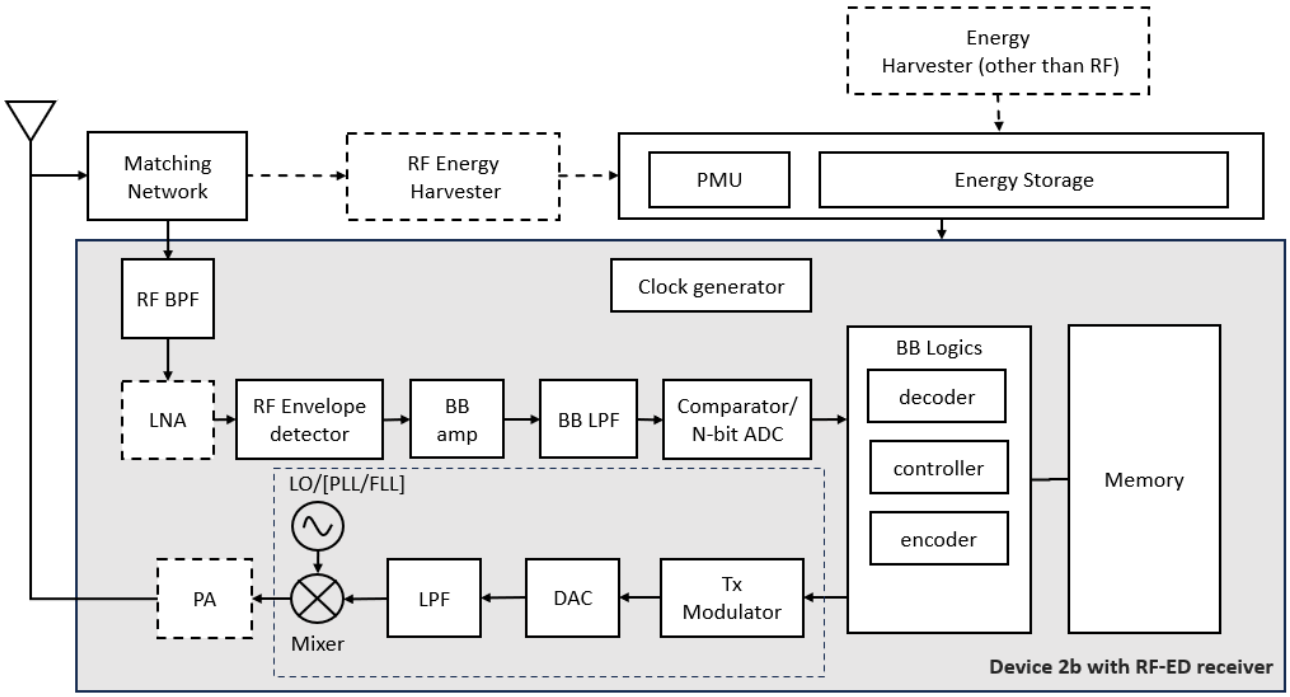


Figure 5.2.2.1-1: Architecture of device 2b with RF-ED receiver

5.2.2.2 IF envelope detector receiver

The architecture of device 2b with an IF envelope detector receiver is summarised in Figure 5.2.2.2-1, with the blocks described as follows.

- **Antenna** could be either shared or separate for RF energy harvester (if present) and receiver/transmitter
- **Matching network** is to match impedance between antenna and other components (including RF energy harvester (if present) and receiver related blocks)
- **Energy harvester** for harvesting energy from e.g., RF signal, solar, vibration/movement, temperature difference, etc.
- **Energy storage** (e.g., capacitor) stores harvested energy from energy harvester
- **Power management unit (PMU)** manages storing energy to energy storage from energy harvester and suppling power to active component blocks which needs power supply
- **Digital BB logic** includes functional blocks like encoder, decoder, controller, etc.
- **Memory** can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing device ID, etc, and 2) registers for temporarily keeping any information required for its operation only - while energy is available in energy storage

Clock generator provides required clock signal(s).

- **Local oscillator (LO)** for generating carrier frequency for Tx, or for generating carrier frequency offset by the IF for Rx
  - FLL(/PLL) can be used for frequency synthesis
  - One LO or separate LOs for Tx and Rx
- **Reception related blocks**
  - **RF BPF** filter for improving selectivity
    - Depending on implementation, it may not exist. RAN4 RF requirement (if any, e.g., ACS) and peak power consumption target also need to be considered

- **LNA** for improving signal strength and sensitivity of receiver, if present
- **Mixer** down converts RF signal to IF stage
  - Depending on implementation, there could be one or two mixers for Rx and Tx
- **IF amplifier** amplifies IF signal
- **IF filter** for filtering out unwanted RF and LO signals
- **IF envelope detector (IF-ED)** detects envelope from IF signal.
- **BB amplifier**
  - Depending on implementation, one or both of IF amplifier and BB amplifier may exist
- **BB LPF** can filter out harmonics and high frequency components to improve input signal quality to comparator/ADC
  - Depending on implementation, it may not exist
- Comparator or N-bit ADC
- Note: image rejection is required

- **Transmission related blocks**
  - **Tx Modulator**: baseband bits are modulated according to modulation scheme. This block could be the part of BB logic
  - **Digital to Analog Converter (DAC)** converts digital signal to analog signal
  - **Low pass filter** for filtering out undesired signal
  - **Mixer** performs up converting baseband signal to RF range
  - **Power amplifier (PA)** amplifies transmitted signal, if present
  - Details of transmitter related blocks depends on e.g., waveform/modulation, etc.

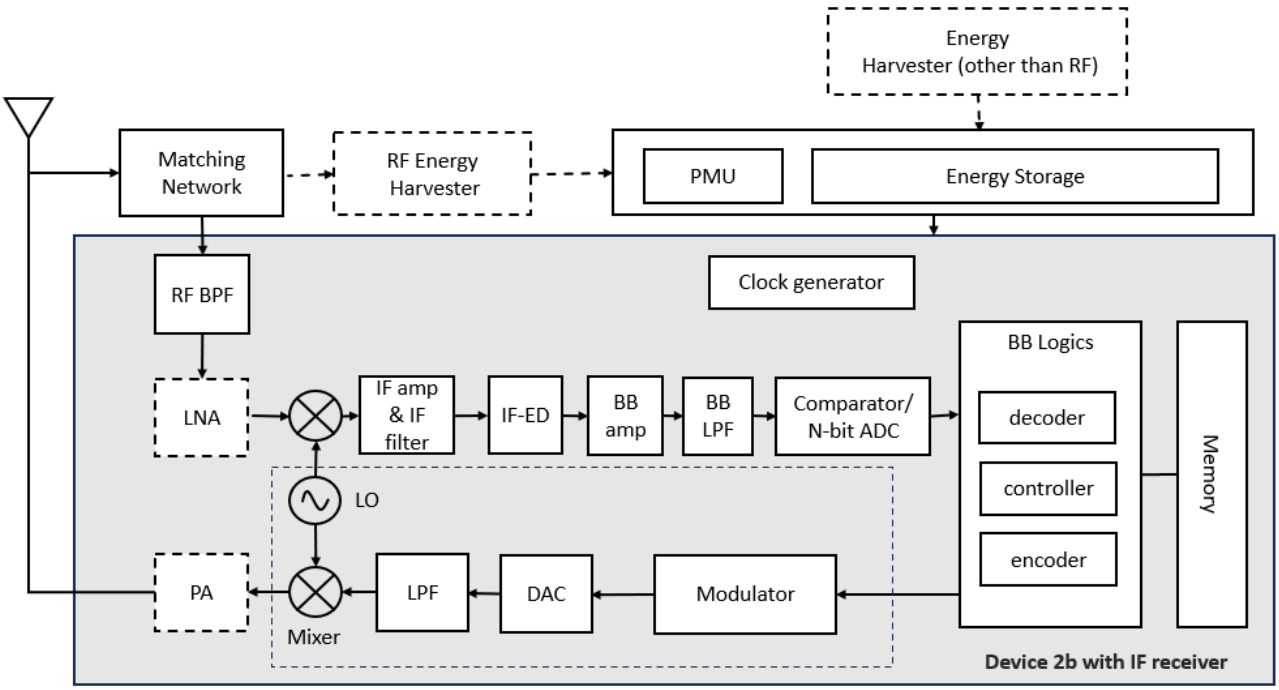


Figure 5.2.2.1-1: Architecture of device 2b with IF-ED receiver

5.2.2.3 ZIF receiver

The architecture of device 2b with a ZIF receiver is summarised in Figure 5.2.2.3-1, with the blocks described as follows.

- **Antenna** could be either shared or separate for RF energy harvester (if present) and receiver/transmitter
- **Matching network** is to match impedance between antenna and other components (including RF energy harvester (if present) and receiver related blocks)
- **Energy harvester** for harvesting energy from e.g., RF signal, solar, vibration/movement, temperature difference, etc
- **Energy storage** (e.g., capacitor) stores harvested energy from energy harvester.
- **Power management unit (PMU)** manages storing energy to energy storage from energy harvester and suppling power to active component blocks which needs power supply
- **Digital BB logic** includes functional blocks like encoder, detector, decoder, controller, etc.
- **Memory** can include two types of memory: 1) Non-Volatile Memory (NVM) such as EEPROM for permanently storing device ID, etc, and 2) registers for temporarily keeping any information required for its operation only while energy is available in energy storage
- **Clock generator** provides required clock signal(s).
- **Local oscillator (LO)** for generating carrier frequency for Tx and Rx
  - FLL(/PLL) can be used for frequency synthesis
  - One LO or separate LOs for Tx and Rx
- **Reception related blocks**
  - **RF BPF** filter for improving selectivity
    - Depending on implementation, it may not exist. RAN4 RF requirement (if any, e.g., ACS) and peak power consumption target also need to be considered

- **LNA** for improving signal strength and sensitivity of receiver, if present
- **Mixer** down converts RF signal to BB stage
  - Depending on implementation, there could be one or two mixers for Rx and Tx
- **BB amplifier** amplifies BB signal
- **BB LPF** can filter out undesired frequency components to improve input signal quality to comparator/ADC.
  - Depending on implementation, it may not exist
- Comparator or N-bit ADC

- **Transmission related blocks**
  - **Tx Modulator**: baseband bits are modulated according to modulation scheme. This block could be the part of BB logic.
  - **Digital to Analog Converter (DAC)** converts digital signal to analog signal.
  - **Low pass filter** for filtering out undesired signal
  - **Mixer** performs up converting baseband signal to RF range.
  - **Power amplifier (PA)** amplifies transmitted signal, if present
  - Details of transmitter related blocks depend on e.g., waveform/modulation, etc.

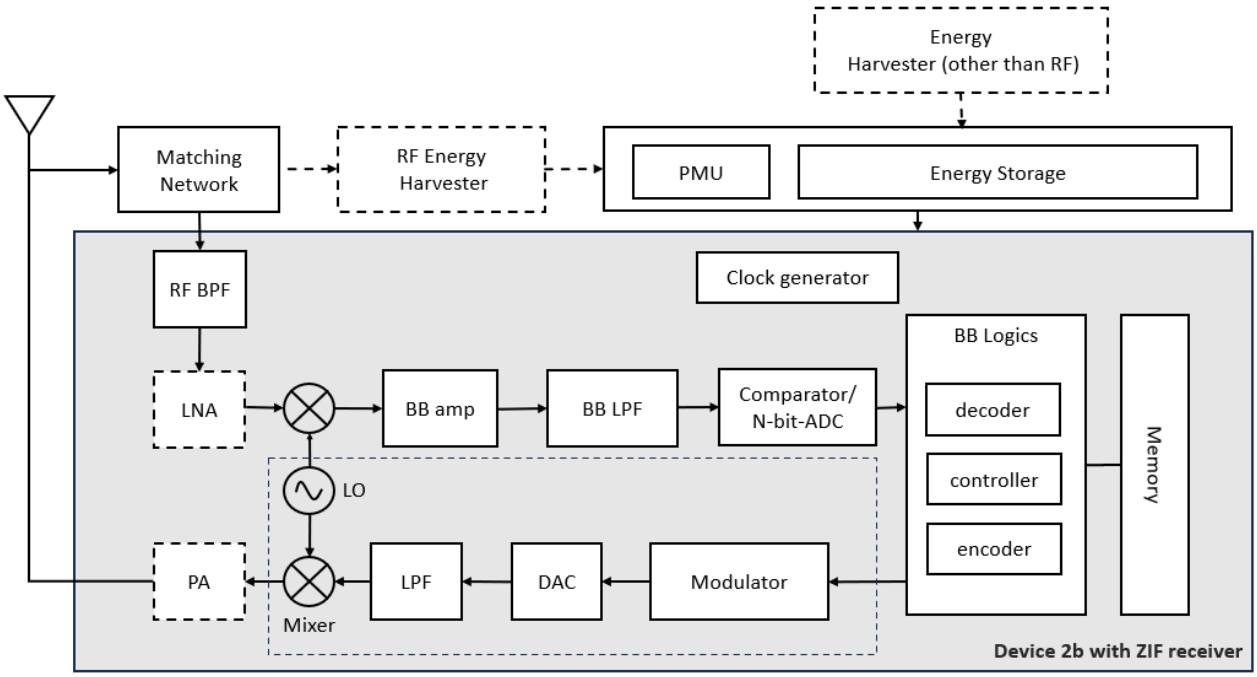


Figure 5.2.2.1-1: Architecture of device 2b with ZIF receiver

5.2A ≤ 1 mW to ≤ 10 mW devices (Device C)

Device C is assumed to have similar device architectures as Device 2b. Values between -3 dBm and 5 dBm are feasible maximum transmit power for Device C, from device power consumption perspective.

5.3 Clock(s)

For A-IoT device clock, 9 sources [7], [9], [8], [16], [10], [12], [11], [15], [17] have reported that clock calibration based on at least clock counting for a known duration is feasible for A-IoT devices.

Table 5.3-1: Descriptions of clocks/LOs

Clock purpose #	Description	Clock speed	Applicable devices	Power consumption	Initial clock Accuracy [ppm]	Accuracy after clock sync / calibration at device side [ppm]
1	Sampling	A few MHz	1 (Note 3)	<1 μW		10^5 (Note 2)
						10^4 (Note 2) the study did not determine whether or not this accuracy can be achieved with < 1 μW power consumption at least for device 1
			2a, 2b	<10 μW	10^3 - 10^4	10^3 (Note 2)
2	Small frequency shift	A few MHz	1	<1 μW		10^5 (Note 2)
						10^4 (Note 2) the study did not determine whether or not this accuracy can be achieved with < 1 μW power consumption at least for device 1
			2a, 2b (if applicable)	<10 μW		10^3 (Note 2)
3	Time counting (if supported)	e.g. tens of kHz	1 (if applicable) 2a, 2b	<0.1 μW	10^4 - 10^5	Calibration may not always be feasible for this clock purpose
4	Large frequency shift (if supported for device 2a)	10s of MHz (e.g. 50 MHz)	2a	10s of μW	10^3 - 10^4	10^3 (Note 2) Whether better accuracy can be achieved has not been determined.
5	LO for carrier frequency (for up/down conversion)	According to carrier frequency e.g., 900 MHz	2b	10s - 100s of μW	10^3 - 10^4	Within 10s – 200 (Note 2), there is no need to determine a single value for the study

Note 1: This table does not necessarily imply that different purposes of LOs/clocks correspond to separate discrete LOs/clocks, which is up to implementation.
Note 2: No drastic temperature change is assumed during calibration.
Note 3: Device 2a, 2b could also use similar implementation.

Several sources [10], [15], [13], [14], [16], [9] think that for clock purpose #1 (sampling), accuracy after clock sync / calibration at device side of 10^4 ppm is achievable with < 1 μW power consumption for device 1, and that 10^5 ppm may not be sufficient for certain designs.

- Source [10] mentioned that device 1 can support basic calibration operation within a reasonable power consumption or complexity, achieving accuracy of 10^4 ppm.

- Source [15] mentioned that with clock calibration based on a clock acquisition part design, 10^4 ppm could be achieved after clock calibration.
- Source [13] reported that clock purpose #1 could be applied to all device types. But, considering complexity and power, device 1 and 2 could have different power, accuracy. For device1, the power consumption should be <1 μW; the timing accuracy should be 10^4 - 10^5 ppm; the calibration can reach to 10^4 ppm.
- Source [14] proposed device 1 with accuracy of 10^4 ppm after clock sync/calibration.
- Source [16] reported that using simple clock counting based calibration method, 1% could be achieved with power consumption <1 μW. It was reported that 1% is the sweet spot of target accuracy considering performance, complexity, signal design, etc.
- Source [9] mentioned that device can use recommended clock calibration method to get high accuracy of 10^3 - 10^4 ppm for device 1 and 10^2 - 10^3 ppm for device 2.

6 Solutions for Ambient IoT

6.1 Physical layer

6.1.0 Aspects in common to R2D and D2R

6.1.0.1 General

The names of physical channels and signals used in this TR are for the sake of the study.

The study assumes that an A-IoT device has a single antenna for both communication (transmission/reception) and RF energy harvesting purposes.

The study defines repetition types as follows:

Block level: All the bits received from higher layers and/or physical layer (according to what is present) after CRC attachment (if used) are blockwise repeated Rblock times

Bit level type 1: Each bit after CRC attachment (if used) is repeated Rbit times

Bit level type 2: Each bit after both CRC attachment (if used) and FEC (if used) is repeated Rbit times

Chip level: Each chip after line coding (if used) or after square wave modulation (if used) is repeated Rchip time. NOTE: This is equivalent to extending the duration of each chip by Rchip times.

6.1.0.2 CRC

Sources as follows recommend that the length of the CRC used for A-IoT depends on:

- Block size or TBS-like size, according to sources [22], [23], [19], [26], [25], [28], [49], [44], [51]. Furthermore:
  - Source [22] states that specifically for short messages, the CRC length can directly impact the transmission efficiency due to the overhead caused by the CRC.
  - Source [49] state that CRC length depends on the TBS due to its impact on the transmission overhead and FAR performance. When the TBS is too small, the overhead of CRC-6 becomes quite significant.
  - Source [51] states that CRC length is fundamentally proportional to the size of the message.
- Target false alarm rate or error detection performance, according to sources [22], [23], [19], [26], [28], [41], [51]. Furthermore:
  - Source [22] evaluated that the probability of undetected errors is higher for CRC-6 as compared to CRC-16, and transmissions with CRC-16 performs better, for a given TBS=24, and source [41] say that a longer CRC would mean a better error detection performance.

- Type of message, such as a command/reply function, to determine when no CRC is used, according to sources [22], [23], [19], [26], [25], [41], [44]. Furthermore:
  - Sources [22] and [23] say that similar to RFID, only important messages, decided based on their functionality, such as configuration/control/command messages, are appended with CRC to increase their reliability, while other messages are not protected by CRC to save the overhead.
  - Source [19] proposes different CRC lengths for Paging-like/select/challenge messages that may or may not contain the device ID/group ID.
  - Source [26] states that some messages such as a QueryRep-like command do not result in serious FAR issues if mis-detected, and hence the functionality of a message/command has to be considered.
- Overhead caused by CRC, according to sources [22], [23], [26], [49], [41], [45]
  - Sources [22], [23], [26], [41] mention that longer CRCs have high overhead but enjoy better error detection performance, affecting the transmission efficiency. In order to balance the overhead and performance, Hence, short CRC can be used for short commands to reduce the false alarm rate and make the system more robust as compared to no CRC.
  - Source [26] states that adding CRC to QueryRep-like commands which are sent repeatedly with a small payload would waste network resources, and hence the overhead should be considered.
  - Sources [49], [45] states that the overhead of CRC decreases as the TBS increases, and that CRC-16 has higher overhead than CRC-6.

For the CRC generator polynomials:

- Sources [24], [29], [27], [28], [38] recommend that the same polynomials from TS 38.212 are reused, giving justifications:
  - Sources [28], [29] state that the polynomials from TS 38.212 were already carefully and thoroughly evaluated and ensured in the NR channel coding design, so there is no need of considering other polynomials.
  - Source [27] states that the device complexity is increased when different polynomials are introduced for the D2R transmission.
  - Source [24] states that link performance is significantly impacted by the CRC lengths, and not the CRC polynomials, hence the polynomials from TS 38.212 should be used.
- On the other hand, sources [49], [45], [54] and [71] recommend that the same polynomial for CRC-6 $\left(\boldsymbol{g_{CRC6}(D) = \left[D^6 + D^5 + 1\right]}\right)$ from TS 38.212 is reused, but a new polynomial for CRC-16 is introduced, $\boldsymbol{g_{CRC16}(D) = \left[D^{16} + D^{11} + D^6 + D^5 + 1\right]}$, incorporating the CRC-6 polynomial, giving justifications:
  - The proposed new CRC-16 polynomial nests the existing CRC-6 within its structure.
  - Using separate sets of circuits to implement the polynomials for CRC-6 and CRC-16 as per TS 38.212 requires a higher number of registers and XOR gates when compared to using the new CRC-16 polynomial.
  - FAR performance is the same for the legacy and proposed new polynomial for CRC-16, but the new polynomial reduces the complexity and power consumption of the device.

The following evaluation results of various CRC designs were provided during the study:

NOTE 1: P_{ud} = Probability of undetected error according to [75].

NOTE 2: Overhead = CRC length [bits] / (TBS [bits] + CRC length [bits]).

Table 6.1.0.2-1: CRC evaluations

	CRC-6			CRC-16		
TBS	**Source**	**CRC overhead**	**P_{FA} or P_{ud}**	**Source**	**CRC overhead**	**P_{FA} or P_{ud}**
8 bits	[22] [72]	43%	~10^(-8) P_{ud} @10^{-3} BER	[22] [72]	67%	~10^(-11) P_{ud} @10^{-3} BER

12 bits	[20] [72]	33% 33%	~10^(-2) P_{FA} ~10^(-8) P_{ud} @10^{-3} BER	[20] [72]	57% 57%	~10^(-5) P_{FA} ~10^(-11) P_{ud} @10^{-3} BER
16 bits	[22] [49] [72]	27% 27% 27%	 ~10^(-7) P_{ud} @10^{-3} BER	[22] [49] [72]	50% 50% 50%	 ~10^(-10) P_{ud} @10^{-3} BER
24 bits	[22] [20] [49] [72]	20% 20% 20% 20%	~10^(-6) Pud @ 10% BLER and 2.7dB SNR ~10^(-2) P_{FA} ~10^(-7) P_{ud} @10^{-3} BER	[22] [20] [72]	40% 40% 40%	~10^(-9) P_{ud} @10% BLER and 2.8dB SNR ~10^(-5) P_{FA} ~10^(-10) P_{ud} @10^{-3} BER
32 bits	[22] [49] [72]	16% 16% 16%	 ~10^(-5) P_{ud} @10^{-3} BER	[22] [49] [72]	33% 33% 33%	 ~10^(-10) P_{ud} @10^{-3} BER
40 bits	[22] [49] [72]	13% 13% 13%	 ~10^(-5) P_{FA} @10% BLER ~10^(-4) P_{ud} @10^{-3} BER	[22] [49] [72]	29% 29% 29%	 ~10^(-10) P_{ud} @10^{-3} BER
48 bits	[20] [49] [72]	11% 11% 11%	~10^(-2) P_{FA} ~10^(-5) P_{FA} @10% BLER ~10^(-4) P_{ud} @10^{-3} BER	[20] [49] [72]	25% 25% 25%	~10^(-5) P_{FA} ~10^(-10) P_{ud} @10^{-3} BER
96 bits	[22] [20] [49] [72]	6% 5% 6% 6%	 ~10^(-2) P_{FA} ~10^(-4) P_{ud} @10^{-3} BER	[22] [20] [49] [72]	14% 14% 14% 14%	 ~10^(-5) P_{FA} ~10^(-9) Pud @10^{-3} BER
128 bits	[49] [72]	4% 4%	 ~10^(-3) P_{ud} @10^{-3} BER	[49] [72]	11% 11%	 ~10^(-9) P_{ud} @10^{-3} BER
256 bits	[22] [72]	2% 2%	 ~10^(-3) P_{ud} @10^{-3} BER	[22] [72]	6% 6%	 ~10^(-9) P_{u}d @10^{-3} BER

6.1.1 R2D

A dedicated physical broadcast channel, e.g. PBCH-like, and reference signals including at least DMRS, PTRS, CSI-RS/TRS, are not considered for R2D. R2D transmission without a midamble is studied as the baseline, if Manchester line coding is used; the necessity of a midamble for the R2D transmission requires further study if PIE line coding is used.

6.1.1.1 R2D waveform, modulation and numerology

An OFDM-based OOK waveform with subcarrier spacing of 15 kHz is studied for R2D, with OOK-1 for single-chip per OFDM symbol transmission, and OOK-4 for *M*-chip per OFDM symbol transmission, starting from the definitions in TR 38.869 [4]. For this waveform, the start of R2D transmission from the reader perspective is assumed to be aligned with the boundary of an NR OFDM symbol (including the CP) for in-band/guard-band operation. For the generation of this waveform, CP-OFDM and DFT-s-OFDM are both feasible. For M=1, i.e. use of OOK-1 or OOK-4 for single-chip per OFDM symbol transmission, CP-OFDM and DFT-s-OFDM are both feasible. For M>1, i.e. use of OOK-4 for *M*-chip per OFDM symbol transmission, DFT-s-OFDM is feasible. For R2D waveform generation using a DFT-s-OFDM based transmitter, from the transmitter perspective, when the generated number of chips for the R2D transmission does not fully occupy the last OFDM symbol, at least padding can be used.

R2D chip duration is a consequence of CP handling design.

For CP handling, the following candidate methods are studied, on the basis of e.g., CP impact on R2D timing acquisition, and decoding & performance of PRDCH, reader and device implementation complexities, interference between R2D and NR DL/UL if in the same NR band, spectrum efficiency.

Method Type 1: Removal of CP at device without specified transmit-side.

Method Type 2: Ensure the CP insertion of OFDM-based waveform will not introduce false rising/falling edge between the last OOK chip in OFDM symbol (n-1) and the first OOK chip in OFDM symbol n.

For Method Type 1, two ways that CP location/length can be determined are studied:

Alt M1-1: Device assumes same CP length for each OFDM symbol, i.e. does not distinguish exact CP length among different OFDM symbols

Alt M1-2: Duration between transition edges is utilized by device to determine CP location/length, i.e. if the duration appears to be invalid based on known chip duration

For Method Type 1, at least for Alt M1-1, device needs to be aware of or determine the boundary of an OFDM symbol (i.e. the beginning of an OFDM symbol) to determine CP location, including the start and the length of CP. Alt M1-1 and/or Alt M1-2 (if both are supported) can be up to device implementation which may depend on M values.

- Some sources [18], [21], [20], [32], [31], [42], [45], [39], [48] [23], [38], [44], [74] report device needs to be aware of or determine the boundary of an OFDM symbol (i.e. the beginning of an OFDM symbol) to determine CP location

- Some sources [25], [22], [26], [18] report the device implementation may depend on *M* values

For Alt M1-1, the potential impacts are discussed as follows:

- Some sources [18], [22], [23], [25], [42], [45] report that CP handling of Alt M1-1 can be used for both small and large *M* values for OOK-4, while [25] reports that for large M values Alt M1-1 is used in combination with Alt M1-2.

- Some sources [20], [49], [74] report that CP handling of Alt M1-1 is challenging to be used for large *M* values for OOK-4 considering large SFO and [25], [35] report that CP handling of Alt M1-1 may not completely remove CP samples due SFO impact.

- Among them, [22] show that the performance loss of PRDCH carrying 20 bits due CP handling is negligible at 10% BLER even for large *M* values (e.g. *M*=24) under large SFO (e.g. 10^4-10^5 ppm). Sources [25], [49] show some performance loss due CP handling for both small (M=4) and large *M* values (*M*=24) under large SFO (e.g. 10^4-10^5 ppm) while [49] shows [1~2 dB] loss compare to no CP case for *M*<24, and an error floor at BLER=10% for *M*=24.

- Some sources [26], [35] report that the device needs additional complexity to handle CP, while other sources [22], [42] reports that it is feasible in terms of implementation complexity based on transition edge detection.

- One source [65] reports that the device might remove the wrong portion of the CP part of the OFDM symbol due to timing error, which could introduce the false rising/falling edge for the subsequent OOK demodulation.

For Alt M1-2, the potential impacts are discussed as follows:

- Some sources [18], [21], [26], [20], [22], [23], [25], [49], [35], [42], [44], [65], [74] report that CP handling of Alt M1-2 cannot be used for large M values, e.g. *M*>8, while [25] reports that for large *M* values Alt M1-2 is used in combination with Alt M1-1.

- One source [39] report that CP handling of Alt M1-2 can be used for both small and large *M* values (e.g. *M*>8) if with the knowledge of OFDM symbol boundaries.

- Among of them, [25] show that the performance of Alt M1-2 is not applicable for large M values (e.g. *M*=24) under large SFO (e.g. 10^4 ppm).

For Method Type 2, two approaches regarding subcarrier orthogonality are studied:

Alt M2-1: Method Type 2 retains subcarrier orthogonality, i.e. CP is copied from the end of an OFDM symbol.

Alt M2-1-1: The first OOK chip(s) and the last OOK chip(s) in an OFDM symbol are the same.

Alt M2-1-2: Ensure a transition edge occurs only at the start or only at the end of the CP, and no transition edge occurs during the CP.

Alt M2-2: Method Type 2 does not retain subcarrier orthogonality.

For Method Type 2, depending on the design, the chip duration generation of OOK-4 for *M*-chip per OFDM symbol transmission could possibly be determined by:

- M, and the length of OFDM symbol with CP

- M, and the length of OFDM symbol without CP

- Depending on detailed solutions, chip duration may or may not be constant.

- Some sources [45], [57] report that non-constant OOK chip duration may impact performance, while some other source [49] report that non-constant OOK chip duration does not impact performance.

For Alt M2-1, the potential impacts are discussed as follows,

- Some sources [22], [26], [25], [50], [38], [28], [35], [20], [68] report that CP handling of Alt M2-1 cannot be used for large M values (e.g. M>8). Source [56] reports that for M>8, the CP size becomes comparable to that of the normal OOK chip, and hence it would be challenging to identify the invalid transition caused by CP. Sources [65] reports that if chip duration is comparable to CP duration, CP could not be identified as the invalid chip by the A-IoT device, e.g., M>8.

- Some sources [18], [23], [45], [49] report that CP handling of Alt M2-1 can be used for both small and large M values.

  - Among of them, some sources [23], [49] show the performance of Alt M2-1 for small (M=4) and large M values (M=24) under large SFO (e.g. 10^5 ppm).

- Some sources [45], [26], [49] report that CP handling of Alt M2-1 may result in non-constant OOK chip duration around CP. Source [57] report that due to non-constant OOK chip duration around CP, Alt M2-1 has ~1 dB worse performance than Alt M1-1 at BLER 10% and BLER 1% when it used for small M value (e.g., M = 6).

- Some sources [20], [22], [28], [49], [39], [21], [44], [74] report that CP handling of Alt M2-1-1 would increase the overhead and reduce spectral efficiency.

- Some sources [42], [18], [26] report that CP handling of Alt M2-1-1 may not be completely transparent to the device thus add additional complexity.

- Source [65] report that if chip duration is significantly different from CP length, M2-1-2 would be complicated to be used for removing false transition edge occurring at the end of the CP. And M2-1-2 would require high complexity of A-IoT device implementation if it is used for the R2D preamble.

For Alt M2-2, the solutions and potential impacts are discussed as follows,

- [25], [29], [28], [38], [71] report solutions for Alt M2-2 (e.g. CP is copied from the start of OFDM symbol or do not insert CP to OFDM symbol).

- [20], [22], [23], [27], [49], [42], [45], [21], [26], [39], [44] report that CP handling of Alt M2-2 would cause interference to NR, while [25] reports single PRB guard band would be sufficient to handle interference. [71] reports the guard band would anyway be needed when SCS is different between R2D and other NR signal.

- Sources [22], [42], [45], [48], [26], [39], [44], [74] report that CP handling of Alt M2-2 would increase the transmitter complexity.

6.1.1.2 R2D line coding

The line codes studied for R2D are Manchester encoding and PIE. For R2D reception, the device can use line codes to achieve at least chip-level time tracking by using the transition(s) of the line code. A chip corresponds to one modulated symbol.

For Manchester encoding, the bit-to-chip mapping is: bit 0→chips{10}, bit 1→chips{01}.

It is reported by sources [22], [26], [32], [28], [49], [33], [38], [36], [48], [35], [39] that PIE has a higher ratio of high to low voltage, which enables it to provide stable instantaneous RF energy to the device. Sources [26], [38], [39], [32] and [48] consider this a justification for studying PIE. Sources [22], [28], [49], [36] and [35] acknowledge this aspect of PIE, but question its necessity.

It is stated by sources [21], [20], [25], [28], [49], [35], [44], [47] that, due to variable codeword lengths in PIE, scheduling time domain resources would be different to transmit the same amount of data due to the number of 0 and 1, as compared to the constant codeword lengths in Manchester codes.

It is stated by sources [22], [23], [28], [49], [33], [35] that, due to variable codeword lengths in PIE, the error detection of one information bit can result in error propagation for the detection of subsequent bits and therefore negatively affecting the detection of the overall data length, as compared to Manchester codes.

Source [20] says that variable codeword lengths in PIE complicates A-IoT resource management, especially in in-band deployments when a reader needs to multiplex both NR and A-IoT communication.

Source [26] says that equal codeword length can overcome the cons of variable codeword length in PIE.

According to sources [21], [20], [28], [49], [33], [38], [35], [44], transitions during Manchester code transmissions can be utilized for recovering the signal's clock.

Sources [21], [20] say that the signal does not lose information or distort when fed to a DC blocker because each Manchester encoded symbol maintains a constant DC level, i.e. always the same average power, unlike if there was a long sequence of ones or zeros.

Evaluations are reported showing that Manchester codes perform better than PIE by 1dB to 6dB @10% BLER – [22], [26], [28], [49], [47]. Source [26] claims only a 1dB gain when comparing Manchester over equal length codeword in PIE, stating that the gain is negligible since the downlink coverage bottleneck is the receiver sensitivity and not the decoding performance.

6.1.1.3 R2D channel coding, repetition, CRC

6.1.1.3.1 Channel coding and CRC

PRDCH without FEC is studied as the baseline, with evaluations performed by comparison to this baseline. The study assumes PRDCH can attach a CRC, where the baseline design is using a 6-bit or 16-bit CRC with polynomials as per TS 38.212 [5]. A baseline of no CRC attachment is also included. For the study of CRC designs, see Clause 6.1.0.2.

Sources [57], [54], [62], [59], [60], [66], [58], [63], [67] and [64] provide justifications for not having R2D FEC beyond the baseline, with the following observations:

- Sources [57], [59], [66], [63] and [67] state that FEC decoders require complicated arithmetic or logical operations which are too complicated to be implemented in device 1.
- Sources [59] and [60] state that it would be difficult for a device to implement a FEC decoder due to its low power consumption.
- Source [57] and [67] state that FEC decoder procedures such as the de-interleaving operation or route metric caching require volatile memory of a certain size with a certain reading/writing throughput, which cannot be supported by device 1.
- They also mention that the received signal power at the device can be relatively high (e.g., >-60 dBm), making the receiver sensitivity not the bottleneck of the link budget for target coverage, even for device 2b, thus questioning the necessity of R2D FEC.

Sources [56] and [72] provide the following justifications for using FEC in R2D for device 2b:

- Source [56] claim that CC with small constraint lengths (e.g., 3 or less) offer a substantial performance gain over uncoded transmission, especially in a fading environment, with reasonable complexity. CC with explicit tail-biting transmission to aid decoding may be suitable for R2D.
- Source [72] claim that even simple block code (e.g., Golay, RM) with hard decisions can significantly reduce the required SNR for achieving a target BLER e.g., 1%.

6.1.1.3.2 Repetition

Regarding R2D repetitions, it is reported by sources [28] (only for R2D control, if supported), [29], [49], [30], [38], [36], [45] and [47] that R2D repetitions should be supported. The following are observations regarding the different types of repetition that should be supported.

Bit-level repetition

Positive observations:

- Source [20] state that bit level repetition can be studied if coverage enhancement of the R2D link is required.

- Source [29] state that bit level repetition where every input bit repeated for 8 times before Manchester coding could have ~4dB gain when compared with no repetition. They claim using Manchester codes with repetitions require a simple structure and consumes extremely low power.

- Source [45] state that bit level repetitions with scrambling is required since the former would improve the link budget and the latter would add extra randomness to the information bits, providing gain by suppressing the interference. They also claim that repetitions can be used in devices that cannot soft combine the repetitions, and majority-based detection would offer gain for these devices.

Negative observations:

- Source [26] state that since envelope detection is used for R2D reception, bit level repetition may not provide expected gain for the reception.

- Source [25] state that though it may be feasible, it increases the device's processing complexity for reception, e.g., combination, repetition parameters determination.

- Source [67] state that repetition gain of a bit-level repetition, which requires additional standardization effort to define necessary control information, mainly comes from the energy accumulation of the signal, and should be similar with the achievable gain by directly lowering the chip rate/reducing the M value, which does not require this additional effort.

Block-level repetition

Positive observations:

- Source [49] state that at least for large TBs, repeatedly transmitting the TB multiple times consecutively provides time diversity gain and increases the probability that at least one of the repetitions can be successfully decoded.

- Source [59] further state that the device can perform the block-wise detection without chase combination of the repeated blocks so that block-level repetition may not need additional buffer and increase the complexity and cost.

- Source [67] state that block-level repetition can obtain a bigger repetition gain than that achieved by bit- or chip-level repetition, and can enjoy both the time diversity gain and the gain of energy accumulation.

Negative observations

- Source [25] state that considering limited capability and cost for an A-IoT device, block level repetition for R2D should be excluded.

- Source [67] state that block-level repetition additionally requires a very large volatile memory to store all received repetitions of one block.

Chip-level repetition

Positive observations:

- Source [26] state that it may be useful for R2D transmission coverage and can be considered to generate a lower data rate than 7kbps.

- Source [47] state that chip-level repetition increases the chip duration, improving the edge detection at the receiver, thereby having a ~2dB performance increase when compared to bit level repetitions.

Negative observations:

- Sources [20], [25] and [28] state that chip-level repetition is equivalent to long chip transmission, i.e., by using a smaller modulation index, and therefore, there is no need to support this option.

- Source [67] state that repetition gain of a chip-level repetition, which requires additional standardization effort to define necessary control information, mainly comes from the energy accumulation of the signal, and should be similar with the achievable gain by directly lowering the chip rate/reducing the M value, which does not require this additional effort.

On the other hand, it is reported by sources [55], [57], [58], [54] and [62] that R2D repetitions should not be supported, giving justifications:

- Source [55] mentions that the transmission power of a R2D transmission is typically much greater than its corresponding D2R transmissions, and if the R2D transmission has coverage issues, then the corresponding D2R transmission would not reach the reader. Hence it should be considered for D2R transmissions alone.
- Source [54] say that not supporting R2D repetition can be the baseline.
- Sources [58], [69] and [63] include that the decision to support R2D repetitions can be based on whether the activation threshold is a bottleneck according to the coverage evaluations.
- Sources [58], [63] and [57] also comment that from a device perspective, especially device 1 with low complexity and memory storage, it is not possible to combine multiple repetitions.

6.1.1.4 R2D bandwidths

The study defines the following bandwidths for R2D:

- Transmission bandwidth, $B_{tx,R2D}$ from a reader perspective: The frequency resources used for transmitting R2D. For an OFDM-based waveform with subcarrier spacing of 15 kHz, $B_{tx,R2D} \leq$ twelve PRBs, and would be down-selected among:
  - Alt 1: Including 180 kHz, 360 kHz
  - Alt 2: Integer multiple(s) of 180 kHz
  - Alt 3: Integer multiple(s) of the subcarrier spacing
- Occupied bandwidth, $B_{occ,R2D}$ from a reader perspective: The frequency resources used for transmitting R2D, and potential guard band.
- $B_{occ,R2D} \geq B_{tx,R2D}$.

Table 6.1.1.4-1 is a starting point for study of M values and the associated minimum $B_{tx,R2D}$ value. The reader can use any transmission bandwidth greater than or equal to the minimum $B_{tx,R2D}$ value.

Note: Depending on further study, the maximum value of M may be less than 32.

Note: The performance can be better when transmission bandwidth greater than the minimum $B_{tx,R2D}$, depending on device processing and transmit power constraint.

Table 6.1.1.4-1: Starting point for *M* values and the associated minimum $B_{tx,R2D}$ value

M	Minimum $B_{tx,R2D}$ # of PRBs
1	1
2	1
4	1
6	1
8	2
12	2
16	2
24	3
32	4

6.1.1.5 PRDCH

For R2D, the only physical channel is PRDCH, which carries any higher-layer payload (including system information, if defined), and L1 R2D control information, if defined. The following design options have been studied for PRDCH:

Option 1: If there is no L1 R2D control information carried by PRDCH, then PRDCH transmission with only R2D data is considered.

Note: In this option, R2D control information (if any) maybe carried by higher-layer signalling.

Option 2: If any L1 R2D control information is supported & carried by PRDCH, then for the CRC attachment (appended if there is non-zero length CRC), following two options have been studied.

Option 2-1: Joint CRC is attached to L1 R2D control information and R2D data.

Option 2-2: Separate CRC is attached to L1 R2D control information and R2D data, respectively.

PRDCH is studied via the blocks shown in Figure 6.1.1.5-1, where other clauses give their detailed descriptions.

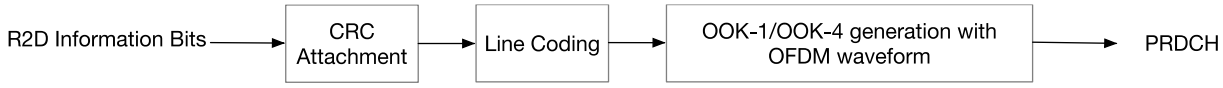


Figure 6.1.1.5-1: PRDCH generation

6.1.1.6 R2D timing

6.1.1.6.1 R2D timing acquisition signal

An R2D timing acquisition signal (R-TAS), immediately preceding the transmission of PRDCH, is included at least for timing acquisition and indicating the start of R2D transmission in the time domain. An R-TAS structure using a preamble is studied, in which a start-indicator part provides the start of the R2D transmission, and immediately precedes a clock-acquisition part which is used to determine the OOK chip duration of the subsequent PRDCH transmission. The preamble is not part of PRDCH.

The R-TAS start-indicator part is not included in T_{D2R_min}, and an ON/OFF pattern i.e., high/low voltage transmission, is applied. The following options have been studied for the start-indicator part:

- Option 1: ON-OFF transmission is considered based on energy/edge detection, and multiple alternatives have been studied including
  - Alt 1: A single ON-OFF transmission, i.e. one high-voltage transmission followed by one low-voltage transmission, where ON and OFF may have same or different durations
  - Alt 2: A multi-ON-OFF transmission, where different ON and different OFF may have same or different durations and different parts may have same or different duration
- Option 2: ON-OFF sequence-based design is considered which consists of a pre-defined sequence for detection of start-indicator part based on digital correlation
- For both the options, it is observed that a fixed duration for the start-indicator part can be considered, regardless of the value of M used for PRDCH transmissions.

MDR, FAR, and detection complexity have been evaluated for the design of the R2D start indicator part as follows:

- It is observed by one source [119] that for an FAR of ~0%, the MDR of less than 1% can be achieved with Alt 2 of Option 1 (considering 2 ON-OFF transmissions with different durations) and it is also observed that low-complexity and reduced power consumption can be achieved.
- One source [122] evaluated Alt 1 of Option 1 (considering same duration for ON and OFF) and Alt 2 of option 1 (considering multiple ON-OFF transmissions with same duration) and observed that for an FAR of ~0%, the MDR of less than 1% can be achieved and Alt 1 of Option 1 performs better than Alt 2 of Option 1.
- One source [121] observed with ON-OFF pattern, that for an FAR of ~0%, the MDR of less than 1% can be achieved with a duration of at least 1 OFDM symbol.
- One source [126] compares the performance between option 1 and option 2. It shows almost similar coverage range (SNR requirement) for target MDR of 1%. For MDR of 10%, it shows that sequence-based design provides better performance, and it is observed that during the available time, it is feasible for all devices to detect the start-indicator sequence within the power budget. It is further observed that the FAR with sequence-based design can be improved in case of interference scenarios when compared with pattern-based design.

For both the options, it may be beneficial that the start-indicator part is distinguishable at least from other parts of the R2D transmissions.

The clock-acquisition part is based on OOK without line coding, and includes rising/falling edges, including at least two rising or at least two falling edges for the device to determine the OOK chip duration. The following options for design of the clock-acquisition part are studied:

- Option 1: Duration of the clock-acquisition part is variable for different *M* values, i.e. the duration becomes shorter with increasing value of *M.*
- Option 2: Duration of the clock-acquisition part is constant for different *M* values based on repetition, i.e. repetition factor is increased with increasing value of *M* to keep the duration constant.

On impact/restriction of *M* values for the clock-acquisition part (CAP):

- The following 9 sources provided observations on the impact/restriction of M values for the clock-acquisition part design requirements:
  - 1 source [131] observed that increasing value of M, while retaining the same transmission duration, improves the auto-/cross- correlation properties of the sequence due to increase in sequence length and use of $M \geq 4$ provides better timing estimation accuracy even in the presence of SFO as the sequence length spans only over a shorter duration.
  - 2 sources [130], [132] observed for option 1 of the clock-acquisition part design that no restriction is required to be placed on the *M* values. Furthermore, 1 source [132] observed that the same 2 ON-OFF voltage (with the same duration) satisfies the FDR performance metric of less than 1% for different *M* values, e.g., $M = 2$, 6 and 24, where FDR is the False detection ratio, i.e. incorrectly calculating *M*, is the performance metric.
  - 1 source [133] observed that pattern of the clock-acquisition part is related to *M* chips per OFDM symbol and when *M* is small, the clock-acquisition part may cross multiple OFDM symbols, and the CP insertion may degrade the timing acquisition performance.
  - 1 source [134] observed that with option 2, the duration of the clock-acquisition part remains consistent across all *M* values, at least three OFDM symbols maybe required for clock-acquisition part and it maybe not as efficient as option.1.
  - 1 source [138] observed that among the two options studied for the clock-acquisition part, option 2 provides increased robustness, especially in case of large value of *M*, when compared to option 1 and potentially increase the detection performance of the clock-acquisition part.
  - 1 source [140] observed that if the chip duration is variable based on the *M* value used for OOK-4 waveform, the detection performance would be limited by the received SINR of the CAP with clear transition of the rising and falling edges.
  - 1 source [148] observed that the option 1 with $M>1$ has shorter duration of clock acquisition part than $M=1$ and worse timing acquisition accuracy. At least part of PRDCH following the clock acquisition part may need to be used to improve the timing acquisition. Furthermore, the larger *M* (e.g., $M>4$) with small chip duration is more sensitive to the SFO accuracy and the restriction of *M* for the clock acquisition part may be needed.
  - 1 source [149] further observed that different *M* values may impact the chip accuracy obtained by the clock acquisition part.

On impact of CP insertion/handling on the clock-acquisition part:

- The following 10 sources observed that the CP insertion/handling may impact the design requirements of the clock-acquisition part:
  - 1 source [133] further observed that when the clock-acquisition part occupies more than one OFDM symbol, ON-OFF state transition around CP can avoid the error rising or falling edges due to the CP insertion.
  - 1 source [134] further observed that to mitigate the impact of the CP in the clock-acquisition part for large M values, it can reuse the CP handling method for PRDCH
  - 1 source [135] further observed that CP insertion/handling on the clock-acquisition part can cause false rising/falling transition and, therefore, the clock acquisition part should be designed such that it does not incur a false rising or falling edges due to CP insertion when CP-OFDM is used for OOK signal generation.

- 1 source [137] further observed that CP insertion/handling on the clock acquisition part will impact the chip duration estimation accuracy. It is further observed that for CP handling, device may not be able to count the clock and estimate OFDM symbol duration accurately until the clock acquisition part if the start indicator only includes a single ON-OFF transmission.
- 1 source [140] further observed that the SER will be degraded due to uneven chip interval when the CP is inserted within an OFDM symbol, where SER refers to the number of samples which is mismatched for comparing to the total number of samples in a chip.
- 1 source [147] further observed if CP insertion would cause false rising/falling edges, accuracy of timing acquisition may be impacted.
- 1 source [149] further observed that the issues of chip extension, false raising/falling transition, and additional raising/falling transition caused by CP insertion/handling considering different M values will impact the chip accuracy obtained by the clock acquisition part.
- 1 source [136] further observed that the design of clock acquisition part should consider that CP insertion does not cause a false rising or falling edges and does not cause different length of multiple high / low voltages within the clock acquisition part when the clock acquisition spans multiple OFDM symbols.

- 1 source [132] observed CP insertion/handling may not impact the design requirements of the clock-acquisition part.

On purpose of SFO estimation/correction based on the clock-acquisition part

- The following 3 sources provided observations on the applicability of clock-acquisition part for frequency synchronization:
  - 1 source [131] observed that the length of preamble sequence may need to consider also the robustness against SFO
  - 1 source [140] observed that device 2a/2b may require higher synchronization accuracy for signal transmission or backscattering and therefore, the design of CAP may be required to accommodate the requirement of additional frequency synchronization and clock calibration for Device 2a/2b.
  - 1 source [148] further observed for Option 1, as the CAP duration with high M is decreased, only CAP may not be sufficient for SFO correction and for Option 2, as the CAP duration is fixed and independent from M, the CAP with long enough duration can support SFO correction.

On purpose of CFO estimation/correction based on the clock-acquisition part

- The following 2 sources provided observations on the applicability of clock-acquisition part for frequency synchronization:
  - 1 source [144] observes that the clock-acquisition part can be utilized to solve the frequency synchronization problem without impacting the time-domain sequence, for example by transmitting in some frequency resources and it can be a harmonized solution for both chip duration indication and device frequency synchronization. However, it is further observed that if the time interval between an R2D transmission and the corresponding D2R transmission and if the device loses the timing obtained from the R2D timing acquisition signal due to timing drift at the time for the D2R transmission, then an additional synchronization signal is needed
  - 1 source [148] further observed for Option 1, as the CAP duration with high M is decreased, only CAP may not be sufficient for CFO correction and for Option 2, as the CAP duration is fixed and independent from M, the CAP with long enough duration can support CFO correction.

6.1.1.6.2 R2D end timing

To determine or derive the end of PRDCH transmission, the following options are studied:

- Option 1: TBS information via implicit/explicit L1 R2D control information.
- Option 2: Postamble at the end of PRDCH.

14 sources [131], [132], [134], [133], [135], [144], [143], [146], [148], [136], [149], [150], [144], [137] provided following observations on the above two options for determining the end of PRDCH:

- 3 sources [131], [132], [134] observed that option 2 provide two benefits, namely, the variable payload length and to provide timing acquisition before the subsequent transmission of either PDRCH or PRDCH, thus improving the detectability at both reader and the device, respectively. Furthermore, 1 source [132] observed that R2D postamble indicates the TBS with high efficiency for small packets by avoiding a large padding overhead, unlike option 1, which may require devices to perform blind detection of different PRDCH formats (if supported) and the overhead caused by the inclusion of a R2D postamble does not exceed 20% for even the smallest of message sizes and may be less than the signaling overhead caused by using a dedicated TBS indicator.

- 1 source [133] observed for option 2, that for small payload size with only a few bits, the presence of long postamble generates large resource overhead, while for large payload size with more bits, the resource overhead of postamble is smaller.

- 1 source [137] observed for option 2, that for small payload size with only a few bits, the presence of long postamble generates large resource overhead.

- 1 source [135] observed option 2 is not strictly required, however, given the possible clock drift at a device, it may be still beneficial to also attach postamble at least for the determination of the end of PRDCH at a device.

- 3 sources [143], [136], [150] observed that with option 2, the false detection may be higher for shorter postamble. Source [143] observed that in contrast to option 2, it is more reliable and efficient to indicate TBS with control information in option 1.

- 2 sources [146], [137] observed that if a message type or a command ID is included in L1 control information and implicitly indicates a known size of a fixed TB, then there is no need for either option 1 or option 2.

- 2 sources [148], [137] observed that option 1 has the advantages of avoiding blind detection of postamble and providing the power saving for non-target devices to skip the R2D detection.

- 1 source [149] observed that option 1 is feasible for the device to avoid the unnecessary reception of a TB with a specific size and thus enable power saving, e.g., when the TB has a size exceeding the allowance of the device remaining power.

- 1 source [144] observed option 2 is not strictly required if the end of PRDCH can be explicitly indicated by R2D control information, and it is subject to the miss-detection rate. It may be beneficial if a PRDCH postamble can serve as an additional timing acquisition signal prior to a PDRCH transmission.

6.1.1.7 CFO calibration signal for device 2b

For device 2b, a signal for CFO calibration should be provided to synchronize / calibrate the device clock for LO for carrier frequency (Clock purpose #5) to achieve the accuracy after clock sync / calibration at device side captured in Table 5.3-1. Frequency calibration at device 2b is beneficial at least to reduce the guard-bandwidth of D2R transmission.

For the CFO calibration signal, which is required only for device 2b to reduce the frequency offset range and the guard-bandwidth of D2R transmission, the following observations are captured in TR 38.769:

- Sources [99] report that a single-tone RF signal is used as the CFO calibration signal, it is not a part of time acquisition signal and can be transmitted as an optional R2D signal after the PRDCH transmission.

- Sources [98], [110] and [111] report that additional synchronization signal is needed.

  - [111] state the R2D timing acquisition signal may not be sufficient or may not be usable for CFO calibration since a reference frequency is needed when separate LOs are used for Tx and Rx in device 2b.

- Sources [103], [105], [114], [106] report that additional synchronization signal is needed if the synchronization for carrier frequency using R2D signal/channel does not provide required functionalities for device 2b.

- Sources [101], [115] report that it may not be possible to achieve enough frequency accuracy (0.01 ppm) even after CFO calibration based on R2D time acquisition signals for coherent detection at reader especially when the D2R data rate is low.

6.1.1.8 R2D multiplexing

For R2D, time-domain multiplexing is the baseline. Code-domain multiplexing is not considered for device 1/2a/2b. Frequency-domain multiplexing is not considered for the devices with an RF-ED receiver (see Clause 5). For device 2b with IF-ED or ZIF receivers, the study considered the following technical aspects:

Table 6.1.1.8-1: Observations on the feasibility and necessity of FDM for Device 2b

Aspects to be considered for feasibility/benefit	Observations
Inventory completion time	Sources [20], [39], [21], [29] state that FDM is beneficial to reduce the inventory completion time, especially considering more devices per reader due to the larger maximum distance for Device 2b. Source [62] state that inventory latency reduction would be limited, due to difficulty of allocating frequency resources efficiently for Msg2 by reader with uncertainty of number of successful Msg 1, and difficulty of informing an A-IoT device R2D frequency location other than Msg2.
Device implementation	Sources [35], [44], [20], [39], [29] state that channel selection may be performed by a narrowband filter (IF filter or BB filter) after the mixer for Device 2b, if the LO accuracy is sufficiently good. Source [71] states that narrowband RF filtering at device side to realize R2D FDM would be challenging considering reception performance and complexity, while such filtering would also limit the deployment scenario supported by device. Sources [51], [41], [23], [19], [22] state that it would be challenging for a device using an RF-ED receiver architecture to distinguish the different incoming signal fall into the RF BW without narrowband RF filtering which may cause increasing device implementation complexity and power consumption. Source [20] states that the larger R2D responses are harder for the devices to process in the case of TDM+FDM/TDM only for D2R/R2D, respectively.
Spectrum utilization	Source [51] state that the spectral efficiency may be impacted by the guard band across the FDMed R2D transmissions to multiple devices. Source [56] state that the spectrum utilization can still be higher for non-RF-ED based devices if FDM is used, despite guard bands. Source [62] state that spectrum efficiency improvement would be limited, due to difficulty of allocating frequency resources efficiently for Msg2 by reader with uncertainty of number of successful Msg 1, and difficulty of informing an A-IoT device R2D frequency location other than Msg2.
Coverage (in the case of single reader)	Source [22] states the R2D link budget of a reader is decreased due to the power splitting between the parallel R2D channels. Source [20] states the coverage target of Device 2b is still larger than that of Device 1 (with RF-ED architecture).
Reader implementation (in the case of single reader)	Source [22] states that additional interference suppression may be needed to deal with the intermodulation between the parallel R2D transmissions.

Harmonized design for all devices	Sources [43], [26], [36], [61] state that it is not appropriate to include FDM only for Device 2b, while Device 1 and 2a cannot support it. Source [62] state that non-harmonized resource allocation for different device types complicates the system design. Source [56] state that a deployment supporting a combination of RF-ED devices and non-RF ED devices can be harmonized by TDMA'ing different types of R2D time slots, where some time slots can support only a single frequency occasion while other time slots support multiple frequency occasions.

6.1.1.9 Scheduling of R2D

For R2D reception, the following information potentially can be explicitly/implicitly indicated to the device via the corresponding PRDCH:

- ID associated with device(s) intended for the reception of R2D, potentially including all devices (if supported)

For each information, it is for further study whether higher-layer signalling and/or L1 R2D control signalling is used - see Clause 6.1.1.10.

6.1.1.10 R2D control information

For transmission of R2D control information for R2D reception and D2R scheduling, following two options are studied and captured in the TR 38.769:

- Option 1: L1-control signaling
- Option 2: Higher-layer signaling

The following 12 sources provided following observations on the above two options for signaling R2D control information:

- 1 source [132] observed that option 2 avoids physical layer design, while at the same time offering flexibility in terms of the size of the grant. On the other hand, option 1 may require the device to perform blind detection of different PRDCH formats (if supported), while at the same time, reducing the flexibility and resource utilization efficiency. Option 1 would also require a separate CRC, which would result in breaking the pipelining of the CRC decoding due to the added latency and increases overhead.
- 3 sources [133], [134], [141] observed that for D2R scheduling information, as the PDRCH transmission only occurs when the device can successfully decode the preceding PRDCH transmission, therefore no benefit is observed for option 1 compared to option 2. Furthermore, one source [133] also observed that for R2D reception, if TBS of PRDCH is indicated by R2D control information, such R2D control information should be carried via L1 signaling as the device needs to understand the packet size of PRDCH before decoding.
- 1 source [135] observed that for D2R scheduling information indication, option 1 has an advantage as the decoding of D2R scheduling information is decoupled from that of R2D payload and the device can early start the preparation of the PDRCH. For R2D scheduling information indication, Option 1 has an advantage as the device can early detect whether the corresponding R2D transmission is addressed to the device or not, which also saves device's power consumption.
- 1 source [143] observed that D2R scheduling information can be transmitted as physical layer signalling or higher layer signalling as there is no fundamental difference between the 2 options.
- 1 source [148] observed that option 1 allows the decoding of R2D and D2R scheduling information to occur before the decoding of the R2D payload. The D2R scheduling in L1 control enables the device to provide feedback on the (un)successful detection of the R2D payload, e.g., due to insufficient energy/power or channel/interference variation. Additionally, 6 sources [134], [148], [133], [135], [143], [137] observed that the R2D scheduling information in L1 control enables early identification of target devices and power saving for non-target devices by allowing them to skip the following R2D detection.

- 1 source [136] observed that device can decode L1 control information first which can allow device to obtain the scheduling information (e.g. TBS rated information) before data part reception in option 1 and if device knows from control information that the following data is not for it, the data reception can be skipped.

- 1 source [147] observed that option 1 allows the decoding of R2D control information separately from the subsequent R2D reception, which enables devices to feedback when subsequent R2D reception is failed and also enables identification of target devices before the subsequent R2D reception.

- 1 source [150] observed that Option 1 enables the separate decoding of R2D control information before R2D data reception, which supports feedback if R2D reception fails. Additionally, L1 control can carry D2R scheduling info, allowing devices to avoid the non-intended R2D reception.

- 1 source [144] observed that with Option 1, L1 R2D control information may include important information needed to decode the data portion of the PRDCH signal. This requires L1 R2D control information part to have a fixed payload size and self-decodable with separate CRC.

6.1.2 D2R

Reference signals including DMRS, PTRS, SRS, are not considered for D2R. CSI feedback and autonomous SR are not considered for L1 D2R control information.

6.1.2.1 Waveform and modulation

For D2R by backscattering, the waveform is provided by the carrier wave, see Clause 6.7. The D2R baseband signal (as distinct from the internal or external carrier wave) is non-OFDM.

For all devices, the following D2R baseband modulations are studied:

- OOK

- Binary PSK

- Binary FSK, as MSK (and not GMSK)

OOK and BPSK for baseband modulation are feasible for D2R for all devices.

- Source [73] reports that BPSK performance with coherent receiver highly depends on the CFO value and D2R transmission duration after a D2R amble.

Sources [20], [28], [45], [33] report that MSK is feasible in some way:

- [20], [28], [53], [53] say it is feasible for all devices, for example when it is implemented with multiple impedances switching

- [45] say that it would be implemented as square-wave MSK for devices 1 and 2a, and sine-wave MSK for device 2b

- For device 1 and 2a this type of MSK does not have continuous phase

- [20] say that benefits include lower sidelobes than OOK and BPSK, and lower BER than OOK and same BER as BPSK

Sources [22], [19], [26], [24], [25], [27], [40] report that MSK is either infeasible or should be deprioritized for all devices.

- [22], [26], [24], [25], [19], [27], [40] say that MSK is less spectrally efficient than OOK and BPSK because there are issues due to poor phase accuracy in the device

- [22], [24], [19], [25], [27] say that MSK would increase reader and device complexity

- [25] say that MSK performance for device 2b would materially degrade due to CFO

- [54] say that it is difficult to modulate MSK signal using impedance switching due to the implementation complexity, including frequency mapping and phase continuation. [63] say that if multiple impedances switching are applied to maintain the phase continuity, it violates the principle of low device cost.

2SB modulation is feasible for D2R transmission for all devices. Feasibility and necessity of 1SB modulation would depend on impacts due to issues including: device-side filtering requirements (i.e. image suppression), RF resource usage / spectral efficiency, etc.

Device 2b internally generates a carrier wave with only a single tone. Device 2b does not use a '2 single tone' carrier wave. See Clause 6.8.2 relating to these terms.

6.1.2.2 D2R line coding

The line codes studied for D2R are Manchester encoding, FM0 encoding, Miller encoding, and no line coding. FM0 encoding is deprioritized. A chip corresponds to one modulated symbol at least for OOK and BPSK.

For Manchester encoding, the bit-to-chip mapping is: bit 0→chips{10}, bit 1→chips{01}. For FM0 encoding, according to Figure 6-8 and Figure 6-9 of [6]. For Miller encoding, according to Figure 6-12 of [6].

Sources [22], [26], [25], [28], [49], [38] report that Manchester codes perform better than Miller codes by 0.5dB to 5.5dB @10% BLER.

Sources [22], [26] [49] report that Manchester codes perform better than FM0 by 0.2dB to 2.5dB @10% BLER. According to sources [25], [28], Manchester codes have a similar performance as FM0.

Sources [22], [33] state that the power of Miller coding is more concentrated in the low-frequency region which is adjacent to the carrier wave frequency, as compared with FM0 code and Manchester code which leads to worse interference handling capability for Miller codes.

Source [45] state that performance of BPSK square wave is better than Miller/FM0 by ~3-4 dB @10%BLER with coherent demodulation.

6.1.2.3 D2R channel coding, repetition, CRC

6.1.2.3.1 Channel coding

For D2R, convolutional codes are studied, with comparisons to the case of no FEC. The LTE convolutional code polynomials are a reference, and other designs studied subject to:

- Constraint length, K = 8 or K = 7 or K = 6 or K = 4.
- Mother code-rate, R = 1/6, 1/4, 1/3, 1/2.

As shown in Table 6.1.2.3.1-1, it is observed that:

- Performance of other {k, m} pairs decreased 0.15 – 1.4 dB when k<7, with reported benefit of reduced reader and device complexity due to shorter shift registers.
- Performance of other {k, m} pairs increased 0.22 – 1.8 dB when k>7, with an increase in the number of shift register resulting in increased device complexity.
- Performance of other {k, m} pairs decreased 0.3 – 2.5 dB when m<3.
- Performance of other {k, m} pairs increased 0.2 – 1.8 dB when m>3.

where k = constraint length of the code, and m = 1/code-rate.

Table 6.1.2.3.1-1: Summary of study on D2R FEC

Option #	CC Design	Pros	Cons
Baseline	Constraint length 7 Code rate 1/3	[25] Decoding performance is increased by ~3dB@10% BLER, when compared to no CC but with repetitions. [25] Decoding performance is increased by ~7dB@10% BLER, when compared to no CC or repetitions.	

		[44] Decoding performance is increased by 6.23dB@10% BLER with 2RX, when compared to no CC or repetitions. [44] Decoding performance is increased by 6.42dB@10% BLER with 4RX, when compared to no CC or repetitions. [28] Decoding performance is increased by ~2dB@10% BLER, when compared to LTE CC-TBCC with code rate 1/2. [33] Decoding performance is increased by ~2.5dB@1% BER, when compared to code rate 1/2. [27] Decoding performance is increased by ~1.5dB@1% BLER, when compared to constraint length 4, code rate 1/3 [27] Decoding performance is increased by ~2.5dB@1% BLER, when compared to constraint length 6, code rate 1/3 [55] Decoding performance is increased by 3 dB@ 10% BLER with 2 RX, when compared to no CC or repetitions	
1	Constraint length 4 Code rate 1/2 – 1/4	[20] Code rate 1/2: Detection performance is increased by 3dB@10% BLER, when compared to no CC or line codes.	[26] Code rate 1/2: Decoding performance is decreased by ~0.86dB@10% BLER, when compared to constraint length 7, code rate 1/2. [49] Code rate 1/2: Decoding performance is decreased by ~1dB@10% BLER, when compared to constraint length 7, code rate 1/2. [49] Code rate 1/4: Decoding performance is decreased by ~1.4dB@10% BLER, when compared to constraint length 7, code rate 1/4. [65] Code rate 1/2, 1/3, TBCC: Decoding performance is decreased by ~1dB@10% BLER, when compared to baseline with TBCC.
2	Constraint length 6 Code rate 1/2 – 1/6	[45] Reduced reader and device complexity due to smaller number of shift registers.	[45] Code rate 1/2: Decoding performance is decreased by 0.6dB@1% BLER, when compared to baseline. [49] Code rate 1/2: Decoding performance is decreased by ~0.4dB@10% BLER, when compared to constraint length 7, code rate 1/2. [45] Code rate 1/3: Decoding performance is decreased by

			0.4dB@1% BLER, when compared to baseline. [45] Code rate 1/4: Decoding performance is decreased by 0.2dB@1% BLER, when compared to baseline. [49] Code rate 1/4: Decoding performance is decreased by ~0.4dB@10% BLER, when compared to constraint length 7, code rate 1/4. [45] Code rate 1/6: Decoding performance is decreased by 0.15dB@1% BLER, when compared to baseline.
3	Constraint length 7 Code rate 1/2 – 1/4	[28] Code rate 1/2, TBCC: Decoding performance is increased by ~9dB@10% BLER, when compared to no CC but with Manchester line codes. [28] Code rate 1/4: Decoding performance is increased by ~1.4dB@10% BLER, when compared to baseline. [49] Code rate 1/4: Decoding performance is increased by 0.5dB@10% BLER, when compared to CC with code rate 1/2. [45] Code rate 1/4: Decoding performance is increased by 0.2dB@1% BLER, when compared to baseline.	
4	Constraint length 8 Code rate 1/2 – 1/6	[26] Code rate 1/2: Decoding performance is increased by ~0.22dB@10% BLER, when compared to constraint length 7, code rate 1/2. [49] Mother code rate 1/4, code rate 1/6: Decoding performance is increased by ~1.8@10% BLER, when compared to constraint length 8, mother code rate 1/4, code rate 1/4. [49] Mother code rate 1/3, code rate 1/6: Decoding performance is increased by ~1.8@10% BLER, when compared to constraint length 8, mother code rate 1/3, code rate 1/4. [49] Mother code rate 1/2, code rate 1/6: Decoding performance is increased by ~1.7@10% BLER, when compared to constraint length 8, mother code rate 1/2, code rate 1/4.	

6.1.2.3.2 CRC

The study assumes PDRCH can attach a CRC, where the baseline design is using a 6-bit or 16-bit CRC with polynomials as per TS 38.212 [5]. A baseline of no CRC attachment is also included. For the study of CRC designs, see Clause 6.1.0.2.

6.1.2.3.3 Repetition

For definitions of repetition types, see Clause 6.1.0. For D2R, at least block-level and bit-level repetition type 1 and type 2 are studied.

Block-level repetition

For D2R repetitions, it is reported by sources [22], [21], [19], [25], [26], [27], [28], [49], [38] that block level repetitions should be supported. The following are observations regarding the different types of repetition.

General observations

- Source [22] state that detection of D2R can obtain combining gain and time diversity gain by the block repetition
- Sources [22] and [25] state that it can enable BS to terminate reception early if it successfully decodes the block.
- Source [28] state that with small D2R information block size, memory size of A-IoT devices is comparable for bit level repetition and block level repetition, but compared to block level repetition, bit level repetition would degrade power and BLER performance of D2R transmission.
- Source [21] and [19] state that block level repetition would be a TB level repetition, while [21] goes on to state that the transmission of a PDRCH multiple times may go through multiple channels showing the diversity gain/ channel averaging impact of a specific binary data. Its implementation/specification impact would be minimum.
- Source [25], [49] state that it may require larger buffer than bit level repetition at device side, depending on device implementation.

Performance comparisons

- Source [22] state that block level repetition yields ~2.5 dB performance gain compared with bit level type 2 due to the additional time diversity gain for the combination of decoding.
- Source [27] state that block level repetition provides ~4dB performance gain @1% BLER compared with bit level type 1.
- Source [49] state that block level repetition provides ~6dB performance gain @10% BLER compared with no repetitions and the performance between block level repetition and bit level repetition type 2 is the same.
- Source [25] state that the performance difference between block level repetition and bit level repetition without CW hopping is minor, while block level repetition outperforms bit level repetition with CW hopping.
- Sources [28] and [49] state that bit level repetition and block level repetition have similar performance in the AWGN channel but block level repetition could achieve more time diversity gain than that of bit level type 2 in a fading channel.
- Source [64] state that for the no FEC case, with 3 times repetition, block level repetition provides ~5dB gain at 1% BLER when compared with bit level type 1 repetition.

Bit-level repetition

It is reported by sources [49], and [38] that bit level repetitions should be supported.

- Source [22] state that for bit level type 1, the repetition data bit is decoded after FEC detection, and hence, the digital data bit cannot be combined with combining gain.
- Source [49] state that the performance of bit level type 1 repetition is degraded by ~3.5dB @ 10% BLER under the cases of AWGN and TDL-A channel when compared to bit level type 2.

- Source [22] state that since the bit level type 2 repetition is performed after FEC at the transmitter, at the receiver side, each bit would have a log-likelihood ratio (LLR) that can be combined and provided as the input for FEC decoding, thereby providing a better performance as compared to bit level type 1.

6.1.2.4 D2R bandwidths

The following bandwidths for D2R are defined for the purpose of the study:

- Transmission bandwidth, $B_{tx,D2R}$: The frequency resources scheduled by a reader for a D2R transmission from one device.
- Occupied bandwidth, $B_{occ,D2R}$: The transmission bandwidth plus the potential associated intra A-IoT guard-bands totalling $B_{guard,D2R}$. Note: this guard band is not for coexistence with NR/LTE.
- $B_{occ,D2R} \geq B_{tx,D2R}$.

For $B_{tx,D2R}$ of the D2R transmissions associated with one/each single-tone of a carrier-wave, the following figures depict small frequency-shift, with reference to Clause 6.1.2.7.1, by:

- In Figures 6.1.2.4-1 and 6.1.2.4-3, Manchester line code with option 1, and if no D2R line-code is used (i.e., by using a square-wave corresponding to the small frequency-shift).
- In Figures 6.1.2.4-2 and 6.1.2.4-4, Manchester line code with option 2, and by Miller line code.

where:

- F_c is the carrier-wave frequency, at least when externally generated.
- FS_x depict examples of amount of small frequency shift.
- For 2SB transmission, both $+FS_x$ and $-FS_x$ parts are included in Figures 6.1.2.4-1 and 6.1.2.4-2.
- For 1SB transmission, one of the upper part or lower part applies, as appropriate, in Figures 6.1.2.4-3 and 6.1.2.4-4.

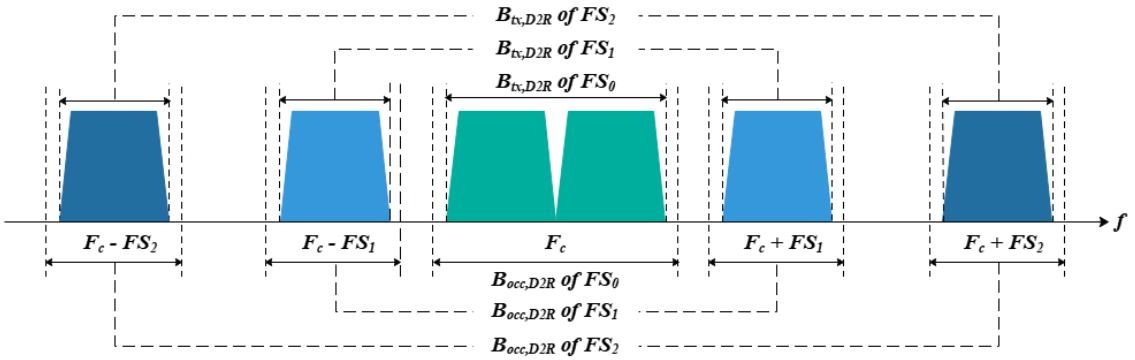


Figure 6.1.2.4-1: $B_{tx,D2R}$ and $B_{occ,D2R}$ for 2SB transmission of Manchester line code with option 1, and if no D2R line code is used (i.e., by using a square-wave corresponding to the small frequency-shift)

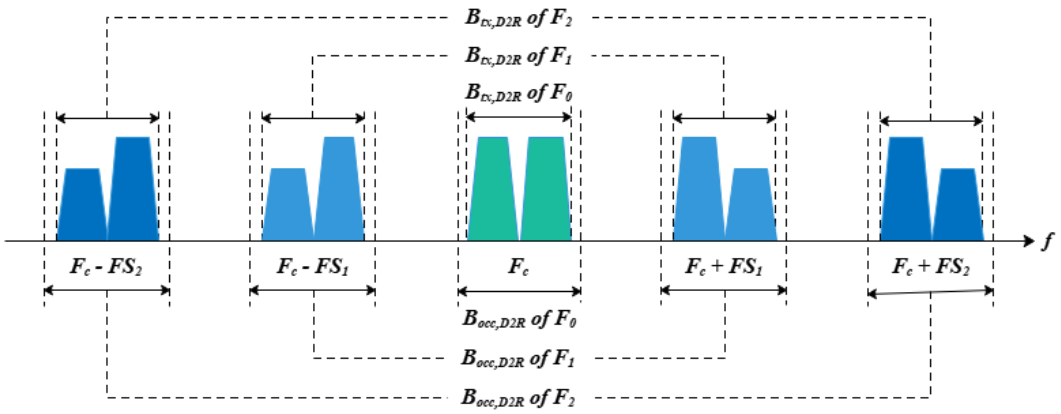


Figure 6.1.2.4-2: $B_{tx,D2R}$ and $B_{occ,D2R}$ for 2SB transmission of Manchester line code with option 2, and by Miller line code

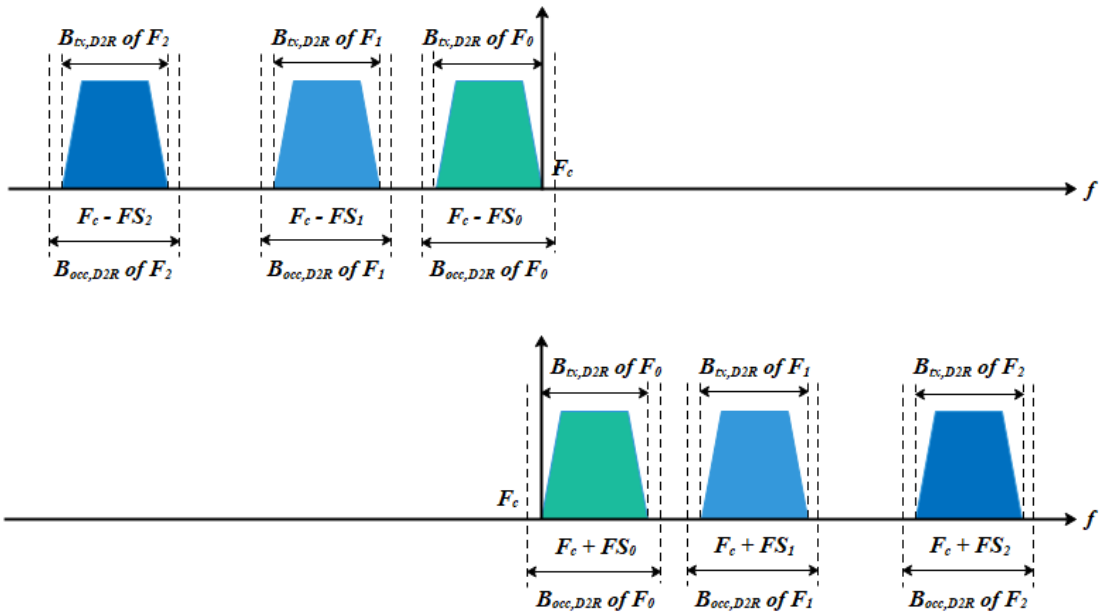


Figure 6.1.2.4-3: $B_{tx,D2R}$ and $B_{occ,D2R}$ for 1SB transmission of Manchester line code with option 1, and if no D2R line code is used (i.e., by using a square-wave corresponding to the small frequency-shift)

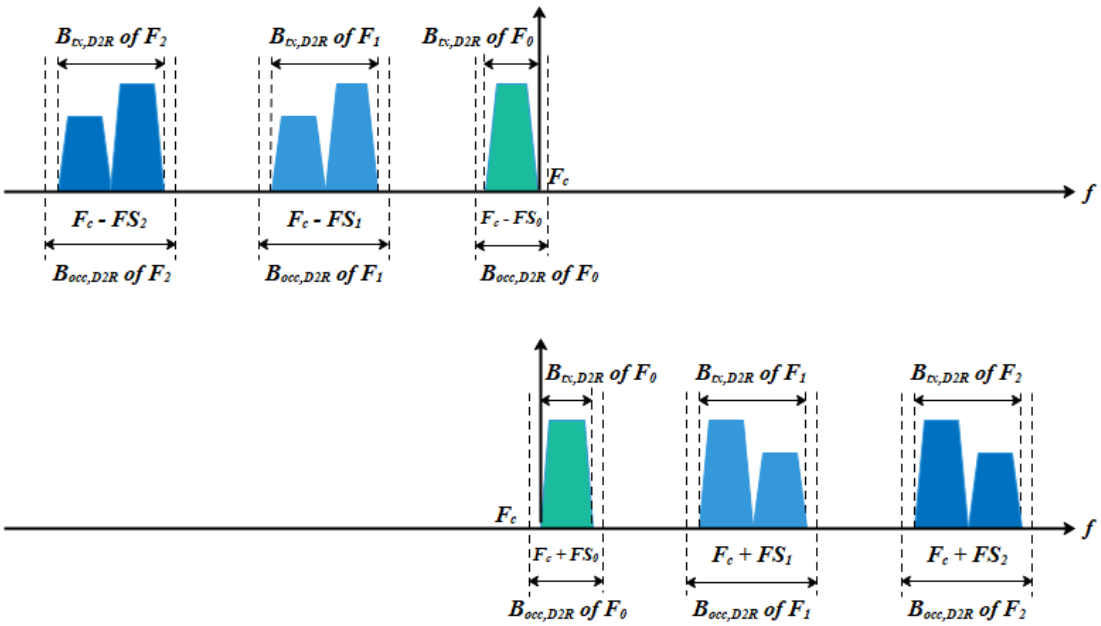


Figure 6.1.2.4-4: $B_{tx,D2R}$ and $B_{occ,D2R}$ for 1SB transmission of Manchester line code with option 2, and by Miller line code

$B_{occ,D2R}$ of the D2R transmission associated with one/each single-tone of a carrier wave would be impacted by at least the following aspects:

- A possible guard band $B_{guard,D2R}$ due to at least SFO.
- A possible guard band $B_{guard,D2R}$ due to CFO for Device 2b.

Since device 2b has internal carrier-wave generation, the 2SB and/or 1SB $B_{tx,D2R}$ and $B_{occ,D2R}$ can be affected by how it performs D2R modulation and baseband waveform generation.

6.1.2.5 PDRCH

For D2R, a physical channel PDRCH carries any higher-layer payload, the response transmitted from device to reader during the contention-based access procedure, and L1 D2R control information, if defined. PDRCH is studied via the blocks shown in Figures 6.1.2.5-1 and 6.1.2.5-2, where other clauses give their detailed descriptions and the location of the "Repetition" block depends on the repetition types described in Clause 6.1.0.1.

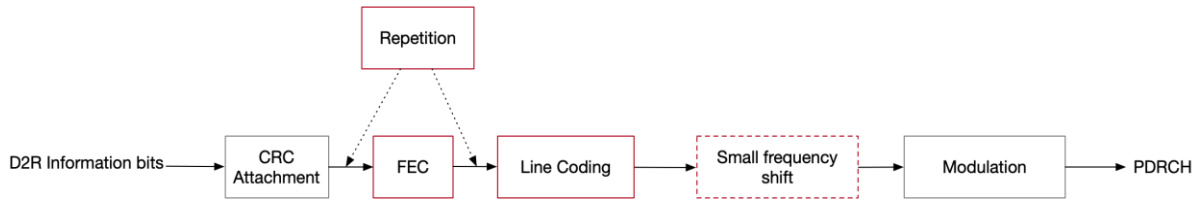


Figure 6.1.2.5-1: Single PDRCH generation at the device with line coding, for the case where FEC and repetition are present

NOTE: Presence of the block "small frequency shift" depends on the options for small frequency shift described in clause 6.1.2.7.1.

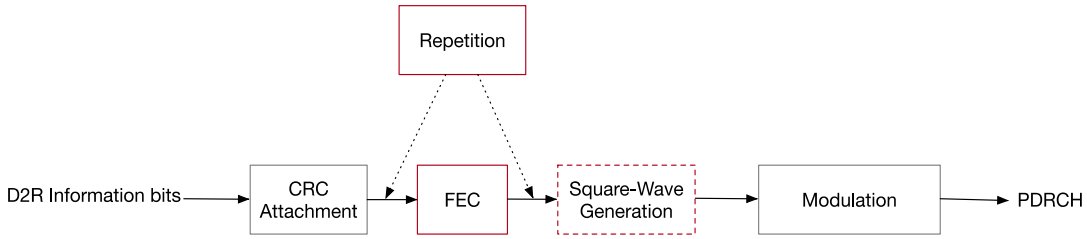


Figure 6.1.2.5-2: Single PDRCH generation at the device without line coding, for the case where FEC and repetition are present.

Scheduling information of PDRCH transmission is provided by a corresponding PRDCH.

6.1.2.6 D2R timing

6.1.2.6.1 D2R timing acquisition signal

A D2R timing acquisition signal (D-TAS), preceding each PDRCH, is included at least for timing acquisition, indicating the start of the D2R transmission in the time domain, and studied potentially for SFO estimation, CFO estimation, channel estimation, and interference estimation. A D-TAS structure using a preamble is studied, for which a binary signal is considered. The preamble is not part of PDRCH.

The following aspects have been studied for D2R preamble design:

- Autocorrelation property
  - 10 sources [131], [132], [133], [139], [140], [143], [144], [147], [148], [134] observed that the signal should have good autocorrelation properties for accurate peak detection based on the signal correlation at the reader
- Cross-correlation property
  - 7 sources [131], [133], [143], [144], [148], [134], [140] observed that the signal should have good cross-correlation properties if multiple D2R preamble sequences are considered (e.g. for multiple access schemes (if supported) for D2R transmissions).
- Line coding
  - 1 source [131] observed that line coding may impact the autocorrelation property of the sequence.
  - 1 source [132] observed that for D2R preamble, to apply backscattering, line coding can help improve the detection performance based on shifting the D2R signal's frequency location away from the carrier wave
- Sequence types (not limited to below types only)
  - M-sequence
    - 3 sources [131], [137], [139] observed that m-sequence can be considered for D2R preamble mainly owing to good correlation properties.
  - Golay sequence
    - 4 sources [133], [137], [139], [135] observed that Golay sequence can be considered for D2R preamble mainly owing to good correlation properties and availability of large number of distinct sequences and complementary pairs.
  - Walsh sequence
    - 1 source [143] observed that Walsh sequence can be considered as a candidate for D2R preamble thanks to its good auto/cross-correlation property and flexible length

There are also the following general observations:

- 1 source [132] observed can achieve 0.97% residual SFO with 98% probability under -2.5dB SNR and 0.1% MDR with [-1/8, 1/8] chip timing error with 99.05% probability under -2.5dB SNR with D2R preamble

including 2-parts with clock-like sampling frequency signal and timing-acquisition signal, having 32-length '1' sequence (encoded to 64-chip Manchester code) and 32-length sequence (encoded to 64-chip Manchester code), respectively.

- 4 sources [130], [133], [134], [137] observed that for D2R preamble with binary signal, the timing synchronization performance is highly related to the sequence length of the preamble. Furthermore, 1 source [133] observed that to achieve a BLER performance at 10%, the timing synchronization error should be less than 10%. Furthermore, 1 source [134] observed that the channel estimation performance is also highly related to sequence length. 1 source [134] observed that using a 32 bits preamble provides ~8 dB, ~5 dB performance gain than using 8 bits, 16 bits preamble, respectively. And using a 64 bits preamble provides ~2.5dB performance gain than using a 32 bits preamble.
- 1 source [144] observed that for D2R preamble with binary signal, normalized SFO estimation error of less than 10% can be achieved with a training sequence length 64 or longer. The simulated D2R preamble consisting of a Golay complementary pair can tolerate SFO up to 1% (AWGN) with up to 1 dB loss in performance for a sufficiently long preamble sequence length (32 or greater).

6.1.2.6.2 D2R end timing

For the reader to acquire the end of PDRCH transmission, the following options are studied:

- Option 1: D2R postamble immediately follows the PDRCH
- Option 2: Based on control information

6.1.2.6.3 D2R midamble

The necessity of a midamble is studied at least for the purpose of performing timing/frequency tracking or channel estimation or interference estimation.

6.1.2.6.4 Trade-offs analysis

The study analysed the trade-offs among the following D2R amble(s) options:

- Option 1: D2R preamble only
- Option 2: D2R preamble + X midamble(s), where $X \geq 1$
- Option 3: D2R preamble + postamble
- Option 4: D2R preamble + Y midamble(s) + postamble, where $Y \geq 1$

For coherent detection of PDRCH with a payload of 16 bits or 20 bits with 6-bit or 16-bit CRC, using 1/2 Manchester coding and 1/3 or 1/2 convolutional code:

- Source [102] observed that with up to 10% SFO, ~1kbps data rate, when the same amble(s) overhead is maintained, Option 3 provides comparable performance results to Option 1.
- Source [103] observed that with up to 10% SFO, ~5kbps data rate, for device 1 and with up to 1% SFO for device 2, the decoding performance with/without midamble are similar
- Source [105] observed that Option 1 is sufficient to achieve 10% and 1% BLER, with no more than 8 SFO hypotheses tested at the reader side.
  - With up to 10% SFO, ~ 5kbps data rate, the SNR needed to achieve 10% and 1% BLER is similar (~ -2dB and 4 dB) for Option 1, Option 2 of D2R preamble+1midamble and Option 3.
  - With up to 1% SFO, ~ 5kbps data rate, the SNR needed to achieve 10% and 1% BLER is similar (~ -2.8dB and 3.3dB) for Option 1, Option 2 of D2R preamble+1 midamble and Option 3.
- Source [111] observed that with perfect SFO estimation, 1kbps data rate and OOK modulation, there is no noticeable performance gain from using midamble(s) and/or postamble for PDRCH transmission, regardless of the payload size.
- Source [114] observed that the required SFO estimation accuracy to achieve 1% and 10% BLER is $<5\times10^{3}$ ppm

- With up to 10% SFO, achieving the required accuracy necessitates more than 20 SFO hypotheses at the reader side for Option 1 and 10 SFO hypotheses are sufficient for Option 3 of D2R preamble + postamble. But for Option 3 reader has to store the received samples and wait for the postamble that is after the end of PDRCH for any of SFO/channel/interference estimation, demodulation, decoding, etc.

- With up to 1% SFO, 4 SFO hypotheses are sufficient for Option 1 to achieve the required accuracy.

For coherent detection of PDRCH with a payload of 96bits with 16-bit CRC (or 6-bit CRC [108]), using 1/2 Manchester coding and 1/3 or 1/2 convolutional code,

- Sources [99], [101] and [108] observed that Option 1 cannot achieve 10% BLER.

- Sources [102], [103], [105], [111] and [114] observed that Option 1 can achieve 10% BLER.

- Sources [99], [101], [102], [103], [105], [108], [117] observed that adding additional amble improves the performance.

  - Source [99] observed that with up to 10% SFO,

    - Option 2 of D2R preamble+ 1 midamble achieves 10% BLER at SNR around -3dB, but cannot achieve 1% BLER.

    - Option 3 of D2R preamble+ postamble achieves 10% BLER at SNR around -4dB, and can achieve 1% BLER at SNR around 4dB.

  - Source [101] observed that with up to 10% SFO, Option 3 allows reader to precisely search and detect the SFO with 0.03% residual SFO at -3dB SNR TDL-A channel, achieving 10% BLER -2.44dB SNR for ~1 kbps data rate and -2.17 dB for ~6 kbps data rate. Source [101] further observed that when the reader adopts same number of SFO hypothesis based on preamble, with 1% SFO, Option 3 can achieve 10% BLER at -4.27 dB SNR for ~1 kbps and at -4.29 dB SNR for ~6 kbps, which provides 1~2 dB performance gain when compared to 10% SFO.

  - Source [108] observed that with up to 10% SFO, Option 2 of D2R preamble + 1 midamble, option 3, and option 4 of D2R preamble + 1 midamble+postamble achieve basically the same performance, the SNR for 10% BLER is 5dB for 1.25 kbps data rate.

  - Source [102] observed that with up to 10% SFO, ~1kbps data rate, and the same amble(s) overhead, Option 3 can provide 1~2 dB, 5dB performance gain for 10% BLER, 1% BLER, respectively, compared to Option 1. Additionally, Option 3 can provide ~1dB, 2dB performance gain for 10% BLER, 1% BLER, respectively, compared to Option 2 of D2R preamble +1 midamble.

  - Source [103] observed that for ~5kbps data rate, compared to option 1,

    - For device 1 with up to 10% SFO, Option 2 of D2R preamble + 1 midamble provides ~0.5 dB SNR gain at 10% BLER with TDL-A channel and ~0.9 dB SNR gain with TDL-D channel.

    - For device 2 with up to 1% SFO, Option 2 of D2R preamble + 1midamble provides ~1 dB SNR gain at 10% BLER with TDL-A channel and ~1.4 dB SNR gain with TDL-D channel.

    - Moreover, the additional ambles i.e., midamble(s) and/or postamble introduces additional overhead and postamble may prevent pipelined processing of the reception.

  - Source [105] observed that,

    - With up to 10% SFO, ~5.5kbps data rate, Option 1 achieves 10% and 1% BLER at SNR ~0.7dB and 10dB, respectively; Additionally, maintaining the same amble overhead, Option 2 (D2R preamble + 1 midamble) and Option 3 demonstrate similar performance, achieving 10% and 1% BLER at SNR around -1.7dB and 5.2dB, respectively.

    - With up to 1% SFO, ~5.5kbps data rate, Option 1 achieves 10% and 1% BLER at SNR ~ -1.3dB and 11dB, respectively. Additionally, with the same amble overhead, the SNR difference between Option 2 (D2R preamble+1midamble) and Option 3 for 10% and 1% BLER is less than 1dB, with SNRs ~ -3.1dB to -2.5dB for 10% BLER and ~3.6dB to 4.5dB for 1% BLER.

- Source [117] observed that with up to 10% SFO, ~7.5kbps data rate, there is ~6~7dB performance gap at 10% BLER and ~10.5~11.5dB performance gap at 1% BLER between option 2 of D2R preamble+111 midambles and option 1. Note that Source [117] does not use any convolutional code.

- Source [111] observed that with perfect SFO estimation, 1kbps data rate and OOK modulation, there is no noticeable performance gain from using midamble(s) and/or postamble for PDRCH transmission, regardless of the payload size.

- Source [114] observed that the required SFO estimation accuracy to achieve 1% and 10% BLER is <10^3 ppm. To achieve the required accuracy,

  - For Option 1, more than 50 SFO hypotheses at reader side are necessary for device with up to 10% SFO and 6 SFO hypotheses are sufficient at reader side for device with up to 1% SFO.

  - For Option 3, 10 SFO hypotheses are sufficient for device with up to 10% SFO, but reader has to store the received samples and wait for the postamble that is after the end of PDRCH for any of SFO/channel/interference estimation, demodulation, decoding, etc.

For coherent detection of PDRCH with a payload of 400bits with 16-bit CRC, using 1/2 Manchester coding and 1/3 or 1/2 convolutional code,

- For option 1 of D2R preamble only,

  - Sources [99], [101], [102], [104], [105], [108] observed that with up to 10% SFO, 10% BLER cannot be achieved.

  - Source [111] observed that with perfect SFO estimation, 1kbps data rate and OOK modulation, there is no noticeable performance gain from using midamble(s) and/or postamble for PDRCH transmission.

- For other amble options,

  - Source [99] observed that:

    - With accurate SFO estimation, Option 2 of D2R preamble + 4 midambles can achieve 10% BLER at SNR ~ 2.7dB but cannot achieve 1% BLER.

    - With up to 10% SFO, Option 3 cannot achieve 10% BLER.

    - With up to 10% SFO, Option 4 of D2R preamble+2 midambles+postamble achieves 10% BLER at SNR of ~0.25dB; But it cannot achieve 1% BLER. Option 4 of D2R preamble+3 or 4 midambles+postamble, achieves a 10% BLER at an SNR of around -0.2 dB, and achieves 1% BLER at SNR around 9dB or 8dB, respectively.

  - Source [101] observed that with up to 10% SFO, Option 4 of D2R preamble combined with 1 to 4 midambles + postamble, achieves 10% BLER at SNR of 2.5 dB, 1 dB, 0.8 dB, or 0.5 dB, respectively, for a data rate of around 1 kbps.

  - Source [102] observed that with up to 10% SFO,

    - Option 3 can provide ~5.5 dB performance gain compared to option 2 of D2R preamble+1midamble for 10% BLER, with the same amble(s) overhead for ~1kbps data rate.

    - Option 2 of D2R preamble+1midamble cannot achieve 1% BLER for ~1kbps data rate.

    - Option 4 of the D2R preamble+1 or 2 midamble(s)+postamble, has similar performance, it can achieves a 10% BLER at SNR of -1dB and achieves a 1% BLER at SNR of 6dB and 5dB respectively for ~1kbps data rate.

  - Source [104] observed that with up to 10% SFO,

    - Option 3 of D2R preamble+ postamble cannot achieve 10% BLER for ~7kpbs.

    - Option 4 of D2R preamble + 1 midamble + postamble can achieve 10% BLER and 1% BLER at SNR around -6dB and 0 dB, respectively for ~7kpbs data rate.

  - Source [105] observed that

    - With up to 10% SFO, maintaining the same amble overhead, both Option 2 of D2R preamble+5 midambles and Option 4 of D2R preamble+4 midambles+1 postamble demonstrate similar performance, achieving 10% BLER at SNR ~0.2dB. While for 1% BLER, the SNR for Option 2 and Option 4 is ~ 9.2dB and 12.8dB, respectively for ~5.5kpbs data rate.

    - With up to 1% SFO, maintaining the same amble overhead, both Option 2 of D2R preamble+5 midambles and Option 4 of D2R preamble+4 midambles+1postamble demonstrate similar performance, achieving 10% BLER at SNR around -1.2dB. While for 1% BLER, the SNR for Option 2 and Option 4 is ~7.8dB and 9.1dB, respectively for ~5.5kpbs data rate.

- Source [108] observed that with up to 10% SFO, Option 2 of D2R preamble+3 midambles and Option 4 of D2R preamble+3 midambles+postamble can achieve 10% BLER when the SNR is within the range of [15, 25] dB for 1.25 kbps data rate.

- Source [114] observed that the required SFO estimation accuracy to achieve 1% and 10% BLER is much smaller than 10^3 ppm. To achieve the required accuracy,

    - For Option 2 of D2R preamble+X midamble(s) where midamble inserted per every certain number of PDRCH bits (e.g., 192 bits),

    - For SFO estimation using each amble for the subsequent PDRCH bits (e.g., 192 bits), with up to 10% SFO, more than 50 SFO hypotheses are necessary at the reader side and with up to 1% SFO, 6 SFO hypotheses are sufficient at the reader side.

    - For SFO estimation based on the time gap between preamble and midamble, with up to 10% SFO, 10 SFO hypotheses are used, but reader has to store the received samples and wait for the midamble to start SFO/channel/interference estimation, demodulation, decoding, etc.

    - For Option 3 of D2R preamble+postamble, SFO estimation is based on the time gap between preamble and postamble, with up to 10% device SFO, 10 SFO hypotheses are used for reader, but reader has to store the received samples and wait for the postamble that is after the end of PDRCH for any of SFO/channel/interference estimation, demodulation, decoding, etc.

- Source [103] observes that the additional ambles i.e., midamble(s) and/or postamble introduces additional overhead and postamble may prevent pipelined processing of the reception.

For the synchronization and timing tracking of D2R transmission,

- Source [101] report that with up to 10% SFO, option 1 is not sufficient for D2R reception since the residual SFO at reader side is larger than 1%. While with option 3, the reader can precisely search and detect the SFO with a residual SFO of 0.03% at -3dB SNR TDL-A channel.

- Source [108] report that

    - For packet size of 96bits, when the SNR is increased from -4dB to 20dB, the ratio of device residual SFO over 100ppm decreases to 6% for Option 2, 3 and 4, but remains at 95% for Option 1.

    - For packet size of 400bits, when the SNR is increased from -4dB to 20dB, the ratio of device residual SFO larger than 10ppm decreases to 5% for Option 2, 3, and 4, but is still 99.6% for Option 1.

- Sources [105], [109] report that SFO estimation based on D2R preamble can achieve accurate estimation without additional ambles (midamble or postamble).

    - Source [105], [103] observed that for non-coherent detection of PDRCH, the number of SFO hypotheses and the SNR needed for 10% and 1% BLER cannot significantly be reduced for option 2, 3 and 4 compared to the option 1. Moreover, the additional ambles i.e., midamble(s) and/or postamble introduces additional overhead and postamble may prevent pipelined processing of the reception.

- Source [109] observed that

    - The coarse estimation of SFO based on the D2R preamble indicates that the SFO estimation error is less than 1% with a probability of 99.3%, and less than 0.1% with a probability of 49.9%.

    - The fine estimation of SFO based on the D2R preamble shows that the SFO estimation error is less than 1% with a probability of 99.5%, and less than 0.1% with a probability of 90.8%.

  - Reader/gNB can achieve a probability of not less than 99.5% for SFO estimation error below 1%, and 90.8% for SFO estimation error below 0.1% by receiving D2R preamble signals.

- Source [114] report that for D2R with coherent demodulation at reader, the reader needs to estimate the device clock frequency with the accuracy of 0.5% (5 * 10^3 ppm) or lower for a short message (e.g., 72 bits after CRC/coding) and 0.1% (10^3 ppm) or lower for a long message (e.g., 224 bits after CRC/coding). The source further reports that design of D2R amble(s) (e.g., overhead) and the correspondingly required number of SFO hypothesis for the estimation depend on the sampling clock accuracy that the device uses for D2R.

- Source [115] reports that transmitting 96-bit packet size with 16-bit CRC requires residue SFO after reader compensation to be 1000 ppm, and transmitting 1000-bit packet size with 16-bit CRC requires residue SFO after reader compensation to be 100 ppm.

Note: In the observations above where coherent detection is used, sources that evaluated option 3 and option 4 assumed that the postamble is used at least for time/frequency tracking and for channel estimation.

6.1.2.7 D2R multiple access

6.1.2.7.1 Multiple access schemes

Time-domain multiple access, and frequency domain multiple access at least by using a small frequency shift in baseband are studied. For device 2b, FDMA in D2R can be achieved by direct modulation of its internally generated carrier wave at the desired frequency. Whether code-domain multiple access is feasible and necessary for all devices would require further analysis which was not carried out in this study.

Time-domain multiple access is the baseline. Sources [26], [28], [20], [18], [55], state that the benefit of TDMA is the low implementation complexity for both device and reader, while the inventory efficiency may be relatively low for TDMA only, and sources [44], [39], [41], [65], [55], [72], state that the guard interval, if supported, between consecutive D2R transmissions from different devices depends on the SFO after clock calibration.

According to sources [20], [26], [28], [52], [44], [18], [22], [42], [55], the potential benefit of frequency-domain multiple access is to increase the transmission efficiency and reduce collisions, while the cons include more complicated frequency resource management and reception processing at reader according to source [26], and potentially increased power consumption for devices according to sources [28], [34], [29], [55]. It is observed that the performance of FDMA may be impacted by the following aspects.

- Large SFO of device

  - Sources [45], [49] state that large SFO (e.g. up to 10^5 ppm) produces higher BLER degradation due to inter-device interference than a smaller (e.g. up to 10^4 ppm) or the ideal case of zero SFO. Source [72] states that under the case of the large SFO (e.g. up to 10^5 ppm), two among four devices using small frequency shifts have BLER floor and cannot achieve BLER 1%. Source [49] state that under the case of the large SFO (e.g. up to 10^5 ppm), two FDMA-ed devices induce about 0.6~1dB performance loss compared to single device.

  - Source [22] state that the large SFO (e.g. up to 10^5 ppm) has little impact (e.g., ≤1dB) on the performance of FDMA between multiple devices.

  - Sources [25] think that sufficient gap between D2R transmissions should be reserved to accommodate frequency error caused by SFO/CFO

  - Sources [34] think that the required guard band size increases for higher switching frequencies for passive devices.

  - Source [58] observed that the performance gap among 10% SFO and 1% SFO (residual) is similar compared FDMA with no FDMA, e.g., about 0.5dB @ 10% BLER, i.e., the performance gap among 10% SFO and 1% SFO (residual) is irrelevant to whether FDMA scheme is used or not.

  - Source [57] state that large SFO (e.g. up to 10^5 ppm) has little impact on the link performance of FDMA, as it is observed that the performance gap between 10% and 1% SFO is negligible in the case of FDMA among 4 devices.

- Timing offset between devices

    - Sources [42] state that timing offset results in a degradation of up to ~4 dB and the loss varies for different devices depending on the level of experienced interference

- Maximum range of small frequency shift

    - Sources [41] think that the frequency gap among devices will impact the interference, which is highly depends on the small frequency shift capability, i.e., how large the frequency shift can be via small frequency shift.

- Harmonics in the backscattered signal

    - Sources [25] state that FDMA is feasible by proper frequency resource allocation not using odd harmonic frequency of FDMed D2R transmissions.

- Potential intermodulation spectral leakage in the backscattered signal

- Frequency resource collision

    - Source [68] thinks that if the guard band size between D2R transmissions is fixed, allocating passive devices with large SFO to frequency shifts closer to the A-IoT carrier frequency and either (1) passive devices with smaller SFO or (2) active devices to frequency shifts further from the A-IoT carrier frequency reduces frequency resource collision.

- Number of multiplexed devices

    - Source [49] reports that performance loss increases with the increase of device number. Besides, for FDMA detection at reader side, there is about 1.5 - 3dB performance loss from 6 FDMA-ed devices compared to single device.

    - Source [68] thinks that the potential number of multiplexed devices depends in the maximum rate of frequency switching.

    - Source [68] thinks that if the guard band size depends on the SFO capability of the device, the number of multiplexed devices can be increased if passive devices with large SFO are allocated frequency shifts closer to the A-IoT carrier frequency and passive / active devices with smaller SFO are allocated frequency shifts further from the A-IoT carrier frequency.

According to sources [28], [IIT], [29], [49], [55], CDMA can improve the resource utilization efficiency without increasing the device complexity significantly. Sources [44] thinks CDMA would help the multiplexing among readers and it can alleviate the cross-link interference. Source [29] thinks CDMA is also beneficial for the latency reduction and success rate improvement of access procedure. Source [65] thinks that the CDMA scheme is mostly used for the signals without carrying information. However, sources [35], [43], [20], [18], [22], [23], [25], [55] show concerns on the necessity and feasibility of CDMA, especially considering the limited capability (e.g., large SFO/CFO) of Ambient IoT devices and the cost (additional memories to store a set of codewords at device) versus benefits. In detail, the observations are as follows.

Impact of SFO on the performance of CDMA

In the case of large SFO (e.g., 10^5 ppm):

- Source [23] think that the orthogonality between different codes/sequences will be severely disrupted, as the large SFO will accumulate an additional sampling error of 10 points of 100 points.

- Source [25] think that the increased inter-device interference would materially degrade D2R performance, e.g., increased false alarm rate and miss detection probability, which in turn reduce spectrum efficiency even lower than the case of simple TDMA.

- Source [18] think that the accurate timing and power control required by CDMA are far-fetched for Ambient IoT devices, referring to the IS-95 CDMA system.

- Source [49] state that CDM-ed MA has comparable or better performance than FDM-ed MA under different SFO assumptions (i.e., $0/10^3$-$10^4/10^4$-10^5) and device numbers (i.e., 1/2/3). Source [49] state that CDMA by mapping Manchester encoded bit or convolutional encoded bit with 4-length orthogonal code is feasible for D2R transmission carrying 20 information bits.

- Source [59] state that convolutional codes and BPSK modulation for the CDMA improve the D2R transmission performance with multiple multiplexing devices. Source [59] state that for D2R transmission with TBS16+CRC0, the performance difference between the CDMA scheme using 4-length orthogonal code for 4 devices and the repetition scheme for single device at 1%BLER is less than 0.5dB. While for D2R transmission with TBS96+CRC16, the performance difference between the CDMA scheme with 4-length orthogonal code for 4 devices and the repetition scheme for single device at 1%BLER is also less than 0.5dB.

- Source [72] state that CDMA for msg-1 enables multiplexing large number of msg-1 sequences even when power variation is [-9dB, +9dB], while quite large number of SFO hypotheses is necessary at reader to achieve reasonable false-alarm and miss detection probabilities with the SFO of [0.1 – 1] * 10^5 ppm.

- Source [22] state that correlation properties of sequences are severely damaged with the SFO of 10^5 ppm. Besides, D2R receiver fails to estimate the SFO of each of the parallel D2R transmissions for the SFO of 10^5 ppm.

- Source [29] state that CDM of RACH preambles using either m-sequences or Gold sequences of length 63 is feasible and preambles from multiple devices can be clearly detected by the reader, even in challenging conditions (SFO = 5%, SNR = 0dB). For 1% missed-detection rate, simulation results showed that m-sequences and Gold sequences are able to achieve this performance level when SNR is about -24dB and -23dB, respectively.

In the case of relatively smaller SFO (e.g., 10^4 ppm),

- Source [72] state that CDMA for msg-1 enables multiplexing large number of msg-1 sequences even when power variation is [-12dB, +12dB]for the SFO of [0.1 – 1] * 10^4 ppm with reasonable number of SFO hypotheses at reader.

Impact of CFO on the performance of CDMA for Device 2b

- Source [22] state that codeword detection at D2R receiver fails considering non-coherent demodulation has to be used due to the quick phase rotation caused by the residual CFO of e.g. 10s or 100s of Hz after CFO estimation and correction.

Impact of timing offset between CDMed D2R transmissions on the performance of CDMA for Device 2b

- Source [48] state that binary modulated orthogonal sequence such as Golay sequence can tolerate timing error by selecting a suitable cyclic shift spacing.

- Source [29] state that it is possible to detect multiple transmitters with timing difference and power difference among devices.

- Source [41] think that poor synchronization performance in time and frequency domain of device would degrade the code orthogonality and thus results in a bad cross-correlation performance.

- Source [38] think that the different propagation delays from devices may also degrades decoding performance.

- Source [49] state that the negative impact of asynchronization can be mitigated with some enhancements, e.g. enhanced synchronization sequence and enhanced detection method at reader/BS side, e.g., sliding window based detection and setting constraints on the start of D2R transmission.

The impact of power variation between devices on the performance of CDMA is studied as follows.

- Source [72] state that the larger power variation of at least up to [-9dB, +9dB] can be addressed by reader receiver and CDMA can achieve significant throughput gain for msg-1.

- Source [49] state that the greater the disparity in received power among multiple devices, the better performance will be obtained by SIC receiver with CDM-based multiple access scheme.

Except the impact of SFO/CFO of devices, whether CDMA provides benefit is also studied as depending on the length of the orthogonal or pseudo-orthogonal code and the number of available codes for parallel D2R transmissions:

- Source [20] think that using spreading sequence can lead to transmitting a larger number of bits which can be extremely inefficient considering that the devices are extremely power inefficient.

- Source [20] think that CDMA might be too complex to implement in A-IoT devices, which might involve complexities with generating orthogonal sequences.

- Source [41] think that a large device density (e.g., 150 devices per 100 m^2 for indoor scenarios per TR) requires a long code sequence, which is challenging for the device with limited buffer size.

- Source [19] think that CDMA leads to higher power consumption and lower data rate.

- Source [25] think that the usable number of binary sequences would be much smaller due to impairment such as timing/frequency error and interference.

- Source [29] state that in comparison to RN16, when Msg. 1 is transmitted using RACH preamble m-sequences or Gold sequences, the number of usable binary sequences that can be used is large since the base sequence design from LTE and NR can be reused.

- Source [29] state RN16 cannot tolerate collision for any one of its bits. Once collided, the bit sequence is changed and became non-detectable. On the other hand, m-sequences and Gold sequences are able to tolerate transmission overlap.

- Source [49] state that CDMA by mapping Manchester encoded bit or convolutional encoded bit with 16-length or 64-length orthogonal code improve the D2R transmission performance and multiplexing capacity compared with using 4-length orthogonal code for mapping. Source [59] state that for D2R transmission with TBS16+CRC0 under the assumption of SFO being 10^4~10^5, the performance difference between the CDMA scheme with 64-length orthogonal code for 4 devices and the repetition scheme for single device at 1%BLER is about 1dB.

- Source [49] state that BPSK modulation and convolutional code for the CDMA further improve the D2R transmission performance with multiple multiplexed devices compared with OOK based modulation. Source [59] state that for D2R transmission with TBS16+CRC0 under the assumption of SFO being 10^4-10^5 and using convolutional codes and BPSK modulation, the performance difference between the CDMA scheme using 16-length or 64-length orthogonal code for 6 devices and the repetition scheme for single device at 1%BLER is less than 1dB.

6.1.2.7.2 Small frequency shifts

For OOK and BPSK, small frequency shifts are studied:

- For applying with Manchester line codes with a repetition number $R \geq 1$:

  - Option 1: Each Manchester codeword is repeated by a codeword repetition number R, within the same time duration T_b corresponding to an information bit, where $R = T_b/(2 \times \text{chip length})$, such that the amount of small frequency shift in Hz is $R/T_b = 1/(2 \times \text{chip length})$.

  - Option 2: By multiplying the Manchester codeword with a square wave corresponding to the small frequency shift, the time duration T_b corresponding to each information bit includes R number of square wave periods, where $R = T_b/(2 * \text{chip length})$, such that the amount of small frequency shift in Hz is $R/T_b = 1/(2 \times \text{chip length})$. The multiplication operation is performed as either an XOR or XNOR operation between a Manchester codeword corresponding to the information bit and the square wave for the small frequency shift.

- For applying with Miller line codes, according to Figure 6-13 of [6].

- For FM0, small frequency shift is not defined

- If no D2R line code is used, by using a square-wave corresponding to the small frequency-shift, the time duration T_b corresponding to each information bit includes R number of square wave periods generated by $2R$ OOK chips [0, 1, 0, 1 …]/[1, 0, 1, 0 ..] or BPSK chips [-1, +1, -1, +1, …]/[+1, -1, +1, -1, …], such that the amount of small frequency shift in Hz is R/Tb.

- Potential purposes include:

  - FDMA of D2R, if supported

  - CW interference avoidance, if supported

Note: Small frequency shifts for D2R are studied for the same potential purposes for MSK.

For study purposes, for BPSK and OOK, the small-frequency-shift factor is defined as the ratio of bit-length to two times the chip length. The value of small-frequency-shift factor depends on details of design, which according to proposals considered could be, e.g.:

- The Manchester codeword repetition number R within duration T_b.
- The M value of Miller in Figure 6-12 of UHF RFID standard.
- The number of square-wave periods within T_b.

Sources [18], [25], [45], [57] state that Manchester codeword repetitions within the same time duration corresponding to an information bit is equivalent to bit-level repetitions within the same duration prior to Manchester encoding. Sources [58] and [62] state that option 1 has a more concentrated spectrum, and requires lesser bandwidth as compared to Option 2. Source [62] further states that while Option 1 and option 2 show similar BLER performance for single device case, Option 1 outperforms Option 2 with FDMA, especially with presence of 10^5 ppm SFO. Option 1 can achieve additional gain for coverage evaluation due to lower effective noise power. Source [70] observed that while Option 1 has a narrower main lobe, it has higher sidelobes than Option 2.

Sources [25], [49] and [30] state that the output waveform for Manchester line codes by Option 2 introduces a phase reversal of the output waveform in the middle of the time duration corresponding to an information bit as compared to Option 1.

Sources [18], [20] [26] [45] [42] [39] state that the output waveform using Manchester line codes by Option 2, and no line codes, is equivalent to using Manchester line codes by repetition of the codewords within the same time duration corresponding to an information bit.

Source [49] shows that small frequency shift by Manchester Option 1 is more sensitive to SFO and the time difference between devices than small frequency shift by Manchester Option 2, and small frequency shift via Miller code has the worst performance compared with by Manchester Option 1 and Option 2.

6.1.2.8 Scheduling of D2R

For D2R scheduling, the following information potentially can be explicitly/implicitly indicated to the device via the corresponding PRDCH:

- Time domain resources
- Frequency domain resources
- MCS-like information
- Chip duration
- ID associated with device(s)
- Repetitions
- Midamble (if supported) related information

For each information, it is for further study whether higher-layer signalling and/or L1 R2D control signalling is used - see Clause 6.1.1.10.

6.1.2.9 D2R control information

The following 14 sources studied feedback information regarding the (un)successful reception of the R2D command at the device and provided following observations:

- 5 sources [132], [141], [134], [139], [145] observed that L1-based ACK/NACK is unnecessary or infeasible for ambient IoT, since all D2R transmission are controlled and activated by the reader. Additionally, 1 source [134] observed that it is infeasible for devices to report NACK since it cannot distinguish whether the R2D transmission is dedicated for it and the resource allocation for D2R NACK if the R2D transmission is not successfully detected. And ACK can be transmitted in higher layer signaling.
- 11 sources [135], [136], [143], [144], [147], [148], [131], [149], [150], [139], [138] observed that ACK/NACK feedback improves transmission reliability, when the BS or intermediate node sends R2D transmission to the

device in the command procedure and if the device does not respond ACK/NACK, the BS or intermediate node cannot determine whether the transmission is successful. 1 source [142] observed that in the use cases of command where some device actions are required and no D2R message needs to be fed back, only D2R ACK feedback in response to the PRDCH is beneficial for the reader to know that the reception of the PRDCH is successful and the required actions have been done by the device.

6.1.3 Timing relationships

A-IoT processing time aspects are studied in terms of the following timing relationships:

T_{R2D_min}: Minimum time between a R2D transmission and the corresponding D2R transmission following it.

T_{D2R_min}: Minimum time between a D2R transmission and the corresponding R2D transmission following it.

T_{D2R_max}: Maximum time between the D2R transmission and the corresponding R2D transmission following it, so that the R2D transmission timing is expected to be within [T_{D2R_min}, T_{D2R_max}], when a R2D transmission in response to a D2R transmission is expected for A-IoT Msg2 response to A-IoT Msg1 for the A-IoT device. See clause 6.3 for message descriptions.

$T_{R2D_R2D_min}$: Minimum time between two different consecutive R2D transmissions to the same A-IoT device.

$T_{D2R_D2R_min}$: Minimum time between two different consecutive D2R transmissions from the same A-IoT device.

For the time interval between a R2D transmission and the corresponding D2R transmission following it, there are two options studied:

Option 1: Define a maximum time T_{R2D_max} between a R2D transmission and the corresponding D2R transmission following it, so that the device transmits D2R transmission within [T_{R2D_min}, T_{R2D_max}].

Option 2: The corresponding D2R transmission timing T_{R2D} following a R2D transmission is determined based on the control information in the R2D transmission, where $T_{R2D} \geq T_{R2D_min}$.

6.1.4 Random access

From the perspective of the physical layer, at least when a response is expected from multiple devices that are intended to be identified, an A-IoT contention-based access procedure initiated by the reader is used, for which at least slotted-ALOHA based access and FDMA, are studied. The study of FDMA includes how the frequency domain resources for Msg1 are allocated, and how a device determines that frequency-domain resource allocation.

The response transmitted from the device to the reader during this procedure is transmitted on PDRCH.

A R2D transmission triggering random access determines X time domain resource(s) for D2R transmission(s) for Msg1, where each D2R transmission for Msg1 occurs in one time domain resource of the X time domain resource(s).

The study includes:

- X=1 and X>1 and X>=1. The maximum value of X>1 should be set considering the device implementation complexity, device power consumption, the resource usage efficiency affected at least by SFO, and inventory latency.
- Size(s) for resource allocation in the time domain.
- Determination of the X time domain resource(s) by the device.
- Addressing timing errors for adjacent time domain resources due to residual SFO of the device.

For Msg2 transmission in response to multiple Msg1 transmissions initiated by a R2D transmission triggering random access, the following options have been identified:

Option 1: A PRDCH for Msg2 transmission corresponds to an A-IoT Msg1 received from one device.

Option a: the starting time for Msg2 monitoring is separate for each Msg1 resource.

Option b: the starting time for Msg2 monitoring is common for multiple Msg1 resources.

Option 2: A PRDCH for Msg2 transmission corresponds to multiple A-IoT Msg1 received from different devices, the starting time for Msg2 monitoring is common for multiple Msg1 resources.

To determine the time duration for monitoring Msg2 in response to multiple Msg1 transmissions initiated by a R2D transmission triggering random access, the following options have been identified:

Option 1: Predefined

Option 2: Indicated in the R2D transmission triggering random access

Note: Option 1 and Option 2 may not be mutually exclusive.

For a device to determine the reference time for the starting time for monitoring Msg2 in response to multiple Msg1 transmissions initiated by a R2D transmission triggering random access, the following options have been identified:

Option 1: End of time domain resource where the device transmitted the Msg1

Option 2: End of the last of X time domain resource(s) determined for Msg1 transmission(s)

Option 3: End of the R2D transmission triggering random access

Option 4: End of a R2D transmission carrying Msg2

Note: The reference time is used to determine the starting time for Msg2 monitoring, it does not mean that the starting time always equals the reference time.

For Msg3 from multiple devices in response to a given set of one or multiple Msg2 transmission(s) during access procedure, FDMA and/or TDMA of D2R transmissions are studied, including how the frequency and time domain resources for Msg3 are allocated.

6.1.5 Topology 2

For resource allocation and/or controlling of intermediate UE in topology 2 for R2D and D2R transmissions via the Uu interface between the BS and intermediate UE, following two options have been studied:

- Option 1: Higher-layer signalling is used.
- Option 2: L1 and higher-layer signalling are used.

From RAN1 perspective, detailed aspects on the two options are left up to work item phase.

The following observations related to the two options have been made:

- Four sources [129], [124], [120], [125] observed that Option 1 can save overhead compared to Option 2.
- One source [119] observed that the timing gap between R2D message and D2R message is not extended, and inventory rate is comparable with UHF RFID and furthermore, it is observed that with Option 1, the standard impact includes no or less impact on L1 signalling for resource allocation and controlling of the intermediate UE.
- Six sources [129], [126], [123], [118], [128], [127] observed that Option 2 with L1 signalling provides better resource utilization compared to Option 1.
- 1 source [137] observed that if UE determines the resource used for A-IoT transmission within A-IoT resources configured by the network, option 1 provides better resource efficiency and option 2 enables faster A-IoT resource adaptation compared to option 1. And it is further observed that if the resource allocation from NW is for each R2D and/or D2R transmission and is indicated by a L1 signaling of option2, option 2 may affect the A-IoT communication timing relations and causes substantial overhead between NW and UE reader and may lead to resource inefficiency as NW lacks direct information of channel condition between A-IoT device and UE reader.
- 1 source [139] observed that option 1 can save signaling overhead b/w gNB and intermediate UE compared to option 2 but may cause resource waste issue. For resource allocation of intermediate UE, Option 2 can provide better resource efficiency compared to Option 1 but requires more signaling overhead b/w gNB and intermediate UE.

- 1 source [140] compared option 1 and option 2 in terms of signaling overhead, signaling latency, flexibility and resource utilization. It is observed that signaling overhead is higher with option 2, but latency is higher with option 1. Option 2 may provide higher flexibility and better resource utilization compared to option 1.
- 1 source [142] observed that option 1 only requires the standardization workload for adding new higher layer signaling and does not impact the current DCI design for UEs, but the resource utilization efficiency of option 1 may be lower than that of option 2. Option 2 can provide better resource usage efficiency than that of option 1, but Option 2 requires standardization effort on renewing DCI design for intermedia nodes.
- 1 source [148] observed that option 2 is too costly in terms of overhead and latency if using DCI-based scheme to trigger each R2D or D2R, but it is beneficial for flexible resource sharing between AIoT communications as well as the coexistence with the legacy transmission.
- 1 source [149] observed that Option 2 would increase the latency of A-IoT transmission. Furthermore, it is observed that the two options are not necessarily exclusive.
- 1 source [136] observed that the signaling overhead of option 2 is higher and the spec impact in RAN1 is lager. If each R2D or D2R transmission resource is scheduled by DCI, option 2 will bring high transmission latency and low transmission efficiency.
- 1 source [134] observed that Option 2 has larger standardization impacts, for example the DCI design. And Option 2 may affect the A-IoT timing definitions.

6.2 Device (un)availability

The following directions, not for down-selection, are studied regarding the potential impact of device unavailability due to energy harvesting:

Direction 1: Reader does not provide information to a device regarding when the device may become available/unavailable.

Direction 2: Reader can provide information to a device based on which the device may become available/unavailable.

6.2.1 Direction 1 solution details

The following solution descriptions, observations or analysis or evaluations, and specification impacts if any, were provided during the study.

Table 6.2.1-1: Details from Source [76]

Source	Details
Source [76]	Solution description By implementation, the reader can provide the device enough time to charge. In addition, the reader can arrange the sequence of messages so that a device can perform its communications / operations spread out over time if possible to allow a device to remain available. Observations or Analysis or Evaluations N/A Specification impacts, if any Additional benefits in availability of "ID associated with device(s) intended for the reception of R2D" when it is located at the front of the PRDCH, using a L1 R2D control information.

Table 6.2.1-2: Details from Source [77]

Source	Details
Source [77]	Solution description The reader does not provide information to a device regarding when the device may become available/unavailable and the device behavior will entirely depend on the available energy in its energy storage and be highly implementation dependent

	Observations or Analysis or Evaluations Observation #1: RF energy harvesting is a suitable way to deploy Device 1 with capacitor size that is sufficient to complete at least an inventory round. Observation #3: RF energy harvesting is not a suitable way to deploy Device 2. Other ambient sources, e.g., indoor lighting, vibration, etc., which are pervasive and continuous, and provide higher energy conversion, should be used. Observation#4: It is expected that Direction 1 has no or less specification impact from both reader and device perspective. Specification impacts, if any There are no direct specification impact unless some mechanisms need to be specified to mitigate the potential drawbacks of Direction 1.

Table 6.2.1-3: Details from Source [97]

Source	Details
Source [97]	Solution description The reader may be able to request the A-IoT devices to provide information on the current energy status and/or the ability on energy harvesting such as charging rate and capacity, which could be different depending on the device types. If needed, the reader can predict the device availability, but it does not mean the reader provide information on device availability to a specific device. Observations or Analysis or Evaluations Basically, the A-IoT device availability/unavailability cannot be controlled by the reader, which is up to its current energy status at the device. The AIoT-devices may have the following operation state. - ON state: transmission, reception, and a clock running - OFF state: energy harvesting only - SLEEP sate: energy harvesting, monitoring, maintaining memory, and a rough clock running The A-IoT devices may switch the state depending on its energy harvesting status. Once it receives a triggering signal, its status may be switched from OFF to ON, but it may not be controllable by the network at least for the passive devices. If the reader wants information relevant to energy harvesting, the device needs to provide information. Specification impacts, if any The spec needs to support necessary procedures and signaling such as indication on the A-IoT device to report the energy status. How to utilize the information is up to the reader implementation.

Table 6.2.1-4: Details from Source [78]

Source	Details
Source [78]	Solution description The reader by implementation maintains the maximum time between the Paging messages as ≤ Tdischarge/2, ensuring that the device can switch to its ON duration after charging completely, receive the Paging message and complete the entire inventory procedure before running out of charge. The device is expected to be fully charged and then switch to an ON duration to be able to receive the Paging message from the reader. The interval of the Paging messages would ensure that the device receives at least one Paging message before it runs out of charge. Once the device completes the inventory procedure, the device maintains a state flag using an ultra-low power latch during the OFF duration such that the device does not perform inventory repeatedly after it completes a charging cycle. If the device runs out of charge during the inventory procedure, or misses out on Paging messages during its OFF or charging duration, the device is simply expected to charge fully and then monitor for the next Paging message.

	Any failed reception of Paging messages during the device charging time is handled by the higher layer re-access procedure. The multi device inventory latency for direction 1 is 18.3 seconds, to reach 100% of the devices. Observations or Analysis or Evaluations The leakage current for a given capacitor size has to be much lesser than the output of the energy harvester in order to achieve satisfactory charging. For capacitor sizes larger than 10 µF, the leakage current is in the order of micro amperes, resulting in a slow charging or continuous discharging using RF energy and hence is an unsuitable energy source for such capacitors. For device 1, the discharging time is in the order of a few seconds, providing the devices with adequate time to complete an inventory round. For device 2, the discharging time is in the order of milli seconds, which is too short a duration for the device to complete an entire inventory procedure. For device 2 to be able to increase its discharge times, capacitors with larger capacitance, in the order of 1000 µF can be used, however, it would be infeasible to use RF energy as a source due to the high leakage current. RF energy harvesting is not a suitable way to deploy devices with short discharge times, e.g. device 2, due to the following reasons: - The discharge time is in the order of milli seconds, which is not enough for the device to complete an entire inventory round. - Increasing the capacitor size results in a high leakage current, resulting in a slow charging or continuous discharging, deeming RF energy as an unsuitable energy source. - Other ambient sources, e.g. indoor lighting, vibration, etc. which are pervasive and continuous, and provide higher energy conversion, should be used. RF energy harvesting is a suitable way to deploy devices with long discharge times, and the reader's scheduling implementation is able to appropriately handle the duration of an inventory round. For calculating the multi device inventory latency, the charging time taken by the first set of devices that receive the first Paging message and are to participate in the first inventory round is taken into account when calculating the latency. Specification impacts, if any No specification impact. Possible with proper implementation of devices with adequate charge/discharge times and capacitor size, as well as of the reader with the required time gap between the Paging messages, there is no specification impact.

Table 6.2.1-5: Details from Source [78]

Source	Details
Source [78]	Solution description The device provides information to the reader pertaining to e.g. the remaining charge or charge status, such that the reader can optimally transmit the Paging messages to ensure that all devices are inventoried in the shortest amount of time. The reader is expected to transmit a request message to the device, responding to which the device would provide the reader with a remaining charge status report. The reader would then group the devices with similar amount of charge left and based on this grouping, the reader can then transmit the Paging message accordingly for each of these groups to participate in different inventory rounds. For devices that did not have enough charge remaining to complete an inventory round, the reader would wait for a favorable report such that it can be inventoried in subsequent rounds. Observations or Analysis or Evaluations Solution 2: The device cannot transmit remaining charge-related information to the reader without the reader requesting for such a report.

	Transmission of the request message and corresponding report from each of the devices multiple times to the reader would cause a substantial overhead. Additional hardware is required for the device to be able to determine the amount of charge remaining in its capacitor. Specification impacts, if any Solution 2: Significant specification impact. Request message and report might require new signals or channels. Definitions of these messages are required. Changes to the architecture to include additional hardware for the device to measure the remaining charge.

Table 6.2.1-6: Details from Source [104]

Source	Details
Source [104]	Solution description The energy harvest and transition between states (ON and OFF) are achieved by device implementation The device indicates to the reader before entering the OFF state. Observations or Analysis or Evaluations For device 1, reader can assume that device 1 with 10uF capacitor is always available during an inventory or command procedure, if the inventory/command procedure is immediately after the device is powered on. For device 2, by implementing a larger capacitor, device 2 can be assumed available during an inventory or command procedure, if the inventory/command procedure is immediately after the device is powered on. RF energy harvesting is not enough for larger capacitors (e.g., 100uF), other energy sources should be considered. Before entering the OFF state, the device may indicate to the reader which may be beneficial for the reader to perform the subsequent scheduling or transmission. Specification impacts, if any no specification impact is needed for energy harvest and transition between states. Indication design (e.g., from device to reader)

Table 6.2.1-7: Details from Source [95]

Source	Details
Source [95]	Solution description: - Signaling of energy status reporting - Energy aware segmentation Observation: - RAN1 is still discussing whether device 1 can support sleep state in addition to the ON and OFF state, however, it is clear that the device 2b can support ON; OFF and sleep state. - The RF based energy harvesting eliminates the negative consequence of device unavailability due to smaller capacitor size, - Even though, reader does not provide any information related to the device availability or unavailability as part of direction#1, reporting of coarse energy status helps reader schedule or not schedule subsequent R2D transmission to the device. - ***For 96 bits payload, the transmission time increases from 14.9ms to 89.6ms when the data rate decreases from 6kbps to 1kbps.*** - With duty-cycle and energy harvesting, the device can be sustainably operated in case of 6kbps with storage smaller of less than 0.5uF, while it requires at least 1.5uF storage to be sustainably operate in case of 1kbps.

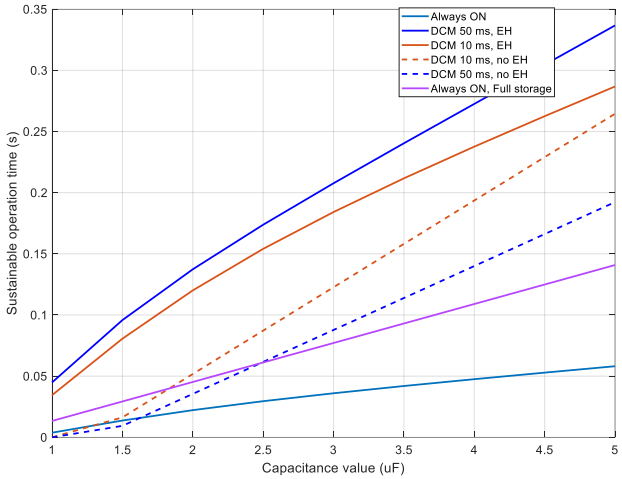
	- With always ON condition, the device can be sustainably operated in case of 6kbps with storage smaller of less than 2uF, while it requires at least 10uF storage to be sustainably operate in case of 1kbps. - ***For 400 bits payload, the transmission time increases from 69.3ms to 388ms when the data rate decreases from 6kbps to 1kbps.*** - With duty-cycle and energy harvesting, the device can be sustainably operated in case of 6kbps with storage smaller of less than 1.5uF, while it requires at least 6uF storage to be sustainably operate in case of 1kbps. - With always ON condition, the device can be sustainably operated in case of 6kbps with storage smaller of less than 6uF, while it requires very high storage capacity storage to be sustainably operate in case of 1kbps.  **Figure 1: Sustainable operation time of device with duty cycle and w/o duty cycle, distance = 12m** - Since device may run out of charge during the transmission of larger payload size, retransmitting the entire packet requires more charging time. Hence, device can transmit part of the packet using segmentation based on the available energy while indicating to the reader about the subsequent packets to be transmitted after charging. Specification impact: - Device signaling its coarse energy status as part of the Msg-1 or Msg-3 transmission to the reader. Reader can schedule or not schedule subsequent R2D transmission to the device. Device behavior when the expected R2D transmission is not received. - RAN2 can discuss the energy-based segmentation.

Table 6.2.1-8: Details from Source [94]

Source	Details
Source [94]	Solution description A-IoT device is assumed to have two states: ON, OFF. • ON state supports transmission, reception for communication and does not support energy harvesting considering device has a single antenna for both communication and RF energy harvesting in TDMed manner. • OFF state supports at least energy harvesting and does not support at least transmission, reception for communication, maintaining a timer. There are wo schemes for a device switching between ON and OFF states: • Scheme 1 (OFF when fully discharged): The device switches to OFF when fully discharged and turns ON when there is stored energy and the incoming RF power is above the activation threshold. • Scheme 2 (OFF when energy < X%): The device switches to OFF when the stored energy falls below a threshold X, e.g., X=90% and turns ON when the stored energy exceeds that threshold and the incoming RF power is above the activation threshold. Scheme 2 has advantages over scheme 1, such as shorter device downtime for energy harvesting and sufficient energy for communication once the target PRDCH is received. However, its implementation complexity and cost may be higher.

	There are two options for a reader to transmit the PRDCH e.g., A-IoT paging triggering the random access (RA). • Option 1 (P-R2D Tx): Reader transmits the PRDCH periodically. • Option 2 (A-R2D Tx): Reader transmits the PRDCH multiple times on demand. Option 1, where the paging triggering the RA occurs periodically, remains transparent to the device as the reader cannot control or be aware of the device's state. Additionally, the reader can use option 2 to decrease the likelihood that the device will miss the paging when it is in the OFF state. Observations or Analysis or Evaluations Evaluations and observations can be found in Table 6.2.2-1 [94]. Specification impacts, if any For both Schemes, it is expected to have no or minimal specification impacts.

Table 6.2.1-9: Details from Source [79]

Source	Details

Source [79]

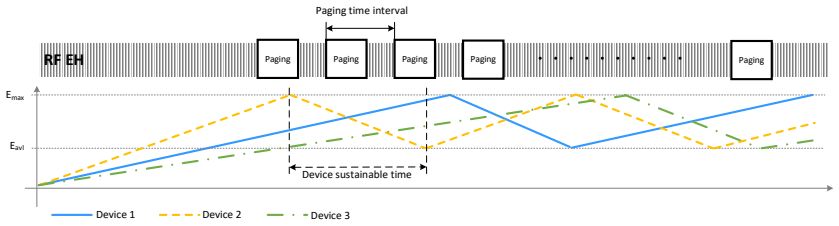
Solution description

In Direction 1, the reader does not provide information to a device nor be aware of a device regarding when the device may become available/unavailable. The device is assumed to have two states:

- ON state: Device can at least perform transmission and reception, run MHz sampling clock, and retain memory.
- OFF state: Device cannot perform transmission and reception, no clock running. Device can harvest energy. Device can retain memory of being inventoried.

From reader perspective, it transmits paging messages with time interval shorter than the device sustainable time. To avoid devices running out of energy before successfully completing the inventory, the paging time interval can be comparable to half of the device sustainable time.

From device perspective, it becomes available when it is fully charged, and then keeps performing Tx/Rx procedure until it runs out of its available energy. Once a device finishes the inventory, it needs to retain a memory of being inventoried so that it does not re-enter the inventory procedure.



Observations or Analysis or Evaluations

A. Results for Device 1

Figure 3.2-2 and Figure 3.2-3 show the evaluation results of percentage of successfully inventoried devices with respect to the inventory time for Device 1, where Figure 3.2-2 provides results assuming {R2D, D2R} data rate of {56, 160} kbps and Figure 3.2-3 provides results assuming {R2D, D2R} data rate of {14, 40} kbps. The inventory completion time considering 99% and 100% of devices successfully complete inventory are summarized in Table 3.2-1. As the sustainable time of Device 1 is 4.75 s, and therefore two paging time interval of 2 s and 4 s are considered in the evaluation.

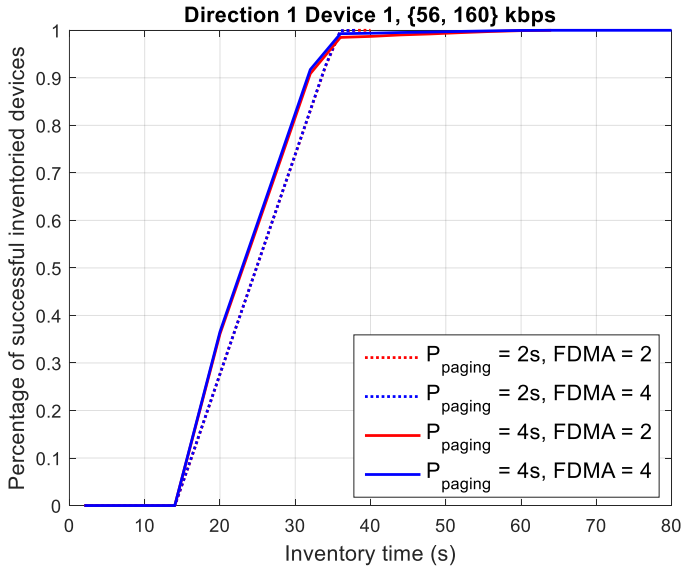


Figure 3.2-2: Percentage of successful inventoried devices with respect to inventory time for Direction 1 Device 1 in case of {R2D, D2R} data rate of {56, 160} kbps.

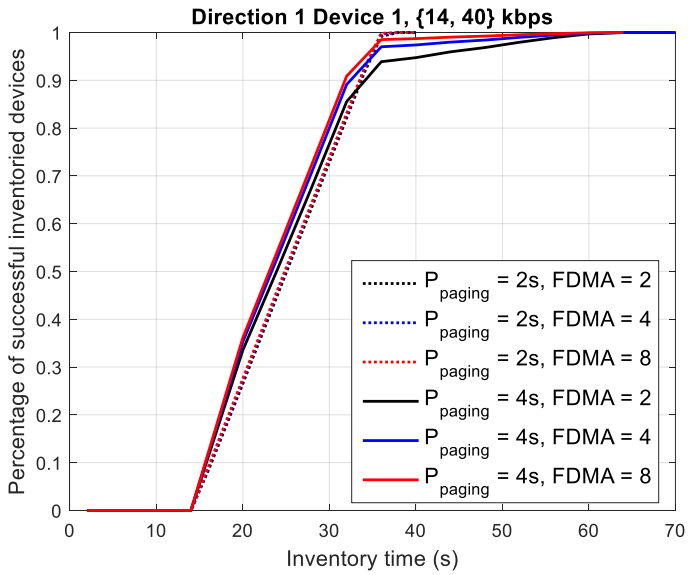


Figure 3.2-3: Percentage of successful inventoried devices with respect to inventory time for Direction 1 Device 1 in case of {R2D, D2R} data rate of {14, 40} kbps.

Table 3.2-1: Inventory completion time for Direction 1 Device 1

Successful rate	{R2D, D2R} Data rate	FDMA resources	Inventory completion time	
			Paging time interval = 2 s	Paging time interval = 4 s
99%	{56,160} kbps	2	36.40 s	46.40 s
		4	36.40 s	48.40 s
	{14,40} kbps	2	36.40 s	62.40 s
		4	36.40 s	58.40 s
		8	36.40 s	46.40 s
100%	{56,160} kbps	2	39.42 s	64.02 s
		4	39.17 s	59.64 s
	{14,40} kbps	2	40.34 s	69.63 s
		4	39.83 s	67.03 s
		8	39.43 s	63.96

B. Results for Device 2

Figure 3.2-4 and Figure 3.2-5 show the evaluation results of percentage of successfully inventoried devices with respect to the inventory time for Device 2, where Figure 3.2-4 provides results assuming {R2D, D2R} data rate of {56, 160} kbps and Figure 3.2-5 provides results assuming {R2D, D2R} data rate of {14, 40} kbps. The inventory completion time considering 99% and 100% of devices successfully complete inventory are summarized in Table 3.2-2. Two paging time interval of 20 ms and 40 ms are considered considering that the sustainable time of Device 2 is only 47.5 ms.

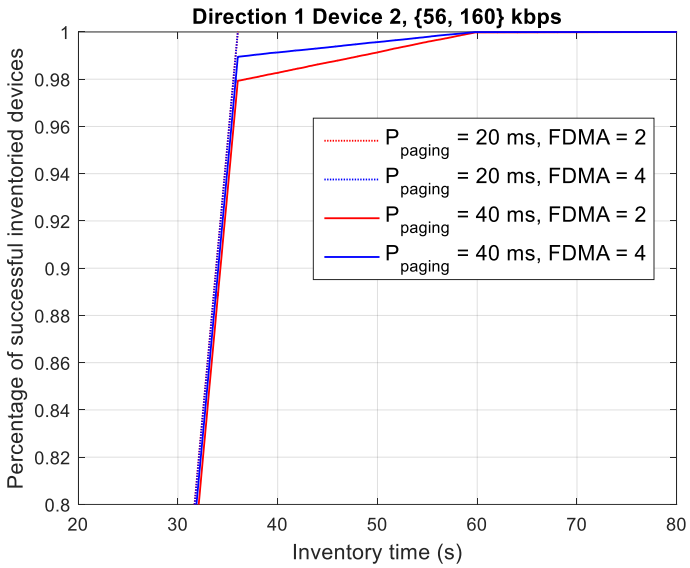


Figure 3.2-4: Percentage of successful inventoried devices with respect to inventory time for Direction 1 Device 2 in case of {R2D, D2R} data rate of {56, 160} kbps.

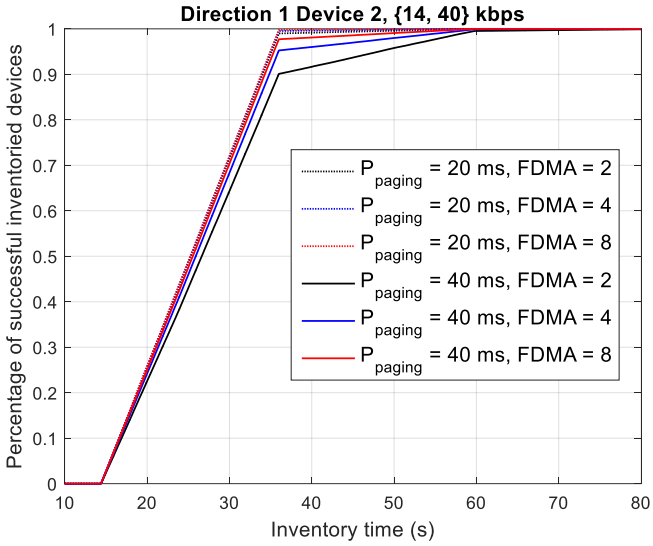


Figure 3.2-5: Percentage of successful inventoried devices with respect to inventory time for Direction 1 Device 2 in case of {R2D, D2R} data rate of {14, 40} kbps.

Table 3.2-2: Inventory completion time for Direction 1 Device 2

Successful rate	{R2D, D2R} Data rate	FDMA resources	Inventory completion time	
			Paging time interval = 20 ms	Paging time interval = 40 ms
99%	{56,160} kbps	2	35.80 s	48.44 s
		4	35.80 s	37.16 s
	{14,40} kbps	2	36.34 s	58.36 s
		4	35.90 s	55.32 s

			8	35.84 s	49.32 s
	100%	{56,160} kbps	2	35.99 s	57.38 s
			4	35.99 s	53.55 s
		{14,40} kbps	2	55.57 s	72.66 s
			4	51.18 s	62.32 s
			8	46.35 s	57.34 s

Observation 3: For Direction 1, when the paging time interval is comparable to half of the sustainable time, at least 99% of devices can be successfully inventoried before running out of energy and no device goes through recharging.

Observation 4: For Direction 1, with the increase of paging time interval, when the paging time interval is comparable to the sustainable time, there are devices cannot complete inventory before running out of energy and therefore the inventory completion time sufficiently increases.

Observation 5: For Direction 1, as the sustainable time of Device 1 can be up to several seconds, it is practical for a reader to transmit paging messages with a time interval comparable to half of device sustainable time, e.g., in the level of seconds.

Observation 6: For Direction 1, as the sustainable time of Device 2 can only last for tens of ms, a reader has to transmit paging messages very densely, e.g., in the level of tens of ms.

Specification impacts, if any
For Direction 1, no specification impact is expected at reader side nor device side.

Table 6.2.1-10: Details from Source [80]

Source	Details
Source [80]	Solution description The reader can set a proper value of the interval in-between two paging messages, to ensure all the devices can complete at least one entire inventory procedure. Observations or Analysis or Evaluations The reader may or may not have the prior information on each device's corresponding discharging time, which may rely on the deployment; The R2D signalling overhead may also need to be considered. Specification impacts, if any No

Table 6.2.1-11: Details from Source [81]

Source	Details
Source [81]	Solution description In Direction 1, an A-IoT device would become available after fully charged and become unavailable at the time of charging ort the energy in the storage running out. A-IoT devices can perform transmission and reception for communication in available state and perform energy harvesting in unavailable state. The switching between device availability and device unavailability is based on implementation without specification impact. In this case, carrier wave should always exist to provide energy for A-IoT devices. Observations or Analysis or Evaluations Observation 3: In Direction 1, carrier wave should always exist on to provide energy for A-IoT devices and is not guaranteed in the A-IoT field deployments for all A-IoT devices. The availability/unavailability of A-IoT devices are not fully controlled by the network during A-IoT operation due to the limitation of device within the carrier wave coverage. Specification impacts, if any In Direction 1, the switching between device availability and device unavailability is based on

	implementation without specification impact.

Table 6.2.1-12: Details from Source [82]

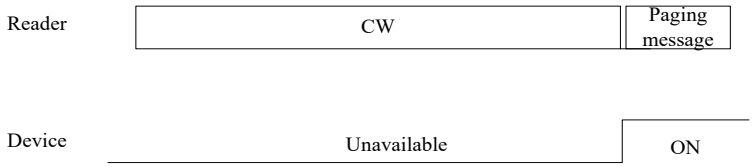
Source	Details
Source [82]	Solution description Method 1: The Reader sends continuous CW before paging message to ensure that all devices achieve a high energy storage state. After the continuous CW, the reader transmits the paging message. Devices can harvest energy from the CW while in unavailable state. Upon the end of the continuous CW through power detection, devices switch to the available state to receive the subsequent paging message and enter the random access process. Figure 8 illustrates the transmission and reception between the reader and the device in Method 1.  **Figure 8 Transmission and reception between reader and device in Method 1** Observations or Analysis or Evaluations Method 1: Before random access, Method 1 based on Direction 1 can ensure that devices receive the paging message at a high power level and enter the random access process Specification impacts, if any Method 1: The energy node transmits the CW before paging message. The device is required to detect the end of continuous CW and receive subsequent paging message.

Table 6.2.1-13: Details from Source [82], [102]

Source	Details
Source [82], [102]	Solution description Method 2: - The reader can transmit multiple paging messages in one access round. Wherein, each paging message includes an indicator that indicates whether the paging message is the initial paging message of the access round. - During one access round, the initial paging message indicates the total number of occasions that the access round contains and other paging messages indicate the remaining number of occasions after them. - If a device receives the initial paging message, it performs random access based on the number of occasions indicated by the initial paging message.

- If a device receives a paging message for the first time, and this paging message is not the initial paging message, it will perform random access based on the number of occasions indicated by this paging message.
- If a device receives a paging message that is neither the first paging message it has received nor the initial paging message, it will decrease its stored occasion counter.

To facilitate the comparison between the existing slotted-ALOHA and Method 2, Figures 9 and 10 respectively illustrate the function of paging message and R2D trigger (i.e. counter decrement command) of the two methods. It can be observed that Option 2 replaces some R2D triggers with paging messages in one access round.

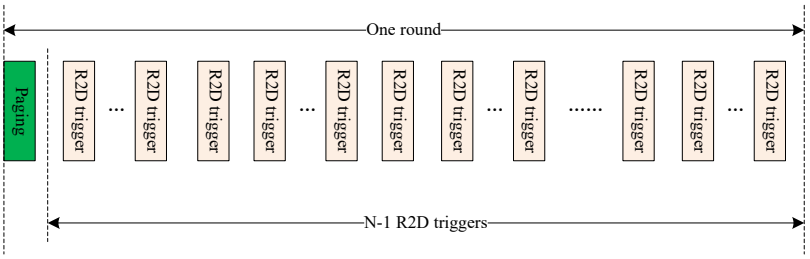


Figure 9 Function of paging message and R2D triggers in slotted-ALOHA process

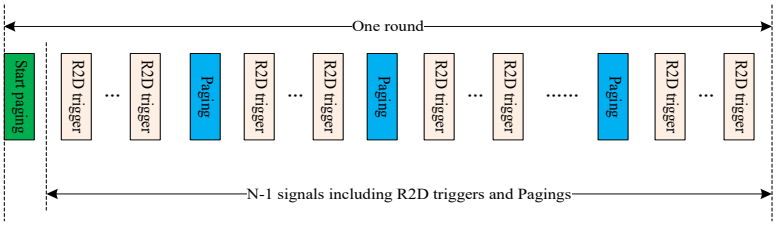
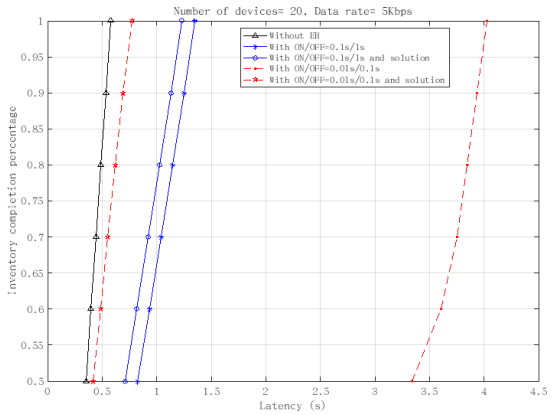


Figure 10 Function of paging messages and R2D triggers in Method 2

Observations or Analysis or Evaluations

Method 2:



- Before random access, Method 2 based on Direction 1 can increase the probability of devices receiving the paging message without incurring additional R2D overhead.
- For the inventory process of 20 devices, device (un)availability leads to an increase in random access latency. The Method 2 based on Direction 1 can significantly mitigate the latency impact caused by device (un)availability.

Specification impacts, if any

Method 2:

	Define the paging message transmission mechanism and relevant signaling indication. Specify the device's behavior after receiving a paging message.

Table 6.2.1-14: Details from Source [83]

Source	Details
Source [83]	Solution description Since the reader does not provide any information (e.g., scheduling or coordination) that can be used by a device for switching from one operating state to another (e.g., for the purpose of power saving or energy harvesting), the state transition in a device is expecting to be based on its implementation choice. For example, - For a device that supports only 2 states (ON / OFF), the state transition is expecting to be based on device energy storage level (i.e., switch to the ON state when the storage level is charged above a X% and switch to the OFF state when the storage level is dropped below a Y%, where X could be 50/80/100 and Y could be 20/10/0 depending on device implementation choice). That is, a device could be power-ON only when it is fully charged and power-OFF only when it is fully dis-charged. Therefore, it is not possible / infeasible for a device to perform energy re-charging in the middle of an inventory or command process, and as such, the device should complete an inventory or command process as soon as power-ON and before it is fully dis-charged. - For a device that supports 3 states (ON / OFF / SLEEP), the timing/determination of state transition between ON and SLEEP will be also sorely base on its implementation choice (since the reader does not provide any information). It is mentioned in the past, a device may employ a duty-cycle based ON / SLEEP state switching/transition. For example, for every duty-cycle period (which is to be determined by the device implementation), the device switches to the ON state for a small amount of time to monitor/receive any paging message from the reader and switches back to the SLEEP state for a longer duration for energy re-charging. The device repeats this duty-cycle state switching operation to keep its energy storage level high, until it detects during a ON period a paging message that triggers an inventory or command process, then it remains in the ON state to complete the process. In order to cater for the above possible state transition behaviours, it is mentioned in one solution that the reader periodically transmits a paging message triggering an inventory or command process with a periodicity equal to ½ of the expected sustainable operating time (ON duration) of a device (where the triggered process also finishes within the periodicity). If the device power-ON or switches to the ON state with a remaining energy level that is or near 100% full, even if the next paging message is transmitted ½ of the expected sustainable operating time later, the remaining energy level should still be able to sustain device operation for the entire process. Observations or Analysis or Evaluations For A-IoT devices that support only 2 states (ON / OFF), the reader should frequently transmit a paging message triggering an inventory or command process in order to ensure a device does not spend a lot of energy monitoring the channel for a paging message. Although the above-described solution is in-principle workable from the view point of a device sustainable operating time, however, this does not take into account when the number of devices trying to access the same process is more than the capacity that can be handled within a process, then TX collision will occur and some devices would not complete the process. Subsequently, these devices may re-attempt in the next period / triggered process, take up time and frequency resources, but not be able to finish the process. Furthermore, these device with failed attempts will take 10's of seconds to be re-charged to 100%, wake-up and try a new process again until they complete one round of inventory or command process. As seen, this type of mechanism is prone / exposed to TX collisions and long latency delays. For A-IoT devices support 3 operating states (ON / OFF / SLEEP), the reader does not need to frequently transmit the paging message as devices are always in a high energy storage level. But the drawback would be the inventory / command process completion time for multiple devices. Even to make an inventory of just one device, the reader may need to transmit the

	paging message multiple times, as the device ON timing and duration is not influence by any information from the reader. Reader transmission of a paging message that falls within the device ON period will be purely coincident. To make an inventory of a large number of devices, at the beginning the paging transmission may easily coincide with one or multiple devices, which is fine. But when only a few devices remained, the coincident may not happen often. Based on the above solution description and analysis, the following observations are reached for Direction 1: - When the device supports only two operating states (ON / OFF), - the reader should frequently transmit a paging message triggering an inventory or command process in order to ensure a device does not spend a lot of energy monitoring the channel for a paging message, and - for a mechanism where a paging trigger is periodically transmitted and the periodicity is ½ of the expected sustainable operating time of the device, this type of mechanism may be prone / exposed to TX collisions and long latency delays to complete a round of inventory or command process with the reader. - When the device supports three operating states (ON / OFF / SLEEP), - when a device employs a duty-cycle based ON / SLEEP state transition scheme, device energy storage could always remain in a high level, but the inventory / command process completion time for multiple devices may be long, since the timing of a paging message transmission that falls within the ON duration of a device is based on coincident. Specification impacts, if any Behaviour(s) and restriction(s) for autonomous state transition in a device may need to be specified (e.g., TS 38.213 or 214) in order to ensure the overall operation of an inventory or command process (A-IoT communication) is not negatively impacted. Detailed behaviour(s) and restriction(s) should be further studied and discussed.

Table 6.2.1-15: Details from Source [84]

Source	Details
Source [84]	Solution description Reader scheme: Transmits PRDCH multiple times. Device scheme: Either continuous On until fully discharged or continuous On until discharged to a threshold. Observations or Analysis or Evaluations Reader does not control the device state Specification impacts, if any no specification impact

Table 6.2.1-16: Details from Source [86]

Source	Details
Source [86]	Solution description The device is assumed to always have sufficient energy to receive commands from the reader at any time. Observations or Analysis or Evaluations The power consumption of the device must be less than the achievable rate of energy harvesting. The range from device to CWE is limited by the power consumption of the device. CWE power consumption and interference are greater for this direction than for direction 2. Specification impacts, if any Specification impacts are minimal.

Table 6.2.1-17: Details from Source [89]

Source	Details
Source [89]	Solution description When CW and communication is same frequency, - To use only this direction can work. When CW and communication are different frequencies, - After one transaction (inventory or command) or multiple of transactions, reader is not required to call the successful communication device again. In such situation, the reader does not provide information to a device regarding when the device may become available/unavailable as next inventory or command are sufficiently future like 1 hour or 1 days later. - On the other hand, during one transaction or multiple of transactions, when communication is possible needs to be aligned between the reader and the devices. If not, the reader is not able to initiate the communication. This operation is controlled by the timer between ON and RETENTION. Observations or Analysis or Evaluations If the device is able to know no more transaction is expected after successful communication, the device can go for OFF. This may be realized by "no intention for the more paging for now" kind of the message by the reader. Or other implicit means. If the device is not sure no more transaction or not even after successful transaction, the device needs to repeat the transition between ON and RETENTION by the timer for possible future paging when CW and communication are different frequencies. Specification impacts, if any What information needs to be kept in volatile Memory and what timers are required to be run during RETENTION. When CW and communication is same frequency, how the device distinguish them using the start-indicator part. When CW and communication is different frequency, in what condition, the device is not required to repeat the transition between ON and RETENTION.

Table 6.2.1-18: Details from Source [90]

Source	Details
Source [90]	Solution description Define a minimum and a maximum time between two PRDCHs providing random access triggering messages such that the device is not required to monitor a potential R2D transmission during the minimum time, if the device does not participate the corresponding random access round. During the minimum time, the device may transition into the PS state. After the minimum time, the device is require to monitor a potential R2D transmission at least until the maximum time from the preceding PRDCH reception. Define a minimum time for a device, who successfully finished the random access procedure, such that the device is not required to monitor a potential R2D transmission during the defined minimum time. The minimum time may be dependent on the count of the current random access round. Define a minimum time for a device to continuously monitor a potential R2D transmission, whenever the device becomes active after unavailable periods from energy harvesting, such that a reachability from a reader to the device is guaranteed. Observations or Analysis or Evaluations N/A Specification impacts, if any N/A

Table 6.2.1-19: Details from Source [114]

Source	Details
Source [114]	Solution description Applicable primarily to Device 1 (ultra-low-power devices) Relies on the device's long availability duration (e.g., >10 seconds) Reuse simple slotted-ALOHA based access mechanism R2D includes end timing information for scheduled D2R or R2D transmission Devices make autonomous decisions based on their remaining energy and the scheduled transmission duration Observations or Analysis or Evaluations Device 1 operates at ultra-low power (~1μW) and can rely solely on RF energy harvesting. Long availability duration (>10 seconds) is possible with a 10 μF capacitor for Device 1. This approach is suitable when the device's available duration is long enough to cover the period of slotted-ALOHA. Devices can estimate their remaining energy using the formula P1 = (M1 – N1)/M1, where M1 is the minimum available duration and N1 is the counted slot number after entering the available state. By providing end timing information, devices can autonomously decide whether to participate in scheduled transmissions based on their energy status, potentially avoiding D2R or R2D failures due to energy shortage. Specification impacts, if any Device: Definition and test requirement on M1, counting behavior for N1, skip and go unavailable if identifying energy shortage for upcoming scheduling, reporting behavior of remaining energy information Reader: Command periodicity for initial contact with the device based on M1 value, end timing information in R2D transmissions for scheduled D2R or R2D transmissions

Table 6.2.1-20: Details from Source [91]

Source	Details
Source [91]	Solution description RAN1 assumes the device has enough energy for at least for one message (e.g., Msg1) then leave it to RAN2 to add signaling from the device to indicate to the reader if it does not have enough energy for follow up messages (e.g., energy status report in Msg1). Observations or Analysis or Evaluations RAN1 can assume the device has enough energy, with a fallback energy status report. Specification impacts, if any RAN2 specification impact, if RAN2 decides to add an energy status report.

Table 6.2.1-21: Details from Source [93]

Source	Details
Source [93]	Solution description Each device reduces a duration of contiguous unavailable time and reader transmits paging frequently Observations or Analysis or Evaluations This direction would mean that each device wakes up and starts monitoring paging just when available energy is sufficient. If unavailable time of a device is long, inventory completion time will be degraded. For example, a paging is transmitted right after a device becomes long unavailable state for energy harvesting. In this case, the inventory session cannot be completed until the device wakes up, notices a re-paging after a lot of re-paging, and detects

satisfaction of response conditions.

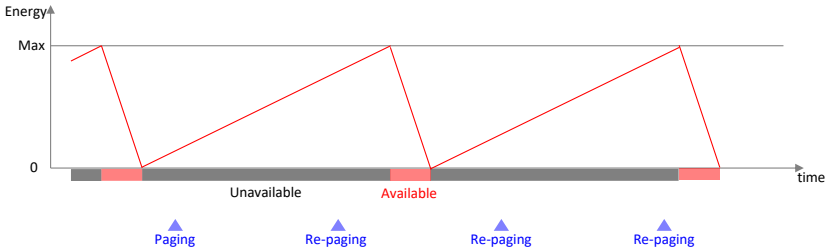


Fig: An example of Direction 1.

To avoid this kind of situation, a possible way is that each device reduces a duration of contiguous unavailable time and reader transmits paging frequently so that device notices existence of an on-going inventory session earlier. For spec impact, such a kind of device behavior is defined with a threshold of energy storage, and it may be better to define device behavior after a paging/re-paging is detected in consideration of missing paging/re-paging a lot of times.

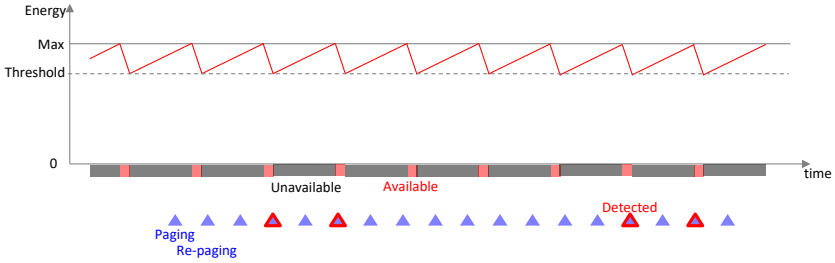


Fig: A possible way of Direction 1.

Specification impacts, if any
E.g., a threshold of energy storage is defined to do state transition for energy harvesting.

E.g., device behavior to monitor paging/re-paging and to perform Msg1 TX is defined in consideration of missing a lot of paging/re-paging.

Table 6.2.1-22: Details from Source [96]

Source	Details
Source [96]	Solution 1 description Reader continues to try reaching A-IoT device until it becomes available Observations or Analysis or Evaluations The simple solution for a reader to communicate with an unavailable A-IoT device is to wait until the device is available after charging. Without the knowledge/controllability of device availability/unavailability, the reader has to repeat/retry the communications over time. For example, suppose the case where a reader wants to do inventory for A-IoT devices. The reader should repeat msg-0 over time so that whenever the target device is available, it can receive the msg-0. If there is no msg-0 received during an available time duration, the device is not aware of the inventory procedure from the reader. Therefore, the reader should send msg-0 at least once per device available time duration. The reader should send msg-0 at least once per these time durations. Unavailable time duration for charging depends on Rx power, RF EH efficiency, capacitor size, etc. Therefore, the reader is not able to identify when and how long each device becomes available. The best thing that a reader can do is to continue the tries as much as possible and as long as possible.

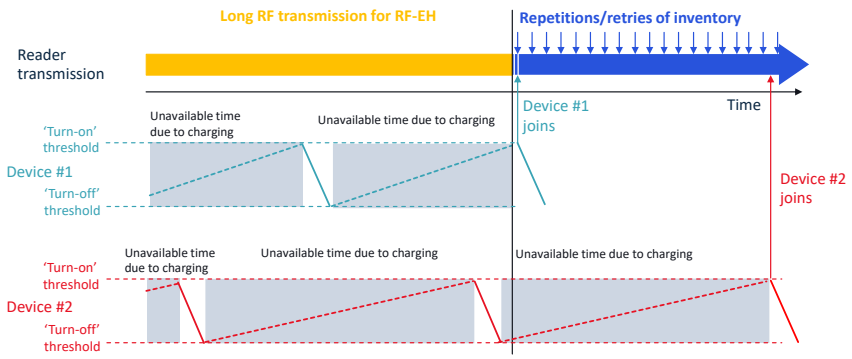


Fig. A1 Direction 1: a reader has no idea and cannot control when a device is available/unavailable

A typical inventory procedure is a slotted-ALOHA based contention-access procedure. If there are a large number of devices, a number of devices compete with each other in the slotted-ALOHA msg-1 transmissions. In such case, the inventory procedure for a device may not be able to be completed within the available time duration. Once the device becomes unavailable, the device will not be available for tens or hundreds of seconds due to charging. As such, congestion of the contention-access procedure would further increase the time duration where a reader has to try.

Observation A1:

- Direction 1 is the case where a reader has no idea and cannot control when a device is available/unavailable due to the amount of energy in the storage.
- As a solution, a reader should try to reach the target device in blind manner.
  - Take an inventory procedure as an example, reader should transmit msg-0 multiple times until msg-1 is received, where the msg-0 transmission should be at least once per device available time duration.
- If an A-IoT communication activity (e.g., message exchanges for contention-based access procedure) cannot be completed within an available time duration of a device for some reason (e.g., congestion of slotted-ALOHA), the device becomes not accessible (unavailable) due to charging again, which leads additional delay for the reader.

Solution 2 description
Improved energy harvesting efficiency by deployment/implementation

Observations or Analysis or Evaluations
The other solution is to improve energy harvesting efficiency by deployment/implementation. For example, network can deploy energizers densely such that all the devices are always fully energized and available. Or, device can implement energy harvesting sources such as solar, vibration, etc, which increases the form factor of the A-IoT devices.

Note that large capacitor size for a device does not resolve the issue. With a large capacitor size, the device available time duration gets longer. However, it also prolongs the device unavailable time duration for charging. There must be cases where some devices has (almost) zero available energy when a reader starts repeat/retry of inventory procedure, and such devices needs longer charging time until it becomes first time available after the start of the reader's inventory.

Observation A2:

- Another solution for direction 1 to improve energy harvesting efficiency by deployment/implementation.
  - A network can deploy energizers densely such that all the devices are always fully energized and available.
  - Devices can implement energy harvesting sources such as solar, vibration, etc, which increases the form factor of the A-IoT devices.
- Using large capacitor size cannot be a solution for direction 1

	- In direction 1, reader cannot make sure all the devices are fully charged when the reader starts the overall inventory procedure Specification impacts, if any No specification impact is envisioned for direction 1.

Table 6.2.1-23: Details from Source [92]

Source	Details
Source [92]	Solution description When the energy storage is full or reaches a certain level, or the device receives a RF power larger than an activation threshold, the device wakes up to detect the R2D signal and then transmit the D2R signal. When the procedure of inventory/command is completed or the energy storage is not enough to process the following communication, the device switches to OFF and only performs the energy harvesting. Observations or Analysis or Evaluations N/A Specification impacts, if any How and when to transmit CW for energy harvesting, for example, controlled by the reader Energy harvesting in Topology 2, for example, when the RF energy source and the UE are the same node, the impacts of different RRC states (IDLE/Inactive/Connected mode) Transmitting power of RF energy source

Table 6.2.1-24: Details from Source [112]

Source	Details
Source [112]	Solution description - Device states are not defined in A-IoT system. - The energy harvesting can be handled by device implementation. That is, if the energy is insufficient, the device power is turned off, and device performs energy harvesting. After that, if the energy is charged, the device power is turned on, and device performs TX/RX. - Reader can provide inventory round triggering command (e.g., Query command) frequently, allowing devices with relatively short available time to perform inventory round. - For example, reader can set a small Q value (which determines the total number of slots) for each inventory round so that each inventory round can be terminated quickly. - When the reader provides the inventory round triggering command, available devices will participate in the inventory round. - In other words, the device determines whether it can remain available during the corresponding inventory round. Observations or Analysis or Evaluations - Small specification impact. - R2D resource overhead may increase because the reader needs to transmit inventory round triggering command frequently. - Device available time is relatively short because the energy harvesting is not considered until the device is turned off. Specification impacts, if any

	- Potential information feedback on the energy state (e.g., remaining energy) of the device.

Table 6.2.1-25: Details from Source [117]

<table>
<tr><th>Source</th><th>Details</th></tr>
<tr><td>Source [117]</td><td>
Solution description for direction 1

State description
<table>
<tr><th>State</th><th>Description</th><th>Transition event</th><th>Device</th></tr>
<tr><td>ON</td><td>Transmission and reception
running a clock for timing
maintaining the memory</td><td>Transit to ‘ON’ state:
· the charging energy is higher than an energy threshold or a certain percentage, and the incoming RF power is higher than an activation threshold
Transit from ‘ON’ state:
· the charging energy is lower than an energy threshold or a certain percentage, or fully discharged
· Receives an indication from Reader to switch to the other state.</td><td>Device 1
Device 2a</td></tr>
<tr><td>OFF</td><td>Energy harvesting</td><td>Transit to ‘OFF’ state:
· the charging energy is lower than an energy threshold or a certain percentage or fully discharged
· Receives an indication from Reader to switch to ‘OFF’ state.
Transit from ‘OFF’ state:
· the charging energy is higher than an energy threshold or a certain percentage</td><td>Device 1
Device 2a</td></tr>
</table>
Scheme:
• Reader transmits the control information based on the PRDCH to the devices.
➢ Option 1: Reader transmits the scheduling information on the PRDCH periodically.
➢ Option 2: Reader transmits the scheduling information on the PRDCH multiple attempts based on the reader’s on-demand scheduling.
• The device switches to ON when the charging energy is higher than an energy threshold and the incoming RF power is higher than an activation threshold, then the device receives the reader’s control information in a certain period or one attempt of the multiple attempts.
• Reader stops its attempt or period transmission when a response has been received by the reader side or a max attempt amount has been achieved.

Observations
• High communication delay due to multiple transmission attempts.
• Low communication efficiency.

Specification impacts, if any
• No or minimal specification impacts.
</td></tr>
</table>

Table 6.2.1-26: Details from Source [116]

Source	Details

Source [116]	Solution description The reader does not provide information to a device nor be aware of a device regarding when the device may become available/unavailable. The device can report the energy status to the reader. In this case, the device is assumed to have two states i.e. ON and OFF state. Observations or Analysis or Evaluations Reduce the time interval between the paging messages based on the device capacitor size and the threshold value of the device The device can report its energy status during the random access Scheduling for R2D/D2R can be based on the reported energy status Specification impacts, if any Define the paging message transmission mechanism and relevant signaling indication. Device behavior for e.g. reporting of energy status Content of R2D/D2R control information

6.2.2 Direction 2 solution details

The following solution descriptions, observations or analysis or evaluations, and specification impacts if any, were provided during the study.

Table 6.2.2-1: Details from Source [76]

Source	Details
Source [76]	Solution description A device is (nearly) fully charged when it first becomes available. A device monitors R2D transmissions to determine whether it should participate in a round of communications. If a device is not among the device or a group of devices scheduled to participate, it can use a next scheduled opportunity field in the R2D transmission to determine when it can/should be available. This field indicates the next round of communications. A device then may become unavailable and can harvest energy. When the device determines it is near the next scheduled opportunity, and it has sufficient energy for a round of communications, a device can become available to process the PRDCH. If a device is (nearly) fully charged, it may become available at any time. Observations or Analysis or Evaluations N/A Specification impacts, if any Availability of "ID associated with device(s) intended for the reception of R2D" is located at the front of the PRDCH, using a L1 R2D control information. Availability of a next scheduled opportunity for another round of communications, using a L1 R2D control information. The next scheduled opportunity field can represent a number of events, such as R2D transmissions.

Table 6.2.2-2: Details from Source [77]

Source	Details
Source [77]	Solution description Support for wake-up signal in the SLEEP state Observations or Analysis or Evaluations Observation #1: There is a need to study a solution for extending the sustainable operation

	time for devices with short discharge times and long charge time by harvesting only RF energy. Observation #2: The state transitions to or from the OFF states can be based on an energy level of the device, which can be on implementation issue. On the other hand, the state transitions between ON and SLEEP states need to be studied as it involves a monitoring behavior of a device, which may be a potential scope of specification. Observation #3: Following benefits are achievable by reader controlling the device's state transition between the ON and SLEEP: - Avoiding very long unavailable time duration after the reader starts inventory: Since the consumed energy in the energy storage in each available time duration is quite limited, the time duration that a device is unavailable for energy harvesting is short. - Keeping large amount of energy in the storage all the time: Whenever a device joins inventory, the energy storage is (almost) full. This avoids the case where a device joins inventory but the remaining amount of energy is not sufficient. Specification impacts, if any At least a wake-up signal needs to be specified. The wake-up signal support should be specified in a way so that it is transparent for (and can be ignored by) devices that do not support it.

Table 6.2.2-3: Details from Source [78]

Source	Details
Source [78]	Solution description The following states and the corresponding functions of the device for each of the states are considered, where their duration is determined by device implementation.: - ON state – Device is fully operational – it can transmit and receive communication messages, run a clock, maintain memory. - OFF state – Device is not operational – it cannot transmit or receive any messages, nor run a clock. In this state, the device performs energy harvesting and maintains an inventory flag. - SLEEP state – Device is semi-operational – it cannot transmit or receive communication messages, but it can run a clock and perform energy harvesting. It may also maintain memory. The objective of this solution is not to force all the devices to adhere to a single configuration to periodically shift between the ON and SLEEP states, but rather to ensure that all devices shift to ON state when the Paging messages are being transmitted. To achieve this, we propose an on-demand shifting of states where the reader controls the device to switch to the ON state from the SLEEP state using an indicator which would inform the device of the duration it has to remain in the SLEEP state. The device is expected to perform energy harvesting and charge itself during the OFF state, and when its fully charged, it would shift to the ON state and listen for the indicator to be transmitted by the reader. The indicator would indicate the time left for the inventory procedure to start and for the device to receive a Paging or Query message. The reader transmits the indicator in an aperiodic manner, controlled by reader implementation, such that the device has to be in an ON state for only a short time, enabling it to switch to the SLEEP state until the next Select or Query message is expected to be received to conserve energy. At the indicated time, the device will shift to the ON state and perform the inventory procedure, ensuring that the device would have enough energy to be able to complete the procedure during the ON state. Within an inventory round, if the device failed to receive a Query message, the device can remain in ON state until it receives the indicator, after which it can go to SLEEP state until the next expected Query message, based on the indicated duration. The following are the possible transitions that the device would be expected to perform.

	- OFF to ON state – Transition based on when the device is fully charged. - ON to OFF state – Transition based on when the device runs out of charge, or is discharged. - ON to SLEEP state – Transition based on when the device receives the indicator. - SLEEP to ON state – Transition based on when the device completes the time indicated. - SLEEP to OFF state - Transition based on when the device runs out of charge, or is discharged. The multi device inventory latency for direction 2 is 18.3 seconds. Observations or Analysis or Evaluations During the SLEEP state, it has to be ensured that the power consumption of the device while running the clock is lower than the charging rate of the capacitor. The on-demand shifting of states does not require the device to maintain any fixed duration of the states or a periodicity to shift between states, thus avoiding any potential need for the device to be preconfigured with any information. Specification impacts, if any Minor specification impact. The indictor needs to be specified to indicate the time left for the inventory procedure to start, which can be a higher layer message carried on the PRDCH.

Table 6.2.2-4: Details from Source [104]

Source	Details
Source [104]	Solution description Duty cycle-liked solution (with sleep state) The device indicates to the reader before entering the OFF state. Observations or Analysis or Evaluations Reader can provide an indication to the device to perform sleep state transition. Energy threshold based transition can be up to implementation. Timer or counter based transition can be considered Before entering the OFF state, the device may indicate to the reader which may be beneficial for the reader to perform the subsequent scheduling or transmission. Specification impacts, if any Sleep-state and state transition conditions Indication or timer/counter design for state transition

Table 6.2.2-5: Details from Source [95], [100]

Source	Details
Source [95], [100]	Solution description: • Wake up signal for the device or configuring device monitoring periods • Configuring the duty cycles and aligning duty cycle among group of devices • Energy aware scheduling and selection of subset of device in each access round within the inventory Observation: • Reader can control the power states of device 2b - ON; OFF and sleep and help to extend the operational time of devices. • The device can be periodically configured to transition from sleep to the ON state as part of the duty cycle for monitoring the trigger from the reader. Contrary to it, wake up signaling indicating an identifier associated to a device or a group of device wakes up selected device for the access round.

- Contrary to the traditional Q based device scheduling implemented in UHF RFID, devices can be selected for each access occasion where each access occasion reserve resources for multiple devices.
- Energy aware selection of device for each access round decrease the storage size as shown in our analysis.
- For 96 bits payload, the transmission time increases from 14.9ms to 89.6ms when the data rate decreases from 6kbps to 1kbps.
  - With duty-cycle and energy harvesting, the device can be sustainably operated in case of 6kbps with storage smaller of less than 0.5uF, while it requires at least 1.5uF storage to be sustainably operate in case of 1kbps.
  - With always ON condition, the device can be sustainably operated in case of 6kbps with storage smaller of less than 2uF, while it requires at least 10uF storage to be sustainably operate in case of 1kbps.
- For 400 bits payload, the transmission time increases from 69.3ms to 388ms when the data rate decreases from 6kbps to 1kbps.
  - With duty-cycle and energy harvesting, the device can be sustainably operated in case of 6kbps with storage smaller of less than 1.5uF, while it requires at least 6uF storage to be sustainably operate in case of 1kbps.
  - With always ON condition, the device can be sustainably operated in case of 6kbps with storage smaller of less than 6uF, while it requires very high storage capacity storage to be sustainably operate in case of 1kbps.

Evaluation:

- Comparing the sustainable operation time of device supporting always ON with that of duty cycle and energy harvesting is provided in Fig. 3.

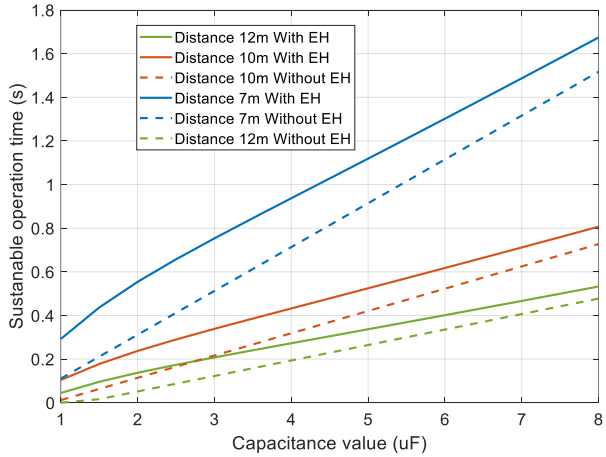


Figure 3: Comparison of operation time of Ambient IoT devices

- Energy aware scheduling benefits illustrated in Fig. 6. As it reduces the required capacitance vale significantly

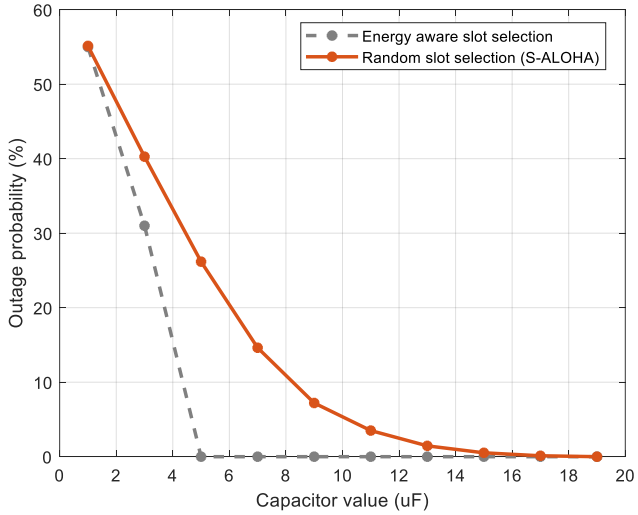
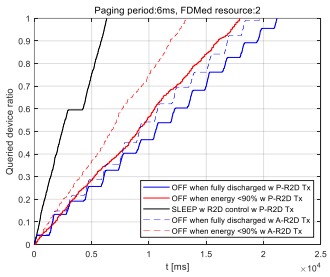


Figure 6 Comparison of outage for different scheduling mechanism for Ambient IoT

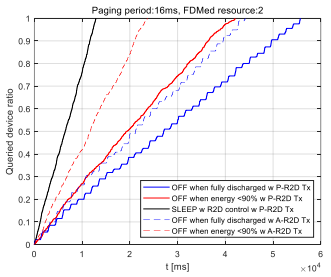
	devices Specification impact: • Wake up signal design and content otherwise device monitoring periodicity • Duty cycle configuration for a group of devices Energy aware scheduling using selecting a subset of devices for each access round.

Table 6.2.2-6: Details from Source [94]

Source	Details
Source [94]	Solution description A-IoT device is assumed to have three states: ON, OFF, SLEEP. - SLEEP state supports maintaining a timer, maintaining a memory content from ON state, energy harvesting and does not support transmission, reception for communication. - The functions for ON and OFF states supported and unsupported are the same as those captured for direction 1. Under direction 2, one scheme for a device switching between ON and SLEEP states: - Scheme 3 (SLEEP w R2D control): Device switches between ON and SLEEP states under reader's control. The reader can provide the time duration for ON or SLEEP and their periodicity for the device to perform ON and SLEEP state transition. Thus, only periodic paging transmission, i.e., Option 1 (P-R2D Tx) is used for reader to imitates RA. Observations or Analysis or Evaluations The detailed evaluation methods and simulation assumptions listed in Table 3-3 can be found in section 3.3 of [94]. Case 1: For 2 RA resources and paging period of 6ms and 16ms



Paging period = 6ms



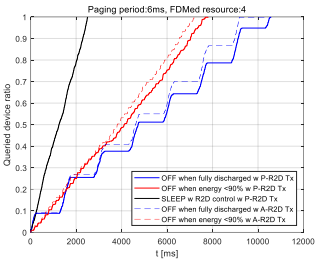
Paging period = 16ms

Figure 1. Time required to inventory 100% of devices for 2 RA resources and 6/16ms paging period

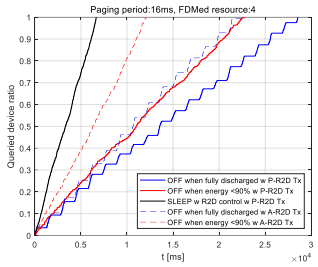
Table 1: Time required to inventory 100% of devices using 2 RA resources per paging message

Schemes	**Totally inventory time [s]**	
	Paging period: 6ms	Paging period: 16ms
OFF when fully discharged w P-R2D Tx	21.16	55.69
OFF when energy <90% w P-R2D Tx	17.918	42.019
SLEEP w R2D control w P-R2D Tx	6.309	12.833
OFF when fully discharged w A-R2D Tx	19.677	44.179
OFF when energy <90% w A-R2D Tx	13.30	23.747

Case 2: For 4 RA resources and paging period of 6ms and 16ms



Paging period = 6ms



Paging period = 16ms

Figure 2. Time required to inventory 100% of devices for 4 RA resources and 6/16ms paging period

Table 2: Time required to inventory 100% of devices using 4 RA resources per paging message

Schemes	**Totally inventory time [s]**	
	Paging period: 6ms	Paging period: 16ms
OFF when fully discharged w P-R2D Tx	10.567	28.538
OFF when energy <90% w P-R2D Tx	7.805	22.732
SLEEP w R2D control w P-R2D Tx	2.50	6.67
OFF when fully discharged w A-R2D Tx	10.441	22.684
OFF when energy <90% w A-R2D Tx	7.284	12.081

Observations based on above Case 1 and Case 2

Observation 1:

- Decreasing the paging period reduces total inventory latency across all schemes.
  - For 2 RA resources scheduled by a single paging message, when the paging period is reduced from 16ms to 6ms,
    - Schemes 1 and 2 reduce the time to inventory all devices by about 62% and 57% for periodic R2D transmission, and by about 55% and 44% for aperiodic R2D transmission.
    - Scheme 3 achieves about 51% reduction for periodic R2D transmission.
  - For 4 RA resources scheduled by a single paging message, when the paging period is reduced from 16ms to 6ms,
    - Schemes 1 and 2 show about 63% and 66% time reduction for periodic R2D transmission, and 54% and 40% for aperiodic R2D transmission.
    - Scheme 3 shows about 62% latency reduction for periodic R2D transmission.

Observation 2:

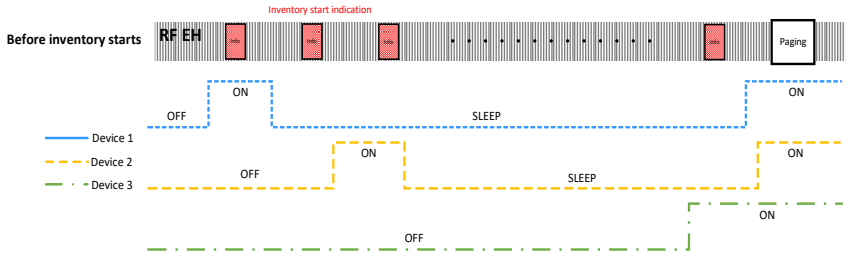
- Increasing the random access (RA) resources for D2R transmission scheduled by a single A-IoT paging reduces total inventory latency significantly across all schemes.
  - For 6ms paging period, when RA resources is increased from 2 to 4,
    - Scheme 1 and 2 reduce the time to inventory all devices by about 50% and 56% for periodic R2D transmission, and 47% and 45% for aperiodic R2D transmission.
    - Scheme 3 show about 60% latency reduction for periodic R2D transmission with the reduced paging period from 16ms to 6ms.

	- For 16ms paging period, when RA resources is increased from 2 to 4, - Schemes 1 and 2 show about 49% and 46% time reduction for periodic R2D transmission, and about 48% and 49% for aperiodic R2D transmission. - Scheme 3 shows about 48% latency reduction for periodic R2D transmission. Observation 3: - For Scheme 1 and Scheme 2, aperiodic R2D transmission can further reduce the inventory time, especially for long paging period, although it results in more paging overhead than the periodic R2D transmission. - For 2 RA resources, compared to 6ms/16ms-period R2D transmissions, Scheme 1 and 2 further reduce the time to inventory all devices by about 7%/21% and 26%/43% with additional aperiodic R2D transmission - For 4 RA resources, compared to 6ms/16ms-period R2D transmissions, Scheme 1 and 2 further reduce the time to inventory all devices by about 1.2%/20% and 6.7%/49% with additional aperiodic R2D transmission Observation 4: - Devices under Scheme 1 needs to recharge more often compared to Scheme 2 and Scheme 3. - Scheme 3 generally offers the shortest inventory time, while Scheme 1 has the longest. - For scheme 1 and scheme 2 that reader cannot control or be aware of the device state, short paging period and/or aperiodic R2D transmission can reduce inventory latency. For scheme 3, to achieve the tradeoff between the inventory latency and R2D overhead, it is advisable to set the paging period and the ON to SLEEP time ratio to allow device harvest enough energy. Specification impacts, if any For scheme 3, it may have following specification impacts: - Define SLEEP, ON, and OFF states for the A-IoT device with their functions. - Include signaling from the reader to the device for transitioning between ON and SLEEP states. - Detail the device's monitoring behavior, transmission, and reception procedures.

Table 6.2.2-7: Details from Source [79]

Source	Details
Source [79]	Solution description In Direction 2, solutions to extend device availability for both stages before the inventory starts and during the inventory procedure are proposed. The device is assumed to have three states: - ON state: Device can at least perform transmission and reception, run MHz sampling clock, and retain memory. - OFF state: Device cannot perform transmission and reception, no clock running. Device can harvest energy. Device can retain memory of being inventoried. - SLEEP state: Device cannot perform transmission and reception. Device can run a low accuracy clock, retain memory, may or may not harvest energy (depends on the relationship of the consumed power and the input power). Solution before inventory starts From reader perspective, it provides indication of the potential starting time or remaining time of the future paging message. By implementation, the reader can provide the indication in a shorter time interval, so that a device does not need to drain its energy to monitor the indication. From device perspective, it becomes available when it is fully charged and monitors the indication of the transmission time of the upcoming paging message. The device transits to

SLEEP state to save or harvest energy, if necessary. During SLEEP state, the device needs to retain a clock for time counting.



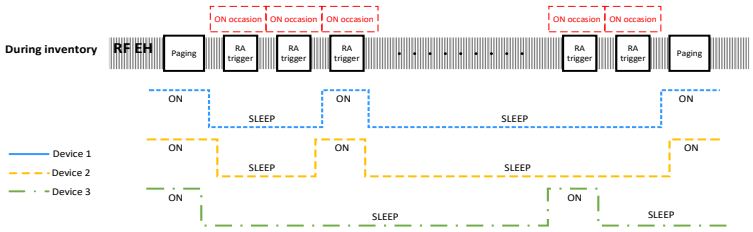
Solutions during inventory procedure

B1: Non-slot-ALOHA based solution

This solution is a proposed new random access solution for Ambient IoT, and is not based on the slot-ALOHA solution specified in UHF RFID.

From reader perspective, it provides duty cycle information in paging message, which includes at least number and duration of ON occasions.

From device perspective, upon receiving a paging message, it randomly selects an ON occasion for random access, and transits to SLEEP state, if necessary, to conserve and harvest energy. During SLEEP state, the device needs to retain memory regarding its ON occasion information and runs a clock for time counting. The device transits to ON state till the selected ON occasion is coming. Within the ON occasion, the device performs random access upon receiving triggering message. If collision occurs, devices transit to SLEEP state and wait for the next paging message. If a device does not successfully access before running out of available energy, it recharges until fully charged again.



B2: Slot-ALOHA based solution

This solution is an optimization based on the slot-ALOHA solution specified in UHF RFID.

From reader perspective, it additionally sends a slot decrement indication to inform the decrement progress to devices.

From device perspective, upon receiving a paging message and acquires the Q value, it first randomly selects an access slot within [0, 2Q-1]. The device transits to SLEEP state before the selected access slot, if necessary. During SLEEP state, the device needs to retain memory regarding its access slot information and runs a clock for time counting with an assuming slot duration. The device transits to ON state when it counts to the selected slot. The device monitors the decrement indication to obtain the actual slot decrement progress and determines whether it re-transits to SLEEP state or keeps monitoring to its upcoming access slot. If a device finds out that its access slot has passed, it responds immediately in the follow-up slot. If collision occurs, devices transit to SLEEP state and wait for the next paging message. If a device does not successfully access before running out of available energy, it recharges until fully charged again.

Observations or Analysis or Evaluations
Necessity of Direction 2 for Device 1

Figure 3.3-4 to Figure 3.3-7 show the comparison results between Direction 1 and Direction 2 for Device 1.

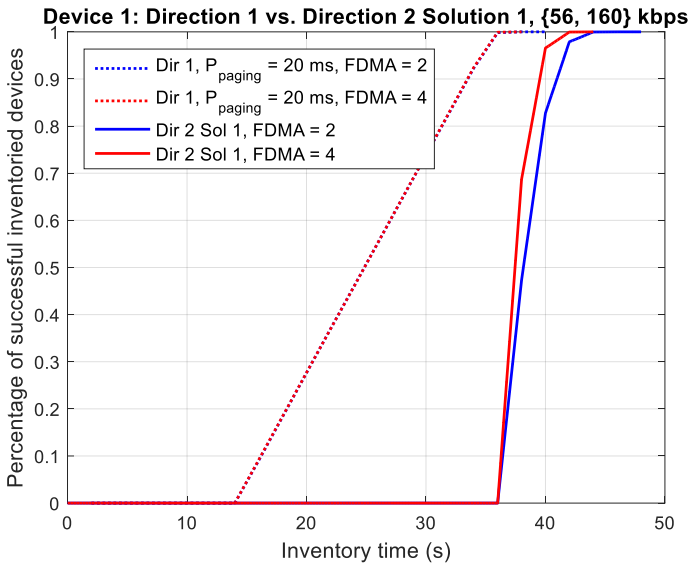


Figure 3.3-4: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 1 Solution 1 versus Direction 1 Device 1 in case of {R2D, D2R} data rate of {56, 160} kbps.

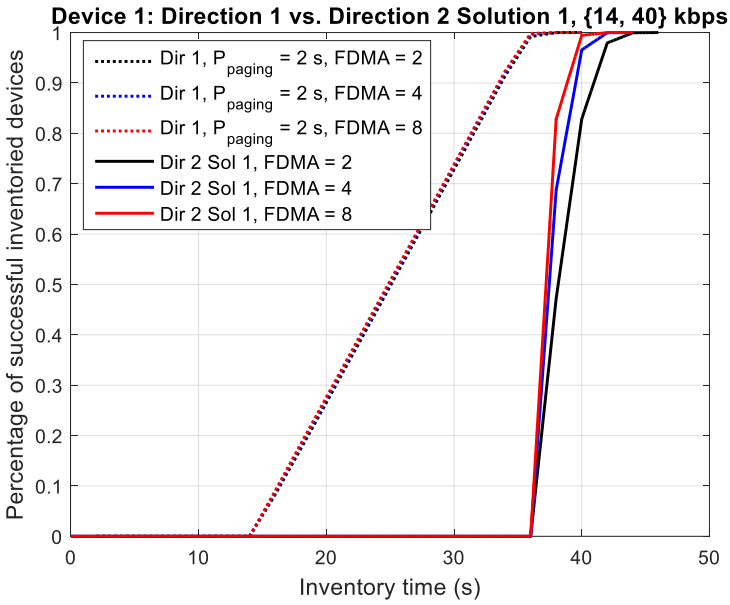


Figure 3.3-5: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 1 Solution 1 versus Direction 1 Device 1 in case of {R2D, D2R} data rate of {14, 440} kbps.

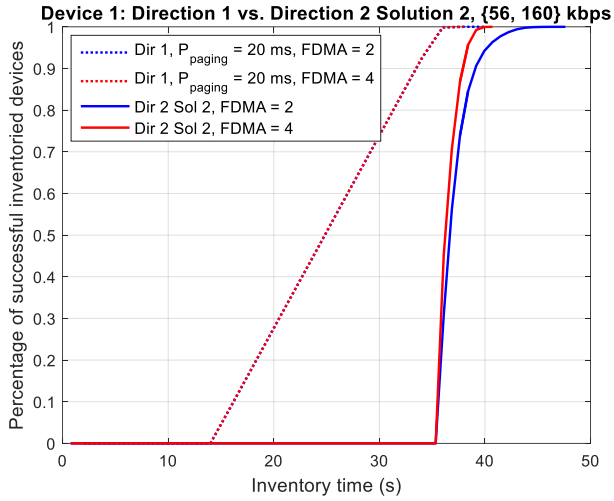


Figure 3.3-6: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 1 Solution 2 versus Direction 1 Device 1 in case of {R2D, D2R} data rate of {56, 160} kbps.

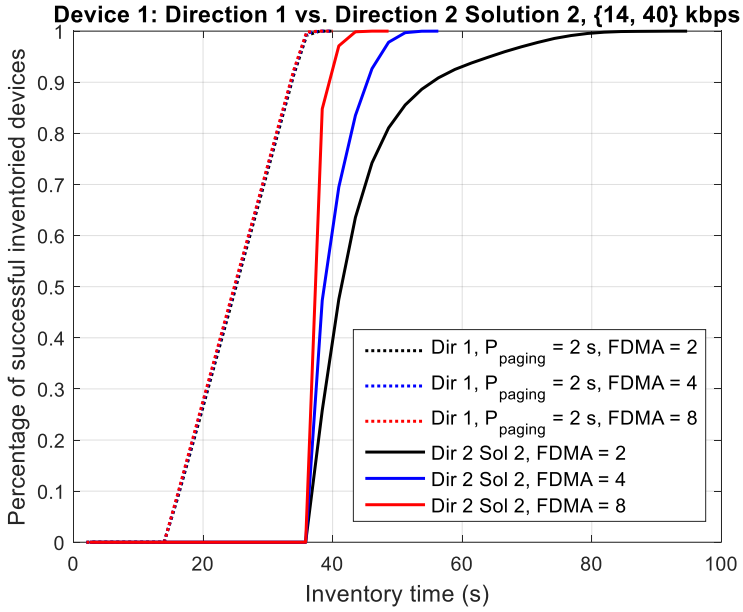


Figure 3.3-7: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 1 Solution 2 versus Direction 1 Device 1 in case of {R2D, D2R} data rate of {14, 440} kbps.

Observation 7: For Device 1, the inventory completion time for Direction 1 is shorter than Direction 2 for all evaluation cases.

Observation 8: At least for Device 1 implemented with large capacitor size (e.g., 10 uF), it seems not necessary to study Direction 2.

Results for Direction 2 Device 2

B1. Results for Solution 1 (Solution A + Solution B1)

Figure 3.3-8 and Figure 3.3-9 show the evaluation results of percentage of successfully inventoried devices with respect to the inventory time for Solution 1, where Figure 3.3-8 provides results assuming {R2D, D2R} data rate of {56, 160} kbps and Figure 3.3-9 provides results assuming {R2D, D2R} data rate of {14, 40} kbps. The inventory completion time considering 99% and 100% of devices successfully complete inventory are summarized in Table 3.3-1. In addition, we also provide comparison between Direction 1 and Direction 2 Solution 1, as shown in Figure 3.3-10 and Figure 3.3-11.

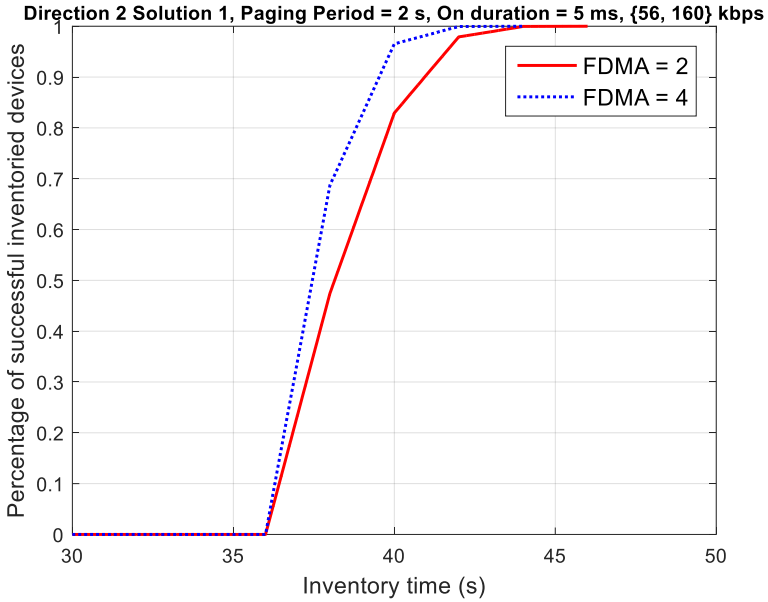


Figure 3.3-8: Percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 1 in case of {R2D, D2R} data rate of {56, 160} kbps.

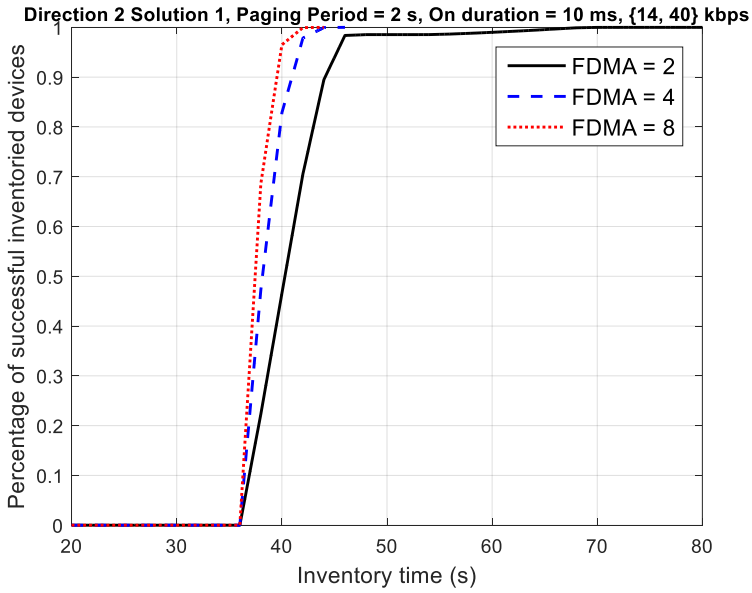


Figure 3.3-9: Percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 1 in case of {R2D, D2R} data rate of {14, 40} kbps.

Table 3.3-1: Inventory completion time for Direction 2 Solution 1

Successful rate	{R2D, D2R} Data rate	FDMA resources	Inventory completion time
99%	{56,160} kbps	2	43.08 s
		4	41.44 s
	{14,40} kbps	2	59.97 s
		4	43.09 s
		8	41.46 s

100%	{56,160} kbps	2	43.96 s
		4	42.09 s
	{14,40} kbps	2	67.11 s
		4	43.69 s
		8	42.10 s

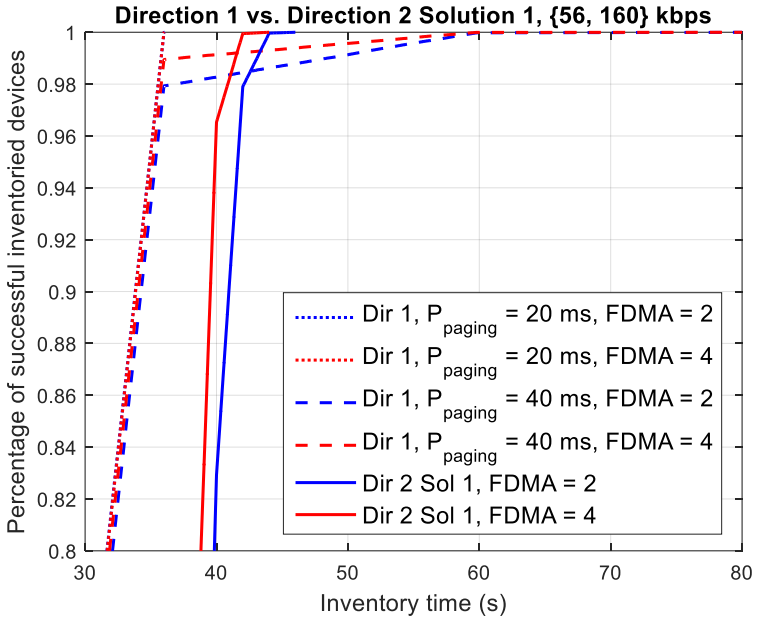


Figure 3.3-10: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 1 versus Direction 1 Device 2 in case of {R2D, D2R} data rate of {56, 160} kbps.

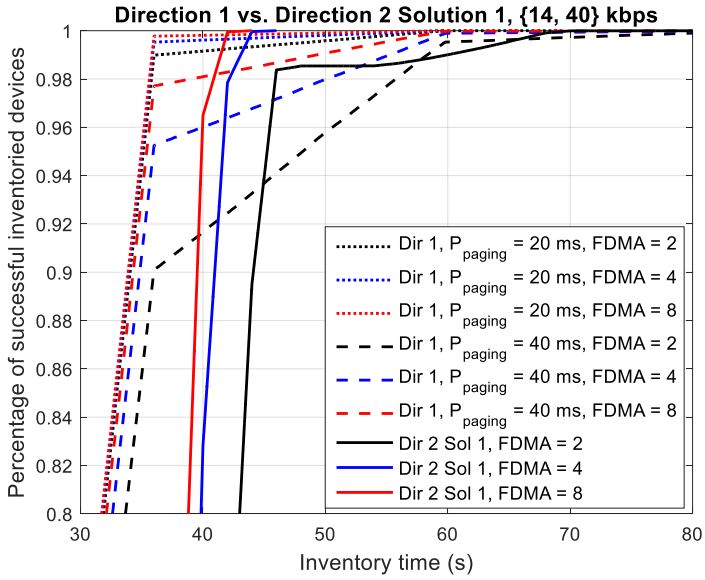


Figure 3.3-11: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 1 versus Direction 1 Device 2 in case of {R2D, D2R} data rate of {14, 40} kbps.

B2. Results for Solution 2 (Solution A + Solution B2)

Figure 3.3-12 and Figure 3.3-13 show the evaluation results of percentage of successfully inventoried devices with respect to the inventory time for Solution 2, where Figure 3.3-12 provides results assuming {R2D, D2R} data rate of {56, 160} kbps and Figure 3.3-13 provides

results assuming {R2D, D2R} data rate of {14, 40} kbps. The inventory completion time considering 99% and 100% of devices successfully complete inventory are summarized in Table 3.3-2. In addition, we also provide comparison between Direction 1 and Direction 2 Solution 2, as shown in Figure 3.3-14 and Figure 3.3-15.

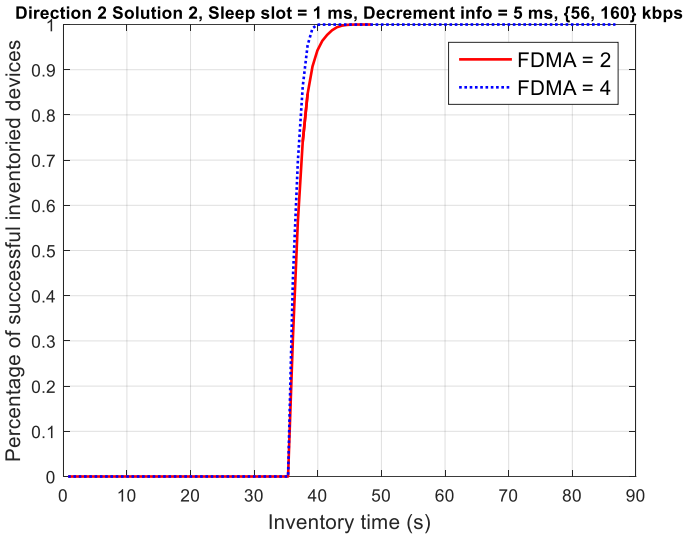


Figure 3.3-12: Percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 2 in case of {R2D, D2R} data rate of {56, 160} kbps.

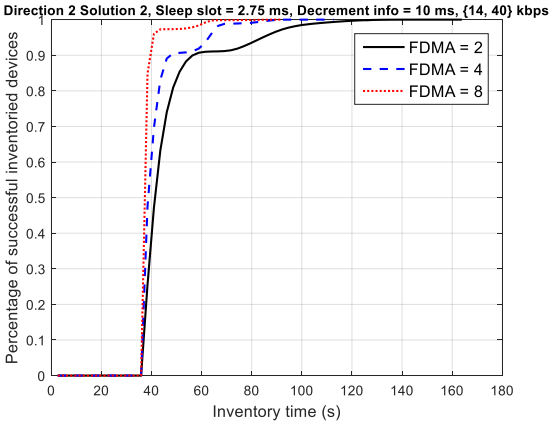


Figure 3.3-13: Percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 2 in case of {R2D, D2R} data rate of {14, 40} kbps.

Table 3.3-2: Inventory completion time for Direction 2 Solution 2

Successful rate	{R2D, D2R} Data rate	FDMA resources	Inventory completion time
99%	{56,160} kbps	2	43.68 s
		4	39.84 s
	{14,40} kbps	2	107.68 s
		4	76.96 s
		8	61.60 s
100%	{56,160} kbps	2	44.06 s
		4	41.05 s

	{14,40} kbps	2	107.68 s
		4	87.73 s
		8	71.87 s

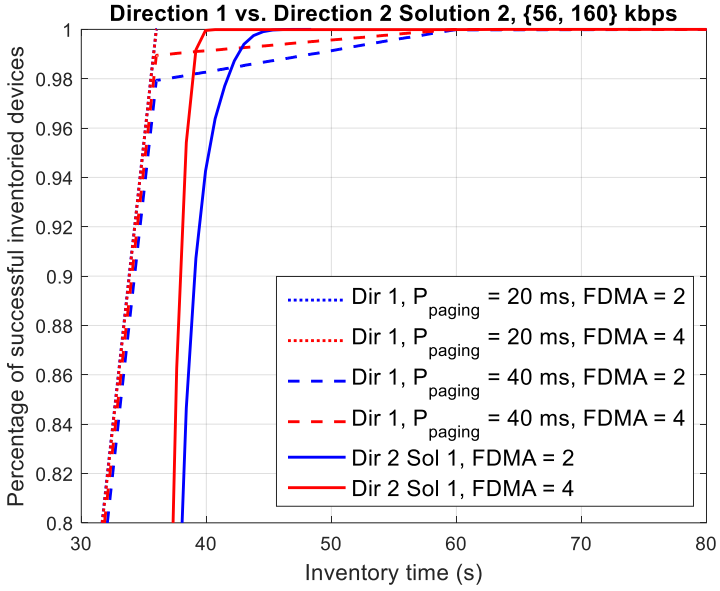


Figure 3.3-14: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 2 versus Direction 1 Device 2 in case of {R2D, D2R} data rate of {56, 160} kbps.

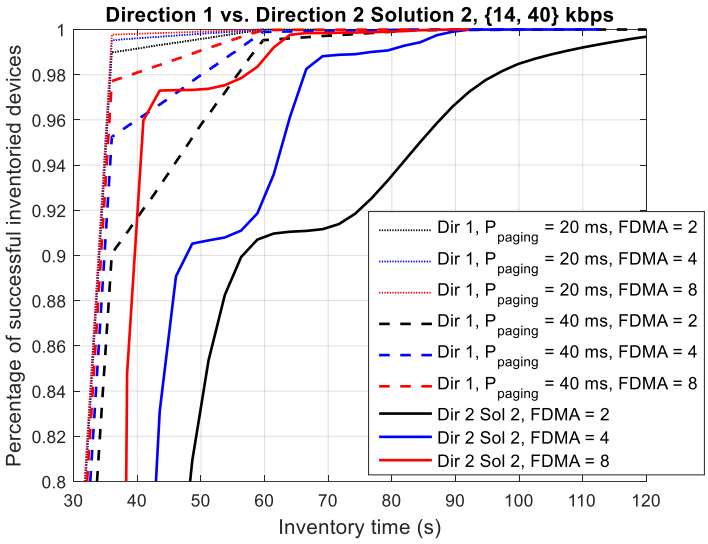


Figure 3.3-15: Comparison of percentage of successful inventoried devices with respect to inventory time for Direction 2 Device 2 Solution 2 versus Direction 1 Device 2 in case of {R2D, D2R} data rate of {14, 40} kbps.

Observation 9: For Direction 2, and for both Solution 1 and Solution 2, when considering higher {R2D, D2R} data rate of {56, 160} kbps, at least 99% of devices can be successfully inventoried before running out of energy and no device goes through recharging.

Observation 10: For Direction 2, and for both Solution 1 and Solution 2, when considering lower {R2D, D2R} data rate of {14, 40} kbps, there are devices cannot complete inventory before running out of energy and therefore the inventory completion time sufficiently increases.

Observation 11: For Direction 2, and for both Solution 1 and Solution 2, when compared to Direction 1 assuming paging time interval is comparable to half of the device sustainable time:

	- In case of higher {R2D, D2R} data rate of {56, 160} kbps, Direction 2 consumes longer inventory completion time when at least 99% of device successfully complete inventory, since the first paging message is sent at the time when the farthest device finish energy harvesting in Direction 2. - In case of lower {R2D, D2R} data rate of {14, 40} kbps, Direction 2 consumes longer inventory completion time when at least 99% of device successfully complete inventory, since there are devices goes through recharging for Direction 2 but there is no device goes through recharging for Direction 1. Observation 12: For Direction 2 Solution 1, when compared to Direction 1 assuming paging time interval is comparable to the device sustainable time, Solution 1 has benefits in the following cases: - Considering 99% devices successfully complete inventory, in case of higher {R2D, D2R} data rate of {56, 160} kbps and 2 FDMA resources; in case of lower {R2D, D2R} data rate of {14, 40} kbps and 4/8 FDMA resources. - Considering 100% devices successfully complete inventory, in case of higher {R2D, D2R} data rate of {56, 160} kbps and 2/4 FDMA resources; in case of lower {R2D, D2R} data rate of {14, 40} kbps and 2/4/8 FDMA resources. Observation 13: For Direction 2 Solution 2, when compared to Direction 1 assuming paging time interval is comparable to the device sustainable time, Solution 2 has benefits in the following cases: - Considering 99% devices successfully complete inventory, in case of higher {R2D, D2R} data rate of {56, 160} kbps and 2/4 FDMA resources. - Considering 100% devices successfully complete inventory, in case of higher {R2D, D2R} data rate of {56, 160} kbps and 2/4 FDMA resources. Specification impacts, if any For Direction 2, potential specification impact is expected at both reader side and device side. From reader perspective, - Before the inventory starts, the reader provides indication of the potential starting time or remaining time of the future paging message. - During the inventory procedure, for Solution 1, the reader provides duty cycle information at least including number and duration of duty cycle ON occasions. - During the inventory procedure, for Solution 2, the reader provides slot decrement indication to inform the slot decrement progress. From device perspective, - Before the inventory starts, the available device monitors the starting time indication of the paging message and can transits to SLEEP state, if necessary. - During the inventory procedure, for Solution 1, the device randomly selects a duty cycle ON occasion for random access. During SLEEP state, the device retains memory regarding its ON occasion information and runs a clock for time counting. During the inventory procedure, for Solution 2, the device randomly selects an access slot and can transits to SLEEP state, if necessary. During SLEEP state, the device retains memory regarding its access slot information and runs a clock for time counting with an assuming slot duration. The device transits to ON state when it counts to the selected slot, monitors the decrement indication to obtain the actual slot decrement progress and determines whether it can re-enter SLEEP state or keeps monitoring to its upcoming access slot.

Table 6.2.2-8: Details from Source [80]

Source	Details
Source [80]	Solution description The device can switch to "SLEEP" state to recharge the power and wake up before the selected access occasion starting time If a device has selected a later access occasion based on the indicated "Q" value Observations or Analysis or Evaluations The SLEEP state may need to be defined in which the device can maintain a timer and the memory content from ON state; The device's SFO may need to be taken into account. Specification impacts, if any Designing a timer to control the time length of "SLEEP" state and/or additional information carried in the R2D command like "QueryRep" [RAN2].

Table 6.2.2-9: Details from Source [81]

Source	Details
Source [81]	Solution description In Direction 2, A-IoT devices perform transmission and reception for communication in available state. The devices would become unavailable during charging or after running out the stored energy. While for A-IoT devices in unavailable state, devices may perform energy harvesting or just be in a fully charged state without the requirement of transmission and reception for communication. The unavailable devices do not become available to perform transmission and reception for communication until receiving the status switching information provided by the reader. Direction 2 has more specification impacts, e.g. specify the content and the form of the information. In addition, carrier wave should not always exist to provide energy for A-IoT devices since the devices may be fully charged in unavailable state. Observations or Analysis or Evaluations Observation 5: In Direction 2, carrier wave should not always exist on to provide energy for A-IoT devices. Observation 6: In Direction 2, A-IoT devices are unavailable when they are in a fully charged state without the requirement of transmission and reception for communication Specification impacts, if any In Direction 2, at least the content and the form of the information used for switching A-IoT devices to become available needs to be specified.

Table 6.2.2-10: Details from Source [83]

Source	Details
Source [83]	Solution description In Direction 2, since a reader provides assistance or coordination information to assist a device to determine whether/when to become available or unavailable, it is assumed the device should support 3 operating states (ON, OFF, SLEEP), such that the device can switch from SLEEP-to-ON based on an internal clock (time counting). Switching from OFF-to-ON is, in general, based on device implementation (e.g., whenever the energy storage level has reached X % full, where X=70/80/100). To switch between ON/SLEEP for energy harvesting (e.g., based on time counting via an internal clock), the main intention is to prolong/extend the sustainable operating time of a device such that the device has sufficient energy to complete at least a round of inventory or command process. Energy harvesting for a device should focus on two stages: before and after an inventory or command process is triggered. Before an inventory or command process is triggered: - A device power-ON once it is sufficiently charged and receives information from the reader relating to the timing/periodicity of future inventory/command processes to be triggered. - Device selects one of the future periodic inventory / command processes, switches to SLEEP state for power saving and energy harvesting and performs time counting

	using an internal clock. To minimize the congestion / collisions of every device trying to wake-up at the same time and attempt to access the same inventor process together, it is beneficial for the reader to indicate multiple future processes and a device selects one of the future processes (e.g., randomly), so that all devices are spread out across the multiple processes. - Before start of the selected inventory / command process, device switches to ON state to receive paging message from the reader. After an inventory or command process is triggered: - After receiving the paging message that triggers an inventory or command process and selecting an access slot (q) as part of slotted-ALOHA, the device switches to SLEEP state for power saving and energy harvesting/re-charging. - The device counts the time (via internal clock) or detects number of paging repeat messages or timing acquisition signals / R2D preambles, and switches to ON state before the start of the selected access slot (q) to carry out a random-access procedure with the reader. Observations or Analysis or Evaluations Based on the solution descriptions, the goal / intention is for a device to be in the ON state only when it is necessary, i.e., for the selected process (before the inventory / command process is triggered) and the selected access slot (after the inventory / command is triggered). In other time, the device switches / remains in the SLEEP state for energy harvesting. Hence, the energy storage level in the device remains in a high level and the sustainable operating time is maximized to ensure the device is able to complete an inventory/command process with the reader. Specification impacts, if any To ensure the energy storage level in a device is maximized for prolonging the sustainable operating time and to ensure a smooth operation of running an inventory or command process, a unified state transition behaviour(s) based on reader signaling or assistance/coordination information should be specified.

Table 6.2.2-11: Details from Source [84]

Source	Details
Source [84]	Solution description Solution 1: Reader scheme: Transmits indications multi times to notify the next access occasion(s). Device scheme: Random On to receive the indication and select the access occasion by device. Observations or Analysis or Evaluations Solution 1: Reader does not control the device state Specification impacts, if any Solution 1: minor specification impact.

Table 6.2.2-12: Details from Source [84]

Source	Details
Source [84]	Solution description Solution 2: Reader scheme: Periodically transmits PRDCH. Device scheme: Periodically On to access according to the reader configuration. Observations or Analysis or Evaluations Solution 2: Reader controls the device state Specification impacts, if any Solution 2: has specification impact for both reader and device.

Table 6.2.2-13: Details from Source [85]

Source	Details
Source [85]	Solution description Each device can be associated with a number of "unavailable time windows", within which the device is not available for communications e.g. due to charging by energy harvesting. The reader can implicitly or explicitly indicate the "unavailable time windows" (or how to determine them) to a device. For example, if a device successfully receives an R2D transmission carrying A-IoT paging message intended for a specific device group NOT including the device, it can be a good indication that the device will not miss anything even if it stops transmissions/receptions for a period of time right after the the R2D transmission, and that period of time can be a "unavailable time window" for the device. Precision in definition of a "unavailable time window" can take the SFO of a device into account. Observations or Analysis or Evaluations "Unavailable time windows" for A-IoT devices can be exploited for energy harvesting, minimizing the negative impacts to communication delays due to charging by energy harvesting. Specification impacts, if any The reader needs to indicate "unavailable time windows" (or how to determine them) to a device, implicitly or explicitly.

Table 6.2.2-14: Details from Source [86]

Source	Details
Source [86]	Solution description The device operates on a duty-cycled basis for channel monitoring and reception. The reader indicates the times at which the device should be ON and when the device may sleep. The reader can only communicate with the device when the device is ON. The device indicates to the reader the maximum ratio of ON to OFF time, based on the power consumption and energy harvesting rate of the device. Observations or Analysis or Evaluations The range from CWE to device is not limited by the power consumption of the device. The range is limited by the threshold at which energy can be harvested. Devices that are further from the CWE operate with a longer duty cycle. Devices that are close to the CWE can operate in "direction 1 mode". CWE power consumption and interference are less than for direction 1. The reader needs to signal the duty cycle timing to the device. The device needs to signal its supported duty cycles to the reader, based on current energy harvesting conditions. Specification impacts, if any Procedures for duty cycle operation need to be defined. Reader needs to be signal the ON / OFF duty cycle to the device. The device needs to signal its desired ON / OFF duty cycle to the reader.

Table 6.2.2-15: Details from Source [87]

Source	Details
Source [87]	Solution description The reader can group the devices in the same "available duration" for scheduling. Reader indicates to the devices the time for activity cycle allowing the devices to monitor R2D transmissions during the same available duration. To adjust timing drift of devices due to clock inaccuracy, the reader transmits a "Sync Align" outside the activity cycle to help devices to align with the same available duration. Optionally, the Reader can indicate a minimum applicable duty cycle (ratio of available to total time) for the device in the Sync Align message. If a device can determine its duty cycle, it follows the indication if its duty cycle is higher than the indicated minimum. This enables grouping devices by their duty cycle. Observations or Analysis or Evaluations By having a common "available duration" among different devices, the reader can schedule multiple devices in the same duration. Broadcast/groupcast message can be then transmitted

	during the common available duration. The solution is suitable for all device types and can achieve better coexistence between device type 1 and type 2a/2b. Specification impacts, if any Introducing R2D broadcast message to indicate information related to activity cycle.

Table 6.2.2-16: Details from Source [88]

Source	Details
Source [88]	Solution description Define a duty-cycle based paging procedure where each cycle include 'monitoring period' and 'nonActive period', where reader transmits paging command within 'monitoring' period to trigger inventory procedure. An A-IOT device maintains a running clock to count the timing location of 'monitoring period' and 'nonActive period'. An A-IOT Device moves to 'ON' state if it wants to involve a paging procedure; Otherwise, it moves to 'Sleep' state for RF energy harvesting. This is one solution in Direction 2. Observations or Analysis or Evaluations N/A Specification impacts, if any Define a duty-cycle based paging procedure where each cycle include 'monitoring period' and 'nonActive period', where reader transmits paging command within 'monitoring' period to trigger inventory procedure. Introduce a signal to indicate the duty-cycle configuration.

Table 6.2.2-17: Details from Source [89]

Source	Details
Source [89]	Solution description When CW and communication is same frequency, - The start-indicator part can be used to switch CW and communication. There is no need to indicate it in advance. When CW and communication are different frequencies, - During one transaction (inventory or command) or multiple of transactions, the reader would inform or specify "when Msg 0 and/or 2 and/or 4" are to be sent. This situation is "reader provide the information to a device on which the device become available. Observations or Analysis or Evaluations Without such some level of (rough) aligned timing operation, reader and device are not able to communicate well. Specification impacts, if any When CW and communication is different frequency,the timer value related to "when Msg 0 and/or 2 and/or 4" are to be sent.

Table 6.2.2-18: Details from Source [90]

Source	Details
Source [90]	Solution description R2D signaling indicating a device to go dormancy and the allowed time duration. R2D wake-up signal to instruct a device to transition into an active state and monitor a potential R2D transmission.

	Direct indication of the one or more timers described under Direction 1. Observations or Analysis or Evaluations N/A Specification impacts, if any N/A

Table 6.2.2-19: Details from Source [114]

Source	Details
Source [114]	Solution description Applicable to both Device 1 and Device 2 (higher power devices) Reader provides counter information in R2D transmission for slotted-ALOHA mechanism Includes periodicity information of potential Reader command occasions (or expected device monitoring occasions) Remaining energy estimation and report of the information by device Scheduling commands provide end timing information for R2D reception or D2R transmission (utilization of remaining energy information) Observations or Analysis or Evaluations Necessary for resolving unavailability impact on slotted-ALOHA mechanism. Helps devices align to periodic occasions of potential Reader commands. Allows devices to estimate remaining energy and adjust behavior accordingly. Enables Reader to optimize communication based on device energy information. Examples to improve successful R2D or D2R transmissions, Small payload: Higher priority for devices with low remaining energy Large payload: Higher priority for devices with high remaining energy Addresses challenges when devices are scattered in a wide region with unaligned availability. Specification impacts, if any Define format for counter information in R2D transmission. Specify periodicity information format for potential Reader command occasions. Define end timing information format in scheduling commands. Explicit: Indication of transmission time/length Implicit: Derived from TBS and data rate indicated Establish protocol for devices to report remaining energy information to Reader. Design of initial R2D periodicity to ensure efficient initial contact with unaligned devices.

Table 6.2.2-20: Details from Source [113]

Source	Details
Source [113]	Solution description Reader provides information on when paging/re-paging are potentially transmitted, and device wakes up based on the information. Observations or Analysis or Evaluations This direction would intend to control device availability/unavailability by reader/NW side. When reader transmits paging/re-paging can be announced to devices, and each device can wake up at the announced timing. In this case, unlikely to direction 1, device can detect paging/re-paging a lot of times or device can sleep until re-paging to which the device can respond. Necessary rule between reader and devices will be what kind of information is provided and how/when, and detailed conditions to wake-up in consideration of energy storage status and the indicated information. In addition, how to achieve accurate synchronization is

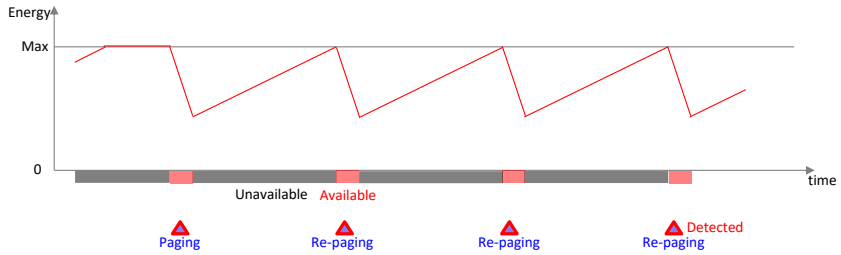
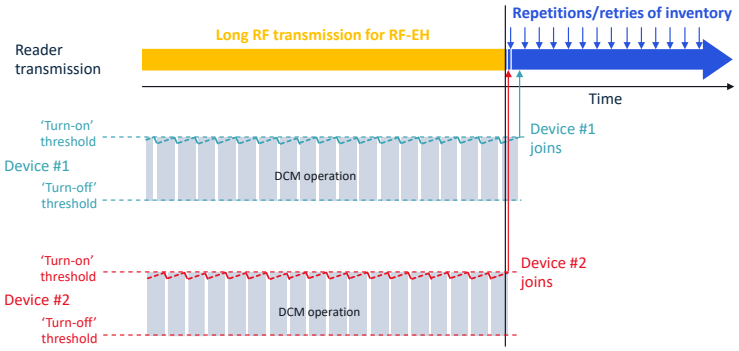
	also important to make the information announcement meaningful. In our view, temporarily periodic R2D sync signal is a straightforward solution for this sync issue.  Fig: A possible way of Direction 2. Specification impacts, if any E.g., what kind of information is provided and how/when. E.g., detailed conditions to wake-up in consideration of energy storage status and the indicated information. E.g., device monitors temporarily periodic R2D sync signals.

Table 6.2.2-21: Details from Source [114]

Source	Details
Source [114]	Solution description Device duty-cycle monitoring (DCM) Observations or Analysis or Evaluations We believe a feasible solution for all A-IoT devices is device duty-cycle monitoring (DCM). The essence of DCM is quite simple: A device shortens/controls the available time duration. Outside the available time duration, the device is unavailable, and harvests energy from RF without communication activities. A benefit of device DCM is that the device unavailable time duration becomes shorter. Figure B1 illustrates an image of how inventory procedure with DCM looks like. Similar to the case of direction 1, different devices may become available at different timings due to different energies. However, by shortening/controlling device available duration, DCM enables much shorter available/unavailable cycles for a device. In case of inventory, devices can receive msg-0 from the reader at very early stage of the inventory procedure that the reader tries.  Fig. B1 Duty-cycle monitoring (DCM) Even after a device joins the inventory procedure running by a reader, the DCM can still work for longer sustainable operation. For example, in an inventory procedure that has multiple contention-access

procedures, if a device has received msg-0 and transmitted msg-1 but failed to receive the corresponding msg-2, the device can consider the current contention-based access has failed and expect another msg-0 reception in future for the retry. If the device knows the reader's msg-0 periodicity/frequency, or the expected available/unavailable cycle for the device, the device can behave such that it can make use of DCM to save the energy consumption as an example being illustrated in Fig. B2.

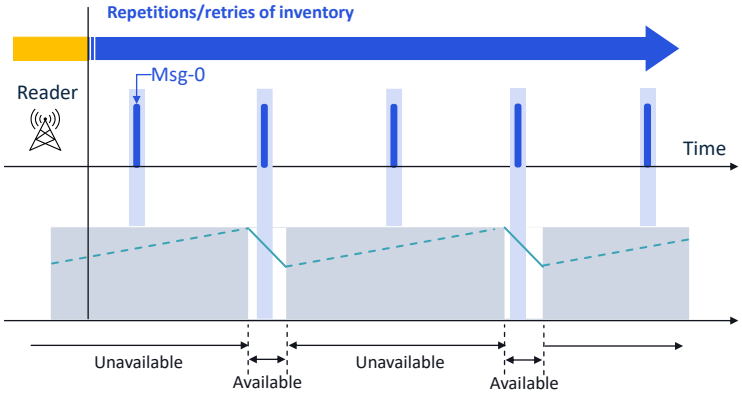


Fig. B2 DCM example after a device joins inventory procedure from a reader

DCM can be realized by a device by counting the time after it becomes available and switches the device state from available state (ON state) to an unavailable state. As for the unavailable state, two options can be considered. The first option is to define a SLEEP state. During the SLEEP state, the device may retain a certain memory and run a low-power sleep clock (e.g., 30kHz) while harvesting energy from RF. With the SLEEP state, the device is able to identify when it should be available (switch to ON state) again.

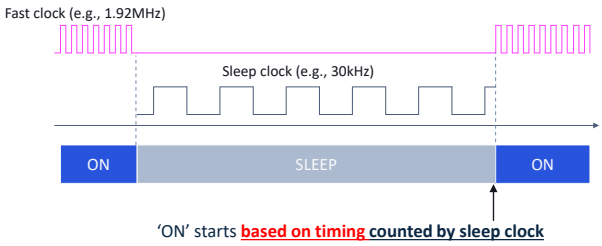


Fig. B3 Switch between ON state and SLEEP state

The other option to realize DCM is to use OFF state as the unavailable state. In the OFF state, the device may not retain a memory and may not run a clock. Therefore, the device is not able to control the unavailable time duration. The unavailable time duration is determined by the time to charge the energy storage until the 'Turn-on' threshold, which depends on Rx power, RF-EH efficiency, and the amount of energy consumed during available time duration.

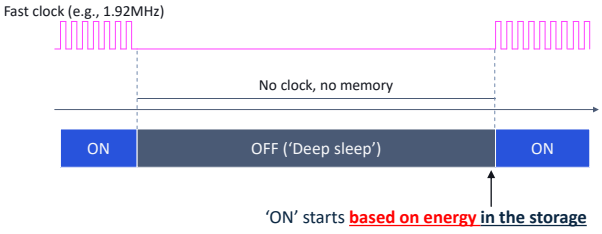


Fig. B4 Switch between ON state and OFF state

Observation B1:

- Direction 2 is the case where a reader has some ideas on when/how a device is available/unavailable, and possibly is able to control the device availability/unavailability.
- As a solution, device duty-cycle monitoring (DCM) is promising.
  - For DCM, a device shortens/controls the available time duration, and for outside the available time duration, the device is unavailable and harvests energy from RF without communication activities.
    - The device counts the time after it becomes available and switches the device state from available state (ON state) to an unavailable state.
    - As for unavailable state, two options can be considered:
      - Opt.1: SLEEP state
        - In the SLEEP state, the device may retain a certain memory and run a low-power sleep clock (e.g., 30kHz) while harvesting energy from RF
        - In the SLEEP state, the device is able to identify when it should be available (switch to ON state) again
      - Opt.2: OFF state
        - In the OFF state, the device may not retain a memory and may not run a clock while harvesting energy from RF
        - In the OFF state, the device is not able to identify when it should be available (switch to ON state) again
        - The device is switched from OFF state to ON state once the amount of energy in the storage reaches the 'Turn-on' threshold
    - An appropriate DCM operation enables efficient energy usage of a device and reduced unavailable time duration for charging

Let us consider the following readers transmission behavior: (1) before starting inventory, the reader transmits RF for energy harvesting as the pre-inventory charging phase; (2) then the reader starts inventory which contains multiple contention-based access procedures, each of which is triggered by a msg-0.



Fig. B5 Inventory model

In each round, reader transmits a msg-0 triggering of up to M msg-1 transmissions from devices. The contention-based access procedure is assumed to contain following steps:

- Msg-0: Reader transmits a msg-0 that triggers msg-1 transmissions from devices
  - Msg-0 indicates a set of msg-1 resources
- Msg-1: A-IoT device that has received a msg-0 selects one of the msg-1 resources and transmit msg-1
  - Multiple A-IoT devices may select a same resource, in which case collision occurs
- Msg-2: Reader transmits a msg-2 as a response to each msg-1
  - In case of msg-1 collision, msg-2 reception fails; then the devices does not proceed this round
- Msg-3: A-IoT device who has received the msg-2 transmits a msg-3 as a response to the msg-2
- Msg-4: Reader transmits a msg-4 as a response to the devices that were successfully identified

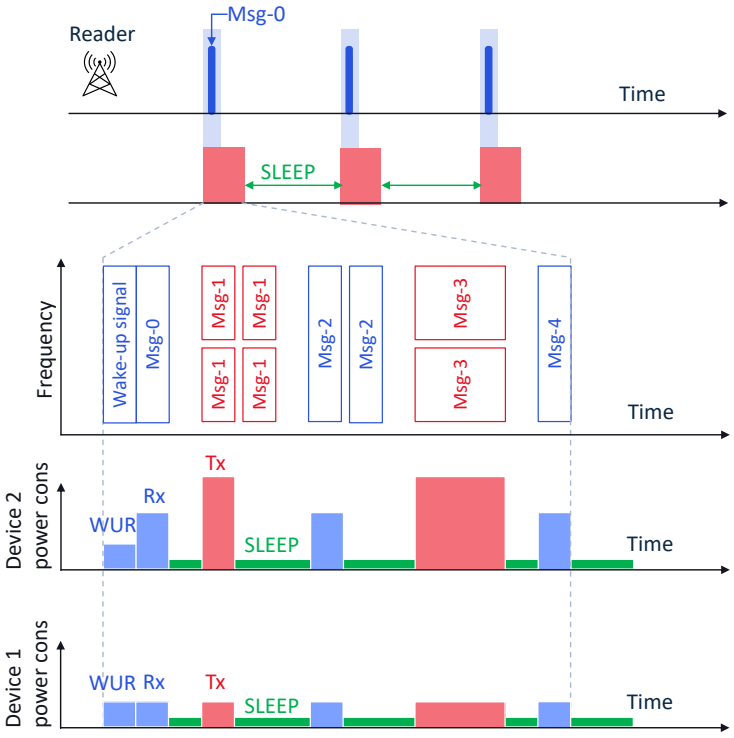


Fig. B6 The model of random-access procedure for each msg-0

Other assumptions/parameters are listed in the table below. For simplicity, we assume following:

- All devices are assumed to have same Rx power, -28dBm or -32dBm, same RF-EH efficiency
- Separate evaluations for device 1 and device 2 (no mixed scenario)

The baseline is no DCM. For DCM, we evaluated two cases.

- Case 1: SLEEP-based msg-0 monitoring before the device joins inventory
  - Device switches ON state and SLEEP state until it detects msg-0 first time
  - Device uses SLEEP state during contention-based access procedures after it detects the first msg-0 whenever possible
  - This enables controlled duty-cycle monitoring all the time
  - However, SLEEP state consumes a certain amount of energy for sleep clock/memory, leading to long unavailable time duration when device cannot sustain SLEEP state due to low Rx power
- Case 2: OFF-based msg-0 monitoring before the device joins inventory
  - Device switches ON state and OFF state until it detects msg-0 first time
  - Device uses SLEEP state during contention-based access procedures after it detects the first msg-0 whenever possible
  - This enables controlled duty-cycle monitoring after the first msg-0 detected
  - Until a device detects msg-0 first time, there is no/negligible power consumption during OFF state

Note that in both cases, if a device consumes all the available energy during ON or SLEEP state, the device is in OFF state until it is fully charged.

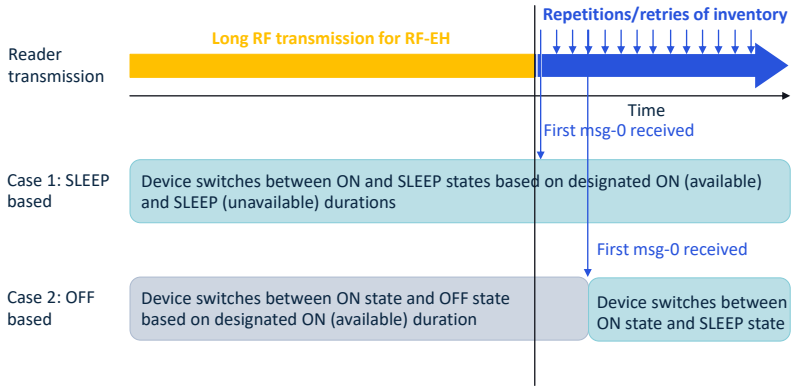


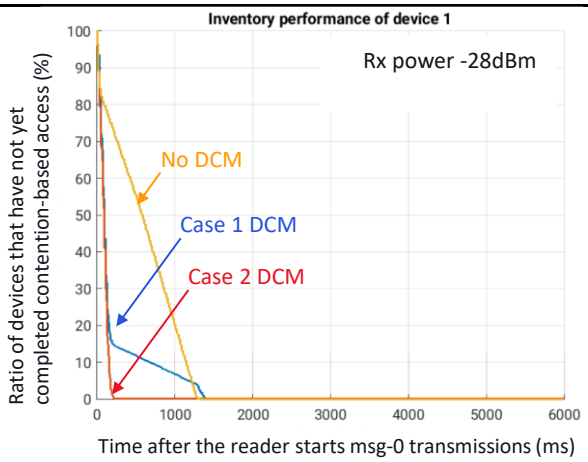
Fig. B7 Case 1 and Case 2 of DCM

Table B1 Evaluation parameters/assumptions

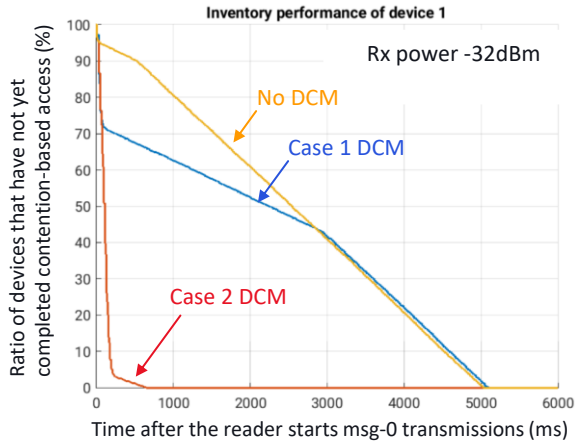
Parameters			Device 1	Device 2
Number of A-IoT devices			20 (all assumed to have same Rx power)	
Rx power			-28dBm, -32dBm	
RF energy harvesting			0.2uW/sec @ Rx power -28dBm 0.05uW/sec @ Rx power -32dBm	
Available energy in storage			0.25uJ (1uF capacitor)	2.5uJ (10uF capacitor)
Power consumption	ON state	Msg-0 monitor	1uW	
		Rx	1uW	50uW
		Tx	1uW	100uW
	SLEEP state		0.1uW	
	OFF state		Negligible	
Inventory msg duration		Msg-0	1ms	
		Msg-1	0.5ms	
		Msg-2	0.5ms	
		Msg-3	3ms	
		Msg-4	0.5ms	
Number of resources for msg-1			8 resources per msg-0	16 resources per msg-0
DCM periodicity before the first msg-0		No DCM	N/A	
		Case 1 DCM	20ms	50ms
		Case 2 DCM	Energy amount dependent	
Msg-0 periodicity			20ms	50ms
Device available duration before the first msg-0		**No DCM**	**250ms**	**2500ms**
		Case 1 DCM	**2ms**	**1ms**
		Case 2 DCM	**30ms**	**75ms**

In Fig. B8, X = 0 is the time that the reader starts inventory (the timing that the first msg-0 is transmitted by the reader), and Y-axis indicates the ratio (%) of A-IoT devices that are not yet completed the contention-based access procedure. Fig. 9 shows that in case no DCM, the ratio of non-inventorized devices linearly decreases as the time from the start of the inventory increases. This is because available time of different devices are distributed in time and hence in each msg-0 timing, some devices are available. However, this implies that when the reader starts inventory, there can be devices that has almost zero available energy. Such devices are accessible only after they are charged until 'Turn-on' threshold. This is observed as the bottleneck, i.e., inventory completion time from the reader perspective cannot be better than the charging time for these unavailable devices.

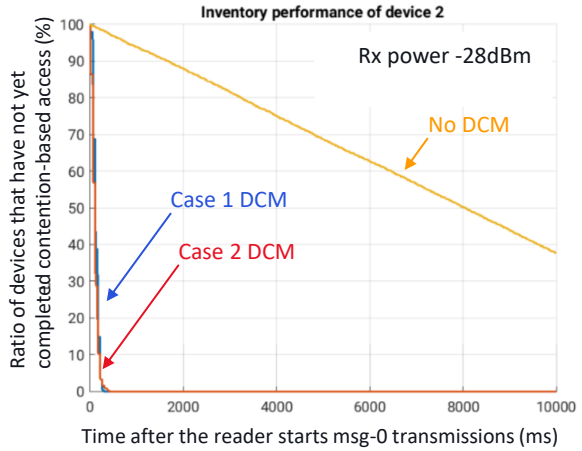
On the other hand, DCM achieves much shorter inventory completion time. This is because the device unavailable time is much shorter than the case of no DCM. Between two cases of DCM, case 1 DCM is sensitive to the Rx power; it can achieve much shorter inventory completion time under relatively high Rx power (e.g., > -28dBm) compared to no DCM, while it requires similar completion time as no DCM with low Rx power (e.g., < -30dBm). This is because SLEEP state is assumed to consume a certain power. The actual gain of Case 1 DCM highly depends on power consumption of SLEEP state. On the other hand, case 2 DCM offers significant performance benefit for all the evaluated cases.



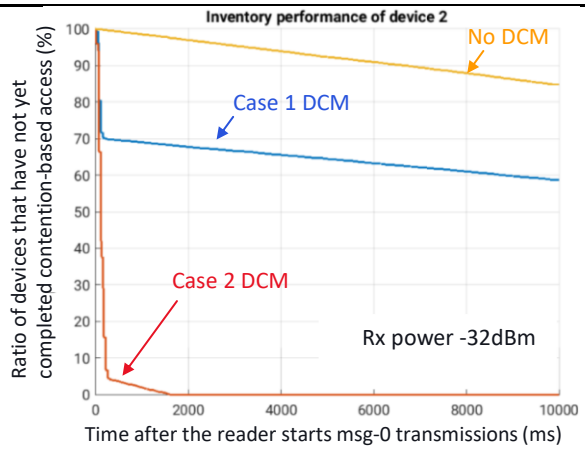
(a) Device 1, Rx power = -28dBm



(b) Device 1, Rx power = -32dBm



(c) Device 2, Rx power = -28dBm



(d) Device 2, Rx power = -32dBm

Fig. B8 Performance evaluation of inventory with and without duty-cycle monitoring

Observation B2:

- Simulation results indicate that duty-cycle monitoring is essential for A-IoT to complete inventory in a reasonable time duration.
  - Case 1 DCM is sensitive to Rx power (or RF EH efficiency) since SLEEP state consumes a certain power; it works if Rx power (or RF EH efficiency) is relatively high.
  - Case 2 DCM works with various range of Rx power (or RF EH efficiency).
  - With no DCM, inventory completion time is limited by device charging time for large amount of energy (from 'Turn-off' threshold to 'Turn-on' threshold)

Further, the analysis of inventory completion time for 600 devices is conducted. Based on the previous results, we focus on 'Case 2 DCM' as the DCM scheme.

- Method 1: Access probability control
  - For each msg-0, there are a number of msg-1 resources. The reader monitors the occupancy of msg-1 resources associated with the i-th msg-0 and indicates the access probability P(i+1) in the (i+1)-th msg-0. The indication of access probability P(i+1) is determined such that the occupancy of msg-1 resources observed by the reader is around 36.8%.
  - A device that receives the (i+1)-th msg-0 select one of the msg-1 resources associated with the msg-0 and transmits the msg-1 with the access probability P(i+1).
  - Note: method 1 is similar to the Q value indication in Query command of UHF RFID in a sense that the probability of msg-1 transmission from a device is controlled by the msg-0 indication.
  - Note: method 1 is applicable to both 'No DCM' and 'Case 2 DCM'.
- Method 2: Device grouping for time-domain distribution of DCM
  - Reader transmits msg-0 with the periodicity Pt from t = 0.
  - A device, after the first msg-0 it receives, monitors msg-0 with periodicity N x Pt based on the reader's instruction, where N is a non-zero integer value. For example, when N = 2, a device that receives (2n+1)-th msg-0 from the reader as the first msg-0 monitors (2n+3)-th, (2n+5)-th, …, msg-0s from the reader using DCM, while another device that receives (2n+2)-th msg-0 from the reader as the first msg-0 monitors (2n+4)-th, (2n+6)-th, …, msg-0s from the reader using DCM, as illustrated in Fig. B9.
  - Note: method 2 can be realized in direction 2 where the reader can provide information to a device based on which the device may become available/unavailable.
  - Note: method 2 is not applicable to 'No DCM'.

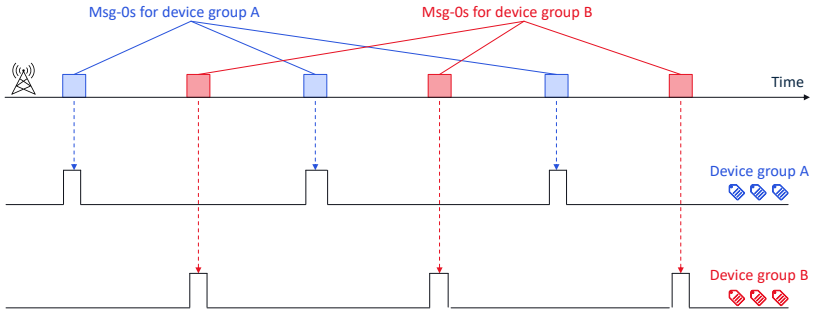


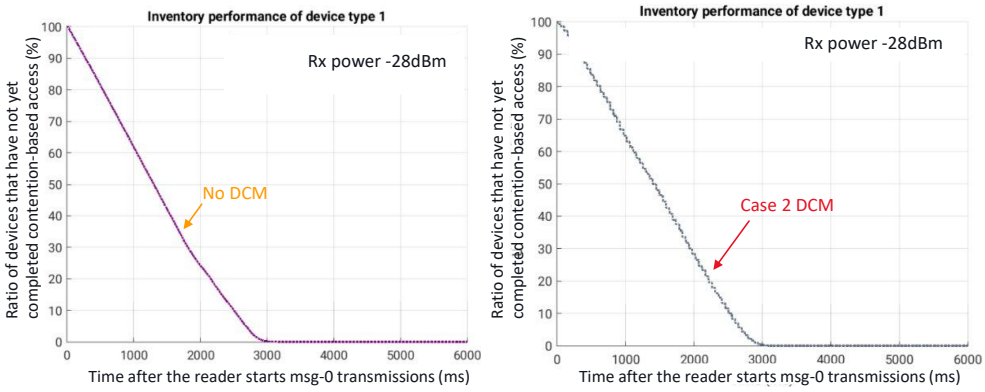
Fig. B9 Device grouping for time-domain distribution of duty-cycle monitoring

To accelerate the inventory for such a large number of devices, msg-0 transmission for the case 'No-DCM' is assumed to be aperiodic, i.e., the reader transmits msg-0 whenever possible and as much as possible. For 'Case 2 DCM', msg-0 periodicity is set to 12ms. In this additional analysis, we added Rx power -36dBm with device energy harvesting efficiency 5% as an additional case. Overall, the assumptions different from Table B1 is summarized in Table B2 below.

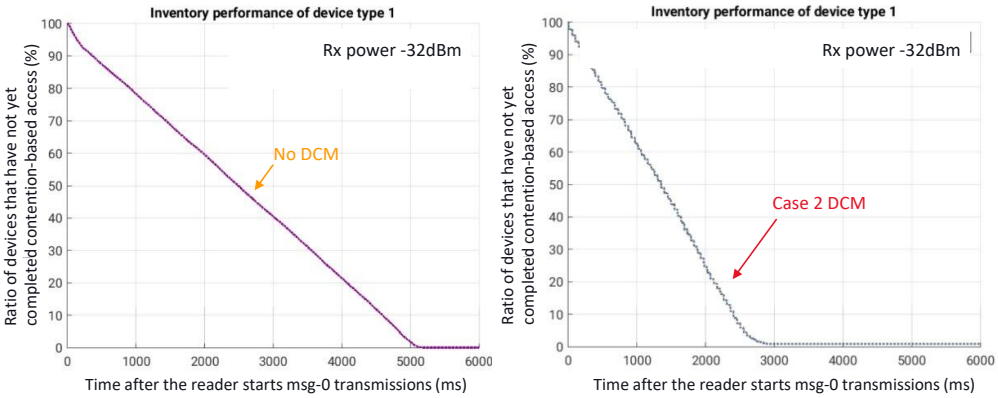
Table B2 Parameters/assumptions of inventory performance analysis

Parameters			Device 1
Number of A-IoT devices			600 (all assumed to have same Rx power)
Rx power			-28dBm, -32dBm, -36dBm
RF energy harvesting			0.2uW/sec @ Rx power -28dBm 0.05uW/sec @ Rx power -32dBm 0.025uW/sec @ Rx power -36dBm
Available energy in storage			0.25uJ (1uF capacitor)
Power consumption	ON state	Msg-0 monitor	1uW
		Rx	1uW
		Tx	1uW
	SLEEP state		0.1uW
	OFF state		Negligible
Inventory msg duration		Msg-0	1ms
		Msg-1	0.5ms
		Msg-2	0.5ms
		Msg-3	3ms
Number of resources for msg-1			8 resources per msg-0
DCM periodicity before the first msg-0		No DCM	N/A
		Case 2 DCM	Energy amount dependent
Msg-0 periodicity		No DCM	Aperiodic (msg-0s as many as possible)

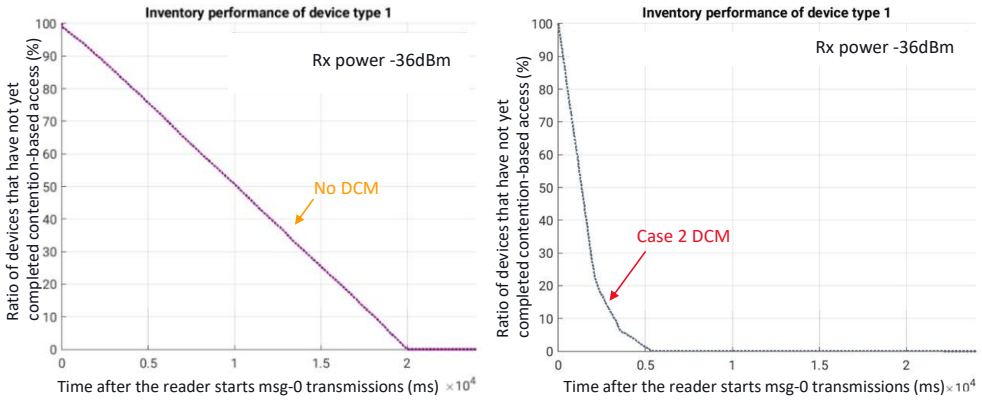
	Case 2 DCM	Periodic (12 ms)
Device available duration before the first msg-0	No DCM	250ms
	Case 2 DCM	30ms



(a) Rx power -28dBm, 'No DCM' and 'Case 2 DCM'



(b) Rx power -32dBm, 'No DCM' and 'Case 2 DCM'



(c) Rx power -36dBm, 'No DCM' and 'Case 2 DCM'

Fig. B10 Inventory completion time for 600 devices

With Rx power = -28dBm, inventory completion time is not much different between 'No DCM' and 'Case 2 DCM'. This is because the charging time from 'turn-off' threshold to 'turn-on' threshold is shorter than the necessary time to complete random access for 600 devices and hence the charging time in 'No DCM' is not visible. With lower Rx power such as -32dBm and -36dBm, the inventory completion time for 'No DCM' significantly increases since some devices need long time to charge from 'turn-off' threshold to 'turn-on' threshold when the reader starts inventory, which takes around 5 seconds with Rx power = -32dBm and 20 seconds with Rx power = -36dBm, respectively. On the other hand, 'Case 2 DCM' guarantees all the

devices have enough energy when the reader starts inventory, regardless of Rx power for each device. It is observed that the inventory completion time of 'Case 2 DCM' with Rx power = -32dBm is almost same as that with Rx power = -28dBm. With Rx power = -36dBm, 'Case 2 DCM' can complete inventory for 600 devices in around 5.5sec, whereas 'No DCM' needs 20sec.

Observation B3:

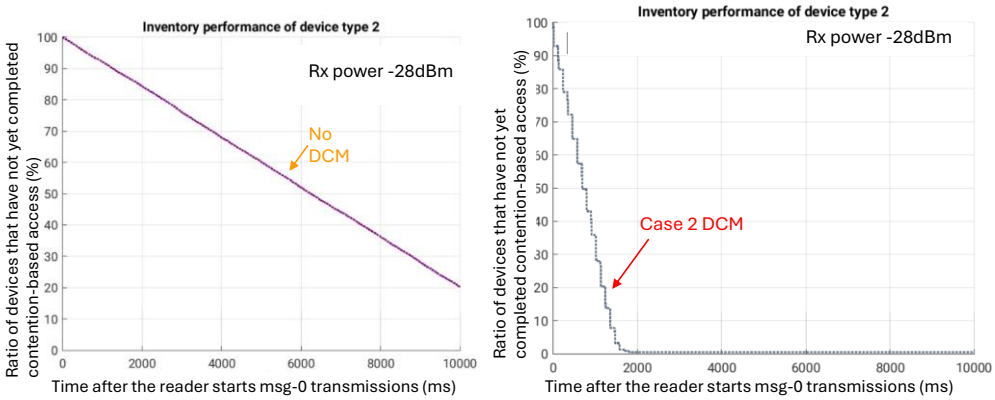
- Simulation results indicate that duty-cycle monitoring is essential for A-IoT to complete inventory in a short time duration even for 600 devices.
  - Case 2 DCM can keep the inventory completion time unchanged between Rx powers -28dBm and -32dBm.
  - Case 2 DCM can complete inventory with around 5.5 sec, whereas 'No DCM' requires around 20 sec.

We have also evaluated inventory completion time of 600 devices assuming device 2 capacitor size and power consumption. The assumptions are captured in Table B3. Except for available energy in storage and power consumption models, most of the parameters are same as those for inventory completion time evaluation for 600 device 1 above.

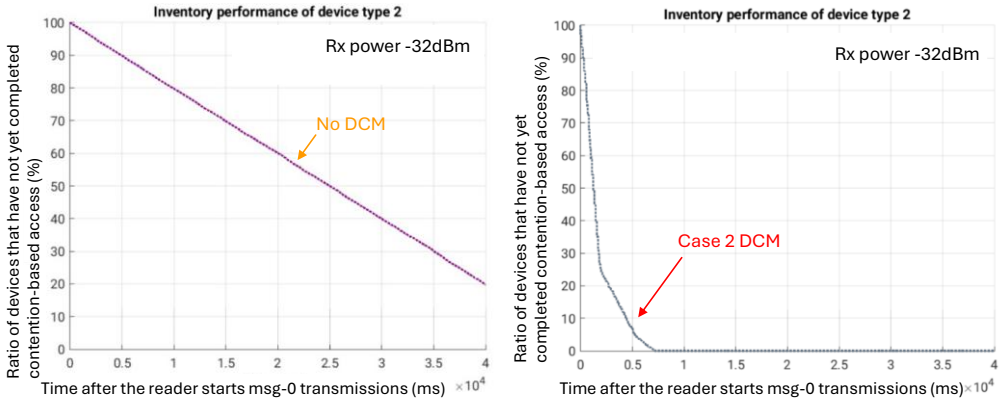
Table B3 Parameters/assumptions of inventory performance analysis

Parameters			Device 2
Number of A-IoT devices			600 (all assumed to have same Rx power)
Rx power			-28dBm, -32dBm, -36dBm
RF energy harvesting			0.2uW/sec @ Rx power -28dBm 0.05uW/sec @ Rx power -32dBm 0.025uW/sec @ Rx power -36dBm
Available energy in storage			2.5uJ (10uF capacitor)
Power consumption	ON state	Msg-0 monitor	1uW (only before the device detects the first msg-0)
		Rx	50uW
		Tx	200uW
	SLEEP state		0.1uW
	OFF state		Negligible
Inventory msg duration		Msg-0	1ms
		Msg-1	1ms
		Msg-2	0.5ms
		Msg-3	3ms
Number of resources for msg-1			16 resources per msg-0
DCM periodicity before the first msg-0		No DCM	N/A
		Case 2 DCM	Energy amount dependent
Msg-0 periodicity		No DCM	Aperiodic (msg-0s as many as possible)
		Case 2 DCM	Periodic (14 ms)
		No DCM	2500ms

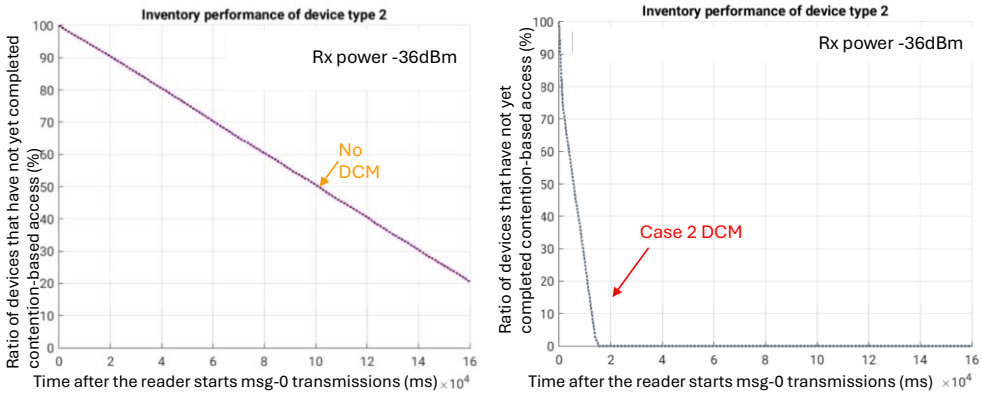
Device available duration before the first msg-0	Case 2 DCM	75ms



(a) Rx power -28dBm, 'No DCM' and 'Case 2 DCM'



(b) Rx power -32dBm, 'No DCM' and 'Case 2 DCM'



(c) Rx power -36dBm, 'No DCM' and 'Case 2 DCM'

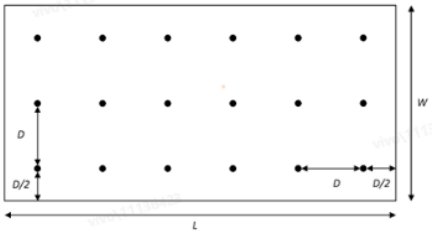
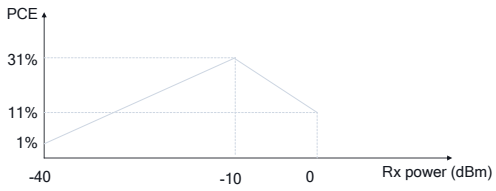
Fig. B11 Inventory completion time for 600 devices (device 2)

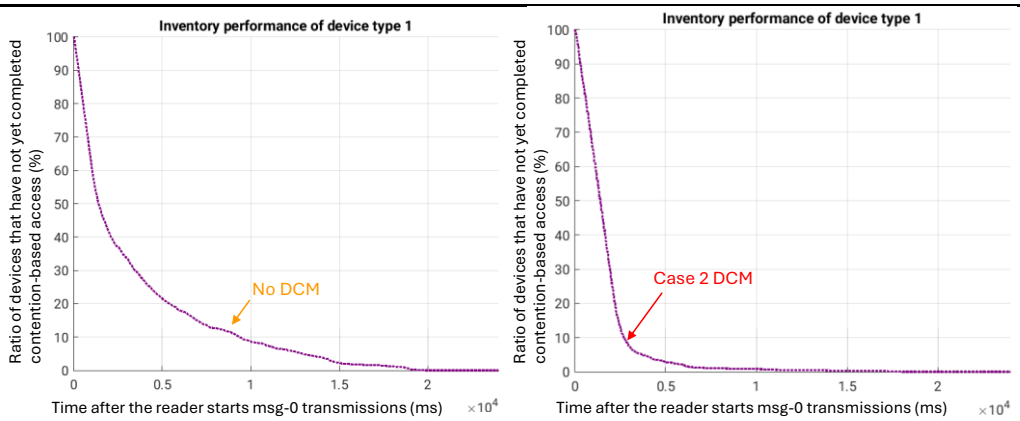
Observation B4:

- Simulation results indicate that duty-cycle monitoring is essential for A-IoT to complete inventory in a short time duration even for 600 devices (device 2).
  - ➢ Case 2 DCM can achieve inventory completion for the given device Rx power roughly within 1/10 of the time that is necessary for the case no DCM for the worst case where all devices have Rx power of -36dBm.

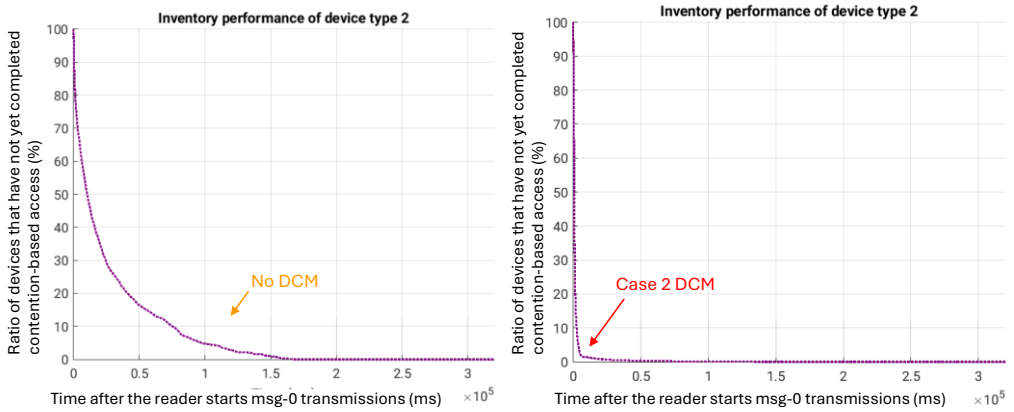
Finally, the results with the device Rx power distribution based on geometry are provided. Device Rx power distribution is assumed to be based on D1T1 as summarized in Table B4. RF energy harvesting efficiency for a given Rx power is modelled as in the table.

Table B4 Assumptions/parameters for additional performance analysis of device duty-cycle monitoring (DCM) for 600 devices with Rx power distribution

Parameter	Value
Scenario	InF-DH
Hall size	120 x 60m
Room height	10m
Sectorization	None
BS deployment	18 BSs on a square lattice with spacing D, located D/2 from the walls. L=120m x W=60m; D=20m BS height = 8 m 
Device distribution	Device Height= 1.5 m A-IoT devices drop uniformly distributed over the horizontal area
Device mobility	3kph
Device receiver sensitivity	-36dBm
RF energy harvesting	- Rx power (dBm) + 41 for Rx power (dBm) < -10 dBm - 2 * Rx power (dBm) + 11 for Rx power (dBm) >= -10dBm 
Other parameters/assumptions are same as in Tables 2.3.2a-1 and 2.3.2b-1	



(a) Device 1



(b) Device 2

Fig. B12 Inventory completion time for 600 devices with geometry-based Rx power distribution

Even with the Rx power distribution among 600 devices, we see a significant gain of DCM over no-DCM. For device 1, inventory completion time for 99% devices is about 18 seconds with "No-DCM", while it is about 8 seconds with "Case 2 DCM". For device 2, inventory completion time for 99% devices is about 150 seconds with "No-DCM", while it is less than 25 seconds with "Case 2 DCM".

Observation B5:

- Simulation results indicate that duty-cycle monitoring is essential for A-IoT to complete inventory in a short time duration even for 600 devices with geometry-based Rx power distribution.
  - For 600 of device 1, case 2 DCM can complete inventory with less than 1/2 of the time that is necessary for the case no DCM.
  - For 600 of device 2, case 2 DCM can complete inventory with less than 1/6 of the time that is necessary for the case no DCM.

Specification impacts, if any

For direction 2, specification impacts can be considered for reader and/or device perspective. For example, if msg-0 transmission periodicity/frequency for contention-access procedure is specified (or is designed as configurable), the device DCM operation can be designed to fit with that. On the other hand, if device DCM related behaviors, such as the duration of ON state or duty-cycle periodicity, are specified (or are designed as configurable), the reader's inventory/communication procedure can be designed to fit with that.

Further, accuracies of device ON state duration and/or SLEEP state duration would need to be understood. If the device clock error is assumed to be quite large, e.g., +/- 10%, for DCM operation, the uncertainty of timings/durations may need to be taken into account in the specifications related to resource/timing.

Observation B4:

- For direction 2, impacts on specifications and designs are envisioned.
  - If msg-0 transmission periodicity/frequency for contention-access procedure is specified (or is designed as configurable), the device DCM operation can be designed to fit with that.

	➢ If device DCM related behaviors, such as the duration of ON state or duty-cycle periodicity, are specified (or are designed as configurable), the reader's inventory/communication procedure can be designed to fit with that. - Accuracies of device ON state duration and/or SLEEP state duration would need to be discussed. ➢ If the device clock error is assumed to be quite large, e.g., +/- 10%, for DCM operation, the uncertainty of timings/durations may need to be taken into account in the specifications related to resource/timing. ➢ SFO correction (clock calibration) framework would offer better clock accuracy such as clock error of < +/-10% for direction 2.

Table 6.2.2-22: Details from Source [92]

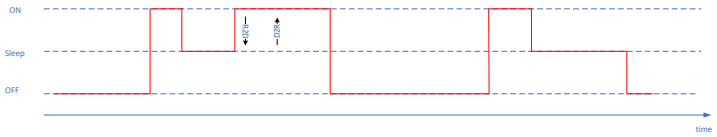
Source	Details
Source [92]	Solution description For direction 2, the reader can control the time of ON and Sleep and the device will switch the states. As shown in Figure 1, when the energy storage is full or reaches a certain level, or the device receives a RF power larger than an activation threshold, the device switches to ON to detect the R2D signal and transmit the D2R signal. By explicit or implicit way, for example, indicated by the reader, the device switches to Sleep state from ON state. After the sleep time, the device switches to ON. Additionally, it is possible that the device switches to OFF state from ON/Sleep state if the energy storage is not enough.  **Figure 1 An illustration of device's states** Observations or Analysis or Evaluations N/A Specification impacts, if any The device's states transition, including the device behavior in each state, the explicit or implicit indication, transition time, etc. The energy status reporting from the device Impacts of random access procedure If the RF energy source is applied, How and when to transmit CW for energy harvesting, for example, controlled by the reader Energy harvesting in Topology 2, for example, when the RF energy source and the UE are the same node, the impacts of different RRC states (IDLE/Inactive/Connected mode) Transmitting power of RF energy source.

Table 6.2.2-23: Details from Source [82], [102]

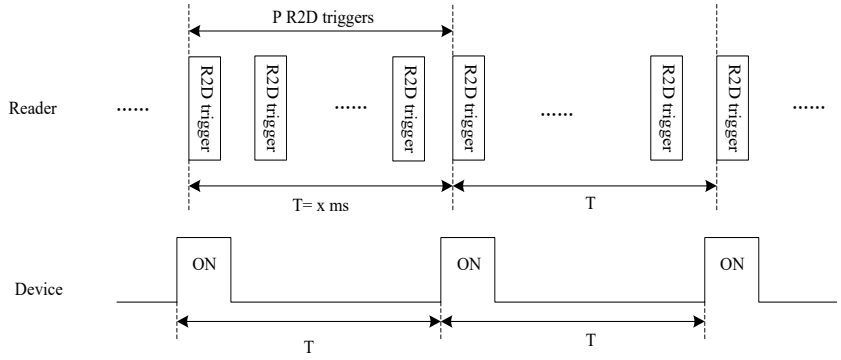
Source	Details
Source [82], [102]	Solution description The reader transmits counter decrement commands in one access round. Wherein, some counter decrement commands are necessarily transmitted on specific periodic time points. The period is equal to an absolute time length, e.g. x ms. If the stored occasion count is larger than or equal to P, the device is required to only detect the periodic decrement commands. Each time the device detects the periodic decrement

command, it reduces the stored occasion counter by P, where P is the number of decrement commands transmitted in a period. The value of P could be fixed in one round or be indicated in the periodic decrement commands.

After receiving the periodic decrement command each time, if the stored occasion counter is larger than or equal to P, the device can switch to sleep state to conserve power.

If the stored occasion counter is less than P, the device maintains ON state and detects the decrement commands sequentially. Each time the device detects a decrement command, it reduces the stored occasion counter by 1.

When the stored occasion counter decreases to 0, the device transmits Msg1.



Observations or Analysis or Evaluations

Table 4 Potential device availability time without the solution

Parameters	Values			
Device type	1		2	
Capacitance (uF)	10		1000	
High operating voltage U_{high} (V)	1	2	1	2
Low operating voltage U_{low} (V)	0.5	1	0.5	1
Available energy in storage (uJ) $C(U_{high}^2 - U_{low}^2)/2$	3.75	15	375	1500
Rx power consumption (uW)	1		150	
Available time for Rx	3.75 s	15 s	1.5 s	10 s

Table 5 Available time with the solution for device 1

Parameters	Values							
Available energy (uJ)	3.75				15			
Rx power consumption during ON state (uW)	1							
Power consumption during SLEEP state (uW)	0.2							
Ratio of ON and SLEEP duration	1:3	1:5	1:10	1:25	1:3	1:5	1:10	1:25

Available time for Rx (s)	9.375	11.25	13.75	16.25	37.5	45	55	65

Table 6 Available time with the solution for device 2

Parameters	Values							
Available energy (uJ)	375				1500			
Rx power consumption during ON state (uW)	150							
Power consumption during SLEEP state (uW)	1							
Ratio of ON and SLEEP duration	1:3	1:5	1:10	1:25	1:3	1:5	1:10	1:25
Available time for Rx (s)	9.8	14.5	25.8	55.7	39	58	103	222

Observation 1: During random access process, the above method based on Direction 2 can reduce the number of counter decrement commands that the device needs to detect and conserve device's power.

Observation 2: The solution in Table 6.2.2-23 of TR38.769 can increase the available time for R2D signal reception to at least 2.5 times for device 1 and 3.9 times for device 2.

Specification impacts, if any

Define the transmission period of the periodic decrement commands.

Define the number of decrement commands transmitted in a period.

Specify the device's behaviour after receiving a decrement command.

Table 6.2.2-24: Details from Source [112]

Source	Details
Source [112]	Solution description - Device states are defined in A-IoT system. - Three states (ON/OFF/SLEEP) or two states (ON/OFF) can be considered. - Reader needs to provide necessary information to enable the device to transition between states (ON/OFF/SLEEP or ON/OFF). - For example, the reader can provide the information that the current slot is for the device that selected slot number counter X (i.e., $0 \le X \le 2^Q-1$, Q value can be provided in inventory round triggering command). - For another example, the reader can provide the information that when the next inventory round would be started (or when the next inventory round triggering command would be sent). Observations or Analysis or Evaluations - Device available time is relatively long because the information provided by the reader allows the device to perform state transition and energy harvesting.

	- R2D resource overhead may not increase because the reader doesn't need to transmit inventory round triggering command frequently. - Large specification impact. Specification impacts, if any - Potential information feedback on the energy state (e.g., remaining energy) of the device. - State definition (ON/OFF/SLEEP or ON/OFF). - State transition mechanism (e.g., timing, threshold). - Definition of additional information to support direction 2.

Table 6.2.2-25: Details from Source [107]

Source	Details
Source [107]	Solution description ON to OFF: Support state transition from ON to OFF directly for 3-state device by indication from Reader. ON to Sleep: Support state transition from ON to Sleep immediately for 3-state device by indication from Reader. Observations or Analysis or Evaluations ON to OFF: ● In some cases, device state transit from ON to OFF directly is necessary. One typical scenario would be Reader envisions no more inventory/command task in a short period for this device after current round of inventory/command task. In this case, regardless of device's power level, state transition directly to OFF state is more power efficient than putting it to Sleep. To achieve this, disabling of Sleep state can be indicated from Reader to guide device state transition to OFF directly. The necessity of this indication is that ON state can transit to either Sleep state or OFF state depends on its power level. ON to Sleep: ● In some cases, device state transit from ON to Sleep instantly is necessary for better power saving of device. One common scenario would be when device finish the inventory/command, device can go to Sleep state immediately for power saving. To achieve this, an end indication can be provided so that device can immediately change to Sleep state upon successful reception of that PRDCH. Compared to existing solutions, an additional monitoring of PRDCH in an extra time window can be saved, thereby further saving the latency of state transition as well as device power consumption for extra monitoring. Specification impacts, if any ON to OFF: At least a disabling Sleep indication needs to be specified for 3-state devices. ON to Sleep: At least an explicit end indication on inventory/command needs to be specified for 3-state devices.

Table 6.2.2-26: Details from Source [117]

Source	Details

Source [117]

Solution description for direction 2

State description

State	Description	Transition event	Device
ON	Transmission and reception running a clock for timing maintaining the memory	Transit to 'ON' state: · the charging energy is higher than an energy threshold or a certain percentage, and the incoming RF power is higher than an activation threshold · 'SLEEP' state duration time has been achieved. Transit from 'ON' state: · the charging energy is lower than an energy threshold or a certain percentage, or fully discharged · the 'ON' state duration time has been achieved. · Receives an indication from Reader to switch to the other state.	Device 1 Device 2a
OFF	Energy harvesting	Transit to 'OFF' state: · the charging energy is lower than an energy threshold or a certain percentage or fully discharged · Receives an indication from Reader to switch to 'OFF' state. Transit from 'OFF' state: · the charging energy is higher than an energy threshold or a certain percentage	Device 1 Device 2a
SLEEP	Energy harvesting running a clock for timing maintaining the memory	Transit to 'SLEEP' state: · the charging energy is lower than an energy threshold or a certain percentage · the 'ON' state duration time has been achieved. · Receives an indication from Reader to switch to the other state. Transit from 'SLEEP' state: · the 'SLEEP' duration time has been achieved. · the charging energy is lower than an energy threshold or a certain percentage or fully discharged	Device 2a

Scheme:

- Reader transmits the control information of the corresponding energy harvesting process based on the PRDCH to the devices, which may include the duty-cycle periodicity and duty-cycle duration related information.[1]
  - Case 1: baseline case, the device can receive the reader's first command successfully.
    - The device switches to ON when the charging energy is higher than an energy threshold and the incoming RF power is higher than an activation threshold, then the device receives the reader's control information and acquires the duty-cycle related information.
    - The device transits to 'SLEEP' state to save power or 'OFF' state to complete this communication process or ON' state again to continue the communication process based on the time duration configuration or the reader's indication.
  - Case 2: enhancement case, the device misses the reader's first command.
    - The device misses the first command and does not receive the periodicity reader-assist timing indication, and switches to 'OFF' or 'SLEEP' state when the energy is lower than a threshold.
    - The device switches to ON when the charging energy is higher than an energy threshold and the incoming RF power is higher than an activation threshold, then the device receives the reader's assist information in one periodicity or one attempt of the multiple attempts.

	✧ The device transits to the 'SLEEP' state to save power and will transit to the 'ON' duration when the next reader command arrives based on the time duration obtained by the reader assist information. - The device executes the corresponding behavior based on the reader's command or indication.[2] Note 1: For enhancement case, reader will start to transmit the assist information in a certain periodicity or multiple attempts, containing the gap information between the current occasion and the next command reception occasion. Note 2: For enhancement case, reader will stop the attempt or period transmission of its assist information when a response has been received by the reader side or a max attempt amount has been achieved. **Observations** • Low communication delay and high communication efficiency. • More resource allocation and more signaling overhead due to configuration or scheduling. • Low power consumption and low energy harvesting requirement due to the introduction of 'SLEEP' state. **Specification impacts, if any** Much more specification impacts including the configuration method or the device behavior definition.

Table 6.2.2-27: Details from Source [116]

Source	Details
Source [116]	Solution description The reader provides control information to assist a device to determine whether/when to become available or unavailable. The device is assumed to have three states i.e. ON, OFF, and sleep state. Observations or Analysis or Evaluations Design the duty cycle for Tx and Rx for the device Increase the operational time for the device The time interval between the paging message can be defined based on the device energy storage Reader can provide the information related to switching between the states by the device Specification impacts, if any Scheduling can be based on the duty cycle Designing of the timer of sleep state Device behavior for monitoring the paging message Signaling indication for switching the states Impacts on random access procedure Potential information feedback on the energy state

6.3 Protocol stack and signalling procedures

6.3.1 General aspects and overall procedure

The study aims that the design on the A-IoT radio interface between reader and A-IoT device is common for Topology 1 and Topology 2. The difference of topologies is transparent to the A-IoT devices and has no impact on A-IoT devices. Unless explicitly stated, the descriptions in clause 6.3 apply to all A-IoT device types and both Topology 1 and Topology 2.

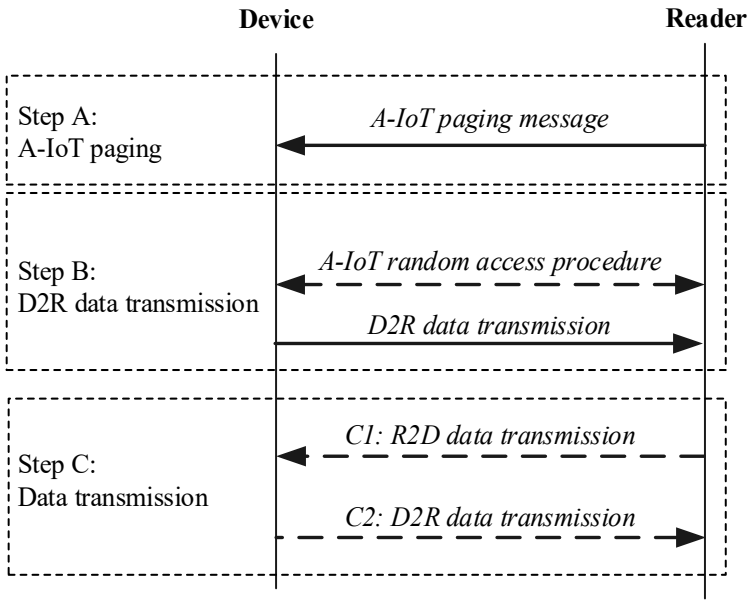


Figure 6.3.1-1 Overall AS procedures between A-IoT device and reader

The overall AS procedures can be formulated as:

- Step A: A-IoT paging. Based on the service request, the reader sends the A-IoT paging message indicating device(s) that need to respond. See clause 6.3.3.
- Step B: D2R data (device ID) transmission. Triggered A-IoT device(s) perform the device ID transmission via the A-IoT random access procedure or without using the A-IoT random access procedure. See clause 6.3.4 (and 6.3.5)..
- Step C1: Possible R2D data transmission (e.g., for sending the command). See clause 6.3.5.
- Step C2: Possible D2R data transmission (e.g., the corresponding response to command). See clause 6.3.5.

Then, above AS procedure can support indoor inventory and indoor command use cases by the following manners:

- For the detailed use case of "inventory-only", it is supported by the procedure with Step A and Step B as baseline.
- For the detailed use case of "inventory and command", it is supported by the procedure with Step A, Step B, Step C1 and Step C2, as baseline.

NOTE 1: For the use case of "inventory and command", it does not imply that the A-IoT paging message includes both the inventory and command and it does not imply the inventory and command are received by the reader at the same time from upper layer.

- For the detailed use case of "command-only":
  - It can be also supported by the baseline procedure with Step A, Step B, Step C1 and Step C2.
  - In addition, another candidate to support this use case is following, whose feasibility still depends on the conclusion from [151] and [152]:
    - Step A': A-IoT paging. Based on the service request, the reader sends the A-IoT paging message including the command, indicating device(s) to process/respond the command.
    - Step C2: Possible D2R data transmission (e.g., the device ID and/or the corresponding response to command), via the A-IoT random access procedure or without using the A-IoT random access procedure.

The following information are considered useful to be visible to the reader from CN:

NOTE 2: It needs to be further discussed on whether following information is mandatory or optional.

- The A-IoT service type (e.g., inventory, command)

NOTE 3: It needs to be further discussed if more information on command type (e.g. read/write/disable) is useful. The reader needs to know whether a D2R response is expected in D2R direction and, if so, the expected D2R message size. It needs to be further discussed on the details on how the reader gets that information (related to the "expected D2R message size" discussion in clause 6.3.5). It needs to be further discussed on whether the command type, if needed, is explicit or inferred from the "expected D2R message size" if available.

- Whether the service is targeted for one or more than one A-IoT devices
- The approximate number of target A-IoT devices of this service

6.3.2 Protocol stack and functionality aspects

For A-IoT, it is assumed the commands (e.g., read/write/disable) and/or inventory information are carried over A-IoT radio interface as the upper layer data. As to the protocol stack for A-IoT radio interface between A-IoT device and reader, it is assumed:

- RRC layer is not supported
- SDAP layer is not supported
- PDCP layer is not supported
- RLC layer is not supported
- A-IoT MAC layer is supported
- A-IoT physical layer is supported
- There is no control plane and user plane differentiation

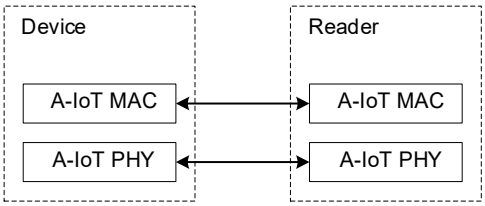


Figure 6.3.2-1 Protocol stack for A-IoT radio interface between A-IoT device and reader

As to the A-IoT required functionalities, the following functionalities are supported:

- A-IoT paging (see clause 6.3.3)
- A-IoT random access procedure (see clause 6.3.4)
- A-IoT data transmission (see clause 6.3.5)

As to the A-IoT required functionalities, at least the following functionalities are NOT supported (see TS 38.300 [153] for references for any legacy NR functionality):

- AS security (The AS layer design assumes no support of AS security, unless the study in [152] further concludes differently.)
- RRC states
- RRC connection management
- RRM L3 measurement reporting
- Mobility

- ASN.1 encoding/decoding
- Periodical system information and MIB
- Tracking/RAN area update procedure
- Per-packet QoS and per-QoS flow at AS level
- HARQ
- RLC ARQ/AM
- AS-layer (above physical layer) RLC-like/ARQ-like retransmission
- AS-layer (above physical layer) repetition

NOTE 1: It is not precluded that the reader and A-IoT device send the "payload" again as new transmission from A-IoT MAC perspective.

- Multiple A-IoT logical channels for upper layer data
- NR SR, as in TS 38.300 [153]
- NR BSR, as in TS 38.300 [153]

6.3.3 A-IoT paging

In A-IoT AS layer, the A-IoT paging functionality is to use A-IoT paging message to indicate device(s) that need to respond.

As to the A-IoT paging message, the identifier may be required to identify the device/group of devices in this trigger message (e.g., for the case of reaching a single or a group of devices). Following cases are studied:

- The A-IoT paging message containing an identifier of a single A-IoT device
- The A-IoT paging message containing a group ID that maps to multiple A-IoT devices
- The A-IoT paging message that does not contain any identifier, i.e., indicating all A-IoT devices that can receive the A-IoT paging message need to respond
- The A-IoT paging message containing multiple identifiers of A-IoT devices. The need for this use case is still to be confirmed/dependent according to the conclusion in [151]. From RAN2 perspective, it is feasible to support paging multiple identifiers of A-IoT devices, pending on TB size and multiplexing design of A-IoT paging message.

NOTE 1: The details of the above identifier and group ID and also the use case/scenario are studied in [151].

As to the A-IoT paging message, it can additionally indicate the information from which the device(s) can determine the resource(s) to be used for D2R response message(s). It can be further considered on more details for the discussion in clause 6.1.

For A-IoT device paging functionality, it is understood that the legacy paging message, legacy paging occasion and legacy DRX from NR are not supported (See TS 38.300 [153] for references for any legacy NR functionality). From RAN2 perspective, it is assumed that the A-IoT device can receive as long as there is enough energy (it can be further considered on the discussion in clause 6.1 and 6.2).

It is supported that the reader can send multiple (subsequent) A-IoT paging messages that are associated with the same service request from the CN. The duplicated response from devices for the same service request should be avoided. The A-IoT paging message can include information to avoid this duplicated response from the device to a reader. It needs to be further discussed on how to design this information in A-IoT paging message (e.g., including stage-3 details and considering the related aspects from other WGs). Then, based on this information, the device determines whether to skip sending the response to A-IoT paging message or not (if the device had successfully responded the same service before). This information should be short and simple. This information is one ID, while it needs to be further discussed

whether the ID is generated by the reader or by the core network. It needs to be further discussed on the size of this information.

It needs to be further discussed for the scenario that different readers may send A-IoT paging messages, which are associated with the same service request from the CN, to the same device for response. If this scenario is in the scope, it needs to be further discussed, by considering the progress from all the WGs.

It needs to be further discussed on whether other information is included in A-IoT paging message to indicate service type/command type or to indicate whether there will be subsequent messages.

6.3.4 A-IoT random access procedure

A-IoT random access procedure is used for the Ambient IoT device(s) to access the network for data transmission.

The A-IoT random access procedure is triggered by the reader, including triggering the access for a single A-IoT device, group of A-IoT devices, or all A-IoT devices under the coverage of the reader.

The slotted-ALOHA is the baseline for A-IoT random access procedure.

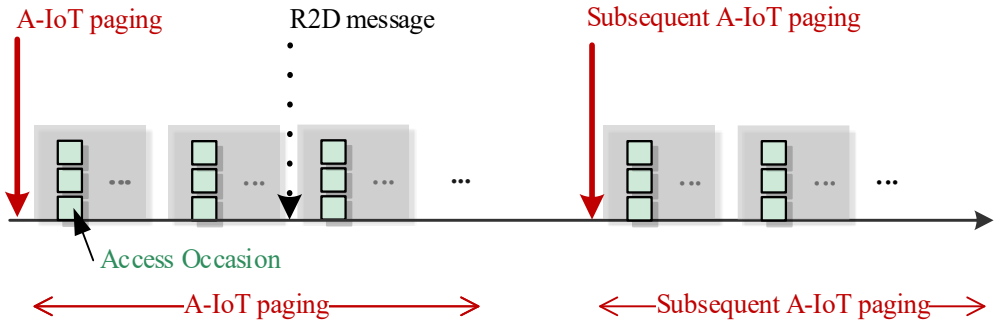


Figure 6.3.4-1 General framework of slotted-ALOHA for A-IoT random access procedure

Access occasion: An opportunity of time-frequency resource for A-IoT device(s) to perform access (e.g., transmitting the A-IoT Msg1 by the device). A set of access occasion(s) for different A-IoT device(s) is scheduled via the R2D message (referring to the "R2D transmission triggering random access" in clause 6.1.4) by the reader.

When the A-IoT device is selected to respond in accordance to the clause 6.3.3, the A-IoT device performs the following procedure:

- **Step 1**: Random access type (i.e., contention-free or contention-based) and access occasion/resource determination:
  - The A-IoT device determines the random access type from the A-IoT paging message, in accordance to clause 6.3.3. The reader can configure either contention-free access or contention-based random access (and the corresponding configurations). It needs to be further discussed on whether it is explicitly or implicitly.
  - If the random access is contention-free access:
    - Selects the indicated D2R occasion/resource;
    - Skips the contention resolution in Step 2 and performs the data transmission in accordance to clause 6.3.5.
  - If the random access is contention-based random access:
    - Performs access occasion/resource selection: as the baseline for CBRA, at least for TDMA case, the device can randomly select one access occasion for A-IoT Msg1 within the access occasions provided/assigned by the reader. It needs to be further discussed if this is applicable to FDMA case. Further enhancement option(s) can be also considered after more physical layer detailed design on TDMA and FDMA;
    - Performs the Step 2 for contention resolution.
- **Step 2**: Contention resolution of contention-based random access:

- There are three candidate solutions being studied for the contention resolution, as below (it needs to be further discussed on the down selection and/or the unified design):
  - ***Solution 1: A-IoT Msg1 without data***
    - A-IoT Msg1: When the A-IoT device identifies the start of its own access occasion, it sends one 16-bit random ID generated by the A-IoT device to the reader.
    - A-IoT Msg2: The reader responds with the successfully received random ID.

      If the A-IoT device receives the A-IoT Msg2 including a random ID, which is the same as the previously transmitted one in A-IoT Msg1, it considers the contention resolution as successful.
  - ***Solution 2: A-IoT Msg1 with data***
    - A-IoT Msg1: When the A-IoT device identifies the start of its own access occasion, it sends the A-IoT Msg1 including the upper layer data, which can be the device ID and/or any other upper layer data, in addition to one 16-bit random ID generated by the A-IoT device to the reader.
    - A-IoT Msg2: The reader may respond with the successfully received random ID.

      If the A-IoT device receives the A-IoT Msg2 including a random ID, which is the same as the previously transmitted one in A-IoT Msg1, it considers the contention resolution as successful. If the A-IoT Msg2 is not received by the device, the re-access is not autonomously performed i.e., the re-access is always controlled by the reader.
  - ***Solution 3: A-IoT Msg1 optionally includes data (unified solution supporting both Solution 1 and 2)***
    - A-IoT Msg1: When the A-IoT device identifies the start of its own access occasion, it sends one 16-bit random ID generated by the A-IoT device to the reader. In addition, it is controlled by the reader on whether to also include the upper layer data, which can be the device ID and/or any other upper layer data.
    - A-IoT Msg2: The reader responds with the successfully received random ID.

      If the A-IoT device receives the A-IoT Msg2 including a random ID, which is the same as the previously transmitted one in A-IoT Msg1, it considers the contention resolution as successful.

NOTE 1: The random ID is randomly generated by the A-IoT device.

NOTE 2: The A-IoT Msg2 is used for contention resolution, since it is assumed that the size of random ID in A-IoT Msg1 should be sufficient for contention resolution purpose. It is assumed that, with the sufficient range for random ID, the probability of multiple A-IoT devices selecting the same access occasion and the same random ID for A-IoT Msg1 transmission will be sufficiently low.

NOTE 3: It is up to the reader implementation whether A-IoT Msg2 is sent in random access *Solution 2*.

NOTE 4: Further information may be included in A-IoT Msg2 by considering the discussion in clause 6.1.

NOTE 5: For the advantage and disadvantage of the above solutions, it is understood that: *Solution 1* can support relatively large coverage for A-IoT Msg1 than *Solution 2*. *Solution 2* has the message number efficiency in case of low probability of A-IoT Msg1 collision (e.g. it can reduce one step for data transmission interaction in case of success).

After the A-IoT device considers the contention resolution as successful if the contention-based random access is used, or if the contention-free access is used, it may perform the upper layer data transmission with the reader, which can be the device ID and/or any other upper layer data, if any, in accordance to clause 6.3.5.

It is also studied for the possible unification between the above contention-free access and contention-based random access, from the device perspective. In this solution, when the A-IoT device identifies the start of its own access occasion, it sends at least one 16-bit random ID generated by the A-IoT device to the reader (regardless of the contention-free or contention-based access occasion). The motivation and disadvantage to include the random ID in contention-free access are still to be further discussed. For instance, one disadvantage is that the reader may not know whether the intended device is responding on the given resources.

It is supported for the A-IoT device to re-access in another opportunity controlled/provided by the reader (i.e., to retry the random access above), in case of D2R data transmission failure and contention resolution failure of contention-based random access.

The A-IoT device is not expected to autonomously re-access. The re-access is always controlled by reader. It is supported for reader to use the optional explicit R2D failure/success feedback indication to determine the re-access of device:

- This indication can be used at least to determine the re-access for addressing the transmisison failure of the first D2R message, which contains the device ID and/or any other upper layer data (i.e. "Msg3");
- This indication can be also used for the following D2R data (as described in clause 6.3.5), to determine the re-access for addressing the transmisison failure.

The R2D message is used by the reader to provide access occasion(s), which can be used for re-access purpose. It needs to be further discussed if additional information is needed in this R2D message to differentiate the re-access purpose.

- A-IoT paging message is one of the options for this R2D message (e.g., see the "subsequent A-IoT paging message" in Figure 6.3.4-1).
- Another option can be some R2D messages between A-IoT paging (e.g., see the other "R2D message" in Figure 6.3.4-1).

6.3.5 A-IoT data transmission

According to the protocol stack and functionalities in clause 6.3.2, more detailed data transmission functionalities are studied in this clause.

The handling of data transmission failure has been studied. It is understood that the subsequent R2D data transmission after the D2R data transmission does not need to be always sent. In case of D2R data transmission failure, the A-IoT device follows the reader subsequent R2D instruction, if any.

- For instance, the reader can repeat the R2D upper layer "command" to trigger the A-IoT device to re-send the same D2R upper layer "response" (i.e., the A-IoT device just follows the received R2D to transmit following D2R).
- For instance, the reader can re-send A-IoT Msg2 to specific device(s) to echo the random ID(s), in case of failure reception of corresponding D2R data transmission (i.e. "Msg3") after the initial A-IoT Msg2, together with the corresponding D2R resource scheduling. This can trigger the A-IoT device to re-send the same D2R data transmision (i.e. "Msg3"). If it is supported to include the echoed random IDs for multiple devices, the re-sent A-IoT Msg2 only includes the random IDs of the devices, whose "Msg3" is not successfully received.

It is studied on the functionality of segmentation. It is understood that segmentation and reassembly would add complexity. Therefore, the simplified segmentation in the A-IoT MAC layer is studied, considering following studies in physical layer:

- In both R2D and D2R directions, there is no lower bound on the minimum TB size, and a maximum TB size of around 1000 bits can be supported.
- How large TB that can be transported at a given time depends on target coverage/data rate, energy consumption/device availability, etc.

The segmentation study focuses on D2R direction (it can be further considered on R2D direction). For the candidate solution for segmentation:

- The sequence number, the segment number and the number of segments are not supported;
- An indication is used to indicate to the reader on whether the data is segmented and whether the MAC PDU is the last segment. It needs to be further discussed on the size of this indication (one or two bits) and the corresponding further details;
- It is beneficial for the reader to be able to trigger a re-transmission of a segment;

- It is assumed that the A-IoT device will not support AS layer buffering for A-IoT segmentation functionalities, i.e., all segment(s) are stored in upper layer(s).

The use of the following assistance information is studied (the need of each is still to be decided):

- The energy status report from the A-IoT device to the reader, which indicates that the A-IoT device, if the A-IoT device can determine this, does not have sufficient energy to perform the remaining/follow-up procedure.
  - The A-IoT device may report a 1-bit energy status indication to the reader in a D2R message. It needs to be further discussed if and how the reader control this indication of such report.
  - The reader may take this indication into account for the remaining/follow-up procedure (up to the reader implementation), e.g., the reader may choose not to transmit the subsequent messages for some time, or the reader may choose to not do anything in particular, etc.
- It is beneficial for the reader to know an estimate of expected D2R message size. This D2R message size information would be beneficial but it is not essential. Two options are studied:
  - Option 1: The estimate of expected following D2R message size from the CN to the reader;
    - Advantages/disadvantages of option 1: If this information is available from CN, it saves the overhead/latency caused by Option 2 in AS layer. It can be also used by the reader as early as for the first D2R message, if needed.
  - Option 2: The simple message size indication from the device to the reader.
    - Advantages/disadvantages of option 2: It is useful in case the CN is not able to provide or does not provide the expected D2R message size. Limited granularity is supported considering the AS layer overhead. It requires D2R message to carry this information in advance, before the reader can actually use this size information for scheduling.

From higher layer perspective, it is assumed that "AS ID" (if defined according to the design in clause 6.1) is used at least for purpose of D2R scheduling and R2D reception. From higher layer perspective, it is assumed that this "AS ID" should be a short AS layer ID, rather than the full upper layer device ID. It needs to be further discussed if this "AS ID" can be based on partial upper layer device ID. It needs to be further discussed on the length of this "AS ID". From higher layer perspective, following options are possible for this "AS ID" (it is aimed to define one common design for all access procedures in clause 6.3.4, if technically possible):

- Option 1: a random ID (if used in first D2R message) can be reused;
- Option 2: the reader assigns this "AS ID" to the device. It needs to be further discussed via which R2D message.
- Option 3: It is up to the reader whether to reuse the random ID (if used in first D2R message) as the "AS ID" or to assign a new "AS ID". It needs to be further discussed via which R2D message.

NOTE 1: The further down-selection for "AS ID" needs to consider the involved message and whether the reader needs to handle the collision.

6.3.6 Topology 2 aspects on the interface between UE reader and RAN

For Topology 2, the architecture/protocol stack options in [151] are studied (also corresponding to the studies in clause 6.4.2.1), while no new AS layer architecture/protocol stack options will be studied:

- RRC based solution: A-IoT upper layer information is explicitly forwarded via UE reader’s NR Uu RRC message.
- NAS based solution: A-IoT upper layer information is transmitted over UE reader's NAS message.
- UP based solution: A-IoT upper layer information is transmitted as UE reader's user plane data.

It is assumed that the intermediate UE authorization is performed by upper layers, according to [151] and [152]. It is not studied, from RAN2 signalling perspective, the scenario to support bistatic mode of operation (e.g., D2T2-A1).

The radio resources used by A-IoT radio interface between the A-IoT device(s) and UE reader are controlled by the network. The radio resources, which are dedicated for a UE reader, are only configured to the UE reader via dedicated signalling. The mechanisms for shared resource pool amongst UE readers are not considered in this release. The UE reader in coverage of BS scenario is supported. The UE reader may perform the A-IoT procedure on A-IoT radio interface between the reader and the device(s), only if the radio resource configuration is valid in the cell, which is under network control. The radio resource validity across multiple cells is not supported in this release. Hence, for all options below, the resources remain valid only in the same cell in which the resources were configured.

For the case when the radio resources are allocated semi-statically by the network, the radio resource validities are studied in the following scenarios with candidate options (down-selection can be decided in the WI phase):

- Scenario 1: In RRC_CONNECTED, the UE reader can perform A-IoT operations using the allocated A-IoT radio resource for A-IoT radio interface:
  - Option 1: UE reader receives the resources configuration in NR Uu dedicated RRC signalling. The radio resources remain valid until the network releases them explicitly. This would be supported by the NR Uu RRC reconfiguration; and/or
  - Option 2: UE reader receives the resources configuration in NR Uu dedicated RRC signalling, which configures a time period in which the corresponding resource can be used. The UE reader considers that the resources remain valid for that time period, unless the resource configuration is explicitly released by the network by the NR Uu RRC reconfiguration.
- Scenario 2: In RRC_CONNECTED, the UE reader can perform A-IoT operations using the allocated A-IoT radio resource for A-IoT radio interface, upon its temporary out of connection scenarios (e.g., RLF and handover cases):
  - Option 1: The UE reader considers the resources as temporarily invalid during the temporary out of connection condition. The resource may become valid again after the UE recovers from the condition.
  - Option 2: The UE reader considers that the resources remain valid for a time period. It needs to be further discussed on whether this time period belongs to the one in Option 2 for Scenario 1 or is a different time period. It needs to be further discussed on whether the duration of this time period is related to the ongoing A-IoT operations.

NOTE 1: The A-IoT radio resource can be (re)configured by UE reader's RRC reconfiguration (including during handover procedure, after re-establishment of the UE reader, etc.)

- Scenario 3: In RRC_ INACTIVE, the UE reader can perform A-IoT operations using the allocated A-IoT radio resource for A-IoT radio interface:
  - Option 1: UE receives the resources configuration in NR Uu dedicated RRC signaling, before moving to RRC_INACTIVE. The radio resources remain valid until the network releases them explicitly. This would be supported by the NR Uu RRC reconfiguration (the UE may need to resume or be paged by the network back to RRC_CONNECTED before that).
  - Option 2: UE reader receives the resources configuration in NR Uu dedicated RRC signalling, before moving to RRC_INACTIVE, which configures a time period in which the corresponding resource can be used. The UE reader can consider that the resources remain valid for that time period.
- Scenario 4: In RRC_ IDLE, the UE reader can perform A-IoT operations using the allocated A-IoT radio resource for A-IoT radio interface.
  - If this scenario 4 is supported, the options of scenario 3 can be the candidates.
  - The support of scenario 4 (UE reader performing A-IoT operations in RRC_IDLE) is down-prioritized. It can be considered in WI phase, if feasible with minimum specification impact.

For the radio resources allocation request, following alternatives are studied:

- Alternative 1: BS configures/allocates a set A-IoT radio resource to the UE reader, in response to a request from the UE reader; and/or
- Alternative 2: BS configures/allocates a set A-IoT radio resource to the UE reader, based on the service request from CN.

NOTE 2: The above two alternatives can co-exist. The configuration for radio resource allocation may also include the validity criteria (if applicable, see the time period in previous option 2 of those scenarios)

For the UE reader selection, it is up to RAN3 and SA2 on whether BS or CN selects the UE reader(s) and what information is shared between BS and CN. From RAN2 perspective, it is assumed, at least for RRC based solution, there may be radio related reasons for the BS to be involved in the UE reader selection.

6.4 RAN architecture aspects

This clause attempts to identify and describe architectural elements necessary to define a RAN architecture for support of Ambient IoT embedded in the overall 5G system architecture in support of topology 1 and topology 2 (as defined in TR 38.848 [2]).

This clause also attempts to identify a functional split between RAN and CN.

The logical system architecture for A-IoT consists of the following architectural elements:

A-IoT device: Equipment with characteristics outlined e.g., in TS 22.369 [154] and TR 38.848 [2].

A-IoT RAN: Hosts certain functions for A-IoT as part of the functional in RAN.

A-IoT radio: Radio interface between A-IoT device and A-IoT RAN node in topology 1 and between A-IoT device and A-IoT-enabled UE in topology 2.

A-IoT CN: Hosts certain functions for A-IoT as of the functional in CN.

NOTE 1: the details of A-IoT CN are subject to SA2.

NOTE 2: Further details regarding A-IoT functions hosted in the A-IoT CN and in the A-IoT RAN and the respective functional split to be decided by RAN2, RAN3 and SA2.

XX interface: Interface between the A-IoT RAN/A-IoT-enabled gNB and the A-IoT CN on which certain A-IoT specific functions are performed.

XX interface is NG interface, NGAP is used between A-IoT RAN and AIoT CN.

NOTE 3: How the functions represented by the XX interfaces are defined in various solutions and A-IoT CN topologies (direct/indirect connection between AIoT RAN and AIOTF).

Common reader function: A function that communicates with the A-IoT device by means of A-IoT radio.

A-IoT RAN node function: A function that contains e.g., the control of the A-IoT radio resources used towards the A-IoT device.

NOTE 4: Aspects concerning coordination of the Upper Layer functions (e.g., Inventory, Command) e.g., in case these functions have to be performed over a multitude of instances of the Common Reader Function, need further discussion.

6.4.1 Support of Topology 1

Figure 6.4.1-1 depicts a logical system architecture for topology 1, where the Common reader function and A-IoT RAN node function are deployed within an A-IoT RAN.

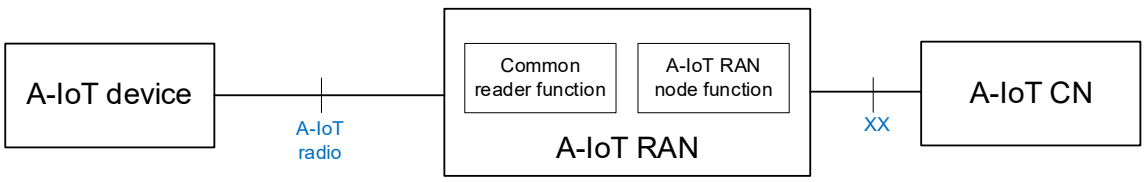


Figure 6.4.1-1 Logical system architecture for topology 1

For Topology 1, architecture and protocol aspects of split RAN architecture are not studied.

In Topology 1, the XX interface is the NG-C interface.

Figure 6.4.1-2 shows the Protocol stack for Topology 1:

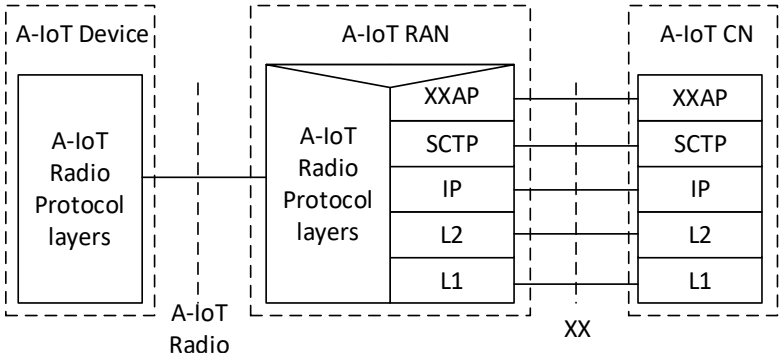


Figure 6.4.1-2. Protocol Stack for Topology 1

NOTE 1: The protocol stack in Figure 6.4.1-2 does not illustrate how A-IoT upper layer information, if any, is transported over XXAP.

For topology 1, the XXAP is terminated at an A-IoT RAN node.

The signalling transport for XXAP at the A-IoT RAN node is SCTP/IP. Other options of signalling transport for XXAP at the A-IoT RAN node (e.g., HTTP/2/TLS/TCP) were discussed, but will not be pursued.

NOTE 2: Whether "XXAP" is realised by including AIoTF information in NGAP or by carrying a new protocol layer over NGAP, needs further discussion and decision.

NOTE 3: The A-IoT CN may include AMF and A-IoTF which is up to SA2 decision.

In Topology 1, an A-IoT RAN node may serve one or more readers.

The A-IoT RAN node should enable the coordination of the usage of the A-IoT radio resources among readers.

NOTE 4: How to perform coordination of the usage of the A-IoT radio resources among readers needs further discussion.

Reader selection may need coordination between A-IoT RAN node and A-IoT CN.

NOTE 5: How to perform reader selection needs further discussions.

NOTE 6: In Topology 1, whether the AIoTF needs to be aware of the readers served by the A-IoT RAN node and their location and how this information is provided to the AIoTF needs further discussion.

6.4.2 Support of Topology 2

Figure 6.4.2-1 depicts a logical system architecture for topology 2, where the Common reader function is located at an A-IoT-enabled UE, and the A-IoT RAN node function is located at an A-IoT-enabled gNB.

The following definitions apply:

A-IoT-enabled gNB: A gNB supporting A-IoT RAN node function in topology 2, which is able to communicate with the A-IoT-enabled UE via NR Uu interface.

A-IoT-enabled UE: A UE supporting Common reader function, which is able to communicate with the A-IoTdevice via the A-IoT radio interface.

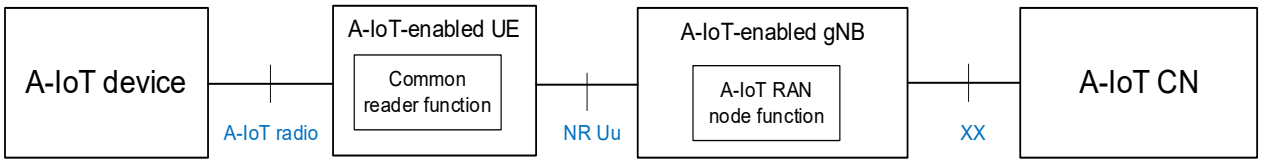


Figure 6.4.2-1 Logical system architecture for topology 2

NOTE 1: Figure 6.4.2-1 doesn't illustrate the protocol between A-IoT enabled UE and A-IoT CN, if needed.

NOTE 2: The A-IoT CN could include AMF, UPF and A-IoTF, which is up to SA2 decision.

NOTE 3: Details of the radio resource management performed by the A-IoT enabled gNB are subject to RAN1 and RAN2 mechanisms.

For Topology 2, architecture and protocol aspects of split RAN architecture are not studied.

In Topology 2, XX interface is NG interface.

In Topology 2, the RAN architecture should enable the coordination of the usage of the A-IoT radio resources among readers.

NOTE 4: How to perform coordination of the usage of the A-IoT radio resources among readers needs further discussion.

An A-IoT-enabled gNB could support both topology 1 and topology 2, this is an implementation matter.

6.4.2.1 Solutions for Topology 2

6.4.2.1.0 General

To support Topology 2, the following solutions are to be studied for conveying A-IoT upper layer information:

- **RRC based solution.** With this solution, A-IoT CN applies A-IoT upper layer information explicitly over XXAP signaling. A-IoT upper layer information is then relayed explicitly to/from the A-IoT-enabled UE via NR Uu RRC.
- **NAS based solution**. With this solution, there is no explicit termination of A-IoT upper layer information at A-IoT-enabled gNB. A-IoT upper layer information is transmitted over A-IoT enabled UE's NAS.
- **UP based solution**. With this solution, there is no explicit termination of A-IoT upper layer information at A-IoT-enabled gNB. A-IoT upper layer information is transmitted as A-IoT-enabled UE's user plane data.

NOTE 1: The protocol stack for each solution option does not illustrate A-IoT CN internal architecture and how A-IoT upper layer information is transported, if any.

No evaluation and no down selection for T2 solutions has been performed in RAN3.

Only if the UE is authorized to perform the A-IoT service, the UE can communicate with the A-IoT device and be configured with AIoT radio resources controlled by the A-IoT enabled gNB.

NOTE 2: The NGAP protocol functions for the A-IoT-enabled UE authorization needs further discussions.

In RRC based solution, Reader selection may need coordination between A-IoT-enabled gNB and A-IoT CN.

NOTE 3: How to perform reader selection needs further discussions.

NOTE 4: In NAS/UP based solution, whether A-IoT CN selects the A-IoT-enabled UE for the selection needs further discussion.

6.4.2.1.1 Solution 1: RRC based solution

Upon receiving XXAP: A-IoT related message from A-IoT CN, the A-IoT-enabled gNB transmits the related information towards the A-IoT-enabled UE via NR Uu RRC, and vice versa.

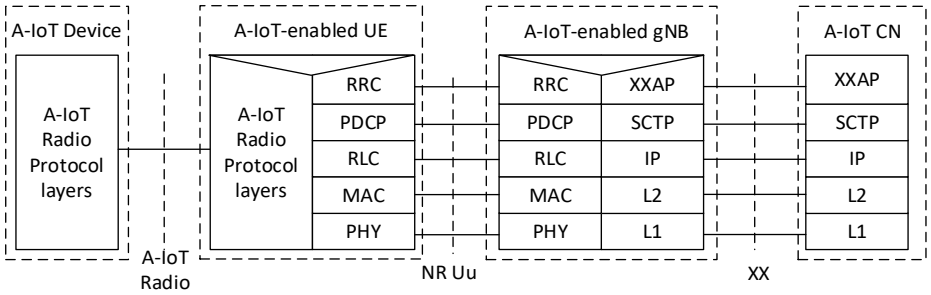


Figure 6.4.2.1.1-1: RRC based solution of Topology 2

In this solution, XX interface is NG-C interface.

NOTE: Whether "XXAP" is realised by including AIoTF information in NGAP or by carrying a new protocol layer over NGAP, needs further discussion and decision.

6.4.2.1.2 Solution 2: NAS based solution

The A-IoT related messages between the AIoTF and the A-IoT-enabled UE are carried via A-IoT-enabled UE's DL/UL NAS packets, the A-IoT-enabled gNB handles the A-IoT-enabled UE's NAS packets as legacy, i.e., using DL NAS Transport and UL NAS Transport procedures over NGAP.

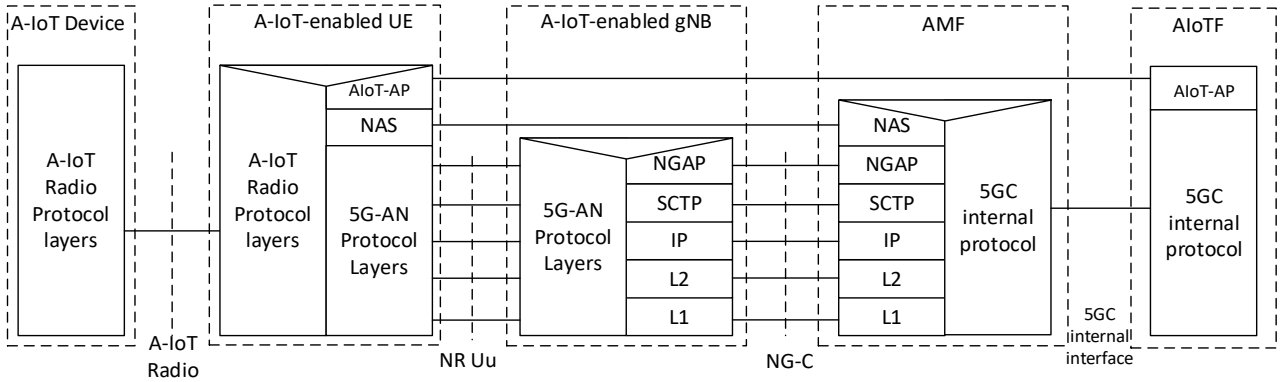


Figure 6.4.2.1.2-1: NAS based solution of Topology 2

NOTE 1: The communication between AIoTF and the A-IoT-enabled UE via A-IoT related messages and the presence of AIoT-AP protocol are being discussed in SA2. If agreed by SA2, the AIoT-AP may carry same/similar information as defined for other solutions over XXAP, this requires coordination between RAN3 and SA2.

NOTE 2: The definition and description of AIoTF and AIoT-AP refers to TR 23.700-13 [151].

There are different ways discussed to achieve A-IoT radio resource coordination and allocation as described in 6.5.1.3. Figure 6.4.2.1.2-2 illustrates one candidate solution to support AIoT session resource control for the NAS based solution by utilising XXAP between A-IoT CN and A-IoT enabled gNB.

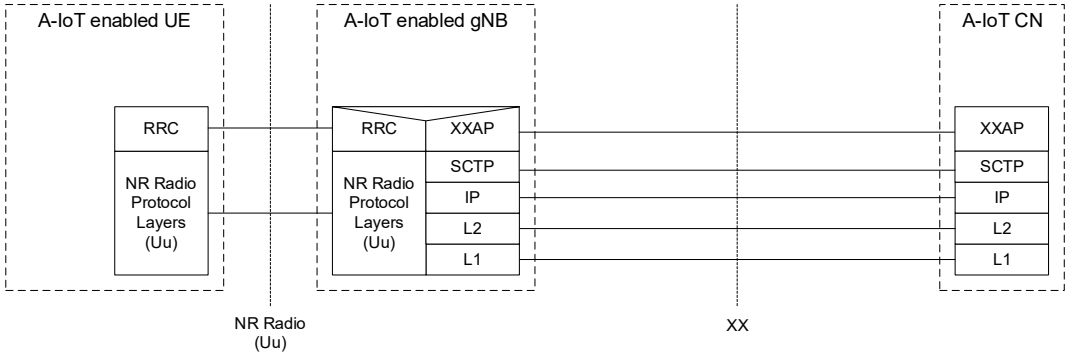


Figure 6.4.2.1.2-2: one candidate solution of Resource control for NAS based solution of Topology 2

For the NAS-based solution there are other candidate solutions discussed (e.g., A-IoT enabled UE request-based resource allocation) described in clause 6.5.1.3 which might not need this additional protocol layer (XXAP) for resource allocation, refer to clause 6.5.3.1.2.

6.4.2.1.3 Solution 3: UP based solution

The A-IoT related messages between the AIoTF and the A-IoT-enabled UE are carried via A-IoT-enabled UE's PDU Session, the A-IoT-enabled gNB handles the A-IoT-enabled UE's user plane data as legacy, i.e., over NG-U GTP-U tunnels.

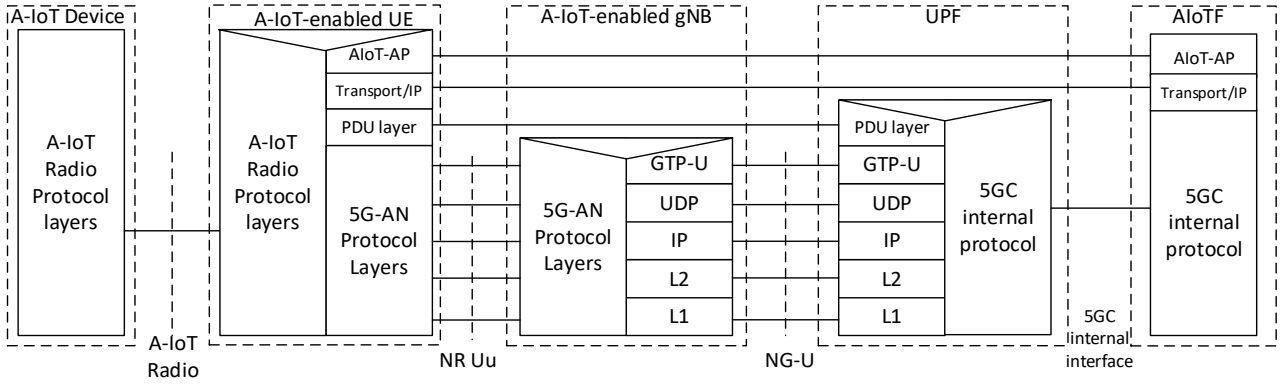


Figure 6.4.2.1.3-1: UP based solution of Topology 2

NOTE 1: The communication between A-IoTF and the A-IoT-enabled UE via A-IoT related messages and the presence of AIoT-AP protocol is being discussed in SA2. As the AIoT-AP may carry same/similar information as defined for other solutions over XXAP, this requires coordination between RAN3 and SA2.

NOTE 2: The definition and description of AIoTF and AIoT-AP/Transport/IP refers to TR 23.700-13 [151].

There are different ways discussed to achieve A-IoT radio resource coordination and allocation as described in 6.5.1.3. Figure 6.4.2.1.3-2 illustrates one candidate solution to support AIoT session resource control for the UP based solution by utilising XXAP between A-IoT CN and A-IoT enabled gNB for resource control.

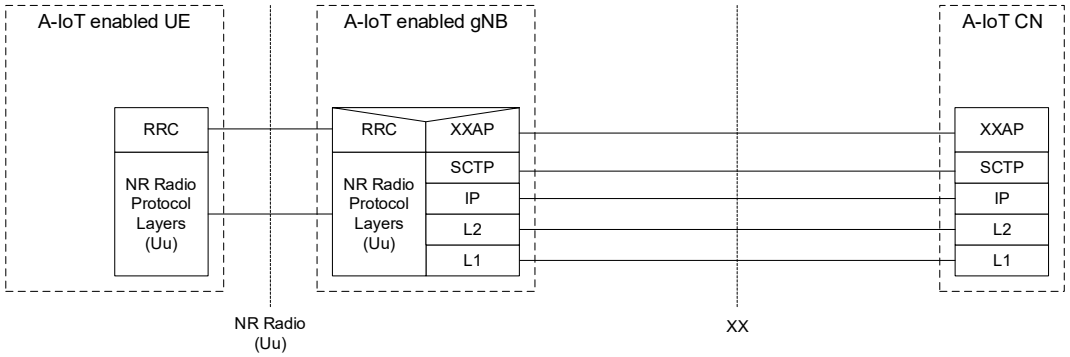


Figure 6.4.2.1.3-2: one candidate solution of Resource control for UP based solution of Topology 2

For the UP based solution there are other candidate solutions discussed (e.g., A-IoT enabled UE request-based resource allocation) described in clause 6.5.1.3 which might not need this additional protocol layer protocol layer (XXAP) for resource allocation, refer to clause 6.5.3.1.2.

6.5 Impacts on CN-RAN interface

6.5.1 Information exchanged between A-IoT CN and A-IoT RAN

6.5.1.1 Inventory

Inventory can be sent by the A-IoT CN for a single device, or a group of devices, or all devices.

The Inventory Request from the A-IoT CN to the A-IoT RAN, may include the following:

(1) A-IoT Device Identification (to find a single device, a group of devices, or all devices)

NOTE 1: The definition of this identification is out of RAN3 scope.

For topology 1, A-IoT RAN needs to store the A-IoT Device Identification received from the inventory request.

NOTE 2: Whether to interpret and process AIoT Device Identification received from the inventory request needs further discussion.

(2) Scope of inventory request (e.g., a certain area in which the inventory is to be triggered)

Multiple individual A-IoT Device IDs (one ID per device) can be provided to the A-IoT CN via a single Inventory Report.

NOTE 3: For topology 1, whether device ID received from the device is sent transparently to the A-IoT CN by the A-IoT RAN depends on RAN2 and SA2.

NOTE 4: Whether the description in this clause apply to A-IoT enabled gNB in topology 2 needs further discussion and is dependent on SA2 and RAN2.

6.5.1.2 Command

Command can be sent by the A-IoT CN for a single device.

For topology 1, A-IoT RAN node should be able to differentiate between command and inventory.

NOTE 1: Whether command can be performed for a group of devices depends on RAN2, SA2 and SA3.

NOTE 2: Whether the A-IoT-enabled-gNB should be able to differentiate between command and inventory needs further discussion, which depends on SA2 and RAN2.

NOTE 3: Whether the description in this clause apply to A-IoT enabled gNB in topology 2 needs further discussion and is dependent on SA2 and RAN2.

6.5.1.3 A-IoT radio resource allocation in case of NAS/UP based solutions

In NAS/UP based solutions, there are different ways to trigger A-IoT session resource allocation, upon CN request or upon UE request.

A-IoT session resources can be requested by the AIoT CN in advance or in parallel to the NAS/UP based communication with the A-IoT device.

NOTE 1: In case of UE request, whether and how the A-IoT enabled gNB is informed about the A-IoT session when A-IoT enabled UE requests for A-IoT radio resources and the interaction with resource allocation needs further discussion.

NOTE 2: There were also discussions about requesting AIoT session resources along with the NAS/UP based communication. Details would need further discussions.

6.5.2 Signaling and Procedures for Topology 1

NOTE: XX communication depicted in the following clauses uses protocol elements (messages and information elements) detailed in clause 6.5.1 and are not repeated unless additional description is necessary.

6.5.2.1 Candidate procedures for A-IoT Inventory for Topology 1

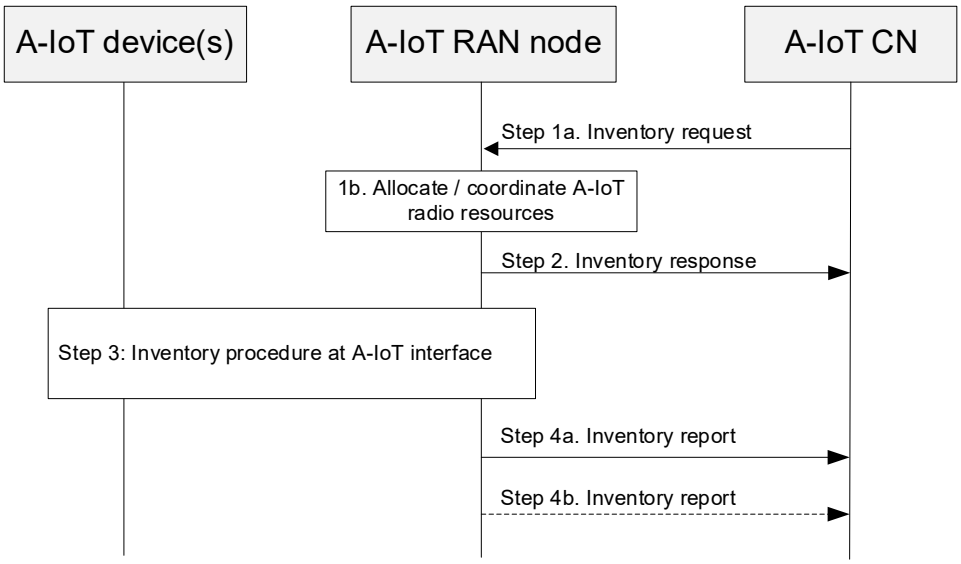


Figure 6.5.2.1-1: Message flow for A-IoT Inventory in Topology 1

1a. The A-IoT CN sends an Inventory request message to the A-IoT RAN node, taking into account, among others, the A-IoT transaction scope.

1b. The A-IoT RAN node allocates and coordinates the usage of A-IoT radio resources.

2. The A-IoT RAN node sends an Inventory response message to the A-IoT CN.

NOTE 1: In step 2, the A-IoT RAN node may instead send an Inventory failure message to the A-IoT CN indicating that the inventory procedure could not be initiated towards the A-IoT device(s).

3. The A-IoT RAN node performs the inventory procedure towards the A-IoT device(s) over the A-IoT radio interface.

4a/4b. After receiving the inventory result from the A-IoT device(s), the A-IoT RAN node may send one or multiple Inventory reports towards the A-IoT CN including the received inventory result.

NOTE 2: Steps 4a/4b may happen in parallel with Step 3 for different A-IoT devices.

6.5.2.2 Candidate procedures for A-IoT Command in Topology 1

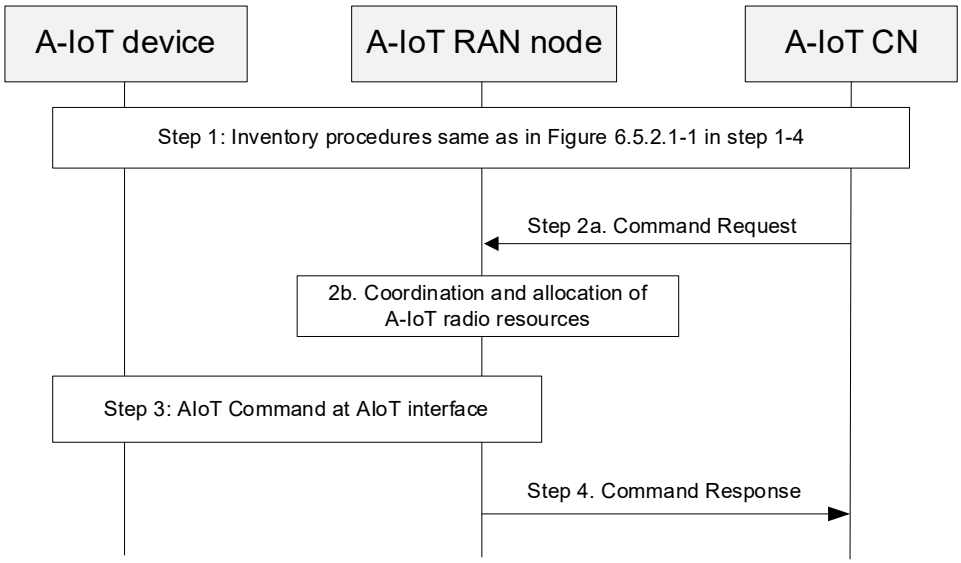


Figure 6.5.2.2-1: Message flow for A-IoT Command in Topology 1

1. The Inventory procedures are performed in step 1, which are the same as in Figure 6.5.2.1-1 in step 1-4.

NOTE 1: Step 1 is performed for the "A-IoT Inventory and Command" case. Whether and under which conditions step 1 may be skipped in case of "Command-only" and which consequences this would have for the overall message flow depends on discussions led by SA2, SA3 and RAN2.

NOTE 2: Whether and which way steps 2a and 2b may be part of step 1 is dependent on discussions led by SA2, SA3 and RAN2.

2a. The A-IoT CN node sends a Command Request message to the A-IoT RAN.

2b. The A-IoT RAN node coordinates the usage of A-IoT radio resources and allocates A-IoT radio resources for the A-IoT command session.

3. The A-IoT RAN node performs A-IoT command procedures towards the A-IoT device over the A-IoT radio interface.

4. The A-IoT RAN node sends a Command Response message to the A-IoT CN, if any command result is received from A-IoT device, the A-IoT RAN node may include the command result in the Command response message.

NOTE 3: In step 4, the A-IoT RAN node may instead send a Command Failure message to the A-IoT CN indicating that the command procedure has failed.

6.5.3 Signaling and Procedures for Topology 2

6.5.3.1 Candidate procedures for A-IoT Inventory for Topology 2

6.5.3.1.1 RRC based solution

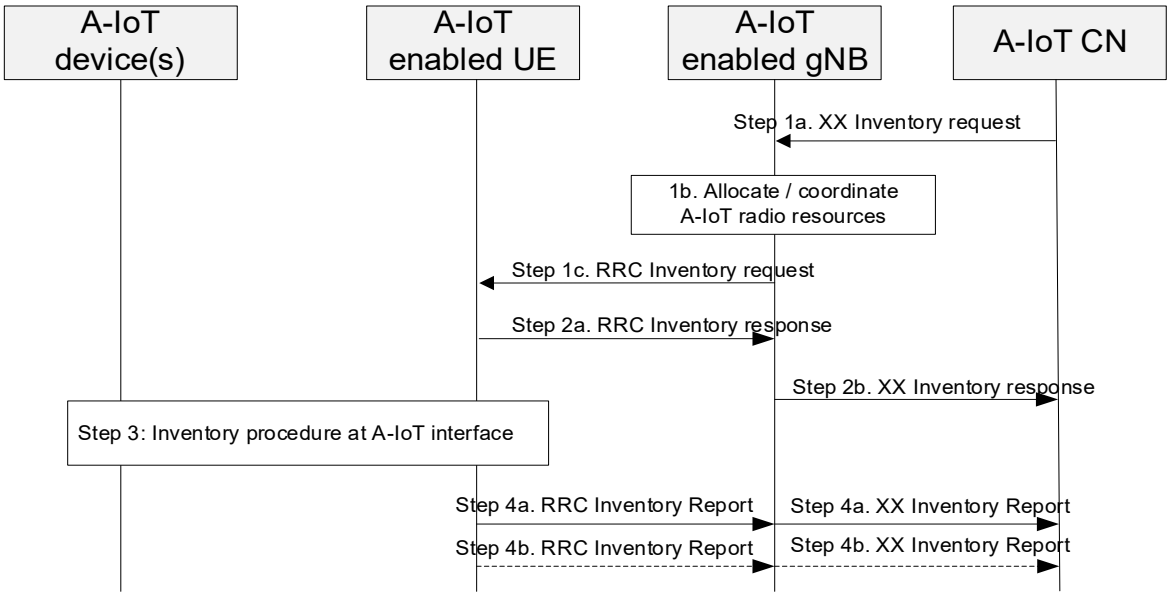


Figure 6.5.3.1.1-1: Message flow for A-IoT Inventory in Topology 2 - RRC-based solution

1a. The A-IoT CN sends an Inventory request message to the A-IoT enabled gNB

1b. The A-IoT enabled gNB allocates and coordinates usage of A-IoT radio resources.

1c/2a RRC communication with the A-IoT enabled UE takes place.

NOTE 1: RRC based communication is only depicted schematically, details subject to RAN2.

2. The A-IoT-enabled gNB sends an Inventory response message to the A-IoT CN.

NOTE 2: In step 2, the A-IoT-enabled gNB may instead send an Inventory failure message to the A-IoT CN indicating that the inventory procedure could not be initiated towards the A-IoT device(s).

3. The A-IoT-enabled gNB requests the A-IoT-enabled UE(s) to trigger inventory procedure towards the A-IoT device(s).

4a/4b. After receiving inventory result reported from the A-IoT enabled UEs, the A-IoT-enabled gNB may send one or multiple Inventory reports towards the A-IoT CN including the received inventory result.

NOTE 3: Steps 4a/4b may happen in parallel with Step 3 for different A-IoT devices.

NOTE 4: Step 4a/4b between A-IoT-enable UE and A-IoT-enabled gNB subject to RAN2.

6.5.3.1.2 NAS/UP based solution

Note: Whether and which of the following options on CN triggered A-IoT session resource allocation and UE triggered A-IoT session resource allocation will be finally considered would need further discussion involving other WGs as needed. This was not done in this study.

NAS/UP based solution – A-IoT CN triggered A-IoT session resource allocation

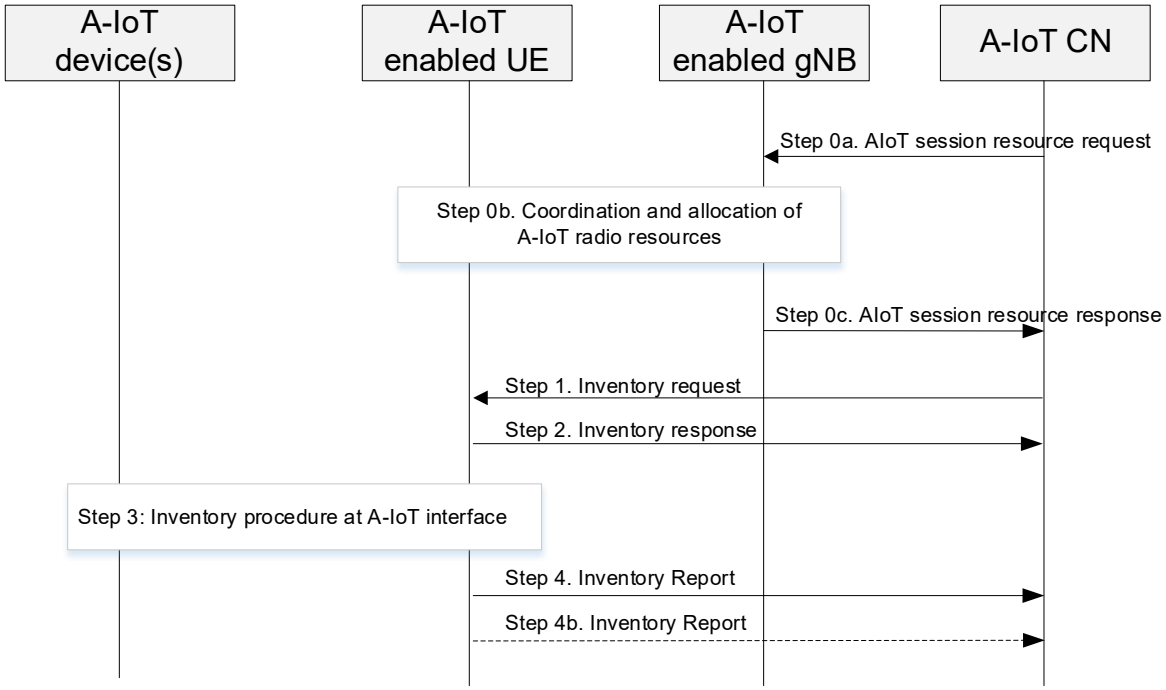


Figure 6.5.3.1.2-1: Message flow A-IoT Inventory in Topology 2 - NAS/UP based solution – A-IoT CN triggered A-IoT session resource allocation

This option is characterised by the A-IoT enabled gNB receiving the request to establish A-IoT session resources by the A-IoT CN.

0a. The A-IoT CN requests A-IoT session resources.

0b. The A-IoT enabled gNB coordinates A-IoT radio resources and allocates A-IoT radio resources to the A-IoT enabled UE accordingly.

0c. The A-IoT enabled gNB confirms the request for A-IoT session resources.

NOTE 1: In step 0c, the A-IoT-enable gNB can reject the request for A-IoT session resource.

1. The A-IoT CN sends an Inventory request message to the A-IoT-enabled UE.

2. The A-IoT-enabled UE(s) sends an Inventory response message to the A-IoT CN.

NOTE 2: In step 2, the A-IoT-enabled UE may instead fail the Inventory request.

3. The A-IoT-enabled UE performs the inventory procedure towards the A-IoT device(s).

4/4b. The A-IoT-enabled UE may send one or multiple Inventory reports towards the A-IoT CN including the received inventory result.

NOTE 3: Steps 4/4b may happen in parallel with Step 3 for different A-IoT devices.

NAS/UP based solution – A-IoT enabled UE triggered A-IoT session resource allocation

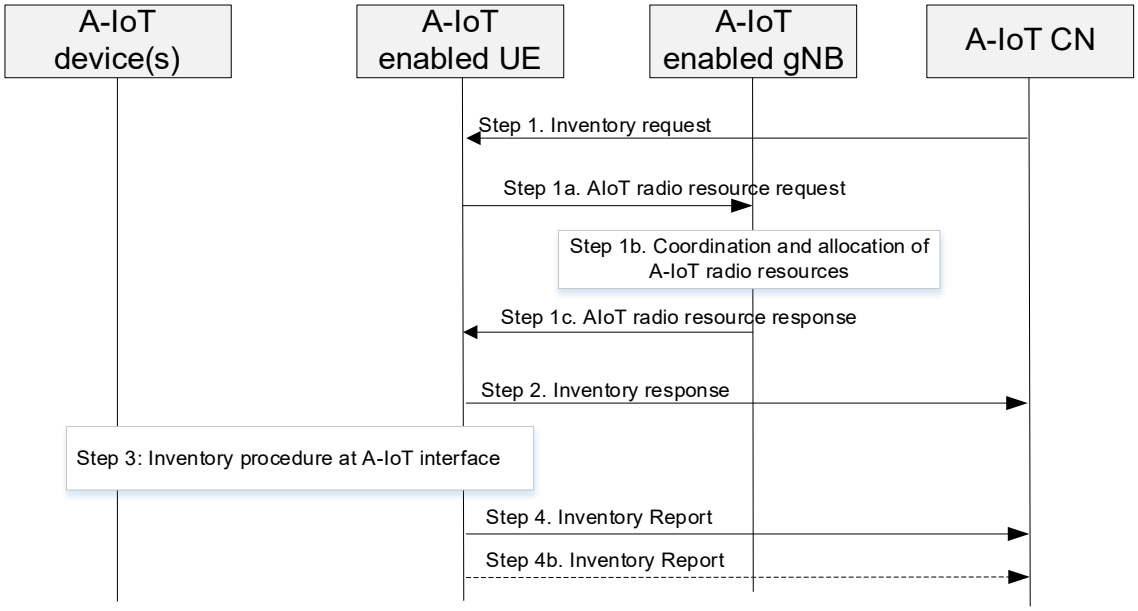


Figure 6.5.3.1.2-2: Message flow for A-IoT Inventory in Topology 2 (if - NAS/UP based solution is used) – A-IoT enabled UE triggered A-IoT session resource allocation

This option is characterised by the A-IoT enabled gNB receiving the request to establish A-IoT session resources by the A-IoT enabled UE.

Additional means to enable the AIoT CN to provide AIoT session related information directly to the AIoT enabled gNB, not shown in Figure 6.5.3.1.2-2, may be needed.

1. The A-IoT CN sends an Inventory request message to the A-IoT-enabled UE.

1a. The A-IoT enabled UE request A-IoT radio resources from the A-IoT enabled gNB.

1b. The A-IoT enabled gNB coordinates A-IoT radio resources and allocates A-IoT radio resources to the A-IoT enabled UE accordingly.

1c. The A-IoT enabled gNB responds the A-IoT radio resources to the A-IoT-enabled UE.

NOTE 4: In step 1c, the A-IoT-enable gNB can reject the A-IoT radio resource request. Details w.r.t. steps 1a and 1c are subject to RAN2 discussions.

2. The A-IoT-enabled UE sends an Inventory response message to the A-IoT CN.

NOTE 5: In step 2, the A-IoT enabled UE may instead send an Inventory failure message to the A-IoT CN indicating that the inventory procedure could not be initiated towards the A-IoT device(s), and the procedure ends.

3. The A-IoT-enabled UE(s) performs the inventory procedure towards the A-IoT device(s) over the A-IoT radio interface.

4/4b. After receiving inventory result reported from the A-IoT device(s), the A-IoT-enabled UE may send one or multiple Inventory reports towards the A-IoT CN including the received inventory result.

NOTE 6: Steps 4/4b may happen in parallel with Step 3 for different A-IoT devices.

6.5.3.2 Candidate procedures for A-IoT Command for Topology 2

6.5.3.2.1 RRC based solution

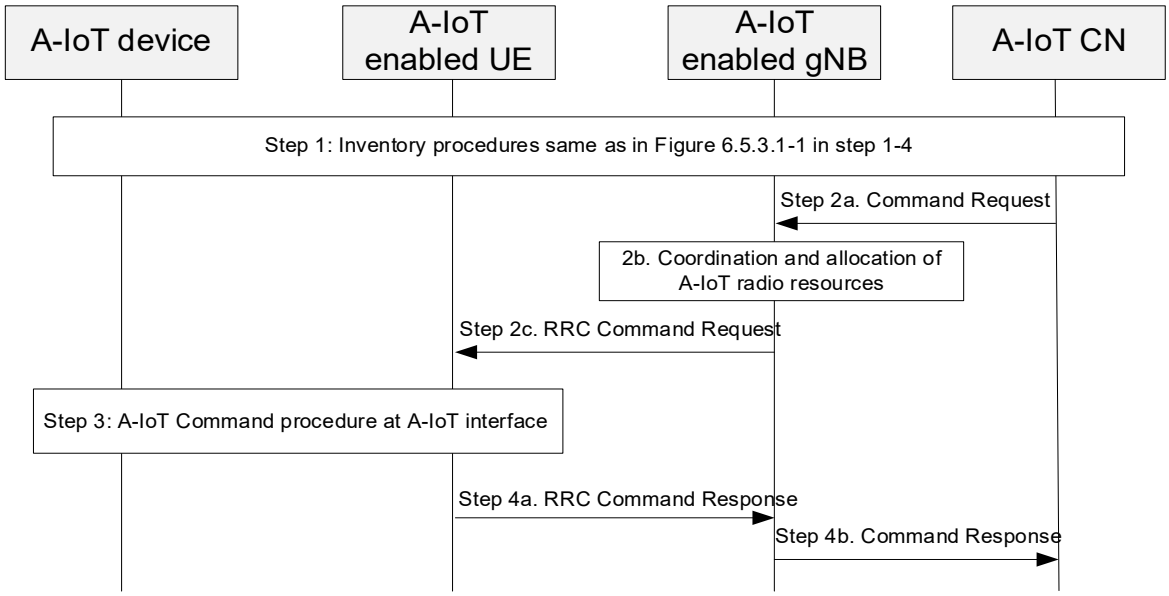


Figure 6.5.3.2-1: Message flow for A-IoT Command in Topology 2 (RRC-based solution)

1. The Inventory procedures are performed in step 1, which are the same as in Figure 6.5.3.1.1-1 in step 1-4.

NOTE 1: Step 1 is performed for the "A-IoT Inventory and Command" case, Whether and under which conditions step 1 may be skipped in case of "Command-only" and which consequences this would have for the overall message flow depends on discussions led by SA2, SA3 and RAN2.

NOTE 2: Whether and which way steps 2a, 2b and 2c may be part of step 1 is dependent on discussions led by SA2, SA3 and RAN2.

2a. The A-IoT CN sends a Command Request message to the A-IoT enabled gNB.

2b. The A-IoT enabled gNB coordinates A-IoT radio resources and allocates A-IoT radio resources to the A-IoT enabled UE accordingly.

2c. The A-IoT enabled gNB sends a RRC Command Request message to the A-IoT enabled UE.

NOTE 3: RRC based communication, i.e., performing the A-IoT Command procedure and allocation of A-IoT radio resources, is only depicted schematically, details subject to RAN2.

3. The A-IoT-enabled UE performs A-IoT command procedures at A-IoT interface towards the A-IoT device over the A-IoT radio interface.

4a. A-IoT enabled UE sends a Command Response message to A-IoT enabled gNB, if the command result is received from A-IoT device, the A-IoT enabled UE may include the command result in the Command Response message.

4b. The A-IoT enabled gNB sends a Command Response message to the A-IoT CN, it may include the command result in the Command Response message, if any.

NOTE 4: In step 4b, the A-IoT enabled gNB may instead send a Command Failure message to the A-IoT CN indicating that the command procedure is failed towards the A-IoT device.

6.5.3.2.2 NAS/UP based solution

The A-IoT Command procedure also applies to the NAS/UP based solutions, and the AIoT Command procedure signallings are exchanged between the A-IoT enabled UE and the AIoTF over UE's NAS/PDU session. The A-IoT session resource allocation steps described in 6.5.3.1.2 also applies to this procedure.

6.6 Coexistence of ambient IoT and NR/LTE

6.6.1 Co-existence scenarios and cases

The coexistence evaluation is conducted considering the different scenarios listed in Table 6.6.1-1

Table 6.6.1-1: Co-existence scenarios

Deployment scenario No. (Case No.)	Topology	Spectrum
1-1(a/b/c/d)	D1T1-A2- NR UE only outdoor	R2D: DL CW2D and D2R: DL
1-2(a/b/c/d)	D1T1-A2- NR UE indoor	R2D: DL CW2D and D2R: DL
2-1(e/f/c/d)	D1T1-B- NR UE only outdoor	R2D: DL CW2D and D2R: UL
2-2(e/f/c/d)	D1T1-B- NR UE indoor	R2D: DL CW2D and D2R: UL
3 (Optional)(a/b/g/h)	D1T1-A2- NR UE indoor	R2D: UL CW2D and D2R: UL
5-1(e/f/g/h)	D2T2-A2- NR UE only outdoor	R2D: UL CW2D and D2R: UL
5-2(e/f/g/h)	D2T2-A2- NR UE indoor	R2D: UL CW2D and D2R: UL
6-1(Optional)(a/b/g/h)	D2T2-B- NR UE only outdoor	R2D: UL CW2D and D2R: DL
6-2(Optional)(a/b/g/h)	D2T2-B- NR UE indoor	R2D: UL CW2D and D2R: DL

The main co-existence scenario considered for ambient IoT is D1T1 and D2T2. The deployment parameters are described in Table 6.6.2.1-1.

The co-existence evaluation captures cases where NR and A-IoT (Reader/device) are both victim and aggressor networks. It is to evaluate impact on legacy NR networks if A-IoT is introduced in indoor scenario, also to understand impact of the legacy NR network on A-IoT system. The co-existence cases are listed in Table 6.6.1-2.

Table 6.6.1-2: Co-existence cases

Case No.	Aggressor	Victim	note
a	Device	NR DL	D2R
b	NR DL	reader	D2R
c	Reader	NR DL	R2D
d	NR DL	device	R2D
e	Device	NR UL	D2R
f	NR UL	reader	D2R
g	Reader	NR UL	R2D
h	NR UL	device	R2D

6.6.2 Co-existence evaluation assumptions

6.6.2.1 Deployment

Simulation assumptions related to network layout is captured for D1T1 and D2T2 in Table 6.6.2.1-1.

Table 6.6.2.1-1. Deployment parameters for D1T1 and D2T2

Parameter	D1T1	D2T2
Carrier frequency	900 MHz (Band n8)	

BW for NR	10MHz with 15KHz SCS	
BW for A-IOT system	180kHz, for 15kHz SCS	
Waveform (R2D)	OOK waveform generated by OFDM modulator	
Waveform (CW)	Unmodulated single tone	
A-IoT DL power control	No	
A-IoT UL power control	No	
Traffic model	Full buffer	
Frequency reuse	1	
NR BS deployment (outdoor), i.e. scenario 1-1 and 1-2	Hexagonal grid, 19 macro sites, 3 sectors per site with wrap around, 1 A-IOT indoor scenario per sector the minimum 2D distance between macro BS and indoor factory center is set as 100m. Inter-NR BS distance: 750m NR BS height: 25m	
NR BS deployment (indoor), i.e. scenario 2-1 and 2-2	NR indoor BS deployed co-site with A-IoT indoor reader; NR indoor BS ISD as 20m; NR indoor Min BS-UE distance: 0m; NR indoor UE uniformly distributed. NR indoor BS Tx power assumed [24] dBm Antenna pattern of A-IoT reader is reused. The self-interference cancellation of R2D in UL for co-located scenario can be reported by interested company.	
A-IoT reader deployment / Intermediate UE dropping	For D1T1-A2 and D1T1-B: 18 A-IoT readers on a square lattice with spacing D, located D/2 from the walls. L=120m x W=60m; D=20m Reader height = 8 m Room height = 10m 2 A-IoT readers are activated in one drop as baseline. Minimum distance between active readers: 60m as baseline	For D2T2-A2 and D2T2-B: The intermediate UEs selected from the fixed positions. L=120m x W=50m; D=20m Intermediate UE height = 1.5 m Room height = 3m Number of intermediate UE for simulation: 2 UE at one drop. Minimum distance between intermediate UEs: 60m as baseline
CW deployment	For D1T1-A2 and D2T2-A2, CW and A-IoT reader are collocated. For D1T1-B and D2T2-B, CW topology layout is the same as A-IoT reader. For each device, the nearest CW node will be activated during the simulation, and CW node is not co-located with any activated reader	
Device distribution	Device Height= 1.5 m A-IoT devices drop uniformly distributed over the horizontal area Number of A-IoTs = Total area × activated density (1.5 A-IOT devices/m²) 1 active A-IoT device under one reader at one drop Minimum distance between reader and device is 1m	
NR UE dropping	For NR UE only outdoor, uniformly distributed outdoor. For NR UE indoor, uniformly distributed, Option1: 10% indoor, 90% outdoor, Option2: 100% indoor UE number: - DL active UE: 1 UE per cell - UL active UE: 3 UE per cell	
NR BS Inter-site distance	MCL of 70 dB	
Pathloss model	$\mathrm{PL} = PL_b + PL_{tw} + PL_{in} + N(0, \sigma_P^2)$ - PL_{in} = 0.5 * d2D-in where d2D-in is the distance to nearest factory/office boundary on the line between Tx and Rx point. Set maximum value of d2D-in as [25m] as optional	
	PL_b : NLOS and LOS in TR38.901 NR BS – NR UE: Uma Outdoor NR UE – A-IoT reader/ device: Umi Device – A-IoT reader: InF-DH Indoor NR UE – device: InH-Office Indoor NR UE –A-IoT reader: InF-DH	PL_b : NLOS and LOS in TR38.901 NR BS – NR UE: Uma Outdoor NR UE – intermediate UE/ device: Umi Device – Intermediate UE: InH-Office Indoor NR UE – device: InH-Office Indoor NR UE – intermediate UE: InH-Office
O2I penetration loss	High penetration loss as in TR 38.901	

6.6.2.2 NR BS/ A-IoT reader/ intermediate UE/ CW RF characteristics

The NR BS and A-IoT reader are defined for two different antenna configurations as illustrated in Table 6.6.2.2-1 and Table 6.6.2.2-2. Assumptions related to CW RF characteristics relevant for different deployment scenarios are captured in Table 6.6.2.2-4.

Table 6.6.2.2-1: NR BS RF parameters

NR BS Parameter	Values for evaluation
Macro-BS Tx power (dBm)	46
BS antenna gain (dBi) and antenna pattern	Antenna Array Geometry: BS point at fixed beam direction: vertical: θtilt + 90°, horizontal: 0, 120, 240 ° Antenna pattern (horizontal) (For 3-sector cell sites with fixed antenna patterns): $A_H(\phi) = -min\left[12\left(\frac{\phi}{\phi_{3dB}}\right)^2, A_m\right]$ φ_{3dB} = 65 degrees, A_m = 25 dB Antenna pattern (vertical) (For 3-sector cell sites with fixed antenna patterns): $A_V(\theta) = -min\left[12\left(\frac{\theta-90°-\theta_{etilt}}{\theta_{3dB}}\right)^2, SLA_v\right]$ θ_{3dB} = 10 degrees, SLA_v = 25 dB, θ_{etilt}= 9 degrees Combining method in 3D antenna pattern: $A(\varphi,\theta) = -\min\{-[A_H(\varphi) + A_V(\theta)], A_m\}$ BS antenna gain (dBi) (including feeder loss:15
Height of macro NR BS (m)	25
NR Macro-BS Noise Figure(dB)	5
Network location	outdoor
ACLR	Option 1: 30dBc, Option 2: 17dBc

Table 6.6.2.2-2: A-IoT reader RF parameters

A-IoT reader parameters	Values for evaluation
A-IoT reader total Tx power	33 dBm
A-IoT reader receiver Noise Figure (dB)	10
A-IoT reader antenna gain (dBi)	6
ACLR	ACLR of legacy NB -IOT gNB (i.e. ACLR1:40dB, ACLR2:50dB)
ACS	Same as legacy NR BS
Antenna configuration	Antenna Array Geometry: equals to omni-directional antenna pattern in GCG in horizontal Antenna pattern (horizontal): $A_H(\phi) = -min\left[12\left(\frac{\phi}{\phi_{3dB}}\right)^2, A_m\right]$, φ_{3dB} = 90°, A_m = 15 dB Antenna pattern (horizontal): $A_V(\theta) = -min\left[12\left(\frac{\theta-90°}{\theta_{3dB}}\right)^2, SLA_v\right]$, θ_{3dB}= 90°, SLA_v = 15 dB Combining method in 3D antenna pattern: $A(\varphi,\theta) = -\min\{-[A_H(\varphi) + A_V(\theta)], A_m\}$ BS antenna gain (dBi) (including feeder loss): 6

Table 6.6.2.2-3: Intermediate UE RF parameters

intermediate UE parameters	Values for evaluation
Intermediate UE total Tx power (dBm)	Baseline: 23 Optional: Tx power control is allowed
Gain of antenna intermediate UE (dBi)	0
Intermediate UE receiver Noise Figure (dB)	9
Antenna configuration	Omni directional antenna

Table 6.6.2.2-4: CW RF parameters

CW parameters	D1T1	D2T2
Tx power (dBm)	If UL spectrum is used, UE Tx power is assumed, i.e. 23dB If DL spectrum is used, A-IoT reader Tx power is assumed, i.e. 33 dBm	Intermediate UE Tx power is assumed.
Antenna gain	Same as A-IoT reader	Same as intermediate UE

6.6.2.3 NR UE/ A-IoT device RF characteristics

Assumptions relevant for modelling the NR UE and A-IoT device RF characteristics are captured in Table 6.6.2.3-1 and Table 6.6.2.3-2.

Table 6.6.2.3-1: NR UE RF parameters

NR UE Parameter	Values for evaluation
UE TX power in dBm	-40 to 23
NR UE Antenna gain (dBi)	0
Height of UE antenna (m)	1.5
NR UE ACLR (dB)	30
NR UE Noise Figure (dB)	9
Antenna configuration	Omni direction antenna

Table 6.6.2.3-2: A-IoT device RF parameters

A-IoT device parameters	Device 1	Device 2a
A-IoT device effective antenna gain per Tx or Rx branch (dBi)	0	0
A-IoT device reflection (backscatter) loss (dB)	OOK: -6 dB	OOK: -6 dB
A-IoT device power gain of reflection amplifier (dB)	N/A	10(M)
A-IoT Device receiver sensitivity (dBm) Use this value to determine whether device can camp on the cell.	-36	-45
A-IoT device noise figure (dB)	24	20
Guard band	0PRB	0PRB

6.6.3 Co-existence simulation methodology

6.6.3.1 Coexistence evaluation methodology

The coexistence evaluation methodology can be summarized as:

1) Aggressor and victim network are generated. NR UEs and A-IoT devices are distributed as described by parameter assumptions.

2) UEs are associated to BS based on coupling loss, and A-IoT devices are associated to A-IoT reader or intermediate UE based on coupling.

   NOTE: For A-IoT devices are associated to A-IoT reader or intermediate UE, there are two ways applied by companies:

   a) The device access to the reader is randomly selected from candidate devices whose R2D receiving power/coupling from one reader is higher than the other. The receiving power of device meets the device receiver sensitivity.

   b) The device can access the reader with best receiving power among of 12 or 18 total readers in one building. The receiving power of device meets the device receiver sensitivity.

3) Once association is done, round robin scheduling is used.

4) For inter-system interference (between A-IoT and NR):

   - If SINR degradation is smaller than and equal to 1 dB, it can be considered that inter-system interference is negligible.

   - Note: For SINR degradation, SINR refers to the 5% and 50% CDF SINR

6.6.3.2 SINR definition

For D1T1-A1and -B /D2T2-A1and -B

SINR definition for D2R:

- SINR includes CW interference is used as the baseline reference for co-existence evaluation for CW reader.

$$baseline\ SINR = \frac{received\ wanted\ signal\ power}{(noise + intra_system\ interference) + (residual\ CW\ interference\ after\ cancellation)}$$

- SINR is calculated as total power ratio:

$$SINR = \frac{received\ wanted\ signal\ power}{(noise + intra_{system} interference + inter_{system} interference) + (residual\ CW\ interference\ after\ cancellation)}$$

For the CW outside scenarios (D1T1-B and D2T2-B), the impact to SINR due to phase noise are also evaluated. The baseline D2R SNR for CW outside can be calculated by

$$baseline\ SINR\ for\ CW\ outside = \frac{received\ wanted\ signal\ power}{(noise + intra_{system interference} + phase\ noise)_{within\ total\ receiver\ bandwidth} + (residual\ CW\ interference\ after\ cancellation)_{in\ linear\ scale}}$$

Phase noise can be modelled as XdBc/Hz, e.g., -112dBc/Hz compared to the received power for D2R.

The noise and intra-system interference are within total receiver bandwidth, and the residual CW interference after cancellation is in linear scale.

SINR definition for R2D:

- Signal power of device to the noise and interference within 10MHz is baseline assumption

- Consider 180KHz noise and interference bandwidth after BB LPF as optional

6.6.3.3 Coupling loss

The Coupling Loss (CL) is defined as the loss in signal between NR BS-to- NR UE, A-IoT reader -to- A-IoT device and intermediate UE -to- A-IoT device. CL is defined as the loss including propagation loss and antenna gains.

6.6.3.4 Impact from CW to NR

In 6.6.3.2, the SINR definition includes the impact from CW to D2R which is treat as intra-system interference. However, Whether and how the CW will influence the NR system also needs to be studied. In [156], the impact of CW signal under different assumptions are investigated and the evaluation results are shown in figure 6.6.3.4-1.

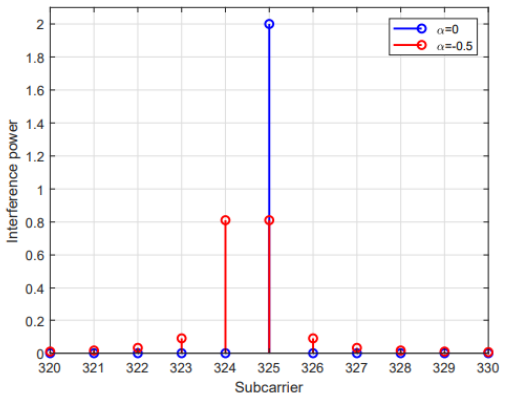


Fig. 2. Interference power distribution after the FFT when the OFDM system is subjected to one CW signal (closest to subcarrier $m = 325$), with separation $\alpha = 0$ and $\alpha = -0.5$ from the subcarrier 325.

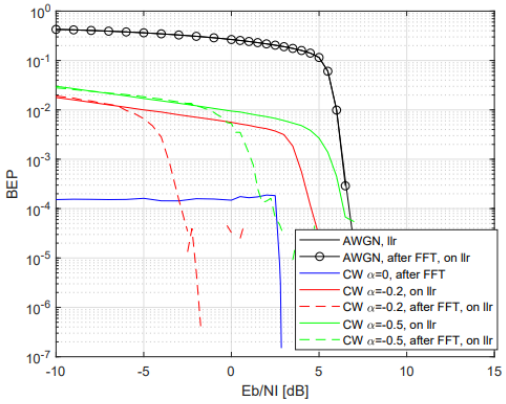


Fig. 6. Resulting BEP as a function of E_b/N_I when limiters are used to mitigate interference consisting of AWGN or one CW signal. Limiters are placed after FFT or on the demodulator soft output LLR.

Figure 6.6.3.4-1: Impact from CW to NR system [156]

The results shows that when the CW is orthogonal with NR subcarrier, i.e., α=0, the influence of interference is much lower and also easier to be removed after FFT, only if the power of CW interference does not exceed the dynamic range of receiver.

Based on the analysis above, the impact from CW to NR is not modelled in the co-existence evaluation for simplicity.

6.6.4 Co-existence evaluation results

The simulation results submitted for the ambient IoT and NR co-existence study from all contributing companies in RAN4 were summarized in [155] attached to this report.

6.6.5 Summary of co-existence evaluation

The clause captures the co-existence simulation results for ambient IoT categorized by scenarios and cases. The conclusions below are derived from the coexistence results.

6.6.5.1 Scenario 1-1: D1T1-A2- NR UE only outdoors (R2D/ CW2D/ D2R: DL)

Scenario 1-1 considers D1T1 topology, all the NR UE outdoors and Readers/CW collocated, with DL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.1-1.

Table 6.6.5.1-1: Scenario 1-1 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
a	Device	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed
b	NR DL	reader	SINR degradation is less than 1dB under the assumption of 30dBc inband leakage for legacy NR gNB. Inter-system interference is negligible. 2 companies observed some SINR degradation under the assumption of 17dBc inband leakage for legacy NR gNB.
c	Reader	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed
d	NR DL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.2 Scenario 1-2: D1T1-A2- NR UE indoor (R2D/ CW2D/ D2R: DL)

Scenario 1-2 considers D1T1 topology, 10% of the NR UE indoors and Reader/CW collocated, with DL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.2-1.

Table 6.6.5.2-1: Scenario 1-2 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
a	Device	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed
b	NR DL	reader	SINR degradation is less than 1dB under the assumption of 30dBc inband leakage for legacy NR gNB. Inter-system interference is negligible. 1 company observed some SINR degradation under the assumption of 17dBc inband leakage for legacy NR gNB.
c	Reader	NR DL	Under the assumption of 10% NR UE indoor, NR DL throughput degradation is observed. Under the assumption of 100% NR UE indoor, significant NR DL throughput degradation is observed.
d	NR DL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.3 Scenario 2-1: D1T1-B- NR UE only outdoor (R2D: DL, CW2D/ D2R: UL)

Scenario 2-1 considers D1T1 topology, all the NR UE outdoors and Readers/CW non-collocated, with DL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.3-1.

Table 6.6.5.3-1: Scenario 2-1 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
e	Device	NR UL	Negligible NR UL throughput degradation for both average throughput and cell edge throughput is observed.
f	NR UL	reader	SINR degradation is less than 1dB. Inter-system interference is negligible.
c	Reader	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed.
d	NR DL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.4 Scenario 2-2: D1T1-B- NR UE indoor (R2D: DL, CW2D/ D2R: UL)

Scenario 2-2 considers D1T1 topology, 10% of the NR UE indoors and Readers/CW non-collocated, with DL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.4-1.

Table 6.6.5.4-1: Scenario 2-2 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
e	Device	NR UL	Negligible NR UL throughput degradation for both average throughput and cell edge throughput is observed.
f	NR UL	reader	Under the assumption of 10% and 100% NR UE indoor, significant SINR degradation is observed
c	Reader	NR DL	Under the assumption of 10% and 100% NR UE indoor, NR DL throughput degradation is observed.
d	NR DL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.5 Scenario 5-1: D2T2-A2- NR UE only outdoor (R2D/ CW2D/ D2R: UL)

Scenario 5-1 considers D2T2 topology, all the NR UE outdoors and Readers/CW collocated, with UL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.5-1.

Table 6.6.5.5-1: Scenario 5-1 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
e	Device	NR UL	Negligible NR UL throughput degradation for both average throughput and cell edge throughput is observed
f	NR UL	reader	SINR degradation is less than 1dB. Inter-system interference is negligible.
g	Reader	NR UL	NR UL throughput degradation is observed
h	NR UL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.6 Scenario 5-2: D2T2-A2- NR UE indoor (R2D/ CW2D/ D2R: UL)

Scenario 5-2 considers D2T2 topology, 10% of the NR UE indoors and Readers/CW collocated, with UL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.6-1.

Table 6.6.5.6-1: Scenario 5-2 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
e	Device	NR UL	Negligible NR UL throughput degradation for both average throughput and cell edge throughput is observed
f	NR UL	reader	Significant SINR degradation is observed for both 10% NR indoor UE and 100% NR indoor UE assumption.
g	Reader	NR UL	NR UL throughput degradation is observed
h	NR UL	device	Under the assumption of 10% NR UE indoor, SINR degradation is larger than 1dB. Under the assumption of 100% NR UE indoor, significant SINR degradation is observed.

6.6.5.7 Scenario 6-1: D2T2-B- NR UE only outdoor (R2D: UL, CW2D/ D2R: DL)

Scenario 6-1 considers D2T2 topology, all the NR UE outdoors and Readers/CW non-collocated, with UL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.7-1.

Table 6.6.5.7-1: Scenario 6-1 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
a	Device	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed.
b	NR DL	reader	SINR degradation is less than 1dB under the assumption of 30dBc inband leakage for legacy NR gNB. Inter-system interference is negligible.
g	Reader	NR UL	NR UL throughput degradation is observed.
h	NR UL	device	SINR degradation is less than 1dB. Inter-system interference is negligible.

6.6.5.8 Scenario 6-2: D2T2-B- NR UE indoor (R2D: UL, CW2D/ D2R: DL)

Scenario 6-2 considers D2T2 topology, 10% of the NR UE indoors and Readers/CW non-collocated, with UL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.8-1.

Table 6.6.5.8-1: Scenario 6-2 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
a	Device	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed
b	NR DL	reader	SINR degradation is less than 1dB under the assumption of 30dBc inband leakage for legacy NR gNB. Inter-system interference is negligible. 1 company observed some SINR degradation under the assumption of 17dBc inband leakage for legacy NR gNB
g	Reader	NR UL	NR UL throughput degradation is observed.
h	NR UL	device	Under the assumption of 10% NR UE indoor, SINR degradation is larger than 1dB. Under the assumption of 100% NR UE indoor, significant SINR degradation is observed.

6.6.5.9 Scenario 3 (Optional): D1T1-A2- NR UE indoor (R2D: UL, CW2D/ D2R: DL)

Scenario 3 optionally considers D1T1 topology, 10% of the NR UE indoors and Readers/CW collocated, with UL spectrum applying to reader-to-device link. The conclusions of each cases are listed per scenario in Table 6.5.6.9-1.

Table 6.6.5.9-1: Scenario 3 co-existence conclusions

Case No.	Aggressor	Victim	Co-existence conclusion
a	Device	NR DL	Negligible NR DL throughput degradation for both average throughput and cell edge throughput is observed
b	NR DL	reader	1 company provides simulation results based on assumption of 30dBc NR gNB inband leakage and observed less than 1dB SINR degradation. 1 company provides simulation results based on assumption of 17dBc NR gNB inband leakage and observed larger than 1 dB SINR degradation.
g	Reader	NR UL	NR UL throughput degradation is observed.
h	NR UL	device	1 company provides simulation results based on assumption of 10% NR UE indoor and observed less than 1dB SINR degradation 1 company provides simulation results based on assumption of 100% NR UE indoor and observed larger than 1 dB SINR degradation.

6.6.5.10 General remarks on coexistence findings

For D1T1 deployment scenario 1-1 (R2D, CW2D/D2R: DL) and 2-1(R2D: DL, CW2D/D2R: UL), which has no legacy NR UE indoor. Negligible performance degradation for NR system is observed. Less than 1 dB SINR degradation due to inter-system interference from NR to ambient IOT system is observed. Co-existence between ambient IOT and NR is feasible based on performance metric of NR throughput degradation and ambient IOT SINR degradation, no additional coexistence measures are required. Ambient IOT intra-system performance evaluated by link level simulation is also one of the key aspects affecting ambient IOT system performance, which will have impact to the outage of ambient IOT system.

For the deployment scenarios with legacy NR UE indoor (scenario 1-2, 2-2, 5-2, 6-2), which either significant NR throughput degradation or significant SINR degradation for ambient IOT system is observed, some existing mechanisms based on gNB implementation can be considered to avoid/mitigate the interference. For example, for RRC idle mode, NR UEs can camp on a frequency different from ambient IOT system based on frequency priority configured by network; for RRC connected mode, network can handover NR UEs to a different cell (e.g. indoor gNB on a different frequency) based on existing measurement report triggering and event. Besides, other interference management solutions to reduce the inter-system interference from ambient IOT to proximity NR UE and vice versa can also be considered.

For the D2T2 deployment scenarios (scenario 5-1, 5-2, 6-1, 6-2), NR UL throughput degradation is observed. The mechanisms to mitigate the interference can be considered. For example, enhanced emission requirements for intermediate UE or power control for intermediate UE [157], [158].

Additionally, RAN4 has also considered using guard band to mitigate the interference between ambient system and NR system.

6.7 RF requirements study

6.7.1 System parameters

6.7.1.1 Operating band

According to the SID (RP-240826), Ambient IoT uses FR1 licensed spectrum in FDD. NR band n8 can be used as an example band.

Table 6.7.1.1-1 summarized the A-IoT system spectrum usage from co-existence scenario. It can be observed that The spectrum usage for A-IoT operation can be flexible, and it depends on several factors, such as whether the reader is A-IoT BS or intermediate UE, and whether CW transmission is deployed inside or outside topology. For the case which different spectrum is used for D2R and R2D, the FDD band definition can be kept. As CW transmission may interfere the R2D reception, the R2D and D2R should be TDM and therefore, for FDD band definition, it should be half duplex mode.

Table 6.7.1.1-1: Spectrum usage for Ambient IoT

	Spectrum for R2D	Spectrum for D2R	Spectrum for CW2D transmission
A-IoT BS	DL spectrum	DL spectrum	DL spectrum
	DL spectrum	UL spectrum	UL spectrum
	UL spectrum	DL spectrum	DL spectrum
	UL spectrum	UL spectrum	UL spectrum
Intermediate UE as reader	UL spectrum	UL spectrum	UL spectrum
	UL spectrum	DL spectrum	DL spectrum

6.7.1.2 Channel bandwidth

The R2D waveform is agreed to be OFDM-based waveform, therefore the channel bandwidth definition can follow the channel bandwidth definition in NR. The R2D channel bandwidth can be defined as one or multiple PRBs with guardband if necessary. The value for this can be determined during the WI phase.

6.7.1.3 Channel arrangement

The channel spacing may be specified for D2R FDM operation for device 1/2a.

For Ambient IoT BS and intermediate UE, channel spacing and channel raster may be needed for deployment purpose, for example, to co-deploy with existing NR network.

6.7.2 Ambient IoT BS

6.7.2.1 General

For Release 19, the RF requirement discussion focus on BS type 1-C for Ambient IoT indoor scenarios. The RF requirement for ambient IoT BS is studied based on the assumption that ambient IoT BS only transmits and receives ambient IoT signal.

Current the coexisting scope focuses on device 1 and device 2a, it includes the scenarios and cases which describes the different factors to be considered for the ambient IoT BS. These factors are listed in Table 6.7.2.1-1 below. CW signal can be transmitted by A-IoT BS if manufacture declares this capability. CW can be transmitted by a separate node, therefore CW related RF requirements can be specified separately for ambient IoT BS or a separate CW node.

Table 6.7.2.1-1: Factors impacting the ambient IoT BS operation to device 1 and device 2a

	Spectrum for R2D	Spectrum for D2R	Spectrum for CW transmission	A-IoT BS transmitting the CW signal
A-IoT BS	DL spectrum	DL spectrum	DL spectrum	Yes
	DL spectrum	DL spectrum	DL spectrum	No
	DL spectrum	UL spectrum	UL spectrum	Yes
	DL spectrum	UL spectrum	UL spectrum	No

For the scenarios only considering device 1 and device 2a, simultaneously CW and R2D transmission will interfere the R2D reception, as such the R2D and D2R transmission should be operated in TDM for device 1 and device 2a.

6.7.2.2 Ambient IoT BS TX requirements

6.7.2.2.1 Base station output power

Base station output power requirement is potentially needed, RAN4 agrees to use current requirements of Micro NR BS type 1-C as the starting point.

6.7.2.2.2 Output power dynamic

Output power dynamic requirement usually includes RE power control dynamic range requirement and total power dynamic range requirement. Power boosting requirement is not needed in R19 with the assumption that ambient IoT BS only transmits and receives AIOT signal.

6.7.2.2.3 Transmit ON/OFF power

Regarding the transmit on/off requirement, it usually includes of transmitter off power requirement and transmitter transient period requirement, RAN4 agrees to further discuss in the WI phase.

6.7.2.2.4 Transmit signal quality

Frequency error and modulation quality (EVM) are potentially needed. TAE is only applicable for BS transmitting from multiple antennas via MIMO, CA, or combination of them. A-IoT BS doesn’t support these features, so TAE requirement is not needed.

6.7.2.2.5 Unwanted emissions

Unwanted emission requirement should be defined because it is the requirement to limit the emission to other channel or system to make this equipment can co-exist with other equipment. OBW requirement, ACLR requirement, OBUE requirement, transmitter spurious emission requirement is potentially needed.

6.7.2.3 Ambient IoT BS Rx requirements

6.7.2.3.1 Reference sensitivity level

Regarding the reference sensitivity level for A-IoT BS, RAN4 reached the following consensus:

For BS type 1-C,

$$P_{REFSENS}(\text{dBm}) = -174\text{dBm} + 10 \times \log10\,(\text{BW}) + N_F + I_M + \text{SNR} + [\text{ desens target}]$$

- BW is the transmission bandwidth for D2R.
- N_F is the noise figure of A-IoT BS, equal to 10 dB for Medium Range BS.
- I_M is the implementation margin, equal to 2dB.
- SNR is the SNR value for D2R decoding, the detailed metrics would be discussed in the WI phase.
- Desens target is considered for CW signal interference cancellation in D1T1 scenario. The desens level for CW signal cancellation might be different for D1T1-A1/D1T1-A2 and D1T1-B. For D1T1-A2, CW signal interference is coming from self-interference and also potential CW interference from other nodes transmitting CW signal. For D1T1-A1, the CW signal interference is coming from other nodes transmitting CW inside the topology. For D1T1-B, the CW signal interference is coming from CW nodes transmitting outside topology.

6.7.2.3.2 Dynamic range

For NR and LTE, receiver dynamic range requirement is to ensure that the base station can receive high throughput also in the presence of increased interference levels. The interference level is derived from simulation of co-channel interference from the same operator. It measures the effects of base station receiver impairments. For Ambient IoT indoor deployment, the co-channel interference comes from co-channel device transmission and CW interference. The necessity and the value can be studied at WI phase.

6.7.2.3.3 In-band selectivity and blocking

Regarding ACS requirement and in-band blocking requirement, RAN4 reached the following consensus:

ACS requirement and the interference level shall be determined by RAN4 co-existence study and wanted signal configuration and its corresponding configuration could be postponed to WI phase.

In-band blocking requirement and the interference level shall be determined by RAN4 co-existence study, the wanted signal configuration and its corresponding configuration could be postponed to WI phase.

6.7.2.3.4 Out-of-band blocking

Regarding Out-of-band blocking requirement, the existing out-of-band blocking requirement may be still applicable for A-IoT BS except for the wanted signal level.

6.7.2.3.5 Receiver spurious emissions

Regarding the receiver spurious emission requirement, this requirement might differ as following:

- For D1T1-A1 and D1T1-B, the receiver spurious emission requirement is still needed due to is half duplex operation mode.
- For D1TA-A2, due to the simultaneous transmission of CW signal and D2R reception, if the spurious emission requirement of CW signal is exceeding the receiver spurious emission requirement, then the receiver spurious

requirement is not needed similar as FDD BS otherwise this requirement is still needed to regulate the receiver RF performance.

6.7.2.3.6 Receiver intermodulation

Regarding the general intermodulation requirement, it is agreed to postpone to WI phase due to the potential relationship between D2R operation assumption e.g. single tone or multiple tone D2R assumption.

Regarding the narrowband intermodulation requirement, it is potentially necessary due to the potential Narrow band blocking at the operating frequency range, however its detailed requirement could be postponed to WI phase.

6.7.2.3.7 In-channel selectivity

Regarding the receiver in-channel selectivity requirement, this is agreed to be discussed in WI phase.

6.7.3 Intermediate node (UE)

6.7.3.1 General

The intermediate UE, under NW control, can send Ambient IoT data and/or signalling to the device, and receive the backscatter signal from the device. The intermediate UE can also operate as a legacy NR UE. For simplicity, non-concurrent operation of Ambient IoT and NR for intermediate UE is assumed, and focus on the case where the intermediate UE acts as an A-IoT Reader.

Considering the intermediate UE need to support both A-IoT and NR operation and if the requirements in NR and A-IoT is the same, it is possible to verify only on the NR side.

6.7.3.2 Intermediate UE TX requirements

6.7.3.2.1 Maximum output power

The goal of the maximum output power is to achieve the DL coverage. To reuse of the legacy NR UE hardware, PC3 NR UE can serve as a baseline for intermediate UE using FDD UL spectrum as R2D spectrum. High power UE is not precluded in the future release for better coverage.

6.7.3.2.2 Output power dynamics

Power control is not designed for A-IoT R2D link currently, and intermediate UE transmit at maximum output power for better coverage, making a minimum output power and power control accuracy requirements unnecessary. However, during the co-existence study, some companies propose that implementing the power control can help intermediate UE to mitigate the interference to NR system, so this requirement will be further discussed in WI phase depending on RAN1 specification.

Transmit OFF power is defined to ensure the UE will not affect the reception of other UE or BS when the transmitter is inactive. Since for A-IoT standalone operation, the HD-FDD mode could be one option for outside topology and the requirement can be further discussed in WI phase.

For the transmit ON/OFF time mask, refer to the RFID specification, a settling time requirement is used to assess signal ripple characteristics. It can be determined whether or not this settling time is necessary in WI phase. For the intermediate node that has capability to transmit CW signal, the transient time between R2D to CW transmission also needs to be further discussed.

6.7.3.2.3 Transmitted signal quality

Frequency error mainly depends on the timing and synchronization performance and PLL performance within transceiver chain. A-IoT intermediate UE is expected to reuse the existing timing and synchronization network directly. Meanwhile, similar PLL performance of A-IoT intermediate UE as NR UE is also expected as they are operating at the same frequency range. With the above considerations, frequency error requirement of NR UE can be considered reusing for A-IoT intermediate UE.

For legacy NR UE，transmit modulation quality includes EVM, carrier leakage and In-band emissions. These requirements may still be needed as R2D waveform is based on OFDM transmitter.

The carrier leakage requirement intends to reduce the DC offset which will distort the constellation for QAM signal. For OOK signal, the DC offset will increase the amplitude of waveform and the carrier leakage should also be low enough to ensure the received signal still within the receiver dynamic range. Whether and how to define this requirement can be further discussed in WI phase

The in-band emission in NR can limit the interference between UEs, but when intermediate UE perform as A-IoT reader, this requirement is unnecessary if all RB within the channel bandwidth always allocated for A-IoT transmission.

6.7.3.2.4 Output RF spectrum emissions

Output RF spectrum emissions for legacy NR UE includes occupied bandwidth, out of band emission (ACLR), spurious emissions and transmit intermodulation.

The occupied bandwidth is the width of a frequency band such that, below the lower and above the upper frequency limits, the mean powers emitted are each equal to a specified percentage β/2 (=0.5%) of the total mean transmitted power. Occupied bandwidth: Potentially needed for intermediate UE. The specific value is dependent on the physical layer design of R2D, which has not yet been finalized at this stage.

ACLR can be defined based on the co-existence study. During SI co-existence evaluation, the IBE of legacy NR is used as emission leakage model, and performance degradation is observed in the results, so the ACLR better than IBE may be required, and the details can be determined in WI phase.

Existing spurious emission requirement for legacy NR UE can be applicable, since this is coming from the ITU-R recommendation SM.329.

The transmit intermodulation in NR is the capability of the transmitter to inhibit the generation of signals in its non-linear elements caused by presence of the wanted signal and an interfering signal reaching the transmitter via the antenna. this requirement can be further discussed in WI.

6.7.3.3 CW for D2T2

6.7.3.3.1 Output power/Output power dynamic range

The output power of CW is needed to ensure the link quality for D2R. For the intermediate node that has capability to transmit CW signal, the MPR/A-MPR needs to be studied as the CW waveform defined with unmodulated sinusoid waveform is new waveform.

Considering the CW and R2D cannot transmitting simultaneously, the output power dynamic related requirements, e.g., transmit OFF power, can be further discussed in WI phase.

6.7.3.3.2 Transmit signal quality

Since the D2R signal transmission for device1/2a closely rely on the CW, and the signal quality may need to be considered e.g., frequency error which needs more discussion.

6.7.3.3.3 RF spectrum emission/Spurious emissions/Phase noise

The emission related requirement needs to be specified. Meanwhile, close-in phase noise may drown the D2R signal. CW cancellation can provide help in the CW inside topology scenario, but the phase noise might play a critical role for CW outside topology. In addition, the regulation is also needed to be considered. Therefore, the details of how to specify RF spectrum emission, and whether and how to specify spurious emission and phase noise can be further discussed in WI phase.

6.7.3.4 Intermediate UE RX requirements

6.7.3.4.1 Reference sensitivity

The reference sensitivity power level REFSENS is the minimum mean power applied to each one of the intermediate UE antenna ports at certain performance metric. Distinguished from the legacy NR UE, Ambient IoT intermediate UE may suffer from residual noise from CW interference, So it should be considered when defining receiver sensitivity. Reference sensitivity requirement for intermediate UE could be derived based on the following equation:

$$P_{REFSENS}(dBm)=-174dBm+10*\log_{10}BW+NF+IM+I_{CW}+SNR$$

Where:

- BW is the maximum transmission bandwidth
- NF is noise figure of intermediate node
- IM is the implementation margin, equal to 2dB.
- For inside topology, Icw is IoT due the residual CW interference, For outside topology, Icw would be less and in this case, receiver to decode the backscattered signal without desensitization of receiver caused by CW leakage. value will be defined in the WI.
- SNR is the SNR value for which we reach 10% BLER target. Each company provided simulation results, and average will be done for each BW.
- The REFSENS requirements of an intermediate UE may be affected when it simultaneously transmits uu UL and A-IoT signals.

For inside topology, the receiver needs to tolerate the high leakage CW power from the transmitter when intermediate node can transmit CW signal and desensitization of receiver may be allowed with a CW cancellation function enabled in such case. For outside topology, the received CW power would be less and in this case, receiver to decode the backscattered signal without desensitization of receiver caused by CW leakage. Therefore, for REFSENS requirements, different requirements can be defined, corresponding to inside topology and outside topology respectively.

6.7.3.4.2 Maximum input power

Maximum input level for legacy NR UE is defined as the maximum mean power received at the UE antenna port. Considering the intermediate UE may need to implement the receiver at UL spectrum when CW/D2R are also located in UL, which is different from legacy NR UE, this requirement can be further discussed in WI phase.

6.7.3.4.3 Adjacent channel selectivity

Adjacent channel selectivity (ACS) is a measure of a receiver's ability to receive an NR signal at its assigned channel frequency in the presence of an adjacent channel signal at a given frequency offset from the center frequency of the assigned channel.

ACS requirement and the interference level depend on co-existence study.

6.7.3.4.4 Blocking characteristics

The blocking characteristic is a measure of the receiver's ability to receive a wanted signal at its assigned channel frequency in the presence of an unwanted interferer on frequencies other than those of the spurious response or the adjacent channels, without this unwanted input signal causing a degradation of the performance of the receiver beyond a specified limit.

In-band blocking requirement and the interference level depend on co-existence study.

The existing out-of-band blocking requirement for legacy NR UE is applicable for intermediate UE.

Narrow band blocking can be discussed in WI phase.

6.7.3.4.5 Spurious response

Spurious response is a measure of the ability of the receiver to receive a wanted signal on its assigned channel frequency without exceeding a given degradation due to the presence of an unwanted CW interfering signal at any other frequency for which a response is obtained. The existing spurious emission requirement for legacy NR UE is applicable for intermediate UE.

6.7.3.4.6 Receiver intermodulation

Intermodulation response rejection is a measure of the receiver's capability to receive a wanted signal on its assigned channel frequency in the presence of two or more interfering signals with a specific frequency relationship to the wanted signal. It depends on the RAN1 design. If multiple-tone carrier wave is supported, intermodulation requirements potentially needed.

6.7.3.4.7 Spurious emissions

The spurious emissions power is the power of emissions generated or amplified in a receiver that appear at the UE antenna connector. The legacy NR UE receiver spurious emission requirement could be applicable.

6.7.4 Ambient IoT Device

6.7.4.1 Device 1

6.7.4.1.1 TX

Transmitter power

For device 1, D2R backscattering is transmitted on a carrier wave provided externally, backscattering power and power backscattering loss are potentially needed to be defined to ensure the coverage range between reader and device. Maximum output power and other transmit output power requirements are potentially not needed.

The minimum backscattered modulated power can be specified for device 1. The backscatter power is the amount of power reflected back from the device when a successful command sequence is completed at the device at a specific sensitivity level.

The backscatter loss due to the modulation can be defined as scattering cross-section or modulation efficiency in equation (4). From equation (4), it can be observed that the modulation efficiency is defined as the ratio of the power radiated by the device to the power density incident on its antenna.

Assume the power density of the incident radio wave on the device antenna is S

$$S = \frac{P_t \cdot G_t}{4\pi r^2} \quad (1)$$

Where the P_t is transmission power of a CW signal and G_t is the transmitter antenna gain, r is the distance between test equipment to the device.

The received power on device antenna is P_a

$$P_a = \frac{\lambda^2 \cdot G_d}{4\pi} \cdot S \quad (2)$$

Where the λ is the radio wavelength and G_d is the device antenna gain.

The power backscattered back in the direction of the transmitter is $P_{backscatter}$

$$P_{backscatter} = K \cdot P_a \cdot G_d = \frac{4 \cdot R_a^2}{|Z_a + Z_{c1,c2}|^2} \cdot \frac{\lambda^2 \cdot G_d^2}{4\pi} \cdot S \quad (3)$$

Where the $Z_a = R_a + jX_a$ is complex antenna impedance and $Z_{c1,c2} = R_{c1,c2} + jX_{c1,c2}$ is complex chip load in different modulation status.

The backscatter loss or cross section of the device 1 can be defined as

$$\sigma = \frac{P_{backscatter}}{S} = \frac{R_a^2}{\left|Z_a + Z_{c1,c2}\right|^2} \cdot \frac{\lambda^2 \cdot G_d^2}{\pi} \quad (4)$$

The received backscattered signal at the test device:

$$P_{received} = \frac{P_{backscatter} \cdot G_r}{4\pi r^2} = \frac{P_t \cdot G_t}{4\pi r^2} \cdot \frac{G_r}{4\pi r^2} \cdot \sigma \quad (5)$$

Output power dynamics

Output power dynamics includes minimum output power, transmit OFF power, transmit ON/OFF time mask and power control.

Power control is not designed for A-IoT device 1 D2R link, and the transmit power of device 1 comes from CW node, making Output power dynamics requirements unnecessary.

Transmitted signal quality

Transmitted signal quality includes frequency error, carrier leakage, in-band emission and EVM.

As D2R backscattering is transmitted on a carrier wave provided externally, the requirements for frequency error, carrier leakage and in-band emission are potentially unnecessary.

The waveform for D2R could be OOK or BPSK. OOK signal is not mapped by the legacy constellation. EVM may not be suitable for D2R OOK waveform, and whether EVM testing is feasible for BPSK waveform needs to be discussed. Requirement to guarantee the modulation quality is potentially needed. How to define modulation quality requirements will be discussed in WI.

Output RF spectrum emissions

Output RF spectrum emissions includes occupied bandwidth, out of band emission (SEM, ACLR), spurious emissions and transmit intermodulation.

Occupied bandwidth is defined as the bandwidth containing 99 % of the total integrated mean power of the transmitted spectrum on the assigned channel. Occupied bandwidth requirement is potentially needed. The specific value is dependent on the physical layer design of D2R, which has not yet been finalized at this stage.

SEM and spurious emissions are related to co-existence study. Whether to define these requirements need to be discussed in WI. Based on the co-existence study, ACLR is not needed.

The transmit intermodulation performance is a measure of the capability of the transmitter to inhibit the generation of signals in its non-linear elements caused by presence of the wanted signal and an interfering signal reaching the transmitter via the antenna. However, device1 is low complexity and low cost, and has no active components at the transmitter. SEM, spurious emissions may cover this function and requirement. Transmit intermodulation requirement is unnecessary.

6.7.4.1.2 RX

Reference sensitivity

Reference sensitivity is strongly related to the coverage of R2D. Reference sensitivity is potentially needed.

Maximum input level

The received power level for device 1 is up to the Reader and the distance between Reader and device. No power control for device. Whether maximum input level to be defined needs to be discussed in WI.

Adjacent channel selectivity(ACS)

Device1 is RF-ED architecture with low complexity and cost, apart from the simple LPF and matching network, other components cannot suppress the interference. Hence, potentially, there is no need to define ACS.

Adjacent Subcarrier selectivity(ASCS)

Same as ACS, there is no need to define ASCS.

In-band blocking

Same as ACS, potentially, there is no need to define in-band blocking.

Out-of-band blocking

Same as ACS, potentially, there is no need to define in-band blocking.

Spurious emissions

The spurious emissions power is the power of emissions generated or amplified in a receiver that appear at the UE antenna connector. The received power comes from R2D signal and device 1 has no LNA. Hence, potentially, there is no need to define spurious emissions.

Spurious response

Same as ACS, potentially, there is no need to define spurious response.

Intermodulation Characteristic

Same as ACS, potentially, there is no need to define spurious response.

Interference rejection

The interference could be wide-band NR signal (e.g. in-band deployment) or CW signal (e.g. from unsynchronized reader/CW node) or D2R signal from other devices. Whether interference rejection to be defined needs to be discussed in WI.

6.7.4.2 Device 2a

6.7.4.2.1 TX

Transmitter power

For device 2a, the transmitter architecture is based on backscattering technology, and the backscattering power or power backscattering loss is potentially needed to be defined to ensure the coverage target. The maximum output power is potentially not needed.

The minimum backscattered modulated power can be specified for device 2a. The backscatter power is the amount of power reflected back from the device 2a when a successful command sequence is completed at the device at a specific sensitivity level.

The backscatter loss due to the modulation can be defined as scattering cross-section or modulation efficiency in equation (1). It can be observed that the modulation efficiency is defined as the ratio of the power radiated by the device to the power density incident on its antenna.

The backscatter loss or cross section of the device 2a can be defined as

$$\sigma = \frac{P_{backscatter}}{S} = \frac{R_a^2}{\left|Z_a + Z_{c1,c2}\right|^2} \cdot \frac{\lambda^2 \cdot G_d^2}{\pi} \quad (1)$$

The received backscattered signal at the test device:

$$P_{received} = \frac{P_{backscatter} \cdot G_r}{4\pi r^2} = \frac{P_t \cdot G_t}{4\pi r^2} \cdot \frac{G_r}{4\pi r^2} \cdot \sigma \quad (2)$$

Where $P_{backscatter}$ is the power backscattered back in the direction of the transmitter defined in clause 6.7.4.1.1, and S is the power density of the incident radio wave on the device antenna defined in clause 6.7.4.1.1.

Output power dynamic

For device 2a, power control is not designed for backscattering transmission, and RAN4 concluded that all the output power dynamic requirements are not relevant for backscattering type devices.

- **Transmit OFF power**: when there is no incoming CW signal, the device 2a does not emit any high-power level. The requirement of transmit OFF power is potentially not needed.

- **Transmit ON/OFF time mask:** The time mask requirement is to test the ON/OFF time period for RF chain. For device 2a, this time period is not crucial and it is potentially not needed.
- **Power control**: Power control is not designed for backscattering transmission, the power control requirement is potentially not needed for device 2a.

Output RF spectrum emissions

- **Occupied bandwidth:** The occupied bandwidth is the width of a frequency band containing 99 % of the total integrated mean power of the transmitted spectrum on the assigned channel. Occupied bandwidth is potentially needed for Device 2a. The specific value is dependent on the physical layer design of D2R, which has not yet been finalized at this stage.
- **ACLR:** According to the co-existence simulation, the ACLR requirement is potentially not needed for device 2a.

6.7.4.2.2 RX

Reference sensitivity

The reference sensitivity is the minimum mean power applied to device antenna port to guarantee the receive performance. The requirement of reference sensitivity for device 2a is potentially needed.

Adjacent channel selectivity (ACS)

For device 2a, the RF BPF is not assumed in RF architecture for the reason of cost and size consideration, So the ACS requirement which relies on digital filter is potentially not needed.

Adjacent Subcarrier selectivity (ASCS)

Same as ACS, potentially, no need to define ASCS.

In-band blocking

Same as ACS, potentially, no need to define in-band blocking requirement.

Out-of-band blocking

Same as ACS, potentially, no need to define out-of-band blocking requirement.

Spurious emissions

The spurious emissions power is the power of emissions generated or amplified in a receiver that appear at the UE antenna connector. The received power comes from R2D signal. For device 2a, LNA is not assumed in RF architecture. Hence, there is potentially no need to define spurious emissions.

Spurious response

Same as ACS, there is potentially no need to define spurious response.

Intermodulation Characteristic

Same as ACS, there is potentially no need to define intermodulation characteristic.

6.7.4.3 Device 2b

6.7.4.3.1 TX

Maximum output power

In the transmission direction (D2R), for device 2b, as it can generate the signal by itself, the device function can be more similar to legacy IoT devices than backscattering-type devices. Therefore, its maximum output power and the corresponding power class can also be specified similarly to legacy IoT devices,

However, due to the nature of energy harvesting, its transmission and reception duty cycle may be limited by the availability of the harvested energy. Moreover, the power constraint on the AIoT devices may further impact the choice of components, e.g., oscillator, amplifier, etc.; thus, a significantly lower performance/capability can be expected.

Transmit OFF power

As device type 2b, assuming the device is fully charged, it can be foreseen that it will work similarly to legacy IoT devices. Therefore, the transmit OFF power should be potentially specified for type 2b to ensure the power level when the RF part is turned off will not exceed a certain range.

Transmit ON/OFF time mask

The time mask requirement is to test the time period between the RF part turned on and off. For A-IoT devices, this time period is not a crucial requirement and hence it is potentially not needed.

Minimum output power

As device type 2b, assuming the device is fully charged, it can be foreseen that it will work similarly to legacy IoT devices. Therefore, the minimum output power should be potentially specified for type 2b to ensure the lower limit of the output power for coverage.

Power control

As device type 2b, assuming the device is fully charged, it can be foreseen that it will work similarly to legacy IoT devices. Therefore, the power control should be potentially specified for type 2b to ensure the output power range.

Frequency error

As device type 2b, assuming the device is fully charged, it can be foreseen that it will work similarly to legacy IoT devices and the LO is assumed to be used. Hence the frequency error is potentially required.

Occupied bandwidth

The transmission bandwidth configuration defines the bandwidth of the R2D/D2R signal in baseband. For the R2D channel, one or multiple PRBs are under consideration in RAN1. For D2R channel, the transmission bandwidth of 15kHz or more is being discussed.

6.7.4.3.2 RX

Reference sensitivity

The receiver sensitivity is to test the receiver capability to certain performance metrics in a given lowest RX power level. For device 2b, this requirement will be potentially left to be further defined in WI phase to consider the RF architecture and the exact RF component capability.

Maximum input power

In NR, the maximum input power is to guarantee the UE can still work normally when it is close to BS. Similarly, this requirement is still potentially needed for AIoT device to ensure the device have reasonable dynamic range.

ACS

The ACS requirements shall be derived based on co-existence simulation, and guard band/guard RB may also be defined. As the BB LPF is proposed, the ACS requirement is potentially supposed to be essential to verify its RF performance

It’s expected that the AIoT system will be deployed within an NR channel from the same operator. The AIoT device should be able to receive in the presence of a wideband NR interferer. Additionally, the device might also suffer from CW interference from a nearby reader/CW node, if it’s not synchronised with the serving reader.

ASCS

The ASCS requirement is supposed to define the selectivity of the in-channel receiver. However as the channel bandwidth of the device is limited to only 1RB, the ASCS is potentially not needed.

Spurious response

Spurious response is a measure of the ability of the receiver to receive a wanted signal on its assigned channel frequency without exceeding a given degradation due to the presence of an unwanted CW interfering signal at any other frequency for which a response is obtained. In the A-IoT deployment, the CW signal is set at a certain frequency for energy harvesting and reflection. Hence the scenario for this test is potentially not needed.

6.7.4.4 Testability

6.7.4.4.1 Testability of device

Based on the architecture described in clause 5, with such highly integrated architectures for device 1/2a/2b, conducted antenna connectors are assumed not to be available. Therefore, OTA testing is considered as the baseline testing methodology for device 1, device 2a and device 2b RF verification. Far-filed measurement system in an anechoic chamber could be considered as the baseline setup.

To be fully functional, A-IoT devices require a dedicated energy source for harvesting to be provided during the test. According to the SI description, the devices under consideration are either battery-less or with limited energy storage capability (i.e., using a capacitor) and the energy is provided through the harvesting of radio waves, light, motion, heat, or any other power source that could be seen suitable. Note that the following description on the energy harvesting is just an example to illustrate the testability issues for Energy Harvesting. It doesn't mean to define the Energy Harvesting procedure. Device goes from OFF state to ON state when it detects energy and has been able to accumulate sufficient amount of energy for the parts to function. This Energy Harvesting Sensitivity (EHS) is the value when device starts to harvest energy. Device behaviour principle with non-constant energy availability is shown in Figure 6.7.4.4.2-1.

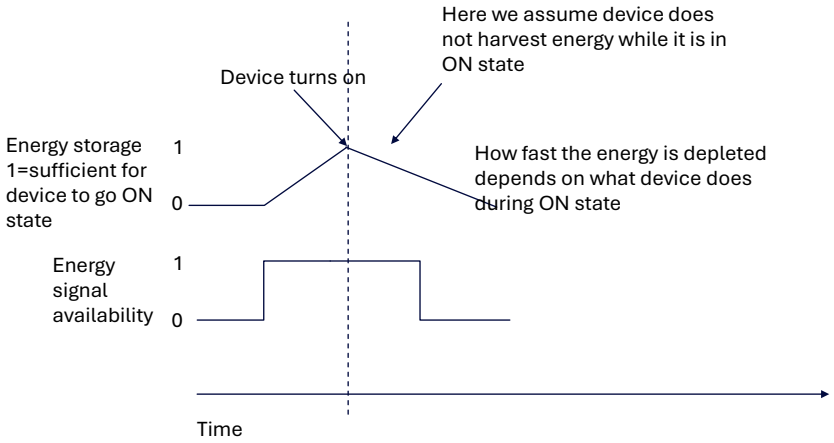


Figure 6.7.4.4.2-1: An example of energy harvesting principle

Energizing signal is a trigger for device to turn on. This is specifically for device type 1 and 2a. After device turns on, it needs to monitor the A-IoT R2D signal and device should respond to network in test environment once the signal from network is detected successfully. Device may also decide to sleep or continue monitoring.

In testing phase, for a device that harvests RF energy, tester must provide sufficient RF waveform electrical field density to the device to ensure it is turned on and time gate the RF testing to the ON periods of the A-IoT device. An external antenna may be needed to provide the CW and also provide RF energy, while a separate testing antenna is also needed, which are illustrated in Figure 6.7.4.1.2-2.

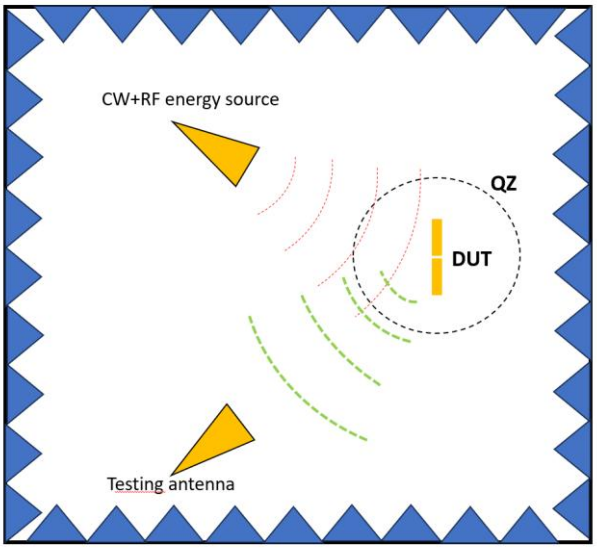


Figure 6.7.4.1.2-2: A possible OTA test setup with CW and RF energy source

Therefore, RAN4 needs to define a list of testable requirements in the WI phase and also the time domain pattern for the tests.

6.7.4.4.2 Summary of testability issues

RAN4 take OTA as the baseline testing methodology for device 1, device 2a and device 2b RF verification. The following test setup issues can be discussed and decided in the WI phase: power level of RF energy source and CW signal, duration time of harvesting, ON-OFF testing pattern, isolation between the CW antenna and the testing antenna, etc.

6.7.5 CW interference cancelation study

6.7.5.1 General

The CW cancellation can be divided into 3 stages sequentially: (1) Spatial isolation for separate TX/RX antenna configuration; (2) Analog cancellation from antenna to ADC and (3) Digital cancellation at baseband.

Spatial isolation for separate TX/RX antenna configuration

For separate TX/RX antenna configuration, the spatial isolation can be achieved by antenna isolation, or coupling loss between CW transmitter and D2R receiver.

For example, if the CW transmission source is separately deployed from the D2R receiver, the spatial isolation relies on the distance and antenna sidelobe suppression between the CW node and A-IOT node.

Analog cancellation

Interference cancellation in the RF front-end can be applied to reduce the desensitization of the receiver. In principle, it can be implemented by reconstructing a local CW with the same amplitude but opposite phase of a CW interferer received from antenna at the input of the LNA, and combine this cancellation signal with the received signal. The cancellation performance depends on the phase and amplitude error between the reconstructed and actual received CW interference.

Additionally in the case of integrated carrier-wave transmitter and D2R receiver in sharing antenna case, the interference suppression could be provided by a circulator/directional coupler.

Digital cancellation

In the digital baseband, suppression can be achieved by filtering, as the received single-tone carrier-wave is shifted to direct current (DC) at baseband. Digital interference cancellation can also be achieved by using a low complexity self-interference active cancellation technique.

6.7.5.2 Interference cancelation analysis

6.7.5.2.1 Ericsson

The min received power could be -74 dBm or lower, the signal with such low level received power should be decoded in the presence of the CW transmission which is either 33 dBm or 23 dBm for monostatic antenna and reference at antenna. The CW transmission can leak to the receiver chain via either reflection from environment or the imperfection of the isolation between transmitter and receiver when CW is transmitted by A-IoT BS or UE as reader. As the receiver has limited dynamic range and linearity, this CW leakage needs to be attenuated/isolated to mitigate the desensitization of the receiver. The CW signal frequency is used to modulate the backscatter signal and the frequency separation of the CW signal and modulated signal is around 100kHz for example as illustrated in Figure 6.7.5.2.1-1 according to the current D2R waveform design. Using an RF filter before the LNA to attenuate the CW signal is not possible. As such, a high-level CW leakage signal will be present together with a low wanted backscattered signal before LNA.

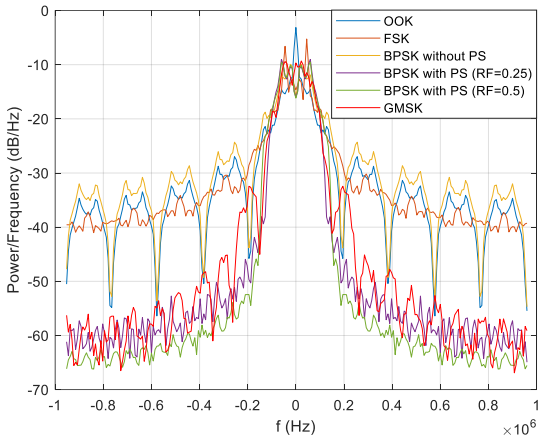


Figure 6.7.5.2.1-1: D2R waveform in frequency domain

To analysis the impact of the receiver desensitization, the backscattered signal is modelled as below [2]:

$$S_b(t) = A_b \cos(\omega_b \cdot t + \varphi_b) \cdot \cos\left(\omega_c \cdot t + \varphi_c\right)$$

with ω_c representing the CW frequency and ω_b is the message frequency.

And the leakage CW signal is modelled as below:

$$S_I(t) = A_I \cdot \cos\left(\omega_c \cdot t + \varphi_I\right)$$

Then the receiver nonlinearity is modelled with a power series below assuming the second-order nonlinear product can be filtered:

$$y_o(t) = a_1 \cdot x(t) + a_3 \cdot x(t)^3$$

Where a_1 is the small signal gain of the receiver and $a_3 < 0$ represents the nonlinear behaviour of the system.

The input signal before LNA is then:

$$x(t) = S_b(t) + S_I(t)$$

Assuming the $\varphi_b = \varphi_c = \varphi_I = 0$, at output of the LNA, only the terms associated with fundamental frequency $\cos(w_b t)\cos(\omega_c t)$ are of interest and they are below:

$$y_{oI}(t) = \left(a_1 A_b + \frac{9}{16} a_3 A_b^3 + \frac{9}{4} a_3 A_b A_I^2\right) \cos(\omega_b \cdot t) \cdot \cos(\omega_c \cdot t)$$

As the CW signal is much larger than backscattered signal, the term $\frac{9}{16} \cdot a_3 \cdot A_b^3$ can be dropped and further simplified with:

$$y_{oI}(t) = \left(a_1 A_b + \frac{9}{4} a_3 A_b A_I^2\right) \cos(\omega_b \cdot t) \cdot \cos(\omega_c \cdot t)$$

The voltage of the fundamental signal at output of LNA can be reformulated as:

$$v(t) = a_1 \cdot A_b \cdot \left(1 + \frac{9}{4} \cdot \frac{a_3}{a_1} \cdot A_I^2\right)$$

Since $a_3 < 0$, we can write:

$$v(t) = a_1 \cdot A_b \cdot \left(1 - \frac{9}{4} \cdot \frac{|a_3|}{|a_1|} \cdot A_I^2\right)$$

It can be observed from the above that the wanted signal level is degraded with scaling factor which directly reduces the SNR, therefore the desensitization factor is:

$$SNR_D = 20log_{10}\left(1 - \frac{9}{4}\frac{|a_3|}{|a_1|} \cdot A_I^2\right)$$

Further the IIP3 of LNA can be expressed with:

$$A_{IIP3_LNA}^2 = \frac{|a_1|}{|a_3|} \cdot \frac{4}{3}$$

Therefore, the SNR degradation can be formulated below:

$$SNR_D = 20log_{10}\left(1 - 3 \cdot \frac{A_I^2}{A_{IIP3_{LNA}}^2}\right) = 20log_{10}\left(1 - 3 \cdot \frac{P_{cw_{leakge}}}{P_{IIP3_{LNA}}}\right)$$

Assuming CW signal is transmitted by A-IoT BS or UE as reader and monostatic antenna configuration, with the return loss of 15 dB, the CW leakage power will be 18 dBm. If the SNR degradation is allowed with 3 dB, the IIP3 of LNA should be 28 dBm. Such high IIP3 is demanding for design. For other combination value for SNR degradation and CW leakage power, it is illustrated in Figure 6.7.5.2.1-2 and the IIP3 value is listed in Table 6.7.5.2.1-1. Referring to TR 38.858, the reported IIP3 for WA BS is from the range of 10 dBm to -27.6 dBm, this in turn demands some CW cancellation capability before LNA.

Table 6.7.5.2.1-1: IIP3 of LNA with different SNR degradation and CW leakage power

		SNR degradation (dB)		
		-1	-3	-10
CW leakage power (dBm)	18	32	28	24
	8	22	18	14
	0	14	10	6
	-10	4	0	-4

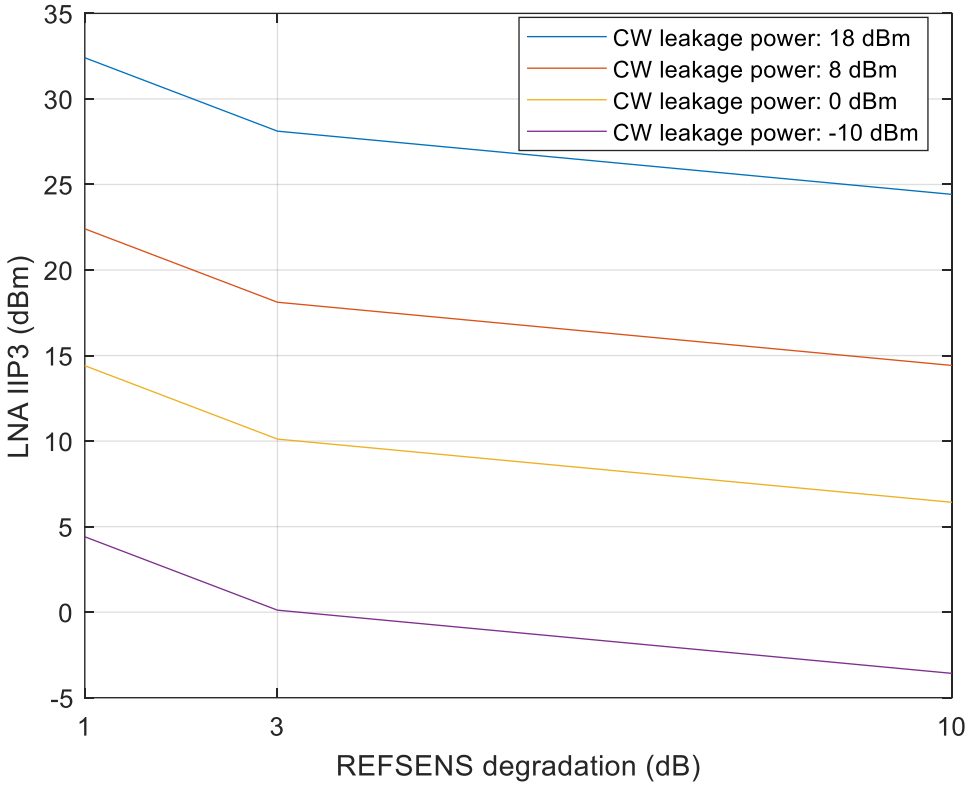


Figure 6.7.5.2.1-2: IIP3 of LNA with different SNR degradation and CW leakage power

For CW cancellation, there are passive or active cancellation schemes existing for RFID reader front-end. In both schemes, a copy of the CW leakage signal is created with same amplitude but opposite phase. The cancellation scheme is illustrated in Figure 6.7.5.2.1-3. For the passive cancellation scheme, the cancellation path needs to be calibrated and for active cancellation scheme, the cancellation path can be tuned to adapt to the leakage CW signal. However, the cancellation is limited by either the hardware imperfection or the detecting signal amplitude/phase accuracy. In [6], it is mentioned that for 40 dB CW cancellation, the amplitude error should be 0.09 dB and phase error to be 0.5 degree. This is quite demanding in design considering the current RRM measurement accuracy for RSRP is +/-2 dB in normal condition.

When the CW and A-IoT signal are transmitted from different hardware, the active CW cancellation scheme should be used in this case. This applies to the case where the CW node is co-located with A-IoT BS for D1T1-A2 or D2T2-A2.

To enable the CW cancellation function, reader needs some time to train the circuit to generate the reverse CW leakage with the same amplitude and the opposite phase. The cancellation training time is implementation specific and also relates to the targeted cancellation performance. Considering this time and also the transient time to transmit the CW signal, the timing relation between the R2D, CW and D2R can be illustrated in Figure 6.7.5.1-4.

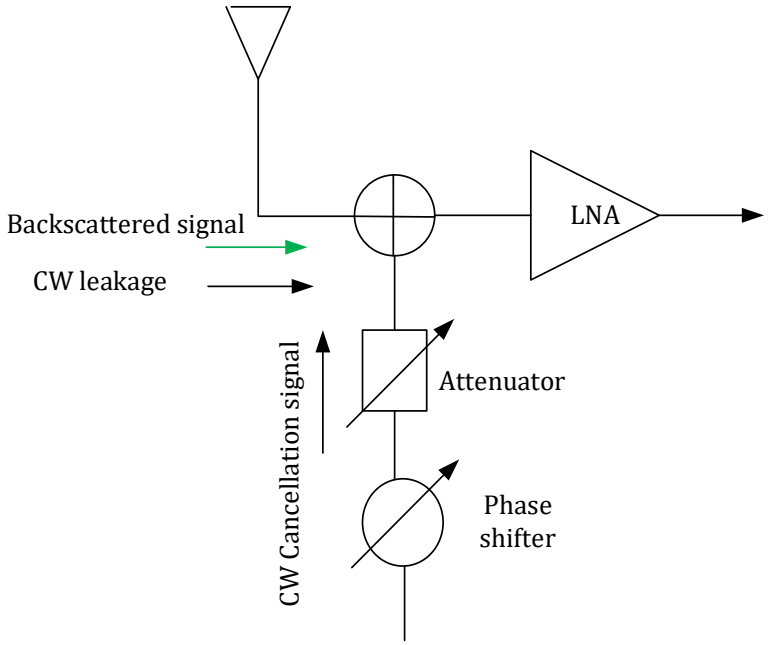


Figure 6.7.5.1-3: CW cancellation scheme

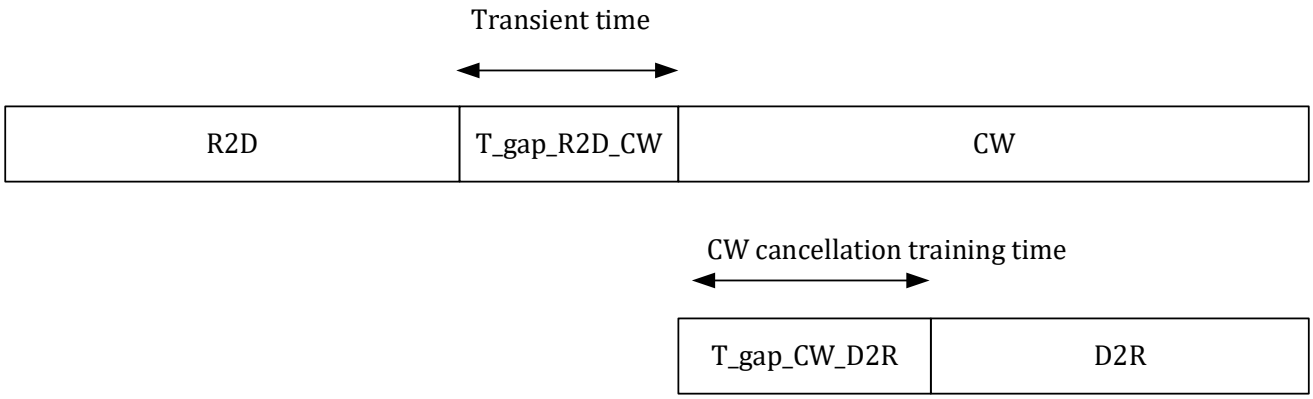


Figure 6.7.5.1-4: Transient time for CW and R2D transmission and CW cancellation training time before D2R reception

6.7.5.2.2 Qualcomm

6.7.5.2.2.1 Feasible values CW cancellation from implementation impact

In principle, the opposite phased CW can be injected to the receiver and then two signal will cancel each other out. The problem is how to know the phase of the CW in the receiver and how to adjust the cancelling signal perfectly. This is especially difficult in the intermediate node since it has smaller dimensions and shared antennas. The circuitry and relevant aspects are shown in Figure 6.7.5.2.2.1-1.

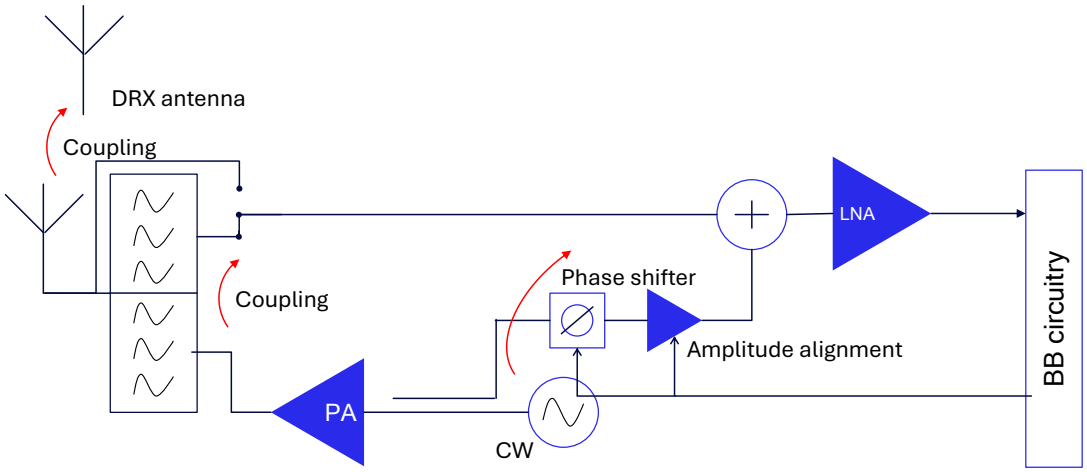


Figure 6.7.5.2.2.1-1. CW cancellation in an Intermediate node

From Figure 6.7.5.2.2.1-1 we can see that CW has many leakage paths and conclude that the cancellation circuitry has to cancel very high-level CW compared to the backscattered signal level.

RAN4 has not discussed techniques for active cancellation but it should be noted that the processing accuracy in a handset receiver typically results in the order of 1.5-4 % Rx EVM [226]. This is the level Intermediate node baseband can provide information to the phase and amplitude cancellation circuits for the analogue cancellation. In the LS [226], the typical UE Rx EVM is 2-4 % for a UE that supports 1024QAM. UE fulfilling minimum requirements and not support 1024QAM would have even smaller minimum implied assumption for Rx EVM. The analogue domain phase and amplitude controller also have finite resolution and 1% accuracy would mean 1.8 degrees phase step assuming signal can be inverted separately. This level of granularity maybe challenging in practice. Intermediate node CW cancellation circuitry processing accuracy is in the order of 1.5-4 %.

Translating this to CW cancellation in dB, we need to split this in to phase accuracy and amplitude accuracy. With a simple mathematical analysis, if we assume that waveform with an amplitude of Y is cancelled with:

$$\text{Cancellation waveform} = Y*(1-\textit{amplitude_error})*\sin(wt+\textit{phase_error})$$

And calculate the resulting CW as

$$Z = Y - \text{Cancellation waveform},$$

We can get cancellation value of Z compared to Y as in Figure 3 below. The EVM % is calculated from the *amplitude_error* and *phase_error*.

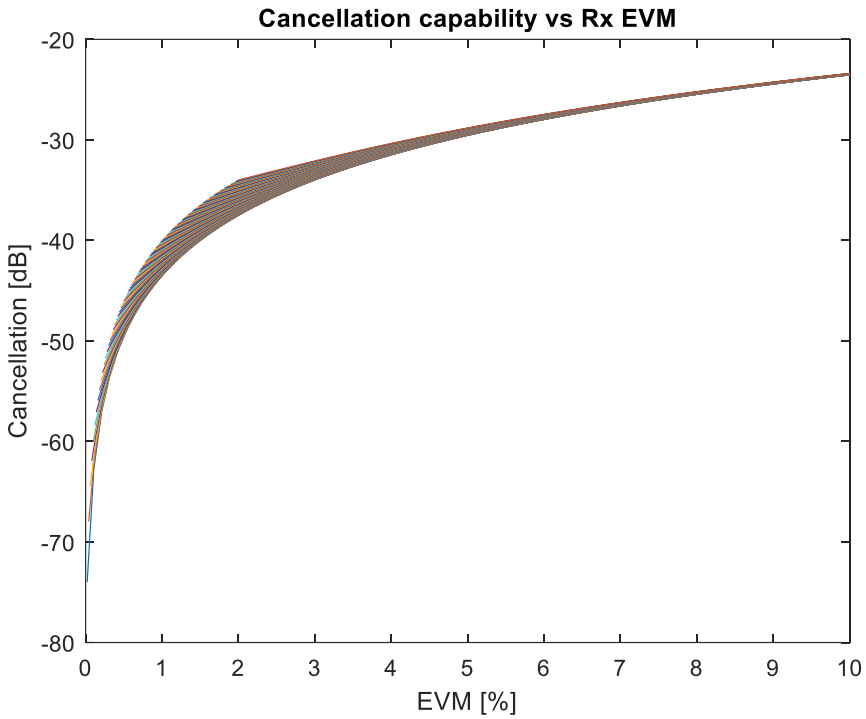


Figure 6.7.5.2.2.1-2: Cancellation as a function of EVM in the cancellation circuitry.

The curve in Figure 6.7.5.2.2.1-2 is not explicitly defined since the contribution of phase error to the EVM % is not same as its contribution to the cancellation. In other words, cancellation is more sensitive to the amplitude error than it is for phase error. In Figure 6.7.5.2.2.1-3 we show the dependency as surface plot for reference.

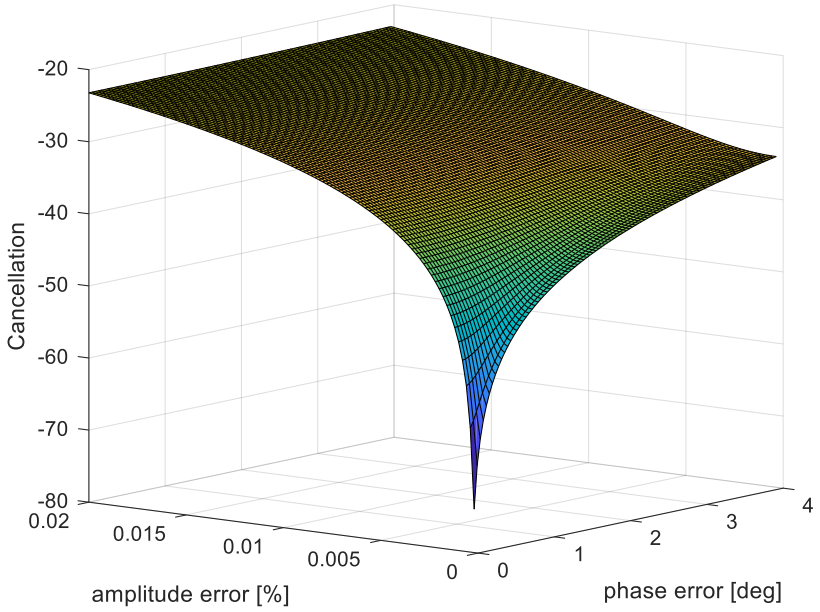


Figure 6.7.5.2.2.1-3: Cancellation sensitivity as a function of amplitude and phase error

We can see that the ampltude error and phase have different impact to cancellation and Intermediate node realistic analogue domain CW cancellation capability is at most 50 dB and more likely typical value is closer to less than 40 dB.

Coming back to the incoming signal characteristics, this means that the UE can cope with the CW of the level:

CW_to_signal= Analogue CW cancellation + Digital domain cancellation.

The digital domain cancellation is same process as DC offset compensation. Normally, the DC offset to be compensated is small since it is a result of non-perfections in the circuitry. In this case, the CW is much larger than the receiver modulated signal so the DC offset cancellation circuitry needs some tuning to work in this new situation. We can assume here that the digital domain cancellation can cope at least with the CW to signal ratio where LNA remain linear enough not to distort the signal.

6.7.5.2.2.2 Needed CW cancellation when CW from reader

In the first phase, to make the analysis a bit more concrete, we can assume that current handheld UE requirements are taken as baseline. Here we consider the maximum input level of -25 dBm and further assume that digital path dynamic range is defined by SNR for 256 QAM and ACS resulting in to 68 dB (12 bits) but in practice some more dynamic range is needed to accommodate some range for the digital AGC in practice. The output power is the one used in co-ex work, 23 dBm, and antenna coupling is 10 dB thus same what is used in earlier and in all other work such as MSD work in RAN4 for handhelds. Additionally, we can assume that the needed D2R signal SNR for successful decoding is 10 dB.

Table 6.7.5.2.2.2-1. Assumptions for the CW cancellation for intermediate node

Parameter	Value	Note
Max input level	-25 dBm	
Output power of CW	23 dBm	
Antenna coupling	-10 dB	Also reflection loss
Digital path dynamic range	12 bit (14 bit)	68 dB (74 dB)
SN(I)R for D2R signal	10 dB	At receiver

With the above numbers, we can calculate that the maximum CW cancellation in digital domain is 68 – 10 = 58 dB. This will depend on number of bits used in the ACD, and for effective 14 bit ADC it would be 74 dB.

Reflected power from the antenna is 10 dB lower than the transmitted power, this means LNA without analogue domain CW cancellation will see 13 dBm CW but since the maximum input level is -25 dBm, the analogue domain cancellation has to be 38 dB for 23 dB output power and to ensure LNA max input level is not exceeded. Even if handheld device uses different antennas, the coupling can be 10 dB and the result is the same conclusion. Analogue domain CW cancellation must be 38 dB for 23 dBm output power and -25 dBm maximum input level for the receiver.

The analysis in previous section we discussed that this is feasible from the control point of view. With 38 dB in analogue domain, the total CW cancellation maximum is 112 dB.

Looking more closely, the range of the reader can be calculated from the above numbers following the logic:

Received D2R power = Max CW power – 2x pathloss – backscattering loss

Minimum received power at lowest SNR = max input level + SNR_{D2R} – DNR_{ADC}

When me make both equations equal:

Max CW power – 2x pathloss – backscattering loss >= max input level + SNR_{D2R} – DNR_{ADC}

Pathloss =< (-max input level - SNR_{D2R} + DNR_{ADC} + Max CW power - backscattering loss)/2

In numbers, assuming 10 dB backscattering loss:

Pathloss =< (-(-25 dBm) – 10 dB + 84 dB + 23 dBm - 10 dB)/2 = 56 dB, which is about 14 meters at 1 GHz with 14 bit ADC.

And this is assuming the analogue domain CW cancellation is as above.

To understand the CW cancellation role to the range, we can first look at the analogue domain cancellation. If the cancellation is less than 38 dB but the maximum input level can not be exceeded, the output power has to be reduced and this reduces the range accordingly. If the CW cancellation is better than the 38 dB, the output power can be increased and this increases the range.

Looking at the digital domain cancellation, firstly we state that the digital domain can cancellation can be very large if a long training period is allowed. When intermediate node is the reader, it can choose when to turn CW or and train its own receiver thus time can be long.

For a fixed output power, the digital domain dynamic range defines the range since more bits there are available, more bits can be used for the CW cancellation with 2 effective bits needed to cover 10 dB SNR needed for the successful decoding, only the training time may need to be increased. However, this analysis did not look in to the noise contribution. With the above values, the received D2R signal power is close to -100 dBm. Without knowing what is the bandwidth of the D2R signal, it is difficult to say exactly what is the noise contribution but at -100 dBm level and e.g. 200 kHz BW when kTB = -120 dBm, there will be some noise contribution to the SINR.

6.7.5.2.2.3 Needed CW cancellation when "CW from outside of the topology"

Case is different when CW comes from outside the topology. The level of the CW at the receiver is lower compared to the D2R signal level but the CW phase noise plays a role in the range since the down conversion will be done with a different signal source with non-correlating phase noise.

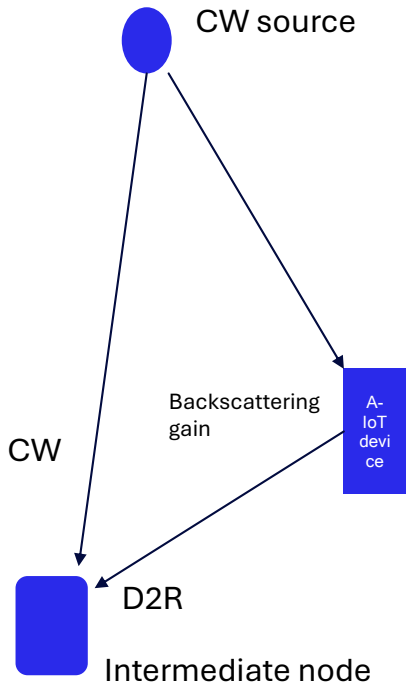


Figure 6.7.5.2.2.3-1: CW not in the reader "CW outside topology"

The way the CW nodes are distributed in the aged topologies result almost constant -39 dBm received CW in the reader. The assumption is to always use closest CW source. Looking at the statistical D2R received power distribution in Figure 2, for 90% of the connections see power higher than -101 dBm for D1T1 and -98 dBm for D2T2.

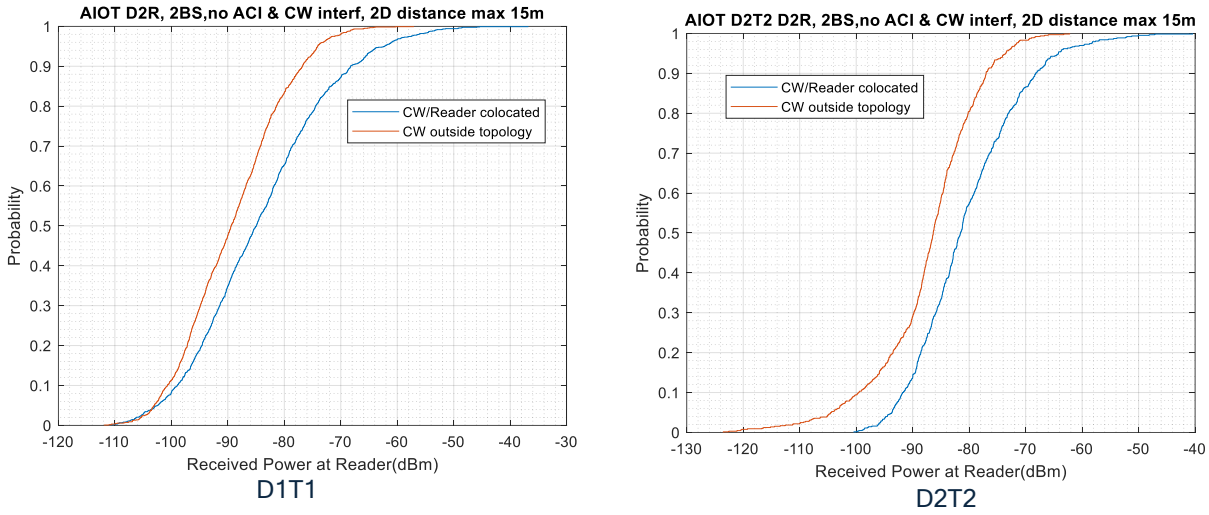


Figure 6.7.5.2.2.3-2: D2R received power distribution in the two topologies. CW level is almost constant at the receiver

Calculating the ratios yields that CW to D2R signal levels for 90 % probability are 62 and 62 dBc in D1T1 and D2T2 when CW comes outside the topology.

In this topology, it is difficult to relate the CW cancellation and range since it depends on how the CW source and reader are positioned. From statistical data in Figure we can see that the cancelation needs to be much lower level than when CW source is in the reader and up to the point that maybe analogue domain cancellation is not needed.

6.7.5.2.2.4 Phase noise impact when CW is outside topology

If the reader will be using a different CW to down convert the D2R signal than the CW used in the backscattering, the phase noise of the incoming CW plays a critical role in system performance.

From the same conclusion, that the CW to D2R power ratio at the reader is approximately 62 dBc, we can conclude that the integrated phase noise over the D2R bandwidth can not be bigger than -72 dBc, to allow for 10 dB SNIR. However, this assumes phase noise is the only noise or interference source and more comfortable specification could be -80 dBc below the CW. Integrated phase noise for the CW coming outside the topology over the D2R signal BW ($B_{\text{tx,D2R}}$) must be better than -80 dBc.

We ran system simulation and impact to SNR because of multiple phase noise values and in Figure 6.7.5.2.2.4-1 present the results. Here we chose 15 kHz D2R bandwidth and assumed it is offset from the CW sufficiently to avoid increase in integrated noise because of high flicker noise close to the CW. In that way, the simulated results show most optimised case. If the bandwidth I large, the integrated noise is also larger.

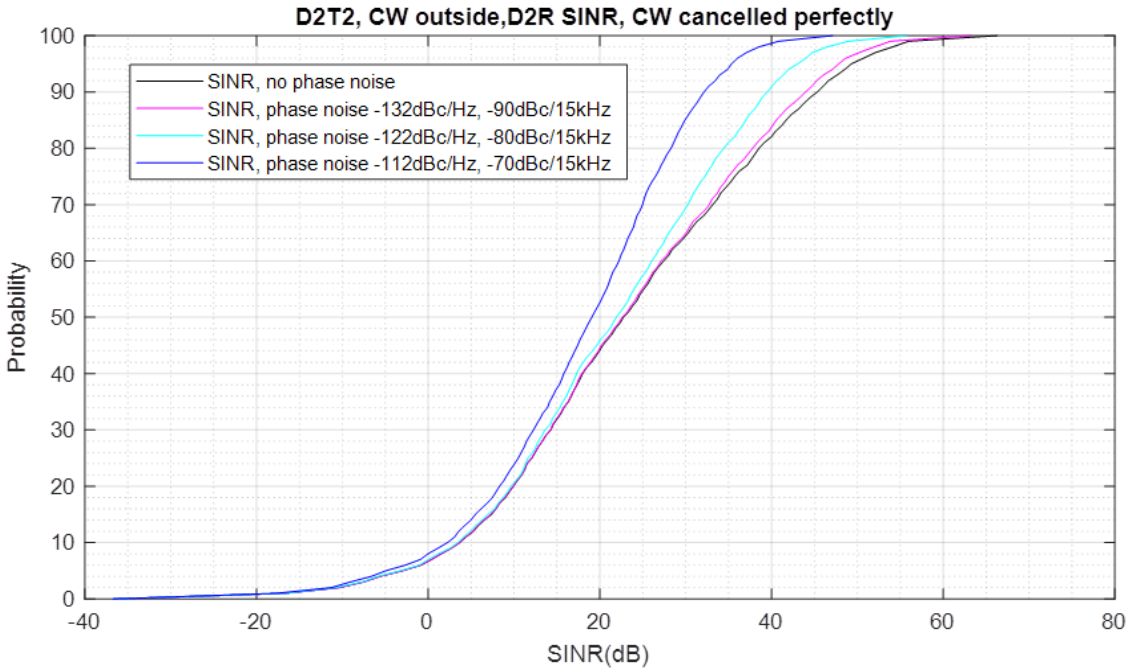


Figure 6.7.5.2.2.4-1: Impact of phase noise to SINR.

Typically the PLLs and oscillators are designed to meet wide BWs and in handheld device also settling time needs to be accounted for and they may not be optimised for close in noise where backscattering signals in A-IoT case are assumed to be positioned with relatively narrow bandwidth.

In practical design, the phase noise is a result of many components, such as VCO, PLL loop and 1/f noise. In Figure 6.7.5.2.2.4-2 these are summarize and the resulting phase noise is the darker red curve.

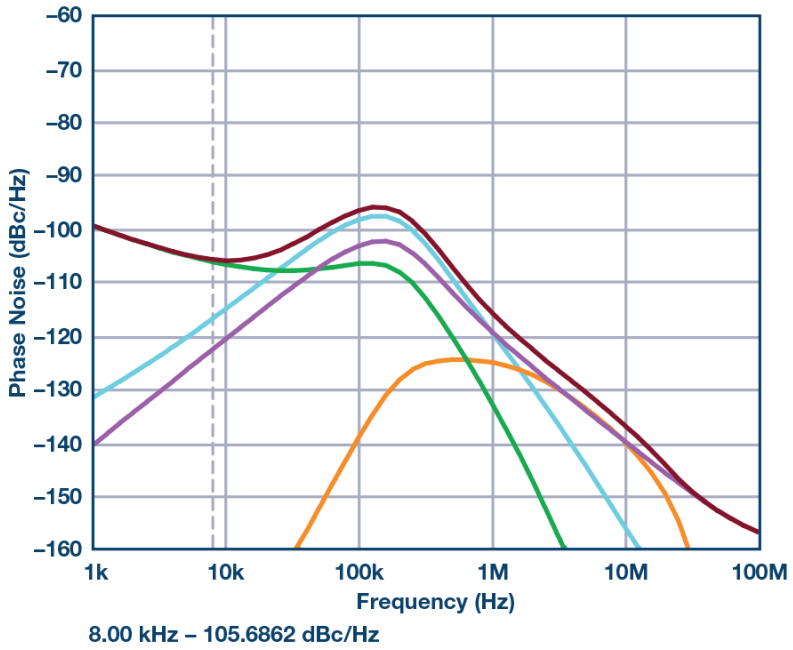


Figure 6.7.5.2.2.4-2: Example of a phase noise characterises and contributing sources ([159])

From Figure 6.7.5.2.2.4-2 we can see that -80 dBc integrated phase noise over 100 kHz specification may require dedicated high-performance design.

RAN4 then should specify this for the component which provides the phase noise to the system but this issue should be firstly understood and documented in the study item and then what kind of specification will be written then discussed in work item.

Understanding phase noise of the CW source may also be important for the radio system design. RFID uses an offset from CW in the D2R link and this helps with the phase noise since as we can observe from Figure 3, offsetting the placement of $B_{tx,D2R}$ further away might be useful in relaxing the CW source phase noise requirements.

Placing $B_{tx,D2R}$ further away from CW in frequency is beneficial for the system throughput and range.

6.7.5.2.3 Huawei

2 aspects are evaluated: (1) Blocking of D2R receiver; (2) Carrier-wave interference cancellation

Blocking of D2R receiver

If the carrier-wave source is separately deployed from the D2R receiver (e.g., D1T1-A1 and D1T1-B scenarios), the spatial isolation relies on the distance between the CW node and A-IOT node. For example, the path loss is 57 dB for an isolation distance of 20 meters, assuming free-space propagation. Normally the CW node and A-IOT node deploy at the same height, the antenna gain is about -5 dB at 90°offset from the boresight for both TX antenna and RX antenna. For D1T1-A1 and D1T1-B scenarios the received power at the antenna connector is 23-57-10= -44 dBm and 23-51-10= -38 dBm for 20 meters and 10 meters distance respectively, which is close to the narrowband blocking requirements for small cell NR base station. Hence there is no blocking issue for these scenarios.

In the case of integrated carrier-wave transmitter and D2R receiver in D1T1-A2, analog interference cancellation in the RF front-end can be applied when the interference power exceeds the blocking threshold of receiver. The performance heavily depends on the similarity between the reconstructed and actual received interference. A suppression of 10-30 dB can usually be expected for RF interference cancellation. Assuming the same reference clock between the carrier-wave transmitter and D2R receiver (e.g., in D1T1-A1/A2 scenario, or D1T1-B scenario with wired connection between the base station and the carrier-wave node), higher interference suppression can be reached, such as 20 dB or higher.

Overall carrier-wave interference

In the case of separated carrier-wave transmitter and D2R receiver in D1T1-A1 or D1T1-B, through spatial isolation, RF interference cancellation and high-pass filtering, the overall suppression to the carrier-wave interference can reach about (61+10+80) = 151 dB at 10 meters distance. For larger spatial isolation corresponding to longer distance between the carrier-wave transmitter and D2R receiver (e.g., ~60 dB @ 25 m), the carrier-wave interference can be even higher (e.g., 70+10+80 = 160 dB).

In the case of integrated carrier-wave transmitter and D2R receiver in D1T1-A2, with ~40 dB interference suppression provided by a circulator/directional coupler and/or the antenna with high isolation between Tx and Rx, the carrier-wave interference suppression can reach about (40+20+80) = 140 dB.

6.7.5.2.4 vivo

Spatial domain

For A1/B scenario, the receiver and transmitter are separated, which means spatial domain isolation for the CW interference can be obtained through this deployment. The amount of spatial isolation can be achieved by pathloss isolation.

For the D1T1-A1/B scenario, the pathloss isolation can reach 61 dB, which assumes the distance between CW emitter and receiver is 20 meters for InH-DH-NLOS path loss model. Meanwhile, for the D2T2-A1/B scenario, the pathloss isolation is 54 dB when the CW emitter and receiver distance is 10 meters and the pathloss model is InH-Office-NLOS.

For A2 scenario, the D2R receiver and CW transmitter are located at same node, so the pathloss can't provide suppression in this case, and the spatial domain suppression capability mainly from Tx/Rx antenna isolation. For D1T1 we assume the antenna isolation is 15 dB, and for D2T2 we assume 10 dB which is a typical value for NR UE.

The suppression capability in spatial domain is summarized below:

Scenario	Part1 Spatial domain suppression
D1T1-A1/B	61 dB
D1T1-A2	15 dB
D2T2-A1/B	54 dB
D2T2-A2	10 dB

Analog domain

This part is the suppression before the LNA. A typical self-interference cancellation architecture is show in Figure 6.7.5.2.4-1

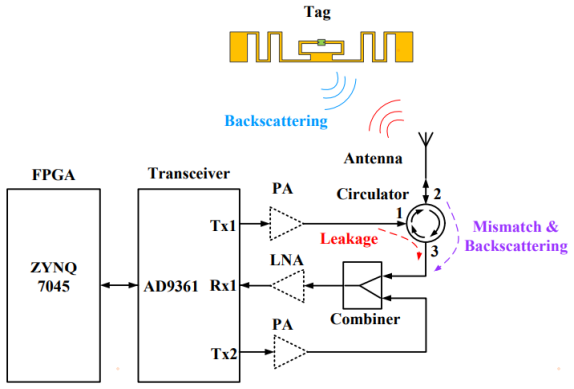


Figure 6.7.5.2.4-1: A typical self-interference cancellation architecture [1]

The basic rationale is to generate a CW that have -180° phase difference compared to the incoming interference and combine them together. According to [1][2], the RF-IC suppression capability between 30dB~ 60dB can be achieved generally, and we assume 45dB here.

For A1/B scenario, since the CW transmitter is separated from D2R receiver, the method above actually hard to be used at receiver side, so we assume 0 dB suppression capability as worst case.

The suppression capability in analog domain is summarized below:

Scenario	Part2 Analog domain suppression
D1T1-A1/B	0 dB
D1T1-A2	45 dB
D2T2-A1/B	0 dB
D2T2-A2	45 dB

IM3 product

After LNA, the CW and D2R will generate intermodulation product, and the IM3 product will overlap with D2R signal in frequency domain, and the BB filter actually cannot help on this. The traditional IM3 calculation procedure is described in the table below

Table 6.7.5.2.4-1: The model of CW total cancellation capability

System configuration									
0A	Scenarios	D2T2-A2	D2T2-B	D2T2-A2	D2T2-B	D1T1-A2	D1T1-B	D1T1-A2	D1T1-B
		InH-LOS	InH-LOS	InH-LOS	InH-LOS	InF-DH-NLOS	InF-DH-NLOS	InF-DH-NLOS	InF-DH-NLOS

0A1	CW case								
0B	Device type	Device1		Device2a		Device1		Device2a	
0C	Center frequency (GHz)	0.9				0.9			
Transmitter									
1E	CW Tx power(dBm)	23				33			
1F	D2R bandwidth (Hz)	160000				160000			
1G	Tx antenna gain(dBi)	0				2			
1N	Cable/connector loss	1				3			
Receiver									
2B	bandwidth used for the evaluated channel (Hz)	160000				160000			
2C	Receiver antenna gain(dBi)	0				2			
2D	Receiver Nosie Figure(dB)	7				5			
2E	Thermal Noise(dBm/Hz)	-174				-174			
2F	Noise Power(dBm)	-114.9588002				-116.9588002			
2G	Tx/Rx spatial isolation capability	10	0	10	0	15	0	15	0
2F	RX cable, connector loss	1				3			
2J	Rx RF-IC suppression capability; For outside topology, this item is pathloss suppression interference	45	54.46	45	54.46	45	61.18	45	61.18
	CW cancellation (dB)	55	54.46	55	54.46	60	61.18	60	61.18
2K1	Remaining CW interference(dB)	-34	-33.46	-34	-33.46	-29	-30.18	-29	-30.18
2K2	D2R power (dBm), for D1T1 scenario, assuming 10m distance between reader and tag; for D2T2 scenario, assuming 5m distance between reader and tag	-72.15	-72.15	-62.15	-62.15	-88.17	-88.17	-78.17	-78.17

2K3	Rx IIP3 capability(dBm)	-6	-6	-6	-6	-10	-10	-10	-10
2K4_1	Rx IM3#1 = 2K1+2*2K2-2*2K3	-166.31	-165.77	-146.31	-145.77	-185.34	-186.52	-165.34	-166.52
2K4_2	Rx IM3#2 = 2K2+2*2K1-2*2K3	-128.15	-127.08	-118.15	-117.08	-126.17	-128.52	-116.17	-118.52
2K4	Rx IM3 = 2K4_1+2K4_2	-128.15	-127.07	-118.15	-117.07	-126.17	-128.52	-116.17	-118.52
2K5	Total CW cancellation capability=1E-2K4	151.15	150.07	141.15	140.07	159.17	161.52	149.17	151.52

Digital domain

From our side, it is not clear how digital domain process can provide benefit on the interference suppression. Since if there is no blocking or non-linear distortion, the single tone CW interference is converted to DC in baseband, which has no impact to detection performance as long as the ratio between desired signal and residual interference is within receiver dynamic range. If blocking happens, it is also hard to overcome it in digital baseband.

6.7.5.3 Summary

CW cancellation performance is implementation specific. Spatial isolation, Analog cancellation and Digital cancellation are studied in clause 6.7.5.2. For RF CW cancellation performance, it is trade-off among the allowed receiver desensitization, receiver linearity and the CW leakage power received at antenna. To enable the CW cancellation, CW cancellation training time should be considered in the implementation. During the CW cancellation training time, D2R reception should be avoided.

6.8 Characteristics of carrier-wave waveform

6.8.1 CW transmission

For the case that D2R backscattering is transmitted in the same carrier as CW for D2R backscattering, and for topology 1, the following cases for CW transmission are studied:

Case 1-1: CW is transmitted from inside the topology, transmitted in DL spectrum

Case 1-2: CW is transmitted from inside the topology, transmitted in UL spectrum

Case 1-4: CW is transmitted from outside the topology, transmitted in UL spectrum

The observations shown in Table 6.8.1-1 are made regarding these cases.

Table 6.8.1-1: Observations on CW transmission cases for topology 1

CW Transmission case	Observations
Case 1-1	- No need for BS to support full-duplex capability in DL spectrum for scenario A1 - Spatial isolation is possible for scenario A1, reducing the received CW interference power at BS side. - Cross-link interference handing for CW at BS side for scenario A1. - BS needs to support full-duplex capability (including self-interference suppression for CW) in DL spectrum for scenario A2. - Higher CW transmission power can be assumed in the DL spectrum than that of in the UL spectrum.
Case 1-2	- No need for BS to support full-duplex capability in UL spectrum for scenario A1 - Spatial isolation is possible for scenario A1, reducing the received CW interference power at BS side. - Cross-link interference handing for CW at BS side for scenario A1. - BS needs to support full-duplex capability (including self-interference suppression for CW) in UL spectrum for scenario A2. - Lower CW transmission power can be assumed in the UL spectrum than that of in the DL spectrum.
Case 1-4	- No need for BS to support full-duplex capability in UL spectrum - Spatial isolation is possible, reducing the received CW interference power at BS side. - Cross-link interference handing for CW at BS side. - Estimation of CW interference may be needed for successful D2R reception (i.e., at BS). - Lower CW transmission power can be assumed in the UL spectrum than that of in the DL spectrum.

For the case that D2R backscattering is transmitted in the same carrier as CW for D2R backscattering, and for topology 2, the following cases for CW transmission are studied:

Case 2-2: CW is transmitted from inside the topology (i.e., intermediate UE), transmitted in UL spectrum

Case 2-3: CW is transmitted from outside the topology, transmitted in DL spectrum

Case 2-4: CW is transmitted from outside the topology, transmitted in UL spectrum

The observations shown in Table 6.8.1-2 are made regarding these cases.

Table 6.8.1-2: Observations on CW transmission cases for topology 2

CW Transmission case	Observations
Case 2-2	- No need for intermediate UE to support full-duplex capability in UL spectrum for scenario A1 - Spatial isolation is possible for scenario A1, reducing the received CW interference power at intermediate UE side. - Cross-link interference handling for CW at intermediate UE side for scenario A1. - Intermediate UE needs to support full-duplex capability (including self-interference suppression for CW) in UL spectrum for scenario A2. - Lower CW transmission power can be assumed in the UL spectrum than that of in the DL spectrum.
Case 2-3	- No need for intermediate UE to support full-duplex capability in DL spectrum - Spatial isolation is possible, reducing the received CW interference power at intermediate UE side. - Cross-link interference handling for CW at intermediate UE side. - Estimation of CW interference may be needed for successful D2R reception (i.e., at intermediate UE). - Higher CW transmission power can be assumed in the DL spectrum than that of in the UL spectrum.
Case 2-4	- No need for intermediate UE to support full-duplex capability in UL spectrum - Spatial isolation is possible, reducing the received CW interference power at intermediate UE side. - Cross-link interference handling for CW at intermediate UE side. - Estimation of CW interference may be needed for successful D2R reception (i.e., at intermediate UE). - Lower CW transmission power can be assumed in the UL spectrum than that of in the DL spectrum.

If large frequency shift is supported by device 2a, the D2R backscattering can be transmitted in a different carrier as CW for D2R backscattering (e.g., CW in DL spectrum and D2R in UL spectrum, or CW in UL spectrum and D2R in DL spectrum), and the followings are observed.

- In-band full-duplex capability is not needed for CW transmission and D2R reception at D2R receiver side, if the D2R receiver only supports device 2a with large frequency shift capability.
- No interference from CW to the corresponding D2R reception at D2R receiver side.
- For CW in DL spectrum and D2R in UL spectrum, higher CW transmission power can be assumed in the DL spectrum than that of in the UL spectrum.

6.8.2 CW characteristics

Candidates for the CW for D2R backscattering are waveforms consisting of:

Waveform 1: A single-tone unmodulated sinusoid, also referred to as 'a single tone'.

Waveform 2: Two single tones.

Table 6.8.2-1 captures observations on the above CW waveform candidates.

Table 6.8.2-1: Observations and/or comparisons of CW waveform candidates

CW waveform characteristics	Waveform 1 compared to Waveform 2 NOTE 1: Waveform 1 without frequency hopping NOTE 2: Waveform 2 with both tones from the same CW node	Waveform 1 with frequency hopping (2-hops) compared to waveform 1 without frequency hopping
D2R reception performance	Waveform 2 provides [0, 8] dB frequency diversity gain compared to waveform 1 at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 1Tx CW transmitter, at least depending on the gap between the two tones and the channel's coherence bandwidth. In a TDL-A fading channel with 30ns delay spread - For the gap between [75KHz, 900KHz], the frequency diversity gains at 1% BLER target observed by 6 sources are within [0, 1.5] dB, and the frequency diversity gains at 10% BLER target observed by 3 sources are almost 0dB. - For the gap between [1.08MHz, 4.2MHz], the frequency diversity gains at 1% BLER target observed by 6 sources are within [3, 5.8] dB, and the frequency diversity gains at 10% BLER target observed by 3 sources are within [0.4, 2.5] dB. - For the gap between [5MHz, 10MHz], the frequency diversity gains at 1% BLER target observed by 5 sources are within [5, 8] dB, and the frequency diversity gains at 10% BLER target observed by 5 sources are within [1.3, 4] dB. In a TDL-D fading channel with 30ns delay spread - For 10MHz gap, 1 source [160] observed 0.7 dB@1%BLER and -0.2dB@10%BLER frequency diversity gain. (Note: loss due to the power split in TDL-D); and 1 source [161] observed 1dB@1%BLER and 0.5dB@10%BLER frequency diversity gain In a TDL-A fading channel with 150ns delay spread - For the gap is 180Khz, the frequency diversity gains at 1% BLER target observed by 2 sources are within [1, 3] dB, and the frequency diversity gains at 10% BLER target observed by 2 sources are within [0, 2.5] dB. - For the gap is 2.16MHz, the frequency diversity gains at 1% BLER target observed by 2 sources are within [7, 8] dB, and the frequency diversity gains at 10% BLER target observed by 2 sources are within [2.5, 5.5] dB. - For the gap is 5MHz, the frequency diversity gains at 1% BLER target observed by 2 sources are within [7, 8] dB, and the frequency diversity gains at 10% BLER target observed by 2	Observations on frequency diversity: - 6 sources observed that waveform 1 with frequency hopping should be used together with bit-level repetition or TB-level repetition. - 3 sources further observed that the unaligned timing (due to SFO at A-IoT device) between reader and devices makes it hard to align the boundary of each hop of the external carrier-wave with a corresponding bit or block repetition of the D2R transmission. - 2 sources observed that for milli-second level CW duration for each frequency hop/block repetition, the diversity gain loss caused by miss-alignment of a few micro-seconds level is marginal, TB level repetition can tolerate some time mis-alignment. - 1 source [162] observed that waveform 1 with frequency hopping achieves frequency diversity gain by using polynomial sweeping-based CC encoding (proposed by [162]) instead of repetitions When bit-level repetition, TB-level repetition or polynomial sweeping-based CC encoding is not used, - Waveform 1 with frequency hopping provides [0, 0.5] dB frequency diversity gain compared to waveform 1 without frequency hopping at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 1Tx CW transmitter. - In a TDL-A fading channel with 30ns delay spread - For 10MHz gap between two hops, 1 source observed 0 dB@10%BLER frequency diversity gain. - In a TDL-A fading channel with 150ns delay spread - For 1.65MHz gap between two hops, 1 source observed 0.5 dB@10%BLER and 0 dB@1%BLER frequency diversity gain. When bit-level repetition, TB-level repetition is used, - Waveform 1 with frequency hopping provides [0.5, 8] dB frequency diversity gain compared to waveform 1 without frequency hopping at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 1Tx CW transmitter, at least depending on the gap between the two hops and the channel's coherence bandwidth. - In a TDL-A fading channel with 30ns delay spread - For the gap of 2.16MHz, 1 source observed 4dB@1% BLER and 2dB@10% BLER frequency diversity gains.

	sources are within [2.5, 3] dB. The following observations were made by one source [161]: - Waveform 1 with antenna hopping (AH) provides [1, 5.75] dB spatial diversity gain compared to Waveform 1 with no AH at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 2Tx CW transmitter. - In a TDL-D fading channel with 30ns delay spread, the spatial diversity gain is 1.6 dB at 1% BLER and 1 dB at 10% BLER. - In a TDL-A fading channel with 30ns delay spread, for the gap of 2.16MHz, the spatial diversity gain is 5.75dB at 1% BLER target and 2.5dB at 10% BLER target. For the gap of 10MHz, the spatial diversity gain is 2.2dB at 1% BLER target and 1.5dB at 10% BLER target. - Waveform 2 with AH provides [1.4, 11.5] dB spatial diversity and frequency diversity gain compared to Waveform 1 with no AH at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 2Tx CW transmitter, at least depending on the gap between the two tones and the channel's coherence bandwidth. - In a TDL-D fading channel with 30ns delay spread, for the gap of 10MHz, the spatial diversity and frequency diversity gain is 2.9 dB at 1% BLER and 1.4 dB at 10% BLER. - In a TDL-A fading channel with 30ns delay spread, for the gap of 2.16MHz, the spatial diversity and frequency diversity gain is 8.25dB at 1% BLER and 5dB at 10% BLER. For the gap of 10MHz, the spatial diversity and frequency diversity gain is 11.5dB at 1% BLER and 7dB at 10% BLER. Note: The total transmission power is assumed the same for both waveforms.	- For the gap between [5MHz, 10MHz], the frequency diversity gains at 1% BLER target observed by 2 sources are within [5.8, 8] dB, and the frequency diversity gains at 10% BLER target observed by 3 sources are within [1.2, 6] dB. The 8 dB gain is achieved under the assumption of ideal channel estimation. - In a TDL-D fading channel with 30ns delay spread - For 10MHz gap, 1 source observed 1 dB@1%BLER and 0.5dB@10%BLER frequency diversity gain. - In a TDL-A fading channel with 150ns delay spread - For the gap between [1.65MHz, 2.16MHz], the frequency diversity gains at 1% BLER target observed by 2 sources are within [5.5, 8] dB, and the frequency diversity gain at 10% BLER target observed by 2 sources is 5 dB. The 8 dB gain is achieved under the assumption of ideal channel estimation. When polynomial sweeping-based CC encoding [162] is used or assumed - In a TDL-A fading channel with 30ns delay spread, and for 10MHz gap between two hops, 1 source [162] observed that waveform 1 with frequency hopping provides 4 dB frequency diversity gain compared to waveform 1 without frequency hopping at 1% BLER for a 1Rx receiver and a 1Tx CW transmitter. The following observations were made by one source ([161]): - Waveform 1 with frequency hopping and antenna hopping provides [1, 9] dB spatial diversity and frequency diversity compared to Waveform 1 with no frequency hopping and no antenna hopping at 1% or 10% BLER in a fading channel for a 1Rx receiver and a 2Tx CW transmitter, at least depending on the gap between the two tones and the channel's coherence bandwidth and assuming the ideal channel estimation. - In a TDL-D fading channel with 30ns delay spread, for the gap of 10MHz, the gain is 2.6 dB at 1% BLER and 1 dB at 10% BLER. - In a TDL-A fading channel with 30ns delay spread, for the gap of 2.16MHz, the gain is 7dB at 1% BLER target and 3.5dB at 10% BLER target. For the gap of 10MHz, the gain is 9dB at 1% BLER target and 5dB at 10% BLER target. Note: The total transmission power is assumed the same for both waveforms. Note: The above evaluations assume the same time domain resources overhead for waveform 1 with or without frequency hopping.
Spectrum utilization of backscattered signal	For the D2R transmission bandwidth corresponding to the CW waveforms, waveform 2 requires twice the frequency domain resources for D2R transmission of waveform 1, if the frequency gap between the two tones is no smaller than the transmission bandwidth of the	4 sources observed that for the D2R transmission BW corresponding to the CW waveforms, compared to waveform 1 without frequency hopping, waveform 1 with frequency hopping utilizes the same amount of frequency domain resources for D2R transmission at a certain time.

corresponding to the CW waveforms	corresponding D2R transmission.	
CW interference suppression at D2R receiver	Waveform 2 requires additional complexity if RF interference cancellation is used at least with CW waveform reconstruction, and requires individual cancellation for each of the tones, e.g. two RF or IF narrow-band bandpass filters. Note: RF interference cancellation is needed when the received CW interference power exceeds the blocking threshold of the receiver.	5 sources observed that waveform 1 with frequency hopping requires additional complexity for CW interference suppression if RF interference cancellation is used. 4 sources observed that waveform 1 with frequency hopping requires individual cancellation for each hop in different time, e.g. one tuneable RF or IF narrow-band bandpass filter, or two RF or IF narrow-band bandpass filters. 2 sources observed that for the RF interference cancellation at the D2R receiver, the frequency hopping of external carrier-wave requires fast re-detection of the parameters of the received carrier-wave interference, which at least doubled the implementation complexity of D2R receiver and may not be supported by the intermediate UE in D2T2. Note: RF interference cancellation is needed when the received CW interference power exceeds the blocking threshold of the receiver.
Relative complexity of CW generation	Waveform 2 leads to higher PAPR of the generated CW, which impacts the implementation of the power amplifier in the CW node.	2 sources observed that waveform 1 with frequency hopping requires additional complexity to construct the single-tone unmodulated sinusoid waveform at different center frequencies in a TDMed manner. 2 sources observed that waveform 1 with frequency hopping doesn't lead to higher PAPR of the generated CW. 1 source observed that waveform 1 with frequency hopping does not require additional complexity of CW generate compared to waveform 1 without frequency hopping.

For the gap between two tones or two frequency hops of a single tone to be able to leverage frequency diversity gain, the bandwidth and spectrum characteristics of the D2R transmission, and the channel coherence bandwidth, should be taken into account.

The following CW waveform characteristics which would need control of the CW node(s) are identified:

- When CW is transmitted or not transmitted
- Transmission Power
- Frequency resources, e.g. frequency location(s) for waveform (e.g. waveform 1, waveform 2)

Other CW waveform characteristics which would need control of the CW node(s), if any, can be further identified in a later stage.

6.9 Locating ambient IoT devices

6.9.1 General

A-IoT device location information may be used for the following purposes:

(a) improving the A-IoT operation itself, e.g., by A-IoT CN sending a Command to one or more readers (e.g., the last reader(s)) associated to the device rather than sending it blindly.

(b) providing location information to the consumer of the A-IoT service.

Locating an Ambient IoT device at "reader ID granularity" is useful for both purposes and is to be supported.

NOTE: "reader ID granularity" is likely not sufficient for purpose (b) and may be refined in the future.

6.9.2 Topology 1

For support of A-IoT device location, the A-IoT RAN node may report the A-IoT RAN node ID to the A-IoT CN in the Inventory Report or Command Response messages.

The A-IoT RAN node may additionally report a Reader ID to the A-IoT CN in the Inventory Report and/or the Command Response messages, which corresponds to the reader to which the A-IoT device responded.

NOTE 1: Whether multiple Reader ID(s) can be included by the A-IoT RAN node in the Inventory Report/Command Response message needs further discussion.

NOTE 2: Definition and format of the A-IoT RAN node ID and Reader ID needs further discussion.

NOTE 3: Whether and how to have awareness of Reader location in A-IoT CN needs further discussion.

6.9.3 Topology 2

For support of A-IoT device location, the A-IoT CN may receive in the Inventory Report and/or Command Response messages the identifier of the A-IoT enabled UE to which the A-IoT device responded.

The A-IoT CN may obtain the A-IoT enabled UE’s location via ULI in NGAP messages. The A-IoT CN may also obtain the A-IoT enabled UE’s location via legacy LCS mechanisms.

NOTE: Definition and format of the identifier of the A-IoT enabled UE needs further discussion.

6.9.4 Proximity determination

Proximity determination is feasible with either of the two following solutions. Potential specification impact or not will not be determined in this study item.

Solution 1: If the reader successfully receives D2R transmission from the device in response to R2D transmission, then the device is determined as near to the reader.

Solution 2: If the reader successfully receives D2R transmission from the device in response to R2D transmission, then the device is determined as near to the reader based on measurements at the reader side.

Proximity determination based on device-side measurements is not considered.

6.10 DO-A assessment

The study included an assessment of whether the harmonized air interface design can address the DO-A use case, only to identify which part(s) of the harmonized air interface design is/are not sufficient for the DO-A use case.

From the RAN1 perspective, at least the following aspect of the air interface for DO-DTT and DT traffic types is not sufficient for the DO-A traffic type: For DO-DTT and DT traffic types, the D2R resource(s) for D2R transmission is/are indicated in a R2D transmission, but this is not applicable at least for the first D2R transmission for DO-A traffic.

From RAN2 perspective, the DO-A traffic type/use case cannot be supported with the current design in the study item. It is assessed that, from RAN2 perspective, at least the A-IoT paging is an aspect/part of the current design which is not sufficient for the DO-A use case.

6A Solutions for Ambient IoT in outdoor scenarios with active devices

6A.1 R2D

6A.1.1 R2D waveform, modulation, bandwidth, arrangement

The R2D waveform described in TS 38.291 [228] is used for the study.

For CP handling for $M_{\text{chip}}^{\text{symb}} \geq 24$, if supported:

- Source [281], [264], [298] state that, with improved SFO accuracy, special CP handling may not be necessary for all supported values of M.
- Source [276] states that, if the CP handling does not impact the R2D reception performance, the additional two chips with "ON" would not be needed. In this case, the transmission efficiency would be further improved for R2D transmission in Rel- 20.
- Source [241] proposes to adopt the same CP handling method used for M=24 to the case of M=32.
- Source [242] states that, for M = 32, CP handling issue should be further considered.
- Source [256] states that, Regarding the large M value, additional CP handling solutions may be required. For example, mapping the last N chips (e.g., N = 3 or 4) of the OOK chip part to ON chips could be considered as a potential solution.

A minimum R2D transmission bandwidth, $B_{tx,R2D}$, of 1 PRB is studied as the baseline.

A-IoT device assumes the centre of the frequency resource used for the transmission of R-TAS, L1 R2D control information and corresponding data payload of PRDCH by the reader are the same.

6A.1.2 $M_{\text{chip}}^{\text{symb}}$ values

6A.1.2.1 Retention, removal, addition of values

Compared to Rel-19, possible addition of $M_{\text{chip}}^{\text{symb}} = 1$ and $M_{\text{chip}}^{\text{symb}} = 32$ to, and the possible removal of $M_{\text{chip}}^{\text{symb}} = 24$ from, the $M_{\text{chip}}^{\text{symb}}$ value set in TS 38.291 [228] was studied, as summarised in Table 6A.1.2.1-1, and the following clauses.

Table 6A.1.2.1-1: Study of *M*-values for R2D

$M_{\text{chip}}^{\text{symb}}$	Study aspect
1	Possible addition
2	As in Rel-19
6	
12	
24	Possible removal
32	Possible addition

6A.1.2.2 $M_{\text{chip}}^{\text{symb}} = 1$

Sources [241], [242], [244], [245], [246], [247], [251], [249], [255], [256], [257], [259], [266], [267], and [269] states that addition of $M_{\text{chip}}^{\text{symb}}$=1 is beneficial to improve R2D coverage.

- Source [246] shows that, $M_{\text{chip}}^{\text{symb}}$ = 1 provides ~3 dB performance gain compared with $M_{\text{chip}}^{\text{symb}}$ = 2 using correlation detection with TBS 20 bits and 96 bits, and ~2.5dB performance gain with TBS 400 bits, when repetition and FEC is not applied, where the CINR/CNR for OOK4 signal, is calculated according to Clause 4.3.2. Source [279] states that longer chip length can enable better coverage performance than M=2 and the performance gain is ~3 dB.

- Source [257] shows that, $M_{\text{chip}}^{\text{symb}}$ = 1 shows performance gain, when repetition and FEC is not applied, where the CINR/CNR for OOK4 signal, is calculated according to Clause 4.3.2.

- Source [249] states that, for $M_{\text{chip}}^{\text{symb}}$=1, the larger chip duration could reduce the DS impact and improve the detection performance for Device 2b/C in outdoor deployment scenario.

Sources [243], [254], [252], [264], [265] and [285] state that addition of $M_{\text{chip}}^{\text{symb}}$=1 is not necessary.

- Sources [243], [254], [252], [265] and [285] state that, OOK with M=2 using together with repetition achieve the same data rates as $M_{\text{chip}}^{\text{symb}}$=1.

- Source [254] further observed that OOK with M=2 using together with repetition can achieve better time-domain diversity gain and power boosting gain than OOK with $M_{\text{chip}}^{\text{symb}}$=1, where the CINR/CNR for OFDM symbol with non-zero power is calculated according to Clause 4.3.2, OFDM symbol with zero power is not counted.

- Source [254] states that, OOK4 with $M_{\text{chip}}^{\text{symb}}$ =2 and $M_{\text{chip}}^{\text{symb}}$=4 have better detection performance than OOK4 with $M_{\text{chip}}^{\text{symb}}$ =1, where the CINR/CNR for OFDM symbol with non-zero power is calculated according to Clause 4.3.2, OFDM symbol with zero power is not counted.

- Source [264] shows that, $M_{\text{chip}}^{\text{symb}}$=1 cannot provide better performance than $M_{\text{chip}}^{\text{symb}}$=2, if the CINR/CNR for OFDM symbol with non-zero power is calculated according to Clause 4.3.2, and OFDM symbol with zero power is not counted. Source [264] also shows that $M_{\text{chip}}^{\text{symb}}$=1 provide less than 2dB performance gain than $M_{\text{chip}}^{\text{symb}}$=2, if the CINR/CNR, for OOK4 signal across all OFDM symbols, is calculated according to Clause 4.3.2.

- Source [292] states that, M=1 does not offer performance improvements relative to M=2, unless power boosting during the ON duration are allowed.

6A.1.2.3 $M_{\text{chip}}^{\text{symb}} = 24$

Sources [246], [247], [257], [251] and [243] state that $M_{\text{chip}}^{\text{symb}}$ = 24 for R2D can be removed.

- Sources [246], [247] and [257] state and provide evaluation results showing that $M_{\text{chip}}^{\text{symb}}$ = 24 cannot work well for outdoor scenarios as its chip length is too short to resist large multipath delay and channel fading.

  - Source [246] shows that $M_{\text{chip}}^{\text{symb}}$ = 24 cannot work for outdoor scenarios with BLER close to 1 under different SNRs

  - Source [247] shows that 10% BLER cannot be achieved with $M_{\text{chip}}^{\text{symb}}$ = 24 (even with repetitions)

  - Source [257] shows that, when $M_{\text{chip}}^{\text{symb}}$ = 24 (and $M_{\text{chip}}^{\text{symb}} > 12$), both Device 2b and Device C cannot achieve 1% BLER and have an error floor around CNR=10 dB or larger (even without SFO/CFO)

- Source [243] states and provides evaluation results show that, R2D transmission with $M_{\text{chip}}^{\text{symb}}$ = 24 cannot achieve 1% BLER even with 2 block-level repetitions, and under similar data rates, the performance of $M_{\text{chip}}^{\text{symb}}$ = 24 (with

2 repetitions) is worse than that of M=12 (without repetition) Source [279] states and provides evaluation results show that, there is an error floor at BLER of 10^{-1} for M=24.

- Source [281] states that CP handling method 2 is not needed for M=24 values in R20 due to better clock accuracy.

Sources [265], [241], [252], [253], [254], [264], [249], [244] and [295] state that it is necessary to keep $M_{\text{chip}}^{\text{symb}}$ = 24 for R2D.

- Source [265] states that $M_{\text{chip}}^{\text{symb}}$ = 24 can be supported for a device which is close to reader and beneficial for device power saving to reduce PRDCH reception duration with higher peak data rate
- Source [241] provides evaluation results show that $M_{\text{chip}}^{\text{symb}}$ = 24 can still achieve 10% BLER with a reasonable SNR, and Source [276] provides evaluation results show that the coverage requirement of 500m at 10% BLER.
- Sources [241], [252], [253] and [276] state that $M_{\text{chip}}^{\text{symb}}$ = 24 is needed to achieve data rates similar as R19.
- Source [244] states that, $M_{\text{chip}}^{\text{symb}}$ = 24 should not be removed, since it is still useful for indoor scenario, and the air interface for Device 2b/C studied for outdoor scenarios will be reused in indoor scenarios.

6A.1.2.4 $M_{\text{chip}}^{\text{symb}} = 32$

Sources [241], [264], [256], [242], [251], [255] state it is beneficial to add $M_{\text{chip}}^{\text{symb}}$= 32 for R2D.

- Source [241] states and provide evaluation results shows that, for the TDL-C 300 ns environment, the BLER of $M_{\text{chip}}^{\text{symb}}$=32 is below 10% when CP handling method is used with 8RB resource allocation;
- Source [241], [264], [256], [251], [255] and [242] state that, the addition of $M_{\text{chip}}^{\text{symb}}$=32 could be considered for the purpose of achieving higher data rate. Source [264] also states that it can increase peak data rate and reduce latency, and active devices can support it for TDL-A fading or LOS channel conditions, with the decision up to reader scheduling.
- Source [251] states that, from the energy consumption perspective, $M_{\text{chip}}^{\text{symb}}$ = 32 can enable short transmission duration to avoid interruption.

Sources [243], [246], [252], [249], [253], [254], [257], [265] and [247] state that it is not necessary to add $M_{\text{chip}}^{\text{symb}}$ for R2D.

- Sources [243], [246], [253], [249], [254], [265], [257], and [252] state concern on feasibility of $M_{\text{chip}}^{\text{symb}}$= 32 due to too short chip duration.
  - Source [246] provide evaluation results show that, $M_{\text{chip}}^{\text{symb}}$= 32 cannot work for outdoor scenarios with BLER close to 1 under different SNRs, as its chip length is too short to resist large multi-path delay and channel fading, so addition of $M_{\text{chip}}^{\text{symb}}$= 32 is not supported.
  - Source [249] provide evaluation results show that, for Device 2b, $M_{\text{chip}}^{\text{symb}}$= 32 cannot satisfy the 500m coverage requirement at 10% BLER, and for Device C, though it can reach 10% BLER, it has an error floor above 1% BLER.
  - Source [243] provide evaluation results show that R2D transmission with $M_{\text{chip}}^{\text{symb}}$= 32 cannot achieve 1% BLER even with 3 block-level repetitions, and under similar data rates, the performance of $M_{\text{chip}}^{\text{symb}}$= 32 (with 3 repetitions) is worse than that of $M_{\text{chip}}^{\text{symb}}$= 12 (without repetition). Source [281] further states that M=32 can not be distinguished from $M_{\text{chip}}^{\text{symb}}$= 24 at A-IoT device. According to RAN4 requirements, when the nominal chip length is Tc, the allowed chip length can be 0.7*Tc to 1.3*Tc, which makes A-IoT device difficult to differentiate $M_{\text{chip}}^{\text{symb}}$ = 32 from $M_{\text{chip}}^{\text{symb}}$ = 24.

- Source [257] provide evaluation results show that, when $M_{\text{chip}}^{\text{symb}}$ = 32, both Device 2b and Device C cannot achieve 1% BLER and have an error floor around CNR=10 dB and larger due to smaller chip duration.
- Source [247] provide evaluation results show that, for Device 2b and Device C, a 10% BLER cannot be achieved with $M_{\text{chip}}^{\text{symb}}$= 32. Source [279] provides evaluation results show that, there is an error floor at BLER of 10^{-1} for $M_{\text{chip}}^{\text{symb}}$ = 32. Source [297] provides evaluation results show that that BLER for $M_{\text{chip}}^{\text{symb}}$ = 32 is close to 1 in outdoor scenarios across SNRs and failing to meet the 10% BLER target even with FEC and block-level repetitions.

- Sources [252] and [253] state that, higher data rate with $M_{\text{chip}}^{\text{symb}}$= 32 lacks motivation.
- Source [265] states that, it would increase device complexity as devices need to blindly detect and differentiate $M_{\text{chip}}^{\text{symb}}$= 24 and $M_{\text{chip}}^{\text{symb}}$= 32 with certain FDR.

6A.1.3 R2D generation

6A.1.3.1 Channel coding

6A.1.3.1.1 Coding schemes

FEC for R2D is studied, with code rates of 1/2, 1/3 and 1/4 and the following candidate coding schemes:

Coding scheme 1: LTE tail biting convolutional code

Coding scheme 2: Tailed convolutional codes with the same polynomials as LTE tail biting convolutional codes

Coding scheme 3: Reed-Muller codes

For comparison between Coding scheme 1 and Coding scheme 2:

- Source [241] states, for the same number of information bits N, a tailed convolutional code encodes L-1 additional bits (tail bits) to ensure the final state returns to 0. With tail bits, a typical decoder can operate on N+L-1 bits. With TBCC, no additional bits are needed but a typical decoder operates on at least 2N bits. For moderate and larger number of information bits, tailed convolutional codes and TBCC have the same performance. For smaller number of information bits, with additional decoding iterations, the performance of TBCC can approach the performance of tailed convolutional codes.
- Source [243] states, compared with LTE TBCC, tailed convolutional code can reduce decoding complexity while increase the overhead.
- Source [246] states tailed convolutional codes with the same polynomial as LTE TBCC has a performance similar to LTE TBCC but suffers rate loss with zero bits padded at the end of the input bits.
- Source [266] states, the decoding for Tailed convolutional code maybe simpler, but the tail bits need to be sent, and the actual effective code rate slightly decreases.
- Source [277] states, TBCC is prioritized over tailed CC, provided that the decoding complexity is acceptable for device 2b/C.
- Source [280] states, tailed CC results in a code rate loss, making the code rate less than 1/n, where n is the number of polynomials, TBCC encoding does not require additional complexity or registers compared with tailed CC. And, TBCC encoding has larger code rate compared with Zero-tail CC.
- Source [285] states, tailed convolutional codes suffer from reduced spectral efficiency due to zero padding.

For comparison between Coding scheme 1 and Coding scheme 3:

- Source [243] states RM code suffers performance degradation and larger complexity at least for larger transport block size, without power reduction benefit
- Source [245] states, RM code is deprioritized due to the restricted operational range

- Source [246] states RM codes have a 4-5 times higher power consumption than convolutional codes due to the complicated fast Hadamard transform calculation for the FEC decoder, and has a 3-11 bit TBS restriction in the NR baseline design.
- Source [266] states the combinations of code length and code rate for RM code are limited, which is less flexible than convolutional codes. And the evaluation for the performance of the RM code is needed under low signal-to-noise ratio conditions.
- Source [264] states RM codes performs worse than TBCC. RM with k=8 or 10, n=32 has lower decoding complexity, memory cost, hardware area cost and lower power consumption than Viterbi decoding for TBCC with 1/2 or 1/3 coding rate.
- Source [265] states decoding complexity of RM code is expected to be lower than Viterbi decoding for TBCC. RM is not supported for payload size larger than 11.
- Source [280] states the RM codes used for PUCCH in 5G can not satisfy the requirement of A-IoT.
- Source [279] states, TBCC is corresponding to high decoding complexity at device side than RM.
- Source [285] states that RM codes provide limited coding gain for long data lengths and incur exponentially increasing decoding complexity as the input length increases.

If FEC is supported for R2D:

- Sources [243], [245], [246], [248], [249], [250], [255], [256], [259], [266], [269], [252], [257], [264], [277], [280], [285] and [296] report LTE TBCC (Coding scheme 1) should be supported. Source [271] report LTE TBCC (Coding scheme 1) can be supported when the number of information bits is small.
- Source[241], [255] reports tailed CC with same polynomial as LTE TBCC (Coding scheme 2) can be supported. Source [271] report tailed CC with same polynomial as LTE TBCC (Coding scheme 2) can be supported when the number of information bits is large.
- Source [264] reports Reed-Muller coding (Coding scheme 3) can be supported. Source [249] reports Reed-Muller coding can be supported for L1 control, if L1 control is no larger than 11 bits, otherwise, TBCC is supported.

6A.1.3.1.2 Code rate

For Code rate R=1/3:

- Sources [243], [244], [245], [246], [248], [256], [257], [259], [254], [264] and [266] state R=1/3 can be supported.
  - Source [245] states that, 1/3 code rate is supported when interleaving is not applied.
  - Source [246] states that, 1/3 code rate is necessary and feasible, and should be supported due to its performance gain and an increased power consumption for active devices.
  - Source [256] states that, it is beneficial to reuse the LTE TBCC with a preferred code rate of 1/3 to ensure reliable transmission and reduce specification complexity
  - Source [257] states that, for Device C, LTE TBCC with a coding rate of at least 1/3 could be beneficial to support for better coverage.
  - Source [264] states that, LTE TBCC with a coding rate of 1/3 could achieve better performance than 3 repetitions.
  - Source [276] states that, the coding rate of TBCC should be selected between R=1/2 and R=1/3 based on the R2D coverage target. The BLER performance of TBCC with a code rate of 1/3 is significantly better than that of TBCC with a code rate of 1/2, with a gain ranging from 1.14 to 2.32 dB at a BLER of 10^{-1}.
  - Source [285] states that, if FEC is supported for R2D, R=1/3 can be supported because LTE TBCC with code rate 1/3 is a well-established coding scheme and is already used in Rel-19 D2R.
  - Source [279] states that additional code rates can be realized by LTE bit collection scheme based on 1/3 TBCC, and the puncturing and repetition ways are not flexible for realizing multiple values of cade rates.

For code rate R=1/2, achieved by puncturing 3rd branch

- For Code rate R=1/2, sources [243], [244], [248], [249] and [259] report that R=1/2 can be supported.
  - Source [243] reports, 1/2 code rate can improve spectrum efficiency and reduce power consumption for decoding.
  - Source [249] states that ,1/2 coding rate is preferable for TBCC, as it reduces device power consumption while achieving the same spectral efficiency as the Manchester coding in Rel-19 R2D. [276] states that, without LTE interleaving, TBCC codes with coding rate of 1/2 have more than 5dB coding gain over the Manchester code in TDL-C channel with 300ns delay spread, for OOK-4 with M=2, 6, 12 at BLER=10-1.
  - Sources [259] and [248] state that, 1/2 and 1/3 coding rate as baseline.
  - Sources [241] state that, the performance difference between R=1/2 and R=1/3 is about 2 dB.
- Source [246] states that, the necessity of supporting other code rates apart from 1/3 for TBCC, which was supported in D2R for Rel-19, is not clear, because a 1/2 code rate performs worse than a 1/3 code rate.
- Source [285] states that, 1/2 coding rate should not be supported because its coding gain is lower than 1/3 and it may even degrade R2D coverage, while its benefit is unclear.

For code rate R=1/4, source [265] reports that Code rate of R=1/4 can be achieved by repetition and puncturing based on LTE TBCC with 1/3 code rate.

For code rate R=1/4, obtained by adding a new polynomial for the fourth branch

- Sources [264], [265] and [259] report that, Code rate R=1/4 obtained by adding a new polynomial for the fourth branch, can be supported.
  - Source [259] states that, 1/4 coding rate can be considered for extreme coverage scenarios.
  - Source [264] states that, 1/4 coding rate can further improve the performance than that of 1/3 coding rate. The extra polynomial [133 171 165 137], nested to the existing LTE CC (K=7 and R=1/3), is used
  - Source [241] states that, the performance difference between R=1/3 and R=1/4 is about 1.25 dB.
- Source [243] states that, 1/4 mother code rate increases power consumption and larger buffer for decoding, and it requires large standard effort to determine new polynomial for the 4th branch. The coverage enhancement can also be achieved by 1/3 mother code rate with block-level repetition. [281] states that, 1/2 code rate with 2 repetitions achieves similar performance of 1/4 mother code rate or even better performance than 1/4 code rate by repetition of bits from virtual circular buffer based on mother code rate 1/3.
- Source [244] states that, 1/4 code rate should be deprioritized since it increases complexity.
- Source [276] states that, due to its higher decoding complexity and negligible coding gain, 1/4 coding rate should not be considered.
- Source [285] states that, 1/4 coding rate derived by repeating TBCC branch outputs should not be supported, because its benefit largely overlaps with R2D block-level repetition. R=1/4 achieved by adding a new coding polynomial should not be supported, because it would impose additional implementation burden at the gNB and require extra specification effort
- Source [283] states that if code rate 1/4 is adopted in both R2D and D2R, same generation method should be applied.

Source [241] provide views for above code rates using tailed convolutional code with the same polynomial as LTE TBCC; while other sources provide views for above code rates using LTE TBCC.

6A.1.3.1.3 Power consumption, complexity, etc.

For LTE TBCC Sources [243], [246], [264], [244], [249], [254] and [259] report that the power consumption for TBCC decoding, as follows:

- Source [243] states power consumption is tens of uW to few hundreds of uW

- Sources [246] and [254] state power consumption is less than 200uW.
- Source [244] states power consumption can be less than 100uW.
- Source [249] states power consumption is about 100uW
- Source [264] states power consumption is tens of percent higher than a few mWs or less, dependent on the clock frequency, with constraint length K=7.
- Source [259] states power consumption is tens of uW.

Sources [241] and [264] also provide complexity and memory cost analysis for LTE TBCC:

- Source [241] states that, the decoding complexity depends on number of bits, constraint length K and number of iterations.
- Source [264] states that, the decoding complexity and memory depend on constraint length K and trackback depth D.

For tailed convolutional code with the same polynomial as LTE TBCC:

- Source [241] states the complexity is no larger than LTE TBCC.
- Source [246] states that its performance is similar to LTE TBCC but suffers rate loss with zero bits padded at the end of the input bits.

For RM:

- Sources [264], [265] and [251] state memory and power consumption for RM is lower than TBCC, and it is feasible.
  - Source [264] further state that, the decoding complexity, power consumption, memory cost and hardware area cost of RM code is lower than that of LTE TBCC, and suitable for ultra-low power A-IoT devices.
- Sources [246] and [255] state that, RM decoding based on fast Hadamard Transform incurs higher power consumption, compared with the short-constraint length convolutional codes. Source [255] states that, the power consumption for RM decoding may not meet power consumption limit of device 2b/C.

6A.1.3.1.4 Overall assessment

Sources [271], [272], [276], [278], [280], [281], [283], [286], [290], [293], [295], [296], [284], [292] and [294] report that it is feasible and necessary to support R2D FEC for outdoor scenario.

- Source [272] provides evaluation results shows that, it is necessary to support R2D enhancements including FEC to ensure R2D transmissions with high reliability, facilitate early termination, and usability for outdoor scenarios considering co-site deployment.
  - With M = 1 and no FEC, interleaving or repetitions, it can only achieve 124.4m - 267.6m for the target BLER 1% with the case of TX power 33dBm and 20 dB penetration loss, which is not sufficient for outdoor scenarios with co-site deployment.
  - Introducing FEC, specifically TBCC, provide ~4 dB performance gain for Rel-20 R2D transmissions, which is necessary for outdoor scenarios and beneficial for coverage enhancement.
- Source [281] provides evaluation results shows that R2D FEC is necessary to achieve desirable coverage, compared with cases without FEC and without R2D repetition.
  - For UMa scenario, to achieve 500-meter coverage, FEC is necessary for the following cases:
    - The coverage is 591.34m with 1/3 FEC, while coverage is 373.46m without FEC, for {96bit payload size, 43dBm Tx power}.
    - The coverage is 705.67m with 1/3 FEC, while coverage is 276.53m without FEC, for {400bit payload size, 43dBm Tx power}.

    - The coverage is 525.6m with 1/3 FEC, while coverage is 205.97m without FEC, for {400bit payload size, 38dBm Tx power}.

  - For UMa scenario, to achieve 334-meter coverage, FEC is necessary for the following cases:

    - The coverage is 440.45m with 1/3 FEC, while coverage is 278.17m without FEC, for {96bit payload size, 38dBm Tx power}

    - The coverage is 391.49m with 1/3 FEC, while coverage is 153.41m without FEC, for {400bit payload size, 33dBm Tx power}

  - For RMa scenario, to achieve 500-meter coverage, FEC is necessary for the following cases:

    - The coverage is 984.03m with 1/3 FEC, while coverage is 472.75m without FEC, for {400bit payload size, 43dBm Tx power}.

    - The coverage is 652.24m with 1/3 FEC, while coverage is 475.57m without FEC, for {96bit payload size, 38dBm Tx power}.

    - The coverage is 730.44m with 1/3 FEC, while coverage is 350.92m without FEC, for {400bit payload size, 38dBm Tx power}.

    - The coverage is 542.2m with 1/3 FEC, while coverage is 260.49m without FEC, for {400bit payload size, 33dBm Tx power}.

- Source [281] further provides evaluation results shows that R2D FEC is necessary to achieve desirable coverage, compared with cases with R2D repetition and without FEC, under the same data rate.

  - For UMa scenario, to achieve 500-meter coverage, both repetition and FEC are necessary for the following cases:

    - The coverage is 560.8m with 2 repetitions and 1/3 FEC, while coverage is 489.73m with 6 repetitions only for {96bit payload size, 38dBm Tx power}

    - The coverage is 538.14m with 2 repetitions and 1/3FEC while coverage is 445.67m with 6 repetitions only for {400bit payload size, 33dBm Tx power}

  - For UMa scenario, to achieve 334-meter coverage, both repetition and FEC are necessary for the following cases:

    - The coverage is 347.97m with 2 repetitions and 1/2 FEC while coverage is 307.47m with 4 repetitions only for {96bit payload size, 33dBm Tx power}

  - For RMa scenario, to achieve 1154-meter coverage, both repetition and FEC are necessary for the following cases:

    - The coverage is 1219.52m with 2 repetitions and 1/2FEC while coverage is 1063.3m with 4 repetitions only for {400bit payload size, 43dBm Tx power}

  - For RMa scenario, to achieve 500-meter coverage, both repetition and FEC are necessary for the following cases:

    - The coverage is 561.94m with 2 repetitions and 1/2FEC while coverage is 487.05m with 4 repetitions only for {96bit payload size, 33dBm Tx power}

- Source [276] provides evaluation results shows that, without LTE interleaving, TBCC codes with coding rate of 1/2 have more than 5dB coding gain over the Manchester code in TDL-C channel with 300ns delay spread, for OOK-4 with M=2, 6, 12 at BLER=10^-1, under the same data rate.

- Source [271] states that, R2D FEC coding gains from convolutional codes far exceed the gains from repetition, and it is necessary to support any channel coding on the R2D link as errors due to long distance propagation, fading, and low SNR.

- Source [278] states that, to ensure the coverage targets are met under practical conditions and to support larger payloads, the adoption of FEC and repetition becomes beneficial and necessary.

- Source [292] states that, FEC offers additional coding gain and generally outperforms repetition at equivalent data rates, and FEC should be prioritized compared to repetition.
- Source [286] states that, FEC based on LTE TBCC is essential for achieving outdoor coverage targets for device 2b/C given the transmission power constraints.
- Source [293] states that, it is necessary to use FEC, interleaver to improve the interference robustness.
- Source [274] states that FEC should only be considered if repetition proves insufficient.

Source [277] states that, it may be beneficial for extending R2D coverage, but the necessity depends also on how R2D coverage compares with D2R coverage.

Source [273], [275], [279] and [297] state that R2D FEC is not necessary, and the distance target can be fulfilled without FEC.

Source [285] states that, R2D FEC requires decoding at the device side, which may significantly increase device complexity and energy consumption, while it provides limited benefit considering R2D is not bottleneck.

6A.1.3.2 Interleaving and bit collection

Interleaving for PRDCH with FEC and the use of the LTE bit collection scheme with the following candidate methods are studied:

Alt 1: Both the LTE sub-block interleaver and the LTE bit collection scheme are used.

- Sources [241], [246], [252], [249], [256] and [257] report LTE sub-block interleaver and LTE bit collection scheme can be supported, if FEC is supported.
  - Source [241] states that, reusing both LTE sub-block interleaver and LTE bit collection scheme enables a device to eliminate the repetition block and associated memory as the bit collection scheme can use the memory for the sub-block interleaver for repeated bits. Another benefit is the various coding rates are achievable without modifying the encoder specification.
  - Source [246] states that, reusing the LTE sub-block interleaver and LTE bit collection scheme should be supported due to its performance gain especially for larger TBS transmissions, providing 1dB and 2.8 dB performance gain compared with Alt 2 for BLER 10% and BLER 1% respectively, for a TBS of 400 bits.
  - Source [256] states that, Alt 1 and Alt 2 are beneficial as they reuse the LTE bit collection scheme and/or the LTE sub-block interleaver, enabling R2D coverage enhancement with minimal specification effort
  - Source [257] states that, LTE sub-block interleaver could be beneficial for R2D to reduce the burst error caused by fading, especially for the longer message size with smaller values of M
  - Source [264] states that, Alt 1 and Alt 2 can achieve similar BLER performance, and both require buffering full-length coded bits.
  - Source [276] states that, R2D interleaving performance is correlated with both M (for OOK-4) and K (information bit length) values; the smaller the M and the larger the K, the better the interleaving performance. Alt 1 significantly outperforms Alt 2 and Alt 3, especially for smaller M values and larger information bit lengths.
- Sources [243], [246] and [265] state it is feasible to support LTE subblock interleaving and LTE bit collection.
  - Source [243] report with references to show that the power consumption for LTE de-interleaving for turbo decoding is tens of uW, even with multiple parallel (de)interleaving operation, and each operation with max number of bits of 6144. It can be expected that the power consumption for TBCC with LTE bit collection and with or without LTE sub-block interleaving can be even simpler. [281] also report that, LTE bit-collection w/ or w/o sub-block interleaving can be implemented with low power consumption (~10 uW), low latency (a few micro seconds), and limited complexity (area of hundreds μm2). Hence, both LTE TBCC with bit collection with or without LTE sub-block interleaving are feasible.
  - Source [246] and [265] report that the additional power consumption can be supported by active devices due to higher peak power consumption, allowing them to support larger memory. Source [246] further

report that larger memory would enable these devices to support the use of a LTE de-interleaver, and can follow the same hardware as used for the LTE sub-block interleaver with LTE bit collection scheme in D2R.

Alt 2: Only the LTE bit collection scheme is used.

- Sources [243], [248], [254], [256], [259] and [264] report LTE bit collection scheme without LTE sub-block interleaver can be supported, if FEC is supported.
  - Source [243] states that, Alt 1 and Alt 2 presents similar performance; thus Alt 2 is sufficient. Source [281] report that, LTE bit-collection w/o sub-block interleaving can be implemented with low power consumption (~10 uW), low latency (a few micro seconds), and limited complexity (area of hundreds μm2). Hence, LTE TBCC without LTE sub-block interleaving is feasible.
  - Source [254] states that, since bit collection does not require interleaving, the complexity of Alt 2 is lower than interleaving. FEC with bit collection and FEC with sub-block interleaver has similar performance compared with FEC with bit collection.
  - Source [256] states that, Alt 1 and Alt 2 are beneficial as they reuse the LTE bit collection scheme and/or the LTE sub-block interleaver, enabling R2D coverage enhancement with minimal specification effort
  - Source [259] states that, Alt 2 provides a good balance between performance and complexity for device 2b/C, reusing existing 3GPP designs while minimizing memory requirements.
  - Source [264] states that, for small number of coded bits, Alt2 can be used to improve interleaving gain

Alt 3: Only an updated bit collection scheme based on the LTE bit collection scheme is used.

- Source [264] and [248] report that, LTE bit collection scheme with segment can be supported, if FEC is supported.
  - Source [248] states that, Alt 3 can be adopted if memory size limitation exists in devices
  - Source [264] states that, Alt3 with two segmentations requires half of the buffering size versus Alt2 at price of 2.5dB performance loss, but still outperforms the no-FEC approach using three block-level repetitions. For large number of coded bits, Alt3 can be applied to reduce buffering requirements
- Source [243], [246] and [276] state that, Alt 3 degrades performance due to lower time diversity gain compared with Alt 1/2, and [243] observes that, Alt 3 with two segments for 1000 bits TBS also suffers loss compared with block-level repetition.
- Sources [243] and [246] state that, since memory/power consumption by Alt 1 and Alt 2 is feasible for AIoT device, it is unnecessary to further reduce memory/power consumption by Alt 3.
- Source [256] states that, Alt 3 is potentially advantageous in terms of memory reduction for A-IoT devices, but is considered less favourable due to the extensive discussion required to evaluate the performance of various bit collection schemes

Sources [243], [246], [264], [265] and [254] state that it is beneficial to support interleaving to improve coverage by obtaining time diversity gain.

Source [272] states that it is necessary to support the sub-block interleaver and bit collection scheme to overcome the significant performance loss for large TBS transmissions due to the lack of time diversity gain. With M = 1 and only 1/3 TBCC, it can achieve 160.3m - 319.3m for the target BLER 1% with the case of TX power 33dBm and 20 dB penetration loss. While with M = 1, 1/3 TBCC, sub-block interleaver and LTE bit collection, it can achieve 334.7m - 431.3m for the target BLER 1% with the case of TX power 33dBm and 20 dB penetration loss, which shows significant gain for larger TBS transmissions.

6A.1.3.3 Repetition

6A.1.3.3.1 Types

Block-level repetition as defined in Clause 6.1.0.1 is considered for the study; other types of repetition are not considered.

6A.1.3.3.2 Feasibility

It is reported by sources [241], [243], [245], [246], [248], [249], [250], [252], [255], [256], [259], [264], [268], [269], [257], [263], [265], [267], [285] and [280] that R2D repetition is feasible and can be considered.

Sources [243], [246], [245], [252], [249], [255], [259], [267], [285] and [280] report that repetition schemes including block-level repetition are feasible.

Source [275] and [274] state that, block-level repetition requires a large volatile memory to store received repetitions of a block, Source [274] further states that block-level repetition should be feasible for small TBS. whether block-level repetition(s) can be applied to large TBS e.g., thousands of bits, should be further studied.

6A.1.3.3.3 Overall assessment

Sources [271], [272], [278], [280], [274], [281], [283], [284], [286], [293], [290], [295], [294], [296], and [297] report that it is necessary or beneficial to support repetition for outdoor scenario.

- Source [272] provides evaluation results shows that, it is necessary to support R2D enhancements including block repetitions to ensure R2D transmissions with high reliability, facilitate early termination, and usability for outdoor scenarios considering co-site deployment.
  - With M = 1, 1/3 TBCC, sub-block interleaver and LTE bit collection, it can achieve 334.7m - 431.3m for the target BLER 1% with the case of TX power 33dBm and 20 dB penetration loss, which shows significant gain for larger TBS transmissions, but still requires further enhancement for outdoor co-site deployment.
  - With M = 1, 1/3 TBCC and LTE bit collection & sub-block interleaver, block repetition number 2, it can achieve 439.0m - 589.3m for the target BLER 1% with the case of TX power 33dBm and 20 dB penetration loss, which can approximately satisfy the outdoor co-site deployment requirement.
- Source [281] provides evaluation results shows that R2D Block level repetition is necessary to achieve desirable coverage
  - For UMa scenario, to achieve 500-meter coverage, repetition is necessary for the following cases:
    - The coverage is 513.36m with 3 repetitions, while coverage is 373.46m without repetition, for {96bit payload size, 43dBm Tx power}
    - The coverage is 616.24m with 3 repetitions, while coverage is 276.53m without repetition, for {400bit payload size, 43dBm Tx power}
  - For UMa scenario, to achieve 334-meter coverage, repetition is necessary for the following cases:
    - The coverage is 382.37m with 3 repetitions, while coverage is 278.17m without repetition, for {96bit payload size, 38dBm Tx power}
    - The coverage is 458.99m with 3 repetitions, while coverage is 205.97m without repetition, for {400bit payload size, 38dBm Tx power}
    - The coverage is 341.87m with 3 repetitions, while coverage is 153.41m without repetition, for {400bit payload size, 33dBm Tx power}
  - For RMa scenario, to achieve 1154-meter coverage, repetition is necessary for the following cases:
    - The coverage is 1205.07m with 6 repetitions, while coverage is 472.75m without repetition, for {400bit payload size, 43dBm Tx power}
  - For RMa scenario, to achieve 500-meter coverage, repetition is necessary for the following cases:
    - The coverage is 934.3m with 3 repetitions, while coverage is 472.75m without repetition, for {400bit payload size, 43dBm Tx power}
    - The coverage is 595.03m with 3 repetitions, while coverage is 475.57m without repetition, for {96bit payload size, 38dBm Tx power}
    - The coverage is 693.53m with 3 repetitions, while coverage is 350.92m without repetition, for {400bit payload size, 38dBm Tx power}

  - The coverage is 514.81m with 3 repetitions, while coverage is 260.49m without repetition, for {400bit payload size, 33dBm Tx power}

- Source [280] provide evaluation results shows that 2 times repetition provides 2-3 dB performance gain @BLER = 0.1 compared with baseline. 3 times repetition provides 2-4 dB performance gain @BLER = 0.1 compared with baseline.

  - The candidate values of the repetition number can be [1, 2, 4, 8] if FEC is supported for PRDCH, and [1, 2, 4, 6, 8, 12, 16, 32] is recommended if FEC is not supported for PRDCH.

- Source [278] provide evaluation results shows that, with 1/3 CC but without repetition, coverage of Device 2b degrades significantly when a practical 20dB penetration loss is considered, resulting in coverage distances of 712m (UMa) and 1231m (RMa), which do not provide a sufficient reliability margin.

- Source [271] states that, for the R2D link, block repetition will complement convolutional coding.

- Source [274] states that Block-level repetition(s) offers a straightforward mechanism to improve coverage without introducing substantial processing overhead, and it should be considered for R2D transmission at least for small TBS.

- Source [284] states Block level repetition would be applied for data of any size although more benefit for small TBS.

- Source [286] states that Block-level repetition provides additional coverage enhancement beyond FEC alone through time diversity and energy accumulation at the receiver.

- Source [293] states that, it is necessary to use block-level repetitions to improve the interference robustness.

- Source [295] states that, R2D Block level repetition can be considered together with FEC encoding to further increase the performance.

- Source [296] states that block level repetitions offers gains via non-coherent combining at the reader and provides time/frequency diversity while keeping device complexity and occupancy low, and low (e.g., <=4) number of repetitions can be supported.

- Source [297] States that Block-level repetition provides the highest diversity gain for R2D by enabling combining of multiple repetitions, and remains feasible for Device 2b/C despite increased memory and latency requirements.

Source [277] states that, it may be beneficial for extending R2D coverage, but the necessity depends also on how R2D coverage compares with D2R coverage.

Source [285] states that, the necessity of R2D repetition depends on whether evaluation results can satisfy the coverage requirements.

Source [273], [275] and [279] state that R2D Block level repetition is not necessary, and the distance target can be fulfilled without repetition.

6A.1.3.4 Line coding

The Manchester line code of Clause 6.1.1.2 is retained.

6A.1.3.5 Scrambling

Necessity and feasibility of a scrambling step during PRDCH was studied.

The following positive observations were reported:

- Source [252], [265], [266], [253], [264] and [297] state that, scrambling should be applied to R2D transmissions for mitigating interference among R2D transmissions from different readers.

- Source [264] states that, it is feasible and necessary to apply PRDCH scrambling for interference randomization and inter-cell interference reduction, especially for dense deployments with frequency reuse. Scrambling sequences can be initialized using parameters like reader ID, device ID or partial ID for PRDCH, which can be

implemented efficiently even in constrained devices. The scrambling can be applied to the bits after coding and repetitions.

- Source [247] states that, R2D scrambling is necessary to reduce inter-cell interference and ensure statistical uniformity, preventing long runs of identical bits and improving clock recovery. R2D scrambling is feasible, requiring only a small LFSR and one XOR per bit , leading to negligible hardware complexity.

The following negative observations were reported:

- Source [243] states that, it is unnecessary to support scrambling for PRDCH, as interference randomization is limited for OOK modulation with Manchester coding.
- Source [243] states that, continuous 1s or 0s cannot be effectively reduced by scrambling, while Manchester coding can guarantee no larger than 2 continuous 1s or 0s and simplify gNB transmission power adjustment. Source [281] states that Manchester coding already guarantees no larger than 2 continuous 1s or 0s.
- Source [246] states that, R2D scrambling should not be supported since it cannot work for intra-frequency interference handling as the receiver cannot detect the signals received from multiple readers of the same frequency locations.
- Source [249] and [251] state that, for PRDCH, scrambling should not be supported to avoid increasing A-IoT complexity and power consumption.
- Source [279] states that, the scrambling for PRDCH is unnecessary for interference randomization and continuous 1s or 0s avoidance, and should be deprioritized.

6A.1.3.6 Order and inclusion of generation steps

Sources [243], [246], [248], [249], [254], [241], [259], [264], [284], [297] and [290] report that channel coding before block-level repetition should be supported, as shown in Figure 6A.1.3.6-1, with following justifications:

- Sources [243], [248], [249] and [264] state that less memory is required by the channel coding before block-level repetition, compared with channel coding after block-level repetition.
- Source [246] states that the order of the channel coding first, followed by block-level repetition should be supported because of the diversity gain achieved due to the increased distance of a given parity bit in the output stream when compared to the reverse order.
- Source [249] states that block-level repetition before FEC scheme requires Rblock times more interleaver memory as well as higher implementation complexity and processing delay for both devices and reader.
- Source [284] and [297] state, block-level repetition after channel coding enables time diversity and the soft-combining gain.

Sources [252] and [256] state that channel coding after block-level repetition should be supported.

It is feasible to support R2D FEC and block-level repetition for Rel-20 A-IoT. It is necessary to support R2D FEC and/or block level repetitions for A-IoT device in R2D to achieve maximum coverage (e.g., 200 m with penetration loss for Device 2b, 500 m with penetration loss for Device C) as observed in Clause 6A.1.3.1.4, and it can be enabled under certain condition. This does not preclude reader to disable FEC and/or block repetition.

For R2D, channel coding (with interleaving, if supported) is before block-level repetition (if supported). Figure 6A.1.3.6-1 shows the overall generation of PRDCH, and subsequent clauses report the study of certain blocks.

NOTE: Interleaving includes potential sub-block interleaving (if supported) and bit collection.

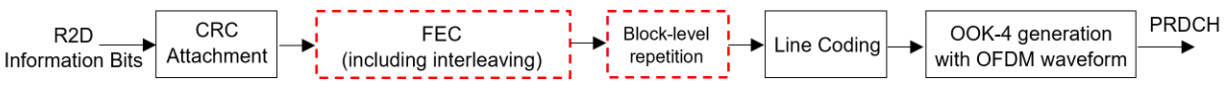


Figure 6A.1.3.6-1: Generation of PRDCH. Dashed boxes indicate location if the step is supported.

6A.1.4 R2D timing

6A.1.4.1 Start of R2D

For the functionality of indicating the start of the R2D transmission containing PRDCH, the following options are studied.

Option 1: A binary sequence-based SIP, where the candidate sequence types are as follows:

Option 1-1: m-sequence

- Sources [243], [252], [259], [260], [264], [256], [279] and [283] report that m-sequence can be considered as binary sequence for R2D SIP, with the following observations:
  - Source [243] states that m-sequences with same length but different cyclic shifts/initial states can be considered to achieve good cross-correlation between different sequences.
  - Sources [260], [256] and [264] state that m-sequence design can be expected to be implemented with lower complexity than a Golay sequence. Source [264] further state that m-sequences can be generated by storing the initialization state of the polynomial, reducing memory requirements.

Option 1-2: Golay sequence

- Sources [246], [264], [265], [256], and [283] report that Golay sequence can be considered as binary sequence for R2D SIP, with the following observations:
  - Source [246] states that Golay sequence achieves lower operational and correlation complexity compared with m-sequence.
  - Source [265] states that Golay sequence with flexible sequence length may be beneficial for alignment with the OFDM symbol boundary.
  - Sources [264] and [256] states that Golay-sequence offers slightly better cross-correlation than m-sequence, and source [264] further states cross-correlation depends on the choice of Golay pairs, and Golay sequences may be programmed into devices.

- Sources [264] and [260] provide evaluation results showing that m-sequence and Golay sequence shows similar MDR/FAR and correlation property. [265] states that both m-sequence and Golay sequence have good auto-correlation property. Source [280] report that Different sequences with same length have similar performance @BLER = 0.01.
- Sources [241], [243], [244], [246], [251], [257], [258], [264]and [282] report that multiple sequences can be used for a given length for R2D SIP.
  - Sources [243], [257], [258], [291] and [282] report the purpose of multiple sequences is for reader differentiation, Sources [243] and [257] suggest to consider 3 sequences.
  - Source [246] report that different length sequences are mapped to different coverage distances, in order to maximize the performance for each of the distances for outdoor scenarios while maintaining MDR and FAR to be ≤ 1%. Source [291] state that multiple sequences allow for different required coverage range.
  - Sources [251], [258] report the purpose of multiple sequences is to indicate parameters of upcoming R2D transmission. [251], [258]report that presence of CFO calibration signal can be indicated by SIP sequences, and [258] also report that inclusion or exclusion of L1 control information, synchronization signal, or PRDCH, can be indicated by SIP sequences.
- Source [248] and [280] report that single length and single sequence is beneficial for lower complexity and power consumption.
- Source [280] report that due to the use of OOK modulation in the downlink, multiple sequences for SIP will not reduce inter cell interference.
- Source [265] report that device complexity on correlation-based detection for multiple sequences should be considered.

Regarding whether to apply Manchester coding to the binary sequence-based SIP, sources [243], [250], [264] and [270] report that Manchester coding can be applied to binary sequence for R2D SIP

- Source [243] and [264] state that sequence Manchester coding achieves similar MDR and timing performance.
  - Source [243] provide evaluation results show that for 31-bit m-sequence with Manchester coding, 31-bit m- sequence without Manchester coding, and 63-bit without Manchester coding, these three cases have comparable durations (~15 OFDM symbols), and shows similar MDR performance.
  - Source [264] provide evaluation results show that the 15-bit m-sequence with Manchester coding and the 31-bit m sequence without Manchester coding have comparable durations (~7.5 OFDM symbol), and both show similar MD performance.
- Source [270] states that applying Manchester coding to the sequence is important in OOK modulation with low data rate where a long number of zeros and ones may result in the clock drift.
- Source [243] also states that timing performance for sequence with Manchester coding is slightly better compared with longer sequence w/o Manchester coding (same overall duration), due to narrower main lobe of auto-correlation can be achieved with Manchester coding.

Sources [272], [279], [282] and [280] report that Manchester coding is not applied to binary sequence for R2D SIP

- Source [272] state that Golay sequence with Manchester encoding with M = 2 yield ~1 dB performance loss when compared with same length Golay sequence without Manchester encoding with M = 1; and Golay sequence with Manchester encoding with M = 2 has very close residual timing error performance with the same length Golay sequence without Manchester encoding with M = 1.
- Source [272] state that removing Manchester encoding from the R2D SIP before performing correlation with the original sequence will increase the complexity of the device.
- [279] states that for a same sequence, applying Manchester coding to the sequence-based SIP will consume twice the time domain resources to without Manchester.
- [280] states that, if correlation-based detection is used for SIP, Manchester coding provides no benefit in decoding performance and limits the selection of sequences, although 8-length sequence with Manchester coding has similar performance as 16-length sequence without Manchester coding.
- Sources [241] and [282] state that the disadvantages of applying Manchester coding include increasing complexity and decreasing the number of sequences.

For sequence length for binary sequence based R2D SIP

- Source [243] report that:
  - At least 31-length m-sequence with M=2 and with Manchester coding (i.e., 31 OFDM symbols), is needed for R2D SIP, assuming 38 dBm Tx power and 20 dB penetration.
  - At least 127-length m-sequence with M=2 and with Manchester coding (i.e., 127 OFDM symbols), is needed for R2D SIP, assuming 33 dBm Tx power and 20 dB penetration.
- Source [246] report that:
  - 256, 64 and 16-length Golay sequence-based SIP can achieve an SNR performance of 4.0 dB, 8.2 dB and 12.0 dB, respectively, with the MDR ≤ 1%, FAR ≤ 1% and M=1 without Manchester coding.
- Source [259] report that:
  - Consider m-sequence based SIP with length 15 or 31 to improve detection performance while maintaining reasonable overhead
- Source [264] report that SIP duration with at least 7 OFDM symbols by using 15-length and M=2 without Manchester coding or 31-length and M=2 with Manchester coding may be needed to achieve required SINR for PRDCH.

- Source [248] report that single length and single sequence is beneficial for lower complexity and power consumption.

- Source [254] report that the Release 19 SIP sequence with a length of 8 can achieve a BLER of less than 10% at an SNR of 4 dB, and BLER of less than 1% at an SNR of 10 dB. Source [280] report that 16-length sequence has 2-4dB performance gain @BLER = 0.1 compared with 8-length sequence, and the maximum sequence length of R2D SIP is determined based on R20 coverage target.

- Source [279] report that the sequence type can be m-sequence and the sequence length can be 7.

Source [291] states that binary sequence-based SIP design allows for a lower complexity of the preamble design and lower processing and therefore lower power consumption at the device side

- Binary sequence-based SIP, m-sequence or Golay sequence has an advantage over release 19 SIP design as it can be designed such that it can also provide clock acquisition and is used for synchronization purposes, allowing for a one-part low-complexity preamble design.

Source [292] states that the sequence of SIP should be distinguishable from other R2D in case the sequence length is short.

Option 1 was found to be feasible and it can be used for timing synchronization/tracking.

- Source [285] states, due to CFO/SFO drift, sequence-based SIP may not always achieve ideal detection performance.

- Source [276] states for Option 1, the CP handling issue should be considered for the design of R-TAS SIP. The insertion of CP in the binary sequence will seriously affect the detection performance of binary sequence-based SIP if the CP is not removed before correlation-based SIP detector. When the device detects the binary sequence-based SIP, it does not know the location of CP and cannot remove it. For binary sequence-based SIP, it occupies more than a dozen or even dozens of OFDM symbols, it will seriously affect the detection performance of the binary sequence without CP removal and leading to a degradation in the detection performance of the start of the R2D transmission.

Option 2: A SIP as per TS 38.291 [228]

- Sources [276] provides evaluation results show that R-TAS SIP defined in Rel-19 A-IoT can satisfy the detection requirements of target MDR of 1% for the FAR up to 1% in outdoor scenarios

  - Reusing Rel-19 SIP will not introduce the CP handling issue. For binary sequence-based SIP, the insertion of CP in the binary sequence will seriously affect the detection performance of SIP.

- Source [273] states that, the Rel-19 SIP can be reused in Rel-20 without any enhancement. Sequence-based design will increase the complexity for device to perform coherent detection.

- Source [296] states that, it is reasonable to expect equal or better detection performance by re-using the existing SIP design, considering that Rel-20 Device 2b/C types will feature improved oscillator stability and processing capability

- Sources [243], [246], [264], [254], [279] and [281] provide evaluation results to show R19 SIP cannot be reliably detected.

  - Sources [243] and [281] show that R19 SIP does not outperform sequence-based SIP in MDR, even if R19 SIP uses sequence correlation detection instead of edge detection in R19. Source [281] show that R19 SIP cannot achieve coverage of 500 meters in UMa Scenario. MDR of Rel-19 SIP using edge detection method (with LNA, mixer, IF amplifier/filter, BB amplifier enabled, no power saving benefits) is not even close to meet coverage of 500 meters at 10% MDR. Even if sequence correlation is used for Rel-19 SIP detection, the coverage of 500meters cannot be met at the 10% MDR when BS Tx power is 33dBm.

  - Source [264] shows that, for Rel19 SIP using energy detection (Option 2), the required CNR at BLER=10% is about 8dB higher than PRDCH with 20bit+6bit CRC, M=2, TBCC 1/3 (e.g., for R2D L1 control), indicating that Rel19 SIP cannot be reused and enhancement is needed.

- Source [246] shows that if the same R-TAS design is used for a device C using correlation detection, the required SNR is 17.0 dB to achieve MDR ≤ 1% and FAR ≤ 1%, which is also not sufficient for the coverage enhancement for outdoor scenarios.

- Source [254] provides evaluation results shows that the Release 19 SIP sequence performance at BLER=1% cannot satisfy the coverage requirements, while the performance at BLER=10% can satisfy the coverage requirements.

- Source [279] states that Sequence-based SIP outperforms Rel-19 SIP with ~6 dB gain at BLER of 10% and ~3dB gain at BLER of 1%.

- Source [283], [286], [287] and [277] states that, binary sequence-based SIP is beneficial and necessary to provide better coverage performance for device 2b/C.

- Source [276] states that energy/edge detection-based Rel-19 SIP can significantly reduce the complexity and power consumptions of SIP detection, compared with sequence-based SIP. Source [285] states, sequence-based SIP may introduce increased complexity and higher power consumption compared to Rel-19 SIP.

- Source [281] states that using Rel-19 SIP detection at A-IoT device cannot provide power saving benefits.

  - In R19, the reason of low power consumption for SIP detection originated from the inherent low power consumption for device 1, in which there is no LNA, no mixer, no IF Amplifier, no IF-filter, and no BB amplifier in envelop detector, thus the SIP can be detection with low power consumption.

  - For R20 Active device, only (Z)IF-ED receiver is considered, in which LNA, mixer, IF amplifier, IF filters and BB amplifiers are enabled in detector of active device, and these components are most power consuming components in receiver. Device should enable all these power-hungry components to obtain the transition edge for Rel-19 SIP detection.

  - Complexity of sequence correlation in binary-sequence based detection is limited, and the multiplications in sequence correlation are simplified to add operations thanks to binary sequence.

6A.1.4.2 Chip duration determination

6A.1.4.2.1 Chip duration of data payload

The chip duration determination of the data payload in the PRDCH and the L1 control information are studied. The following options are studied for the functionality of the chip duration determination of the data payload in the PRDCH:

Option 1-data: The chip duration of the corresponding L1 R2D control information is used.

- Sources [242], [246], [247], [253], [256], [259] and [269] report that, the use of the same chip duration for L1 R2D control information and data payload in the PRDCH, have the following benefits:

  - Sources [242], [246], [253], [269] and [259] state that the use of the same chip duration simplifies device implementation, and [246] further states that the L1 R2D control information can avoid any decoding delay and avoid the need for a time gap between the L1 R2D control information and the data payload of the PRDCH.

  - Sources [247] states that, for broadcast PRDCH and paging PRDCH, the R2D parameters, including chip length for data payload of PRDCH, can be fixed for cater to the cell-edge scenarios.

  - Sources [253], [269] and [256] state that using the same M value for the L1 R2D control part and the data payload avoids additional signaling overhead, and [253], [269] further states that this ensures consistent coverage for both control and data, and no clear benefit to use different chip length for L1 control part and data part.

Option 2-data: The chip duration indicated by the corresponding L1 R2D control information is used.

- Sources [241], [243], [245], [247], [248], [250], [251], [254], [256], [257],[265], [259], [264], [268] and [276] report that, chip duration of the data payload in the PRDCH indicated by the corresponding L1 R2D control information, have the following benefits:

- Sources [241], [243], [245], [248], [250], [251], [256], [254], [257], [265] and [268] state that chip duration indication through L1 control information provides better flexibility to meet diverse reliability and data rate needs. [254] further states that it better coordinates with TB size, repetition rate, and code rate for improved link adaptation; [241] states that it allows tailoring data rates for specific devices and can improve efficiency;

- Source [243] state that with better coverage for L1 control part using longer chip length, unnecessary data reception can be avoided due to more robust L1 control;

- Sources [259] and [247] state that this option provides flexibility to adapt to different coverage conditions or message types, with [247] states this option provides benefits for PRDCH like Msg2 and Msg4.

Source [280] state that the chip duration of an R2D data payload without L1 control (if supported), is a fixed value.

6A.1.4.2.2 Chip duration of L1 R2D control information

The following options are identified for the functionality of the chip duration determination of the L1 R2D control information:

Option 1-control: A fixed chip duration is used, with no use of CAP.

Option 2-control: A set of pre-defined chip durations, blindly detected by a device with no use of CAP.

Option 3-control: A CAP is used, where multiple alternatives are studied:

Option 3-1-control: Using the pattern of CAP defined in TS 38.291 [228].

Option 3-2-control: Enhancing CAP by the following options:

Option 3-2a-control: Using the pattern of CAP defined in TS 38.291 [228] with repetitions,

Option 3-2b-control: Using a pattern that associates the chip duration of the clock acquisition part to different chip durations of the L1 R2D control information.

Option 4-control: A set of binary sequences with fixed or variable length is used, where multiple alternatives are studied:

Option 4-1-control: Use of the binary sequence-based SIP, with no use of CAP.

Option 4-2-control: Use of binary sequences for the CAP.

Option 5-control: Broadcast information is used to indicate the chip duration.

For Option 1-control, the following points were reported as advantages:

- Sources [247], [249], [256] and [250] states that fixed chip duration for L1 R2D control information lowers the complexity and power consumption of chip duration detection.

- Source [249] states that fixed chip duration for L1 R2D control information minimize the ambiguity of the chip duration detection, and thus enhance the success rate of L1 R2D control information reception.

- Source [247] states that using fixed chip duration (using the smallest M value for optimal coverage) can achieve higher reliability for L1 R2D control information.

- Source [280] states that the chip duration of L1 control part for broadcast information, is a fixed value.

For Option 1-control, the following points were reported as disadvantages:

- Source [264] and [253] states that to achieve the largest coverage, L1 control would need to use the lowest M, which increases overhead for each R2D transmission.

- Sources [241], [253], [254], [246], [243], [265], [245], [291] and [279] state that, if chip duration for L1 control information is fixed, the flexibility or transmission efficiency is restricted.

For Option 2-control, the following points were reported as advantages:

- Sources [245] and [257] state that blind detection within configured set provide better flexibility for chip length for L1 R2D control information.

- Source [245] states that blind detection is viable if the number of predefined chip duration candidates is restricted. Additionally, [245] states that Option 2 can be combined with device-specific configuration signaling to enhance flexibility without excessive complexity.

For Option 2-control, the following points were reported as disadvantages:

- Sources [241], [253], [243], [265], [247], [254], [283], [291] and [279] state that Option 2 increases detection complexity.

- Source [264] states that it may be impractical due to constraints on power consumption and memory needed for buffering the incoming R2D L1 control information samples.

- Sources [246] and [256] state that option 2 would require additional time gap between the L1 control information and the data payload of the PRDCH to allow for decoding delay.

- Sources [243] and [247] state that reliability is doubtful due to blind detection.

- Source [243] states that repetition and FEC parameters should be indicated anyway to achieve flexibilities in coverage and resource overhead, no need to leave chip duration for blind detection.

For Option 3-control, the following points were reported as advantages:

- For Option 3-1 Rel-19 CAP pattern, sources [253] and [269] state that Rel-19 CAP pattern to determine chip length of L1 R2D control part offer a balance between detection complexity and performance.

- For Option 3-2b, CAP pattern and the association between chip durations for CAP and chip durations of L1 R2D control information, sources [256], [259] and [264] report that this option can achieve better detection reliability compared with R19 CAP. Source [259] further states that this option maintains consistency with Release-19 design principles. Source [264] report that R19 CAP with repetition using correlation-based detection improves reliability, with low detection flexibility. Source [293] also provides simulation results that the CAP with 2bits and fixed length (fixed M=2 chip duration) can be detected with 10% false detection at an SNR lower than PRDCH with 20bit+6bit CRC, M=2, TBCC 1/3 (e.g., for R2D L1 control).

- Source [292] state that options 3 have an advantage as flexible chip duration of L1 R2D control.

For Option 3-control, the following points were reported as disadvantages:

- Source [264] states option 3 requires blind detection; and MD/FA rates and CAP overhead needs to be considered.

- Source [254] states option 3 results in additional overhead due to transmission of CAP.

- Sources [245], [243], [246] and [283] state that Option 3 is based on edge detection to determine the chip duration by using CAP, so the performance is limited or unfeasible due to the use of edge detection to detect the CAP pattern. Sources [254] and [247] also report the performance of this option should be further evaluated.

- Source [281] provide evaluation result show that it is not feasible to use R19 CAP (Alt 3-1) and R19 CAP with repetitions (Option 3-2a) to determine chip length for L1 control part. The BLER performance is not even close (more than 10dB performance gap) to meet coverage of 500 meters at 10% BLER in UMa scenario.

- Source [247] states that Option 3 would make the decoding of the L1 R2D control information more complicated for the device.

- Source [243] states that repetition and FEC parameters should be indicated anyway, no need to leave chip duration for blind detection using CAP.

- Source [291] state that using a CAP for chip length determination requires a two-part preamble design which is not necessary with a binary sequence based SIP design.

For Option 4-control, the following points were reported as advantages:

For Option 4-1, using binary sequences for SIP. No use of CAP for chip duration determination

- Sources [244], [246], [248], [256], [265], [257] and [279] state that this method can provide flexibility for R2D chip duration.

- Source [246] state that a restricted set of binary sequences of differing lengths are mapped to different chip durations and coverage distances, which can be detected with limited blind detection complexity. Parallel processing for the blind detection of different sequence lengths avoids and processing delay. Besides, R-TAS design is simplified, which consist only sequence-based SIP.

- Source [277] state that sequence-based SIP can be effectively used to indicate chip duration of a PRDCH for L1 control with limited indication flexibility to keep low device complexity, and only limited flexibility is needed, e.g., up to 2 sequence is sufficient.

- Source [291] state that single-part preamble where a sequence-based signal can not only provide start and synchronization functionality, but it can also be used to carry information with regard to chip length of the subsequent control and data payload, has an advantage over the other approaches in terms of low-complexity and power consumption.

For Option 4-2: Using binary sequences for CAP.

- Sources [244] and [257] report that flexibility on chip length of L1 R2D control can be achieved.

For Option 4-control, the following points were reported as disadvantages:

- Sources [243], [247], [264] and [292] state that option 4 increases detection complexity. And [247] states that option 4 would make the decoding of the L1 R2D control information more complicated for the device.

- Sources [254] and [247] report that the performance of this option should be further evaluated.

- Source [243] states that repetition and FEC parameters should be indicated anyway, no need to leave chip duration for blind detection; source [243] further reports that it leader more challenges on preamble design, requiring larger number of sequences to indicate four M values in addition to reader/cell ID differentiation;

For Option 5-control, the following points were reported as advantages:

- Sources [243], [254] and [264] state that better flexibility can be achieved without blind detection, and sources [243] and [254] further state that the chip duration for L1 can be indicated together with other parameters for control information, e.g., repetition and FEC related parameters. Source [245] state that this method can be used for L1 R2D control information common for devices, for example, the L1 R2D control information for scheduling paging message.

- Source [252] state that the chip length should be pre-known by device to minimize complexity and power consumption at AIoT device.

For Option 5-control, the following points were reported as disadvantages:

- Source [264] states that it requires system info acquisition and memorization.

- Source [246] and [253] states that only to indicate chip duration in broadcast information limits flexible time resource allocation from system perspective.

- Source [247] states that this option is a simple solution but not applicable for paging and broadcast PRDCH.

- Source [265] states that state that option 5 has less flexibility on chip duration of L1 R2D control compared to option 2/3/4.

- Source [279] states that, as a common indication method, option 5 is an efficient way for chip duration indication of the L1 R2D control information but loses some flexibility for the L1 R2D control information transmission towards different Devices.

The following options are not considered for normative specification:

- Option 2-control

- Option 3-2a-control.

6A.1.4.3 SFO calibration

It is reported by sources [241], [242], [243], [246], [250], [249], [253], [254], [266], [264], [265], [263] and [282] that SFO calibration is necessary for device 2b. And it is reported by sources [241], [242], [246], [249], [266], [264], [265], [263] and [282] that SFO calibration is necessary for device C.

The functionality of SFO calibration using the following options is studied:

Option 1: CAP defined in TS 38.291 [228] and/or Manchester encoding.

- Sources [249], [253], [259], [263], [266], [282] and [275] report that SFO is feasible and can be achieved by reusing CAP.
- Sources [242], [246], [252], [253], [254], [259], [263], [282] and [275] state that the SFO calibration is feasible and can be achieved by Manchester coding for PRDCH. Additionally, source [246] state that this method applies to device 2b.

Option 2: A sequence-based SIP.

- Sources [243], [248], [250], [254] and [282] report that SFO calibration is feasible and can be achieved by receiving sequence-based SIP.

Option 3: A CFO calibration signal.

- Sources [241], [242], [246], [249], [252], [259], [264], [263], [270] and [265] report that SFO calibration is feasible and can be achieved by CFO calibration signal. Additionally, source [246] state that this method applies to device C.

6A.1.4.4 Periodic synchronization signal

6A.1.4.4.1 Functionality

A periodic synchronization signal with a default pre-defined transmission periodicity is necessary at least for the functionality of initial frequency synchronization, including frequency acquisition, and initial timing synchronization.

6A.1.4.4.2 Signal design

6A.1.4.4.2.1 Sequence characteristics

It is a binary signal in the time domain, reusing the Rel-19 waveform and having a pre-defined chip length. The number of binary sequence(s) is less-than-or-equal to 4 and if more than 1 sequence is supported, the purpose of doing so is for potential reader identification/differentiation.

A device assumes that the binary sequence has a pre-defined length.

It was concluded that an overlaid sequence on the R2D periodic synchronization signal is not considered in Rel-20 A-IoT.

6A.1.4.4.2.2 Sequence design

Regarding sequence type for periodic synchronization signal:

- Source [281] and [276] state, that m sequence can be considered as sequence type for periodic sync signal.
- Source [275] states, a set of unique patterns of the binary sequences can be used for periodic sync signal
- Source [286] states that, M-sequence with length 31 should be considered as baseline for periodic synchronization signal, with Manchester coding applied for SFO tracking.
- Source [292] states that, for sequence type of R2D synchronization signal for frequency synchronization, m sequence or Golay sequence with good auto-correlation property should be considered.

- Source [246] state that the periodic synchronization signal is binary sequence-based, using a Golay sequence with good auto-correlation characteristic and low correlation complexity, with a fixed maximum length, in order to reduce the blind detection effort by the device for the time and frequency synchronization.

Regarding sequence length for periodic synchronization signal:

- Source [281] states that, for sequence length of periodic sync sequence, to achieve target coverage for outdoor scenario.
  - The m-sequence with length 31 with Manchester coding using M=2, occupied 31 OFDM symbols can be considered for periodic sync sequence, assuming 38 dBm Tx power and 20 dB penetration.
  - The m-sequence with length 127 with Manchester coding using M=2, occupied 127 OFDM symbols can be considered for periodic sync sequence, assuming 33 dBm Tx power and 20 dB penetration.
- Source [293] states that, for R2D synchronization signal (SS), Device 2b requires a longer length than Device C because of its larger initial SFO/CFO.
  - For Device C with initial SFO/CFO of 50ppm
    - The SS sequence with 7.5 symbols (with 63-length m-sequence and M=2) can achieve MD of 10% at an SNR lower than PRDCH with 20bit+6bit CRC, M=2, TBCC 1/3 (e.g., for R2D L1 control).
  - For Device 2b with an initial SFO/CFO of 1000ppm,
    - The SS sequence with 31.5 symbols (with 15-length m-sequence and M=2) can achieve MD of 10% at an SNR lower than PRDCH with 20bit+6bit CRC, M=2, TBCC 1/3 (e.g., for R2D L1 control).
- Source [246] state that the periodic synchronization signal should be the maximum length e.g., 256 for the binary sequence to guarantee all the devices with the maximum coverage distance can access the network.

Regarding the number of sequences for periodic synchronization signal:

- Sources [243], [245], [247], [249] and [264] report that a predefined number, e.g., 3~4 sequences can be considered for reader differentiation
  - Source [276] states that, the reader identification information, which indicates the identifier of the reader, should be carried by R2D synchronization signal. The number of binary sequence(s) of R2D synchronization signal should be equal to 4 for reader identification.
  - Source [281] states that, it is benificial to consider m-sequence with ~3 sequences for reader differentiation
- Sources [246], [248], [245] and [261] report that Single sequence can be considered.
  - Source [272] further state that if multiple sequences of the periodic synchronization signal are used, it will increase the complexity and power consumption of the device.
- Source [280] states that, the blind decoding using multiple 8-length sequences has ~2dB performance loss compared with single-sequence detection. The necessity of synchronization signal with multiple sequences needs to be discussed.
- Source [292] states that, for the number of sequences of R2D synchronization signal for frequency synchronization, device complexity on correlation-based detection should be considered assuming that device performs correlation-based detection to detect the signal on multiple candidate frequencies at least for initial frequency acquisition.

6A.1.4.4.2.3 Manchester coding

Regarding whether to apply Manchester coding to the periodic synchronization signal:

The following were negative:

- Source [272] states that, applying Manchester coding to the sequence-based synchronization signal would result in a marginal performance loss with similar residual timing error performance, while increasing the complexity of the device if it is expected to remove the Manchester encoding before performing correlation with the original sequence.

- Source [293] states that, applying Manchester coding to synchronization signal could lead to an increased false alarm rate due to its resemblance to the PRDCH.

The following were positive:

- Source [274] states that, applying Manchester coding to the binary sequence of synchronization signal should be considered, and the application of Manchester coding to the binary sequence ensures continuous clock embedding, which mitigates timing drift, reduces DC offset issues, and facilitates accurate signal detection with lower complexity.

- Source [276] states that, in order to prevent continuous 1s or 0s during the transmission of R2D synchronization signal and improving clock recovery, Manchester coding should be applied to the binary sequence of R2D periodic synchronization signal.

- Source [281] provide evaluation results show that for 31-bit m-sequence with Manchester coding, 31-bit m-sequence without Manchester coding, and 63-bit without Manchester coding, these three cases have comparable durations (~15 OFDM symbols), and shows similar MDR performance. Besides, timing performance for sequence with Manchester coding is slightly better compared with longer sequence w/o Manchester coding (same overall duration), due to narrower main lobe of auto-correlation can be achieved with Manchester coding.

- Source [293] states that, there is no major impact on miss detection and timing performance if Manchester coding to synchronization signal, while this approach helps prevent consecutive 1s or 0s.

Source [275] states that, before applying Manchester coding to the binary sequence, it needs to be verified/evaluated whether it outperforms or not the performance of a binary sequence designed with good auto-correlation and cross-correlation properties.

6A.1.4.4.3 Periodical nature of transmission

Regarding the periodicity of the synchronization signal, the following observations were made:

- Source [281] states that it is beneficial to consider a long periodicity for periodic sync signal block (AIoT-SSB) to avoid collision between R2D/D2R transmissions and AIoT-SSB. The periodicity can be in between [320ms, 1.28s], and short periodicity may lead to collision with other R2D transmissions.

- Source [278] states that frequency acquisition and reader identification latency, the timing drift due to SFO within the periodicity of the synchronization signal and the overhead of synchronization signal should be considered when determine the periodicity of the R2D periodic synchronization signal. And periodicity can be hundreds of ms, e.g., 160ms, as a starting point.

- Source [275] states that dense periodic synchronization signal can accelerate reader/cell search and improve timing synchronization at the cost of increased resource consumption (and energy consumption at the device side) and signaling overhead, whereas sparse transmission may reduce resource usage but require extra preamble overhead for reliable PRDCH demodulation. The periodicity can be 160ms, and other value can be studied.

- Source [296] states that, periodicity in the order of 100s of ms can be considered, to support calibration for DO-A traffic case (i.e., device initiates D2R transmission after fast fix from such periodic R2D) without introducing large latency penalty.

- Source [280] states that 40ms can be a starting point , and the Periodic R2D synchronization signal can be transmitted at a specific frame boundary.

- Source [292] states that for time synchronization/tracking, additional synchronization signal without corresponding broadcast information can be beneficial for reducing accumulated time error due to SFO while avoiding additional overhead of broadcast information

- Source [276] states that the transmission period of the periodic synchronization signals should be longer than 160 ms.

If the synchronization signal transmission is strictly periodic at the reader (Option 1):

The following positive observations were made:

- Source [272] state that if the synchronization signal is periodic with respect to the reader, it will be transmitted with a pre-defined period and the device would be able to detect it periodically, enabling it to perform the timing synchronization every time in order to access to the network and maintain the timing tracking to obtain the accurate timeline. The synchronization signal is periodic in order to ensure that the device can perform time and frequency synchronization periodically in order to be able to access the network in a timely manner.

- Source [273] states that for the reader, there is no additional cost/complexity for transmitting the sync signal strictly periodic, and for the device, it is beneficial for monitoring and energy-saving operations.

- Source [275] states that the sync-signal should be strictly periodic to handle new devices attempting access at any time, which is a non-deterministic event.

- Source [278] states that strictly periodic transmission is necessary to avoid ambiguity in determining system timing, which is critical for RACH resources and DRX configurations. The overhead is low from the network perspective.

- Source [291] and [292] states that periodic transmission should be considered as baseline for frequency acquisition and timing synchronization/tracking.

- Source [281] states that strictly periodic occasions help avoid blind searching within a transmission window, which would increase device power consumption.

- Source [289] states that R20 A-IoT systems require precise synchronization, with periodic synchronization signals playing a critical role.

The following negative observations were made:

- Source [271] states that it imposes unnecessary scheduling constraints for the reader and devices may not benefit from it.

- Source [285] states that strict periodicity could lead to overlaps with other signals and introduce blocking/collisions based on current resource determination methods.

If the synchronization signal transmission is not strictly periodic at the reader (Option 2) e.g., it is window-based or flexible:

The following positive observations were made:

- Source [285] states that prefer up to reader determination of resource position of periodic synchronization signal, which ensure at least one synchronization signal being transmitted within the time interval corresponding to synchronization periodicity. It is necessary to support reader side flexibility to adjust the position of periodic synchronization signal. If other signals which can be utilized for synchronization functionality are transmitted before the expected time position of periodic synchronization signal, it is feasible to cancel/postpone the transmission of periodic synchronization signal with no noticeable performance loss.

- Source [276] and [284] states that R2D synchronization signal transmission should not be strictly periodic at reader to avoid collisions.

The following negative observations were made:

- Source [272] states that the use of a window-based periodic transmission (not strictly periodic) will have a negative impact on the device's ability to perform initial frequency acquisition due to timing uncertainty since it cannot know when to expect the signals.

- Source [281] states that a flexible time position within a window requires blind searching, increasing power consumption, and requires additional indication of the actual location.

- Source [281] states that the flexibility to omit transmissions (while keeping occasions strictly periodic) is consistent with NR specification logic where BS is not mandated to transmit SSB periodically.

- Source [278] states that if the R2D synchronization signal is not strictly periodic, it will lead to ambiguity for a device to determine the timing information of the system. Without timing information, Rel-20 A-IoT system is difficult to configure RACH resources to support DO-A traffic, and DRX-like configuration for device power saving, etc.

- Source [281] states that if transmission periodicity for periodic sync signal is long enough, e.g., longer than several hundred ms, there is no clear motivation to have flexibility for BS to adjust the transmission occasion within a window or omit for periodic sync signal.

Source [283] and [293] state that, RAN1 may only need to specify a default periodicity assumption from device perspective for the R2D sync signal. It is unnecessary to further discuss whether R2D sync signal transmission is strictly periodic at reader in RAN1, it can be totally up to reader's implementation.

6A.1.4.4.4 Overall assessment

At least for the functionality of initial frequency synchronization (including frequency acquisition) and initial timing synchronization, source [243], [245], [246], [249], [251], [254], [255], [256], [257], [259], [261], [263], [264], [265], [268] , [247] and [268] state that the periodic sync signal is needed, following details are stated by companies

- Sources [243] and [246] report that the synchronization signal allows the device to acquire the frequency in which the reader is intending to transmit, and thereby attains coarse frequency synchronization. [246], and [256] state that periodic sync signal ensures that the device can perform time and frequency synchronization periodically in order to be able to access the network in a timely manner.

- Source [247] state that the sync-signal should be strictly periodic because new devices can attempt to access the reader anytime, therefore it is an unpredictable event that requires a synchronization signal to be transmitted periodically as to deal with this non-deterministic event.

- Sources [243], [251], [254] and [259] states that periodic sync signal is needed to determine time domain resource for DOA transmission. Sources [245], [261], [254], [263], [264], [268] and [268] states that periodic sync signal is needed to for time and frequency synchronization before DO-A transmission.

- Sources [243], [245] and [246] states that, with timing alignment between reader and A-IoT device achieved through periodic sync signal, DRX monitoring can be performed.

- Sources [243] and [265] states that, periodic sync signal is needed for A-IoT device to obtain broadcast information to access to reader within certain duration.

- Sources [249] and [257] states that, both periodic and aperiodic R2D synchronization signals could be supported with the presumption that the A-IoT devices do not require to maintain synchronization with the reader.

- Source [264] states that additional periodic sync signal can be configured which is associated with D2R transmission.

6A.1.4.5 Aperiodic synchronization signal

There was study of an aperiodic synchronization signal, with the following positive observations reported:

- Source [256] states that aperiodic synchronization signal, selectively skipping some of the configured periodic signal is beneficial in terms of resource utilization and reader power consumption, and aperiodic synchronization signal can be considered with periodic synchronization signal as baseline.

- Source [259] states that, aperiodic synchronization signal can be additionally supported for reader-triggered scenarios, with periodic synchronization signal as baseline.

- Source [263] states that the aperiodic sync signal is request by A-IoT device, and aperiodic synchronization signal can be considered as lower priority.

- Source [280] states that aperiodic R2D synchronization signal can also be considered to increase the synchronization accuracy for DO-A traffic transmission.

- Source [276] and [284] state that both periodic and aperiodic R2D synchronization signals could be supported with the presumption that the A-IoT devices do not require to maintain synchronization with the reader in Rel-20 to support A-IoT device synchronization to the reader.

The following negative observations were reported:

- Source [246] states that, with the aperiodic synchronization signal, the device can only perform the timing synchronization and maintain timing tracking at a specific time range. If the device fails to detect the synchronization signal at this time range, then it will fail to access to the network.

- Sources [243] and [245] state that, if the synchronization signal is aperiodic, or there is no default assumption of a periodicity from a device perspective to monitor the synchronization signal, a device has to keep monitoring, which is not feasible for Device 2b/C with limited energy storage.

6A.1.4.6 R2D end timing

Use of L1 R2D control information to indicate the TBS to provide the functionality of indicating the end of the PRDCH is studied.

6A.1.5 CFO calibration signal

6A.1.5.1 Functionality

A CFO calibration signal for the functionality of CFO estimation/LO calibration at the device is studied.

6A.1.5.2 Signal design

For the signal design, the following options are studied:

Option A: Unmodulated single tone sinusoid.

- Sources [241], [242], [243], [245], [246], [247], [248], [250], [251], [253], [254], [256], [258], [259], [261], [262], [263], [264], [265], [266], [267], [268], [269], [260], [270] and [277] and [284] state that Option A is feasible, and can be considered for CFO calibration signal.

The following design principles for this option are described according to sources:

- Sources [243], [246] and [265], report with references, and state that, the CFO calibration is based on injection lock mechanism, and the working principle of injection lock is to leverage the phase interaction between an external single-tone signal and the internal oscillation signal of an oscillator to lead the oscillator’s frequency lock to that of the unmodulated sinusoid single-tone signal.

- Source [245] report with simulation result, and states that, CFO calibration using unmodulated sinusoid is performed at intermediate frequency, where the reader transmits the CFO calibration signal with a pre-defined duration, the device mixes the CFO calibration signal with its internally generated wave, can then count the number of periodicities of the residual frequency within the duration to estimate and calibrate the CFO.

The following sources provide reasons to support this option:

- Source [245] report that with a duration of 1 OFDM symbol, the CFO after calibration can be reduced to less than 10 ppm.

- Sources [241], [247], [250], [253], [254], [256], [264], [258], [267], [279], [277], [286], [287], [284], [295], [296] and [294] state that unmodulated sinusoid single tone can achieve CFO calibration with low complexity. Source [280] states that it allows CFO correction to be implemented using hardware circuits (e.g., AFC loop), making it well-suited for A-IoT devices. And Source [296] states that the constant-amplitude single-tone allows simple phase-based CFO estimation without correlation or symbol detection.

- Source [243] states that single tone signal is already supported in NB-IoT, (e.g., NPRACH), there is no co-existence issue.

- Source [293] provide evaluation results and show that Device 2b requires a higher CNR than Device C because of its larger SFO/CFO before/after calibration, e.g., by using Option a, the CFO calibration signal with 2 OFDM symbol can achieve CFO calibrated from 1000ppm to 100ppm for Device 2b with 10% error at SNR=8dB and 1% error at SNR=18dB, and to achieve CFO calibrated from 50ppm to 10ppm for Device C with 10% error at SNR=0dB and 1% error at SNR=10dB.

- Source [294] states that the envelope of unmodulated signal for CFO calibration is constant, which provides higher calibration precision than modulated signal.

The following sources provide reasons to not support this option:

- Source [249] states that it will create the inter-channel interference to neighboring NR sub-bands.
- Sources [264] and [276] state that this method may be vulnerable to deep fading, reducing reliability.
- Source [276] states that injection lock mechanism will consume huge energy of the device, which is unacceptable even for active devices.

Option B: Unmodulated multiple single tone sinusoids.

- Sources [256], [264], [266] and [284] report that Option B can be considered for CFO calibration

The following reasons were provided to support this option:

- Source [256] states that, Option B provides frequency diversity gain while maintaining low complexity; considered as optional feature. It is preferable that the tones are sufficiently separated in frequency, e.g., by at least the Rx filter bandwidth.
- Source [264] states that, using multiple sinusoidal tones can improve robustness against fading by providing frequency diversity.
- Source [266] states that, although Option B is slightly more complex than a single tone, it is still a very simple waveform.

The following reasons were provided to not support this option:

- Sources [241], [243], [242], [270], [266], [263], [273], [280], [283] and [294] state that Option B increases in detection complexity, and it may also complicate frequency resource allocation.
- Sources [243] and [246] raised feasibility concerns on Option B
  - Source [243] states that even with increased complexity, it is not clear how does it work, since frequency of each tone cannot be accurately determined, due to only frequency range of each can be determined if captured by Rx filter, and the target frequency for LO calibration cannot be accurately determined.
  - Sources [246], [248], [279] and [294] state that multiple frequency components would occur when mixed with the local RF frequency signal and it is difficult to lock the LO frequency.
- Source [269] state that multiple unmodulated tones increase PAPR and require FFT-based tone separation
- Source [280] states that the benefit of using multiple sinusoid single tones for CFO calibration is unclear compared with single tone.
- Source [281] states that Option b (unmodulated multiple sinusoid single tones), based on counting of number of cycles of envelop of frequency components of $|f1 – f2|$, is not feasible for achieving 10 ppm accuracy. It requires either an extremely long duration or an excessive transmission bandwidth, both of which are impractical for R20 A-IoT.
- Source [276] states that, for Options b sinusoidal signals for CFO calibration signals transmitted on NR DL spectrum, it will create the inter-channel interference to neighbouring NR sub-bands.

This Option B is not considered for normative specification.

Option C: Multiple modulated tones, where the CFO calibration signal may be the same as or different from the synchronization signal.

The following three different design principles for this option are described according to sources:

- Source [247] state that the unmodulated multiple tones are transmitted in different frequency locations in different time resources

The reader transmits the synchronization signal using a centered bandwidth sweeping method, where each centered bandwidth covers a portion of the overall synchronization signal bandwidth

The device detects the synchronization signal power at different strengths on each of the subsets and based on the subset with the strongest synchronization signal power, the device applies a frequency correction

- Sources [249] and [261] report that, CFO calibration is done in baseband domain, and CFO Calibration is OOK sequence (not R19 SIP or CAP). The CFO pre-compensation method is given as follows:

  Step 1: The received signals with pre-compensated CFO ±Δf is added in the received R2D signals to limited the frequency offset within ±1/2 Δf.

  Step 2: The correlation processing of the CFO calibration signal with the received signals, the precompensated received signals with additional Δf, and the pre-compensated received signals with additional -Δf.

  Step 3: The peak of the correlation values from the three correlation outputs would be selected as the received signals for the R2D signal processing.

- Source [265] states that, the CFO calibration can be achieved by measuring chip length, and that one possible implementation is devices calibrate CFO based on SFO calibration by counting the number of samples of 0s and 1s and comparing with the defined duration of a pattern.

The following reasons were provided to support this option:

- Source [247] state that using the synchronization signal to perform the CFO calibration of device 2b is foreseen to be suitable (residual CFO of 100 ppm)

- Source [249] state that the OFDM waveform-based CFO calibration signals would be orthogonal to neighboring NR channels

- Sources [249] and [261] state that R2D synchronization signal can be used for CFO calibration signal

- Source [265] state that may be feasible if same clock (LO) is used for both carrier frequency generation and sampling.

- Source [276] states that, modulated multiple tones, can be used for CFO calibration based on pre-compensation, R2D synchronization signal and M-sequence can be used for CFO calibration signal.

  - With the CFO pre-compensation method, if R2D CFO calibration signal adopts M-sequence with M value equals to 4 and its length equals to 7 bits, it will occupy only two OFDM symbols, the 90% CDF residual CFO after calibration at the device side is no more than 5ppm for Device C and 20ppm for Device 2b.

  - The pre-compensated CFO received signals should be used for CFO calibration to reduce the degradation of received signal detection and decoding performance due to frequency errors for Device 2b and Device C.

The following reasons were provided to not support this option:

- Source [247] state that the CFO calibration of device C may require using unmodulated sinusoid single tone to meet a residual CFO of 10 ppm.

- Sources [241], [243], [245] and [268] state that the CFO hypothesis at AIoT receiver cannot be constructed in baseband due to phase information for received signal is not kept in envelop detector, and no I/Q path in receiver baseband.

- Sources [254] and [281] report that, in multi-tone LP-SS discussion, the LP-WUR doesn't have the capability to acquire good frequency calibration, similarly this method is not reliable. And source [281] further states that, according to agreements for LP-SS OOK sequence in LP-WUS WI (RAN1#118 meeting), the design metric only include time synchronization accuracy and RRM measurement accuracy, no design consideration and metric for frequency error calibration in LP-SS design.

- Sources [246] and [248] state that the LO frequency calibration cannot rely on the baseband clock to fulfil the calibration procedure and the calibration signal should be utilized for the RF-end clock.

- Source [243] state that using ON chip duration to calibration Clock it not feasible, since according to RAN4 requirements on pulse width, Table 6.4.2-1 in section 6.4.2 of Ts 38.194, the pulse of a chip, the allowed chip length can be <=1.3 Tc when the nominal chip length is Tc, in this case, it is not feasible to use chip length/pulse width for CFO calibration.

- Source [258] states that pulse width distortion analysis is possible to be used for A-IoT device due to simplicity, but timer resolution at device is required to be much less than the difference between estimated pulse width and the transmitted pulse width.

- Source [281] states that it is not feasible to use option c, modulated multiple tones, i.e., OOK sequence for R2D CFO calibration, as that in D2R receiver.

    - Each CFO hypothesis or CFO compensation should be constructed by multiplying a phase rotation sequence to local sequence, and each phase rotation sequence corresponds to a CFO hypothesis. However, only envelop/amplitude of the Rx signal can be obtained in IF-ED detector, and there is no I/Q path in receiver, phase information which reflects frequency offset is lost in receiver.

    - Estimation or compensation using R2D OOK sequence is still not feasible, since the R2D waveform generation is not specified, which means the sample level amplitude (with ripples) and phase at transmitter is unknown to receiver when constructing local sequence, because OOK signal generation totally up to Reader implementation.

    - It is not feasible to use RSRP-like metric to determine CFO at receiver. To achieve 10ppm calibration accuracy, several filter covering different frequency point should be used which is too complex for A-IoT devices. Even if using so many IF filter, the received power captured into each filter would be very close. The right hypothesis cannot be reliably determined, the accuracy of CFO calibration cannot be guaranteed.

- Source [283] states that, since the CFO calibration is mainly applied to the LO in RF level, which cannot be based on the baseband clock, for the signal design, Option c is not feasible.

6A.1.5.3 Transmission

6A.1.5.3.1 Periodic

Sources [241], [243], [245], [246], [249], [250], [251], [254], [259], [260], [261], [263], [269], [259], [264], [266], [258] and [268] state that the CFO calibration signal is periodically transmitted.

- Sources [243], [245], [250] and [270] state that, CFO calibration signal with hundreds ms or 1 second periodicity is already sufficient. Aperiodic CFO calibration signal, e.g., on demand transmission with PRDCH is not needed due to slow frequency drift. Source [247] also states that the CFO calibration signal can be transmitted in very sparse manner.

- Sources [245], [246], [250], [258] and [263] states that, to support DO-A traffic, periodic CFO calibration signal is needed.

    - Source [245] states that, periodic CFO calibration is needed to avoid a device from keep monitoring the CFO calibration signal before transmitting the very first D2R transmission for DOA traffic.

- Source [261] and [242] states that it is beneficial to provide regular opportunities for devices to perform frequency calibration and maintain accurate synchronization with the reader.

- Source [285] suggest to study the condition that CFO calibration signal transmitted with a specific periodicity.

Source [278] states that, the periodicity of the CFO calibration signal can be the same as the periodic synchronization signal, i.e., use 160 ms as a starting point.

Source [280] states that, the periodicity of CFO calibration signal should be same as or multiple of the periodicity of DO-A resources.

Source [292] states that, considering low CFO drift rate, i.e., 1ppm/s or 0.1 ppm/s, and hundreds of ms periodicity, CFO calibration via periodic CFO calibration signal can be sufficient.

Source [279] states that for the purpose of power saving at reader side, a large periodicity can be considered for CFO calibration signal transmission.

Source [296] states that, support periodic transmission with sparse periodicity for CFO calibration signal.

6A.1.5.3.2 Aperiodic

Sources [246], [249], [251], [254], [256], [261], [263], [242], [258], [259], [262], [264], [267], [253] and [268] state that the CFO calibration signal can be transmitted aperiodically. In which, Sources [246], [250], and [263] state that an aperiodic CFO calibration signal can be considered with periodic CFO calibration signal as baseline.

- Source [246] state that with the DRX configuration, the device will miss several instances of the periodic CFO calibration signal. In this case, the CFO calibration signal can be considered to be transmitted as needed, e.g., transmitted after the PRDCH carrying the paging message. Source [285] suggest to study the condition that CFO calibration signal can be together with specific R2D message types (e.g.,paging).

- Source [249] states that the R2D calibration signal should be transmitted before R2D reception and D2R transmission. Since R2D reception and D2R transmission may be either periodic or aperiodic, both periodic and aperiodic CFO calibration signals should be supported.

- Sources [254], [256], [258], and [261] states that, at least for D2R triggered by PRDCH, the on-demand transmission of CFO calibration signals is beneficial to keep the interval between the CFO calibration signal and the subsequent D2R transmission short for better CFO calibration accuracy. Source [293] states that aperiodic CFO calibration signal may be needed in addition to periodic CFO calibration to ensure the device's CFO calibration accuracy for D2R transmission.

- Source [243] states that if CFO calibration totally relying on aperiodic signal, the CFO calibration accuracy cannot be guaranteed, which depends on traffic.

- Source [285] suggest to study the condition that CFO calibration signal transmitted in every R2D transmission.

Source [284] states that, the synchronization signal transmission can be quasi periodic at reader if these are within the limit on variance and the device is assumed not to use the relation between synchronization signals.

6A.1.5.3.3 Presence/absence

Sources [272], [279] and [285] state that the presence of the CFO calibration signal can be indicated e.g., in L1 control signal, for flexibility and overhead reduction. Source [272] additionally state that this indication is used in the case of aperiodic CFO calibration signals.

6A.1.5.4 Time location

The time location of transmitting the CFO calibration signal is studied in relation to the following candidate time references:

Option 1: Preceding L1 R2D control information

The following positive observations were made:

- Source [276] and [282] states that R2D CFO calibration signal should be located before the L1 R2D control information to enable frequency offset correction before reception of control and data, enhancing decoding performance.

- Source [282] states that option 1 provides benefits for potential R2D FDM operation, as the device can perform R2D processing with reduced residual CFO.

- Source [284] states that if the PRDCH / L1 R2D control information is device specific, option 1 could be easier as other devices are not required to determine the TBS comparing with option 2.

- Source [292] states that CFO calibration signal transmitted before L1 R2D control can be applied for CFO calibration for L1 R2D control (if needed), PRDCH (if needed), aperiodic D2R Tx triggered by the R2D, and first X occasions of periodic D2R Tx triggered by the R2D where X depending on periodicity and CFO drift.

The following negative observations were made:

- Source [272] states that it is not necessary to transmit an CFO calibration signal prior to an R2D transmission after the periodic block since the device has already calibrated its LO using the CFO calibration signal transmitted along with the periodic synchronization signal and the MIB-like broadcast information, and can take advantage of the residual CFO and the minimal frequency drift.
- Source [281] states that the start of CFO calibration signal cannot be reliably determined due to time drift if it is located before L1 control information and before SIP.
- Source [281] states that CFO calibration signal immediately preceding L1 control could cause interruption to subsequent control or data payload reception due to calibration/RF switching delay.
- Source [295] states that option 1 may require the device's RF/baseband to have a high-speed frequency offset estimation loop that can be adjusted within a few symbol times, which would be a challenge for A-IoT devices.
- Source [281] states that the CFO calibration accuracy is unstable if PRDCH with L1 control transmission is aperiodic.

Option 2: Data payload of the PRDCH

Option 2-1: Preceding the data payload of PRDCH

The following positive observations were made:

- Source [282] states that transmitting the CFO calibration signal before PRDCH reception allows the device to update its CFO estimate immediately prior to the R2D transmission, and provides benefits for potential R2D FDM operation, as the device can perform R2D processing with reduced residual CFO, when CFO calibration signal is transmitted before the data payload of the PRDCH.
- Source [292] states that CFO calibration signal transmitted after PRDCH can be applied for CFO calibration for aperiodic D2R Tx triggered by the R2D, and first X occasions of periodic D2R Tx triggered by the R2D where X depending on periodicity and CFO drift.

The following negative observations were made:

- Source [276] states that R2D CFO calibration signal will not be able to apply to the detection of L1 R2D control information prior to time location of CFO calibration signal.
- Source [281] states that CFO calibration signal immediately before R2D data payload would cause interruption to the payload reception.

Option 2-2: After the data payload of PRDCH at the end of a R2D transmission

The following positive observations were made:

- Source [285] states that the CFO calibration signal be placed right after the R2D data payload. This allows devices not requiring calibration to ignore the signal and enter sleep earlier, while avoiding interference with ongoing payload decoding. Besides, the impact of padding can be further studied.
- Source [272], [285] and [288] state that an aperiodic CFO calibration signal transmitted after the PRDCH (e.g., paging message), with its presence indicated by L1 control.
- Source [295] states that placing the calibration signal after receiving the PRDCH has minor impact on timeliness and does not delay acquisition of control info.
- Source [281] observes that interruption of R2D transmission can be avoided if CFO calibration signal is transmitted at the end of the R2D block (after the data payload).
- Source [293] supports aperiodic LO calibration using CFO calibration signal transmitted together with the PRDCH.
- Source [296] and [288] state that it is efficient to include CFO calibration signal together with PRDCH transmission.

- Source [289], [283], [273] and [279] states that it is beneficial to place the CFO calibration signal after the PRDCH to improve performance of PDRCH.

The following negative observations were made:

- Source [281] states that the availability of CFO calibration signal depends on traffic if not transmitted along with broadcast information.

Option 3: Prior to the PDRCH (D2R transmission)

The following positive observations were made:

- Source [280], [282] and [292] state that the periodic CFO calibration signal is transmitted before PDRCH/DO-A transmission, which is beneficial for ensuring frequency accuracy for D2R transmissions.
- Source [286] notes that independent transmission of the calibration signal (Option 3) can be considered as a viable but it requires additional device wakeup.

The following negative observations were made:

- Sources [276] states that R2D CFO calibration signal is transmitted before the PDRCH, and it cannot be used for the calibration of the device clock for LO for carrier frequency for the reception of PRDCH.
- Source [281] states that, standalone CFO calibration signal in relation to PDRCH (not transmitted with R2D) causes resource fragmentation, and leads to high overhead of transmitted in relation to all PDRCH transmissions.
- Source [281] and [295] states that the start of CFO calibration signal cannot be reliably determined due to accumulated time drift after the previous synchronization.
- Source [280] states that device power consumption may increase in this option, since the device needs to wake up twice to receive common signals.
- Source [296] states that having a separate CFO calibration signal transmission will be inefficient for ultra-low-power A-IoT devices due to extra monitoring and wake/sleep overhead.
- Sources [272], [275], [281] and [278] find that this location offers no remarkable difference or gain for D2R reception performance due to slow CFO drift. Source [272] further states that periodic CFO calibration signal with a predefined period will always help to improve the D2R performance by maintaining the residual CFO level with very small additional frequency drift.

Option 4: Synchronization signal

Option 4-1: Preceding the synchronization signal

The following negative observations were made:

- Source [281] states that CFO calibration signal preceding synchronization signal would cause interruption to synchronization signal reception.

Option 4-2: After the synchronization signal in a R2D block

The following positive observations were made:

- Source [281] states that CFO calibration signal can be transmitted immediately after the periodic synchronization signal (if not with broadcast). This ensure a stable accuracy due to periodic transmission.
- Source [281], [280] and [288] state that this option minimizes resource overhead, and achieves higher resource efficiency.
- Source [280] state that placing common signals within the same time period is beneficial for device power saving.
- Source [272] states that periodic CFO calibration signals can be transmitted immediately after a periodic synchronization signal and MIB-like part.

- Source [286] recommends that the CFO calibration signal be transmitted at the end of the periodic synchronization signal as a baseline, efficiently utilizing device wake-up periods.
- Source [293], [282] and [295]states that CFO calibration signal as part of the periodic synchronization signal. Source [295] states that periodic synchronization signals are very suitable for also performing frequency offset calibration at the same time.
- Source [278] states that A-IoT device can first search the periodic synchronization signal for time synchronization and frequency acquisition due to it is robust to CFO, and then receive the CFO calibration signal.
- Source [290] states that periodic R2D synchronization signal contains at least one sequence for timing synchronization and one sequence for CFO calibration, which would reduce monitoring complexity.
- Source [283] and [288] state that the CFO calibration is transmitted immediately after periodic synchronization signal is beneficial to D2R performance.
- Source [292] and [274] state that CFO calibration signal is transmitted after synchronization signal and can be applied for CFO calibration for broadcast information Rx (if needed), aperiodic D2R Tx and periodic D2R Tx.

The following negative observations were made:

- Source [280] states that the periodicity of CFO calibration signal along with periodic synchronization signal may not suitable for D2R.

Option 5: Broadcast information

Option 5-1: Preceding broadcast information

The following positive observations were made:

- Source [292] states that R2D CFO calibration signal be received before broadcast information for better detection/decoding of PRDCH signals.
- Source [280] states that periodic CFO calibration signal is scheduled after the synchronization signal and before the broadcast information, is beneficial for reliable reception of broadcast information, and also beneficial for power saving. And there is a gap between the synchronization signal and the CFO calibration signal, as well as another gap between the CFO calibration signal and the broadcast information.

The following negative observations were made:

- Source [280] states that CFO calibration signal is scheduled after the synchronization signal and before the broadcast information limits the flexibility for transmission of CFO calibration signal.
- Source [280] states that the periodicity of CFO calibration signal along with broadcast information may not suitable for D2R.
- Source [272] states that it is not necessary to transmit an CFO calibration signal prior to an MIB-like transmission of the periodic block since the device may need additional delay to calibrate its LO frequency and it will potentially introduce time gap between the CFO calibration signal, which will increase the resource overhead of the system and the implementation complexity of the device.

Option 5-2: After broadcast information at the end of the R2D block

The following positive observations were made:

- Source [278] states that the synchronization signal, CFO calibration signal, and MIB-like broadcast information can be transmitted as a block, with CFO calibration signal immediately after the synchronization signal or the MIB part, A-IoT device can first search the periodic synchronization signal for time synchronization and frequency acquisition due to it is robust to CFO, and then receive the CFO calibration signal.

- Source [272] states that periodic CFO calibration signals can be transmitted immediately after a periodic synchronization signal and MIB-like part, this enable the device calibrated its LO on receiving the CFO calibration signal initially along with the synchronization signal.
- Source [281] supports CFO calibration signal transmitted at the end of the R2D block with broadcast information (MIB-like or SIB1-like), ensuring stable calibration accuracy, and it provides common reference accessible to all devices, and resource overhead is minimized.
- Source [283] and [292] state that the CFO calibration is transmitted along and immediately after broadcast information is beneficial to D2R performance.

6A.1.5.5 Frequency location

Source [283] states that exact frequency of CFO calibration signal may need to be explicitly indicated from reader and then determined by the device.

Source [281] states that in predefined frequency location within the transmission BW of the associated R2D transmission, e.g., center frequency of the periodic sync signal (AIoT-SSB) or one subcarrier adjacent to the center frequency of AIoT-SSB.

Source [285] states that, it is preferred that frequency positions of the CFO calibration signal start with same frequency domain resource as of R2D transmission.

Source [289] suggests to study the following frequency-allocation alternatives for the CFO calibration signal, Alt 1. Transmit on the initial frequency used to provide initial access to A-IoT devices, and Alt 2. Transmit on the frequency that provides resources for D2R data payload transmission.

6A.1.6 Broadcast information

6A.1.6.1 Functionality

Broadcast information is studied, including the information included, the time and frequency resources for its transmission, whether it is carried on the PRDCH, and whether it is transmitted periodically or aperiodically. The functionality of broadcast information is to provide the device with at least necessary information for subsequent transmission and reception, for which it is necessary to be periodic. It would need to be decided whether the transmission of broadcast information is strictly periodic at reader, and the periodicity of the transmission.

6A.1.6.2 Content

On the potential necessary information included in the broadcast information, following observations were provided during the study:

- [364] Broadcast information can contain time/frequency resource for L1 R2D control monitoring, time and frequency resources and related configurations for the initial DO-A transmission, etc.
- [365] Time offset (and possibly frequency resource) of the random-access occasions (RO) for the first D2R transmission (Msg1) for all traffic types (DO-A, DO-DTT, and DT). Cell-specific power control parameters. Time offset (and possibly frequency resource) of the paging occasions for Msg0 transmission of DT and DO-DTT traffic. Reader/cell selection information including certain RSRP level for reader selection/re-selection. D2R transmission configurations/parameters, e.g., size of preambles/midambles. Reader/network related information. Other information deemed necessary by RAN2.
- [366] It’s not necessary to includse reader identification information, SFN, DO-A resource configuration, power control related parameters, device (un)availability related parameters and transmission parameters for L1 control information in broadcast information.
- [367] Broadcast information, if supported, includes power control parameters such as target received power, allowed increment of transmission power, one or more chip duration values for a PRDCH for L1 control
- [368] Study the following information to be included in broadcast information. Timing information, e.g. SFN-like info. Periodic random access resource configuration, e.g. periodicity, offset, available resource in each

periodicity, and frequency resources for FDM transmission. DRX configuration for paging monitoring. Configurations of cell/reader-common L1 R2D control information. Cell ID related information.

- [369] The number of Cell/Reader ID can be larger than 4, e.g., 1008, which may consider acquisition method based on synchronization signal sequence(s) in addition to MIB-like and/or SIB-like information. Timing information can be contained in broadcast information. Whether to use SFN or timing offset indication can be study by avoiding over-designing. Broadcast information containing L1 R2D control related information can provide configuration/updating flexibility.

- [370] The following information can be considered included in the broadcast information: Paging-related information, timing information, e.g., SFN, frequency information for DL FDM, UL reference frequency position, DO-A resource related information, i.e., time and frequency resource information, parameters of R2D L1 control, e.g., chip duration, repetition number, code rate, cell/Reader/Transaction ID, SIP length information.

- [371] Study Case B2: Two-step acquisition of the broadcast information including both MIB-like and SIB1-like: The information can be carried by MIB-like: SFN information, A-IoT SIB scheduling information. The information can be carried by SIB1-like: DOA resource configuration, L1 control related information configuration(s), R2D preamble configuration, The frequency information of A-IOT carrier, DRX configuration for device power saving, Criteria for cell-(re)selection, Reader ID related information, Power control related information, Periodicity of sync signal (and the payload i.e., MIB). Note: other information is subject to RAN2 study.

- [372] Regarding the purpose and necessary information of broadcast information, it is observed that following information is needed: Cell/reader ID-related information to enable reader identification. Resource-allocation information for DO-A transmissions. Timing information indicating the allocation timing of the D2R resources. D2R power-control parameters for device transmit-power control.

- [373] At least the following information can be carried in the A-IoT broadcast information. Scheduling Info for the first DO-A D2R transmission; Reader ID-related information; FFS: other information to be carried in A-IoT broadcast information.

- [374] In Rel-20, study the necessity and feasibility of the following scheduling information for device 2b/C: T/F resource configuration for SIB-like message and/or MSG1, SFN value, repetition index or period index of synchronization signal, time unit index (e.g., half-frame index), frequency offset between the synchronization signal and the SIB-like message or MSG1.

- [376] Broadcast information is transmitted for the following purposes: Purpose #1: Triggering the inventory procedure (i.e., paging-like message for device discovery/presence checking), Purpose #2: Providing DO-A resources so that a device can initiate RA by itself, and Purpose #3: Providing reference information for open-loop power control. Broadcast information can carry at least one of the following information sets, depending on the purpose: Information for the inventory procedure trigger, DO-A resource information (e.g., periodicity, access occasion configuration, resource pool information, etc.) and Reference information related to R2D Tx power for power control.

- [379] The broadcast information should include at least the following information elements: Cell/reader ID-related information, power control-related information, periodic D2R resource configuration for DO-A, and timing information (e.g., SFN-like information).

- [383] For broadcast information, following information should be considered. Information for reader differentiation/identification such as reader ID. Parameters for first D2R transmission for DO-A such as D2R resource, physical layer design related parameters such as chip duration, FEC, repetition, etc., and power control information. Parameters for subsequent R2D reception such as potential R2D reception resources of R2D sync signal, L1 R2D control, A-IoT paging, R2D command and corresponding physical layer design related parameters such as M value, FEC, repetition, etc.

- [385] Periodic system information (SIB) can provide necessary information including at least: Reader ID-related information (e.g., higher layer reader ID), R2D-related configuration (frequency location(s) for R2D carriers, configuration of periodic R2D monitoring occasions for SS and paging, L1 control information per coverage level), D2R-related configuration (frequency location(s) for D2R carriers, configuration of periodic DO-A D2R resources, power control per coverage level).

- [380] states a two-step method for acquiring broadcast information. In the first step, the broadcast provides a list of available frequencies for device access. In the second step, further time–frequency resources are supplied for device transmission/reception, along with possible reader ID information.

- [362] For the broadcast information, the MIB-like transmission contains the reader identifier, SFN and the time and frequency resource allocation information for the SIB-like transmission. The SIB-like broadcast transmission contains the time and frequency resource allocation of DO-A, power control parameters, ARFCN-like frequency location information, RSRP threshold/level for the R2D measurement and other necessary parameters for DO-A and the active device.

6A.1.6.3 Acquisition

For A-IoT devices to acquire the broadcast information, two cases are studied:

Case 1: Single-step acquisition of the broadcast information, which is carried on PRDCH.

Sources [363], [365], [364], [370], [375], [376], [385], [386], [391] state that Case 1 should be supported.

- [363] For broadcast information, single-step acquisition of the broadcast information is supported. PRDCH is used to carry the broadcast information.
- [365] It is simpler and more efficient to transmit all the necessary information in a single step using a single PRDCH. Case 1 may also be better from reader and A-IoT device energy consumption perspectives.
- [364] Due to the limited battery capacity of A-IoT devices, it is desirable to provide all necessary system information to A-IoT devices at one time rather than introducing extra dedicated message for system information delivering (e.g., SIB1 like message), unless the payload size of the broadcast information identified at the end may result in an unacceptable A-IoT SSB length.
- [370] For Case B1-1, the broadcast information includes L1 control, broadcast information scheduling is more flexible. For Case B1-2, the broadcast information does not include L1 control, the design is simpler and makes it easier for the device to detect the broadcast information.
- [375] A single-step acquisition of broadcast signal using PRDCH should be considered.
- [376] A single-step approach (i.e., Case 1) should be used for broadcast information, and introducing a two-step structure (i.e., Case 2) is not strongly motivated from an information-size perspective.
- [385] For Case 1, devices may perform SIB monitoring with a predefined periodicity and detect the SIB directly. This approach is more straightforward and potentially more efficient.
- [386] For the acquisition method of broadcast information, study single-step acquisition, i.e., Case 1 as baseline. Single-step acquisition of broadcast is simpler and costs less standardization efforts than two-step acquisition method. MIB-like in Ambient IoT is not necessary since ambient IoT devices do not require precise synchronization like SFN information and carrier offset. Another reason is the message size of broadcast information may be fixed and is not as large as SIB information.
- [391] For broadcast information acquisition prioritize Case 1. Single-step acquisition minimizes the number of decoding tasks, and therefore, reducing the device activation time and avoiding the need to coordinate multiple acquisition stages. On the contrary, Case 2 introduces additional operational steps, requiring devices to decode two separate layers of system information, potentially from different physical channels. While this structure allows for modularity and independent periodicities for MIB-like and SIB1-like information, it also increases decoding complexity and extends the time during which devices must remain active. Therefore, Case 1 should be prioritized given the energy constrains of A-IoT devices.

For Case 1, following neutral or negative observations were also provided during the study.

- [372] The single-step acquisition allows simple and fast acquisition at the device side but incurs signaling overhead and may introduce reader-identification timing delay.
- [378] For Case 1, the broadcast PRDCH should always be preceded by a synchronizing signal to facilitate detection and acquisition.
- [361] Potential for high system overhead with frequent broadcast information transmission. Possible increase of device unavailability as a device may need to perform blind decoding of the L1 R2D control information before obtaining broadcast information.

- [371] If the periodicity of single broadcast information is same as that of periodic sync signal, larger resource overhead will be caused since some of the information is only used after initial reader search and don't need to be transmission as frequent as periodic sync signal. If the periodicity of single broadcast information is larger than that of periodic sync signal, during initial search, a device needs to do more blind detection to find the potential broadcast information to obtain the essential information for completing the initial search e.g., SFN for timing alignment. This will increase the initial search complexity and latency. If the single broadcast information is SIB1-like information, the starting time location of SIB1-like broadcast information has to be fixed, which is less flexible for reader resource allocation.

- [362] Since the necessary information for the DO-A transmission includes many parameters, e.g., time and frequency resource, power control, etc., using a single-step acquisition of the broadcast information would result in a large overhead which would degrade the transmission efficiency.

Case 2: Two-step acquisition of the broadcast information e.g., MIB-like and then SIB1-like broadcast information. PRDCH is used to carry the SIB1-like broadcast information, and a separate PRDCH or a new dedicated physical channel used to carry the MIB-like information

On whether to use a separate PRDCH or a new physical channel to carry the MIB-like broadcast information, sources [362], [371], [374], [372] state that PRDCH should be used to carry the MIB-like broadcast information.

- [362] PRDCH is used to carry the MIB-like broadcast information as there is no additional motivation to define a separate or new channel like NR SSB, in which 8 information bits are carried in the physical layer payload. Secondly, carrying the MIB-like broadcast information on an additional channel would increase the workload in terms of specification work for the WI.

- [374] If Case 2 is selected for the broadcast information transmission scheme, support to reuse a separate PRDCH for the MIB-like broadcast information. While using PRDCH may require a new R2D message format for MIB-like information, defining an entirely new broadcast channel would be significantly a larger workload. Given the limited TU before the SID is completed, reusing the PRDCH to carry MIB-like information would be more feasible and simpler to achieve.

- [371] Use PRDCH as the transmission channel for both MIB-like and SIB1-like broadcast information. Do not introduce a new channel to carry MIB-like broadcast information.

- [372] If the introduction of a new physical channel for the partial broadcast information is considered, the specification work may significantly increase.

Sources [368], [361] are also open to use a new physical channel dedicated to MIB-like information.

- [368] MIB-like broadcast information can be carried by a separate PRDCH or a new physical channel dedicated to MIB-like information. For MIB-like part, the R2D parameters such as TBS, payload size, FEC or repetition number can be predefined.

- [361] Whether the MIB-like information is carried in its own channel can follow the concept of the PBCH, which is a separate channel for the MIB. This separate channel can simplify the specifications.

Sources [361], [362], [368], [371], [380] state that Case 2 (Two-step acquisition of the broadcast information) should be supported. Following observations were provided during the study.

- [361] Allows the PRDCH containing broadcast information to be similar to other PRDCH by providing parameters to decode L1 R2D control information. Allows a device to know whether to process PRDCH containing broadcast information and whether the PRDCH containing broadcast information is scheduled. Allows a reader to coordinate device availability of all devices it serves.

- [362] The two-step split acquisition of the broadcast information is to facilitate an efficient method to provide the required information to the device at the required time - the system information carried by the MIB-like transmission is required by the device more frequently than the DO-A resource allocation information carried by the SIB-like broadcast information since it does not need to change frequently. The overhead of the resource allocation information for DO-A is larger than the other system information, and transmitting this in the SIB-like transmission with a longer periodic interval as compared to the MIB-like broadcast information can avoid a degradation of transmission efficiency. The impact of overhead is reduced when compared with Case 1 where a single-step acquisition of the broadcast information is used.

- [368] The payload to be carried in the broadcast information is expected to be large, it is more reasonable to split the broadcast information into two parts, i.e., a MIB-like part and a SIB1-like part.
- [371] MIB-like is regarded as the payload of periodic sync signal to carry the minimum required information for device to complete the initial reader search, which has the same periodicity as that of periodic sync signal. Thus, it is less complex for a device to finish the initial reader search within the periodic sync signal cycle. With two-step acquisition of the broadcast information, the periodicity of SIB1-like can be more flexible e.g., larger than that of MIB-like to avoid unnecessary resource overhead. MIB-like can flexibly indicate the scheduling information of SIB1 e.g., the start time location of SIB1, which is benefit for reader resource allocation.
- [380] Support two-step acquisition of the broadcast information (i.e., Case 2). In the first step, broadcast information transmitted on initial frequency provides a list of available frequencies for devices to access. In the second step, broadcast information transmitted on the available frequencies provides further resource occasions for devices to access.

For Case B2, following neutral or negative observations were also provided during the study.

- [370] For Case 2, the design is more complex compared with Case 1, and incurs higher signaling overhead, which increases downlink blockage probability and consequently degrades inventory efficiency.
- [372] The two-step acquisition reduces signaling overhead and allows early reader identification but may cause acquisition delay and may require additional specification work for a new physical channel.
- [378] For Case 2, the MIB-like channel should be designed to be as simple as possible, in order to minimize device complexity.
- [385] For Case 2, devices may perform MIB monitoring with a predefined periodicity. Devices first detect the MIB to obtain the SIB monitoring configuration and then acquire the SIB.
- [386] For case 2, i.e., two step acquisition of the broadcast information results in more complexity comparing with single step acquisition.

For Case 1 and Case 2, following observations were also provided during the study.

- [369] If the payload size of broadcast information is small and fixed, there is no need to split it into two parts.
- [373] RAN1 can further decide whether single-step or two-step acquisition is required for broadcast information in Rel-20 A-IoT after all the information carried by broadcast information and corresponding size have been clearly identified.
- [378] The down-selection between Case 1 and Case 2 may depend on how much broadcast information needs to be carried, as well as the resulting impact on device complexity, coverage, and power consumption.
- [379] The introduction of an MIB/SIB-based structure for broadcast information should be decided after the contents of the broadcast information are agreed.
- [383] Whether the broadcast information acquisition is single-step or two-step depends on the payload size of whole broadcast information, i.e., two-step is necessary if it would be expected to be large, otherwise, single-step is sufficient.

6A.1.6.4 Time location

Source [272], [273], [274], [279], [281], [286], [278], [288], [285], [289], [280] and [292] state that broadcast information can be transmitted immediately after periodic synchronization signal.

- Source [272], [273] and [281] states that this ensures the device can obtain system information (e.g., reader ID) as part of the reader identification/differentiation procedure. Source [272] and [281] state that this applies to MIB-like broadcast information, and [278] state that this can be one potential option.
- Source [279] states that the broadcast information can be transmitted at the end of the periodic synchronization signal with fixed time domain resources to simplify device implementation.

- Source [281] states that it is beneficial to support periodic R2D sync signal block, aka., AIoT-SSB, which can be composed of periodic sync sequence, PRDCH for MIB-like broadcast information, and CFO calibration signal, and these parts are continuous in time.

- Source [280] states that periodic CFO calibration signal and broadcast information is placed after different synchronization signals.

Source [275], [285], [278], [280] and [292] states that, the broadcast information can be transmitted not immediately after periodic synchronization signal.

- Source [275] states that broadcast information can be transmitted with a different periodicity from the periodic sync signal and with pre-defined offset to the sync signal when transmitted. Source [278] also state that, MIB-like part is transmitted separately with the synchronization signal

- Source [285] states that time resource can be derived by periodic synchronization signal with a fixed gap.

- Source [292] states that time resource of the broadcast information can be immediately after the R2D signal or with an L1 R2D control in between.

- Source [280] states that periodic CFO calibration signal is scheduled after the synchronization signal and before the broadcast information. There is a gap between the synchronization signal and the CFO calibration signal, as well as another gap between the CFO calibration signal and the broadcast information.

Source [272], [278] and [281] further provide views on time domain resource for SIB1-like broadcast information

- Source [272] states that, for time resources to receive the SIB-like broadcast information, the start time resource information can either be indicated in the MIB-like broadcast information, or be fixed in relation to the periodic synchronization signal/MIB-like broadcast transmission.

- Source [278] states that, for the SIB1-like broadcast information, the time resources can be indicated by MIB-like part or scheduled by the corresponding L1 R2D control information

- Source [281] states that, it is beneficial to support periodic SIB1-like transmitted separately from MIB-like broadcast information. The periodicity of SIB1-like broadcast information can be equal to or larger than that for AIoT-SSB.

6A.1.6.5 Frequency location

Sources [271], [272], [273], [275], [279], [281], [274], [292], [293], [285] and [286] state that the same frequency resource is used for broadcast information as the periodic synchronization signal.

- Source [272] states that the device has only acquired the frequency synchronization of the sync raster on which it detected the periodic synchronization signal, and same frequency resource as periodic synchronization signal applies to MIB-like broadcast information.

- Source [274] states that a device can assume the same center frequency for receiving both broadcast information and the periodic sync signal, as long as IF/ZIF receiver requirements are met.

- Source [281] state that M value should be $\leq$6 for periodic sync signal and PRDCH for broadcast to ensure applicability to deployment in 1 RB Tx BW.

Source [285] suggests to study whether system supports multiple frequency positions for sync signal (FDM among sync signals), and whether system supports multiple frequency positions for broadcast information (FDM among broadcast information), e.g. to support a raster-based search like sync/MIB acquisition.

Source [272] states that, for frequency resources to receive the SIB-like broadcast information, the frequency resource information can either be indicated in the MIB-like broadcast information, or be fixed in relation to the periodic synchronization signal/MIB-like broadcast transmission.

Source [289] states that, upon synchronizing via signals broadcast on initial frequency, devices acquire system information that enumerates available frequencies for further reception/transmission.

6A.1.6.6 Periodical nature of transmission

Source [281] states that the transmission occasion for broadcast information is strictly periodic.

Source [292] states that periodic transmission should be considered as baseline, and aperiodic transmission can provide flexibility for reader while it may lead to additional workload, e.g., on how much time tolerance can be allowed for broadcast information transmission. Source [292] further state that the periodicity of broadcast information, up to a few hundreds of ms can be feasible depending on energy storage assumption.

Source [285] states that it is preferred not to limit the broadcast information being strictly periodic at reader side.

Source [293] states that RAN1 may only need to specify a default periodicity assumption from device perspective for the R2D broadcast information. It is unnecessary to further discuss whether R2D broadcast information is strictly periodic at reader in RAN1.

6A.1.6.7 Transmission parameters

Source [271], [272], [273], [274], [279], [280], [281], [286], [288], [289], [285] and [284] state that L1 R2D control information is not needed for broadcast information transmission, and parameters (e.g., M value, payload size, repetition, FEC related parameters) can be fixed and predefined. Hence, L1 control information is not needed.

- Source [272], [281] and [288] state that fixed and predefined parameter applies for MIB-like broadcast information for the case broadcast information is split into MIB and SIB.
- Source [273] and [279] state that fixed and predefined parameter simplifies device implementation.
- Source [275] also state that the M value, the number of repetitions, and the code rate are strictly not necessary and can cater to coverage-edge cases, i.e., corresponding to the values that provide the best coverage.
- Source [275] state that the need for variable TBS can be discussed, e.g., considering potential needs to increase the contents of broadcast information in a future release. If needed, the TBS can be indicated via L1 R2D control information or postamble. Otherwise, the TBS can be fixed and predefined.

For parameters for SIB1-like broadcast information when broadcast information is split into MIB-like broadcast information and SIB1-like broadcast information

- Source [272] state that the L1 R2D control information for the PRDCH carrying the SIB-like broadcast information is used to indicate information such as the TBS, FEC code rate and repetition number to save the indication overhead of the MIB-like broadcast information.
- Source [281] state that L1 R2D control information is needed for SIB1-like broadcast information, and the parameters (including parameters of L1 control information of SIB1-like broadcast information, the start time offset and periodicity) can be predefined and/or indicated by MIB-like broadcast information.
- Source [288] state that if broadcast information is split into MIB and SIB, the transmission parameters of SIB may be fixed and predefined or indicated via MIB.

Source [293] state that for system information broadcast messages, L1 control is useful to indicate the transmission parameters, such as TBS and relevant fields for subsequent system information (SIB) within the same PRDCH. Device can perform SIB monitoring with a predefined periodicity and detect SIB scheduled by the L1 control in the same PRDCH, which is more straightforward and potentially more efficient than using a separate MIB.

Source [292] state that for each parameter of the PRDCH with broadcast information, such as M value, TBS, etc., how much flexibility should be ensured can be further studied. Opt.1) fixed value is applied, Opt.2) indicated by R2D L1 control with/without restriction on applicable values.

Source [280] state that both broadcast information with or without L1 control can be studied. The code method, code rate, chip duration, and repetition can be a fixed and L1 control is not needed. The flexibility can be improved if L1 control is available for broadcast information, the chip duration of L1 control part in broadcast information, the chip duration of L1 control part is a fixed value.

6A.1.7 R2D multiplexing

FDM for R2D from the same reader is studied, considering the complexity of a device required to support FDM, and how a device determines the frequency resource to receive the FDMed R2D transmissions.

In assessing feasibility of such FDM:

- Sources [241], [242], [246], [265], [280], [299] state that R2D FDM is feasible for Device 2b and for Device C, considering device equipped with IF/ZIF filters at the receiver.

- Source [247] states that the total number of PRBs required to multiplex devices using FDM needs to be defined. Source [275] further states that if multiplexing UEs using FDM requires a larger transmission bandwidth (ergo channel bandwidth) than in Rel-19, then to minimize the specification impacts, the smallest channel bandwidths available in NR i.e., 3 MHz CBW (15-PRBs) and 5 MHz CBW (25-PRBs) can be considered.

- Source [281] states when A-IoT R2D bandwidth is less than 4 RBs, it is not feasible to support R2D FDM for device 2b with large residual CFO, i.e., 100ppm.

In assessing necessity of such FDM:

- Source [243] reports that the transmission resource efficiency by using R2D FDM cannot be improved compared with R2D non-FDMA, when AIoT R2D bandwidth is no larger than 4 RBs.

- Source [256] states that:

  - If it allows all devices to receive the initial paging message without FDM, while enabling efficient sub-band utilization using FDM for following messages, it will introduce complexity in managing mode transitions between FDM and non-FDM, and coordinating sub-band usage.

  - If it enables device-specific optimization and parallel reception across sub-bands, it will increase transmission resource usage due to duplication of paging messages.

- Sources [243], [247], [248], [253] and [266] state that guard band between FDMed transmissions need to be allocated to compensate the impact of CFO.

- Source [265] states that for R2D, FDM can be beneficial to extend multiplexing capacity. Sources [254] and [264] state that in A-IoT systems, the capacity bottleneck lies on the R2D link. And to enhance system capacity, FDM is a potential solution.

Regarding device complexity to support R2D FDM from the same reader for Device 2b and for Device C

- Sources [245], [249], [251] and [259] state that R2D FDM will increase the device complexity since the adjustment of filter bandwidth and/or IF center frequency is required.

- Source [253] raises the related issue that whether different FDMed frequency resources can be supported by the same device, and if so, how the Rx filter architecture of IF/ZIF receivers can support such flexible determination.

- Sources [254], [246] and [248] state that additional complexity is limited since device with IF/ZIF receiver can already support to distinguish different frequency resources.

Regarding how a device determines the frequency resource for receiving one of the FDMed R2D transmissions:

- Sources [241], [246], [242] and [264] state that the frequency resource of synchronization signal, CFO calibration signal or broadcast information need to be fixed or predefined.

- Sources [246], [242], [264], [265], [248], [254], [252] and [279] state that some of R2D transmissions e.g., L1 control information or a R2D transmission other than sync signal or broadcast information can determine their frequency resources based on the indication.

- Source [252] state that the frequency resource for data payload is indicated by corresponding L1 R2D control.

- Source [248] and [265] state that the frequency resource of a R2D transmission can be associated with the previous corresponding D2R transmission.

- Source [248] state that the exact frequency resource/channel for a R2D transmission can be associated with a specific coverage level from a set of candidates R2D frequency resources preconfigured.

- Source [287] states that a Device 2b or Device C is not required to receive more than one of the FDM'ed R2D transmissions.

6A.1.8 R2D control information

6A.1.8.1 Content

In addition to TBS for PRDCH data payload, the following L1 R2D control information, which potentially can be indicated to the device, is studied:

- M value for PRDCH data payload chip duration, as reported by [243], [264], [245], [252], [241], [247], [265], [244], [248], [249], [250], [251], [254], [255], [256], [257], [259], [260], [266] and [268].
  - On the other hand, sources [242], [246], [247] and [269] suggest to use same chip duration between L1 R2D control information and data payload, hence M value indication is not needed in L1 R2D control.
- Repetition/FEC related parameters for PRDCH data payload, as reported by sources [243], [264], [246], [245], [252], [241], [242], [247], [265], [248], [249], [250], [254], [255], [256], [259], [260], [266], [268] and [294], if Repetition/FEC is supported for R2D.
- Timing related information for PRDCH data payload, as reported by sources [264], [245], [252], [244], [255], [256], [260] and [268], considering potential need of gap between L1 R2D control and corresponding data payload.
  - On the other hand, sources [243], [246], [253], [242], [265], [248] and [254] state that, this information is not included in L1 R2D control, because of no need of gap between L1 R2D control and corresponding data payload.
- Timing gap between R2D and the corresponding D2R transmission, as reported by [264]
- Frequency resource for PRDCH data payload, as reported by sources [247], [263] and [268], if FDM for R2D is supported.
- Message types/Functional indication, as reported by Sources [264], [243], [252], [253], [241], [244], [251], [254] and [245], for the purpose of early termination of R2D detection. Source [293] state that it can be used to indicate which system information is transmitted. Besides, source [285] state that if message type is included in L1 R2D control information, TBS is not necessary to be explicitly indicated in L1 R2D control information.
- ID related information, which potentially include at least one of reader ID, device ID or Transaction ID related information
  - ID related information associated with device(s)
    - Sources [241], [243], [264], [252], [253] and [266] state that, ID related information associated with device(s), indicating for which device or group of devices the upcoming data is intended, can be included in L1 R2D control for early termination.
  - Reader ID related information, if needed, as reported by source [264].
- Indication of next R2D transmission, as reported by source [241], [294] and [284], used to indicate when the next R2D transmission, can help non-intended devices have the opportunity to harvest energy. Source [284] states that starting time for PRDCH for multi PRDCH scheduling should be indicated.
- D2R resources to send ACK/NACK feedback for unicast R2D command reception, as reported by source [293].
- Frequency adjustment information indicated to A-IoT device for CFO calibration, as reported by source [281].
- Indicator relevant to DO-A access, as reported by source [277]. And Indicator relevant to first D2R of scheduling request for DO-A, as reported by source [289].
- Indication of round number in access trigger, as reported by [293].

6A.1.8.2 Time location

L1 control information precedes the data payload of the PRDCH.

Option 1: L1 R2D control information and data payload of PRDCH are contiguous in time.

- Sources [253] and [259] state that, transmit L1 R2D control information and data payload in the same PRDCH with contiguous time resources and same frequency resource is beneficial for low device complexity. [259] further report that this simplifies the timeline design.

- Sources [243] and [248] report that the time gap is not long enough to introduce gap between L1 R2D control information and data payload. Sources [246], [248] further states that decoding of L1 R2D control information the device can parallelly store the data payload in buffer, and receive L1 R2D control information, hence time gap is not needed. Source [281] provide the analysis with reference show that the decoding latency of L1 control signaling (assuming ~30bits payload size) is a few tens of us, less than one OFDM symbol. The memory required for buffer signal during one OFDM symbol is limited(~2000bits), and feasible even for passive device, according to product datasheet, and also feasible for active device.

- Sources [243], [254], [273] and [276] report that L1 R2D control transmitted in the same frequency as data payload in PRDCH, and time gap is not needed.

- Source [256] report that no time gap needed if the chip length for L1 R2D control information is the same as that for data payload of PRDCH. Source [281] state that different chip length for L1 control and data part does not require clock rate change.

Option 2: L1 R2D control information and data payload of PRDCH are not contiguous in time.

- Sources [241], [247] and [251] report that a short time gap is needed if the device is not capable of decoding L1 R2D control and buffering data payload of PRDCH.

- Source [245] state that, L1 R2D control and data payload are transmitted in separated physical channel, hence they are not contiguous.

- Source [252] state that, time gap is needed to switch the frequency if L1 R2D control is transmitted in different frequency from data payload.

- Source [241], [256] report that time gap maybe needed if the chip length for L1 R2D control information is different from that for data payload of PRDCH.

- Source [264] report that, if there is a gap, midamble can be inserted between L1 R2D control information and data payload.

- Source [290] and [294] time gap is needed for flexibility for R2D data payload transmission in time.

Factors affecting the choice between the above options were reported as follows:

- Whether device buffers data part before finishing L1 control decoding, as reported by sources [243], [246], [253], [242], [265], [245], [252], [247], [264], [248], [244], [251], [254] and [259].

  - Sources [243], [246], [253], [242], [265], [248] and [254] state that gap between L1 R2D control and data is not needed, assuming device can buffer data part before finishing L1 R2D control decoding. Source [246] further state that the time gap would increase the power consumption of the device.

  - Sources [245], [252], [251], [277], [290], [291] and [294] state that time gap is needed to reduce/avoid buffer at AIoT device.

- Whether frequency resource is different for L1 R2D control information and data payload, as reported by sources [252], [244], and [294].

- Whether chip duration (M value) is different for L1 R2D control information and data payload, which is reported by sources [241], [246] and [256]

- Whether L1 R2D control information and data payload are separate channel, as reported by source [245]

- Whether CFO calibration signal is placed between L1 R2D control information and data payload, as reported by [258].

- Whether flexibility for time location for PRDCH data payload in relation to L1 R2D control information is necessary, as reported by [258], [281], [290] and [294].

6A.1.8.3 Frequency location

Sources [241], [242], [243], [246], [253], [248], [254], [259], [264], [279], [282], [290] and [276] report that the same frequency resource for R2D preamble, L1 control and data part should be used.

- Source [241] states that different frequency resource will cause additional latency as the transmission carrying control information must be first received followed by receiving the transmission carrying the payload. And unclear of the necessity of this option if FDM is not supported

- Sources [243], [246], [279] and [276] state that frequency retuning can be avoided if L1 control information and the corresponding data payload are located with same frequency locations. Otherwise, it will lead to additional adjustment delay and power consumption.

- Sources [242] and [243] state that the motivation of transmitting L1 control and data payload in different frequency is not clear. Source [290] states that little benefit is expected in a PRDCH center frequency different from the center frequency of the L1 R2D control channel that configured it.

Source [252] reports that frequency resource for R2D preamble, L1 control and data part can be different such as to reduce the load in the frequency resource used for L1 control.

6A.1.8.4 FEC and repetition

Regarding FEC and repetition for L1 R2D control part, sources [243], [252], [245], [241], [248], [249], [250], [254], [264] and [272] report that FEC can also be applied to L1 R2D control information. Source [241] and [249] report that FEC can be applied to L1 R2D control information.[246]

Sources [243], [245], [248], [252], [250], [254] and [264] report methods to determine FEC, repetition related parameters for L1 R2D control information, if FEC/repetition is supported for L1 control.

- Sources [243], [245], [248], [254] and [264] report that the parameters for repetition and FEC for L1 R2D control information can be pre-configured, e.g., by broadcast information. [252] report that these parameters should be pre-known by device.

- Sources [248] and [250] states that parameters for repetition and FEC for L1 R2D control information can be fixed and pre-defined. Source [248] also state that these parameters can be indicated by a set of different binary sequences with fixed or variable length for indication. Source [248] report that a separate interleaving is needed for L1 R2D control.

- Source [272] states that The FEC and repetition parameters of the L1 R2D control information of the SIB-like broadcast information is predefined or indicated by the MIB-like broadcast information. The FEC and repetition parameters of the L1 R2D control information after the SIB-like broadcast information is indicated by the SIB-like broadcast information.

6A.1.8.5 Payload size of control

Sources [243], [252], [247], [259] and [282] report that device only detects one payload size at a given time. And [243] also report that up to two payload sizes can be considered and detected by preamble. Source [293] state that two different sizes for L1 control can be considered for unicast and non-unicast. Source [279] states that only one payload size for L1 control information should be considered to reduce the reception complexity.

Source [272] state that the size of the L1 control information is 10~12 bits considering at least the FEC code rate, repetition number and TBS. Source [285] state that R2D L1 control information is variable, and within 24 bits for 6-bit CRC attachment. Source [280] state that L1 control occupies one or multiple full OFDM symbols.

6A.1.8.6 Physical channel

No new R2D physical channel for L1 R2D control information is introduced.

Sources [243], [264], [246], [253], [241], [265], [248], [256], [259], [260] and [273] state that L1 R2D control is transmitted in the same PRDCH carries data payload for simplicity for standard and device.

- [246] states if L1 control and data are in different PRDCH, the time relationship design is complicated and the device needs to buffer the L1 control information for a long time which is hard for the device to implement.

- Source [283] state that the coding/repetition scheme for the control information is like the data payload, and also no specific reference RS would be dedicatedly introduced for the control information.

Sources [254], [245], [255], [257], [261] and [279] state that L1 R2D control information is transmitted in a separate PRDCH

- Source [254] reports that, L1 R2D control information is encoded separately, it is more appropriate to transmit L1 R2D control information on a separate PRDCH

- Sources [245] and [255] state that, defining a new PRDCH format to carry L1 R2D control information only is more efficient for scheduling and resource utilization.

- Source [257] states that, L1 control information is transmitted in separated PRDCH since control part would require higher reliability, and one control information could be used for multiple PRDCH transmissions.

- Source [279] state that separated transmission should be adopted if option 2(non-contiguous) for time location of L1 R2D control information and R2D data payload.

6A.1.8.7 CRC

R2D control information with a separate CRC from data is studied.

Source [241] states that, for separate CRC on the L1 R2D control information, when the size of the L1 R2D control information is 24 bits or less, the existing CRC6 specification can be reused, if the size of the L1 R2D control information is between 25 and 57 bits, the existing CRC6 polynomial can be reused without impact to the undetected error rate.

Sources [251], [259], [272] and [282] state that how to attach the separate CRC for L1 R2D control information needs to be considered, and a fixed length (equal to or larger than 16) of CRC is required to ensure the reliability and avoid unnecessary blind detection for L1 R2D control information.

Source [275], [281], [293] and [295] suggest to use the same rule as R19, 6 bits for < 25 TBS and 16 bits otherwise. Source [281] state that this provides sufficient CRC protection and reasonable overhead. Source [285] suggest to always use 6 bits, assuming no larger than 24 bits payload for L1 control information.

6A.2 D2R

6A.2.1 D2R waveform, modulation and bandwidth

6A.2.1.1 Waveform

A single-carrier waveform is assumed for the study. The Rel-19 small frequency shift described in Clause 6.1.2.7.2 of the present document, and Clause 6.1.4 of TS 38.291 [228] is not necessary in Rel-20.

1SB and 2SB waveform options are studied, with the following descriptions and analyses, illustrated in Figure 6A.2.1.1-1. The Rel-20 A-IoT device supports 2SB waveform.

1SB

- Source [301] reported that 1SB signal has only one side band, i.e., lower side band or upper side band, on either side of center frequency.

- Sources [303], [301], [300], [335], [312]described 1SB signal generation methods based on phase shift, i.e., Hilbert transform of 2SB baseband signal and filtering of one side band signal from 2SB signal.

- Sources [300], [301], [305], [321] reported that filtering method could be challenging since two sides are adjacent to each other and it is hard to realize filter with steep roll off.

- Sources [305], [301], [312], [321] reported that phase shift method could be challenging because it requires a precise 90-degree phase-shift network, which is non-ideal and highly sensitive to circuit matching and stability.

- Sources [306], [304] reported that Tx LO re-tuning to different frequency points may require additional LO retuning/stabilization time and/or power, especially when frequent change is needed.
- Source [305] reported that BLER performance is more sensitive to phase errors and frequency offset.
- Sources [313], [335], [336], [325], [324], [318] reported 1SB is spectrum efficient than 2SB, but it is more complex requiring precise filtering.
- Sources [327] reported that single sideband output might require the device to have an I/Q modulator or a filtering mechanism to suppress the image frequency.
- Source [330] reported that implementing such a [328] filter for generating 1SB signal would not be practical for A-IoT devices.
- Source [314] reported that based on the ability of Device 2b/C, the 1SB modulation should be considered to improve the spectrum utilization.

2SB

- Source [301] reported that 2SB signal has two sidebands, i.e., lower sideband and upper sideband, on either side of center frequency.
- Sources [306], [305], [300], [302], [307], [303], [335] pointed out that the transmission signal bandwidth of 2SB signal is two times that for 1SB signal in transmitting the same number of bits; 2SB has lower spectral efficiency than 1SB.
- Source [305] reported that 2SB modulation is more tolerant to phase error and CFO, offering relatively more robust demodulation performance, due to its symmetric spectrum structure.
- Source [306], [304] reported that Tx LO re-tuning to different frequency points may require additional LO retuning/stabilization time and/or power, especially when frequent change is needed.
- Sources [313], [335], [336], [324] reported that 2SB is simpler to implement/demodulate, yet requires larger bandwidth resulting lower spectral efficiency compared to 1SB.
- Source [325] reported that 2SB transmission provides optimal balance between implementation complexity and spectral efficiency.

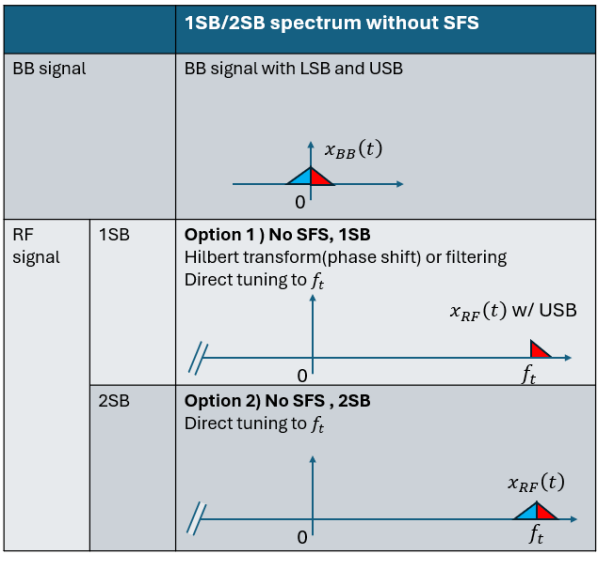


Figure 6A.2.1.1-1: Illustration of 1SB and 2SB spectrum characteristics of D2R waveform without small frequency shift.

6A.2.1.2 Modulation

6A.2.1.2.1 Schemes

The reuse, modification or enhancement of the D2R modulation schemes defined in TS 38.291 [228], including down-selection among them, and the potential introduction of new modulation schemes are studied. The following are the candidate modulation schemes:

BPSK: The input bit "1" is mapped to the real-valued modulation symbol "1" and input bit "0" is mapped to the real-valued modulation symbol "-1" in the baseband.

OOK: The input bit "1" is mapped to the real-valued modulation symbol "1" and input bit "0" is mapped to the real-valued modulation symbol "0" in the baseband.

DBPSK: The n-th input bit b_n is differentially encoded as $d_n = b_n \oplus d_{n-1}$, where d_n and d_{n-1} are the n-th and $(n-1)$th differentially encoded input bit and $\oplus$ is XOR operation. Then, the differentially encoded bits d_n are modulated by BPSK.

MSK: The input bit "1" is mapped to frequency f_1 and input bit "0" is mapped to frequency f_2, with $f_2 - f_1 = \frac{1}{2T}$, where T is a symbol duration.

6A.2.1.2.2 Performance

The following observations were reported regarding the performance of each modulation scheme.

BPSK

- [317], [319] reported that BPSK provides superior detection performance with its coherent detection mechanism.
- [327] BPSK provides lowest SNR for target BLER.
- [330] The BLER performance of the BPSK and DBPSK is not meaningfully different when proper CFO estimation/mitigation is applied at the reader.
- [325] reported that BPSK can provide better coverage than OOK.
- [336] reported that BPSK provide better performance yet at the cost of high overhead for preamble and midamble.
- [320] reported that the required SNR is about -7.7 dB @ 10% BLER and -4.8 dB @ 1% BLER
- [321] reported that the required SNR to achieve the target BLER of 10% and 1% of BPSK modulation was observed as follows: 10%: 1.7 dB and 1%: 6 dB.
- [309] The performance of DBPSK is worse than BPSK with the same reference data rate especially for lower data rate and low SNR region. In detail, the performance gap is ~ 6 dB with reference data rate 0.1 kbps for both BLER 10% and BLER 1%.

OOK

- [319] reported that OOK incurs an inherent theoretical link performance penalty of approximately 6 dB compared to BPSK under the same transmit power.
- [327] reported that Needs ~1–2 dB higher SNR than BPSK under typical conditions. Non-coherent OOK is slightly less power-efficient due to "off" symbols.
- [320] reported that, if Manchester coding is not used, for 96 bits payload size, OOK has more than 5 dB performance loss @1% BLER compared with the OOK case with Manchester coding, and for 400 bits payload size, there is an error floor and 1% BLER cannot be achieved. If Manchester coding is used, the required SNR is about -2.1 dB @ 10% BLER and 1.2 dB @ 1% BLER.
- [318] reported that, for device 2b/C, SNR performance of OOK is worst among all three modulation schemes, and OOK shows obvious performance loss (>3 dB) than BPSK and DBPSK.

- [321] reported that the required SNR to achieve the target BLER of 10% and 1% of OOK modulation was observed as follows: 10%: 11.5 dB and 1%: 15.7 dB.

- [309] reported that the performance of OOK with non-coherent receiver is much worse than BPSK, i.e., ~ 17 dB performance gap with reference data rate 0.1 kbps.

DBPSK

- [317] reported that the convenience of non-coherent detection in DBPSK comes at a cost: the differential detection mechanism is inherently sub-optimal because each decision is based on comparing two noisy symbols, leading to a degradation in detection performance.

- [319] reported DBPSK is more sensitive to noise and requires a higher SNR compared to BPSK for successful demodulation.

- [319] reported the issue of error propagation, which can lead to further performance deterioration of D2R transmission. Theoretically, under high CFO estimation accuracy, DBPSK exhibits a BLER performance degradation of more than 3 dB compared to BPSK.

- [327] reported that DBPSK needs ~3 dB higher SNR than BPSK for the same error rate in theory.

- [320] reported the required SNR is about -5 dB @ 10% BLER and -1.7 dB @ 1% BLER.

- [318] reported that, for device 2b, DBPSK without 1-bit midamble (slightly higher data rate) outperforms BPSK with 1-bit midamble (slightly lower data rate) in which there is ~5dB gain for BLER of 10^-1 and ~1.5dB gain for BLER of 10^-2. For device C, it was reported that SNR performance of DBPSK without 1-bit midamble (slightly higher data rate) is quite close to that of BPSK with 1-bit midamble (slightly lower data rate) and BPSK outperforms than DBPSK ~1 dB at BLER of 10^-1.

MSK

- [308], [319], [327] reported that MSK can have the same/similar detection performance as BPSK under good CFO estimation with coherent detection.

- [314] reported that when the SFO and CFO estimation and compensation are both considered, the BLER performance of MSK would be similar as that of BPSK.

- [320] reported that, for MSK with non-coherent detection, the required SNR is about 0.7 dB @ 10% BLER and 3.7 dB @ 1% BLER.

6A.2.1.2.3 Sensitivity to CFO

The following observations were reported regarding the sensitivity to CFO of each modulation scheme.

BPSK

- [308], [336] reported that BPSK suffers detection performance degradation when there is a carrier frequency offset (CFO) between the actual carrier frequency and expected carrier frequency.

- [319], [327] reported that BPSK is highly sensitive to CFO and its CFO impact can be mitigated by preamble/midamble based CFO estimation and compensation.

- [320] reported that BPSK requires residual CFO $\leqslant$0.3 ppm for about ~1 ms midamble interval is considered to achieve 1dB degradation.

- [318] reported that BPSK is sensitive to phase error (CFO impact) and requires accurate CFO estimation ($\leqslant$ 0.1 ppm).

- [332] reported that BPSK is sensitive to CFO, e.g., frequency error tolerance can be up to 0.5ppm or less. Sensitivity to CFO leads the performance impact and demodulation complexity at reader.

- [309] reported that enhancements on preamble and midamble can ensure good performance with coherent detection and address the impact of CFO.

OOK

- [308], [320] reported that OOK modulation provides robust performance against CFO due to its non-coherent envelope detection method.
- [319], [336], [332] reported that OOK is more robust to CFO than (D)BPSK and MSK.
- [318] reported that OOK is more robust to phase error (CFO impact) and the tolerable residual CFO can be up 100 ppm.

DBPSK

- [319], [327], [325], [336] reported the medium sensitivity; it is more tolerant to CFO than BPSK but less than OOK.
- [317] reported that given that the phase rotation caused by CFO between adjacent symbols is relatively small, DBPSK can tolerate large CFO by using non-coherent detection.
- [320] reported that DBPSK is not sensitive to CFO and tolerance can be increased by decreasing bit duration.
- [318] reported that DBPSK is robust to phase error (CFO impact) and the tolerable residual CFO can be up 1.5 ppm.

MSK

- [308], [336] reported that detection performance with large CFO is unclear since it is challenging differentiate two tones at receiver side with CFO.
- [319] reported that MSK has high sensitivity to CFO. The CFO impact can be significantly mitigated through X-amble-based D2R CFO estimation.
- [327] reported that CFO must be calibrated to achieve the ideal performance.

6A.2.1.2.4 Preamble/midamble design and overhead

The following observations were reported regarding the preamble/midamble design and overhead of each modulation scheme.

BPSK

- [308] The D2R preamble and midamble sequences should be designed to support CFO estimation at the receiver of the reader to maintain BPSK reception performance.
- [317] BPSK demands additional overhead in the amble design to enable accurate CFO estimation and accurate channel estimation.
- [319] reported that large X-amble overhead is needed to ensure high precision CFO estimation
- [327] reported that for a 96-bit payload plus 16-bit CRC (112 bits total), a preamble on the order of 8–16 bits might be used, corresponding to roughly 7–15% overhead.
- [336] reported that BPSK requires overhead of around 30% to provide reasonable performance
- [320] reported that the required x-amble overhead is 25.58% for 1kbps data rate, 96bits of payload.
- [309] reported that BPSK requires 39% to 21% of overhead for data rate of 0.1kbps to 5~7kbps.
- [318] reported amble overhead of 9%~17% for ~1 kbps and 96bits of payload.

OOK

- [319], [327], [336] reported that overhead for OOK can be low (e.g., on the order of ~5–10% of the payload bits) since no complex CFO/channel estimation is needed.
- [320] reported that amble overhead of 2.18% for ~1 kbps and 96bits of payload.
- [309] reported that OOK requires 4% of overhead (preamble only, no midamble) for data rate of 0.1kbps.
- [318] reported amble overhead of 8%~16% for ~1 kbps and 96bits of payload.

DBPSK

- [319] reported that DBPSK needs medium X-amble overhead CFO estimation.
- [327], [336] reported that the overhead for DBPSK is mainly for packet detection and timing sync, similar to OOK. This might be on the order of ~5–10% overhead.
- [320] reported that required amble overhead is 2.18% for 1kbps of data rate and 96bits of payload.
- [318] reported amble overhead of 8%~16% for ~1 kbps and 96bits of payload.

MSK

- [317] noted that accurate CFO estimation and channel estimation are not needed and amble design overhead is reduced.
- [319] reported that MSK requires large X-amble overhead to ensure high precision CFO estimation.
- [327] reported that MSK's overhead could be higher to confidently calibrate CFO and phase before demodulation, e.g., on the order of around 15–20% due to its high sensitivity to CFO.
- [320] reported that the phase of midamble depends on previous data bit, which is unknown before decoding and coherent detection based on channel estimation with midamble cannot be performed for MSK because of midamble phase ambiguity in MSK modulation. The MSK's amble overhead is 2.18%.
- [309] reported that it's not clear for MSK how the channel estimation can be performed based on midamble due to the nature of phase continuity between data and midamble bit.
- [314] reported that reader could detect the midamble at the specified location in PDRCH with the compensation of timing, frequency and channel response using the estimates obtained from preamble. The midamble can be coherently demodulated to obtain the phase shift of MSK from the previous PDRCH chip after channel compensation. The estimated phase offset contains the possible initial phase offset and phase offset caused by non-ideal factors.

6A.2.1.2.5 Transmitter complexity

The following observations were reported regarding the transmitter complexity of each modulation scheme.

BPSK

- [332] reported that BPSK has lower PAPR than OOK and its PA efficiency can be improved.

OOK

- [308], [319], [336] reported that the OOK transmitter offers lower complexity and low power consumption.
- [314] reported that, from the A-IoT device implementation point of view, the OOK modulation could be the better choice than BPSK modulation for its lower power/energy consumption and lower complexity.

DBPSK

- [319], [336] reported that DBPSK transmitters require additional XOR operations, slightly increasing complexity compared to BPSK.

MSK

- [308] reported that the higher complexity in transmitter implementation makes it an unattractive solution for low power A-IoT devices.
- [317] reported that optimal performance of MSK depends on a specific modulation index of exactly 0.5, which ensures the two frequencies are perfectly distinguishable and phase continuity is maintained. However, maintaining this exact 0.5 value is challenging for low-cost IoT devices like Device 2b/C due to inherent hardware limitations.
- [319], [325] reported that transmitters require high-precision frequency shift and phase continuity control, leading to a high complexity.

- [314] reported that MSK modulation can be designed to avoid the need for a power consuming external or explicit RF mixer by using direct digital frequency synthesis or other techniques.

- [320], [309] reported that OQPSK based MSK transmitter with square pulse replaced by half-cycle sinusoid pulse in I/Q path has following challenges; I/Q imbalance (intermodulation and degradation of EVM), inaccurate control of T/2 delay (phase continuity), quadrature receiver.

6A.2.1.2.6 Receiver complexity

The following observations were reported regarding the receiver complexity of each modulation scheme.

BPSK

- [317], [319], [336] reported that BPSK requires coherent detection and channel estimation to correct the phase rotation induced by CFO resulting in higher receiver complexity.

- [309] report that receiver complexity caused by the CFO estimation can be reduced with e.g., FFT-based method.

OOK

- [319], [336] reported that OOK has lower receiver complexity due to relatively simple CFO estimation and non-coherent detection compared to BPSK, DBPSK, and MSK.

- [320] reported that OOK modulation with non-coherent detection requries Manchester coding for reliable threshold training.

DBPSK

- [319], [336] reported that medium receiver complexity is required mainly due to the need for moderately accurate CFO estimation and compensation.

- [320] reported that the CFO estimation complexity for BPSK modulation is much higher than DBPSK; BPSK requires 63bits preamble with 106/1053 hypotheses for device C/2b and DBPSK requires 15bits preamble with 28/274 hypotheses for device C/2b.

MSK

- [319] High receiver complexity due to the need for precise CFO estimation and compensation

- [309] reports that MSK has not been implemented in NR BSs, hence, the implementation complexity at the reader side is high.

- [314] reported that MSK modulation has better BLER performance than BPSK modulation for the less SFO sensitivity, when the SFO is considered. BPSK requires more precise SFO estimation than MSK.

6A.2.1.2.7 Spectrum

The following observations were reported regarding the spectrum of each modulation scheme.

BPSK

- [319], [336] reported that The D2R transmission bandwidth is 1/Tb for single-sideband (1SB) and 2/Tb for double-sideband (2SB). A Tx filtering for spectrum is required to suppress out-of-band emission.

- [327] reported that BPSK's occupied bandwidth is relatively tight but its sidelobes are higher than MSK's unless additional filtering is used.

- [309] reported that filter can be further applied to reduce the leakage.

OOK

- [319] reported that the D2R transmission bandwidth is 1/Tb for 1SB and 2/Tb for 2SB. A Tx filter is required to suppress out-of-band emission.

- [327] reported that the on/off pattern can create spectral lines at the symbol rate and substantial side lobes and its main lobe bandwidth is roughly similar to BPSK for a given bit rate, but the side lobes are typically even higher.
- [309] reported that filter can be further applied to reduce the leakage.

DBPSK

- [327] reported that similar spectral characteristics to BPSK.
- [319], [336] reported that DBPSK requires a Tx filter to suppress out-of-band emission.
- [309] reported that filter can be further applied to reduce the leakage.

MSK

- [327] reported that MSK inherently produces a narrow power spectrum concentrating energy in a very compact main lobe.
- [336] reported that MSK has the lowest sidelobe in its spectrum and may need less stringent tx filtering.
- [314] reported that MSK has narrower spectral occupancy than that of BPSK.
- [320] reported that tx filter may be needed anyway to meet RF requirement.
- [309] reported that MSK has the best performance.
- [319] reported that the mainlobe bandwidth of MSK is similar to that of BPSK/OOK/DBPSK due to its phase continuity. A Tx filter is required to suppress out-of-band emission.

6A.2.1.2.8 Summary assessments

The following summary assessments of the modulation schemes were provided by companies.

Table 6A.2.1.2.7-1: Source [308].

	BPSK	OOK	DBPSK	MSK
Sensitivity to CFO	High	Not sensitive	Medium	High
Required SNR Performance	Theoretical BER performance is 6 dB better than OOK	Required SNR = 14.3dB @ 10% BLER Required SNR = 3.0 dB @ 1% BLER	Theoretical BER performance is 3 dB worse than BPSK	Theoretical BER performance is about same as BPSK
Receiver detection type	Coherent	Non-coherent	Non-coherent	Coherent
Overhead	Medium	Low	Medium	Unclear
Spectrum	The power spectral density of BPSK has higher side lobes	The power spectral density of OOK has higher side lobes	DBPSK requires a larger bandwidth to transmit	The power spectral density of MSK has lower side lobes
Transmitter complexity	Medium	Low	Medium	High
Receiver complexity	Medium	Low	Medium	Medium
Note 1: X-amble overhead is computed for (96+16) bits payload				

Table 6A.2.1.2.7-2: Source [317].

	BPSK	OOK	DBPSK	MSK
Sensitivity to CFO	sensitive	roubust	robust	sensitive
Required SNR Performance		Theoretical BER performance is 6 dB worse than BPSK	Theoretical BER performance is 3 dB worse than BPSK The differential detection mechanism is inherently sub-optimal compared with BPSK, because each	

			decision is based on comparing two noisy symbols, leading to a degradation in detection performance.	
Receiver detection type	Coherent only	Non-coherent	Non-coherent	Coherent/non-coherent
Transmitter complexity	Medium complexity. [Differential LO.]	Lower complexity	Medium complexity [Differential encoding. Differential LO]	Maintaining a specific modulation index of exactly 0.5 is challenging for low-cost IoT devices like Device 2b/C due to inherent hardware limitations.
Receiver complexity	Additional overhead in the amble design to enable accurate CFO estimation and accurate channel estimation. Supporting coherent detection is not so complex for reader.	Lower complexity for non-coherent detection.	Accurate CFO estimation and channel estimation are not needed, and amble design overhead is reduced. Phase estimation is still required for differential detection.	Accurate CFO estimation / compensation required for coherent detection increasing receiver complexity e.g., large # of hypothesis needed
Note 1: X-amble overhead is computed for (96+16) bits payload				

Table 6A.2.1.2.7-3: Source [319].

	BPSK	**OOK**	**DBPSK**	**MSK**
Sensitivity to CFO	High	Low	Medium	High
Receiver detection type	Coherent	Non-coherent	Non-coherent	Coherent
Required SNR Performance	Required SNR = [-3.5] dB @ 10% BLER, [2.5] dB @ 1% BLER for TBS 96 bits	Theoretical BER performance is 6 dB worse than BPSK	Theoretical BLER performance is >3 dB worse than BPSK	Similar to BPSK
X-amble overhead	High	Low	Medium	High
Spectrum	Need Tx filter; D2R transmission bandwidth: 1/Tb for 1SB, 2/Tb for 2SB	Need Tx filter; D2R transmission bandwidth: 1/Tb for 1SB, 2/Tb for 2SB	Need Tx filter; D2R transmission bandwidth: 1/Tb for 1SB, 2/Tb for 2SB	The mainlobe bandwidth of MSK is similar to that of BPSK/OOK/DBPSK due to its phase continuity. A Tx filter is required to suppress out-of-band emission.
Transmitter complexity	Low	Low	Higher than BPSK due to additional XOR operations	High complexity caused by high-precision frequency shift and phase continuity control
Receiver complexity	High complexity due to high CFO estimation accuracy requirement. Supporting coherent detection is not so complex for reader.	Low complexity for low CFO estimation accuracy requirement and non-coherent detection.	Medium complexity for CFO estimation. Supporting coherent detection is not so complex for reader.	High complexity due to high CFO estimation accuracy requirement. Supporting coherent detection is not so complex for reader.

Table 6A.2.1.2.7-4: Source [327].

	BPSK	**OOK**	**DBPSK**	**MSK**
Sensitivity to CFO	High	Low	Medium	High

Required SNR Performance	Baseline and serves as 0dB reference	Required SNR = [1-2] dB @ 10% BLER Theoretical BER performance is 6 dB worse than BPSK	Required SNR = [3] dB @ 10% BLER Theoretical BER performance is 3 dB worse than BPSK	Required SNR = [0] dB @ 10% BLER for coherent detection Required SNR = [2-3] dB @ 10% BLER for non-coherent detection
Receiver detection type	Coherent only	Non-coherent	Non-coherent	Coherent/non-coherent
Overhead	[10-15] % (Note 1)	[5-10] % (Note 1)	[5-10] % (Note 1)	[15-20] % (Note 1)
Spectrum	Moderate bandwidth (main lobe ≈ 2× bit rate).	Broad spectrum due to on/off transients.	Similar to BPSK in spectral shape.	High spectral containment.
Transmitter complexity	Medium complexity. [Differential LO.]	Lower complexity	Medium complexity [Differential encoding. Differential LO]	Higher transmitter complexity (modulation index, rapid frequency shift or IQ branches needed)
Receiver complexity	Higher CFO estimation / compensation complexity (e.g., large # of hypothesis needed). Supporting coherent detection is not so complex for reader.	Lower complexity for non-coherent detection.	Medium complexity for CFO estimation / non-coherent detection. Phase estimation is still required for differential detection.	Accurate CFO estimation / compensation required for coherent detection increasing receiver complexity e.g., large # of hypothesis needed
Note 1: X-amble overhead is computed for (96+16) bits payload				

Table 6A.2.1.2.7-5: Source [336].

	BPSK	**OOK**	**DBPSK**	**MSK**
Sensitivity to CFO	BPSK is quite sensitive to CFO. CFO estimation / compensation is necessary for successful decoding. Residual CFO should be handled by channel estimation	Very robust. With non-coherent reception, performance is not sensitive to CFO.	Robust Due to relative phase detection, detection performance is not sensitive to CFO.	Very sensitive (for coherent).
Required SNR Performance	Required SNR = [8] dB @ 10% BLER Required SNR = [18.5] dB @ 1% BLER Note 2 Note 3 noise BW = 60kHz Required SNR = [X] dB @ 10% BLER Required SNR = [X] dB @ 1% BLER	Required SNR = [17.5] dB @ 10% BLER Required SNR = [28] dB @ 1% BLER Note 2 Note 3 noise BW = 60kHz Required SNR = [X+3.5] dB @ 10% BLER Required SNR = [X+6.5] dB @ 1% BLER	Required SNR = [13.5] dB @ 10% BLER Required SNR = [23] dB @ 1% BLER Note 2 Note 3 noise BW = 60kHz Required SNR = [X-1] dB @ 10% BLER Required SNR = [X]dB @ 1% BLER	TBD
Receiver detection type	Coherent only	Non-coherent	Non-coherent	Coherent
Overhead	[ ]% for midamble (Note 1) Preamble = 63 Midamble length = 4 Midamble interval = 12	[ ]% for midamble (Note 1) Preamble = 63 Midamble length = 2 Midamble interval = 48	[ ]% for midamble (Note 1) Preamble = 63 # of Midamble = 0 Midamble overhead = 0%	[X-Y] % (Note 1)

	Midamble overhead = 25% # of DMRS = 56 OH = (63 + 4*56) / (63 + 4*56 + 672) = 30%	Midamble overhead = 4% # of DMRS = 14 OH = (63 + 4*14) / (63 + 4*14 + 672) = 15%	OH = 63 / (63 + 672) = 8.5%	
Spectrum	Pulse shaping filter is necessary for spectrum shaping.	Pulse shaping filter is necessary for spectrum shaping.	Pulse shaping filter is necessary for spectrum shaping.	MSK has the smallest sidelobe among these four modulations.
Transmitter complexity	Medium complexity LO to generate carrier waves out of phase. Single differential LO can generate them together.	Lower complexity LO does not need to generate out of phase signal	Medium complexity Almost the same as BPSK, yet there is additional XOR processing required.	Higher complexity (modulation index, rapid frequency shift or IQ branches needed)
Receiver complexity	Higher CFO estimation / compensation complexity (e.g., large # of hypothesis needed). Supporting coherent detection is not so complex for reader.	Lower complexity for non-coherent detection. For the detection of OOK symbol, receiver needs to maintain detection threshold, which needs to be trained.	Medium complexity for CFO estimation and non-coherent detection. Phase estimation is still required for differential detection. But requirement is loose compared to that of BPSK.	Accurate CFO estimation / compensation required for coherent detection increasing receiver complexity. (Due to CFO non-coherent detection seems difficult.)
Note 1: X-amble overhead is computed for (96+16) bits payload Note 2: M=4 # of coded bits = 336, # of chips after MC = 672,				

Table 6A.2.1.2.7-6: Source [320].

	BPSK	**OOK**	**DBPSK**	**MSK**
Sensitivity to CFO	≤0.3 ppm for about ~1 ms midamble interval irrespective of bit duration	not sensitive to CFO	tolerance for CFO increases as the bit duration decreases, e.g., ≤1 ppm for bit duration of 133.33μs, ≤2 ppm for bit duration of 66.67μs	tolerance for CFO increases as the bit duration decreases, e.g., ≤0.5 ppm for bit duration of 133.33μs, ≤1 ppm for bit duration of 66.67μs
Required SNR Performance	Required SNR=-7.7 dB @10% BLER Required SNR= -4.8 dB @1% BLER	Required SNR=-2.1dB @10% BLER Required SNR=1.2dB @ 1% BLER	Required SNR = -5 dB @ 10% BLER Required SNR = -1.7 dB @ 1% BLER	Required SNR = 0.7 dB @ 10% BLER Required SNR = 3.7 dB @ 1% BLER
Receiver detection type	Coherent	Non-coherent	Non-coherent	Non-coherent
Overhead	25.58 %	2.18 %	2.18 %	2.18 %
Spectrum	/	/	/	negligible as the transmission filter is anyway needed

Transmitter complexity	Medium complexity.	Lower complexity	Medium complexity	Higher complexity (modulation index, or I/Q branches needed)
Receiver complexity	High complexity for CFO estimation and coherent detection based on channel estimation	Lower complexity for non-coherent detection.	Medium complexity for CFO estimation / non-coherent detection.	Medium complexity for CFO estimation / non-coherent detection.
Note 1: X-amble overhead is computed assuming (96+16) bits payload				

Table 6A.2.1.2.7-7: Source [318].

	BPSK	**OOK**	**DBPSK**
Sensitivity to CFO	BPSK is sensitive to phase error (CFO impact) and requires accurate CFO estimation (≤ 0.1 ppm).	OOK is more robust to phase error (CFO impact) and the tolerable residual CFO can be up 100 ppm.	DBPSK is robust to phase error (CFO impact) and the tolerable residual CFO can be up 1.5 ppm.
Required SNR Performance	For Device 2b, required SNR = 10.5 dB @ 10% BLER, required SNR = 20 dB @ 1% BLER. For Device C, required SNR = 4.8 dB @ 10% BLER, required SNR = 12.5 dB @ 1% BLER.	For Device 2b, required SNR = 14.5 dB @ 10% BLER, required SNR = 22.5 dB @ 1% BLER. For Device C, required SNR = 10.2 dB @ 10% BLER, required SNR = 15 dB @ 1% BLER. Theoretical BER performance is 6 dB worse than BPSK	For Device 2b, required SNR = 5.5 dB @ 10% BLER, required SNR = 17 dB @ 1% BLER. For Device C, required SNR = 5.8 dB @ 10% BLER, required SNR = 13 dB @ 1% BLER. Theoretical BER performance is 3 dB worse than BPSK
Receiver detection type	Coherent only	Non-coherent	Non-coherent
Overhead	For Device 2b, 16.6% For Device C, 9.7% (Note 1)	For Device 2b, 15.6% For Device C, 8.4% (Note 1)	For Device 2b, 15.6% For Device C, 8.4% (Note 1)
Note 1: X-amble overhead is computed for (96+16) bits payload			

Table 6A.2.1.2.7-8: Source [323].

	BPSK	**OOK**	**DBPSK**	**MSK**
Sensitivity to CFO	Some, phase changes due to CFO over time.	Robust	Robust	Some
Receiver detection type	Coherent only	Non-coherent	Non-coherent	Coherent/non-coherent
Transmitter complexity	Medium complexity.	Lower complexity	Medium complexity	Higher transmitter complexity
Receiver complexity	Higher CFO estimation / compensation complexity.	Lower complexity for non-coherent detection.	Medium complexity for CFO estimation / non-coherent detection. Phase estimation is still required but not as sophisticated as BPSK.	Accurate CFO estimation / compensation required for coherent detection increasing receiver complexity

Table 6A.2.1.2.7-9: Source [313].

	BPSK	OOK	DBPSK	MSK
Required SNR Performance	Theoretical BER performance is the same as MSK	Theoretical BER performance is 6 dB worse than BPSK	Theoretical BER performance is 3 dB worse than BPSK	Theoretical BER performance is the same as BPSK
Receiver detection type	Coherent only	Non-coherent	Non-coherent	Coherent/non-coherent
Spectrum	Pulse shaping is necessary to reduce spectral leakage.		Pulse shaping is necessary to reduce spectral leakage.	The constant envelope and continuous phase property of MSK making it more spectrum efficient.
Transmitter complexity	Medium complexity. [Differential LO.]	Lower complexity	Medium complexity [Differential encoding. Differential LO]	
Receiver complexity	Higher CFO estimation / compensation complexity (e.g., large # of hypothesis needed). Supporting coherent detection is not so complex for reader.	Lower complexity for non-coherent detection.	Medium complexity for CFO estimation / non-coherent detection. Phase estimation is still required for differential detection.	Accurate CFO estimation / compensation required for coherent detection increasing receiver complexity e.g., large # of hypothesis needed

Table 6A.2.1.2.7-10: Source [309].

	BPSK	OOK	DBPSK	MSK
Sensitivity to CFO	Sensitive to CFO for phase modulation but achieve better performance.	Robust to CFO for amplitude modulation but achieve worse performance.	Robust to CFO for differential modulation but achieve worse performance.	Sensitive to CFO for phase modulation.
Required SNR Performance	For reference data rate ~ 0.1 kbps, required SNR = -17.3 dB @ 10% BLER, and -15.9 dB @ 1% BLER For reference data rate ~ 1 kbps, required SNR = -10.1 dB @ 10% BLER, and -7.2 dB @ 1% BLER For reference data rate ~ 5~7 kbps, required SNR = -5.0 dB @ 10% BLER, and -1.0 dB @ 1% BLER	For reference data rate ~ 0.1 kbps, required SNR = -0.3 dB @ 10% BLER, and 2.2 dB @ 1% BLER For reference data rate ~ 1 kbps, required SNR = 4.3 dB @ 10% BLER, and 7.1 dB @ 1% BLER For reference data rate ~ 5~7 kbps, required SNR = 5.9 dB @ 10% BLER, and 9.3 dB @ 1% BLER Theoretical BER performance is 6 dB worse than BPSK	For reference data rate ~ 0.1 kbps, required SNR = -11.7 dB @ 10% BLER, and -9.9 dB @ 1% BLER For reference data rate ~ 1 kbps, required SNR = -7.0 dB @ 10% BLER, and -4.1 dB @ 1% BLER For reference data rate ~ 5~7 kbps, required SNR = -5.0 dB @ 10% BLER, and -1.5 dB @ 1% BLER Theoretical BER performance is 3 dB worse than BPSK	The feasibility with coherent detection is not clear with midamble phase ambiguity. The performance with non-coherent detection is poor.
Receiver detection type	Coherent only	Non-coherent	Non-coherent	Coherent/non-coherent
Overhead	39% for reference data rate ~ 0.1 kbps 27% for reference data rate ~ 1 kbps 21% for reference data rate ~ 5 ~ 7 kbps (Note 1)	4% for reference data rate ~ 0.1 kbps 5% for reference data rate ~ 1 kbps 9% for reference data rate ~ 5 ~ 7 kbps (Note 1)	4% for reference data rate ~ 0.1 kbps 5% for reference data rate ~ 1 kbps 9% for reference data rate ~ 5 ~ 7 kbps (Note 1)	
Spectrum	The power leaked outside the transmission bandwidth is acceptable.	The power leaked outside the transmission bandwidth is acceptable.	The power leaked outside the transmission bandwidth is acceptable.	The power leaked outside the transmission bandwidth is acceptable.
Transmitter complexity	Medium complexity.	Lower complexity.	Medium complexity. Differential encoding.	Higher transmitter complexity (modulation index, rapid frequency shift or IQ branches needed)
Receiver complexity	Higher CFO estimation / compensation complexity (e.g., large # of hypothesis needed). Supporting coherent detection is not so complex for reader.	Lower complexity for non-coherent detection.	Medium complexity for CFO estimation / non-coherent detection. Phase estimation is still required for differential detection.	Accurate CFO estimation / compensation required for coherent detection increasing receiver complexity e.g., large # of hypothesis needed
Note 1: X-amble overhead is computed for (96+16) bits payload and for same data rate for different modulations				

In conclusion, for D2R modulation scheme for Rel-20, most companies other than [314] do not prefer MSK.

6A.2.1.3 Bandwidth definition

Companies reported the various aspects of bandwidth related definitions including transmission bandwidth, guard band, occupied bandwidth, and channel bandwidth.

- [311], [312], and [336] suggest that the definition of D2R bandwidth in Rel-19 should be the baseline/starting point for Rel-20.
- [328] suggest to reuse the Rel-19 bit-duration definition for D2R channel bandwidth indication in Rel-20 A-IoT.
- [309] suggest that the detailed values of transmission bandwidth can be discussed by RAN1 in WI phase.
- [311], and [314] suggest that the detailed values of guard band should be discussed by RAN4 in WI phase.
- [309], [317], and [323] suggest that guard band is required for D2R transmissions due to CFO.
- [314] suggests considering guard bands between A-IoT and NR in both DL and UL spectrum for outdoor scenarios.
- [330] proposes that guard bands for different device types should be considered in D2R frequency domain resource allocation.
- [314] suggests that the occupied BW of each D2R transmission should consider the guard band in the allocated BW as (1+a) * BW_signal, where a is the effect factor related to the SFO and CFO and BW_signal is the bandwidth of the D2R signal.
- [317] and [323] suggest that the occupied bandwidth should include both the transmission bandwidth and guard band due to CFO.
- [318] states that guard band is required due to the impact of CFO and SFO for Device 2b, and guard band is required due to the impact of CFO for Device C.
- [319] states that the D2R transmission bandwidth should be defined in integer multiples of 15 kHz, with a minimum bandwidth of 15KHz. The maximum bandwidth is to be decided in WI.

6A.2.2 D2R generation

6A.2.2.1 Channel coding

Enhancements on the D2R channel coding scheme defined in TS 38.291 [228], considering additional FEC code rates and without any change to the polynomials defined in TS 38.291 [228], is studied.

Coding scheme 1: Rate 1/2 by puncturing the third polynomial bits

Coding scheme 2: Mother code rate 1/4 based on an additional polynomial

Coding scheme 3: Repetition of bits from virtual circular buffer based on mother code rate 1/3.

Following sources reported that the introduction of coding rate ½ is necessary.

- [313] reported that, in addition to the mother code rate of 1/3, a coding rate of 1/2 can be natively supported by Rel-19 A-IoT (or LTE) TBCC, offering additional benefits improving resource efficiency while providing coding gain.
- [319] reported that using FEC at a code rate of 1/2 with 1 repetition achieves approximately 2.5 dB gain for a TBS of 20 bits and 3.5 dB gain for a TBS of 96 bits, under the same data rate compared to the existing configuration with no FEC and 2 repetitions.
- [325] reported that rate 1/2 through puncturing trades approximately 1.5 dB of coding gain for 50% improvement in spectral efficiency. This option is valuable when devices are closer to the reader.
- [320] reported that 1/2 code rate by puncturing the third polynomial bits can improve spectrum efficiency compared with 2 repetitions or 1/3 code rate.
- [308] proposes supporting different coding rates using the bit collection procedure.

Following sources reported that coding rate ½ is not necessary.

- [324] reported that the CC with code rate 1/3 has already been validated and deployed for a long period of time. Furthermore, since the performance gap between code rate 1/3 and 1/2 is marginal, there is little motivation to support additional code rates beyond 1/3.
- [309] note that the necessity of coding rate 1/2 is not clear for outdoor scenario.
- [314] states that since TBCC with rate 1/2 cannot satisfy D2R coverage requirements and different D2R chip durations can provide flexibility in D2R data rate, the justification for supporting R=1/2 is unclear and unnecessary.

Supporting new coding rate 1/4 was proposed by companies with or without preference to specific option on how to realize lower coding rate.

Following sources mentioned general positive support of lower coding rate.

- [311] reported that D2R FEC with 1/4 code rate can be considered if needed.
- [327] reported that FEC lower coding rate can be realized through block level repetitions; sending the entire encoded packet multiple times.
- [323] proposes that additional lower FEC code rate (e.g., 1/4) can be supported only when the bottleneck still has been identified for the D2R coverage based on the Rel-19 FEC code rate and repetition number and Option 2 is preferred to reduce extra workload; the 4th sub-block could be generated by equally number of bits from the current three sub-blocks.
- [335] reported that further lowering the code rate to 1/4 can be achieved by enabling circular buffer-based repetition schemes as used in legacy NR systems.

Following sources reported performance evaluation results of Option 1) and Option 2) in terms of SNR gain.

- [309] reported that the coverage for outdoor scenarios is much larger than that in Rel-19, therefore, lower FEC code rate, i.e., 1/4, is necessary and performance of Option 1 is 0.9dB better than Option 2 at 10% BLER.
- [336] reported that for packet size of 96bits, and 400bits, there is no significant performance difference between Option 1 and Option 2. For packet size of 1000 bits, Option 1 has 0.5dB gain at 10% BLER and [~1]dB at 1% BLER.
- [319] states for a code rate of 1/4, mother code rate 1/4 provides more than 1 dB performance gain compared to mother code rate 1/3 with repetition of bits. at 10% BLER.
- [318] reported that for packet size of 96bits, compared with repeating the existing third polynomial (165), the simulation results for all the candidates (141,117,145,111,125) of option 1 show no significant performance gain (~0.5 dB), and all the results are similar. Option 2 is preferred for lower complexity.

Following sources reported not to introduce lower coding rate.

- [313] reported that a code rate of 1/4 is not natively supported by TBCC as specified in TS 36.212 and proposed not to support code rate of ¼.
- [324] proposed not to consider additional coding rates due to reduce implementation burden.
- [332] states that for D2R, lower code rate than 1/3 is not necessary unless meaningful gain compared to 1/3 code rate is observed; e.g., it was observed that 0.2dB gain is obtained for 1/4 code rate compared to 1/3 code rate @1% BLER according to TR38.769.
- [314] reported that with four polynomials for TBCC with coding rate 1/4, its complexity and memory of encoder is about 1/4 higher than that of Rel-19 TBCC with coding rate 1/3. As block repetition could provide good BLER performance for TBCC with rate 1/3, TBCC with a coding rate 1/4 is unnecessary.
- [320] reported that 1/2 code rate with 2 repetitions achieves similar performance of 1/4 code rate with less standard effort.

For Rel-20 D2R FEC, support Rel-19 TBCC with coding rate of, at least, 1/3.

6A.2.2.2 Interleaving and bit collection

Interleaving for PDRCH with FEC and the use of the LTE bit collection scheme with the following candidate methods are studied.

Alt 1: Both the LTE sub-block interleaver and the LTE bit collection scheme are used.

Alt 2: Only the LTE bit collection scheme is used.

Alt 3: Only an updated bit collection scheme based on the LTE bit collection scheme is used.

- Each sub-block is divided into one or two segments. For one segment case, it is the same as Alt 2. For two segments case, the bit collection is performed such that 1st segment bits from each sub-block are collected first and then 2nd segment bits from each sub-block are collected.

The following observations were reported regarding the performance and memory requirements of the candidate methods.

Table 6A.2.2.2-1: Observations on performance

Source	Alt1 (LTE sub-block interleaver + LTE bit collection)	Alt2 (LTE bit collection without interleaver)	Alt3 (Segment-based bit collection)
[336]	• 96-bit: 0.1 dB @ 10% BLER, 4 dB @ 1% BLER • 400-bit: 2 dB @ 10% BLER, 9 dB @ 1% BLER (vs. Rel-19 no FEC, 3 repetitions)	• 96-bit: 0.1 dB @ 10% BLER, 4 dB @ 1% BLER • 400-bit: 2 dB @ 10% BLER, 9 dB @ 1% BLER (vs. Rel-19 no FEC, 3 repetitions)	• 96-bit: 0.1 dB @ 10% BLER, 4 dB @ 1% BLER • 400-bit: 0.1 dB @ 10% BLER, 4 dB @ 1% BLER (vs. Rel-19 no FEC, 3 repetitions) • Provides less latency reduction
[332]	• Can provide robust performance against bursty error scenarios • More beneficial for D2R given longer coherent time, low data rate, and large TBS • Provides time diversity gain	• Can provide similar functionality as interleaving for interference randomization • Different performance under burst error compared to Alt1	• Needs more clarification on memory reduction details
[318]	1.2 dB over Rel-19 scheme	0.5 dB over Rel-19 scheme	0.9 dB over Rel-19 scheme
[317]	• 96-bit @ 10% BLER: ~1 dB gain • 400-bit @ 10% BLER: SNR = 5 dB (Rel-19 fails) (vs. Rel-19 scheme)	• 96-bit @ 10% BLER: ~1 dB gain • 400-bit @ 10% BLER: SNR = 8.5 dB (Rel-19 fails) (vs. Rel-19 scheme)	• 96-bit @ 10% BLER: ~0 dB gain • 400-bit @ 10% BLER: SNR = 9.5 dB (Rel-19 fails) (vs. Rel-19 scheme)
[308]	• Provides two levels of distribution of errors: - Within encoder output stream - Between streams • Best performance from both complexity and performance perspective	• Provides distribution between encoder output streams • Slightly worse than Alt1 because localized errors are not distributed before decoding	• Provides less distribution than Alt2 • Less effective error distribution
[314]	• (16.67 μs, 96 bits): 0.1 dB gain • (16.67 μs, 400 bits): 1.25 dB gain • (133.33 μs, 96 bits): 1.79 dB gain • (133.33 μs, 400 bits): 6.46 dB gain (vs. Rel-19 scheme)	• (16.67 μs, 96 bits): 0.15 dB gain • (16.67 μs, 400 bits): 1.2 dB gain • (133.33 μs, 96 bits): 1.61 dB gain • (133.33 μs, 400 bits): 5.91 dB gain (vs. Rel-19 scheme)	• (16.67 μs, 96 bits): 0.05 dB gain • (16.67 μs, 400 bits): 0.57 dB gain • (133.33 μs, 96 bits): 0.61 dB gain • (133.33 μs, 400 bits): 3.71 dB gain (vs. Rel-19 scheme)
[319]	• 0.2~2.4 dB gain over Rel-19 • Higher complexity but does not provide significant performance gains compared to Alt2 or Alt3	• 0.2~2.3 dB gain over Rel-19 • Lowest computational complexity	• 0.2~2.3 dB gain over Rel-19 • Lowest memory requirement
[320]	• 96-bit: 1~4.7 dB gain over Rel-19 • 400-bit: 1~4.7 dB gain over Rel-19 • Outperforms Rel-19 FEC	• 96-bit: 1~4.1 dB gain over Rel-19 • 400-bit: 1~4.1 dB gain over Rel-19 • Negligible difference (~0.5 dB) from Alt1 • Outperforms Rel-19 FEC	• 96-bit: 0~2.8 dB gain over Rel-19 • 400-bit: 0~2.8 dB gain over Rel-19 • Same performance as Rel-19 for 96-bit • At least 1 dB less than Alt1/2

Source	Alt1 (LTE sub-block interleaver + LTE bit collection)	Alt2 (LTE bit collection without interleaver)	Alt3 (Segment-based bit collection)
[309]	• 1/3 TBCC, 96/400 bits @ 10% BLER: 0.8~2.7 dB gain • 1/3 TBCC, 96/400 bits @ 1% BLER: 1.2~3.6 dB gain • 1/4 TBCC, 8 block repetitions @ 10% BLER: 0.4~1.4 dB gain • 1/4 TBCC, 8 block repetitions @ 1% BLER: 0.3~1.4 dB gain • Beneficial to improve coverage (vs. Rel-19 scheme)	• 1/3 TBCC, 96/400 bits @ 10% BLER: 0.6~2.2 dB gain • 1/3 TBCC, 96/400 bits @ 1% BLER: 0.9~3.2 dB gain • 1/4 TBCC, 8 block repetitions @ 10% BLER: 0.1~0.6 dB gain • 1/4 TBCC, 8 block repetitions @ 1% BLER: 0.1~0.6 dB gain (vs. Rel-19 scheme)	• 1/3 TBCC, 96/400 bits @ 10% BLER: 0.3~1.3 dB gain • 1/3 TBCC, 96/400 bits @ 1% BLER: 0.4~1.9 dB gain • 1/4 TBCC, 8 block repetitions @ 10% BLER: 0.1~0.4 dB gain • 1/4 TBCC, 8 block repetitions @ 1% BLER: 0.1~0.3 dB gain (vs. Rel-19 scheme)

Table 6A.2.2.2-2: Observations on memory requirements

Source	Alt1 (LTE sub-block interleaver + LTE bit collection)	Alt2 (LTE bit collection without interleaver)	Alt3 (Segment-based bit collection)
[336]	• 2TP = 6× max TBS (T = max TBS, P = 3 polynomials) • 96-bit TBS: 576 bits • 400-bit TBS: 2400 bits • Device requires full size buffer	• TP = 3× max TBS • 96-bit TBS: 288 bits • 400-bit TBS: 1200 bits • 50% reduction vs. Alt1	• TP/2 = 1.5× max TBS • 96-bit TBS: 144 bits • 400-bit TBS: 600 bits • Device requires smaller size buffer
[318]	6 times of max TBS	3 times of max TBS	4.5 times of max TBS
[317]	• >2× max TBS	• ≥2× max TBS	• ≥1× max TBS • Lowest memory requirement among three alternatives
[314]	• At least 3× max TBS for rate 1/3 TBCC	• At least 3× max TBS for rate 1/3 TBCC	• Theoretical minimum: 3/2× max TBS • Preferred: 3× max TBS to avoid timing conflicts
[320]	2 times of max TBS	0 times of max TBS, i.e., memory-free	at least 6 bits of memory corresponding to the tail bits of the first TB segment
[319]	• Highest complexity and memory • Requires logic circuits for interleaved addresses • Non-sequential memory operations • Less favorable for power consumption and cost control	• 3× max TBS • Significant memory requirements • Lowest computational complexity	• 1.5× max TBS (with two segments) • Lowest memory requirement • Less favorable for power consumption and cost control vs. Rel-19
[309]	• ~3× max TBS when block repetition is larger than 1	• ~3× max TBS when block repetition is larger than 1	• ~3× max TBS when block repetition is larger than 1
[308]	• Supports Alt1 from complexity perspective • Requires memory for two-level error distribution	• Requires memory for single-level error distribution	• Same memory as Alt2 but provides less error distribution

Regarding analysis of what to should be supported:

- [308] proposes supporting Alt 1 for encoding from both a complexity and performance perspective, with ordering: CRC encoding, convolutional encoding, LTE sub-block interleaving, and bit collection/repetition.
- [327], [330], [309], and [335] support Alt 1: reuse both LTE sub-block interleaver and LTE bit collection scheme.
- [336] proposes considering Alt 2 interleaving with lower required memory and improved performance.
- [320] proposes considering Alt 2 interleaving: The performance of Alt 2 is comparable to Alt 1, better than Alt 3, and the buffer size requirement is smaller than Alt 1 and Alt 3, i.e., memory free for Alt 2 by polynomial sweeping implementation.
- [323] proposes that when memory size is limited at device side, Alt 3 (LTE bit collection with two segments and without LTE sub-block interleaver) should be preferred.

- [317] proposes further studying additional alternatives: Alt. 4 (Rel-19 bit collection with interleaving) and Alt. 5 (Rel-19 bit collection with interleaving and row permutation).

- [314] proposes two options for interleaving for PDRCH with TBCC: either reuse both LTE sub-block interleaver and LTE bit collection scheme, or no interleaver for D2R transmission with FEC.

- [319] reported that For D2R FEC interleaving and bit collection, Alt 2 has the lowest computational complexity and Alt 3 the lowest memory requirement among the three alternatives. Alt 1 exhibits higher complexity but does not provide significant performance gains compared to Alt 2 or Alt 3.

Regarding not supporting interleaving and bit collection:

- [311] propose that considering the limited power consumption budget and cost, it should not support the interleaver and LTE bit collection scheme for TBCC.

- [313] proposes not supporting interleaver or bit collection.

- [324] proposes no study on the interleaving for PDRCH with FEC for Device 2b and Device C. But, if it is supported, then, Alt1 is preferred.

In D2R interleaving, one among Alt 1, Alt 2, and Alt 3 is supported.

6A.2.2.3 Repetition

For the definitions of repetitions, see Clause 6.1.0.1. For D2R, a number of repetitions larger than 2, as supported in Rel-19, for block-level repetition is studied.

Following sources reported that larger number of D2R repetitions are necessary to support larger coverage for device 2b and C than 2 which is the maximum number supported in Rel-19.

- [314] reported that block repetition number of D2R could be selected from {1, 2, 4, 8} or {1, 2, 4, 8, 16} to meet the target coverage requirements in outdoor deployment scenarios.

- [323] reported that larger repetition number (e.g., 4 or 8) can be supported only when the bottleneck still has been identified for the D2R coverage based on the Rel-19 FEC code rate and repetition number.

- [325] reported that extended repetition factors are crucial for outdoor coverage which may require up to 16 repetitions to close link budget gap for 500m.

- [320] reported that it is necessary to support larger number of repetitions, e.g., up to 16 for coverage, and sustainable time can be up to 1s, sufficient for tens of repetitions from storage and power consumption perspective.

- [336] stated that additional numbers (3, 4) are necessary to improve the limited D2R coverage.

Proposed candidate numbers for Rel-20 includes 4, 8, 10, and 16, which can provide additional gain of about 1 to 3dB for every 2x increase in repetition numbers.

- [309] reported that performance with repetition factor 8 has ~ 2.3 dB gain compared with that of repetition factor 4 at BLER 10% and has ~ 2.4 dB gain at BLER 1% and it is feasible for device C to support 8 repetitions.

- [318] reported that, repetition number of 4 has ~3dB and 1 dB gain at BLER of 10^-1 compared with no repetition and 2x repetition, respectively. Larger repetition numbers, e.g., 4, 8, to enlarge the coverage distance for PDRCH were recommended.

- [319] reported that 1) 4 repetitions provide 2.7 dB gain for TBS 96 bits and 4.5 dB for 400 bits relative to the existing 2 repetitions, 2) 8 repetitions provide additional 1.5 dB gain for TBS 96 bits and 1 dB for 400 bits relative to 4 repetitions, and 3) 16 repetitions provide additional 0.5 dB gain for TBS 96 bits and 0.2 dB for 400 bits relative to 8 repetitions.

- [317] reported that, for device C, 10 repetitions are required to meet the coverage requirement under 20dB penetration loss. For device 2b, except for the UMa scenario with 0dB penetration loss, it fails to achieve coverage target in the remaining three scenarios, even with the maximum of 16 repetitions. The results indicate that the link budget gain provided by minimal repetition (e.g., 2 repetitions) is insufficient, and larger repetition numbers are required (>2, typically 10 or 16 as shown).

Three sources pointed out that too long repetitions will require larger energy storage size, which is not desirable for energy harvesting A-IoT devices.

- [336] reported that long transmission duration makes devices spend power for longer duration with larger energy storage size. Hence, it was proposed to support additional D2R repetition numbers up to 4.
- [312] reported that larger number of repetitions directly extends D2R transmission duration, thereby leading to higher energy consumption at the device side. Since A-IoT devices are power-constrained, compared to FEC, too larger number of repetitions is not an efficient use of the limited energy storage proposing the maximum repetition number of up to 4.
- [332] reported that considering data rate at cell edge and corresponding transmission duration of PDRCH, at least 4 or 8 times of repetition is feasible while large number of repetitions such as 32 would not be feasible from energy assumption perspective

Two sources presented the benefit of non-continuous repetitions which can potentially improve performance with increased time diversity.

- [335] reported that enabling non-contiguous repetitions improves time diversity and allows for energy harvesting gaps, offering superior robustness for active devices compared to continuous transmission. It was proposed to support a repetition value of up to 4 to enhance reliability.
- [324] reported that studying D2R repetitions with time-separated transmissions for active devices and D2R segmentation for improved device transmission sustainability.

One source mentioned the joint considerations between FEC coding rate and repetition.

- [311] reported that larger repetition number of block-level repetition can be considered if needed, and that supporting code rate for FEC and repetition number for D2R block-level repetition can be considered jointly for extending the coverage.
- [308] For D2R, for channel coding, support different coding rates using the bit collection procedure.

6A.2.2.4 Scrambling

Scrambling for D2R is necessary for randomizing interference and preventing long runs of identical bits when Manchester code is not applied, with the following options for scrambling sequence type.

Option 1: Gold sequence

Following sources support Gold sequence for PDRCH scrambling due to its good pseudo-random properties, better cross-correlation characteristics compared to m-sequences, and the benefit of reusing existing NR/LTE designs:

- [312], [316], [320], [319], [336], and [309] propose using (length-31) Gold sequence as in NR/LTE.
- [314] proposes that scrambling sequence can be pseudo-random sequence, e.g., Gold sequence.
- [323] proposes that Gold sequence with length shorter than 2^{31}-1 should be applied, and Nc with smaller value can be equipped with the Gold sequence to reduce complexity/computation burden of scrambling.

Option 2: m-sequence

Regarding initialization/generation complexity, it was also reported by three sources that NR/LTE sequence initialization requires 1600 operations for initializing sequence generator, which could be a burden for low complexity A-IoT device. One source reported that sequence generation based on XOR operation is simple.

- [308] reported that NR scrambling sequence initialization requires advancing 1600 additional operations, which is preferable to be reduced/eliminated. It was also pointed out that there are mathematical approaches to advance the 1600 clocks using a lookup table and ~30 exclusive ORs.
- [336] reported that advancing the initial Nc = 1600 steps could be a burden for A-IoT device can be avoided by storing the initial state of shift registers for sequence generations. The number of scrambling sequences could be limited by limiting the number of possible initialization seeds.

- [323] By setting Nc=1600, at least 1600 computation burden is required.

- [309] reported that the generation of scrambling sequences is very simple, e.g., just some bit-level XOR operation, and the complexity is acceptable.

Two different views on sequence generation shift register length or polynomial order were reported; first approach is to use the length-31 as NR/LTE and the second approach is to reduce the length to (e.g., 16) given that A-IoT packet size is limited by 1000bits. The motivation for the second approach is to reduce complexity.

- Sources including [312], [316], [320], [319], [336], and [309] propose using length-31 Gold sequence as in NR/LTE.

- [308] reported that it is preferable to reduce the degree of the generator polynomial (e.g., the degree should be log2 of the maximum number of bits transmitted) for complexity considerations. For 1000 bits and an extreme value for the coding rate of 1/16, a degree 16 generator polynomial is more than adequate.

- [323] reported that Gold sequence with length shorter than 2^31-1 should be applied, and Nc with smaller value than 1600 can be equipped with the Gold sequence to reduce complexity/computation burden of scrambling.

Following sources reported that the sequence generator should be initialized based on Device ID and Reader/Cell ID, with some also suggesting the inclusion of time-based parameters (e.g., SFN, parameter which varies based on periodic R2D signal or parameter signaled by control information).

- [314], [319] proposes that the scrambling sequence generator can be initialized with the device ID and/or the reader ID.

- [316], [320], [321] proposes initializing the scrambling sequence generator with a value which depends on reader identity, device identity and/or time.

- [325] proposes supporting scrambling for PDRCH using device-specific scrambling sequences initialized with device ID.

- [328] proposes that for scrambling of PDRCH, the scrambling sequence initialization is controlled by the Reader, e.g., on a per-higher-layer-session basis.

Companies provided views on whether to apply scrambling sequence to midamble. Sources [309], [312], [320], [319] and [336] suggested not to apply scrambling sequence since it could degrade its correlation property which may be needed for timing estimation and/or frequency offset estimation. One source [314] suggested applying scrambling for interference randomization.

A Gold sequence is used for D2R scrambling.

6A.2.2.5 Line coding

Companies have investigated pros and cons of applying linecoding for D2R.

Following observation was made showing potential performance benefit of line coding.

- [308] reported that with Manchester line coding, decoding can be viewed as a relative amplitude comparison of two consecutive chips. This comparison is much more robust to the fading and noise effects. The evaluation was done in D2R configuration, 96 bits as the message size, 16-bit CRC, no FEC coding. It is observed that there is about 6 dB detection performance gain with Manchester line coding at 10% BLER.

However, multiple companies noted that the original reasons for adopting Manchester (line) coding in Rel-19—such as aiding backscatter carrier suppression, providing timing robustness under severe SFO, ensuring transitions for symbol clocking, and avoiding long runs of identical bits - do not apply to Rel-20. In Rel-20, carrier waves are internally generated, scrambling can prevent long bit runs, and backscatter-related benefits are irrelevant for active A-IoT devices. Companies also highlighted that Manchester coding doubles the occupied bandwidth, reducing spectral efficiency and multiplexing capability.

- [321] reported that while Rel-19 Manchester encoding which increases the required signal bandwidth by a factor of two aids carrier suppression in backscattering systems by introducing a spectral gap near the carrier, this advantage is irrelevant for active A IoT devices that do not perform backscattering.

- [324] reported that Manchester line coding was adopted mainly to facilitate timing/synchronization robustness under severe SFO, since a transition is guaranteed within each bit and can assist timing/bit-boundary tracking. However, line coding increases the occupied bandwidth, which can reduce multiplexing capability.

- [332] reported that the main motivation to support this Manchester coding in Rel-19 was 1) to shift frequency from carrier-wave externally provided and 2) to avoid contiguous 0s or 1s. However, for Rel-20, carrier-waves are internally generated, and externally generated carrier-waves no longer exist. In addition, contiguous 0s or 1s can be avoided by scrambling in Rel-20.

- [336] reported that the main reasons of having linecoding in Rel-19, which are to providing symbol clocking information, improving detection performance for backscatter communication does not apply to Rel-20. Furthermore, it was reported that linecoding lowers spectral efficiency by factor of 2 due to expanded bandwidth.

- [318] observed that Manchester line code without repetition shows a performance similar to that of bit-level repetition without Manchester, but slightly worse.

6A.2.2.6 Order and inclusion of generation steps

For PDRCH, FEC encoding is done before block level repetition, if Alt1/Alt2/Alt3 of interleaving and bit collection (see Clause 6A.2.2.2) is used. Scrambling is applied after block-level repetition and channel coding (if any). Scrambling in D2R is applicable to PDRCH and is not applied to midamble.

If LTE bit collection and interleaving scheme are assumed in Rel-20, then the required memory size and time domain diversity gain for the following two cases become different: case 1) repetition followed by FEC (Rel-19), 2) FEC followed by repetition. The memory impact is captured in [308] observation.

- [308] For the D2R link, when the repetition block is before TBCC encoding:
  - Alt 1: memory estimates and processing times are proportional to the number of repetitions.
  - Alt 2 and Alt 3: memory estimates and processing times are invariant to the number of repetitions. Each alternative has the same memory needs and processing times.

Following companies suggested changing the order of repetition and FEC for Rel-20 active device.

- [309] proposed that performing channel coding before block-level repetition is feasible and necessary.

- [314] observes that compared with block-level repetition applied after FEC for D2R, block-level repetition applied before FEC scheme requires Rblock times more interleaver memory as well as higher implementation complexity and processing delay, and proposes that if interleaving schemes are supported for D2R with FEC, block-level repetition should be performed after FEC and interleaving.

- [317] recommends the order of channel coding first, followed by block-level repetition.

- [320] proposes considering block-level repetition after channel coding (with LTE bit collection) for D2R transmission.

- [336] proposes that for PDRCH, block repetition is done after channel coding and bit collection.

- [324] proposes that D2R block-level repetition after channel coding should be a baseline for device-side complexity reduction and energy sustainability.

- [335] proposes supporting block-level after FEC encoding for D2R transmissions.

- [318] proposes that for PDRCH generation in Rel-20, the channel coding will be followed by block repetition, and the block repetition will be followed by scrambling.

6A.2.3 D2R timing

6A.2.3.1 D2R preamble

6A.2.3.1.1 Functionality

The D2R preamble length, the sequence type used, the number of sequences using the same length and other factors affecting the preamble design are studied. The D2R preamble supports at least the functionalities of D2R timing acquisition and D2R CFO estimation considering the modulation scheme.

Sources [309], [312], [313], [318], [329], [330] reported CFO estimation as one of important functionalities for D2R preamble together with timing acquisition.

- [309] observe that D2R SFO estimation is not necessary for Device C, D2R CFO estimation is feasible for reader to ensure coherent detection with BPSK, and one-part structure preamble is enough for both CFO estimation and timing acquisition.
- [312] proposes that if preamble with length longer than those supported in Rel-19 is introduced for CFO estimation, CFO estimation is not considered for midamble design again as the CFO drift rate of device 2b/C is slow.
- [313] proposes D2R preamble can be used for fine CFO calibration and re-using Rel-19 preamble design for timing acquisition, indicating the start of D2R transmission, for SFO estimation, channel and interference estimation. It was also noted that there is no need to differentiate the traffic types based on physical layer aspects, rather it should be left to higher layer.
- [318] proposes that D2R preamble for PDRCH should support functionalities of D2R timing acquisition, D2R SFO estimation, and D2R CFO estimation.
- [329] proposed D2R CFO calibration signal (CSS) for reader side CFO estimation, where CSS follows preamble.
- [330] proposes studying whether to support a two-part D2R preamble (one part for CFO estimation and one part for timing/channel estimation).

There are diverging views on SFO estimation – whether it is necessary [314], [318], [332] or not [309], [317], [320].

- [314] proposes D2R SFO estimation should be done by D2R preamble and D2R midamble.
- [318] Regarding D2R preamble for PDRCH, the functionalities of D2R timing acquisition, D2R SFO estimation, and D2R CFO estimation are supported.
- [332] suggests D2R preamble and midamble should support SFO estimation functionality at least for device 2b.
- [317] suggests SFO estimation is not necessary for D2R preamble, especially for Device C.
- [320] reported that for active device 2b with up to 1000 ppm SFO and device C with up to 10 ppm SFO, SFO can be hardly estimated more accurately by preamble and the timing drift caused by SFO can be eliminated by re-synchronization based on midamble.

Two sources [311], [319] mentioned that preamble could be also used for D2R measurement purposes.

- [311] reported that D2R X-amble(s) could be used for measurement, (e.g., signal strength, interference estimation) at reader side.
- [319] reported that for the D2R preamble and midamble, the use of multiple sequences per sequence length is beneficial to mitigate cross-cell/inter-reader detection ambiguity and avoid inter-cell interference.

6A.2.3.1.2 Structure

Two sources proposed to study two parts preamble structure to support different functionality. One source reported that one part structure is enough for multiple functionalities.

- [329] proposes studying a two-part D2R preamble structure where the first part indicates the start of D2R transmission and the second part carries information for quick identification of the device or D2R transmission structure.

- [330] proposes studying whether to support a two-part D2R preamble (one part for CFO estimation and one part for timing/channel estimation).

- [309] observed that one-part structure preamble is enough for both CFO estimation and timing acquisition.

6A.2.3.1.3 Differentiation by preamble

Multiple sources mentioned that multiple (e.g., 3) preambles can be used for reader differentiation or device identification, etc. The reported example method for generating multiple sequences includes using multiple cyclic shifts.

- [309], [313] reported that preamble differentiation across cells or for reader identification, can be considered based on e.g., different cyclic shifts, e.g., 3 for three sectors.

- [319] proposes that considering the limited complexity and storage capability of A-IoT devices, it is proposed that three distinct preamble sequences be supported for each sequence length.

- [321] observes that since M-sequence exhibits good correlation properties, multiple cyclic-shifted versions of the D2R preamble can be utilized to distinguish different readers/devices.

- [323] proposes the number of sequences for differentiation should be identified first, since it may affect the choice of sequence type of D2R preamble.

- [327] suggests considering the impact of different D2R Preamble sequence on reader differentiation, identifying device types, or CDMA.

- [336] proposes for D2R preamble, N different sequences are considered to differentiate devices to different readers, where N>=3.

- [332] reports that multiple sequences for D2R preamble using same length can have potential usage for multi-reader scenario that multiple sequences for reader differentiation can be used for early termination at reader, i.e., if unexpected D2R preamble is detected, reader can skip D2R Rx, and for interference alleviation.

6A.2.3.1.4 Sequence design

6A.2.3.1.4.1 Sequence type

The following options for sequence type are studied, where the length of a sequence can be an additional longer D2R preamble length than Rel-19, or can taken multiple length values depending on modulation scheme, coverage level, etc.

Option 1: m-sequence

Option 2: Gold sequence

Sequence differentiation by using 3 or more sequences is considered for differentiating between multiple readers, devices, etc.

Multiple sources mentioned m-sequence and Gold sequence for preamble. There is strong preference on m-sequence due to its good auto-correlation performance. One source mentioned that the number of required sequences for sequence differentiation can affect the sequence type selection.

- [311] propose m-sequence for D2R X-ambles, including preamble and midamble.

- [319] suggest M-sequence for D2R preamble and midamble, with three sequences supported per sequence length.

- [321] reported that M-sequence can be considered for preamble since the M-sequence exhibits good correlation properties, multiple cyclic-shifted versions of the D2R preamble can be utilized to distinguish different readers/devices.

- [327] suggested longer Preamble (e.g., 16 bit or 32 bit) and sequence with good autocorrelation (e.g., m-sequence or Gold sequence).

- [323] reported that the number of sequences for differentiation should be identified first, since the number may affect the choice of sequence type of D2R preamble. For example, length-31 sequence, if the required sequence for differentiation is larger than 31, only Gold sequence can fulfill the requirement. However, if only 3 or 4 sequences are required, both m-sequence and Gold sequence can fulfill the task, and m-sequence offer slightly low computation burden.

6A.2.3.1.4.2 Sequence length

Multiple sources have provided inputs on preamble length from 31bits up to 511 bits to support larger coverage. Following observations capture the necessities / motivations for longer preamble.

- [325] proposed supporting m-sequence preamble with length 31, 63, or 127 chips for BPSK for outdoor scenario.
- [309] propose candidate preamble lengths of {31-bit, 127-bit, 511-bit} corresponding to different coverage levels.
- [318] proposes additional longer preamble lengths than Rel-19 can be 63 and 127 considering required SNR performance and sequence overhead.
- [335] supported a longer preamble sequence for outdoor devices in D2R transmission.
- [323] proposed that longer sequence or denser interval for preamble/ midamble can be supported when coverage bottleneck is identified using the Rel-19 preamble and midamble design. It was noted that multiple lengths of D2R preamble can be considered for different coverage target.

However, different sources have slightly different views regarding the maximum length due to its dependency on considered target data rate, presence of midamble, modulation type, target functionality, etc. Following observations were reported from companies.

- [317] reported that that the residual CFO of 0.1ppm can be achieved at SNR around 0dB / -13.5dB / -12.5dB / -15dB for 31, 63, 127, >127-bit length m-sequence, respectively. [317] proposed multiple D2R preamble sequence lengths for different coverage levels with three sets: short sequence: {31-bit, smaller than 31-bit}, medium sequence: {64-bit, 127-bit}, and long sequence: {255-bit, 511-bit}.
- [320] reported that the required preamble length for BPSK is longer than DBPSK to ensure the performance degradation is less than 1 dB; e.g., for ~5 kbps data rate configuration with bit duration of 66.67 us, the required preamble length is 31 bits for BPSK and 15 bits for DBPSK; for ~1 kbps data rate configuration with bit duration of 133.33 us, the required preamble length is 63 bits for BPSK and 15 bits for DBPSK; for ~0.1 kbps data rate configuration with bit duration of 266.67 us, the required preamble length is 255 bits for BPSK and 127 bits for DBPSK. It was suggested that m-sequence with 63, 127 and 255 bits for additional longer preamble length than Rel-19.
- [336] reported that, with midamble (interval = 266us, length=16.67us), preamble length 31bits provides similar performance as no CFO case and preamble length 127bits does not provide additional gain. It was reported that preamble length 7bits does not provide good CFO estimation resulting in poor performance. It was suggested considering multiple preamble lengths including 31, [63], [127], with shorter values used with midamble and longer values without midamble, if necessary.
- [311] reported that whether to introduce the longer X-ambles in Rel-20 compared with those in Rel-19 should be decided based on the evaluation result for the target performance in outdoor scenarios.
- [321] reported that depending on the D2R modulation scheme (e.g. OOK or BPSK) or whether coherent detection is required at the receiver, the length and sequence type of the D2R preamble can be configured differently.
- [314] reported that SFO estimation should be done by preamble and since the initial SFO of Device 2b/C is much smaller than that of Device 1, the existing Rel-19 D2R ambles should be sufficient for SFO estimation and PDRCH decoding.
- [326] suggests the correlation between D2R coverage and R2D coverage could be considered in the design of D2R preamble sequence lengths and R2D SIP sequence lengths.
- [323] Considering same chip duration/bit duration, length-255, 127, 63 sequence offers about 9dB, 6dB, 3dB gain over length-31 sequence.

6A.2.3.1.5 Overall assessment

For Rel-20 active device, D2R preamble supports at least following aspects.

- Multiple lengths (same and/or longer length than Rel-19)
- Sequence type: m-sequence
- Number of sequences: at least 3 per length
- Structure: one part

6A.2.3.2 D2R midambles

6A.2.3.2.1 Functionality

A D2R midamble signal is studied, considering the midambles pattern(s), the pattern length(s) and interval(s), the sequence type used and whether it is different from the sequence type used for the D2R preamble in Clause 6A.2.3.1, and other factors affecting the midamble design. The supported functionalities are among channel estimation, time tracking, and CFO estimation.

Multiple sources [318], [336], [309], [317], [320], [326] reported that midamble could be used for channel estimation/tracking purpose which handles residual CFO. [321], [326], [313], and [320] reported time tracking functionality. According to [329], [309], [311], [314], interference estimation is yet another functionality which can be supported by midamble.

Regarding SFO estimation, there are diverging views; sources [314], [332] reported that midamble could be used for SFO estimation. Whereas sources [309], [317], [320] reported that SFO estimation is not necessary since SFO has negligible impact on performance. [312]'s view is conditional depending on the assumed preamble length.

Companies' detailed observations are captured below.

- [312] reported that BSPK requires a higher density of midamble in PDRCH to track the channel. In this case, a short midamble type optimized primarily for channel estimation is necessary. This type of midamble should be designed to minimize overhead while maintaining BPSK demodulation performance. It was reported that the necessity of SFO estimation/time tracking depends on preamble length; long preamble might not require midamble based SFO estimation/time tracking, whereas short preamble might require that.
- [309] observe that SFO estimation is not necessary for Device C, CFO estimation can be realized by channel estimation, and interference estimation can be realized by midambles with channel estimation.
- [314] proposes interference estimation can be done by D2R midamble.
- [315] Network Limited proposes the midamble of the PDRCH for Msg1 may contain a few bits of known reference signals decoded by the reader to estimate the channel.
- [317] suggests SFO estimation is not necessary for D2R midamble, especially for Device C.
- [320] proposes for BPSK modulation with coherent detection, two types of midamble are needed:
  - Type-1 midamble: for channel estimation purpose, which is about 0.25 ms time duration and about 1 ms interval, required for both device 2b and device C
  - Type-2 midamble:
    - For device 2b: for time tracking and CFO estimation purposes
    - For device C: for CFO estimation purpose when frequency drift rate is 1 ppm/s

  It was also suggested that for DBPSK modulation with non-coherent detection, only one type of midamble is needed for time tracking for device 2b.
- [326] proposes introducing different types of D2R midamble to support different functionalities.

- [327] suggests considering same functions of D2R Midamble like Rel-19 and studying the impact of modulation on D2R Midamble, e.g., MSK or DBPSK.

- [329] proposes studying using D2R midamble resource for interference estimation and suggested, when multiple functionalities are supported on the D2R midamble resources, explicit/implicit indication may be needed for functionality identification/differentiation.

- [318] states that sequence-based midamble is needed for Device 2b to perform timing tracking and SFO estimation, while it is not needed for Device C. 1-bit midamble can be used for finer CFO estimation and channel estimation when coherent demodulation is performed. Otherwise, 1-bit midamble is not required.

6A.2.3.2.2 Sequence design

6A.2.3.2.2.1 Sequence type

Multiple types of sequences for midamble were reported, as below, with diverging views. The length of a sequence and interval between midambles could be shorter/longer than in Rel-19, and may consider different functionalities for the midamble.

Option 1: m-sequence

- Source [321] suggested M-sequence for D2R midamble and proposed multiple cyclic-shifted versions of the same m-sequence for D2R midamble.

- Source [319] proposes that M-sequence is used for D2R midamble, with three distinct sequences supported per sequence length.

- Source [323] suggested that All 1 sequence and Hadamard sequence should be de-prioritized and preferred using the same methodology as Rel-19 A-IoT.

- Source [314] reported that SFO estimation should be done by preamble and since the initial SFO of Device 2b/C is much smaller than that of Device 1, the existing Rel-19 D2R ambles should be sufficient for SFO estimation and PDRCH decoding.

Option 2: All-1 sequence

- Source [315] proposes multiple options for D2R midamble sequence; long strings of 1's, long strings of 0's, m-sequence, or Golay sequence.

- Source [320] proposed that all-1 sequence for midamble design for channel estimation purpose and reusing 7 bits or 31 bits m-sequence used in Rel-19 amble design for midamble design for time tracking and CFO estimation in Rel-20.

Option 3: Hadamard sequence

- Source [309] reported that orthogonal sequences, e.g., Hadamard sequences, can be considered for midambles, with sequence differentiation considered with at least 3 sequences.

6A.2.3.2.2.2 Sequence length

For both midamble length and interval, three different types of values were suggested: values shorter than Rel-19 (to increase the density of midamble), same values as Rel-19 (to reuse previous design), or longer values than Rel-19.

Sources [309], [317], [320], [336], [312], [319], [330] mentioned shorter mid-amble length and interval for channel estimation/channel tracking functionality.

- [309] proposed candidate intervals of {4-bit, 6-bit, 8-bit} and candidate midamble length of 1 bit.

- [317] proposed studying enhancements on D2R midamble including shorter midamble length than Rel-19 and shorter interval than Rel-19.

- [320] suggested for BPSK modulation with coherent detection, 1 ms amble interval and 0.25 ms time duration of midamble (e.g., 16 bits interval and 4 bits length for bit duration of 66.67 μs) can be considered for D2R midamble design. And for time tracking and CFO estimation, reuse 7 bits or 31 bits m-sequence used in Rel-19 midamble design and the interval depends on the SFO and frequency drift rate.

- [336] proposed multiple values for D2R midamble interval and considering at least 2, 4 bits for midamble length.

- [312] reported that short midamble type optimized primarily for channel estimation can be used in midamble design for BPSK.

- [319] reported that, for a calibrated CFO of 10 ppm, reducing the D2R midamble insertion interval from 48 bits to 20 bits provides an approximate 5 dB improvement in D2R BLER performance for a TBS of 96 bits at 10% BLER proposing smaller midamble insertion interval for CFO estimation in D2R transmission.

- [330] supports a mid-amble placement scheme in D2R with denser, and shorter mid-ambles in Rel-20 for frequent estimate of phase rotation due to CFO.

Sources [311], [313], [314] proposed to reuse Rel-19 design.

- [311] reported that the value of interval between X-ambles (i.e., interval between preamble and first midamble, and between midambles) defined in Rel-19 could be the baseline for that in Rel-20.

- [313] reported that the D2R midamble design for Rel-20 A-IoT can re-use the same configurations (sequence type "m-sequence", explicit indications of midamble presence via R2D control information, separation between preamble and midamble, and consecutive midambles) as Rel-19 A-IoT devices.

- [314] reported that SFO estimation should be done by preamble and since the initial SFO of Device 2b/C is much smaller than that of Device 1, the existing Rel-19 D2R ambles should be sufficient for SFO estimation and PDRCH decoding.

Source [323], [325] suggested to allow longer interval to allow for accumulated timing error for correction.

- [323] suggested regarding D2R midamble of PDRCH, additional longer length and/or longer intervals than those of Rel-19 should be supported.

- [325] proposes supporting midamble with length 31 or 63 chips, noting midamble length does not need to match preamble length.

Source [321] mentions midamble configuration could depend on D2R modulation scheme.

- [321] proposed that depending on the D2R modulation scheme (e.g. OOK or BPSK) or whether coherent detection is required at the receiver, the length and sequence type of the D2R midamble can be configured differently.

6A.2.3.2.3 Overall assessment

For Rel-20 active device, D2R midamble supports at least following aspects:

- Length(s) not longer than Rel-19

- Multiple Intervals

6A.2.4 D2R multiplexing

6A.2.4.1 TDMA

For TDMA, for contention-based random access and DT and DO-DTT traffic types, enhancements to increase the number of time domain resources indicated by the corresponding R2D message for the *Access Random ID* message (Msg1) to greater than 2 and for the *D2R Upper Layer Data Transfer* message (Msg3) to greater than 1, where the messages are defined in TS 38.391 [229], are studied.

On enhancements to increase the number of time domain resources indicated by the corresponding R2D message for the *Access Random ID* message (Msg1) to greater than 2, sources [361], [372], [379], [383], [387], [381], [385], [376], [375], [373]. [366], [378] state that it is feasible. Sources [361], [379], [372], [383], [387], [373]. [378] further state that it is feasible and necessary, and sources [371], [370], [368] state that it is feasible but not necessary. Sources [362], [369], [365] report that it is not necessary.

Following observations were provided during the study:

- [361] mentions that increasing X1 for Msg1 transmission is necessary, as the larger PRDCH overhead can be mitigated by allocating more time-domain resources.
- [372] notes that the smaller residual SFO of Device 2b/C may enable the use of more than two TDM resources for Msg1. (e.g., X1 > 2).
- [379] states that supporting more than two Msg1 time-domain access occasions (X1 > 2) is feasible and necessary for device 2b and device C.
- [383] states that supporting X1 > 2 for Msg1 can enhance multiplexing capacity and shorten the overall random-access procedure.
- [387] states that supporting X1 = {1, 2, 4, 8} for Msg1 is reasonable given the larger coverage and improved SFO capability of active devices.
- [376] notes that with the improved timing accuracy of Device 2b/C, X1 for Msg1 transmission can be increased (e.g., 4) though it must remain limited due to A-IoT device constraints.
- [381] states that supporting X1 > 2 for Msg1 is feasible in Rel-20 because active devices have higher capability and more accurate clocks.
- [385] states that supporting X1 > 2 for Msg1 is feasible because device 2b/C can use more accurate clocks and calibration to reduce the required time gap between TDMed D2Rs.
- [370] notes that increasing X1 is feasible due to smaller SFO and reduced guard time for Device 2b/C, but its necessity remains unclear because a larger X1 introduces multiple drawbacks and therefore requires further evaluation.
- [371] notes that although X1 > 2 for Msg1 is feasible due to the reduced SFO in Rel-20, its necessity is low because the resulting latency-reduction gain is very limited.
- [362] mentions that larger values of X1 for Msg1 are unnecessary because the performance gain is limited and further decreases as X1 increase.
- [369] notes that it is not necessary to support X1 > 2 for Msg1, as the 16 Msg1 transmission resources enabled by 8 FDM + 2 TDM D2R in Rel-19 are already sufficient.
- [366] notes that the maximum number of time domain resources for Msg1 transmission should be limited to 11 for Device 2b/C, as larger values provide limited performance gain.
- [368] notes that supporting X1 > 2 for Msg1 can be studied only if Rel-19 Random access procedure is reused.
- [375] notes that supporting X1 > 2 for Msg1 transmissions could be considered due to the lower SFO of Device 2b/C.
- [377] states that configurable X1 values such as 2, 4, and 8 help reduce Msg1 collision probability in high-density environments.
- [365] mentions that it should be determined whether X1 = 2 is sufficient for outdoor capacity and latency requirements, and that any need for enhancement beyond X1 = 2 for Msg1 should be justified with supporting evaluations.
- [375] states that the Device types 2b and C have more accurate oscillators with lower SFO. To reduce the signaling overhead for Msg1 trigger signal and Msg2, larger transmission occasions could be considered, i.e., X1>2 and X3>1.
- [373] states that X1 values such as 4 and 8 to provide more efficient access.

On enhancements to increase the number of time domain resources indicated by the corresponding R2D message for the *D2R Upper Layer Data Transfer* message (Msg3) to greater than 1, sources [361], [372], [379], [383], [387], [376], [381], [385], [371], [368], [375], [373] state that it is feasible. Sources [361], [372], [379], [383], [387] further state that it is feasible and necessary, and source [371] states that it is feasible but not necessary. Source [362] reports that it is not necessary, source [369] states that it is not feasible, and source [370] states that is neither feasible nor necessary. Following observations were provided during the study.

- [361] mentions that supporting X3 > 1 is feasible due to the higher clock accuracy of Device 2b/C, although increasing X3 beyond 2 is unlikely because of Msg3 length and clock drift.
- [372] notes that the smaller residual SFO of Device 2b/C may enable the use of multiple TDM resources for Msg3 (e.g., X3 > 1).
- [379] states that supporting more than one Msg3 time-domain access occasion (X3 > 1) is feasible and necessary for device 2b and device C.
- [383] states that supporting X3 > 1 for Msg3 can improve random-access efficiency and reduce total access time.
- [387] states that enhancements on X3 > 1 should be limited to X3 = 2, since supporting higher X3 adds design and signaling complexity without clear benefit.
- [376] notes that the X3 capacity for Msg3 transmission should match that of Msg1 and can likewise be increased (e.g., 4) rather than being restricted to half as in Rel-19.
- [381] states that supporting X3 > 1 for Msg3 is feasible in Rel-20 with shorter gaps between time resources compared to Rel-19.
- [385] states that supporting X3 > 1 for Msg3 is feasible since improved clock accuracy of device 2b/C makes the time-gap overhead marginal even for longer Msg3 durations.
- [371] notes that although X3 > 1 for Msg3 is feasible, its necessity is low given the similarly limited latency-reduction benefit.
- [362] mentions that larger values of X3 for Msg3 are unnecessary because the performance gain is limited and further decreases as X3 increase.
- [369] notes that it is not feasible to support X3 > 1 for Msg3, as the device ID can be several hundred bits long.
- [370] notes that supporting X3 > 1 is technically feasible but brings significant performance and scheduling penalties, so its necessity is weak and requires further study.
- [366] notes that TDMA support for Msg3 can reduce transmission latency, though the latency-reduction benefit diminishes as the TDMA size increases, and at least X3 = 2 should be supported for Device 2b/C.
- [368] notes that supporting X3 > 1 for Msg3 can be studied.
- [375] notes that supporting X3 > 1 for Msg3 transmissions could also be considered given the improved oscillator accuracy of Device 2b/C.
- [377] states that configurable X3 values greater than 1 allow parallel Msg3 processing and improve random-access performance.
- [365] mentions that from the device perspective, X3 should be 1.
- [375] states that the Device types 2b and C have more accurate oscillators with lower SFO. To reduce the signaling overhead for Msg1 trigger signal and Msg2, larger transmission occasions could be considered, i.e., X1>2 and X3>1.
- [373] states that X3 values such as 2 and 4 to provide more efficient access.

6A.2.4.2 FDMA

For FDMA, enhancements considering the D2R frequency resources used and how the device determines these resources, whether baseband small frequency shift is supported and whether FDMA is applied to other D2R transmissions apart from the *Access Random ID* message (Msg1) and *D2R Upper Layer Data Transfer* message (Msg3) defined in TS 38.391 [229], are studied.

Multiple sources proposed the support of FDM(A) for Rel-20 D2R by LO tuning. One source suggested FDM support for device C only. One source suggested supporting FDM for D2R transmission beyond Msg1 and Msg3 to improve efficiency.

- [324] specifies that D2R FDMA should be supported only by adjusting the LO frequency (LO re-tuning).

- [335] proposes enabling FDM through a large frequency shift (LFS) for D2R transmission by active device types by adjusting the LO.
- [325] specifically proposes to support FDM for device C, while relying primarily on TDM for device 2b.
- [319] propose that extending FDM to D2R transmission types beyond Msg1 and Msg3 can enhance A-IoT system capacity and efficiency.
- [320] reported that it is feasible and necessary to support D2R FDMed transmission by adjusting LO toward a target frequency.
- [332] reports that other than Msg1 and Msg3, FDMA for contention based D2R transmission in DO-A procedure is beneficial to ensure the multiplexing capacity in outdoor scenario, and it is beneficial for reducing collision probability of D2R transmission and reducing latency of the DO-A procedure.

Multiple sources have provided views on sinewave-based SFS for Rel-20 in aspects including implementation, LO tuning, relation to 1SB with SFS, etc.:

- [320] reported that, for "2SB without SFS" and "1SB with SFS", whether to have SFS or not is an implementation issue and 2SB with SFS has the worst spectrum efficiency.
- [311] reported that SFS followed by up-conversion is more friendly for low complexity and cost devices since it does not need frequent retuning.
- [336] reported that sinewave-based SFS allows device LO tuning to common carrier frequency, whereas devices without sinewave-based SFS should tune to device-specific carrier frequency.
- [332] reported that 2SB without SFS is feasible.
- [318] Replacing target frequency fc with (fc-fSFS) in (1SB RF signal without SFS) can generate almost same 1SB RF signal as (1SB RF signal with SFS) but requires lower implementation complexity.
- [334] reported that the reference frequency can be commonly provided to all targeted A-IoT active devices, e.g., via broadcast information or paging-like R2D message. The small frequency shift can be provided by D2R scheduling information in corresponding R2D transmission.
- [324] reported that supporting SFS (especially Rel-20 SFS) may introduce additional signal processing blocks beyond LO re-tuning, potentially increasing device-side complexity. Based on it, we state that, in Rel-20, D2R FDMA is supported only by adjusting the LO frequency (i.e., LO re-tuning).
- [323] reported that the SFS operation for Rel-19 device is not required for device 2b and C, since the baseband signal can be directly adjusted to the intended frequency by up-conversion.
- [321] reported that sinewave based SFS is more feasible than without SFS because without SFS method requires frequency retuning and may experience increased CFO.

Sinewave-based SFS is not supported in Rel-20 A-IoT.

Multiple sources proposed to define basic frequency domain resource unit(s) considering tx signal bandwidth and required guard band, which are uniformly or equally spaced in frequency domain.

- [327] proposes considering larger number of frequency domain resources for MSG1/Msg3 transmission for Device 2b/C.
- [336] suggests that frequency domain resource unit and spacing are determined considering tx signal bandwidth and required guard band (e.g., due to LO uncertainty).
- [319] propose that:
  - D2R transmission bandwidth should be defined in integer multiples of 15 kHz, with a minimum bandwidth of 15KHz.
  - D2R frequency-domain resource bandwidth is larger than D2D transmission bandwidth and is set to an integer multiple of 15 kHz.
  - A uniform frequency-domain resource arrangement within the A-IoT bandwidth is recommended.

- [312] proposes to define a frequency resource as the transmission bandwidth and two-sides predefined guard band for D2R FDMA.
- [326] proposes to define D2R FDM resources across two levels: the RB level and the D2R channel level.
- [314] proposes two options for FDMed channel allocation: equally spaced center frequencies with fixed bandwidth, or multiple sub-bands allocated with a number of resource units associated with the data rate.
- [318] proposed that the frequency resource unit can be defined by, e.g., the number of sub-channels or the size of guard band.

Multiple sources reported the method for frequency resource determination.

- [309], [312], [324], [328], [330], [332], [319] propose that the frequency location should be derived based on indication from reader, such as a reference frequency and offset.
- [311] proposes that the D2R carrier frequency information should be indicated by reader directly.
- [319] noted two options for frequency-domain resource arrangement: symmetrically distributed on both sides of the reference frequency, or distributed on one side of the reference frequency.
- [323] proposes two options for frequency resource indication: offset represented by k*f_offset indicated by reader, or (sub)-channels pre-defined according to the transmission BW indicated by reader.
- [317] proposes that broadcast information can be used to indicate FDMed frequency resources at least for Msg1 and Msg3, by providing reference frequency point, gap between FDMed frequency resources and the number of frequency resources.
- [318] proposed that the available frequency resources, the number of sub-channels or the size of guard band can be pre-configured, and the index of sub-channel or transmission bandwidth can be indicated dynamically.
- [320] reported that for Msg1, a device determines the target frequency resource by reusing i.e., slot-aloha of Rel-19 A-IoT design. And for other D2R transmissions, the frequency resource occasion for a device can follow the frequency resource selected for Msg1 or be indicated by the scheduling information of the corresponding R2D.
- [314] reported that the frequency offset or the index of the pre-defined sub-channel could be indicated to device via the control information in PRDCH.

In D2R, FDM(A) is supported at least for Msg1 and Msg3.

6A.2.4.3 CDMA

6A.2.4.3.1 Design and analysis

CDMA is studied, considering the sequence type used, the length of the sequence (31 or 63), and the number of sequences (<64, 64, 128).

For sequence-based Msg1, the following types of sequence are considered:

Option 1: m-sequence

Option 2: Gold sequence

Performance aspects compared to Rel-19 are e.g. impact of CFO and SFO on sequence-based Msg1 detection performance, impact of power control, MDR/FAR, modulation, impact to random access procedure, access efficiency gain (e.g., inventory procedure, Msg1).

Sources [312], [317], [327], [336], [332], [329] support the introduction of sequence based Msg1 for Rel-20 to improve access efficiency of CBRA. Source [313], [324], [328] do not see necessity of introducing CDM for Rel-20. [308], [309], [319], [334], [320] suggest further study on technical issues including synchronization, rx power variation, etc.

Companies supporting CDMA:

- [312] proposes that D2R CDM(A) for Device 2b and for Device C with sequence-based Msg1 transmission should be considered in Rel-20 A-IoT. They mentioned that device’s potential lack of ability for accurate

measurement and power adjustment may result in the deviation of the received power at reader side from the target received power, which can affect the feasibility of CDM-based Msg 1 transmission.

- [317] reported that sequence-based Msg1 transmission, with CDMA (code domain multiplexing capability is 4), the collision probability is reduced to 10% and 2% from 70% and 30% of Rel-19 random access based on RN16, if 10% and 50% of time domain resources are reserved for Msg1 transmission, respectively. Hence, D2R sequence-based Msg1 with new physical random-access chancel is recommended for Rel-20 A-IoT.

- [327] proposed to consider CDMA for D2R Msg1 transmission at least for Device C and study if LTE PRACH with ZC sequences could be used as baseline for further study.

- [336] provided evaluation results showing the improvement of the Msg1 access efficiency of CBRA up to 18 times compared to Rel-19 Msg1 under the assumption of +-6dB of rx power variation, length 63 Gold sequence, and modulation scheme not limited to BPSK.

- [332] reported that D2R CDMA can be necessary to ensure the multiplexing capacity in outdoor scenario, and it is beneficial for reducing collision probability of Msg1 transmission and reducing the latency of random access procedure. CDMA can be feasible considering improved SFO for device 2b/C from device 1 and potential support of power control.

- [329] stated that applying CDM to Msg1 transmission increases access capacity by allowing more devices to transmit simultaneously.

Companies with positions against CDMA:

- [313] states there is no strong motivation to introduce CDMA from capacity perspective for Rel-20 A-IoT devices due to increased complexity at receiver side and high specification workload.

- [324] observe that accurate CFO/SFO compensation and tight power control are difficult, so CDMA may not operate reliably due to synchronization errors and near-far effects.

- [328] reported that CDMA has stringent synchronization requirements which may work with large CFO, and Rel-20 FDM/TDM could be further improved to provide sufficient multiplexing capability.

Following companies suggest further study potential benefits, sensitivity to power variation and SFO/CFO, etc.

- [308] observes that CDMA could increase the number of supported Msg1 transmissions. However, CDMA is not necessary for contention-based access procedure, as the bottleneck is Msg3 transmission, not Msg1. The sensitivity of rx power variation caused by near-far and CFO along with modified procedure needs additional study.

- [309] reported that the necessity of sequence-based Msg1 needs further study since access efficiency gain is limited in some cases (e.g., high data rate). They propose studying m-sequence for better autocorrelation performance, considering 255-bit-sequence, addressing near-far effect with power control, and updating random access procedures compared to Rel-19.

- [319] proposed evaluating the feasibility of D2R sequence-based Msg1, including sequence detection performance and complete inventory time for multiple devices.

- [334] proposes studying whether it's necessary to support both Msg1 FDMA and sequence-based Msg1.

- [320] provided input in following aspects:

  - Feasibility of CDM through evaluation: In case of CDMed Msg1 with m-sequence and CDMed sequences are generated by different cyclic shifts with same m-sequence, for 31 bits m-sequence, 2 sequences can be CDMed without error floor in MDR performance; for 63 bits m-sequence, 3 sequences can be CDMed without error floor in MDR performance; for 127 bits m-sequence, 3 or 4 sequences can be CDMed without error floor in MDR performance.

  - Required sequence length: to achieve similar performance as RN-16 based Msg1 with 1/3 TBCC and block level repetition 2, the required sequence length for sequence based Msg1 is 127 bits, and sequence based Msg1 from 3~4 devices can be CDM multiplexed.

  - Considerations on the access efficiency gain evaluation:

- For CDMed sequence based Msg1, unnecessary Msg2 and Msg3 transmission resources will be caused by Msg1 collision, since reader can still successfully decode collided Msg1 sequence in some cases.
- Msg4 is needed to transmit for collision resolution for CDMed sequence based Msg1, but not needed for Rel-19 RN-16 based Msg1.

Following sources reported sequence characteristics such as length, sequence type, available sequence numbers, etc.

- [314] proposes that sequence length for sequence-based Msg1 should not be shorter than 31 and that NR preamble number could be a starting point for Device 2b/C in outdoor scenarios for the number of sequence-based Msg1.
- [312] proposes limiting the number of binary sequences for Msg1 to 64 or fewer, with the exact number being configurable by the reader.
- [323] suggests that Gold sequence offers more choices than m-sequence, and the number of supported sequences should be 2x (e.g., 16, 32, 64, 128).
- [332] reports that for sequence type for sequence-based Msg.1 transmission, both m-sequence and Gold sequence have good auto-correlation property. Gold sequence has better cross correlation property and can provide larger number of sequences with low cross correlation than m-sequence, which should be considered for Msg1 with CDMA.

6A.2.4.3.2 Potential impact on random access procedure

Potential impact of D2R CDM(A) for Msg1 on random access procedure for Device 2b and for Device C is studied and following observations were provided during the study.

- [361] mentions that although CDMA can increase the number of supported Msg1 transmissions in the contention-based access procedure, it would modify the handling of Msg2 and the scheduling of Msg3, while the dominant bottleneck remains the much larger Msg3 transmission.
- [362] mentions that if sequence-based Msg1 is introduced, the RA procedure becomes very similar to the legacy NR 4-step RACH procedure and can be handled by RAN2 during the WI phase.
- [372] notes that the necessity of Msg1 CDMA remains unclear given existing Msg1 TDMA/FDMA designs, but if required, binary sequence–based Msg1 and related procedural enhancements—including sequence indication via Msg2 and predefined D2R modulation—should be considered.
- [373] notes that D2R CDM(A) for sequence-based Msg1 may impact the random-access procedure, requiring consideration of a four-step RA process.
- [383] states that collision handling for sequence-based Msg1 should be studied and can be resolved with minimal enhancements in the RAN2 random-access procedure.
- [385] states that explicit or implicit contention resolution using Msg3 and Msg4 should be considered when sequence-based CDMA is used for Msg1 in the CBRA procedure.
- [371] states that if msg1 is sequence-based, instead of RN16, contention resolution i.e., msg4 is needed for random access procedure, aiming to deal with the msg1 collision.
- [368] states that contention resolution using Msg3 and Msg4 should be considered when sequence-based CDMA is used for Msg1 in the CBRA procedure.

6A.3 L1 procedures

6A.3.1 D2R power control

6A.3.1.1 Purposes and necessity

Power control for D2R is studied, considering open loop and/or closed loop power control, the required measurements, the required information at the reader and device, the signalling overhead and power consumption.

The necessity of D2R Tx power control for Device 2b and Device C for outdoor scenarios are studied and the following observations were provided:

- 22 sources ([337], [351], [338], [339], [340], [355], [341], [360], [352], [344], [354], [359], [353], [346], [350], [348], [342], [358], [343], [347], [349], [357]) reported that the D2R Tx power control for Device 2b and Device C is necessary for outdoor scenarios.
  - 20 sources ([337], [351], [338], [339], [340], [355], [341], [360], [352], [356], [354], [359], [353], [348], [342], [358], [343], [347], [349], [350]) state that it can reduce the interference between different devices (i.e., the Near-Far problem).
  - 14 sources ([337], [351], [338], [339], [355], [341], [344], [354], [359], [353], [348], [347], [349], [350]) reported that it helps devices remain available longer.
  - 4 sources ([337], [355], [353], [358]) reported that the interference due to a transmission from a device to a reader can be reduced at an unintended reader
  - 1 source ([357]) added that depending on device's peak power consumption and PA support, the D2R Tx power control may not be feasible.
- 1 source ([345]) reported that the D2R Tx power control for Device 2b and Device C is not necessary mentioning in its contribution that the inter-channel interference from different devices is negligible, D2R coverage is limited, and the complexity and the power consumption required at the device side is high.

6A.3.1.2 Open-loop power control

The following methods for open-loop power control are studied, considering the device complexity:

Option 1: Transmit power is calculated based on the measured pathloss using a predefined formula and configured parameters from the reader, with the device requiring information on at least the reader's transmit power and the received target power at the reader.

- The following observations were provided during the study.
  - [364] mentions that in Option 1 the device needs to estimate the pathloss using the reader-indicated Tx power and an R2D measurement, and that limited measurement accuracy and power-adjustment capability may cause the received power at the reader to deviate from the configured P0, which can affect CDM-based Msg1 transmission.
  - [365] mentions that under an open-loop power-control mechanism, a simplified NB-IoT–like approach can be used as a baseline for Rel-20 A-IoT.
  - [367] notes that Option 1 should cover cases where the device cannot use the required or indicated transmit power.
  - [370] mentions that Option 1 provides smooth and adaptive power control based on real-time pathloss and a predefined formula, is therefore the preferred approach.
  - [371] notes that Option 1 achieves better power-control accuracy because each device derives its own transmit power using pathloss calculated from RSRP.
  - [373] notes that Option 1 for open-loop power control follows a traditional pathloss-based approach and can be studied for Rel-20.
  - [375] notes that Option 1 can provide more precise transmit-power control but increases the computational complexity of Device 2b/C.
  - [376] notes that Option 1 enables open-loop D2R power control using pathloss estimation based on comparing the average ON-state power of known R2D signals with the configured ON-state power.
  - [377] notes that Option 1 enables continuous Tx power adjustment and can be applied for device C.
  - [379] states that Option 1 provides higher-accuracy D2R transmit power adjustment based on pathloss and is therefore more suitable for A-IoT operation.

- [385] states that Option 1 provides more precise pathloss-based power control but may not be essential for ultra-low-power devices given their large transmit-power tolerance and higher complexity.
- [386] states that Option 1 provides higher-precision D2R transmit-power control through pathloss-based calculation, and the additional complexity is acceptable for device 2b/C.
- [387] states that Option 1 allows pathloss-based D2R transmit-power calculation using reader-configured parameters and is slightly preferred due to lower specification complexity.
- [389] states that Option 1 provides finer-grained D2R transmit-power control through pathloss-based calculation.
- [391] states that Option 1 performs pathloss-based open-loop power control but requires explicit pathloss calculation and higher device complexity.

- Sources [361], [369], [372], [383], [388], [371] report that Option 1 is feasible.
  - [361] mentions that Option 1 is feasible, indicating that a device can measure the received R2D power at RF and derive its D2R transmit power based on a pathloss estimate obtained from reader-provided transmit power and target power level.
  - [369] considers Option 1 feasible and prefers to support it due to its higher adjustment accuracy based on actual pathloss.
  - [372] notes that Option 1 provides higher-accuracy and more adaptive open-loop power control based on pathloss, and is therefore the preferred approach.
  - [383] states that Option 1 offers finer-granularity and more accurate D2R transmit-power control, giving it an advantage for open-loop operation.
  - [388] states that Option 1 provides precise pathloss-based D2R transmit-power control that improves reliability, reduces interference, and enables energy-efficient operation.

Option 2: Transmit power is determined based on the measured RSRP using an RSRP range based on RSRP thresholds that are predefined or configured by the reader, with the device requiring information on at least the RSRP threshold(s) and the corresponding D2R transmit power-related information for each RSRP range defined by the RSRP threshold(s).

- The following observations were provided during the study
  - [364] mentions that in Option 2 the device maps a measured RSRP to a D2R transmit power, but frequent updates of the mapping between RSRP ranges and D2R transmit powers are required when the reader's Tx power changes, resulting in increased R2D signaling overhead.
  - [365] mentions that within the open-loop power-control framework, Option 2 can be considered alongside Option 1.
  - [367] notes that Option 2 assumes that the corresponding D2R transmit power includes zero power.
  - [370] notes that Option 2 uses an RSRP-based mapping with complexity comparable to Option 1.
  - [371] notes that Option 2 provides worse power-control accuracy, as devices rely only on RSRP ranges and may select identical transmit power within the same range, by which the power control is unachievable within each RSRP range. And suitable RSRP thresholds are difficult for the reader to configure. For example, the reader cannot be aware of the certain range of RSRP for devices. To reach all the devices, reader has to consider a very wide total RSRP range, leading to the inventory or service latency.
  - [373] notes that Option 2 can also be studied, but it relies on the underlying assumption that the reader's transmit power remains constant.
  - [375] notes that Option 2 offers a simpler RSRP-based implementation and is therefore the preferred approach for Device 2b/C.
  - [377] notes that Option 2 offers a simpler implementation suitable for device 2b, making it a more practical choice.

- [379] states that Option 2 reduces complexity but offers only coarse power control due to its RSRP-range-based design.
- [385] states that Option 2 offers a simpler RSRP-range-based approach that aligns well with the practicality of ultra-low-power devices and is generally more suitable for such deployments.
- [386] states that Option 2 avoids pathloss calculation but offers only lower-precision power control despite having similar measurement requirements.
- [387] states that Option 2 uses RSRP-range-based power control with lower calculation complexity but involves a trade-off between required information overhead and power-control performance.
- [389] states that Option 2 offers a simpler RSRP-range-based method that aligns well with low-complexity A-IoT devices and can be enhanced through refined RSRP range design.
- [391] states that Option 2 provides a simpler RSRP-range-based open-loop power control method with configurable power-ramping behavior, making it suitable for A-IoT devices with intermittent availability.

- Sources [361], [369], [372], [383], [388], [371] report that Option 2 is feasible, while source [376] reports that Option 2 is not feasible.
  - [361] mentions that Option 2 is feasible, indicating that a device can perform an RSRP-like baseband measurement that includes AGC effects and determine its D2R transmit power using a lookup table provided by the reader.
  - [369] notes that Option 2 is feasible due to its low complexity.
  - [372] notes that Option 2 offers lower complexity through RSRP-based power selection but provides coarser and less adaptive power control.
  - [383] states that Option 2 avoids device-side formula computation, though the impact of such complexity remains unclear.
  - [388] states that Option 2 offers simple RSRP-range-based selection but may cause under- or over-powering due to coarse power levels.
  - [376] notes that Option 2 relies on an RSRP-to-power lookup table with coarse or dense quantization and is therefore not considered feasible for low-complexity device operation.

Between Option 1 and Option 2, sources [369], [370], [372], [379], [387], [386], [388], [371], [364] report that they prefer Option 1 over Option 2, while sources [375], [377], [385], [389], [391] report that they prefer Option 2 over Option 1.

6A.3.1.3 Closed-loop power control

Necessity and feasibility of closed-loop Tx power control for D2R are studied. Sources [383], [363], [364], [367], [370], [376], [389] indicate that closed-loop Tx power control is necessary and source [383] further states that it is feasible and necessary.

- [383] states that closed-loop power control can effectively mitigate interference in multi-reader scenarios, with only minimal additional R2D overhead.
- [363] mentions that because device-side measurement is difficult, at least closed-loop D2R power control should be supported, with the reader measuring the D2R signal, determining the appropriate transmit power, and indicating it to the device for power control.
- [364] mentions that closed-loop power control is suitable for Msg 3 and other D2R data transmissions, where the reader measures the preceding D2R transmission and indicates a power-control command to the device through the D2R scheduling message.
- [367] observes that in a stationary environment, closed-loop power control can achieve significantly higher accuracy than open-loop power control due to device measurement errors, device power-setting tolerance, and FDD uplink/downlink pathloss differences.

- [370] notes that closed-loop power control enables devices to transmit at a configured power level through R2D control signaling.

- [376] notes that closed-loop D2R power control should be studied, as reader feedback can refine device transmit power beyond OLPC accuracy.

- [389] states that closed-loop power control should be studied for D2R transmission, as it enables real-time power adjustment based on reader feedback.

Source [372] states that closed-loop Tx power control is feasible but not necessary.

- [372] notes that closed-loop power control is feasible but not essential compared to open-loop power control.

Source [385] states that closed-loop Tx power control is feasible and can be beneficial.

- [385] states that the closed-loop power control is feasible and can be beneficial for interference management of device-specific D2R transmission at the price of higher overhead and complexity.

Sources [361], [369] state that closed-loop Tx power control is not feasible.

- [361] mentions that closed-loop power control is not feasible, noting that it introduces additional signaling overhead for power adjustment, is limited by device availability, and still requires support for open-loop power control for the initial D2R transmission.

- [369] notes that closed-loop power control increases device power consumption due to continuous monitoring and is therefore not feasible for A-IoT devices with limited power and low complexity.

Sources [371], [379] state that closed-loop Tx power control is not necessary.

- [371] notes that closed-loop power control is unnecessary for R20 A-IoT devices if open-loop power control is introduced, given the added design complexity and unclear HARQ-ACK feedback mechanism.

- [379] states that closed-loop power control is not considered for Rel-20 A-IoT due to its signaling and energy overhead.

Source [375] states that the closed-loop power control would increase the signaling overhead and the device complexity, but a device/group-specific correction parameter could be considered along with the open-loop power control for better power control flexibility.

Source [365] states that the benefit of having in place a power control mechanism for A-IoT devices must be pondered with respect to any measurements, power consumption and signalling required for it to work, which are aspects that may go against the necessity of it.

6A.3.1.4 Provisioning

On the required information for D2R Tx power control at the reader and device, sources [364], [376], [385], [383], [375] state that Tx-power–related parameters can be considered.

- Sources [364], [376], [385] state that the Tx power of the reader can be set by the reader.

- Source [383] states that coefficient of pathloss can be beneficial for balancing between inter-cell interference and performance at cell edge for Option 1.

- Sources [385], [375] state that the target received power at the reader can be configured by the reader for Option 1.

- Sources [385], [375] state that the RSRP thresholds and the Tx-power mapping per RSRP range can be configured by the reader for Option 2.

Sources [367], [381] state that device power/energy-related information is needed for closed-loop Tx power control for D2R.

- Source [367] states that at least the available power of an A-IoT device can be reported by the device.

- Source [381] states that D2R power control depends on the device energy level.

6A.3.2 Reader identification/differentiation

Reader identification/differentiation is necessary for Device 2b and for Device C in outdoor scenarios, to differentiate R2D among different readers. A device acquires the reader ID-related information by at least one of the following options:

Option 1: From the broadcast information.

Sources [362], [363], [368], [370], [373]. [374], [378], [379], [383], [385], [391] state that Option 1 should be supported.

- [362] The device can receive reader ID-related information in the initial part of the access procedure, enabling it to be able to use the reader ID for both reader differentiation and for the initial D2R transmission for DO-A. In the case of 2-step broadcast information acquisition, the reader ID can be included in the first step itself to ensure that the device receives the reader ID-related information in time to perform reader differentiation and DO-A transmissions, while not impacting the overhead significantly.

- [363] Considering reducing the complexity of sequence detection for device, only one sequence of R2D periodic synchronization signal is supported. It's not suitable to carry reader-ID using sync signal, so option 1 should be supported that Reader ID-related information is contained in the broadcast information.

- [368] This relaxes synchronization signal detection complexity, but since the device cannot identify the cell before decoding the channel for broadcast, cell-specific parameters cannot be applied to physical channel carrying broadcast information, so inter-Reader interference randomization cannot be used.

- [370] Considering outdoor networking requirements, additional information such as the Cell or reader ID and Transaction ID also needs to be included in the broadcast information to help the device identify the serving cell or reader.

- [373] Due to the poor anti-interference capability of OOK and Manchester coding, it is difficult to enable different readers to be deployed on the same frequency point. Then, if in a certain area, there can only be one reader at a specific frequency point, i.e., there will be no adjacent readers deployed on the same frequency point, reader ID is unnecessary to be indicated along with each of the R2D transmission, it is enough to only indicate it during the initial frequency acquisition procedure, i.e., indicated by broadcast information.

- [374] Placing the reader ID outside the broadcast information would affect measurement and camp operations, which means measurement is only performed after synchronization procedure and waiting for the related signaling. Therefore, the reader ID should be included in the broadcast information.

- [378] Option 1 (broadcast-based reader ID) should be adopted as a baseline mechanism for reader identification, ensuring that all devices can obtain basic reader identification information with minimal complexity. Options 2, 3, and 4 may be additionally supported and used by devices to further identify and differentiate the corresponding reader, especially in outdoor scenarios where multiple readers coexist and enhanced differentiation is required.

- [379], [378] Option 1 (broadcast-based reader ID) should be adopted as a baseline mechanism for reader identification, ensuring that all devices can obtain basic reader identification information with minimal complexity. Options 2, 3, and 4 may be additionally supported and used by devices to further identify and differentiate the corresponding reader, especially in outdoor scenarios where multiple readers coexist and enhanced differentiation is required.

- [383] For initial broadcast information acquisition, reader ID-related information is contained in the broadcast information or carried by L1 R2D control information, i.e., option 1 or option 3 in the agreement should be considered.

- [385] Periodic system information (SIB) can provide necessary information including at least: Reader ID-related information, e.g., higher layer reader ID, R2D-related configuration, D2R-related configuration, etc.

- [391] Mechanisms that provide reader-ID-related information during the discovery phase are more suitable than those relying on later-stage control or higher-layer signaling. Options 1 and 2, which convey or derive reader identification from broadcast information or periodic synchronization signals,

respectively, align naturally with this requirement. Embedding or deriving reader-ID-related information from these early signals therefore minimizes delay, reduces energy consumption, and supports rapid reader selection in multi-reader and mobility scenarios.

On the other hand, source [366] states that Option 1 should not be supported.

- [366] Reader identification information is best stored in non-volatile memory to ensure A-IoT device can use the information in all the subsequent transmissions after the first reception of reader identification information. The only option is NOT feasible is including in the broadcast information in Option 1, which requires to store the reader identification.

Following observations were also provided for Option 1:

- [361] A device obtains reader ID-related information in a higher layer message without physical layer impact at the rate of broadcast information transmission.

- [369] According to the discussion on broadcast acquisition (i.e., single-step or two-step), MIB-like information and/or SIB-like information can be the potential broadcast information that contains reader ID. The complexity of option 1 is low since device can obtain the reader ID directly by decoding broadcast information without any blind detection on the sequence. If the reader ID is contained only by MIB-like information, a device can simply get it by one-step decoding. However, when the total number of reader IDs is large by referring to PCID (1008), the payload size of MIB-like information will be large. Considering about the limited payload size for MIB-like information, the reader ID can be divided into two parts. One part of reader ID-related information is contained by MIB-like information, and the other part is contained by SIB-like information (if defined). By referring to the legacy design, MIB along with PSS/SSS and SIB are separately transmitted. The transmission gap between MIB-like information and SIB-like information can be large, and their transmission periodicities may also be different. A device can obtain the whole reader ID until it receives a MIB and a SIB-like information, which leads to higher latency and processing time compared to single-step.

- [372] Reader identification via broadcast information requires no additional signal design but forces the device to decode broadcast information even for undesired readers.

- [376] Options 1, 2, and 4 are not necessarily transmitted with every R2D transmission. Therefore, they cannot provide reader identification/differentiation for every R2D transmission. To support reader differentiation, for every R2D transmission, it should be considered to either assume that Option 3 is always included in all R2D transmissions, apply an association rule when reader information is obtained via Options 1/2/4, or leverage the always-present R2D signal (e.g., SIP or CAP) to carry a reader-specific signature for differentiation.

- [275] If the Reader ID were not too long, "Option 2: Reader ID-related information is derived from the periodic sync signal sequence(s)" can be used aiming at performing an earlier reader identification. Otherwise, "Option 1: Reader ID-related information is contained in the broadcast information" can be considered.

Option 2: Derived from the periodic sync signal sequence(s).

Sources [366], [368], [372], [375], [391] state that Option 2 should be supported.

- [366] To facilitate the device in identifying readers and selecting the one with the strongest signal for access, the reader identification information, which indicates the identifier of the reader, should be carried by R2D synchronization signal.

- [368] This is similar as NR. But detecting different sequences during initial cell search increases device energy consumption. So, the number of IDs should be limited for this option.

- [372] Reader identification via a periodic sync signal sequence helps avoiding unnecessary broadcast decoding but increases detection complexity as the number of reader IDs grows.

- [391] Mechanisms that provide reader-ID-related information during the discovery phase are more suitable than those relying on later-stage control or higher-layer signaling. Options 1 and 2, which convey or derive reader identification from broadcast information or periodic synchronization signals, respectively, align naturally with this requirement. Embedding or deriving reader-ID-related information from these early signals therefore minimizes delay, reduces energy consumption, and supports rapid reader selection in multi-reader and mobility scenarios.

On the other hand, sources [362], [363], [373]. [374] state that Option 2 should not be supported.

- [362] Option 2 would increase the device complexity and power consumption since the device would have to blindly detect multiple sequences when performing the initial frequency acquisition procedure to determine the frequency location and reader ID-related information simultaneously.
- [373] For reader identification/differentiation, Option 2 is precluded because the number of sequences of R2D sync signal should be limited.
- [374] If only the reader ID is carried in the sync signal without accompanying broadcast information containing the SFN or time-unit index, the device would be unable to report measurements accurately, and the reader could not perform proper camp control. This would make measurements become meaningless. Consequently, it is preferable to include reader ID within the broadcast information.

Following observations were also provided for Option 2.

- [361] A device may have to perform detection of the periodic sync signal sequence to derive the reader ID-related information. It is unclear how a device can determine the better-quality sync signal sequence when sequences from multiple readers are received. This reader ID-related information is provided at the rate of periodic sync signal transmission.
- [369] This is similar to NR PCID acquisition method based on PSS and SSS detection. Option 2 helps a device to get reader ID at the very beginning step before following steps such as system-like information acquisition and PRDCH reception. As it is agreed, R2D periodic synchronization signal is a binary sequence in time domain. If the number of binary sequence(s) is larger than 1 and no larger than 4, the sequence(s) can be used for potential reader identification/differentiation. However, 4 binary sequence(s) are too limited to identify different readers by considering that outdoor scenario may exist even larger number of cells/readers. In order to support large number of reader ID, two-level indication can be considered by combining Option 2 and Option 1. A device needs to derive the first-level of reader ID by detecting the sequence of periodic synchronization signal and obtain the second-level of reader ID from MIB-like or SIB-like information. Compared to Option 2 only, this combination method requires the device to process two steps to get the reader ID.
- [376] Options 1, 2, and 4 are not necessarily transmitted with every R2D transmission. Therefore, they cannot provide reader identification/differentiation for every R2D transmission. To support reader differentiation, for every R2D transmission, it should be considered to either assume that Option 3 is always included in all R2D transmissions, apply an association rule when reader information is obtained via Options 1/2/4, or leverage the always-present R2D signal (e.g., SIP or CAP) to carry a reader-specific signature for differentiation.
- [378] Option 1 (broadcast-based reader ID) should be adopted as a baseline mechanism for reader identification, ensuring that all devices can obtain basic reader identification information with minimal complexity. Options 2, 3, and 4 may be additionally supported and used by devices to further identify and differentiate the corresponding reader, especially in outdoor scenarios where multiple readers coexist and enhanced differentiation is required.
- [383] For initial broadcast information acquisition, reader identification/differentiation based on periodic sync signal which device needs to blindly search on multiple candidate frequencies increases device complexity considering correlation-based detection. After broadcast information acquisition, reader ID-related information is derived from periodic sync signal or carried by L1 R2D control information.
- [275] If the Reader ID were not too long, "Option 2: Reader ID-related information is derived from the periodic sync signal sequence(s)" can be used aiming at performing an earlier reader identification. Otherwise, "Option 1: Reader ID-related information is contained in the broadcast information" can be considered.

Option 3: Carried by L1 R2D control information.

Sources [366], [375], [376], [383], [385] state that Option 3 should be supported.

- [376] To support reader differentiation, the study on reader identification should proceed by selecting one of the following directions. Direction 1: Assume that Option 3 (L1 R2D control information) is always included in every R2D transmission. Direction 2: Acquire reader-related information via at least one of

Options 1/2/4, and handle R2D transmissions without reader ID-related information based on an association rule (i.e., assume they originate from the same reader based on prior information).

- [383] For initial broadcast information acquisition, reader ID-related information is contained in the broadcast information or carried by L1 R2D control information, i.e., option 1 or option 3 in the agreement should be considered. After broadcast information acquisition, reader ID-related information is derived from periodic sync signal or carried by L1 R2D control information.

- [385] For flexible and reliable broadcast transmission, SIB-L1 control can be used to indicate the physical reader ID-related information, types of broadcast information, TBS, coding/repetitions for different coverage and device types, etc.

- [366] Since the reader identification information is only used in the scenario of the A-IoT receiving the PRDCH information from more than one reader, the retrieval of the reader identification is not critical whether is included in the periodic synchronization signal of Option 2, the L1-control information of Option 3, or the paging-like message of Option 4.

On the other hand, sources [371], [373]. [362] state that Option 3 should not be supported.

- [371] At least for initial reader search, a device anyway needs to acquire all the information to finish the reader search and camp on it, so no big different to carry reader ID information via L1 control information or payload part. It will also increase the payload of L1 control information and result in the loss of L1 control detection performance.

- [373] For reader identification/differentiation, there is no need to further indicating the Reader ID-related information after initial frequency acquisition procedure because there cannot be multiple adjacent readers in a same frequency point. Therefore, Option 3 is precluded.

- [362] The device would receive the reader ID only during an R2D transmission, which would mean that the device would not be able to use the reader ID for reader differentiation, nor for the D2R transmissions for DO-A. It would also increase the overhead of the control information as it is not necessary to be transmitted along with every R2D transmission.

Following observations were also provided for Option 3.

- [361] A device obtains reader ID-related information in the L1 R2D control information, either as a specific field or as a mask applied to the CRC. The mask in combination of option 1, option 2, or option 4 can be used to accept / reject the PRDCH.

- [369] Potential R2D signals with L1 R2D control information are SIB-like information, paging-like information, Msg2, R2D command (if these signals are supported). It is simple for a device to obtain the reader ID from L1 R2D control. When a device gets the reader ID, the device can determine whether to decode the corresponding data in PRDCH that is scheduled by this L1 R2D control. To support Option 3, a pre-condition is that the periodic synchronization signal and/or MIB-like information should also contain the reader ID. For example, a device should be synchronized to a reader and acquired its MIB-like information. If the reader ID in L1 R2D control of the following up R2D transmission(s) is not matched/same with the reader ID of the synchronized reader, the device can determine not to decode the corresponding PRDCH data part scheduled by this L1 R2D control.

- [372] Conveying the reader ID in L1 R2D control information enables explicit identification but introduces signaling overhead when applied to every PRDCH transmission. For Option 3, including the cell/reader ID in the L1 R2D control information is beneficial only when the broadcast information is transmitted on PRDCH and the L1 R2D control information is transmitted prior to the PRDCH. However, transmitting the cell/reader ID in the L1 R2D control information for every PRDCH used for inventory/command procedures is not desirable due to signaling overhead.

- [378] Option 1 (broadcast-based reader ID) should be adopted as a baseline mechanism for reader identification, ensuring that all devices can obtain basic reader identification information with minimal complexity. Options 2, 3, and 4 may be additionally supported and used by devices to further identify and differentiate the corresponding reader, especially in outdoor scenarios where multiple readers coexist and enhanced differentiation is required.

- [391] Options 3 and 4 inherently assume that the device has already achieved synchronization and is actively monitoring reader-specific control or paging occasions. As a result, reader identification becomes

available only after the device has already committed resources to a particular reader context, which increases both latency and energy expenditure.

Option 4: Contained in the higher-layer information in the paging-like message.

Source [366] states that Option 4 should be supported.

- [366] Since the reader identification information is only used in the scenario of the A-IoT receiving the PRDCH information from more than one reader, the retrieval of the reader identification is not critical whether is included in the periodic synchronization signal of Option 2, the L1-control information of Option 3, or the paging-like message of Option 4.

On the other hand, sources [371], [373]. [362] state that Option 4 should not be supported.

- [371] It is infeasible since it cannot provide reader ID-related information during initial reader search. At least Option4 cannot operate as a standalone solution.

- [373] For reader identification/differentiation, there is no need to further indicating the Reader ID-related information after initial frequency acquisition procedure because there cannot be multiple adjacent readers in a same frequency point. Therefore, Option 4 is precluded.

- [362] The device would not be able to use the reader ID for reader differentiation, nor for the D2R transmissions for DO-A. It will result in devices that want to transmit D2R DO-A traffic missing out on receiving the reader-ID related information and not being able to transmit the first D2R transmission for DO-A.

Following observations were also provided for Option 4.

- [361] A device obtains reader ID-related information in a higher layer message without physical layer impact at the rate of paging-like messages.

- [369] If paging-like message is supported, all of paging related information should be contained by higher-layer (e.g., MAC CE) and mapping to the data part of PRDCH in PHY-layer. Not like Option 3 that device can get the reader ID in L1 R2D control before decoding data part, Option 4 requires the device to decode both L1 R2D control and data part, and then the device can get the reader ID. Furthermore, Option 4 also prerequisites that the reader ID should be contained in periodic synchronization signal and/or MIB-like information.

- [372] Reader identification via higher-layer information simplifies physical-layer operations but prevents early identification prior to acquiring higher-layer signaling.

- [376] Options 1, 2, and 4 are not necessarily transmitted with every R2D transmission. Therefore, they cannot provide reader identification/differentiation for every R2D transmission. To support reader differentiation, for every R2D transmission, it should be considered to either assume that Option 3 is always included in all R2D transmissions, apply an association rule when reader information is obtained via Options 1/2/4, or leverage the always-present R2D signal (e.g., SIP or CAP) to carry a reader-specific signature for differentiation.

- [378] Option 1 (broadcast-based reader ID) should be adopted as a baseline mechanism for reader identification, ensuring that all devices can obtain basic reader identification information with minimal complexity. Options 2, 3, and 4 may be additionally supported and used by devices to further identify and differentiate the corresponding reader, especially in outdoor scenarios where multiple readers coexist and enhanced differentiation is required.

- [391] Options 3 and 4 inherently assume that the device has already achieved synchronization and is actively monitoring reader-specific control or paging occasions. As a result, reader identification becomes available only after the device has already committed resources to a particular reader context, which increases both latency and energy expenditure.

Sources [368], [371], [372], [374] provided observations that combination of Option 1 and 2 should be supported.

- [368] This option offers a trade-off between synchronization signal detection complexity and subsequent broadcast decoding performance. Finial determination may depend on the number of Reader IDs and sequence design of synchronization signal. If only a few IDs can be carried by sync signal sequence(s), the remaining can be contained in broadcast information.

- [371] For Option1+Option2, i.e., partial reader identification information is derived from the periodic sync signal sequence (3~4 candidate sequences) and the remaining information is carried in SIB1 broadcast information, the following benefits are observed. Compared to adopting option 1 only, after the initial search, a device can differentiate readers only via synchronization signal without decoding the following broadcast information, thereby reducing power consumption. And different periodic sync signal sequences can also reduce interference from different readers. Compared to adopting option 2 only, due to the limited number (e.g., 3~4) of candidate sequences of periodic synchronization signal, device blind detection complexity and design complexity can be reduced, and also the detection performance loss caused by large number of candidate sync sequences can be avoided. Based on these observations, it is proposed to support Option1+Option2 i.e., partial reader identification information is derived from the periodic sync signal sequence (3~4 candidate sequences) and the remaining information is carried by SIB1 broadcast information.

- [372] When determining the sequence length of the periodic sync signal, the number of cell/reader IDs that the device is required to distinguish needs to be considered. However, if the number of cell/reader IDs increases, either the number of periodic sync signal sequences or the sequence length increases, which results in higher detection complexity at the device side. Therefore, it may be necessary to consider Option 1 or Option 4 together with Option 2. In this case, the periodic sync signal may be used to distinguish cell/reader ID groups, and the actual cell/reader ID within the group may then be identified through broadcast information and/or higher-layer information.

- [374] If the synchronization signal and broadcast information are transmitted together, the reader ID could potentially be derived from the periodic synchronization signal sequences.

Sources [361], [375] provided observations that combination of Option1 and 3 should be supported.

- [361] If the goal is to identify PRDCH from intended readers, then option 3 plus option 1 provides a check on each message using information provided in a broadcast while having limited impact on processing complexity.

- [375] To reduce the device's monitoring time, the device needs to be synchronized with the serving reader(s). For this purpose, the periodic sync signal can carry reader ID-related information (e.g., Option 2). The sync signal does not need to carry the actual reader ID. For instance, 3 or 4 different sync signal sequences could be considered, while each sequence is used by one reader. The sequence can be assigned by NW to the reader. Once the device is synchronized with the serving reader, it will follow the reader for the PRDCH transmissions. Since the subcarrier might be used by other reader(s), the PRDCH should also carry the reader ID-related information. Although such information can be carried by the L1 control information or higher-layer information, using the L1 control information enables the device to only decode the control part to determine whether the message is from the serving reader. The L1 control information can carry the reader ID or a shorter ID dedicated to the reader assigned by the NW.

Sources [378], [385] provided observations that combination of Option1 and 2/3/4 should be supported.

- [378] Option 1 (broadcast-based reader ID) should be adopted as a baseline mechanism for reader identification, ensuring that all devices can obtain basic reader identification information with minimal complexity. Options 2, 3, and 4 may be additionally supported and used by devices to further identify and differentiate the corresponding reader, especially in outdoor scenarios where multiple readers coexist and enhanced differentiation is required.

- [385] All Options could be needed and not contradictory to each other. For Option 1 and Option 4, the reader-related information, i.e., higher-layer reader ID with long length used to uniquely identify a reader, could be included in the MAC IE. For Option 2 and Option 3, the reader-related information, i.e., PHY reader ID with short length used to differentiate the readers for RAN-level network management, could be included in the PHY signal/channel. Partial information can be carried by periodic SS (Option 2) and remaining information can be included in L1 R2D control per PRDCH.

Source [372] provided observations that combination of Option1 and 4 should be supported.

- [372] When determining the sequence length of the periodic sync signal, the number of cell/reader IDs that the device is required to distinguish needs to be considered. However, if the number of cell/reader IDs increases, either the number of periodic sync signal sequences or the sequence length increases, which results in higher detection complexity at the device side. Therefore, it may be necessary to consider Option 1 or Option 4 together with Option 2. In this case, the periodic sync signal may be used to distinguish cell/reader ID groups, and the actual cell/reader ID within the group may then be identified through broadcast information and/or higher-layer information.

6A.3.3 Initial frequency acquisition

An initial frequency acquisition procedure is necessary for Device 2b and for Device C, for flexible deployment of A-IoT frequency location(s) for outdoor scenarios. A device is expected to determine a frequency point from predefined synchronization raster points for reception of the periodic synchronization signal. Details of the raster design belong to RAN4.

6A.4 DO-A

6A.4.1 First D2R transmission for DO-A

6A.4.1.1 Delivery of configuration

The following options on how the necessary information for the first D2R transmission for DO-A traffic is provided to the device are studied. Options 1, 2 and 3 are feasible.

Option 1: Provided by broadcast information

Sources [362], [363], [364], [365], [368], [379], [383], [385], [386], [391], [371] support Option 1 based on the following observations.

- [362] Broadcast information should be used to provide the necessary information for the first D2R transmission for DO-A because it was agreed to be periodically transmitted, thereby ensuring that the device can receive this information regardless of the availability status of the device and the type of DO-A transmission (event-triggered or periodic) and guarantees the device to have the opportunity to transmit the first D2R transmission for DO-A. In the case of a two-step acquisition procedure, the necessary information is included in the SIB-like broadcast information, and the device can use the last received broadcast information to determine the necessary information.

- [364] Candidate D2R resources for the first D2R transmission of DO-A traffic should be indicated by the broadcast information in the A-IoT SSB rather than paging like aperiodic R2D signal (i.e., Option 2). Because whether to transmit DO-A data is fully up to device, which is different from DT or DO-DTT where the device should transmit data whenever triggered by R2D. Thus, aperiodic D2R resource allocation for DO-A cannot timely adapt to the real DO-A transmission demand and will exacerbate the issue that the traffic of DO-A is available but lack of allocated D2R candidate resources.

- [365] For providing necessary information for the first D2R transmission for DO-A for Device 2b and for Device C, the necessary information is provided by broadcast information.

- [368] For Option 1, as long as device acquires cell synchronization and reads broadcast information, it can get the access resource allocation and initiate random access any time it has D2R traffic to report, this option is simple and more straightforward.

- [379] From an initial DO-A transmission perspective, it is desirable for a device to obtain the required resource allocation and related parameters in a simple and deterministic manner, without relying on device-specific signaling or additional interaction with the reader. In this context, broadcast information naturally serves as a suitable mechanism for conveying such information. Providing the necessary information via broadcast information enables the device to acquire all required parameters after synchronization, without waiting for a paging-like message or performing additional decoding procedures. This approach is particularly beneficial for periodic DO-A transmission, where predictable transmission behavior and low latency are desired.

- [383] Necessary information for the first D2R transmission for DO-A provided by periodic broadcast information should be considered.

- [385] For providing necessary information for the first D2R transmission for DO-A, combining Option 1/3 with Option 2 is feasible, where Option 1/3 can be considered for periodic D2R resources and Option 2 can be used to adjust the D2R resources.

- [386] The resource allocation information of the first D2R transmission for DO-A is included in paging-like message or broadcast information.

- [391] Select between Option 1 and Option 3 to provide necessary information for the first D2R transmission for DO-A. Among the options under study for determining the initial set of resources for DO-A, Option 1 and 3 should be prioritized. These approaches not only minimize signaling overhead but also provides greater flexibility.

- [371] As agreed in RAN1, broadcast information is necessary for subsequent transmission and reception by all devices, so it is more signaling efficiency for periodic broadcast information to provide information of D2R transmission for both periodic DoA and event-triggered DoA traffics, compared to option2. For event-triggered DoA traffics, Option1 provide more guaranteed and stable DoA traffic service compared to option2 aperiodic paging-like triggering. Option 1 will not impact on the detection performance of periodic sync signal, compared to option3 that since necessary information derived from a synchronization signal e.g., by increase the number of sync signal sequences.

Following neutral or negative observations were also provided for Option 1.

- [372] Delivering the necessary information through broadcast information is simple and straightforward, but it increases signaling overhead when the broadcast information is transmitted together with a short-period sync signal.

- [361] Broadcast information contains at least one resource allocation. After a device obtains an ID for a DO-A group, the device can select the resource allocation associated with the group. This option offers the reader control to set the allocation globally.

- [370] Option 1 is a periodic DO-A resource allocation scheme that provides predictable and reliable access opportunities with bounded access latency.

- [366] For Options 1 and 3, the resource overhead and energy consumption are high with periodic transmission of the broadcast information. The A-IoT device also needs to store the information in non-volatile memory and count the time interval with the decoded broadcast information.

Option 2: Provided by a paging-like R2D message

Sources [366], [369], [381], [385], [386], [387] support Option 2 based on the following observations:

- [366] For Option 2, a paging-like indicator can be used as a polling message to query the A-IoT devices on the need of resource for first D2R transmission. This mechanism triggers device response to indicate whether DO-A message transmission is required.

- [369] R2D message carrying DO-A indication information can be sent periodically. Between two adjacent R2D messages, several DO-A occasions can be pre-allocated by the adjacent R2D message. For the device who has DO-A data to transmit, it can choose one of the DO-A occasions to send DO-A transmission. For the DO-A occasion, it can be interpreted as a direct transmission occasion, or a transmission occasion for transmission request. For example, the R2D message can be paging, and the occasion is used for Msg1 transmission.

- [381] Since A-IoT devices may have significant SFO, over long intervals, timing synchronization would be degraded. Therefore, to locate the resources accurately, a reference point such as a synchronization signal or a paging-like R2D message would be beneficial. A paging-like R2D message may incur higher overhead depending on the signal design but it would provide higher flexibility in resource allocation.

- [385] For providing necessary information for the first D2R transmission for DO-A, combining Option 1/3 with Option 2 is feasible, where Option 1/3 can be considered for periodic D2R resources and Option 2 can be used to adjust the D2R resources. For Option 2: periodic broadcast information to indicate periodic monitoring occasions for paging message(s) and the paging message can indicate one-time resources for first D2R.

- [386] The resource allocation information of the first D2R transmission for DO-A is included in paging-like message or broadcast information.

- [387] For providing necessary information for the first D2R transmission for DO-A for Device 2b and for Device C, at least support Option 2 and Option 4. For the first D2R transmission autonomously triggered by active device, reader implements CBRA triggering which targets devices with pending DO-A data.

Following neutral or negative observations were also provided for Option 2:

- [361] Paging-like information also contains at least one resource allocation. After a device obtains an ID for a DO-A group, the device can select the resource allocation associated with the group. This option offers the reader control to set the allocation globally.

- [370] Option 2 supports periodic and aperiodic DO-A resource allocation. For the aperiodic allocation, the network can dynamically configure DO-A resources to enable more flexible scheduling and reducing both signaling and resource overhead. Through network coordination, Option 2 can effectively avoid resource conflicts between DO-A traffic and DT traffic and mitigate network congestion.

- [372] Delivering the necessary information through a paging-like R2D message reduces signaling overhead at the reader side, but it introduces transmission delay because DO-A transmission can only occur after the device receives the paging message.

- [362] If the necessary information is provided by a paging-like R2D message (Option 2), if this message is used for polling, there would be an increase in the transmission overhead due to the additional DO-A scheduling request and another paging message to provide resource allocation information for the first D2R transmission. If the message is intended to be triggered by the device, it is unclear how the device can do so without any resource allocation information for such a trigger to be transmitted.

- [368] "Option2: necessary information provided by a paging-like R2D message" means more energy consumption before DOA traffic report than option 1 and has a risk of not enough energy for traffic report.

- [371] Since broadcast information is necessary for subsequent transmission and reception of all devices, if the necessary information is provided by the paging-like R2D message, additional signaling overhead and more power consumption will be caused for a device to get the first D2R resource, compared to option1. For periodic DoA traffic, using option2 aperiodic paging-like R2D message to trigger D2R resource of every periodicity will obviously cause signaling waste. For event-triggered DoA traffic, a reader cannot know when to do the first D2R resource scheduling via paging-like R2D message, thereby resulting in D2R transmission latency and maybe discontinuity of services.

- [379] Paging-like R2D messages introduce additional signaling steps and may not be well suited for devices that have not yet established any access context.

Option 3: Derived from the synchronization signal and provided by broadcast information

Sources [373]. [381], [382], [385], [391] support Option 3 based on the following observations.

- [373] For providing necessary information for the first D2R transmission for DO-A for Device 2b and for Device C, Option 3 should be adopted as following: TR->D for time resource determination is pre-defined and the end of the corresponding R2D sync signal or the PRDCH carrying broadcast information is used for the device to determine the time resource for the first D2R transmission.

- [381] Since A-IoT devices may have significant SFO, over long intervals, timing synchronization would be degraded. Therefore, to locate the resources accurately, a reference point such as a synchronization signal or a paging-like R2D message would be beneficial.

- [382] Option 3 has advantage over the others. When DO-A resources are conveyed via broadcast information, waiting for the next broadcast to detect changes may introduce unnecessary delays. Frequent monitoring of broadcast information to detect updates is inefficient for both devices and the system. Therefore, indicating updates to broadcast information via a field associated with a periodic synchronization signal can reduce unnecessary monitoring.

- [385] For providing necessary information for the first D2R transmission for DO-A, combining Option 1/3 with Option 2 is feasible, where Option 1/3 can be considered for periodic D2R resources and Option 2 can be used to adjust the D2R resources.

- [391] Select between Option 1 and Option 3 to provide necessary information for the first D2R transmission for DO-A. Among the options under study for determining the initial set of resources for DO-A, Option 1 and 3 should be prioritized. These approaches not only minimize signaling overhead but also provides greater flexibility.

Following neutral or negative observations were also provided for Option 3.

- [372] Delivering the resource information through synchronization signals is less desirable because blind detection of synchronization signals is required, which increases device complexity.
- [370] Option 3 supports periodic and aperiodic DO-A resource allocation.
- [361] This option can use MIB-like information and broadcast information. It combines control as well as allocation.
- [368] Option 3 is still based on option 1, and details can be further studied.
- [371] If the necessary information is derived by sync signal e.g., by increasing the number of sync signal sequences, it will have impact on the detection performance of sync signal. What necessary information for the first D2R transmission for DO-A should be carried by periodic synchronization signal and also the corresponding benefit are unclear.
- [379] Deriving information from synchronization signals may impose constraints on signal design and limit flexibility in conveying resource configuration details.
- [362] If the necessary information is derived from a synchronization signal and provided by broadcast information (Option 3), it would require multiple sequences to be used to design the synchronization signal and the device to blindly detect these multiple sequences when performing the initial frequency acquisition procedure, which will add to the complexity and power consumption of the device.
- [366] For Options 1 and 3, the resource overhead and energy consumption are high with periodic transmission of the broadcast information. The A-IoT device also needs to store the information in non-volatile memory and count the time interval with the decoded broadcast information.
- [366] For Options 1 and 3, the resource overhead and energy consumption are high with periodic transmission of the broadcast information. The A-IoT device also needs to store the information in non-volatile memory and count the time interval with the decoded broadcast information.

Option 4:Provided by a device-specific R2D transmission

On the feasibility and applicability of Option 4 for providing necessary information for the first D2R transmission for DO-A, sources [372], [376] state that Option 4 is feasible and applicable after inventory procedure/registration.

- [372] Option 4 is feasible and considered applicable when the device completes the inventory procedure prior to the DO-A data transmission.
- [376] Study should include cases where a device autonomously triggers registration and/or data transmission. In practical deployments, a reader generally collects sensing data only from devices already associated/paired/registered with it, and requiring random access for each transmission would severely impact device energy consumption. Hence, enabling autonomous transmissions via device-specific resource configuration after registration (Option 4) is feasible and necessary.

On the other hand, sources [366], [380], [371] state that it is or may be infeasible.

- [366] Reader cannot know which device would require resource for DO-A transmission. Thus, it is very unlikely that Reader can provide the necessary information by device-specific R2D transmission.
- [380] Not support Option 4. In event-triggered resource allocation, resources are assigned only when a specific predefined event occurs - for example, when a device has DO-A data ready for transmission. Such allocation must be initiated through prior signaling, meaning it cannot be applied until after the first D2R transmission within the DO-A process. Consequently, this approach is not suitable for the very first D2R transmission of DO-A. Moreover, device-specific R2D transmissions are unavailable at this stage, so they cannot supply the necessary information for the first DO A D2R transmission.
- [371] For option4, if the first D2R transmission for DoA is provided by device-specific R2D transmission, it means that DoA traffic can only be served for a device after it is identified/inventoried by a reader, which is one kind of Do-DTT operation in fact. That is, option 4 is infeasible to support DoA traffic to some extent. For devices only oriented to DoA traffic, performing this device identification before the first D2R transmission of DoA traffic is also unnecessary, since it confers no concrete benefit and instead incurs unnecessary resource overhead and power consumption.

Source [387] supports Option 4.

- [387] For providing necessary information for the first D2R transmission for DO-A for Device 2b and for Device C, at least support Option 2 and Option 4. For the first D2R transmission autonomously triggered by active device, reader implements CBRA triggering which targets devices with pending DO-A data.

Following neutral or negative observations were also provided for Option 4:

- [370] The DO-A traffic/resource periodicity is indicated to a connected device via the dedicated device-specific R2D transmission. The device then transmits its DO-A data payload directly based on the period. This option is appropriate for connected devices that have periodic DO-A reporting requirements. For example, when a connected device is configured to periodically report sensor data or status updates.

- [385] For Option 4: the first D2R transmission may be interpreted differently but the resource allocation for the first D2R transmission can be based on paging message in Option 2. If a device needs to go through inventory procedure before DO-A data reporting, the paging message with CBRA indication (as in Option 2) is used to trigger the inventory procedure (Msg1 is the first D2R), and after inventory procedure the device-specific R2D message can trigger DO-A reporting. If a device is known by the reader and does not need to go through inventory procedure, the paging message with CFA indication (as in Option 2) can be used to trigger the DO-A data reporting (DO-A data is the first DO-A).

- [371] For option4, if the first D2R transmission for DoA is provided by device-specific R2D transmission, it means that DoA traffic can only be served for a device after it is identified/inventoried by a reader, which is one kind of Do-DTT operation in fact. That is, option 4 is infeasible to support DoA traffic to some extent. For devices only oriented to DoA traffic, performing this device identification before the first D2R transmission of DoA traffic is also unnecessary, since it confers no concrete benefit and instead incurs unnecessary resource overhead and power consumption.

- [380] Not support Option 4. In event-triggered resource allocation, resources are assigned only when a specific predefined event occurs - for example, when a device has DO-A data ready for transmission. Such allocation must be initiated through prior signaling, meaning it cannot be applied until after the first D2R transmission within the DO-A process. Consequently, this approach is not suitable for the very first D2R transmission of DO-A. Moreover, device-specific R2D transmissions are unavailable at this stage, so they cannot supply the necessary information for the first DO A D2R transmission.

- [362] If the necessary information is contained in a device-specific R2D transmission (Option 4), it is necessary for the device to transmit an additional scheduling request which also increases the transmission overhead.

- [366] Reader cannot know which device would require resource for DO-A transmission. Thus, it is very unlikely that Reader can provide the necessary information by device-specific R2D transmission.

- [368] Option 4 is based on RACH procedure ahead, and is not the first D2R transmission for DOA.

On whether the reader provides the necessary information for the first D2R transmission for DO-A traffic periodically and/or aperiodically, sources [363], [364] state that it is provided periodically without mentioning specific option, and source [376] states that it is provided either periodically or aperiodically. Sources [362], [370], [371], [372], [383], [385], [374] state that it is periodic for Option 1. For Option 2, sources [369], [370], [374] state that it is periodic, while sources [370], [371], [372], [385] state that it is aperiodic. For Option 3, sources [370], [371], [372], [385] state that it is periodic, while source [370] states that it is aperiodic. For Option 4, source [370] states that it is provided periodically. Following options were provided during the study.

- [369] R2D message carrying DO-A indication information can be sent periodically. For example, the R2D message can be paging, and the occasion is used for Msg1 transmission.

- [370] Option 1 is a periodic DO-A resource allocation scheme that provides predictable and reliable access opportunities with bounded access latency. Option 2 supports periodic and aperiodic DO-A resource allocation. For the aperiodic allocation, the network can dynamically configure DO-A resources to enable more flexible scheduling and reducing both signaling and resource overhead. Through network coordination, Option 2 can effectively avoid resource conflicts between DO-A traffic and DT traffic and mitigate network congestion. Option 3 supports periodic and aperiodic DO-A resource allocation. For Option 4, the DO-A traffic/resource periodicity is indicated to a connected device via the dedicated device-specific R2D transmission. The device then transmits its DO-A data payload directly based on the period.

- [371] By using option 1 or option3, necessary information is periodically provided by broadcast information. And for option2, a paging-like R2D message is more like an aperiodic transmitted polling message.

- [385] For providing necessary information for the first D2R transmission for DO-A, combining Option 1/3 with Option 2 is feasible, where Option 1/3 can be considered for periodic D2R resources and Option 2 can be used to adjust the D2R resources. For Option 1/3: periodic broadcast information to indicate periodic resources for first D2R. For Option 2: periodic broadcast information to indicate periodic monitoring occasions for paging message(s) and the paging message can indicate one-time resources for first D2R.

- [374] A strictly periodic broadcast information transmission scheme is supported at the reader side, and a paging-like message is used as the broadcast information.

- [376] The necessary information for the D2R transmission(s) for DO-A for Deivce 2b and for Device C is provided by the reader either periodically or aperiodically. Considering that the necessary information for DO-A can be conveyed via broadcast information, paging-like signaling, or device specific R2D message, such information can be delivered by the reader either periodically or aperiodically.

6A.4.1.2 Resource allocation

On whether the resource for the first D2R transmission for DO-A traffic should be periodic or aperiodic, sources [362], [364], [365], [368], [371], [373]. [375], [376], [380], [383], [386], [390], [391], [363] report that it should be periodic.

- [362] The resource allocation information provided in the SIB-like broadcast information is periodic for both periodic and event-triggered D2R transmissions for DO-A. If the resource allocation information is provided aperiodically, it is possible that the number of resources indicated is less than the actual number of devices wanting to transmit D2R transmissions for DO-A, and is not aware of when the next SIB-like transmission containing the resource allocation information can be expected, resulting in a decreased access capability due to the limited resources.

- [364] Because whether to transmit DO-A data is fully up to device, which is different from DT or DO-DTT where the device should transmit data whenever triggered by R2D. Thus, aperiodic D2R resource allocation for DO-A cannot timely adapt to the real DO-A transmission demand and will exacerbate the issue that the traffic of DO-A is available but lack of allocated D2R candidate resources.

- [365] For resource allocation method for Option 1 (Msg1-like content) for DO-A for Device 2b and for Device C, consider only periodic Msg1 resource allocation.

- [368] For resource allocation method for the first D2R transmission for DO-A for Device 2b and for Device C, support Option 1: Periodic D2R resource allocation. For Option 1, as long as device acquires cell synchronization and reads broadcast information, it can get the access resource allocation and initiate random access any time it has D2R traffic to report, this option is simple and more straightforward. Option 2: Aperiodic D2R resource allocation for the first D2R transmission for DO-A requires more standard effort and consumes more energy before device can send DOA traffic.

- [371] Pros of Option 1 (Periodic D2R resource allocation): 1) For periodic DoA traffic, periodic D2R resource is naturally needed. Otherwise, by using R2D message to trigger every D2R resource will cause heavy signaling overhead. 2) For event trigger DoA traffic type, since reader cannot be aware of when the urgent event will happen, providing a periodic D2R resource occasions will be more guaranteed. Cons of Option 2 (Aperiodic D2R resource allocation): 1) Aperiodic D2R resource cannot support periodic DoA traffic type. 2) A reader cannot know when the need for event-triggered DoA traffic will happen, so using aperiodic D2R resource will cause either traffic transmission latency or resource overhead.

- [373] For resource allocation/determination method for DO-A transmission, at least a subset of periodic signals can have associated PRDCH to carry the scheduling information for the first D2R transmission.

- [375] Support periodic resources for DO-A transmission within a valid time period, indicated by the reader or preconfigured for the device.

- [376] For resource allocation method for the D2R transmission(s) for DO-A for Device 2b and for Device C, periodic D2R resource allocation is feasible. For event-triggered DO-A transmission, the device randomly selects one resource from the configured periodic resource and initiates DO-A transmission via RA. For periodic DO-A transmission, the device transmits DO-A data directly using the configured periodic resource, without RA.

- [380] Periodic resource allocation is recommended. In aperiodic resource allocation, resources are assigned only when a specific predefined event occurs - for example, when a device has DO-A data ready for transmission. Such allocation must be initiated through prior signaling, meaning it cannot be applied until after the first D2R transmission within the DO-A process. Consequently, this approach is not suitable for the very first D2R transmission of DO-A. Dedicated band or broadcasting a set of resource can be further considered for resource allocation in supporting DO-A traffic.

- [383] For resource allocation method for the first D2R transmission for DO-A, periodic D2R resource allocation has an advantage of reducing R2D signalling overhead.

- [390] The resources required for transmitting the first DO-A message can be scheduled after the periodic synchronization signal. These resources do not require explicit scheduling information; instead, they can be implicitly indicated through a defined timing and frequency relationship with the periodic synchronization signal.

- [391] From the device's perspective, an aperiodic resource allocation may impose a significant burden, as devices would need to continuously monitor R2D signals and wait for dynamically indicated transmission opportunities. In our view, a periodic resource allocation is preferred, as it provides predictable transmission opportunities that both the reader and devices can rely on.

- [363] For supporting DO-A, at least following information should be provided by reader: Time domain resource information for DO-A, timeline (for determining the exact location in time domain, including the timeline for determining the first time domain resource), interval between two time domain resource, length of a time domain resource, number of DO-A resource (i.e., number of TDMed DO-A resource), frequency domain resource information for DO-A, frequency location of a typical resource (i.e., a reference frequency resource which can be used for determining other frequency domain resources), number of frequency domain resource (i.e., number of FDMed DO-A resource).

On the other hand, source [366] states that it should be aperiodic.

- [366] The polling-based mechanism for aperiodic resource allocation is an on-demand resource allocation and would allocate the resource efficiently by the reader polling the A-IoT device for query of any DO-A traffic arrival. Compared to aperiodic resource allocation, the periodic resource allocation for DO-A transmission would have a lot of resources not used and wasteful. Most of periodically configured resources for DO-A are not used when the device has no data to transmit.

Sources [370], [377], [381], [385], [369] state that both periodic and aperiodic resource allocation should be supported.

- [370] Periodic resource allocation provides predictable and reliable access opportunities with bounded access latency. Aperiodic resource allocation can dynamically configure DO-A resources to enable more flexible scheduling and reducing both signaling and resource overhead. Through network coordination, Option 2a/2b can effectively avoid resource conflicts between DO-A traffic and DT traffic, and mitigate network congestion.

- [377] Support both periodic and aperiodic resource allocation for DO-A. Periodic allocation for devices with regular reporting patterns and aperiodic allocation using contention-based access for event-triggered traffic.

- [381] For resource allocation method for the first D2R transmission for DO-A for Device 2b and for Device C, both periodic D2R resource allocation and aperiodic D2R resource allocation are supported with further study on: 1) Periodic resource allocation with reference to an R2D periodic sync signal. 2) Aperiodic resource allocation multiplexed with other D2R transmissions. Periodic resource allocation may be useful to accommodate periodic traffic while aperiodic resource allocation may be useful for transmission of event-triggered traffic.

- [385] Periodic resources are needed at least for the first D2R transmission for DO-A. Periodic Access occasion (AO) resources for device random access to support periodic DO-A traffic. Periodic Access request (AR) resources for device common request to support for event-triggered DO-A traffic. Aperiodic resources can be additionally configured by on-demand paging for the following DO-A transmissions. Consider resource sharing for DO-A and DO-DTT/DT to improve resource utilization.

- [369] For DO-A resources determination in Rel-20, resource pool-based method and Msg3-associated method can be studied. For resource pool-based method, R2D message carrying DO-A indication information can be sent periodically. Between two adjacent R2D messages, several DO-A occasions can be pre-allocated by the adjacent R2D message. For the device who has DO-A data to transmit, it can choose one of the DO-A occasions to send DO-A transmission. For Msg3-associated method, Msg0 can be sent periodically, and each DO-A

procedure following an inventory procedure. After the inventory procedure, the R2D message carrying DO-A indication can be sent to allocate several DO-A occasions. For the device who has DO-A data to transmit, it can choose one of the DO-A occasions to send DO-A transmission.

Sources [372], [374] submitted observations without mentioning their preferences.

- [372] For the resource allocation method for the first D2R transmission for DO-A, periodic allocation provides predictable resource availability but may waste resources when device activity is low, whereas polling-based aperiodic allocation improves efficiency by assigning resources only when DO-A traffic is detected.
- [374] Support a SR scheduling mechanism for the device to report its data actively in the DO-A service. Support a DO-A data payload D2R transmission mechanism for the device to report its data actively in the DO-A service. Support a SR mechanism without access step (SR without MSG1) for the device to report its data actively in the DO-A service.

6A.4.1.3 Content of transmission

The following options for the content of the first D2R transmission for DO-A traffic are studied in terms of their recommended scenarios, i.e. event-triggered or periodic, and the subsequent transmissions/receptions/procedures, if any:

Content type 1: Access Random ID message (Msg1) like content

Content type 2: DO-A data payload

Content type 3: DO-A scheduling request

Study results in terms of recommended scenarios, i.e. event-triggered or periodic, and the subsequent transmissions/receptions/procedures (if any), and other aspects are summarized in Table 6A.4.1.3-1.

Table 6A.4.1.3-1: Study of the content of the first D2R transmission for DO-A traffic

	Recommended scenarios (event-triggered or periodic)	Subsequent transmission(s)/reception(s)/procedure(s)	Other aspects
Option 1 Msg1-like content	Both event-triggered and periodic: [361], [362], [365], [367], [368], [369], [370], [371], [385] Event-triggered: [372], [379], [383], [376] Periodic: [380]	[361] Procedure after first transmission: The reader transmits a Msg2-like transmission in response to successful reception of the first transmission. The Msg2-like transmission can provide scheduling parameters (e.g., size of payload). [362] The subsequent procedure follows the CBRA procedure defined in Rel-19, where a random ID response message transmitted by the reader echoes the RN16 for contention resolution. [363] Device needs to perform random access first and then send DO-A data payload according to the reader's schedule. [364] The reader transmits a response to each of identified the Msg1-like transmission and allocates subsequent D2R transmission resource to convey DO-A data and/or further identification information of the A-IoT device accordingly. The A-IoT device transmits the DO-A data and/or further identification information on the allocated D2R transmission resource. Reader transmits an echoed identification to the corresponding A-IoT device for acknowledgement. [365] Option 1 is essentially reusing the random access (RA) messages specified in Rel-19 for DT and DO-DTT also for DO-A, but without being triggered by a preceding R2D transmission (paging message/Msg0), i.e., the device accesses the reader through Msg1 triggered by DO-A traffic.	[361] Other aspects: Mimicking contention-based access can simplify the procedure after first transmission. [362] Reusing the random-access procedure defined in Rel-19 without triggering message reduces the standardization effort. If the resource allocation information is also periodic, the device can determine the required resources to transmit Msg1 irrespective of the transmission being periodic or event triggered without any delay or increased latency. This periodicity can also be configurable based on the use cases to avoid unnecessary wastage of resources. [368] Pre-step: Periodic random access resource is configured by broadcast information. This option is anyway needed either for option3 after paging message or for option2 before periodic D2R resource is configured. [371] The first D2R transmission resources is indicated by broadcast information for Msg 1 transmission. A CBRA based procedure by using sequence-based msg1 as the first D2R transmission content.

		[366] The reader will respond to the detected Msg1-like transmission (containing DO-A SR) and send out the resource allocation on PRDCH for DO-A traffic transmission. The A-IoT device transmits the DO-A traffic according to the resource allocation from reader. [367] Contention-based random access procedure, can be based on the Rel-19 procedure. [368] Random access procedure, e.g.Msg.2/3/4, and DOA data transmission [369] A random access procedure is needed before the transmission of exact sensing result(s). [370] CBRA defined in Rel-19 [371] (1) msg2 response to msg1 transmission, (2) msg3 transmission (3) msg4 for contention resolution. [372] 1) Devices transmit Msg1 like content in a contention-based manner. 2) Successful devices are granted D2R resources and transmit DO-A payload [373] Devices monitor Msg2 message for resource allocation to transmit D2R Upper Layer Data which contains DO-A traffic. [374] The device acquire ACK message contained in the MSG2 as well as other necessary information, such as the scheduling information for next D2R transmission. [377] Reuses random access procedure, device transmits preamble to request resources. [378] Rel-19 CBRA [379] The data payload is transmitted in the form of DO-DTT after the random access procedure. [380] CBRA-like procedure. Applying CDM to Msg1 transmission increases access capacity by allowing more devices to transmit simultaneously. [383] 1) Msg.2 in response to Msg.1. 2) Msg.3 with DO-A data, and ID for contention resolution (if sequence-based Msg.1 is applied). 3) contention resolution (if sequence-based Msg.1 is applied). [385] To receive Msg2 for the following Msg3, Msg4 similar as CBRA [386] The same procedure for random access in Rel-19 can be adopted, that device receives Msg2 and transmits Msg3 carrying DO-A data payload. [376] RA-based access/registration procedure	Compared to R19 CBRA based random access procedure with additional contention resolution msg4, less specification effort will be. [372] Inventory follows a slotted ALOHA based procedure Higher latency due to inventory before DO-A No need for prior inventory [373] For Option 1, DO-A and DO-DTT can use unified procedure, i.e., starting with CBRA and then performing D2R transmission for the traffic. It is possible to share resources for the two types of traffic. [380] The collision between Msg1 transmissions can be resolved by echoing Msg1 content within Msg2. Applying CDM to Msg1 transmission increases access capacity by allowing more devices to transmit simultaneously. [385] The reader does not know which device to access and can configure a set of Msg1-like resources for random access to support large capacity. The CBRA supported in Rel19 can be used as starting point with enhanced collision handling. However, for event-triggered DO-A, which may have tighter latency requirements than periodic DO-A, the reader must allocate resources with shorter periodicity. This approach may lead to significant overhead and inefficient resource utilization, particularly when events occur infrequently. [376] The device initiates the procedure using DO-A traffic. Contention-based resources are assumed. [371] For procedure 1 as provided in the column of Subsequent transmission(s)/reception(s)/procedure(s) for option1, it is a reliable way for a device to access to reader and then perform device-specific transmission. And compared with procedure 1, procedure 4 (i.e., SR as the first D2R content to trigger R2D command and the R2D command schedules msg1-like resource for CBRA) which has additional two steps (scheduling request (SR) and R2D command) is unnecessary and overdesigned. [383] It requires the highest R2D signaling overhead.
Option 2 DO-A data payload	Both event-triggered and periodic: [361], [369], [376], [369] Periodic: [368], [370], [371], [372], [379], [380], [383], [385] None: [362]	[361] Procedure after first transmission: The reader transmits a Msg2-like transmission in response to successful reception of the first transmission. Instead of the random number, the CRC of the first transmission can be used. The Msg2-like transmission can provide scheduling parameters (e.g., size of additional payload resources). [362] CFA-like procedure	[361] Other aspects: Procedure can mix physical layer and upper layer elements are the reader and device. Large payload sizes can lead to unused resources when no device has DO-A content to transmit while small payload sizes can lead to unequal error detection if multiple segments are needed. [362] Low resource utilization and resource wastage due to the reader having to

		[366] The content of subsequent transmission/reception after the first D2R transmission is to indicate the desired resource payload of DO-A data. [367] None [368] Possible feedback and re-transmission [369] None [370] CFA defined in Rel-19 [371] None [372] 1) Reader allocates device specific D2R resource. 2) Device directly transmits DO-A data payload (contention free). [379] A collision resolution mechanism may be considered [380] CFA-like procedure. Acknowledgment mechanism to mitigate collisions among DO-A data payloads and ensure reliability. [383] Contention resolution (ID for contention resolution is provided in the first D2R transmission for DO-A.) [385] If Option 2 is for device-specific resources, to receive the acknowledgement for D2R reception, similar as for paging with CFA indication. If Option 2 is for non-device-specific resources, to receive the acknowledgement for D2R reception, similar as for paging with CFA indication.	allocate a large number of resources in the absence of a scheduling request, to attempt to minimize the collision probability and factor in the large DO-A payload sizes. System performance deteriorates in the absence of a collision resolution mechanism when there are more devices wanting to transmit than available resources. [367] Option 2 is suitable and can be very efficient if the following specific conditions are met: 1. Payload size is small and has an upper bound that is known beforehand (e.g. fixed size for a sensor measurement report). 2. Predictable message rate (e.g. periodic traffic generated by a known number of devices) If the conditions are not met, then this is likely to be extremely inefficient since the network would have to vastly over-dimension the resources for the first D2R transmission. [368] Pre-step: Periodic D2R resource is configured to device. If device common D2R resource is configured, collision may happen for DOA reporting. If device dedicated D2R resource is configured, the resource overhead is large. For this case, the device has to keep a connection state to maintain D2R synchronization, which means high power consumption for long interval, e.g. 5~15 minutes traffic interval in TR22.840. [371] The first D2R transmission resources is indicated by broadcast information for CG resource for DOA. A CFRA based procedure by using DoA data as the first D2R transmission content. For whether device ID should be transmitted together with DoA data, for a reader to distinguish the DoA data from multiple devices, device ID needs to be transmitted together with DoA data. [372] Suitable for low latency or urgent DO-A delivery Requires prior inventory [380] To enhance access capacity, the number of resources indicated by the R2D paging message must be increased. Applying acknowledgment mechanism to mitigate collisions among DO-A data payloads and ensure reliability. [382] Supporting a grant-free channel access for DO-A traffic for active devices in Rel-20 reduces overhead and latency. In our view, a grant-free D2R, where the device is allowed to initiate its transmission on resources it chooses itself is a viable choice. [385] If Option 2 is for device-specific resources: The reader already knows the device and can use device-specific R2D

			transmission to trigger DO-A data reporting. CFA with fewer steps reduces latency and prevents collisions, but it also results in large overhead and low capacity. The CFA supported in Rel19 can be used as starting point. If Option 2 is for non-device-specific resources: The reader does not know the device and configure a set of MsgA-like resources for random access to support large capacity. Two-step CBRA reduces latency but has higher probability of collisions than Msg1-like random access. Two-step CBRA, not supported in Rel19 A-IoT, may have specification impact. [376] Depending on whether the DO-A traffic is periodic data transmission or event-triggered transmission, the resources may be contention-free or contention-based. For periodic reporting, DO-A traffic is transmitted over configured (dedicated) D2R resources. For event-triggered reporting, DO-A traffic is transmitted using contention-based resources. [371] For procedure 2 as provided in the column of Subsequent transmission(s)/reception(s)/procedure(s) for option 2, the target scenario is latency-insensitive periodic DoA traffic with sparse transmission period e.g., in minute level. Due to the low data collision probability, using broadcast information to directly schedule CG-like resource for data transmissions will be efficient. Regarding device ID under option 2, for a reader to distinguish the DoA data from multiple devices, device ID needs to be transmitted together with DoA data. [369] None For Option 2, DO-A data payload is transmitted by the first D2R. It is simple that device can directly use the configured resource(s) for sensing result transmission whenever it has, and it simplifies the whole procedure by eliminating random access from DO-A. [383] It requires the lowest R2D signal overhead and shortest latency of DO-A procedure. On the other hand, it requires the largest size of pre-allocated periodic D2R resources and may lead to waste of the pre-allocated resources if device does not have frequent DO-A traffic to transmit considering event-triggered DO-A traffic.
Option 3 DO-A scheduling request	Both event-triggered and periodic: [362], [367], [368], [369], [370], [378] Event-triggered: [361], [371], [372], [383], [388], [385], [373] Periodic: [380]	[361] Procedure after first transmission: Procedure after first transmission: The reader transmits a paging-like transmission for inventory in response to determining the first transmission contains a scheduling request. Instead of the random number, the CRC of the first transmission can be used. [362] An additional D2R message, i.e., DO-A scheduling request, and a corresponding additional R2D paging message, i.e., DO-A specific paging message carrying the resource	[361] Other aspects: Procedure can mix physical layer and upper layer elements are the reader and device. Large payload sizes can lead to unused resources when no device has DO-A content to transmit while small payload sizes can lead to unequal error detection if multiple segments are needed. A small number of devices can improve resource efficiency for contention. [362]

		allocation information, are required, following which CBRA procedure can be used. [363] Device needs to perform random access first and then send DO-A data payload according to the reader's schedule. [367] It depends on exact usage of the resource for the first D2R transmission. If that resource is e.g., dedicated to a device then no contention resolution is needed, the reader provides a resource grant for a subsequent D2R transmission. Otherwise, contention resolution is required (e.g. reader transmits a paging message which indicates paging only for devices with DO-A to transmit, followed by the legacy CBRA). [368] Paging-like R2D transmission to allocate random access resource, random access procedure, e.g., Msg.2/3/4, and DOA data transmission. [369] It requires additional procedures and extending the latency to transmit sensing result. [370] DO-A specific Paging + Msg1 + CBRA defined in Rel-19 [371] (1) An R2D command to trigger CG-like D2R resource for DOA data transmission, then, (2) following DoA data transmission. [372] 1) Reader allocates common D2R resource for SR. 2) Upon SR reception, reader grants D2R resource. 3) Device transmits DO-A data payload. [373] Devices monitor A-IOT paging message for Msg1 resource allocation and then follow the Rel-19 CBRA procedure to transmit D2R Upper Layer Data which contains DO-A traffic. [375] Once the reader detects the SR, it can initiate the random-access procedure, and data would be transmitted like the DO-DTT procedure. [378] A-IoT paging triggers CBRA by mostly reusing the design in Rel-19. Collision of the "first D2R transmission" from different devices does not matter as the "first D2R transmission" does not serve any purpose other than triggering A-IoT paging. [380] The R2D paging message triggers the subsequent D2R transmission for scheduling requests. Following this, the R2D response message is used to indicate the resources allocated for the DO-A data payload. [383] 1) R2D schedules contention-based D2R data Tx, 2) DO-A data and ID for contention resolution. 3) Contention resolution. [388] After the reader receives the SR-like, the DO-A procedure falls back to the inventory procedure. [385] To receive paging for the following Msg1, Msg2, Msg3, Msg4 similar as CBRA	Increase in signalling overhead due to the additional DO-A scheduling request and the corresponding DO-A specific paging message required to be provided by the reader to each device that requested resources, as compared to using Msg1. [368] Pre-step: Periodic scheduling request resource is configured by broadcast information. If SR resource is device common, it is hard for Reader to determine how many RO resource to be allocated by paging. If the scheduling request rate is high for devices, following paging overhead will be large, the overhead reduction benefit compared to option1 will be reduced. The delay is larger than option1. [371] The SR resources is indicated by broadcast information to avoid the CG-like resource overhead for directly data transmission. [372] Only small D2R resource needed for SR Multiple SRs may cause DO-A collisions Randomized SR resource or post grant inventory may mitigate collisions [373] Multiple devices can share resource for the scheduling request transmission, which can improve the resource efficiency for DO-A first D2R transmission and more applicable to the event-trigger type of traffic. [380] Study the design of R2D paging message, D2R for scheduling request, and R2D response message. Frequency separation between periodic and aperiodic transmissions helps reduce both the interval for periodic transmissions and their associated transmission latency [371] To avoid the CG-like resource overhead of procedure 2 as provided in the column of Subsequent transmission(s)/reception(s)/procedure(s) for option 2, procedure 3 as provided in the column of Subsequent transmission(s)/reception(s)/procedure(s) for option 3 with broadcast information to schedule SR resource instead of CG-like resource could be useful. Once SR is detected by reader, it can schedule the CG-like resource for data transmission by using R2D command. [385] The reader does not know which device to access and can use shared common resources for different devices to send same SR without random access. Only if the reader detects the SR, it will send paging to configure additional resources for any devices who transmits the DO-A SR, where the CBRA supported in Rel19 can be used as starting point after receiving the SR. For event-triggered DO-A, it can save the overhead than Msg1-like D2R (Option 1) when events occur infrequently. [383] It requires the smallest size of pre-allocated periodic D2R resources.

6A.4.2 Synchronization for DO-A

For the time and frequency synchronization required for the device to transmit DO-A traffic, sources [369], [378], [379], [390], [372], [388], [371], [385], [380], [362] report that the time and frequency synchronization can be achieved by at least R2D periodic synchronization signal. Sources [372], [388], [371], [362] further state that in addition to the R2D periodic synchronization signal, CFO calibration signal can also be used for frequency synchronization. Source [361] reports that the time and frequency synchronization can be achieved by PRDCH transmission preceding the DO-A resources, while source [376], [374] states that it can be achieved by periodic or semi-periodic PRDCH transmission. Following observations were provided during the study.

- [369] To decrease the residual SFO, R2D periodic synchronization signal can be studied. For example, the R2D periodic synchronization signal can be inserted between adjacent DO-A indication messages or Msg0s.
- [378] For Device 2b and Device C with periodic DO-A traffic, a periodic synchronization signal is expected to precede the DO-A transmission.
- [379] The device acquires time and frequency synchronization for DO-A by using the periodic synchronization signal.
- [390] A periodic synchronization signal should be used to provide timing and frequency synchronization for DO-A traffic for device 2b and device C.
- [372] A CFO calibration signal may need to be transmitted close to the DO-A transmission occasion depending on whether the periodic sync signal includes a CFO calibration signal and on the period of the periodic sync signal.
- [388] Considering periodic synchronization signal/calibration signal to provide R2D synchronization and timing for DO-A traffic.
- [371] The timing and frequency synchronization for DoA traffic is provided by periodic sync signal and CFO calibration signal.
- [361] A PRDCH transmission precedes the DO-A resources providing timing and frequency synchronization for the first D2R transmission and can be used to mark the beginning of DO-A resources.
- [376] A PRDCH is periodically transmitted, which provides a basis for synchronization for DO-A transmission. This periodic PRDCH provides synchronization for DO-A transmission, system-information-like information for DO-A transmission, etc. Synchronization for DO-A can be obtained using at least one of the elements included in a PRDCH transmission, such as a preamble, a synchronization signal, or broadcast information.
- [362] For initial frequency acquisition, the device uses a set of pre-defined sync rasters to blindly detect the periodic synchronization signal. On detecting the synchronization signal, it obtains coarse frequency synchronization and calibrates its clock accuracy, and on decoding the following MIB-like broadcast information carried on the PRDCH, it obtains the reader ID and SFN information. The device then detects the following CFO calibration signal to obtain fine frequency synchronization and calibrates the CFO/SFO to the residual level, e.g., 10 ppm, hence acquiring time and frequency synchronization.
- [374] A period PRDCH transmission that contains broadcast information will indicate the common MSG1 resource that allow all types of devices to occupy and sent the first D2R transmission for DO-A.

6A.5 Timing relationships

6A.5.1 R2D → D2R

At least for CBRA for DT and DO-DTT, regarding the changes to existing timing offset value b/w R2D and corresponding subsequent D2R (e.g., Toffset1/2/3/4) for Device 2b and for Device C, the following observations were reported:

- [361] mentions that the values of Toffset1, Toffset3, and Toffset4 may vary depending on the PRDCH and PDRCH payload lengths, while Toffset2 is expected to be smaller due to improved SFO but may still reflect propagation-delay effects.

- [366] observes that the timing-offset configuration should be updated in Rel-20 to reflect the revised maximum numbers of time-domain resources for Msg1/Msg3, the improved SFO accuracy of Device 2b/C, and the FEC scheme for R2D transmission.

- [368] notes that the corresponding Toffset values need to be updated for occasions with X1 > 2 and X3 > 1.

- [369] notes that the existing timing-offset values (e.g., Toffset1/2) need to be updated for Device 2b/C, given their smaller SFO values and the potential additional R2D decoding time when R2D FEC is considered.

- [370] notes that new values of Toffset1/2/3/4 need to be considered for Device 2b/C, as their R2D/D2R processing characteristics and SFO values differ from those of Device 1.

- [372] notes that supporting multiple TDM resources may require increasing TR→D for reliable reception and synchronization, and that introducing R2D FEC may further increase the minimum TR→D due to device-side decoding time and the minimum TD→R due to reader-side PRDCH encoding time.

- [373] notes that timing coefficients and related parameters should be refined in Rel-20 due to improved SFO for Device 2b, enhanced processing capability of active devices, the use of multiple Msg1/Msg3 transmissions, and the additional delay introduced by R2D FEC decoding.

- [379] states that Toffset2 as well as Toffset1, Toffset3, and Toffset4 should be revised in Rel-20 A-IoT to reflect improved device SFO performance and updated R2D/D2R air-interface assumptions.

- [388] states that the timing offset TR→D for MSG1, MSG3, and upper-layer data should be updated for device 2b/C due to different processing time compared with device 1.

- [383] states that timing offsets between R2D and the corresponding D2R should be revisited in Rel-20 considering improved device capabilities, potential new processing requirements, updated chip duration, and RTT/SFO impacts, and that using reader-provided timing offsets can offer better scheduling flexibility.

- [364] states that if the same self-calculated approach is reused for device 2b/C, at least Toffset1/2/3/4 can be redefined to optimize both the latency and capacity. However, when the X number of time domain resources is increased (e.g., X=4) for Msg1 or Msg3 transmission for device 2b/C, additional timing offset would be needed for the 3rd and 4th time domain resources.

- [365] states that for Device 2b/C in Rel-20, a more flexible and dynamic time interval between R2D transmission and D2R reception, and vice-versa, are necessary.

Sources [362], [371] state that existing timing offset values Toffset1/2/3/4 should not be supported. Following observations were provided.

- [362] mentions that if the number of time-domain resources for Msg1 or Msg3 exceeds that in Rel-19, additional timing offsets would be required, making the use of predefined timing offsets overly complex and therefore not supportable.

- [371] notes that active A-IoT devices should operate in half-duplex mode, and the minimum timing offset between R2D and the subsequent D2R (e.g., Toffset1/2/3/4) for Device 2b/C should account for Rx–Tx switching delay—around 1 ms as referenced from NB-IoT—and any R2D FEC processing delay if supported.

6A.5.2 D2R → R2D

At least for CBRA for DT and DO-DTT, regarding the changes to existing timing offset value b/w D2R and corresponding subsequent R2D (e.g., T_D2R_min) for Device 2b and for Device C, the following observations were reported.

Sources [361], [364], [371], [388] state that changes to existing timing offset T_D2R_min are necessary.

- [361] mentions that TD2Rmin is expected to change due to the RF switch occurring.

- [364] provides to RAN4 that TD2R_min may be impacted by increased D2R resource usage, the introduction of FDM for R2D, higher R2D-generation complexity (e.g., FEC/repetition), and device frequency switching between UL and DL.

- [371] notes that the Tx–Rx switching delay imposed by half-duplex operation should be an additional factor affecting TD2R_min for Device 2b/C.

- [388] states that the minimum timing offset TD2R_min should also be changed for device 2b/C relative to device 1.

On the other hand, sources [370], [383] state that there is no need to change. Following observations were provided.

- [370] notes that the value of TD2R_min for Device 2b/C is determined by RAN4, as no additional RAN1-side factors affecting this parameter are introduced in Rel-20.

- [383] states that the value of T_D2R_min for device 2b/C does not need to change from Rel-19.

6A.5.3 R2D → R2D

At least for DT and DO-DTT, regarding the necessity and feasibility of introduction of new timing offsets for Device 2b and for Device C, the following observations were reported.

On the timing offset between two consecutive R2D transmissions for the same device (TR2D_R2D_min), sources [361], [371], [372], [364], [370], [383] state that introduction of the new timing offset (TR2D_R2D_min) is feasible and necessary, while source [379] states that it is not necessary. Sources [369], [365] state that the necessity depends on the detailed procedure design. Following observations on the new timing offset (TR2D_R2D_min) were provided during the study.

- [361] mentions that TR2D_R2D_min is needed when consecutive R2D transmissions for the same device are supported, as the device requires time to process the first R2D transmission before receiving the next, and that the corresponding value functions as a scheduling constraint at the reader side that is not provided to the device.

- [371] notes that introducing a minimum timing offset TR2D_R2D_min between two consecutive R2D transmissions for the same device is both feasible and necessary, given that devices cannot decode the latter R2D transmission if the interval is shorter than the required processing time.

- [372] notes that the timing offset between two consecutive R2D transmissions (TR2D_R2D) should be studied to support scenarios where multiple Msg2 PRDCHs are transmitted in succession.

- [364] mentions that a minimum time gap between L1 R2D control information and the corresponding R2D data should be defined if they are not always consecutive.

- [370] notes that a time interval TR2D_R2D between two consecutive R2D transmissions, along with the associated transmission details, should be considered in Rel-20 A-IoT.

- [383] states that the timing offset between two consecutive R2Ds should be studied for cases where R2D control information is encoded with a separate CRC from R2D data.

- [365] mentions that the timing relationship between the R2D sync signal and the R2D broadcast information can be hardcoded in the specification or derived from a predefined rule, and that the start of the paging/Msg0 monitoring window can be defined relative to the R2D sync signal and indicated to devices through the broadcast information.

- [369] states that whether to introduce new timing offset(s) between two consecutive R2D or D2R transmissions for the same device is depending on the definition of Rel-20 A-IoT procedures.

- [379] states that defining a new minimum timing offset T_R2D_R2D_min is not considered because no use cases require consecutive R2D transmissions.

6A.5.4 Whether provided by reader

At least for DT and DO-DTT, regarding the necessity and feasibility of timing offset provided by a reader for Device 2b and for Device C, the following observations were reported.

Sources [361], [362], [371], [379], [385] state that the timing offset provided by a reader is both feasible and necessary.

- [361] mentions that determining the timing values is better handled at the reader side due to their dependence on PRDCH and PDRCH payload lengths, and that the reader-computed value(s) can be included in the PRDCH payload portion and stored for contention-based procedures.

- [362] mentions that indicating the time offset between R2D and the corresponding D2R at the reader side is feasible, beneficial, and should be supported, whereas providing the timing offset between a D2R transmission and the subsequent R2D transmission is unnecessary since the device is expected to continuously receive the corresponding R2D message after transmitting the D2R message subject to processing-delay requirements.

- [371] notes that a reader-indicated timing offset between R2D and the subsequent D2R transmission is both feasible, given achievable timing alignment in Rel-20 A-IoT, and necessary to accommodate new R2D transmission types (e.g., periodic sync signal) and improve scheduling flexibility.

- [379] states that reader-indicated dynamic timing offsets between R2D and the corresponding D2R transmission are feasible in Rel-20 A-IoT.

- [385] states that a configurable time offset between R2D and the corresponding D2R, and between D2R and the corresponding R2D, should be considered, as delayed scheduling with such an offset is noted as feasible and beneficial for supporting active devices.

Following observations were also provided during the study.

- [365] mentions that the reader can indicate Toffset1 through broadcast information and can also indicate Toffset3 and Toffset4 via higher layer or L1 control information in the corresponding PRDCH.

- [368] states that the Rel-20 timing design between R2D and the subsequent D2R transmission should support time-offset indication in the R2D message.

- [376] notes that indicating TR2D/TD2R timing relationships, including the timing of each access occasion in TDMA-based access, in the preceding PRDCH should be studied to provide flexibility beyond the fixed and overly pessimistic timing defined in Rel-19.

- [383] states that device 2b/C can save energy if the R2D reception timing is indicated by the reader.

Sources [363], [366], [372] state that the timing offset provided by a reader is not necessary.

- [363] mentions not supporting the indication of existing timing offsets between R2D and the corresponding D2R, or between D2R and the corresponding R2D, for Device 2b/C.

- [366] considers that dynamic signaling for indicating the timing offset for R2D and D2R transmission is unnecessary in Rel-20.

- [372] notes that providing the timing offset between R2D and the subsequent D2R (or vice versa) helps the device anticipate the next event for timely CFO calibration and synchronization, but a long interval may introduce frequency drift and degrade the accuracy of the subsequent transmission or reception.

6A.6 Device localization

6A.6.1 Applicable signal(s)/channel(s)

D2R signal(s)/channel(s) which is/are feasible and applicable for A-IoT device localization is/are studied. For the D2R signal(s)/channel(s) which is/are feasible and applicable for A-IoT device localization, sources [361], [377], [371] mention D2R preamble.

- [361] For D2R RSRP-like measurement, the reader can use the D2R preamble for the reference signal and measure the power between the end of R2D transmission and start of the D2R transmission to estimate the noise and interference.

- [377] D2R preamble is the applicable signal for RSRP-like measurement. D2R RSRP-like measurement can be performed on existing D2R preamble without requiring dedicated positioning signals, enabling proximity determination for outdoor device localization.

- [371] D2R preamble can be used for the positioning purpose.

Sources [362], [365], [383] mention D2R preamble and midamble(s) for the D2R signal(s)/channel(s) which is/are feasible and applicable for A-IoT device localization.

- [362] D2R preamble and midamble signals are used for D2R RSRP-like measurement for device localization.

- [365] D2R-amble-RSRP is defined as the linear average over the power contributions (in [W] of chips that carry the D2R preamble over PDRCH. A secondary measurement can be defined as the linear averaging over the power contributions of chips that carry D2R preamble and D2R midambles over PDRCH.

- [383] For positioning, at least D2R RSRP measurement on D2R preamble and D2R midamble when m sequence is used for midamble are feasible. In the previous meetings, it was agreed to consider m sequence and Gold sequence for D2R preamble, and it was agreed to consider m sequence, all 1 sequence and Hadamard sequence for D2R midamble. From our perspective, m sequence and Gold sequence can be used for RSRP measurement at reader thanks to good auto-correlation property, and RSRP measurement based on m sequence and Gold sequence has been widely used in NR and should be feasible.

Sources [364], [369], [370] indicate D2R preamble and/or midamble(s).

- [364] D2R preamble and/or mid-amble, which is applicable for all of D2R transmissions even for sequence-based Msg 1 without associated PDRCH, can be considered for D2R RSRP measurement.

- [369] The current D2R signal structure including a preamble, midamble(s), and PDRCH, where preamble and midamble(s) are sequence based on the current agreements. Therefore, for RSRP-like measurement, it is feasible for Reader to measure the D2R preamble or midamble(s).

- [370] Based on introduction of open loop or closed loop power control for D2R, reader is feasible to use D2R preamble or midamble to obtain more accurate localization information.

Source [366] states D2R signals (sequence embedded in the PDRCH) and D2R preamble, and source [385] states D2R preamble and/or midamble(s), and PDRCH.

- [366] A sequence with length 127, which is designed for the A-IoT device positioning, was embedded in the PDRCH for the D2R RSRP-like measurements. For the study of sequence length and type, we conducted the following length-31, length-63 and length-127 of M-sequence and Gold sequence with evaluation results. From the evaluation results, it can be seen length-127 M-sequence or Gold-sequence show a lower residual SFO error comparing to that of length-31 M-sequences or Gold-sequence and length-63 M-sequences or Gold-sequence. Since SFO is one of the main factors to measurement accuracy of RSRP, gold sequences or m-sequences with a length of 127 can be used as candidates.

- [385] For D2R RSRP-measurement based positioning and proximity determination, PDRCH and D2R preamble/midamble with sufficient time duration and stable transmit power can be considered for D2R measurement. Using multiple measurements or extending D2R transmission increases accuracy. For D2R measurement, unicast D2R transmission can be used: Alt1: D2R Preamble/Midamble, Alt2: PDRCH. For Alt.1, the preamble and/or midamble can be used for RSRP measurement, provided they have sufficient transmission time and combined appropriately. For Alt.2, the PDRCH carries D2R control or data payload and repetitions for reliability can be used to improve power measurement accuracy.

6A.6.2 RSRP-like measurement

Regarding the D2R RSRP-like measurement for A-IoT device localization, following observations were provided during the study:

- [361] Use NR measurements (SL PRS-RSRP and UL-SRS-RSRP) as a reference. For D2R RSRP-like measurement, the reader can use the D2R preamble for the reference signal and measure the power between the end of R2D transmission and start of the D2R transmission to estimate the noise and interference. To estimate proximity, a reader can also consider the effects of propagation delay based on when the D2R transmission is received.

- [362] D2R RSRP-like measurement is defined as the time domain linear average over the power contributions (in [W] of preamble and midamble signals within the signal transmission bandwidth during the D2R transmission.

- [365] D2R-amble-RSRP is defined as the linear average over the power contributions (in [W] of chips that carry the D2R preamble over PDRCH. A secondary measurement can be defined as the linear averaging over the power contributions of chips that carry D2R preamble and D2R midambles over PDRCH.

- [366] The definition of RSRP-like and AOA measurement quantity can refer to TR 38.215(NR; Physical layer measurements).

- [369] The RSRP-like definition in A-IoT is the linear average of the received power in occupied chips and over the frequency resources that carry the measured signal. For example, if D2R is applied with OOK modulation, referring to the definition of LP-RSRP where measured signal is also an OOK waveform, RSRP-like definition in A-IoT is the linear average of the received power in OOK ON chips over the frequency resources that carry the measured signal.

- [385] At reader side, how to measure the RSRP may be up to reader implementation. However, it may require the averaging of the D2R measurements based on D2R signals/channels with sufficient time duration and stable transmit power from device side.

- [373] D2R RSRP based measurement and positioning is feasible. R20 A-IoT should support D2R RSRP based measurement.

6A.6.3 Localization methods

On A-IoT device localization for Device 2b/Device C for more accurate outdoor Device localization than based on Reader-ID, following observations on device localization methods were reported:

- [361] A reader can combine timing and power measurements for proximity determination. A reader can estimate the distance to the device based on the delay measurement and accounting for the SFO.

- [362] D2R RSRP measurement based on preamble and midamble, and fingerprint-based positioning scheme is adopted. The positioning algorithm is Weighted K-Nearest Neighbor (WKNN).

- [365] Assuming Tx power of the device is known by the reader and LOS between A-IoT and the reader, one approach for having a more granular outdoor device localization than the reader-based is that the reader can compare the RSRP of D2R signal with different RSRP threshold values, and estimate the zone that the device is located in.

- [366] Joint estimation of distance from RSRP-like measurements and AOA estimation with reader-based beam sweeping in the implementation is proposed.

- [369] Reader determines whether A-IoT device is far from or close to itself by comparing the measured RSRP value with an RSRP threshold. Reader determines a corresponding distance range when the measured RSRP value falls in an RSRP range.

- [373] D2R RSRP is a feasible approach at the current stage. Base on D2R RSRP value, at least device could be grouped into near group or far group roughly.

- [377] Proximity determination based on D2R RSRP measurement is sufficient for outdoor use-cases requiring several tens of meters accuracy, without needing dedicated positioning architecture. Further consider more finer granularity of proximity including, very near, near and far depending up on the measurement strength.

- [391] Introduce an explicit mechanism to set a context for positioning-related signaling. Even when high-accuracy positioning is not required, establishing a coarse spatial relationship such as a coverage zone is beneficial. To enable this, it is important that measurements and signaling exchanges are interpreted within a clear operational context. In particular, devices should be able to distinguish whether a reception instance is intended for regular inventory or command operation, or for positioning-related measurement purposes. Such contextual awareness will prevent unnecessary activity or energy consumption at the device.

- [370] In Rel-20 Ambient IoT, based on D2R preamble or midamble, it is possible for reader to use these ambles to do the RSRP-like measurement and obtain the localization information if the power control including open loop or closed loop power control is introduced.

- [385] Multi-reader positioning based on RSRP measurement is evaluated using weighted-centroid algorithm. The position estimate is given by a weighted average of BS reader locations, wherein the weights are a function of the RSRP measurement at each of the BS/readers and the same D2R transmission using a 63-length m-sequence is measured per reader.

On A-IoT device localization for Device 2b/Device C for more accurate outdoor Device localization than based on Reader-ID, sources [362], [366], [385] provided evaluation results.

- [362] D2R RSRP-like measurement is feasible for Device C to achieve more accurate outdoor Device localization than based on Reader-ID. D2R RSRP measurement based on preamble and midamble, and

fingerprint-based positioning scheme is adopted. The positioning algorithm is Weighted K-Nearest Neighbor (WKNN). Achieved localization performance, i.e., 82m@90%.

- [366] After SFO/CFO calibration, the positioning accuracy of RSRP+AOA method is about 17m (90% CDF) for Device 2b and is about 13m (90% CDF) for Device C.

- [385] Multi-reader positioning based on RSRP measurement is evaluated using weighted-centroid algorithm. The position estimate is given by a weighted average of BS reader locations, wherein the weights are a function of the RSRP measurement at each of the BS/readers and the same D2R transmission using a 63-length m-sequence is measured per reader. For outdoor positioning, Outdoor BS/reader can achieve position estimation about 72~114% of cell radius at 80~90% percentage error for Device 2b in UMi layout, and about 84~110% of cell radius estimation at 80~90% percentage error for Device C in UMa layout. For proximity determination to identify a unique device relative to a reader, in Outdoor scenario, UE/reader can reach 95~98% success rate with 2 to 6 nearby devices; BS reader can achieve 48~80%. In Indoor scenario, UE/reader can reach 77~92% success rate with 2 to 6 nearby devices; BS/reader can achieve 56~83%. BS/readers generally perform worse, primarily because fixed BS/reader cannot approach the devices as flexible as UE/reader.

6A.7 Coexistence of ambient IoT and NR/LTE

6A.7.1 Co-existence scenarios and cases

The coexistence evaluation is conducted considering the different scenarios listed in Table 6A.7.1-1

Table 6A.7.1-1: Co-existence scenarios

Operation mode	Device type	Evaluation case	Aggressor	Victim	Direction
Standalone	Device 2b	1	Device	NR UL	D2R
		2	NR UL	reader	D2R
		3	Reader	NR DL	R2D
		4	NR DL	device	R2D
	Device C	5	Device	NR UL	D2R
		6	NR UL	reader	D2R
		7	NR DL	device	R2D
In-band	Device 2b	8	NR UL	reader	D2R
		9	NR DL	device	R2D
	Device C	10	NR UL	reader	D2R
		11	NR DL	device	R2D

6A.7.2 Co-existence evaluation assumptions

6A.7.2.1 Deployment

Simulation assumptions related to network layout is captured for D4T1 in Table 6A.7.2.1-1.

Table 6A.7.2.1-1. Deployment parameters

Parameter	Value
Carrier frequency	900MHz (Band 8)
BW for NR	10MHz with 15KHz SCS
R2D BW	180kHz, for 15kHz SCS
D2R BW	15kHz
Waveform	R2D: OFDM-based OOK D2R: BPSK
A-IoT DL power control	No
A-IoT UL power control	Option 1: Use fixed Tx power without power control - Device 2b :-10dBm - Device C: 5dBm Option 2: Use power control as following equation - Company report CL_{x-ile}, P_{Max} and R_{min} - Target SNR:15dB(starting point) or other values to be reported $P_t = P_{\max} \times \min\left\{1, \max\left[R_{\min}, \left(\frac{CL}{CL_{x-ile}}\right)^{\gamma}\right]\right\}$
Traffic model	Full buffer
NR BS/A-IoT reader deployment (outdoor)	Hexagonal grid, 19 macro sites, 3 sectors per site with wrap around Inter-NR BS/A-IoT reader distance: [X]m - X= 750m for device 2b/C NR BS height: 25m the distance between macro NR BS/A-IoT reader and device 2b/C as 50-500 m
Frequency reuse	Frequency reuse - 3(Baseline) - other options are not precluded
Device distribution	Device Height= 1.5 m A-IoT devices drop uniformly distributed over the horizontal area - 100% outdoor 0 % in cargo trucks (baseline) - Not preclude to provide simulation result in X% in cargo trucks - For example, X=80 Number of A-IoTs = Total area × activated density (1.5 A-IOT devices/m²) 1 active A-IoT device under one reader at one drop Minimum distance between reader and device is 35m
NR UE dropping	For NR UE only outdoor, uniformly distributed outdoor. UE number: - DL active UE: 1 UE per cell - UL active UE: 3 UE per cell
MCL for NR BS to NR UE	MCL of 70 dB
Pathloss model	Use UMa pathloss model defined in TR38.901
penetration loss	20dB
Shadow fading margin	6dB for UMa NLOS

6A.7.2.2 NR BS/ A-IoT reader RF characteristics

The NR BS and A-IoT reader are defined for two different antenna configurations as illustrated in Table 6A.7.2.2-1 and Table 6A.7.2.2-2.

Table 6A.7.2.2-1: NR BS RF parameters

NR BS Parameter	Value
Macro-BS Tx power (dBm)	46dBm
BS antenna pattern	Antenna Array Geometry: BS point at fixed beam direction: vertical: θtilt + 90°, horizontal: 0, 120, 240 ° Antenna pattern (horizontal) (For 3-sector cell sites with fixed antenna patterns): $A_H(\phi) = -min\left[12\left(\frac{\phi}{\phi_{3dB}}\right)^2, A_m\right]$ φ_{3dB} = 65 degrees, A_m = 25 dB Antenna pattern (vertical) (For 3-sector cell sites with fixed antenna patterns): $A_V(\theta) = -min\left[12\left(\frac{\theta-90°-\theta_{etilt}}{\theta_{3dB}}\right)^2, SLA_v\right]$ θ_{3dB} = 10 degrees, SLA_v = 25 dB, θ_{etilt}= 9 degrees Combining method in 3D antenna pattern: $\mathrm{A}(\varphi,\theta) = -\min\{-[A_H(\varphi) + A_V(\theta)], A_m\}$
BS antenna gain (dBi)	15dBi including feeder loss
Height of macro NR BS (m)	25
NR Macro-BS Noise Figure(dB)	5
Network location	Outdoor
ACLR (in-band)	30 dBc
ACLR (standalone)	Same with legacy NR BS
ACS (in-band)	No interference caused by ACS impact
ACS(standalone)	Same with legacy NR BS

Table 6A.7.2.2-2: A-IoT reader RF parameters

A-IoT reader parameters		Value
A-IoT reader total Tx power	**Standalone**	43dBm
	In-band	38dBm
A-IoT reader receiver Noise Figure (dB)		5dB
A-IoT reader antenna gain (dBi)		15dBi including feeder loss
Antenna configuration		Antenna Array Geometry: BS point at fixed beam direction: vertical: θtilt + 90°, horizontal: 0, 120, 240 ° Antenna pattern (horizontal) (For 3-sector cell sites with fixed antenna patterns): $A_H(\phi) = -min\left[12\left(\frac{\phi}{\phi_{3dB}}\right)^2, A_m\right]$ φ_{3dB} = 65 degrees, A_m = 25 dB Antenna pattern (vertical) (For 3-sector cell sites with fixed antenna patterns): $A_V(\theta) = -min\left[12\left(\frac{\theta-90°-\theta_{etilt}}{\theta_{3dB}}\right)^2, SLA_v\right]$ θ_{3dB} = 10 degrees, SLA_v = 25 dB, θ_{etilt}= 9 degrees Combining method in 3D antenna pattern: $\mathrm{A}(\varphi,\theta) = -\min\{-[A_H(\varphi) + A_V(\theta)], A_m\}$

6A.7.2.3 NR UE/ A-IoT device RF characteristics

Assumptions relevant for modelling the NR UE and A-IoT device RF characteristics are captured in Table 6A.7.2.3-1 and Table 6A.7.2.3-2.

Table 6A.7.2.3-1: NR UE RF parameters

NR UE Parameter	Value
UE TX power in dBm	-40 to 23 dBm
NR UE Antenna gain (dBi)	0dBi
Height of UE antenna (m)	1.5m
NR UE Tx emission（dB）(standalone)	Same with NR UE ACLR
NR UE Tx emission (dB) (inband)	Same with NR UE IBE
NR UE ACS（dB） (Standalone)	33dB
NR UE ACS（dB） (In-band)	No interference caused by ACS impact
NR UE Noise Figure（dB）	9dB
Antenna configuration	Omni direction antenna

Table 6A.7.2.3-2: A-IoT device RF parameters

A-IoT device parameters	Value	
	Device 2b	Device C
A-IoT device effective antenna gain per Tx or Rx branch (dBi)	0	0
A-IoT device max output power（dBm）	-10	5
A-IoT device noise figure (dB)	15	13
Guard band	0 kHz	0 kHz
Antenna configuration	Omni direction antenna	Omni direction antenna

6A.7.3 Co-existence simulation methodology

6A.7.3.1a Coexistence evaluation methodology 1

The coexistence evaluation methodology can be summarized as:

1) Aggressor and victim network are generated. NR UEs and A-IoT devices are distributed as described by parameter assumptions.

2) UEs are associated to BS based on coupling loss, and A-IoT devices are associated to A-IoT reader based on coupling loss.

   a) The device access to the reader is randomly selected from candidate devices whose R2D receiving power/coupling from one reader is higher than the other.

b) The device can access the reader with best receiving power among all readers.

3) Once association is done, round robin scheduling is used.

4) For inter-system interference (between A-IoT and NR):

- When AIoT as victim, SINR with ACI define as following:

$$SINR = \frac{received\ wanted\ signal\ power}{noise + intra_{system}interference + inter_{system}interference}$$

The noise and intra/inter-system interference are within the wanted signal bandwidth.

The SINR correspoinding to the 5 percentile and 50 percentile point of the SINR CDF curve with ACI are obtained. If the SINR is above the decoding threshold, it can be considered that inter-system interference is negligible.

- When NR as victim, if throughput loss is smaller than and equal to 5%, it can be considered that inter-system interference is negligible.

5) For A-IoT as victim, collect the data of SINR with and without ACI.

6) Repeat Step 1-5 for different ACIR values.

6A.7.3.1b Coexistence evaluation methodology 2

The simulation procedure when using the outage as performance metric:

1) The baseline performance is achieved by generating the CDF curve of users versus SINR without enabling the aggressor system.

2) Choosing one SINR decoding threshold to be used for baseline performance corresponding to a certain outage ratio: If a certain outage ratio (e.g. 5%) is decided to be used for baseline performance, the UE percentile point corresponding to the SNR decoding threshold in SINR CDF curve should be equal or lower than this outage ratio. the outage ratio means that the percentage of the UE in the CDF curve below the SINR decoding threshold representing the percentage of UE experiencing outage as these UE are below the SINR decoding threshold.

3) Turning on the aggressor system and generating the new CDF curve of users versus SINR, multiple CDF curves can be generated with different ACIR.

4) By comparing the CDF curve of baseline generated in step 3 and CDF curve with a certain ACIR generated in step 4, an outage degradation data point can be obtained by observing the percentage difference of UEs corresponding to the SINR decoding threshold in these two curves, i.e. the data point is chosen to compare is the percentage in baseline CDF curve and CDF curve corresponding to the SINR decoding threshold when aggressor system is enabled for the same SINR decoding threshold in step 3.

5) Generating an outage degradation curve versus ACIR.

6A.7.3.2 Coupling loss

The Coupling Loss (CL) is defined as the loss in signal between NR BS-to- NR UE, A-IoT reader -to- A-IoT device. CL is defined as the loss including propagation loss and antenna gains.

6A.7.4 Co-existence evaluation results

The simulation results submitted for the ambient IoT and NR co-existence study from all contributing companies in RAN4 were summarized in [393].

6A.7.5 Summary of co-existence evaluation

In the Rel-20 SI, RAN4 evaluated standalone deployment mode involves the non-colocation of A-IoT BS and NR BS, while inband deployment mode involves the colocation of A-IoT BS and NR BS. This chapter presents the co-existence simulation results for ambient IoT based on baseline assumptions in clause 6A.7.2, categorized by operation mode and

device type. The conclusions below are based on the coexistence results. ACLRe (Efficient ACLR) and ACSe (Efficient ACS) were used for evaluation parameter. The details of ACLRe and ACSe deviation can refer to [394].

6A.7.5.1 Standalone with device 2b

For standalone operation mode of device 2b, the observations of each case are listed in Table 6A.6.5.1-1.

Table 6A.7.5.1-1: Standalone device 2b co-existence observations

Operation mode	Device type	Case No.	Aggressor	Victim	Co-existence observation
Standalone	Device 2b	1	Device	NR UL	Negligible NR UL throughput degradation is observed if the device ACLRe is ≥ 20dB (flat model) when the Tx power is fixed at -10dBm.
		2	NR UL	reader	The average SINR (with ACI) @5%-ile based on the collected results is -4.8dB with the standard deviation of 2.3dB. The average SINR (with ACI) @50%-ile based on the collected results is 12.9dB with the standard deviation of 1.5dB. The applied ACSe for the reader ranges from 18dB to 20dB.
		3	Reader	NR DL	Across the reported results, contributions indicate that the throughput loss@50%-ile is ≤ 5% (negligible degradation), whereas at 5% -ile most contributions show more than 10% throughput loss. Several contributions further report that applying reader-side mitigation schemes (including reader Tx power reduction, adjusting reader antenna tilt angle, reducing ISD) result in throughput loss ≤ 5% at both 5%-ile and 50%-ile.
		4	NR DL	device	The average SINR (with ACI) @5%-ile based on the collected results is 3.5dB with the standard deviation of 2.5dB. The average SINR (with ACI) @50%-ile based on the collected results is 17.4dB with the standard deviation of 0.6dB. The applied ACSe for the device ranges from 40dB to 53dB.

6A.7.5.2 Standalone with device C

For standalone operation mode of device C, the observations of each case are listed in Table 6A.7.5.2-1.

Table 6A.7.5.2-1: Standalone device C co-existence observations

Operation mode	Device type	Case No.	Aggressor	Victim	Co-existence observation
Standalone	Device C	5	Device	NR UL	Negligible NR UL throughput degradation can be achieved if the device ACLRe(flat model) is ≥34dB when the Tx power is fixed at 5dBm, or the device ACLRe(flat model) is ≥20dB when Tx power control is applied. One company also observed that increaseing the frequency reuse factor also reduced the NR UL throughput degradation.
		6	NR UL	reader	The average SINR (with ACI) @5%-ile based on the collected results is -0.2dB with the standard deviation of 2.2dB. The average SINR (with ACI) @50%-ile based on the collected results is 16.2dB with the standard deviation of 1.1dB The applied ACSe for the reader ranges from 45dB to 58dB. One company observed 4% outage without ACI based on frequency reuse 9 and SNR decoding threshold 5dB and observed 5% additional outage degradation with ACI. The company also observed 17% outage without ACI based on frequency reuse 3 and SNR decoding threshold 5dB.
		7	NR DL	device	The average SINR (with ACI) @5%-ile based on the collected results is 1.5dB with the standard deviation of 3.7dB. The average SINR (with ACI) @50%-ile based on the collected results is 16.8dB with the standard deviation of 1.5dB. The applied ACSe for the device ranges from 33dB to 50dB. One company observed 4% outage without ACI based on frequency reuse 9 and SNR decoding threshold 5dB and observe with 5% additional outage with ACI. The company also observed 17% outage without ACI based on frequency reuse 3 and SNR decoding threshold 5dB

6A.7.5.3 Inband with device 2b

For inband operation mode of device 2b, the observations of each case are listed in Table 6A.7.5.3-1.

Table 6A.6.5.3-1: Inband device 2b observations

Operation mode	Device type	Case No.	Aggressor	Victim	Co-existence observation
Inband	Device 2b	8	NR UL	reader	The average SINR (with ACI) @5%-ile based on the collected results is -6.1dB with the standard deviation of 2.6dB. The average SINR (with ACI) @50%-ile based on the collected results is 10.7dB with the standard deviation of 1.9dB The applied ACSe for the reader ranges from 45dB to 46dB.
		9	NR DL	device	The average SINR (with ACI) @5%-ile based on the collected results is 3.0dB with the standard deviation of 2.6dB. The average SINR (with ACI) @50%-ile based on the collected results is 16.0dB with the standard deviation of 1.1dB The applied ACSe for the device ranges from 33dB to 50dB.

6A.7.5.4 Inband with device C

For inband operation mode of device C, the observations of each case are listed in Table 6A.7.5.4-1.

Table 6A.7.5.4-1: Inband device C co-existence Observtions

Operation mode	Device type	Case No.	Aggressor	Victim	Co-existence observation
Inband	Device C	10	NR UL	reader	The average SINR (with ACI) @5%-ile based on the collected results is -0.1dB with the standard deviation of 1.1dB. The average SINR (with ACI) @50%-ile based on the collected results is 15.9dB with the standard deviation of 1.0dB The applied ACSe for the reader ranges from 31dB to 51dB. One company observed 4% outage without ACI based on frequency reuse 9 and SNR decoding threshold 5dB and 17% outage without ACI based on frequency reuse 3 and SNR decoding threshold 5dB. For outage with ACI, it is observed with 5% additional outage.
		11	NR DL	device	The average SINR (with ACI) @5%-ile based on the collected results is 3.2dB with the standard deviation of 2.8dB. The average SINR (with ACI) @50%-ile based on the collected results is 16.2dB with the standard deviation of 2.0dB The applied ACSe for the device ranges from 31dB to 48dB. One company observed 4% outage without ACI based on frequency reuse 9 and SNR decoding threshold 5dB and 17% outage without ACI based on frequency reuse 3 and SNR decoding threshold 5dB. For outage with ACI, it is observed with 5% additional outage.

6A.7.5.5 General remarks on coexistence findings

Across the assessed cases, the results consistently show that R2D and D2R demodulation thresholds are critical to coexistence performance and therefore represent a key design driver for the A-IoT system in the outdoor scenario.

When NR DL is the victim, some mitigation schemes are helpful to reduce the A-IoT system interference, e.g., increasing the frequency reuse factor, lowering the A-IoT reader power, reducing the network grid shift and adjusting the AIoT reader antenna tilt, otherwise, the throughput degradation of NR is observed.

When NR UL is the victim, applying Tx power control for device C is beneficial. Otherwise, stringent ACLR requirement of the device might be required.

For both device 2b and C, when A-IoT is the victim system, the performance degradation on the A-IoT D2R and R2D links is observed. Whether this degradation will have impact on A-IoT system depends on the SINR decoding thresholds.

Lower SINR threshold for R2D/D2R decoding could increase the resilience of the A-IoT system to the interference from NR as well as relax the RF design parameters such as ACLR and ACS.

The ISD of 750 meters is assumed in the coexistence simulations. The results show that lower SINR(@5%-ile and @50-ile) is obtained at the reader side for device 2b compared with device C, due to the difference in max Tx power. Whether SINR difference will have impact on A-IoT system also depends on the SINR decoding thresholds and the choice of ISD. For device C, its higher output power compared to device 2b results in higher ACLR requirement assuming fixed maximum power is applied.

When taking the above analysis into account, the A-IoT system can coexist with the NR system. In addition, the evaluation results are dependent of the assumed baseline parameters and device/reader characteristics (e.g., RF parameters ACLR/ACS). Frequency reuse 3 is used in simulation, and higher reuse factor is also considered in RAN4 coexistence study to improve the A-IoT intra-system performance.

7 Evaluations

7.1 Coverage evaluations

For an evaluation scenario:

- For each link *i*,
  - Step 1: Obtain the required SINR for the physical channels under target scenarios and service/reliability requirements if Budget-Alt2 is used for this link *i*.
  - Step 2: Obtain the receiver sensitivity using the method Budget-Alt1 (if a predefined threshold is assumed to derive the receiver sensitivity) or Budget-Alt2 (if no predefined threshold is assumed to derive the receiver sensitivity). See Clause 4.3.1 for the Budget definition.
  - Step 3: Obtain the coverage performance for link *i* based on the receiver sensitivity from step 2 and link budget template.
- The coverage results for each link are provided.

RAN1 has evaluated the coverage for InF-DH NLOS scenarios for D1T1 and InH-Office LOS/InF-DL NLOS scenarios for D2T2, across all devices 1/2a/2b. Coverage evaluation results comparing various R2D Tx power/EIRP value, R2D Rx sensitivity, BLER 1%/10% target, data rate ~1 kbps/~5-7 kbps, D2R coherent receiver/non-coherent receiver, R2D budget-Alt1 and Alt2, CW cancellation, backscatter loss and modulation schemes, D2R with/without channel coding schemes, etc, have been provided. The observations for coverage evaluation results can be found in Clauses 7.1.1.1.2, 7.1.1.2.2, 7.1.1.3.2, 7.1.2.1.2, 7.1.2.2.2, and 7.1.2.3.2.

All the evaluation results are contained in the attached spreadsheet file [189].

7.1.1 Link budget for D1T1

See associated file.

7.1.2 Link budget for D2T2

See associated file.

7.1.3 Link performance for D2R

See associated file.

7.1.4 Link performance for R2D

See associated file.

7.2 Latency evaluations

7.2.1 Single device latency

According to the definitions of clause 4.1, the following are the assessment results for single-device latency.

Table 7.2.1-1: Summary for single-device latency for "inventory-only" case

Source	Evaluation assumptions				Latency [ms]
	Data rate [kbps]	Re-attempt probability (X)	Message size [bits]	Gap [ms]	
[163]	5	10%	Paging: 22 Msg 1: 16 Msg 2: 18 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.1	30.90
[164]	7	10%	Paging: 22 QueryRep: 4 Msg 1/2: 16 Msg 3: 96	Gap A = 0.0875 Gap B1 = 0.0875	24.09
[166]	1	10%	Paging: 128 Msg 3: 96	Gap A = 0.24 Gap B1 = 0.24	249.16
			Paging: 288 Msg 3: 256		604.71
	7		Paging: 128 Msg 3: 96		35.82
			Paging: 288 Msg 3: 256		86.62
[167]	7	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.1	24.30
[169]	7	10%	Paging: 24 Msg 1: 16 Msg 2: 24 Msg 3: 96	Gap A = 1.5 Gap B1 = 1.5	32.07
[186]	1	10%	Paging: 62 Msg 1: 16 Msg 2: 18 Msg 3: 96	Gap A = 10 Gap B1 = 20	246.70
	7			Gap A = 1.4 Gap B1 = 2.9	35.20
[171]	7	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96	Gap A = 1.4 Gap B1 = 1.4	28.30
[172]	1	10%	Paging: 16 Msg 3: 96	Gap A = 0.2	124.70
			Paging: 24 Msg 3: 96		133.50
			Paging: 32 Msg 3: 96		142.40
	7		Paging: 16 Msg 3: 96		18.00
			Paging: 24 Msg 3: 96		19.27
			Paging: 32 Msg 3: 96		20.53
[173]	1	10%	Paging: 20 QueryRep = 20 Msg 1/2: 20 Msg 3: 96	Gap A = 0.25 Gap B1 = 0.25	173.97
	7				24.40
[188]	1	10%	Paging: 20 Msg 1/2: 20 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.2	174.00
			Paging: 20 Msg 3: 96	Gap A = 0.2	129.10
	7		Paging: 20 Msg 1/2: 20 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.2	25.42
			Paging: 20 Msg 3: 96	Gap A = 0.2	18.63
[178]	1	10%	Paging: 22 Msg 1: 16 Msg 2: 18 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.1	169.44
	7				24.68

[180]	1	10%	Paging: 22 Msg 1/2: 16 Msg 3: 96	Gap A = 0.1 Gap B1 = 0.1	166.90
	7				24.00
[181]	7	10%	Paging: 30 Msg 1/2: 24 Msg 3: 200	Gap A = 0.5 Gap B1 = 0.5	45.72
	56				7.18
	640				2.14
[213]	1	10%	Paging: 16 Msg 1/2: 16 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.2	160.56
			Paging: 24 Msg 1/2: 16 Msg 3: 96		169.44
			Paging: 32 Msg 1/2: 16 Msg 3: 96		178.33
	7		Paging: 16 Msg 1/2: 16 Msg 3: 96		23.41
			Paging: 24 Msg 1/2: 16 Msg 3: 96		24.68
			Paging: 32 Msg 1/2: 16 Msg 3: 96		25.94
[194]	1	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.3	167
[198]	1	10%	Paging: 22 Msg 1/2: 16 Msg 3: 96	Gap A = 0.025 Gap B1 = 0.025	166.75
[199]	1	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96	Gap A = 10 Gap B1 = 10	197.8
[204]	7	10%	Paging: 26 Msg 1/2: 16 Msg 3: 96	Gap A = 0.2 Gap B1 = 0.2	25.11
[212]	7	10%	Paging: 26 Msg 1/2: 16 Msg 3: 96	Gap A = 1 Gap B1 = 1	25.7

Table 7.2.1-2: Summary for single-device latency for "inventory and command" case

Source	Evaluation assumptions				Latency [ms]
	Data rate [kbps]	Re-attempt probability (X)	Message size [bits]	Gap [ms]	
[164]	7	10%	Paging: 22 QueryRep: 4 Msg 1/2: 16 Msg 3: 96 Step C R2D/Step D D2R: 16	Gap B = 0.075 Gap C =0.0875	29.35
[166]	1	10%	Paging: 128 Msg 3: 96 Step C R2D: 16 Step D D2R: 800	Gap A = 0.24 Gap B1 = 0.24 Gap C = 0.24	1156.64
			Paging: 288 Msg 3: 256 Step C R2D: 16		1512.20

			Step D D2R: 800		
	7		Paging: 128 Msg 3: 96 Step C R2D: 16 Step D D2R: 800		166.17
			Paging: 288 Msg 3: 256 Step C R2D: 16 Step D D2R: 800		216.96
[167]	7	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96 Step C R2D: 400	Gap A = 0.2 Gap B1 = 0.1	87.90
[169]	7	10%	Paging: 24 Msg 1: 16 Msg 2: 24 Msg 3: 96 Step C R2D: 48 Step D D2R: 96	Gap A = 1.5 Gap B1 = 1.5 Gap B = 3 Gap C = 1.5	59.93
[186]	1	10%	Paging: 62 Msg 1: 16 Msg 2: 18 Msg 3: 96 Step C R2D: 60 Step D D2R: 33	Gap A = 10 Gap B1 = 20 Gap B = 5 Gap C = 20	377.80
	7			Gap A = 1.4 Gap B1 = 2.9 Gap B = 5 Gap C = 20	77.80
[171]	7	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96 Step C R2D: 128 Step D D2R: 128	Gap A = 1.4 Gap B1 = 1.4 Gap B = 1.4 Gap C = 1.4	72.10
[172]	1	10%	Paging: 24 Msg 3: 96 Step C R2D: 64	Gap A = 0.2 Gap B = 0.1	204.80
			Paging: 24 Msg 3: 96 Step C R2D: 256		418.10
			Paging: 24 Msg 3: 96 Step C R2D: 400		578.10
	7		Paging: 24 Msg 3: 96 Step C R2D: 64		40.34
			Paging: 24 Msg 3: 96 Step C R2D: 256		59.71
			Paging: 24 Msg 3: 96 Step C R2D: 400		82.57
[173]	1	10%	Paging: 20 QueryRep = 20 Msg 1/2: 20 Msg 3: 96 Step C R2D: 20	Gap A = 0.25 Gap B1 = 0.25 Gap B = 0.075	196.28
	7				28.66
[180]	1	10%	Paging: 22 Msg 1/2: 16	Gap A = 0.1 Gap B1 = 0.1	291.60

	7		Msg 3: 96 Step C R2D: 96 Step D D2R: 16	Gap B = 0.1 Gap C = 0.1	42.00
[181]	7	10%	Paging: 30 Msg 1/2: 24 Msg 3: 200 Step C R2D: 200 Step D D2R: 10	Gap A = 0.5 Gap B1 = 0.5 Gap B = 0.5 Gap C = 0.5	80.28
	56				12.46
	640				3.62
[213]	1	10%	Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 64 Step D: 4	Gap A = 0.2 Gap B1 = 0.2 Gap B = 0.1 Gap C = 0.2	245.33
			Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 256 Step D: 4		458.67
			Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 400 Step D: 4		618.67
	7		Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 64 Step D: 4		46.61
			Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 256 Step D: 4		65.98
			Paging: 24 Msg 1/2: 16 Msg 3: 96 Step C R2D: 400 Step D: 4		88.83
[214]	1	10%	Paging: 20 Msg 1/2: 20 Msg 3: 96 Step C R2D: 96	Gap A = 0.2 Gap B1 = 0.2 Gap B = 0.2	280.89
			Paging: 20 Msg 3: 96 Step C R2D: 96		236
	7		Paging: 20 Msg 1/2: 20 Msg 3: 96 Step C R2D: 96		40.88
			Paging: 20 Msg 3: 96 Step C R2D: 96		34.1
[194]	1	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96 Step C R2D: 400	Gap A = 0.2 Gap B1 = 0.3 Gap B = 0.1	612
[198]	1	10%	Paging: 22 Msg 1/2: 16 Msg 3: 96 Step C R2D: 20 Step D D2R: 16	Gap A = 0.025 Gap B1 = 0.025 Gap B = 0.025 Gap C = 0.025	206.81

[199]	1	10%	Paging: 20 Msg 1/2: 16 Msg 3: 96 Step C R2D: 128 Step D D2R: 128	Gap A = 10 Gap B1 = 10 Gap B = 10 Gap C = 10	504.4
[204]	7	10%	Paging: 26 Msg 1/2: 16 Msg 3: 96 Step C R2D: 96 Step D D2R: 16	Gap A = 0.2 Gap B1 = 0.2 Gap B = 0.4 Gap C = 0.2	43.56
[207]	1	10%	Paging: 22 Msg 1/2: 16 Msg 3: 96 Step C R2D: 96	Gap A = 0.1 Gap B1 = 0.1 Gap B = 0.1 Gap C = 0.1	273.7
	7				39.4
	1		Paging: 22 Msg 1/2: 16 Msg 3: 96 Step C R2D: 256 Step D D2R: 16		469.3
	7				67.4
	1		Paging: 22 Msg 1/2: 16 Msg 3: 96 Step C R2D: 256		451.4
	7				64.8
[212]	7	10%	Paging: 26 Msg 1/2: 16 Msg 3: 96 Step C R2D: 96 Step D D2R: 96	Gap A = 1 Gap B1 = 1 Gap B = 1	42.1

Observations:

For the "inventory-only" case, considering 10% re-attempt probability:

- Assuming Msg3 size (Device ID size) of 96 bits
  - For data rate of 1 kbps, 10 sources [166], [186], [172], [173], [188], [178], [180], [194], [198], [199] provide results, and the single device latency is [124.70 ms ~ 249.16 ms].
  - For data rate of 5~7 kbps, 14 sources [163], [164], [166], [167], [169], [186], [171], [172], [173], [188], [178], [180], [204], [212] provide results, the single device latency is [18.00 ms ~ 35.82 ms].
- Assuming Msg3 size (Device ID size) of larger than 200 bits
  - For data rate of 1 kbps, 1 source [166] provides results, the single device latency is 604.71 ms.
  - For data rate of 7 kbps, 2 source [166], [181] provide results, the single device latency is [45.72 ms ~ 86.62 ms].
  - For data rate of 56 kbps, 1 source [181] provides results, the single device latency is 7.18 ms.
  - For data rate of 640 kbps, 1 source [181] provides results, the single device latency is 2.14 ms.

For the "inventory and command" case, considering 10% re-attempt probability:

- 11 sources provide results including Step D D2R transmission
  - Assuming Msg3 size (Device ID) of 96 bits and Step D D2R message size small than or equal to 400 bits,
    - For data rate of 1 kbps, 6 sources [186], [180], [213], [198], [199], [207] provide results, and the single device latency is [206.81 ms ~ 618.67 ms].

    - For data rate of 7 kbps, 9 sources [164], [169], [186], [171], [180], [213], [204], [207], [212] provide results, the single device latency is [29.35 ms ~ 88.83 ms].

  - Assuming Msg3 size (Device ID) of 96 bits and Step D D2R message size larger than 400 bits,

    - For data rate of 1 kbps, 1 source [166] provides results, and the single device latency is 1156.64 ms.

    - For data rate of 7 kbps, 1 source [166] provides results, and the single device latency is 166.17 ms.

  - Assuming Msg3 size (Device ID) larger than 200 bits and Step D D2R message size smaller than or equal to 400 bits,

    - For data rate of 56 kbps, 1 source [181] provide results, the single device latency is 12.46 ms.

    - For data rate of 640 kbps, 1 source [181] provide results, the single device latency is 3.62 ms.

  - Assuming Msg3 size (Device ID) larger than 200 bits and Step D D2R message size larger than 400 bits

    - For data rate of 1 kbps, 1 source [166] provides results, and the single device latency is 1512.20 ms.

    - For data rate of 7 kbps, 1 source [166] provides results, and the single device latency is 216.96 ms.

- 5 sources provide results not including Step D D2R transmission

  - Assuming Msg3 size (Device ID) of 96 bits and Step C R2D message size small than or equal to 400 bits,

    - For data rate of 1 kbps, 5 sources [172], [173], [214], [207], [167] provide results, and the single device latency is [196.28 ms ~ 612 ms].

    - For data rate of 7 kbps, 5 sources [167], [172], [173], [214], [207] provide results, the single device latency is [28.66 ms ~ 87.90 ms].

7.2.2 Inventory completion time for multiple devices

For the inventory-only use case, the, inventory completion time for multiple A-IoT devices is defined as the time a reader successfully completed the inventory process for at least 99% of all A-IoT devices within the coverage of the reader, assuming device density of 1.5 devices per m^2.

Note: The study does not define a target for this inventory completion time.

Table 7.2.2-1: Summary of inventory completion time for multiple devices

Source	Evaluation assumption For data rate column, value "X" means same data rate X for both R2D and D2R, value "(X,Y)" means data rate X for R2D and data rate Y for D2R.							Inventory completion time [s]
	No. of devices	Success rate	Random access scheme	Multiplexing resources (TDMA/FDMA/CDMA)	Data rate [kbps]	Energy harvesting consideration	Device type	
[163]	600	90%	Slotted ALOHA	1	5	No	-	17.4 (multi-round) 25.6 (single round)
				2 (FDMA)				14.5
				4 (FDMA)				9.5
				8 (FDMA)				5.3
				12 (FDMA)				3.6
		95%		1				18.6 (multi-round) 37.6 (single round)
				2 (FDMA)				15.5
				4 (FDMA)				10.2
				8 (FDMA)				5.7
		99%		1				19.6 (multi-round) 129.4 (single round)
				2 (FDMA)				16.3
				4 (FDMA)				10.7
				8 (FDMA)				6.0
				12 (FDMA)				4.0
[164]	600	99%	Slotted ALOHA	1	7	No	-	191.1
				30 (FDMA)				15.61
				60(FDMA)				11.56
	6000			1				1910.99
				30 (FDMA)				156
				60 (FDMA)				115.2
[166]	600	100%	Slotted ALOHA	1	(35, 5)	No	-	16.5959
					(35, 16)			6.3974
					(35, 50)			3.2451
					(126, 540)			1.5563

					Distribution based			12.6481
				2 (FDMA)	(35, 5)			12.9208
					(35, 16)			4.8847
					(35, 50)			2.4009
					(126, 540)			1.1014
					Distribution based			9.8101
				4 (FDMA)	(35, 5)			8.6877
					(35, 16)			3.2456
					(35, 50)			1.5635
					(126, 540)			0.6885
					Distribution based			6.5811
				8 (FDMA)	(35, 5)			4.9540
					(35, 16)			1.8434
					(35, 50)			0.8820
					(126, 540)			0.3792
					Distribution based			3.7499
				1	Distribution based	Direction 1	1	18.3
[167]	600	99%	Slotted ALOHA	1	1	No	-	90
					5			18.4
					60			2.2
				2	1			69
					5			14
					60			1.5
				4	1			46.5
					5			9.4
					60			1
				6	1			33
					5			7.1
					60			0.7
				8	1			25
					5			5.5
					60			0.57
[168]	44	99%	Slotted ALOHA	12 (FDMA)	5-7	No	-	1.09~1.46
	275							7.5~9.05
	1111							29.48~35.39
[186]	600	99%	Slotted ALOHA	2 (FDMA)	(56, 160)	Direction 1	1	36.40 (paging interval = 2 s) 46.40 (paging interval = 4 s)
				4 (FDMA)				36.40 (paging interval = 2 s) 48.40 (paging interval = 4 s)
				2 (FDMA)	(14, 40)			36.40 (paging interval = 2 s) 62.40 (paging interval = 4 s)
				4 (FDMA)				36.40 (paging interval = 2 s)

								58.40 (paging interval = 4 s)
				8 (FDMA)				36.40 (paging interval = 2 s) 46.40 (paging interval = 4 s)
		100%		2 (FDMA)	(56, 160)			39.42 (paging interval = 2 s) 64.02 (paging interval = 4 s)
				4 (FDMA)				39.17 (paging interval = 2 s) 59.64 (paging interval = 4 s)
				2 (FDMA)	(14, 40)			40.34 (paging interval = 2 s) 69.63 (paging interval = 4 s)
				4 (FDMA)				39.83 (paging interval = 2 s) 67.03 (paging interval = 4 s)
				8 (FDMA)				39.43 (paging interval = 2 s) 63.96 (paging interval = 4 s)
		99%		2 (FDMA)	(56, 160)		2	35.80 (paging interval = 20 ms) 46.40 (paging interval = 40 ms)
				4 (FDMA)				35.80 (paging interval = 20 ms) 37.16 (paging interval = 40 ms)
				2 (FDMA)	(14, 40)			36.44 (paging interval = 20 ms) 58.36 (paging interval = 40 ms)
				4 (FDMA)				35.90 (paging interval = 20 ms) 55.32 (paging interval = 40 ms)
				8 (FDMA)				35.84 (paging interval = 20 ms) 49.32 (paging interval = 40 ms)
		100%		2 (FDMA)	(56, 160)			35.99 (paging interval = 20 ms) 57.38 (paging interval = 40 ms)
				4 (FDMA)				35.99 (paging interval = 20 ms) 53.55 (paging interval = 40 ms)
				2 (FDMA)	(14, 40)			55.57 (paging interval = 20 ms) 72.66 (paging interval = 40 ms)
				4 (FDMA)				51.18 (paging interval = 20 ms) 62.32 (paging interval = 40 ms)
				8 (FDMA)				46.35 (paging interval = 20 ms) 57.34 (paging interval = 40 ms)
		99%	Non-Slotted ALOHA based solution	2 (FDMA)	(56, 160)	Direction 2	1	43.08
				4 (FDMA)				41.45
				2 (FDMA)	(14, 40)			43.07
				4 (FDMA)				41.45
				8 (FDMA)				39.95
		100%		2 (FDMA)	(56, 160)			43.95
				4 (FDMA)				42.07
				2 (FDMA)	(14, 40)			43.99
				4 (FDMA)				42.07
				8 (FDMA)				41.27

		99%	Slotted ALOHA based solution	2 (FDMA)	(56, 160)			43.68
				4 (FDMA)				39.84
				2 (FDMA)	(14, 40)			75.24
				4 (FDMA)				50.07
				8 (FDMA)				42.21
		100%		2 (FDMA)	(56, 160)			44.01
				4 (FDMA)				39.92
				2 (FDMA)	(14, 40)			76.96
				4 (FDMA)				51.36
				8 (FDMA)				43.68
		99%	Non-Slotted ALOHA based solution	2 (FDMA)	(56, 160)	Direction 2	2	43.08
				4 (FDMA)				41.44
				2 (FDMA)	(14, 40)			59.97
				4 (FDMA)				43.09
				8 (FDMA)				41.46
		100%		2 (FDMA)	(56, 160)			43.96
				4 (FDMA)				42.09
				2 (FDMA)	(14, 40)			67.11
				4 (FDMA)				43.69
				8 (FDMA)				42.10
		99%	Slotted ALOHA based solution	2 (FDMA)	(56, 160)			43.68
				4 (FDMA)				39.84
				2 (FDMA)	(14, 40)			107.68
				4 (FDMA)				76.96
				8 (FDMA)				61.60
		100%		2 (FDMA)	(56, 160)			44.06
				4 (FDMA)				41.05
				2 (FDMA)	(14, 40)			107.68
				4 (FDMA)				87.73
				8 (FDMA)				71.87
[170]	600	99%	Slotted ALOHA	1	1	No	-	33.992 (2-step for step B) 91.306 (3-step for step B)
				2 (FDMA)				26.627 (2-step for step B) 85.966 (3-step for step B)
				3 (FDMA)				24.183 (2-step for step B) 81.539 (3-step for step B)
				4 (FDMA)				22.951 (2-step for step B) 80.347 (3-step for step B)
[171]	389	99%	Slotted ALOHA	1	50	No	-	4.29 (initial TDMA slots = 64) 3.24 (initial TDMA slots = 128)
				2 (FDMA)				2.31 (initial TDMA slots = 32) 1.46 (initial TDMA slots = 64)

								2.04 (initial TDMA slots = 128)
				4 (FDMA)				1.19 (initial TDMA slots = 32) 1.22 (initial TDMA slots = 64) 1.71 (initial TDMA slots = 128)
				8 (FDMA)				1.10 (initial TDMA slots = 32) 1.16 (initial TDMA slots = 64) 1.55 (initial TDMA slots = 128)
[173]	600	99%	Slotted ALOHA	1	1	No	-	132.3
				2 (TDMA+FDMA)				92.0
				4 (TDMA+FDMA)				72.8
				6 (TDMA+FDMA)				65.7
				8 (TDMA+FDMA)				60.0
				1	7			19.7
				2 (TDMA+FDMA)				13.9
				4 (TDMA+FDMA)				10.2
				6 (TDMA+FDMA)				9.3
				8 (TDMA+FDMA)				9.1
				1	50			3.6
				2 (TDMA+FDMA)				2.4
				4 (TDMA+FDMA)				1.7
				6 (TDMA+FDMA)				1.6
				8 (TDMA+FDMA)				1.5
[175]	943	99%	Slotted ALOHA	1	(56, 56)	Direction 1	2	6.32 (22 rounds) 8.16 (5 rounds)
					(56, 160)			4.34 (22 rounds) 5.71 (5 rounds)
					(252, 960)			1.13 (22 rounds) 1.49 (5 rounds)
					(252, 4000)			0.99 (22 rounds) 1.32 (5 rounds)
	11781				(56, 56)			74.07 (22 rounds) 101.87 (5 rounds)
					(56, 160)			50.62 (22 rounds) 71.24 (5 rounds)
					(252, 960)			13.12 (22 rounds) 18.54 (5 rounds)
					(252, 4000)			11.52 (22 rounds) 16.45 (5 rounds)
[178]	600	99%	Slotted ALOHA	1	7	No	-	13.1
[180]	500	99%	Slotted ALOHA	TDMA for R2D and D2R	7	No	-	11.3 (One paging associated with multiple D2R slots)

								13 (Each paging providing a single D2R slot)
				TDMA for R2D, TDMA + FDMA for D2R				2.9 (One paging associated with multiple D2R slots) 3.9 (Each paging providing a single D2R slot)
				TDMA + FDMA for R2D and D2R				1.8 (One paging associated with multiple D2R slots) 2.5 (Each paging providing a single D2R slot)
	10000			TDMA for R2D and D2R				229 (One paging associated with multiple D2R slots) 263 (Each paging providing a single D2R slot)
				TDMA for R2D, TDMA + FDMA for D2R				43 (One paging associated with multiple D2R slots) 50 (Each paging providing a single D2R slot)
				TDMA + FDMA for R2D and D2R				22 (One paging associated with multiple D2R slots) 31 (Each paging providing a single D2R slot)
[181] **[209]**	600	100%	Slotted ALOHA	8 (4 TDM resources, 2 FDM resources)	(28, 70)	Direction 1: No DCM	1	20
						Direction 2: Case 1 DCM (periodic query)		5
						Direction 2: Case 1 DCM (aperiodic query)		2.5
						Direction 2: Case 2 DCM (periodic query)		11 (4 groups)
	600	100%	Slotted ALOHA	16 (4 TDM resources, 4 FDM resources)	(28, 70)	Direction 1: No DCM	2	200
						Direction 2: Case 1 DCM (periodic query)		20
						Direction 2: Case 1 DCM (aperiodic query)		20

						Direction 2: Case 2 DCM (periodic query)		10 (8 groups)
[181]	20	100%	Slotted ALOHA	8 (2 TDMA resources, 4 FDMA resources)	(28, 70)	Direction 1, RF incident = -28 dBm	1	1.3
						Direction 1, RF incident = -32 dBm		5
						Direction 2, Case 1 DCM, RF incident = -28 dBm		1.35
						Direction 2, Case 1 DCM, RF incident = -32 dBm		5.05
						Direction 2, Case 2 DCM, RF incident = -28 dBm		0.2
						Direction 2, Case 2 DCM, RF incident = -32 dBm		0.6
				16 (4 TDMA resources, 4 FDMA resources)		Direction 1, RF incident = -28 dBm	2	>10
						Direction 1, RF incident = -32 dBm		>10
						Direction 2, Case 1 DCM, RF incident = -28 dBm		0.4
						Direction 2, Case 1 DCM, RF incident = -32 dBm		>10
						Direction 2, Case 2 DCM, RF incident = -28 dBm		0.4
						Direction 2, Case 2 DCM,		1.7

						RF incident = -32 dBm		
[191]	600	99%	Slotted ALOHA Collision based re-attempts by re-paging	1	7	No	-	23.90 (rounds=35, TDMA slots per access round=128)
				2(FDMA+TDMA)				11.77 (cycles =9 TDMA slots per access round=128)
				4 (FDMA+TDMA)				6.47 (rounds=5, TDMA slots per access round=128)
				8 (FDMA+TDMA)				3.92 (rounds=3, TDMA slots per access round=128)
			Slotted ALOHA Collision based re-attempts by re-paging	1				17.43 (rounds=5, TDMA slots per access round=512)
				2(FDMA+TDMA)				11.38 (rounds=3 TDMA slots per access round=512)
				4 (FDMA+TDMA)				8.37 (rounds=3, TDMA slots per access round=512)
				8 (FDMA+TDMA)				5.75 (rounds=2, TDMA slots per access round=512)
	6000		Slotted ALOHA Collision based re-attempts by re-paging	1				237.97 (rounds=35, TDMA slots per access round=1280)
				2(FDMA+TDMA)				118.71 (rounds=9 TDMA slots per access round=1280)
				4 (FDMA+TDMA)				65.10 (rounds=5, TDMA slots per access round=1280)
				8 (FDMA+TDMA)				38.59 (rounds=3, TDMA slots per access round=1280)
			Slotted ALOHA Collision based re-attempts by re-paging	1				177.27 (rounds=5, TDMA slots per access round=5120)
				2(FDMA+TDMA)				111.98 (rounds=3 TDMA slots per access round=5120)
				4 (FDMA+TDMA)				85.65 (rounds=3, TDMA slots per access round=5120)
				8 (FDMA+TDMA)				57.10 (rounds=2, TDMA slots per access round=5120)
[213]	600	90%	Slotted ALOHA	1	5	No	-	16.14
		99%						17.53
		90%		4 (FDMA)		No		7.58

		99%	Slotted ALOHA					8.23
		90%	Slotted ALOHA	4 (CDMA)		No		7.58
		99%						8.23
		90%	Slotted ALOHA with anti-collision solution	1		No		14.61
		99%						15.87
		90%	Slotted ALOHA	1		Direction 1 ON/OFF duration = 10ms/100ms		16.35
		99%						17.94
		90%	Slotted ALOHA	1		Direction 1 ON/OFF duration = 100ms/1s		16.36
		99%						18.92
		90%	Slotted ALOHA	1		Direction 1 ON/OFF duration = 500ms/5s		16.36
		99%						19.93
[195]	600	100%	Slotted ALOHA	1 (TDMA)	7	Direction 1: OFF when fully discharged Rx power = -28dBm	1	21.256 (paging period 3ms)
				1 (TDMA)		Direction 1: OFF when energy <90%) Rx power = -28dBm		16.034 (paging period 3ms)
				1 (TDMA)		Direction 2: SLEEP w R2D control Rx power = -28dBm		11.186 (paging period 3ms)
				1 (TDMA)		Direction 1: OFF when fully discharged Rx power = -28dBm		13.546 (paging period 4ms)
				1 (TDMA)		Direction 1: OFF when energy <90% Rx power = -28dBm		9.676 (paging period 4ms)

				1 (TDMA)		Direction 2: SLEEP w R2D control Rx power = -28dBm		5.634 (paging period 4ms)
				1 (TDMA)		Direction 1: OFF when fully discharged Rx power = -28dBm		10.688 (paging period 6.5ms)
				1 (TDMA)		Direction 1: OFF when energy <90% Rx power = -28dBm		7.864 (paging period 6.5ms)
				1 (TDMA)		Direction 2: SLEEP w R2D control Rx power = -28dBm		2.71 (paging period 6.5ms)
				1 (TDMA)		Direction 1: OFF when fully discharged Rx power = -28dBm		69.206 (paging period 10ms)
				1 (TDMA)		Direction 1: OFF when energy <90% Rx power = -28dBm		53.481 (paging period 10ms)
				1 (TDMA)		Direction 2: SLEEP w R2D control Rx power = -28dBm		16.893 (paging period 10ms)
				1 (TDMA)		Direction 1: OFF when fully discharged Rx power = -28dBm		114.45 (paging period 16ms)
				1 (TDMA)		Direction 1: OFF when energy <90% Rx power = -28dBm		84.909 (paging period 16ms)

				1 (TDMA)		Direction 2: SLEEP w R2D control Rx power = -28dBm		27.027 (paging period 16ms)
				2 (TDMA)		Direction 1: OFF when fully discharged Rx power = -28dBm		54.048 (paging period 16ms)
				2 (TDMA)		Direction 1: OFF when energy <90% Rx power = -28dBm		44.627 (paging period 16ms)
				2 (TDMA)		Direction 2: SLEEP w R2D control Rx power = -28dBm		12.884 (paging period 16ms)
				4 (TDMA)		Direction 1: OFF when fully discharged Rx power = -28dBm		25.686 (paging period 16ms)
				4 (TDMA)		Direction 1: OFF when energy <90% Rx power = -28dBm		22.393 (paging period 16ms)
				4 (TDMA)		Direction 2: SLEEP w R2D control Rx power = -28dBm		6.662 (paging period 16ms)
[196]	600	99%	Slotted ALOHA	1	-	No	-	18.42
				2 (FDMA)				16.91
				4 (FDMA)				16.24
				8 (FDMA)				15.91
[199]	389	99%	Slotted ALOHA	1	50	No	-	>10s (Initial TDMA slots=32) 4.35s (Initial TDMA slots=64) 3.43s (Initial TDMA slots=128)
				2 (TDMA+FDMA)				1.91s (Initial TDMA slots=32) 1.10s (Initial TDMA slots=64) 1.77s (Initial TDMA slots=128)

				4 (TDMA+FDMA)				0.62s (Initial TDMA slots=32) 0.67s (Initial TDMA slots=64) 1.18s (Initial TDMA slots=128)
				8 (TDMA+FDMA)				0.44s (Initial TDMA slots=32) 0.50s (Initial TDMA slots=64) 0.89s (Initial TDMA slots=128)
[209]	20	100%	Slotted ALOHA	8 (TDMA+FDMA)	Case 1: (7, 7) Case 2: (28, 70)	No DCM	1	20
						Case 1 DCM (periodic Query)		5
						Case 1 DCM (aperiodic Query)		2.5
						Case 2 DCM (periodic Query)		11 (4 groups)
				16 (TDMA+FDMA)		No DCM	2	200
						Case 1 DCM (periodic Query)		20
						Case 1 DCM (aperiodic Query)		20
						Case 2 DCM (periodic Query)		10 (8 groups)
[212]	600	90	Slotted ALOHA	1	7	No	-	54.43
				2 (FDMA)				27.21
				4 (FDMA)				13.62
		99		1				105.43
				2 (FDMA)				52.72
				4 (FDMA)				26.36

7.3 Evaluations for outdoor scenarios

7.3.1 Methodology

For an evaluation scenario:

- For each link *j*,
  - Step 1: Obtain the required SINR for the physical channels under different reference data rates for this link *j*.
  - Step 2: Obtain the receiver sensitivity using the method Budget-Alt2. See Clause 4.3.1 for the Budget definition.
  - Step 3: Obtain the MPL/distance for link *j* based on the receiver sensitivity from step 2 and link budget template.

- The coverage results for each link are provided.

7.3.2 Observations

All the evaluation results and sources are contained in the attached spreadsheet file [392].

RAN1 has evaluated the coverage for Device 2b and Device C respectively. Evaluation results from companies in terms of different achievable cell edge data rate and achievable maximum distance under different scenarios, penetration margins and Tx power have been observed for R2D and D2R in the tables of following sub-sections.

For R2D coverage, based on evaluation results, the achievable cell edge data rate and achievable maximum distance are observed at least for the following scenarios

- For cell edge data rate around 1 kbps,
  - For UMa NLOS pathloss model, penetration margin 0 dB
    - Device 2b: the achievable maximum distance is at least 500 meters at BLER 10% when BS Tx power is from 33dBm to 43dBm and at BLER 1% when BS Tx power is from 38dBm to 43dBm. The achievable maximum distance can be smaller or larger than 500 meters at BLER 1% (about 370~1800 meters) when BS Tx power is 33dBm
    - Device C: the achievable maximum distance is at least 500 meters at BLER 10% and 1% when BS Tx power is from 33dBm to 43dBm
  - For UMa NLOS pathloss model, penetration margin 20 dB
    - Device 2b: the achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 190~1340 meters) and at BLER 1% (about 110~1000 meters) when BS Tx power is from 33dBm to 43dBm
    - Device C: the achievable maximum distance is at least 500 meters at BLER 10% and 1% when BS Tx power is from 33dBm to 43dBm
- For cell edge data rate around 5~7 kbps,
  - For UMa NLOS pathloss model, penetration margin 0 dB
    - Device 2b: the achievable maximum distance is at least 500 meters at BLER 10% when BS Tx power is from 38dBm to 43dBm and at BLER 1% when BS Tx power is 43dBm. The achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 480~2300 meters) when BS Tx power is 33dBm and at BLER 1% (about 210~1700 meters) when BS Tx power is from 33dBm to 38dBm
    - Device C: the achievable maximum distance is at least 500 meters at BLER 10% when BS Tx power is from 33dBm to 43dBm and at BLER 1% when BS Tx power is from 38dBm to 43dBm. The achievable maximum distance can be smaller or larger than 500 meters at BLER 1% (about 430~1500 meters) when BS Tx power is 33dBm
  - For UMa NLOS pathloss model, penetration margin 20 dB
    - Device 2b: the achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 150~1280 meters) and at BLER 1% (about 70~710 meters) when BS Tx power is from 33dBm to 43dBm

- Device C: the achievable maximum distance is at least 500 meters at BLER 10% when BS Tx power is 43dBm. The achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 280~1070 meters) when BS Tx power is from 33dBm to 38dBm and at BLER 1% (about 130~810 meters) when BS Tx power is from 33dBm to 43dBm

For D2R coverage, based on evaluation results, the achievable cell edge data rate and achievable maximum distance are observed at least for the following scenarios

- For cell edge data rate around 0.1 kbps,
  - For UMa NLOS pathloss model, penetration margin 0 dB
    - Device 2b: the achievable maximum distance is smaller than 500 meters at BLER 10% (about 80~370 meters) and BLER 1% (about 40~300 meters) when device Tx power is -20dBm. The achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 150~660 meters) and at BLER 1% (about 80~540 meters) when device Tx power is -10dBm
    - Device C: the achievable maximum distance is at least 500 meters at BLER 10% and 1% when device Tx power is from -3dBm to 5dBm
  - For UMa NLOS pathloss model, penetration margin 20 dB
    - Device 2b: the achievable maximum distance is smaller than 500 meters at BLER 10% (about 20~200 meters) and at BLER 1% (10~160 meters) when device Tx power is from -20dBm to -10dBm
    - Device C: the achievable maximum distance is at least 500 meters at BLER 10% when device Tx power is 5dBm. The achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 320~680 meters) when device Tx power is from -3dBm to 0dBm and at BLER 1% (about 260~730 meters) when device Tx power is from -3dBm to 5dBm
- For cell edge data rate around 1 kbps,
  - For UMa NLOS pathloss model, penetration margin 0 dB
    - Device 2b: the achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 110~810 meters) and at BLER 1% (about 50~600 meters) when device Tx power is -10dBm. The achievable maximum distance is smaller than 500 meters at BLER 10% (about 60~450 meters) and BLER 1% (about 30~230 meters) when device Tx power is -20dBm.
    - Device C: the achievable maximum distance is at least 500 meters at BLER 10% when device Tx power is from 0dBm to 5dBm and at BLER 1% when device Tx power is 5dBm. The achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 460~1230 meters) when device Tx power is -3dBm and at BLER 1% (about 310~1090 meters) when device Tx power is from -3dBm to 0dBm
  - For UMa NLOS pathloss model, penetration margin 20 dB
    - Device 2b: the achievable maximum distance is smaller than 500 meters at BLER 10% (about 20~250 meters) and at BLER 1% (about 10~180 meters) when device Tx power is from -20dBm to -10dBm
    - Device C: the achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 230~600 meters) when device Tx power is 5dBm. The achievable maximum distance is smaller than 500 meters at BLER 10% (about 140~450 meters) when device Tx power is from -3dBm to 0dBm and at BLER 1% (about 100~450 meters) when device Tx power is from -3dBm to 5dBm.

- For cell edge data rate around 5~7 kbps,

- For UMa NLOS pathloss model, penetration margin 0 dB

  - Device 2b: the achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 100~680 meters) when device Tx power is -10dBm. The achievable maximum distance is smaller than 500 meters at BLER 10% (about 60~270 meters) when device Tx power is -20dBm and at BLER 1% (about 40~390 meters) when device Tx power is from -20dBm to -10dBm

  - Device C: the achievable maximum distance can be smaller or larger than 500 meters at BLER 10% (about 300~1220 meters) and at BLER 1% (about 240~960 meters) when device Tx power is from -3dBm to 5dBm

- For UMa NLOS pathloss model, penetration margin 20 dB

  - Device 2b: the achievable maximum distance is smaller than 500 meters at BLER 10% (about 20~150 meters) and at BLER 1% (10~120 meters) when device Tx power is from -20dBm to -10dBm

  - Device C: the achievable maximum distance is smaller than 500 meters at BLER 10% (about 90~380 meters) and at BLER 1% (70~300 meters) when device Tx power is from -3dBm to 5dBm

For the Rel-20 study, RAN1 has identified the following:

- Device C achieves better coverage than Device 2b for both R2D and D2R.

- Device 2b is applicable to outdoor scenarios for short distance e.g. 50-200m.

- Device C is applicable to outdoor scenarios for both short and large distance e.g. 50-500m.

NOTE: Necessity of Device 2b and C in terms of aspects other than coverage is up to RAN decision.

RAN1 observes 5 dBm as the maximum transmit power for Device C to achieve maximum coverage in outdoor scenarios. Values between -3dBm and 5dBm are feasible to achieve different coverage. This does not preclude RAN to select any value(s) between -3 dBm and 5 dBm as maximum transmit power for Device C.

7.3.3 Link budget for R2D

7.3.3.1 Device 2b

7.3.3.1.1 UMa NLOS

Table 7.3.3.1.1-1: R2D - Device 2b, Reference data rate 1kbps

UMa NLOS, Penetration margin 0 dB			
BS Tx power	**Achievable cell edge data rate**	**Achievable maximum distance**	**Achievable maximum distance**

(dBm)	(kbps)	at BLER 10% (meter)	at BLER 1% (meter)
33	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96	[Source 22]:2150.4~2412.2, [Source 5]:495.6~619.9, [Source 1]:1441.4~1901.3	[Source 22]:1592.3~1802.0, [Source 5]:295.1~369.1, [Source 1]:1024.1~1593.2
38	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 3]:1.84~2.46, [Source 4]:1.79~3.6	[Source 22]:2887.1~3238.6, [Source 5]:665.4~832.3, [Source 1]:1935.2~2552.6, [Source 3]:2303.8~2333.9, [Source 4]:2314.6~2696.4	[Source 22]:2137.8~2419.3, [Source 5]:396.2~495.6, [Source 1]:1375.0~2139.0, [Source 3]:1277.4~1282.6
43	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6	[Source 22]:3876.2~4348.1, [Source 5]:893.3~1117.5, [Source 1]:2598.1~3427.1, [Source 4]:3107.8~3620.1	[Source 22]:2870.2~3248.2, [Source 5]:531.9~665.4, [Source 1]:1846.1~2871.8
UMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96	[Source 22]:661.8~742.4, [Source 5]:152.5~190.8, [Source 1]:443.6~585.2	[Source 22]:490.1~554.6, [Source 5]:90.8~113.6, [Source 1]:315.2~490.3
38	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 3]:1.84~2.46, [Source 4]:1.79~3.6	[Source 22]:888.6~996.8, [Source 5]:204.8~256.2, [Source 1]:595.6~785.6, [Source 3]:709.1~718.3, [Source 4]:712.4~829.9	[Source 22]:657.9~744.6, [Source 5]:121.9~152.5, [Source 1]:423.2~658.3, [Source 3]:393.1~394.8
43	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6	[Source 22]:1193.0~1338.2, [Source 5]:274.9~343.9, [Source 1]:799.6~1054.8, [Source 4]:956.5~1114.2	[Source 22]:883.4~999.7, [Source 5]:163.7~204.8, [Source 1]:568.2~883.9

Table 7.3.3.1.1-2: R2D - Device 2b, Reference data rate 5~7kbps

UMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 11]:8.8~12, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~12.85, [Source 8]:14, [Source 15]:5.25, [Source 18]:7	[Source 12]:1382.1~2307.8, [Source 21]:749.0~909.8, [Source 22]:1611.2~1620.7, [Source 11]:1928.3~2125.2, [Source 5]:412.8~669.3, [Source 1]:868.4, [Source 17]:832.2, [Source 9]:1501.2~1669.1, [Source 8]:1659.3, [Source 15]:1624.3,	[Source 12]:634.7~1272.7, [Source 21]:382.6~541.7, [Source 22]:993.8~1172.1, [Source 11]:1035.7~1144.8, [Source 5]:175.7~352.1, [Source 1]:505.0, [Source 9]:823.1~953.7, [Source 8]:857.7, [Source 18]:214.8

		[Source 18]:478.6	
38	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 6]:7, [Source 10]:6~11.5, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 3]:3.69~7.37, [Source 9]:11.53~12.85, [Source 8]:14, [Source 18]:7	[Source 12]:1855.7~3098.5, [Source 21]:1005.6~1221.4, [Source 22]:2163.1~2175.9, [Source 6]:1488.0~1627.4, [Source 10]:1899.0~1985.0, [Source 5]:554.3~898.6, [Source 1]:1165.9, [Source 3]:1755.9~2168.2, [Source 9]:2015.5~2241.0, [Source 8]:2227.8, [Source 18]:642.6	[Source 12]:852.3~1708.8, [Source 21]:513.7~727.3, [Source 22]:1334.3~1573.6, [Source 5]:235.9~472.8, [Source 1]:678.0, [Source 3]:957.0~1212.8, [Source 9]:1105.0~1280.4, [Source 8]:1151.5, [Source 18]:288.4
43	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 6]:7, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 2]:5.37~10.2, [Source 8]:14, [Source 15]:5.25	[Source 12]:2491.6~4160.1, [Source 21]:1350.1~1639.9, [Source 22]:2904.2~2921.3, [Source 6]:1997.7~2184.9, [Source 5]:744.2~1206.4, [Source 1]:1565.3, [Source 17]:1500.1, [Source 8]:2991.0, [Source 15]:2927.8	[Source 12]:1144.5~2294.3, [Source 21]:689.7~976.4, [Source 22]:1791.4~2112.7, [Source 5]:316.7~634.7, [Source 1]:910.3, [Source 2]:1280.4~1980.1, [Source 8]:1546.1
UMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 11]:8.8, [Source 16]:9.33~13.33, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~12.85, [Source 8]:14, [Source 15]:5.25, [Source 18]:7	[Source 12]:424.8~709.9, [Source 21]:230.5~280.0, [Source 22]:495.9~498.8, [Source 11]:654.1, [Source 16]:440.7~557.9, [Source 5]:127.1~206.0, [Source 1]:267.3, [Source 17]:256.1, [Source 9]:462.0~513.7, [Source 8]:510.7, [Source 15]:499.9, [Source 18]:147.3	[Source 12]:194.0~391.0, [Source 21]:117.8~166.7, [Source 22]:305.9~360.7, [Source 5]:54.1~108.4, [Source 1]:155.4, [Source 9]:253.3~293.5, [Source 8]:264.0, [Source 18]:66.1
38	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 6]:7, [Source 5]:4.12~6.6, [Source 14]:4.8~6.67, [Source 1]:5.25, [Source 3]:3.69~7.37, [Source 9]:11.53~12.85, [Source 8]:14, [Source 18]:7	[Source 12]:570.7~953.4, [Source 21]:309.5~375.9, [Source 22]:665.7~669.7, [Source 6]:458.0~500.9, [Source 5]:170.6~276.6, [Source 14]:298.6~353.0, [Source 1]:358.8, [Source 3]:540.4~667.3, [Source 9]:620.3~689.7, [Source 8]:685.7, [Source 18]:197.8	[Source 12]:261.4~525.4, [Source 21]:158.1~223.8, [Source 22]:410.7~484.3, [Source 5]:72.6~145.5, [Source 14]:260.6~313.2, [Source 1]:208.7, [Source 3]:294.5~373.3, [Source 9]:340.1~394.1, [Source 8]:354.4, [Source 18]:88.8
43	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 6]:7, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 2]:5.37~10.2, [Source 8]:14,	[Source 12]:766.5~1280.2, [Source 21]:415.5~504.7, [Source 22]:893.8~899.1, [Source 6]:614.8~672.4, [Source 5]:229.0~371.3, [Source 1]:481.8, [Source 17]:461.7, [Source	[Source 12]:351.5~705.8, [Source 21]:212.3~300.5, [Source 22]:551.3~650.2, [Source 5]:97.5~195.3, [Source 1]:280.2, [Source 2]:394.1~609.4, [Source 8]:475.8

	[Source 15]:5.25	8]:920.5, [Source 15]:901.1	
UMa NLOS, Penetration margin 80 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 21]:10.16~14.18	[Source 21]:6.7~8.2	[Source 21]:3.4~4.9
38	[Source 21]:10.16~14.18	[Source 21]:9.0~11.0	[Source 21]:4.6~6.5
43	[Source 21]:10.16~14.18	[Source 21]:12.1~14.7	[Source 21]:6.2~8.8

Table 7.3.3.1.1-3: R2D - Device 2b, Reference data rate 48~60kbps

UMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:642.1~784.6, [Source 8]:808.1~1243.2	
38	[Source 8]:42~84	[Source 8]:1085.0~1669.1	
43	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:1157.5~1414.2, [Source 8]:1456.7~2241.0	
UMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84, [Source 8]:84	[Source 17]:197.6~241.5, [Source 8]:248.7	
38	[Source 8]:42~84	[Source 8]:333.9~513.7	
43	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:356.2~435.3, [Source 8]:448.3~689.7	

Table 7.3.3.1.1-4: R2D - Device 2b, Reference data rate other value of 168kbps

UMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:168, [Source 8]:168,	[Source 17]:519.4, [Source 8]:350.9,	
38	[Source 8]:168	[Source 8]:471.1	
43	[Source 17]:168, [Source 8]:168,	[Source 17]:936.3, [Source 8]:632.4,	
UMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:168, [Source 8]:168,	[Source 17]:159.9, [Source 8]:108.0,	

38	[Source 8]:168	[Source 8]:145.0	
43	[Source 17]:168, [Source 8]:168,	[Source 17]:288.2, [Source 8]:194.6,	

7.3.3.1.2 RMa NLOS

Table 7.3.3.1.2-1: R2D - Device 2b, Reference data rate 1kbps

RMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96	[Source 22]:3764.5~4228.4, [Source 5]:853.0~1069.8, [Source 1]:2511.6~3323.6	[Source 22]:2777.8~3148.1, [Source 5]:504.8~633.1, [Source 1]:1777.5~2779.4
38	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 3]:1.84~2.46, [Source 4]:1.79~3.6	[Source 22]:5071.4~5696.4, [Source 5]:1149.1~1441.1, [Source 1]:3383.5~4477.4, [Source 3]:4036.3~4089.5, [Source 4]:4055.6~4732.6	[Source 22]:3742.1~4241.0, [Source 5]:680.1~853.0, [Source 1]:2394.6~3744.3, [Source 3]:2222.7~2232.0
43	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 3]:1.84~2.46, [Source 4]:1.79~3.6	[Source 22]:6832.0~7674.0, [Source 5]:1548.0~1941.5, [Source 1]:4558.2~6031.8, [Source 3]:5437.5~5509.3, [Source 4]:5463.6~6375.6	[Source 22]:5041.2~5713.4, [Source 5]:916.2~1149.1, [Source 1]:3226.0~5044.2, [Source 3]:2994.3~3006.8
RMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96	[Source 22]:1142.9~1283.8, [Source 5]:259.0~324.8, [Source 1]:762.5~1009.1	[Source 22]:843.3~955.8, [Source 5]:153.3~192.2, [Source 1]:539.7~843.9
38	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 3]:1.84~2.46, [Source 4]:1.79~3.6	[Source 22]:1539.7~1729.5, [Source 5]:348.9~437.5, [Source 1]:1027.3~1359.4, [Source 3]:1225.4~1241.6, [Source 4]:1231.3~1436.9	[Source 22]:1136.1~1287.6, [Source 5]:206.5~259.0, [Source 1]:727.0~1136.8, [Source 3]:674.8~677.6
43	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 3]:1.84~2.46, [Source 4]:1.79~3.6	[Source 22]:2074.2~2329.9, [Source 5]:470.0~589.4, [Source 1]:1383.9~1831.3, [Source 3]:1650.9~1672.7, [Source 4]:1658.8~1935.7	[Source 22]:1530.6~1734.6, [Source 5]:278.2~348.9, [Source 1]:979.4~1531.5, [Source 3]:909.1~912.9

Table 7.3.3.1.2-2: R2D - Device 2b, Reference data rate 5~7kbps

RMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~12.85, [Source 8]:14, [Source 18]:7	[Source 12]:2407.3~4043.4, [Source 21]:1295.3~1576.9, [Source 22]:2811.1~2827.9, [Source 5]:709.1~1155.9, [Source 1]:1504.3, [Source 17]:1623.3, [Source 9]:1332.4~1481.4, [Source 8]:2307.0, [Source 18]:823.5	[Source 12]:1095.7~2214.5, [Source 21]:656.6~933.3, [Source 22]:1724.3~2037.5, [Source 5]:298.8~603.7, [Source 1]:869.4, [Source 9]:730.6~846.5, [Source 8]:1192.6, [Source 18]:366.1
38	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 6]:7, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 3]:3.69~7.37, [Source 9]:11.53~12.85, [Source 8]:14, [Source 18]:7	[Source 12]:3243.2~5447.3, [Source 21]:1745.0~2124.3, [Source 22]:3787.0~3809.6, [Source 6]:2594.5~2840.5, [Source 5]:955.2~1557.2, [Source 1]:2026.6, [Source 3]:3066.6~3795.9, [Source 9]:1788.7~1988.8, [Source 8]:3097.2, [Source 18]:1109.4	[Source 12]:1476.4~2983.5, [Source 21]:884.5~1257.3, [Source 22]:2323.0~2744.8, [Source 5]:402.5~813.2, [Source 1]:1171.2, [Source 3]:1659.8~2109.1, [Source 9]:980.8~1136.4, [Source 8]:1601.1, [Source 18]:493.2
43	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 6]:7, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 3]:3.69~7.37, [Source 8]:14	[Source 12]:4369.2~7338.4, [Source 21]:2350.8~2861.8, [Source 22]:5101.7~5132.2, [Source 6]:3495.3~3826.8, [Source 5]:1286.8~2097.9, [Source 1]:2730.2, [Source 17]:2946.1, [Source 3]:4131.2~5113.8, [Source 8]:4158.0	[Source 12]:1989.2~4019.4, [Source 21]:1191.6~1693.8, [Source 22]:3129.4~3697.8, [Source 5]:542.3~1095.6, [Source 1]:1577.8, [Source 3]:2236.0~2841.3, [Source 8]:2149.5
RMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 16]:9.33~13.33, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~12.85, [Source 8]:14, [Source 18]:7	[Source 12]:730.2~1227.2, [Source 21]:393.3~478.7, [Source 22]:853.5~858.6, [Source 16]:757.6~961.5, [Source 5]:215.3~351.0, [Source 1]:456.7, [Source 17]:492.9, [Source 9]:410.1~456.0, [Source 8]:710.2, [Source 18]:250.0	[Source 12]:331.1~671.6, [Source 21]:199.3~283.4, [Source 22]:523.5~618.6, [Source 5]:90.7~183.3, [Source 1]:263.9, [Source 9]:224.9~260.6, [Source 8]:367.1, [Source 18]:111.2
38	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 6]:7, [Source 5]:4.12~6.6, [Source 14]:4.8~6.67, [Source 1]:5.25, [Source 3]:3.69~7.37, [Source	[Source 12]:984.1~1653.5, [Source 21]:529.8~645.0, [Source 22]:1149.8~1156.6, [Source 6]:787.6~862.3, [Source 5]:290.0~472.8, [Source 14]:510.9~605.1, [Source 1]:615.3,	[Source 12]:447.1~905.3, [Source 21]:268.6~381.7, [Source 22]:705.3~833.4, [Source 5]:122.2~246.9, [Source 14]:445.2~536.1, [Source 1]:355.6, [Source 3]:503.9~640.3, [Source

	9]:11.53~12.85, [Source 8]:14, [Source 18]:7	[Source 3]:931.0~1152.5, [Source 9]:550.6~612.2, [Source 8]:953.4, [Source 18]:336.8	9]:301.9~349.8, [Source 8]:492.9, [Source 18]:149.7
43	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 6]:7, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 3]:3.69~7.37, [Source 8]:14	[Source 12]:1326.1~2227.8, [Source 21]:713.7~868.9, [Source 22]:1548.9~1558.2, [Source 6]:1061.1~1161.7, [Source 5]:390.7~636.9, [Source 1]:828.9, [Source 17]:894.5, [Source 3]:1254.3~1552.6, [Source 8]:1280.0	[Source 12]:603.1~1219.9, [Source 21]:361.8~514.2, [Source 22]:950.1~1122.7, [Source 5]:164.6~332.6, [Source 1]:479.0, [Source 3]:678.9~862.6, [Source 8]:661.7
RMa NLOS, Penetration margin 80 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 21]:10.16~14.18	[Source 21]:11.0~13.4	[Source 21]:5.6~7.9
38	[Source 21]:10.16~14.18	[Source 21]:14.8~18.0	[Source 21]:7.5~10.7
43	[Source 21]:10.16~14.18	[Source 21]:20.0~24.3	[Source 21]:10.1~14.4

Table 7.3.3.1.2-3: R2D - Device 2b, Reference data rate 48~60kbps

RMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:1248.9~1529.4, [Source 8]:1123.7~1728.5	
38	[Source 8]:42~84	[Source 8]:1508.6~2320.6	
43	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:2266.5~2775.7, [Source 8]:2025.3~3115.5	
RMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84, [Source 8]:84	[Source 17]:379.2~464.3, [Source 8]:345.9	
38	[Source 8]:42~84	[Source 8]:464.4~714.4	
43	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:688.1~842.7, [Source 8]:623.5~959.0	

Table 7.3.3.1.2-4: R2D - Device 2b, Reference data rate other value of 168kbps

RMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**

33	[Source 17]:168, [Source 8]:168	[Source 17]:1007.7, [Source 8]:487.9	
38	[Source 8]:168	[Source 8]:655.0	
43	[Source 17]:168, [Source 8]:168	[Source 17]:1828.8, [Source 8]:879.4	
RMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:168, [Source 8]:168	[Source 17]:305.9, [Source 8]:150.2	
38	[Source 8]:168	[Source 8]:201.6	
43	[Source 17]:168, [Source 8]:168	[Source 17]:555.3, [Source 8]:270.7	

7.3.3.1.3 UMi NLOS

Table 7.3.3.1.3-1: R2D - Device 2b, Reference data rate 1kbps

UMi NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96	[Source 22]:2446.3~2778.1, [Source 5]:481.8~617.3, [Source 1]:1570.9~2134.5	[Source 22]:1754.0~2011.5, [Source 5]:271.4~347.7, [Source 1]:1076.1~1755.2
38	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96	[Source 22]:3389.6~3849.4, [Source 5]:667.6~855.3, [Source 1]:2176.7~2957.6	[Source 22]:2430.4~2787.2, [Source 5]:376.0~481.8, [Source 1]:1491.1~2432.0
43	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96	[Source 22]:4696.7~5333.8, [Source 5]:925.0~1185.2, [Source 1]:3016.1~4098.2	[Source 22]:3367.6~3862.0, [Source 5]:521.0~667.6, [Source 1]:2066.0~3369.8
UMi NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96	[Source 22]:663.6~753.7, [Source 5]:130.7~167.5, [Source 1]:426.2~579.1	[Source 22]:475.8~545.7, [Source 5]:73.6~94.3, [Source 1]:291.9~476.2
38	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96	[Source 22]:919.6~1044.3, [Source 5]:181.1~232.0, [Source 1]:590.5~802.4	[Source 22]:659.3~756.1, [Source 5]:102.0~130.7, [Source 1]:404.5~659.8
43	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96	[Source 22]:1274.2~1447.0, [Source 5]:250.9~321.5, [Source 1]:818.2~1111.8	[Source 22]:913.6~1047.7, [Source 5]:141.3~181.1, [Source 1]:560.5~914.2

Table 7.3.3.1.3-2: R2D - Device 2b, Reference data rate 5~7kbps

UMi NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~12.85, [Source 18]:7	[Source 21]:761.1~943.9, [Source 22]:1777.1~1788.7, [Source 5]:393.6~671.9, [Source 1]:896.5, [Source 17]:855.2, [Source 9]:1643.3~1848.0, [Source 18]:463.6	[Source 21]:361.8~531.6, [Source 22]:1040.9~1249.5, [Source 5]:152.8~330.0, [Source 1]:491.9, [Source 9]:844.8~994.4, [Source 18]:190.9
38	[Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 10]:6~11.5, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 9]:11.53~12.85, [Source 18]:7	[Source 21]:1054.6~1307.8, [Source 22]:2462.3~2478.4, [Source 10]:2133.0~2240.0, [Source 5]:545.3~931.0, [Source 1]:1242.2, [Source 9]:2276.9~2560.6, [Source 18]:642.4	[Source 21]:501.3~736.7, [Source 22]:1442.3~1731.3, [Source 5]:211.8~457.3, [Source 1]:681.6, [Source 9]:1170.6~1377.9, [Source 18]:264.5
43	[Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14	[Source 21]:1461.2~1812.2, [Source 22]:3411.8~3434.1, [Source 5]:755.6~1290.1, [Source 1]:1721.2, [Source 17]:1641.9	[Source 21]:694.7~1020.7, [Source 22]:1998.4~2398.9, [Source 5]:293.5~633.6, [Source 1]:944.5
UMi NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 16]:9.33~13.33, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~12.85, [Source 18]:7	[Source 21]:206.5~256.1, [Source 22]:482.1~485.2, [Source 16]:440.6~571.9, [Source 5]:106.8~182.3, [Source 1]:243.2, [Source 17]:232.0, [Source 9]:445.8~501.3, [Source 18]:125.8	[Source 21]:98.2~144.2, [Source 22]:282.4~339.0, [Source 1]:133.5, [Source 5]:41.5~89.5, [Source 9]:229.2~269.8, [Source 18]:51.8
38	[Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 5]:4.12~6.6, [Source 14]:4.8~6.67, [Source 1]:5.25, [Source 9]:11.53~12.85, [Source 18]:7	[Source 21]:286.1~354.8, [Source 22]:668.0~672.4, [Source 5]:147.9~252.6, [Source 14]:274.9~330.9, [Source 1]:337.0, [Source 9]:617.7~694.7, [Source 18]:174.3	[Source 21]:136.0~199.8, [Source 22]:391.3~469.7, [Source 5]:57.5~124.1, [Source 14]:236.5~289.8, [Source 1]:184.9, [Source 9]:317.6~373.8, [Source 18]:71.8
43	[Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14	[Source 21]:396.4~491.6, [Source 22]:925.6~931.6, [Source 5]:205.0~350.0, [Source 1]:466.9, [Source 17]:445.4	[Source 21]:188.4~276.9, [Source 22]:542.1~650.8, [Source 5]:79.6~171.9, [Source 1]:256.2
UMi NLOS, Penetration margin 80 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 21]:10.16~14.18	[Source 21]:4.1~5.1	[Source 21]:2.0~2.9

38	[Source 21]:10.16~14.18	[Source 21]:5.7~7.1	[Source 21]:2.7~4.0
43	[Source 21]:10.16~14.18	[Source 21]:7.9~9.8	[Source 21]:3.8~5.5

Table 7.3.3.1.3-3: R2D - Device 2b, Reference data rate 48~60kbps

UMi NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84	[Source 17]:641.8~801.2	
38			
43	[Source 17]:42~84	[Source 17]:1232.3~1538.2	
UMi NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84	[Source 17]:174.1~217.4	
38			
43	[Source 17]:42~84	[Source 17]:334.3~417.3	

Table 7.3.3.1.3-4: R2D - Device 2b, Reference data rate other value of 168kbps

UMi NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:168kbps	[Source 17]:507.5	
38			
43	[Source 17]:168kbps	[Source 17]:974.4	
UMi NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:168kbps	[Source 17]:137.7	
38			
43	[Source 17]:168kbps	[Source 17]:264.3	

7.3.3.2 Device C

7.3.3.2.1 UMa NLOS

Table 7.3.3.2.1-1: R2D - Device C, Reference data rate 1kbps

UMa NLOS, Penetration margin 0 dB

BS Tx power (dBm)	Achievable cell edge data rate (kbps)	Achievable maximum distance at BLER 10% (meter)	Achievable maximum distance at BLER 1% (meter)
33	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~2.25	[Source 22]:2494.4~2919.3, [Source 5]:1682.0~2452.4, [Source 1]:1887.7~2490.0, [Source 19]:1697.7~1932.7	[Source 22]:1648.5~2403.5, [Source 5]:1140.1~1914.8, [Source 1]:1341.3~2086.6, [Source 19]:1600.6~1790.2
38	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6, [Source 19]:1.21~2.25	[Source 22]:3348.9~3919.5, [Source 5]:2258.2~3292.6, [Source 1]:2534.4~3343.0, [Source 4]:2572.0~2996.0, [Source 19]:2279.4~2594.8	[Source 22]:2213.2~3226.9, [Source 5]:1530.7~2570.8, [Source 1]:1800.8~2801.4, [Source 19]:2148.9~2403.5
43	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6, [Source 19]:1.21~2.25	[Source 22]:4496.2~5262.2, [Source 5]:3031.9~4420.6, [Source 1]:3402.6~4488.3, [Source 4]:3453.1~4022.4, [Source 19]:3060.2~3483.8	[Source 22]:2971.4~4332.4, [Source 5]:2055.1~3451.5, [Source 1]:2417.7~3761.1, [Source 19]:2885.1~3226.9
UMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~2.25	[Source 22]:767.7~898.5, [Source 5]:517.7~754.8, [Source 1]:581.0~766.3, [Source 19]:522.5~594.8	[Source 22]:507.3~739.7, [Source 5]:350.9~589.3, [Source 1]:412.8~642.2, [Source 19]:492.6~551.0
38	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6, [Source 19]:1.21~2.25	[Source 22]:1030.7~1206.3, [Source 5]:695.0~1013.4, [Source 1]:780.0~1028.9, [Source 4]:791.6~922.1, [Source 19]:701.5~798.6	[Source 22]:681.2~993.2, [Source 5]:471.1~791.2, [Source 1]:554.2~862.2, [Source 19]:661.4~739.7
43	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6, [Source 19]:1.21~2.25	[Source 22]:1383.8~1619.6, [Source 5]:933.1~1360.5, [Source 1]:1047.2~1381.4, [Source 4]:1062.8~1238.0, [Source 19]:941.9~1072.2	[Source 22]:914.5~1333.4, [Source 5]:632.5~1062.3, [Source 1]:744.1~1157.6, [Source 19]:888.0~993.2
UMa NLOS, Penetration margin 80 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 19]:1.21~2.25	[Source 19]:15.2~17.3	[Source 19]:14.4~16.1
38	[Source 19]:1.21~2.25	[Source 19]:20.5~23.3	[Source 19]:19.3~21.6
43	[Source 19]:1.21~2.25	[Source 19]:27.5~31.3	[Source 19]:25.9~29.0

Table 7.3.3.2.1-2: R2D - Device C, Reference data rate 5~7kbps

UMa NLOS, Penetration margin 0 dB			
BS Tx power	**Achievable cell edge data rate**	**Achievable maximum distance**	**Achievable maximum distance**

(dBm)	(kbps)	at BLER 10% (meter)	at BLER 1% (meter)
33	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 11]:8.8~12, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~14.42, [Source 8]:14, [Source 15]:5.25, [Source 19]:3.59~13.46, [Source 13]:8.24~9.33, [Source 18]:7	[Source 12]:1563.1~2594.7, [Source 21]:872.4~1150.8, [Source 22]:1800.8~1876.8, [Source 11]:2111.3~2361.4, [Source 5]:906.0~1567.2, [Source 1]:1137.3, [Source 17]:936.3, [Source 9]:1610.1~1887.7, [Source 8]:1865.6, [Source 15]:1827.4, [Source 19]:1117.4~1624.3, [Source 13]:2098.9~2817.9, [Source 18]:903.8	[Source 12]:748.1~1491.2, [Source 21]:495.5~634.7, [Source 22]:1104.3~1341.3, [Source 11]:1257.1~1272.0, [Source 5]:323.1~869.4, [Source 1]:661.4, [Source 9]:867.3~1084.9, [Source 8]:1004.9, [Source 19]:1041.1~1465.2, [Source 13]:1431.1, [Source 18]:430.2
38	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 6]:7, [Source 10]:6.2~12, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 9]:11.53~14.42, [Source 8]:14, [Source 19]:3.59~13.46, [Source 13]:8.24~9.33, [Source 18]:7	[Source 12]:2098.8~3483.7, [Source 21]:1171.3~1545.0, [Source 22]:2417.7~2519.5, [Source 6]:1788.1~1956.8, [Source 10]:2136.0~2226.0, [Source 5]:1216.4~2104.1, [Source 1]:1526.9, [Source 9]:2161.6~2534.4, [Source 8]:2504.7, [Source 19]:1500.1~2180.8, [Source 13]:2817.9~3783.3, [Source 18]:1213.4	[Source 12]:1004.6~2002.1, [Source 21]:665.3~852.1, [Source 22]:1482.6~1800.8, [Source 5]:433.8~1167.3, [Source 1]:888.0, [Source 9]:1164.4~1456.6, [Source 8]:1349.2, [Source 19]:1397.7~1967.2, [Source 13]:1921.4, [Source 18]:577.6
43	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 6]:7, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 2]:5.37~10.2, [Source 17]:14, [Source 8]:14, [Source 15]:5.25, [Source 19]:3.59~13.46, [Source 13]:8.24~9.33	[Source 12]:2817.8~4677.2, [Source 21]:1572.6~2074.3, [Source 22]:3246.0~3382.6, [Source 6]:2400.7~2627.1, [Source 5]:1633.2~2824.9, [Source 1]:2050.0, [Source 17]:1687.7, [Source 8]:3362.8, [Source 15]:3294.0, [Source 19]:2014.1~2928.0, [Source 13]:3783.3~5079.5	[Source 12]:1349.0~2688.1, [Source 21]:893.2~1144.0, [Source 22]:1990.5~2417.7, [Source 5]:582.4~1567.2, [Source 1]:1192.2, [Source 2]:1600.6~2475.4, [Source 8]:1811.4, [Source 19]:1876.6~2641.1, [Source 13]:2579.6
UMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 11]:8.8~12, [Source 16]:9.33~13.33, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~14.42, [Source 8]:14, [Source 15]:5.25, [Source 19]:3.59~13.46, [Source	[Source 12]:480.6~798.3, [Source 21]:268.5~354.2, [Source 22]:554.2~577.6, [Source 11]:726.8, [Source 16]:489.7~619.9, [Source 5]:278.9~482.3, [Source 1]:350.0, [Source 17]:288.2, [Source 9]:495.5~581.0, [Source 8]:574.2, [Source 15]:562.4, [Source	[Source 21]:152.5~195.3, [Source 22]:339.9~412.8, [Source 5]:99.4~267.6, [Source 1]:203.6, [Source 9]:266.9~333.9, [Source 8]:309.3, [Source 19]:320.4~451.0, [Source 13]:440.4, [Source 18]:132.4

	13]:8.24~9.33, [Source 18]:7	19]:343.9~499.9, [Source 13]:646.0~867.3, [Source 18]:278.2	
38	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 6]:7, [Source 5]:4.12~6.6, [Source 14]:4.8~6.67, [Source 1]:5.25, [Source 9]:11.53~14.42, [Source 8]:14, [Source 20]:7, [Source 13]:8.24~9.33, [Source 19]:3.59~13.46, [Source 18]:7	[Source 12]:645.6~1072.0, [Source 21]:360.5~475.5, [Source 22]:744.1~775.4, [Source 6]:550.3~602.2, [Source 5]:374.4~647.6, [Source 14]:370.6~438.9, [Source 1]:469.9, [Source 9]:665.3~780.0, [Source 8]:770.9, [Source 20]:556.9, [Source 13]:867.3~1164.4, [Source 19]:461.7~671.2, [Source 18]:373.5	[Source 12]:308.4~615.8, [Source 21]:204.8~262.2, [Source 22]:456.3~554.2, [Source 5]:133.5~359.3, [Source 14]:309.3~404.4, [Source 1]:273.3, [Source 9]:358.4~448.3, [Source 8]:415.2, [Source 19]:430.2~605.4, [Source 13]:591.3, [Source 18]:177.8
43	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 6]:7, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 2]:5.37~10.2, [Source 17]:14, [Source 8]:14, [Source 15]:5.25, [Source 13]:8.24~9.33, [Source 19]:3.59~13.46	[Source 12]:867.0~1439.3, [Source 21]:484.0~638.4, [Source 22]:999.0~1041.1, [Source 6]:738.9~808.6, [Source 5]:502.6~869.4, [Source 1]:630.9, [Source 17]:519.4, [Source 8]:1035.0, [Source 15]:1013.8, [Source 13]:1164.4~1563.3, [Source 19]:619.9~901.1	[Source 21]:274.9~352.1, [Source 22]:612.6~744.1, [Source 5]:179.3~482.3, [Source 1]:366.9, [Source 2]:492.6~761.8, [Source 8]:557.5, [Source 19]:577.6~812.9, [Source 13]:793.9
UMa NLOS, Penetration margin 80 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	Achievable cell edge data rate (kbps)	Achievable maximum distance at BLER 10% (meter)	Achievable maximum distance at BLER 1% (meter)
38	[Source 21]:10.16~14.3, [Source 19]:3.59~13.46	[Source 21]:7.8~10.3, [Source 19]:10.0~14.6	[Source 21]:4.4~5.7, [Source 19]:9.3~13.1
43	[Source 21]:10.16~14.3, [Source 19]:3.59~13.46	[Source 21]:10.5~13.9, [Source 19]:13.5~19.6	[Source 21]:6.0~7.6, [Source 19]:12.5~17.7

Table 7.3.3.2.1-3: R2D - Device C, Reference data rate 48~60kbps

UMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:722.5~882.7, [Source 8]:935.8~1365.2	[Source 8]:415.2
38	[Source 8]:42~84	[Source 8]:1256.3~1832.9	[Source 8]:557.5
43	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:1302.3~1591.1, [Source 8]:1686.7~2460.8	[Source 8]:748.5
UMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**

33	[Source 17]:42~84, [Source 8]:84	[Source 17]:222.4~271.7, [Source 8]:288.0	
38	[Source 8]:42~84	[Source 8]:386.7~564.1	[Source 8]:171.6
43	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:400.8~489.7, [Source 8]:519.1~757.4	[Source 8]:230.4

Table 7.3.3.2.1-4: R2D - Device C, Reference data rate other value of 168kbps

UMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:168, [Source 8]:168,	[Source 17]:584.4, [Source 8]:555.2,	
38	[Source 8]:168	[Source 8]:745.4	
43	[Source 17]:168, [Source 8]:168,	[Source 17]:1053.4, [Source 8]:1000.7,	
UMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:168, [Source 8]:168,	[Source 17]:179.9, [Source 8]:170.9,	
38	[Source 8]:168	[Source 8]:229.4	
43	[Source 17]:168, [Source 8]:168,	[Source 17]:324.2, [Source 8]:308.0,	

7.3.3.2.2 RMa NLOS

Table 7.3.3.2.2-1: R2D - Device C, Reference data rate 1kbps

RMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~2.25	[Source 22]:4374.1~5128.6, [Source 5]:2936.1~4299.7, [Source 1]:3299.6~4366.3, [Source 19]:2964.8~3380.3	[Source 22]:2876.9~4212.9, [Source 5]:1981.3~3347.5, [Source 1]:2335.2~3651.4, [Source 19]:2793.3~3128.2
38	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6, [Source 19]:1.21~2.25	[Source 22]:5892.7~6909.1, [Source 5]:3955.5~5792.4, [Source 1]:4445.1~5882.2, [Source 4]:4511.8~5264.9, [Source 19]:3994.2~4553.9	[Source 22]:3875.7~5675.5, [Source 5]:2669.1~4509.7, [Source 1]:3145.9~4919.1, [Source 19]:3763.1~4214.4
43	[Source 22]:1.14~1.16, [Source	[Source 22]:7938.4~9307.8,	[Source 22]:5221.2~7645.9,

	5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6, [Source 19]:1.21~2.25	[Source 5]:5328.7~7803.3, [Source 1]:5988.3~6132.8, [Source 4]:6078.2~7092.7, [Source 19]:5381.0~6135.0	[Source 5]:3595.7~6075.3, [Source 1]:4238.1~6626.8, [Source 19]:5069.7~5677.6
RMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~2.25	[Source 22]:1328.0~1557.1, [Source 5]:891.4~1305.4, [Source 1]:1001.8~1325.6, [Source 19]:900.1~1026.2	[Source 22]:873.4~1279.1, [Source 5]:601.5~1016.3, [Source 1]:709.0~1108.6, [Source 19]:848.0~949.7
38	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6, [Source 19]:1.21~2.25	[Source 22]:1789.1~2097.7, [Source 5]:1200.9~1758.6, [Source 1]:1349.6~1785.9, [Source 4]:1369.8~1598.5, [Source 19]:1212.6~1382.5	[Source 22]:1176.7~1723.1, [Source 5]:810.4~1016.3, [Source 1]:955.1~1493.5, [Source 19]:1142.4~1279.4
43	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6, [Source 19]:1.21~2.25	[Source 22]:2410.2~2825.9, [Source 1]:1818.1~2405.9, [Source 4]:1845.4~2153.4, [Source 19]:1633.6~1862.4	[Source 22]:1585.2~2321.4, [Source 5]:1091.7~1844.5, [Source 1]:1286.7~2012.0, [Source 19]:1539.0~1723.6
RMa NLOS, Penetration margin 80 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 19]:1.21~2.25	[Source 19]:25.2~28.7	[Source 19]:23.7~26.6
38	[Source 19]:1.21~2.25	[Source 19]:33.9~38.7	[Source 19]:32.0~35.8
43	[Source 19]:1.21~2.25	[Source 19]:45.7~52.1	[Source 19]:43.1~48.2

Table 7.3.3.2.2-2: R2D - Device C, Reference data rate 5~7kbps

RMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~14.42, [Source 8]:14, [Source 19]:3.59~13.46, [Source 18]:7	[Source 12]:2726.4~4552.2, [Source 21]:1511.4~2000.0, [Source 22]:3145.9~3280.0, [Source 5]:1570.3~2733.5, [Source 1]:1976.3, [Source 17]:1828.8, [Source 9]:1429.0~1675.4, [Source 8]:2593.7, [Source 19]:1941.8~2835.2, [Source 18]:1566.4	[Source 12]:1294.0~2599.5, [Source 21]:852.9~1095.5, [Source 22]:1918.3~2335.2, [Source 5]:553.4~1506.2, [Source 1]:1142.1, [Source 9]:769.8~963.0, [Source 8]:1397.3, [Source 19]:1807.8~2554.4, [Source 18]:739.2
38	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 6]:7, [Source	[Source 12]:3673.1~6132.7, [Source 21]:2036.1~2694.3, [Source 22]:4238.1~4418.7,	[Source 12]:1743.5~3502.1, [Source 21]:1149.0~1475.8, [Source 22]:2584.2~3145.9,

	5]:4.12~6.6, [Source 1]:5.25, [Source 9]:11.53~14.42, [Source 8]:14, [Source 19]:3.59~13.46, [Source 18]:7	[Source 6]:3124.5~3422.8, [Source 5]:2115.5~3682.5, [Source 1]:2662.4, [Source 9]:1918.4~2249.2, [Source 8]:3482.0, [Source 19]:2616.0~3819.6, [Source 18]:2110.2	[Source 5]:745.5~2029.1, [Source 1]:1538.6, [Source 9]:1033.5~1292.8, [Source 8]:1875.9, [Source 19]:2435.4~3441.2, [Source 18]:995.8
43	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 6]:7, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 8]:14, [Source 19]:3.59~13.46	[Source 12]:4948.4~8261.8, [Source 21]:2742.9~3629.7, [Source 22]:5709.5~5952.7, [Source 6]:4209.3~4611.3, [Source 5]:2849.9~4960.9, [Source 1]:3586.7, [Source 17]:3319.1, [Source 8]:4674.7, [Source 19]:3524.3~5145.8	[Source 12]:2348.9~4718.0, [Source 21]:1547.8~1988.1, [Source 22]:3481.4~4238.1, [Source 5]:1004.3~2733.5, [Source 1]:2072.8, [Source 8]:2518.4, [Source 19]:3281.0~4636.1
RMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 16]:9.33~13.33, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~14.42, [Source 8]:14, [Source 19]:3.59~13.46, [Source 18]:7	[Source 12]:827.2~1381.7, [Source 21]:458.9~607.2, [Source 22]:955.1~995.8, [Source 16]:842.8~1069.7, [Source 5]:476.8~829.9, [Source 1]:600.0, [Source 17]:555.3, [Source 9]:439.9~515.7, [Source 8]:798.4, [Source 19]:589.5~860.7, [Source 18]:475.6	[Source 12]:391.6~788.6, [Source 21]:258.9~332.6, [Source 22]:582.4~709.0, [Source 5]:168.0~457.3, [Source 1]:346.8, [Source 9]:237.0~296.4, [Source 8]:430.1, [Source 19]:548.8~775.4, [Source 18]:224.4
38	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 6]:7, [Source 5]:4.12~6.6, [Source 14]:4.8~6.67, [Source 1]:5.25, [Source 9]:11.53~14.42, [Source 8]:14, [Source 19]:3.59~13.46, [Source 18]:7	[Source 12]:827.2~1381.7, [Source 21]:618.2~818.0, [Source 22]:1286.7~1341.5, [Source 6]:948.5~1039.1, [Source 5]:642.3~1118.0, [Source 14]:635.7~754.3, [Source 1]:808.3, [Source 9]:590.6~692.4, [Source 8]:1071.9, [Source 19]:794.2~1159.5, [Source 18]:640.7	[Source 12]:391.6~788.6, [Source 21]:348.8~448.1, [Source 22]:784.6~955.1, [Source 5]:226.3~616.0, [Source 14]:529.4~694.4, [Source 1]:467.1, [Source 9]:318.1~398.0, [Source 8]:577.4, [Source 19]:739.3~1044.7, [Source 18]:302.3
43	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 6]:7, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 8]:14, [Source 19]:3.59~13.46	[Source 12]:1502.0~2508.1, [Source 21]:832.8~1102.0, [Source 22]:1733.4~1807.3, [Source 6]:1277.9~1399.9, [Source 5]:865.3~1506.2, [Source 1]:1089.0, [Source 17]:1007.7, [Source 8]:1439.0, [Source 19]:1069.9~1562.1	[Source 12]:712.4~1432.1, [Source 21]:469.9~603.6, [Source 22]:1057.0~1286.7, [Source 5]:304.9~829.9, [Source 1]:629.3, [Source 8]:775.2, [Source 19]:996.0~1407.4
RMa NLOS, Penetration margin 80 dB			

BS Tx power (dBm)	Achievable cell edge data rate (kbps)	Achievable maximum distance at BLER 10% (meter)	Achievable maximum distance at BLER 1% (meter)
33	[Source 21]:10.16~14.3, [Source 19]:3.59~13.46	[Source 21]:12.8~17.0, [Source 19]:16.5~24.1	[Source 21]:7.2~9.3, [Source 19]:15.4~21.7
38	[Source 21]:10.16~14.3, [Source 19]:3.59~13.46	[Source 21]:17.3~22.9, [Source 19]:22.2~32.4	[Source 21]:9.8~12.5, [Source 19]:20.7~29.2
43	[Source 21]:10.16~14.3, [Source 19]:3.59~13.46	[Source 21]:23.3~30.8, [Source 19]:29.9~43.7	[Source 21]:13.2~16.9, [Source 19]:27.9~39.4

Table 7.3.3.2.2-3: R2D - Device C, Reference data rate 48~60kbps

RMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:1407.0~1723.0, [Source 8]:1301.1~1898.1	[Source 8]:577.4
38	[Source 8]:42~84	[Source 8]:1746.8~2548.2	[Source 8]:775.2
43	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:2553.5~3127.1, [Source 8]:2345.1~3421.1	[Source 8]:1040.8
RMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84, [Source 8]:84	[Source 17]:427.2~523.1, [Source 8]:400.5	
38	[Source 8]:42~84	[Source 8]:537.7~784.4	[Source 8]:238.6
43	[Source 17]:42~84, [Source 8]:42~84	[Source 17]:775.3~949.4, [Source 8]:721.9~1053.1	[Source 8]:320.4

Table 7.3.3.2.2-4: R2D - Device C, Reference data rate other value of 168kbps

RMa NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:168, [Source 8]:168	[Source 17]:1135.3, [Source 8]:772.0	
38	[Source 8]:168	[Source 8]:1036.4	
43	[Source 17]:168, [Source 8]:168	[Source 17]:2060.4, [Source 8]:1391.4	
RMa NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:168, [Source 8]:168	[Source 17]:344.7, [Source 8]:237.6	

38	[Source 8]:168	[Source 8]:319.0	
43	[Source 17]:168, [Source 8]:168	[Source 17]:625.5, [Source 8]:428.3	

7.3.3.2.3 UMi NLOS

Table 7.3.3.2.3-1: R2D - Device C, Reference data rate 1kbps

UMi NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~2.25	[Source 22]:2883.1~3431.6, [Source 5]:1863.8~2829.4, [Source 1]:2117.7~2877.4, [Source 19]:1883.1~2173.7	[Source 22]:1822.7~2767, [Source 5]:1211.8~2151.4, [Source 1]:1450.6~2366, [Source 19]:1764.2~1996.9
38	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~2.25	[Source 22]:3994.8~4754.8, [Source 5]:2582.5~3920.5, [Source 1]:2934.3~3987, [Source 19]:2609.2~3011.9	[Source 22]:2525.5~3834, [Source 5]:1679.1~2981.0, [Source 1]:2010~3278.4, [Source 19]:2444.5~2767.0
43	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~2.25	[Source 22]:5535.3~6588.4, [Source 5]:3578.3~5432.3, [Source 1]:4065.8~5524.5, [Source 19]:3615.4~4173.3	[Source 22]:3499.4~5312.4, [Source 5]:2326.5~4130.5, [Source 1]:2785.1~4542.6, [Source 19]:3387.1~3834.0
UMi NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~2.25	[Source 22]:782.1~930.9, [Source 5]:505.6~767.6, [Source 1]:574.5~780.6, [Source 19]:510.9~589.7	[Source 22]:494.5~750.6, [Source 1]:393.5~641.9, [Source 19]:478.6~541.7
38	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~2.25	[Source 22]:1083.7~1289.9, [Source 5]:700.6~1063.6, [Source 1]:796~1081.6, [Source 19]:707.8~817.1	[Source 22]:685.1~1040.1, [Source 5]:455.5~808.7, [Source 1]:545.3~889.4, [Source 19]:663.1~750.6
43	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~2.25	[Source 22]:1501.6~1787.3, [Source 5]:970.7~1473.7, [Source 1]:1103~1498.7, [Source 19]:980.8~1132.2	[Source 22]:949.3~1441.2, [Source 5]:631.2~1120.5, [Source 1]:755.6~1232.3, [Source 19]:918.9~1040.1
UMi NLOS, Penetration margin 80 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 19]:1.21~2.25	[Source 19]:10.2~11.8	[Source 19]:9.6~10.8
38	[Source 19]:1.21~2.25	[Source 19]:14.1~16.3	[Source 19]:13.2~15.0
43	[Source 19]:1.21~2.25	[Source 19]:18.3~20.8	[Source 19]:19.6~22.6

Table 7.3.3.2.3-2: R2D - Device C, Reference data rate 5~7kbps

UMi NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~14.42, [Source 19]:3.59~13.46, [Source 18]:7	[Source 21]:901.1~1224.3, [Source 22]:2010~2103.9, [Source 5]:939.6~1723.4, [Source 1]:1208.5~1208.5, [Source 17]:974.4~974.4, [Source 9]:1775.7~2117.7, [Source 19]:1185.0~1793.2, [Source 18]:937.0	[Source 21]:481.7~633.5, [Source 22]:1169.7~1450.6, [Source 5]:300.0~897.7, [Source 1]:663.1~663.1, [Source 9]:895.2~1147, [Source 19]:1095.8~1599.7, [Source 18]:411.9
38	[Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 10]:6.2~12, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 9]:11.53~14.42, [Source 19]:3.59~13.46, [Source 18]:7	[Source 21]:1248.5~1696.5, [Source 22]:2785.1~2915.2, [Source 10]:2428~2542, [Source 5]:1301.9~2388.0, [Source 1]:1674.5~1674.5, [Source 9]:2460.5~2934.3, [Source 19]:1642.0~2484.6, [Source 18]:1298.4	[Source 21]:667.5~877.9, [Source 22]:1620.7~2010, [Source 5]:415.8~1243.8, [Source 1]:918.9~918.9, [Source 9]:1240.4~1589.3, [Source 19]:1518.4~2216.6, [Source 18]:570.8
43	[Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 19]:3.59~13.46	[Source 21]:1730~2350.6, [Source 22]:3859.1~4039.4, [Source 5]:1804.0~3308.9, [Source 1]:2320.2~2320.2, [Source 17]:1870.7~1870.7, [Source 19]:2275.2~3442.8	[Source 21]:924.9~1216.4, [Source 22]:2245.7~2785.1, [Source 5]:576.1~1723.4, [Source 1]:1273.2~1273.2, [Source 19]:2103.9~3071.4
UMi NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 16]:9.33~13.33, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 17]:14, [Source 9]:11.53~14.42, [Source 19]:3.59~13.46, [Source 18]:7	[Source 21]:244.4~332.1, [Source 22]:545.3~570.8, [Source 16]:495.1~642.7, [Source 5]:254.9~467.5, [Source 1]:327.8~327.8, [Source 17]:264.3~264.3, [Source 9]:481.7~574.5, [Source 19]:321.5~486.5, [Source 18]:254.2	[Source 21]:130.7~171.9, [Source 22]:317.3~393.5, [Source 5]:81.4~243.5, [Source 1]:179.9~179.9, [Source 9]:242.9~311.2, [Source 19]:297.3~434.0, [Source 18]:111.7
38	[Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 5]:4.12~6.6, [Source 14]:4.8~6.67, [Source 1]:5.25, [Source 9]:11.53~14.42, [Source 19]:3.59~13.46, [Source	[Source 21]:338.7~460.2, [Source 22]:755.6~790.9, [Source 5]:353.2~647.8, [Source 14]:349.3~421.2, [Source 1]:454.3~454.3, [Source	[Source 21]:181.1~238.2, [Source 22]:439.7~545.3, [Source 5]:112.8~337.4, [Source 14]:285.9~384.7, [Source 1]:249.3~249.3, [Source

	18]:7	9]:667.5~796, [Source 19]:445.5~674.0, [Source 18]:352.2	9]:336.5~431.2, [Source 19]:411.9~601.3, [Source 18]:154.8
43	[Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 5]:4.12~6.6, [Source 1]:5.25, [Source 19]:3.59~13.46	[Source 21]:469.3~637.7, [Source 22]:1046.9~1095.8, [Source 5]:489.4~897.7, [Source 1]:629.4~629.4, [Source 17]:507.5~507.5, [Source 19]:617.2~934.0	[Source 21]:250.9~330, [Source 22]:609.2~755.6, [Source 5]:156.3~467.5, [Source 1]:345.4~345.4, [Source 19]:570.8~833.2
UMi NLOS, Penetration margin 80 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 21]:10.16~14.3, [Source 19]:3.59~13.46	[Source 21]:4.9~6.6, [Source 19]:6.4~9.7	[Source 21]:2.6~3.4, [Source 19]:5.9~8.7
38	[Source 21]:10.16~14.3, [Source 19]:3.59~13.46	[Source 21]:6.8~9.2, [Source 19]:8.9~13.5	[Source 21]:3.6~4.8, [Source 19]:8.2~12.0
43	[Source 21]:10.16~14.3, [Source 19]:3.59~13.46	[Source 21]:9.4~12.7, [Source 19]:12.3~18.6	[Source 21]:5~6.6, [Source 19]:11.4~16.6

Table 7.3.3.2.3-3: R2D - Device C, Reference data rate 48~60kbps

UMi NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84,	[Source 17]:731.3~912.8,	
38			
43	[Source 17]:42~84,	[Source 17]:1404~1752.6,	
UMi NLOS, Penetration margin 20 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:42~84,	[Source 17]:198.4~247.6,	
38			
43	[Source 17]:42~84,	[Source 17]:380.9~475.5,	

Table 7.3.3.2.3-4: R2D - Device C, Reference data rate other value of 168kbps

UMi NLOS, Penetration margin 0 dB			
BS Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
33	[Source 17]:168	[Source 17]:578.2~578.2,	
38			
43	[Source 17]:168	[Source 17]:1110.1~1110.1,	
UMi NLOS, Penetration margin 20 dB			

BS Tx power (dBm)	Achievable cell edge data rate (kbps)	Achievable maximum distance at BLER 10% (meter)	Achievable maximum distance at BLER 1% (meter)
33	[Source 17]:168	[Source 17]:156.9~156.9,	
38			
43	[Source 17]:168	[Source 17]:301.2~301.2,	

7.3.4 Link budget for D2R

7.3.4.1 Device 2b

7.3.4.1.1 UMa NLOS

Table 7.3.4.1.1-1: D2R - Device 2b, Reference data rate 0.1kbps

UMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:0.077~0.391, [Source 5]:0.1~0.17, [Source 3]:0.28~0.38,	[Source 12]:274.3~366.7, [Source 5]:99.1~130, [Source 3]:76.6~81,	[Source 12]:212.4~298, [Source 5]:76~102.1, [Source 3]:34~43.4,
-10	[Source 12]:0.077~0.391, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104, [Source 3]:0.28~0.38	[Source 12]:495.7~661.2, [Source 5]:178.7~234.3, [Source 4]:477.9~531.4, [Source 3]:138.1~146,	[Source 12]:384.5~538.4, [Source 5]:137~184, [Source 3]:61.2~78.2,
UMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:0.077~0.391, [Source 5]:0.1~0.17, [Source 3]:0.28~0.38,	[Source 12]:81.4~110.6, [Source 5]:30.5~40, [Source 3]:23.6~24.9,	[Source 12]:61.4~89, [Source 5]:23.4~31.4, [Source 3]:10.4~13.4,
-10	[Source 12]:0.077~0.391, [Source 5]:0.1~0.17, [Source 3]:0.28~0.38, [Source 4]:0.077~0.104	[Source 12]:150.9~202.5, [Source 5]:55~72.1, [Source 4]:147.1~163.5, [Source 3]:42.5~44.9,	[Source 12]:116.2~164.2, [Source 5]:42.2~56.6, [Source 3]:18.8~24.1,

Table 7.3.4.1.1-2: D2R - Device 2b, Reference data rate 1kbps

UMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:0.782~0.962, [Source 21]:0.44~1.12, [Source 5]:0.78~1.4,	[Source 12]:221.4~282.6, [Source 21]:57.7~163.8, [Source	[Source 12]:201.3, [Source 21]:20.6~113, [Source

	[Source 1]:0.54~0.96, [Source 9]:0.76~0.84, [Source 3]:0.57~1.14, [Source 17]:1.67~2.5, [Source 18]:1.17	5]:48.9~62.2, [Source 1]:163.5~235.7, [Source 9]:380.9~428.5, [Source 3]:38.5~69.1, [Source 17]:115.7~264.2, [Source 18]:446.0	5]:31.8~41.7, [Source 1]:121.8~201, [Source 9]:300.9~317.3, [Source 3]:14.2~28.3, [Source 18]:228.8
-10	[Source 12]:0.782~0.962, [Source 21]:0.44~1.12, [Source 22]:0.9~1.03, [Source 10]:0.98~1.12 [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 9]:0.76~0.84, [Source 3]:0.57~1.14, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 12]:400.7~510.5, [Source 21]:104.1~295.2, [Source 22]:643.1~662.3, [Source 10]:162~180.8, [Source 5]:88.1~112.2, [Source 1]:294.8~424.8, [Source 4]:269.9, [Source 9]:686.5~772.4, [Source 3]:69.3~124.5, [Source 17]:208.6~476.2, [Source 18]:809.5	[Source 12]:364.5, [Source 21]:37.1~203.7, [Source 22]:532.6~595.7, [Source 5]:57.3~75.1, [Source 1]:219.6~362.3, [Source 9]:542.4~571.9, [Source 3]:25.5~51.1, [Source 18]:415.3
UMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:0.782~0.962, [Source 21]:0.44~1.12, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~0.84, [Source 3]:0.57~1.14, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 12]:64.4~84, [Source 21]:17.8~50.4, [Source 5]:15~19.2, [Source 1]:50.3~72.5, [Source 9]:117.2~131.9, [Source 3]:11.8~21.3, [Source 17]:42.5~81.3, [Source 18]:135.4	[Source 12]:57.8, [Source 21]:6.3~34.8, [Source 5]:9.8~12.8, [Source 1]:37.5~61.9, [Source 9]:92.6~97.6, [Source 3]:4.4~8.7, [Source 18]:69.5
-10	[Source 12]:0.782~0.962, [Source 21]:0.44~1.12, [Source 22]:0.9~1.03, [Source 16]:0.53~0.71, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 9]:0.76~0.84[Source 3]:0.57~1.14, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 12]:121.3~155.5, [Source 21]:32~90.9, [Source 22]:197.9~203.8, [Source 16]:136.6~156.4, [Source 5]:27.1~34.5, [Source 1]:90.7~130.7, [Source 4]:83.1, [Source 9]:211.3~237.7, [Source 3]:21.3~38.3, [Source 17]:76.7~146.6, [Source 18]:245.8	[Source 12]:109.9, [Source 21]:11.4~62.7, [Source 22]:163.9~183.3, [Source 5]:17.6~23.1, [Source 1]:67.6~111.5, [Source 9]:166.9~176, [Source 3]:7.9~15.7, [Source 18]:126.1
UMa NLOS, Penetration margin 80 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 21]:0.44~1.12	[Source 21]:0.5~1.5,	[Source 21]:0.2~1,
-10	[Source 21]:0.44~1.12	[Source 21]:0.9~2.6,	[Source 21]:0.3~1.8,

Table 7.3.4.1.1-3: D2R - Device 2b, Reference data rate 5~7kbps

UMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:7.7~7.83, [Source	[Source 12]:160.3~162.2, [Source	[Source 12]:111.3~114.8, [Source

	6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 2]:5.37~10.2, [Source 17]:5, [Source 8]:5	6]:58.2~85.8, [Source 5]:38.8~56.6, [Source 1]:108.9~108.9, [Source 17]:94.2~174.8, [Source 8]:114.4~273.8	5]:26.6~38.8, [Source 1]:56~56, [Source 2]:121.4~185.5, [Source 8]:74.4~198.0
-10	[Source 12]:7.7~7.83, [Source 22]:3.62~4.1, [Source 6]:3.86~5.49, [Source 11]:11~14, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 8]:5	[Source 22]:470.3~481.5, [Source 6]:104.9~154.7, [Source 11]:523~681.7, [Source 5]:70~101.9, [Source 1]:196.3~196.3, [Source 17]:169.7~315.1, [Source 8]:206.1~493.6	[Source 12]:203.7~209.9, [Source 22]:380.4~394.1, [Source 11]:244.6~378.2, [Source 5]:48~69.9, [Source 1]:100.9~100.9, [Source 8]:134.1~357.0
UMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:7.7~7.83, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 2]:5.37~10.2, [Source 17]:5, [Source 8]:5	[Source 12]:44~44.6, [Source 6]:17.9~26.4, [Source 5]:12~17.4, [Source 1]:33.5~33.5, [Source 17]:29.0~53.8, [Source 8]:35.2~84.3	[Source 12]:26~27.3, [Source 5]:8.2~11.9, [Source 1]:17.2~17.2, [Source 2]:37.4~57.1, [Source 8]:22.9~60.9
-10	[Source 12]:7.7~7.83, [Source 22]:3.62~4.1, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 14]:3.5~4.2, [Source 1]:5.25, [Source 17]:5, [Source 8]:5	[Source 12]:86.7~87.8, [Source 22]:144.8~148.2, [Source 6]:32.3~47.6, [Source 5]:21.5~31.4, [Source 14]:85.7~94.5, [Source 1]:60.4~60.4, [Source 17]:52.2~97.0, [Source 8]:63.4~151.9	[Source 12]:58.6~60.6, [Source 22]:117.1~121.3, [Source 5]:14.8~21.5, [Source 14]:68.9~69.5, [Source 1]:31~31, [Source 8]:41.3~109.9

Table 7.3.4.1.1-4: D2R - Device 2b, Reference data rate other value of 20kbps

UMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 8]:20	[Source 8]:62.3~103.9	[Source 8]:42.2~97
-10	[Source 8]:20	[Source 8]:112.2~241.7	[Source 8]:76.1~174.8
UMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 8]:20	[Source 8]:19.2~41.3	[Source 8]:13.0~29.8
-10	[Source 8]:20	[Source 8]:34.5~74.4	[Source 8]:23.4~53.8

7.3.4.1.2 RMa NLOS

Table 7.3.4.1.2-1: D2R - Device 2b, Reference data rate 0.1kbps

RMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:0.077~0.391, [Source 5]:0.1~0.17, [Source 3]:0.28~0.38	[Source 12]:469.4~629.3, [Source 5]:167.4~220.3, [Source 3]:68.1~72,	[Source 12]:362.7~510.5, [Source 5]:128~172.5, [Source 3]:30.2~38.6,
-10	[Source 12]:0.077~0.391, [Source 5]:0.1~0.17, [Source 3]:0.28~0.38, [Source 4]:0.077~0.104	[Source 12]:853.5~1143.3, [Source 5]:303.9~399.7, [Source 4]:822.2~915.3, [Source 3]:122.7~129.8,	[Source 12]:660.2~927.8, [Source 5]:232.4~313.1, [Source 3]:54.4~69.5,
RMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:0.077~0.391, [Source 5]:0.1~0.17, [Source 3]:0.28~0.38,	[Source 12]:138.9~188.4, [Source 5]:50.8~66.9, [Source 3]:21~22.2,	[Source 12]:105.4~151.7, [Source 5]:38.9~52.4, [Source 3]:9.3~11.9,
-10	[Source 12]:0.077~0.391, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104, [Source 3]:0.28~0.38	[Source 12]:257.1~345.6, [Source 5]:92.3~121.4, [Source 4]:249.6~277.9, [Source 3]:37.8~39.9,	[Source 12]:197.9~279.9, [Source 5]:70.6~95, [Source 3]:16.7~21.4,

Table 7.3.4.1.2-2: D2R - Device 2b, Reference data rate 1kbps

RMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:0.782~0.962, [Source 21]:0.44~1.12, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~0.84, [Source 3]:0.57~1.14, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 12]:378.3~483.7, [Source 21]:96.9~278.3, [Source 5]:81.9~104.6, [Source 1]:277.9~402.1, [Source 9]:338.1~380.4, [Source 3]:34.2~61.4, [Source 17]:264.0~508.5, [Source 18]:395.9	[Source 12]:343.6, [Source 21]:34.2~191.2, [Source 5]:53~69.7, [Source 1]:206.3~342.3, [Source 9]:267.1~281.7, [Source 3]:12.6~25.2, [Source 18]:203.1
-10	[Source 12]:0.782~0.962, [Source 21]:0.44~1.12, [Source 22]:0.9~1.03, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 9]:0.76~0.84, [Source 3]:0.57~1.14, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 12]:688.4~879.3, [Source 21]:175.9~505.1, [Source 22]:1110.2~1143.8, [Source 5]:148.6~189.8, [Source 1]:504.3~729.8, [Source 4]:461.2, [Source 9]:609.4~685.6, [Source 3]:61.6~110.7, [Source 17]:479.1~922.9, [Source 18]:718.5	[Source 12]: 625.6, [Source 21]:62~347, [Source 22]:917.4~1027.4, [Source 5]:96.2~126.5, [Source 1]:374.4~621.3, [Source 9]:481.5~507.7, [Source 3]:22.7~45.4, [Source 18]:368.6

RMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:0.782~0.962, [Source 21]:0.44~1.12, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~0.84, [Source 3]:0.57~1.14, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 12]:110.3~143.3, [Source 21]:29.4~84.5, [Source 5]:24.9~31.7, [Source 1]:84.4~122.1, [Source 9]:104.1~117.1, [Source 3]:10.5~18.9, [Source 17]:80.2~154.4, [Source 18]:120.2	[Source 12]:99.3, [Source 21]:10.4~58, [Source 5]:16.1~21.2, [Source 1]:62.6~103.9, [Source 9]:82.2~86.7, [Source 3]:3.9~7.8, [Source 18]:61.7
-10	[Source 12]:0.782~0.962, [Source 21]:0.44~1.12, [Source 22]:0.9~1.03, [Source 16]:0.53~0.71 [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 9]:0.76~0.84, [Source 3]:0.57~1.14, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 12]:206.5~265.1, [Source 21]:53.4~153.4, [Source 22]:337.1~347.3, [Source 16]:231.6~265.6, [Source 5]:45.1~57.6, [Source 1]:153.1~221.6, [Source 4]:140.0, [Source 9]:187.6~211, [Source 3]:19~34.1, [Source 17]:145.5~280.2, [Source 18]:218.2	[Source 12]:187.2, [Source 21]:18.8~105.3, [Source 22]:278.5~311.9, [Source 5]:29.2~38.4, [Source 1]:113.7~188.6, [Source 9]:148.2~156.3, [Source 3]:7~14, [Source 18]:111.9
RMa NLOS, Penetration margin 80 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 21]:0.44~1.12	[Source 21]:0.8~2.4,	[Source 21]:0.3~1.6,
-10	[Source 21]:0.44~1.12	[Source 21]:1.5~4.3,	[Source 21]:0.5~2.9,

Table 7.3.4.1.2-3: D2R - Device 2b, Reference data rate 5~7kbps

RMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 12]:7.7~7.83, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 8]:5	[Source 12]:273.2~276.5, [Source 6]:97.7~144.8, [Source 5]:64.9~94.9, [Source 1]:184.2, [Source 17]:179.1~334.9, [Source 8]:159.1~380.8	[Source 12]:189.6~195.5, [Source 5]:44.3~64.8, [Source 1]:93.9, [Source 8]:103.5~275.4
-10	[Source 12]:7.7~7.83, [Source 22]:3.62~4.1, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 8]:5	[Source 12]:498.4~504.4, [Source 22]:809.0~828.5, [Source 6]:177.3~262.7, [Source 5]:117.8~172.3, [Source 1]:334.3, [Source 17]:325.0~607.7, [Source 8]:286.7~686.3	[Source 12]:347.7~358.3, [Source 22]:652.8~676.6, [Source 5]:80.4~117.6, [Source 1]:170.5, [Source 8]:186.5~496.4
RMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**

-20	[Source 12]:7.7~7.83, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 8]:5	[Source 12]:76.6~77.6, [Source 6]:29.7~43.9, [Source 5]:19.7~28.8, [Source 1]:55.9, [Source 17]:54.4~101.7, [Source 8]:49.0~117.2	[Source 12]:47.9~50, [Source 5]:13.5~19.7, [Source 1]:28.5, [Source 8]:31.9~84.8
-10	[Source 12]:7.7~7.83, [Source 22]:3.62~4.1, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 14]:3.5~4.2, [Source 1]:5.25, [Source 17]:5, [Source 8]:5	[Source 12]:147.9~149.7, [Source 22]:245.6~251.5, [Source 6]:53.8~79.8, [Source 5]:35.8~52.3, [Source 14]:146.2~161.3, [Source 1]:101.5, [Source 17]:98.7~184.5, [Source 8]:88.3~211.3	[Source 12]:100.6~104, [Source 22]:198.2~205.4, [Source 5]:24.4~35.7, [Source 14]:117.2~118.2, [Source 1]:51.8, [Source 8]:57.4~152.8

Table 7.3.4.1.2-4: D2R - Device 2b, Reference data rate other value of 20kbps

RMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 8]:20	[Source 8]:86.6~186.5	[Source 8]:58.7~134.9
-10	[Source 8]:20	[Source 8]:156.1~336.1	[Source 8]:105.8~243.1
RMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 8]:20	[Source 8]:26.7~57.4	[Source 8]:18.1~41.5
-10	[Source 8]:20	[Source 8]:48.0~103.5	[Source 8]:32.6~74.8

7.3.4.1.3 UMi NLOS

Table 7.3.4.1.3-1: D2R - Device 2b, Reference data rate 0.1kbps

UMi NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 5]:0.1~0.17	[Source 5]:81.1~109.5	[Source 5]:60.5~83.8
-10	[Source 5]:0.1~0.17	[Source 5]:155.7~210.2	[Source 5]:116.1~160.9
UMi NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 5]:0.1~0.17	[Source 5]:22.0~29.7	[Source 5]:16.4~22.7
-10	[Source 5]:0.1~0.17	[Source 5]:42.2~57.0	[Source 5]:31.5~43.6

Table 7.3.4.1.3-2: D2R - Device 2b, Reference data rate 1kbps

UMi NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 21]:0.44~1.12, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~0.84, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 21]:44.6~141.4, [Source 5]:37.1~48.4, [Source 1]:141.2~211.6, [Source 9]:360~410.1, [Source 17]:117.2~240.1, [Source 18]:400.2	[Source 21]:14.2~93.8, [Source 5]:23.0~30.1, [Source 1]:101.9~177.4, [Source 9]:277.3~294.1, [Source 18]:205.3
-10	[Source 21]:0.44~1.12, [Source 22]:0.9~1.03, [Source 10]:0.98~1.12, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~0.84, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 21]:85.6~271.5, [Source 22]:642.9~664.2, [Source 10]:141.4~159, [Source 5]:71.2~93.0, [Source 1]:271.1~406.2, [Source 9]:691.1~787.4, [Source 17]:224.9~461.0, [Source 18]:726.3	[Source 21]:27.3~180, [Source 22]:521.8~590.6, [Source 5]:44.2~57.8, [Source 1]:195.6~340.6, [Source 9]:532.4~564.6, [Source 18]:372.6
UMi NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 21]:0.44~1.12, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~0.84, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 21]:12.1~38.4, [Source 5]:10.1~13.1, [Source 1]:38.3~57.4, [Source 9]:97.7~111.3, [Source 17]:31.8~65.1, [Source 18]:121.5	[Source 21]:3.9~25.4, [Source 5]:6.2~8.2, [Source 1]:27.6~48.1, [Source 9]:75.2~79.8, [Source 18]:62.3
-10	[Source 21]:0.44~1.12, [Source 22]:0.9~1.03, [Source 16]:0.53~0.71, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~0.84, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 21]:23.2~73.7, [Source 22]:174.4~180.2, [Source 16]:115.6~134.4, [Source 5]:19.3~25.2, [Source 1]:73.5~110.2, [Source 9]:187.5~213.6, [Source 17]:61.0~125.1, [Source 18]:220.5	[Source 21]:7.4~48.8, [Source 22]:141.6~160.2, [Source 5]:12.0~15.7, [Source 1]:53.1~92.4, [Source 9]:144.4~153.2, [Source 18]:113.1
UMi NLOS, Penetration margin 80 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 21]:0.44~1.12	[Source 21]:0.2~0.8	[Source 21]:0.1~0.5
-10	[Source 21]:0.44~1.12	[Source 21]:0.5~1.5	[Source 21]:0.1~1

Table 7.3.4.1.3-3: D2R - Device 2b, Reference data rate 5~7kbps

UMi NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5	[Source 5]:28.7~43.6, [Source 1]:90~90, [Source 17]:76.6~152.0	[Source 5]:18.9~28.7, [Source 1]:43.1~43.1,

-10	[Source 22]:3.62~4.1, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5	[Source 22]:454.7~466.7, [Source 5]:55.2~83.7, [Source 1]:172.8~172.8, [Source 17]:147.1~291.8	[Source 22]:359.5~373.9, [Source 5]:36.4~55.1, [Source 1]:82.7~82.7
UMi NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-20	[Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5	[Source 5]:7.8~11.8, [Source 1]:24.4~24.4, [Source 17]:20.8~41.2	[Source 5]:5.1~7.8, [Source 1]:11.7~11.7,
-10	[Source 22]:3.62~4.1, [Source 5]:4.21~5.44, [Source 14]:3.5~4.2, [Source 1]:5.25, [Source 17]:5	[Source 22]:123.3~126.6, [Source 5]:15.0~22.7, [Source 14]:68.3~76, [Source 1]:46.9~46.9, [Source 17]:39.9~79.2	[Source 22]:97.5~101.4, [Source 5]:9.9~15.0, [Source 14]:53.6~54.1, [Source 1]:22.4~22.4,

7.3.4.2 Device C

7.3.4.2.1 UMa NLOS

Table 7.3.4.2.1-1: D2R - Device C, Reference data rate 0.1kbps

UMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 12]:0.15~0.391, [Source 5]:0.1~0.17,	[Source 12]:771.6~1030.1, [Source 5]:1130.6~1572.6,	[Source 12]:538.4~838, [Source 5]:1059.7~1422.7,
0	[Source 12]:0.15~0.391, [Source 22]:0.08~0.11, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104,	[Source 12]:920.9~1229.4, [Source 22]:1858.4~1936.7, [Source 5]:1349.2~1876.6, [Source 4]:2004~2216.4,	[Source 12]:642.7~1000.2, [Source 22]:1622.9~1762.4, [Source 5]:1264.5~1697.8,
5	[Source 12]:0.15~0.391, [Source 22]:0.08~0.11, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104,	[Source 12]:1236.6~1650.7, [Source 22]:2495.1~2600.1, [Source 5]:1811.4~2519.5, [Source 4]:2690.5~2975.7,	[Source 12]:863.1~1343, [Source 22]:2178.9~2366.2, [Source 5]:1697.8~2279.4,
UMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 12]:0.15~0.391, [Source 5]:0.1~0.17	[Source 12]:236.4~316.2, [Source 5]:348~484,	[Source 12]:164.2~256.9, [Source 5]:326.1~437.9,
0	[Source 12]:0.15~0.391, [Source 22]:0.08~0.11, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104, [Source 20]:0.04	[Source 12]:282.6~377.7, [Source 22]:572.0~596.1, [Source 5]:415.2~577.6, [Source 4]:616.8~682.1, [Source	[Source 12]:196.5~307, [Source 22]:499.5~542.4, [Source 5]:389.2~522.5,

		20]:508.8~508.8,	
5	[Source 12]:0.15~0.391, [Source 22]:0.08~0.11, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104,	[Source 12]:379.9~507.5, [Source 22]:767.9~800.3, [Source 5]:557.5~775.4, [Source 4]:828.1~915.8,	[Source 12]:264.7~412.7, [Source 22]:670.6~728.3, [Source 5]:522.5~701.5,

Table 7.3.4.2.1-2: D2R - Device C, Reference data rate 1kbps

UMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 19]:0.57~2.27, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 12]:627.7~790, [Source 21]:166.7~462, [Source 5]:793.9~1016.8, [Source 1]:701.5~1010.9, [Source 19]:380.7~698.4, [Source 9]:977.6~1180.5, [Source 17]:448.7~1024.3, [Source 18]:1228.6	[Source 12]:574.5, [Source 21]:73.5~311.3, [Source 5]:665.3~898.5, [Source 1]:522.5~862.2, [Source 19]:344.4~635.6, [Source 9]:749.9~869, [Source 18]:630.2
0	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 12]:749.2~942.9, [Source 21]:198.9~551.3, [Source 22]:1166.1~1200.9, [Source 5]:947.4~1213.4, [Source 1]:837.2~1206.3, [Source 4]:1132.2~1132.2, [Source 19]:454.3~833.4, [Source 18]:1469.1	[Source 12]:685.8, [Source 21]:87.7~371.5, [Source 22]:971.4~1086.5, [Source 5]:793.9~1072.2, [Source 1]:623.5~1028.9, [Source 19]:411.0~758.5, [Source 18]:753.6
5	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 10]:1.1~1.2, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 12]:1006.1~1266.1, [Source 21]:267.1~740.2, [Source 22]:1565.6~1612.4, [Source 10]:400~443, [Source 5]:1272~1629.2, [Source 1]:1124~1619.6, [Source 4]:1520.1~1520.1, [Source 9]:1566.3~1891.3, [Source 17]:718.9~1641.1, [Source 19]:609.9~1119.0, [Source 18]:1979.2	[Source 12]:920.9, [Source 21]:117.8~498.8, [Source 22]:1304.2~1458.7, [Source 5]:1065.9~1439.6, [Source 1]:837.2~1381.4, [Source 9]:1201.5~1392.2, [Source 19]:551.8~1018.3, [Source 18]:1015.3
UMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source	[Source 12]:191.9~242.1, [Source 21]:51.3~142.2, [Source 5]:244.4~313, [Source	[Source 12]:175.4, [Source 21]:22.6~95.8, [Source 5]:204.8~276.5, [Source

	9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 19]:0.57~2.27, [Source 18]:1.17	1]:215.9~311.1, [Source 9]:300.9~363.3, [Source 17]:164.9~315.3, [Source 19]:117.2~215.0, [Source 18]:373.0	1]:160.8~265.4, [Source 9]:230.8~267.4, [Source 19]:106.0~195.6, [Source 18]:191.3
0	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 12]:229.5~289.3, [Source 21]:61.2~169.7, [Source 22]:358.9~369.6, [Source 5]:291.6~373.5, [Source 1]:257.7~371.3, [Source 4]:348.5~348.5, [Source 19]:139.8~256.5, [Source 18]:446.0	[Source 12]:209.9, [Source 21]:27~114.3, [Source 22]:299~334.4, [Source 5]:244.4~330, [Source 1]:191.9~316.7, [Source 19]:126.5~233.4, [Source 18]:228.8
5	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 16]:0.53~0.71, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 12]:308.8~389, [Source 21]:82.2~227.8, [Source 22]:481.8~496.2, [Source 16]:330.5~378.4, [Source 5]:391.5~501.4, [Source 1]:345.9~498.5, [Source 4]:467.8~467.8, [Source 9]:482.1~582.1, [Source 17]:264.2~505.1, [Source 19]:187.7~344.4, [Source 18]:600.9	[Source 12]:282.6, [Source 21]:36.2~153.5, [Source 22]:401.4~448.9, [Source 5]:328.1~443.1, [Source 1]:257.7~425.2, [Source 9]:369.8~428.5, [Source 19]:169.8~313.4, [Source 18]:308.2
UMa NLOS, Penetration margin 80 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 21]:0.53~1.14, [Source 19]:0.57~2.27	[Source 21]:1.5~4.1, [Source 19]:3.4~6.3	[Source 21]:0.7~2.8, [Source 19]:3.1~5.7
0	[Source 21]:0.53~1.14, [Source 19]:0.57~2.27	[Source 21]:1.8~4.9, [Source 19]:4.1~7.5	[Source 21]:0.8~3.3, [Source 19]:3.7~6.8
5	[Source 21]:0.53~1.14, [Source 19]:0.57~2.27	[Source 21]:2.4~6.6, [Source 19]:5.5~10.0	[Source 21]:1.1~4.5, [Source 19]:5.0~9.1

Table 7.3.4.2.1-3: D2R - Device C, Reference data rate 5~7kbps

UMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 12]:7.7~7.83, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 19]:2.71~5.33, [Source 17]:5, [Source 8]:5, [Source 13]:3.5	[Source 6]:292.3~302.8, [Source 5]:453.6~609, [Source 1]:467.2~467.2, [Source 19]:317.1~443.7, [Source 17]:365.1~677.7, [Source 8]:342.2~763.3, [Source 13]:346.4~617.1	[Source 12]:311.6~321.9, [Source 5]:369.1~498.5, [Source 1]:240.1~240.1, [Source 19]:278.6~399.0, [Source 8]:237.5~548.8, [Source 13]:270.5~479.0
0	[Source 12]:7.7~7.83, [Source	[Source 12]:535.2~538.4, [Source	[Source 12]:372.2~384.5, [Source

	22]:3.92~4.69, [Source 6]:3.86~5.49, [Source 11]:11~14, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 2]:5.37~10.2, [Source 8]:5, [Source 19]:2.71~5.33	22]:857.8~873.1, [Source 6]:348.8~361.4, [Source 11]:942.6~1228.8, [Source 5]:541.3~726.8, [Source 1]:557.5~557.5, [Source 8]:408.3~910.9, [Source 19]:378.4~529.5	22]:693.9~714.6, [Source 1]:286.5~286.5, [Source 11]:498.9~681.7, [Source 5]:440.5~594.8, [Source 1]:286.5~286.5, [Source 2]:459.7~662.3, [Source 8]:283.4~654.9, [Source 19]:332.4~476.2
5	[Source 12]:7.7~7.83, [Source 22]:3.92~4.69, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 15]:4.5, [Source 17]:5, [Source 8]:5, [Source 19]:2.71~5.33, [Source 13]:3.5	[Source 12]:718.9~723.2, [Source 22]:1151.7~1172.3, [Source 6]:468.3~485.2, [Source 5]:726.8~975.8, [Source 1]:748.5~748.5, [Source 15]:1097.7~1097.7, [Source 17]:584.9~1085.8, [Source 8]:548.2~1223.0, [Source 19]:508.1~710.9, [Source 13]:555.0~988.7	[Source 12]:500.1~516.6, [Source 22]:931.6~959.5, [Source 5]:591.3~798.6, [Source 1]:384.6~384.6, [Source 8]:380.4~879.3, [Source 19]:446.3~639.3, [Source 13]:433.4~767.5
UMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 12]:7.7~7.83, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 8]:5, [Source 19]:2.71~5.33, [Source 13]:3.5	[Source 12]:136.2~137, [Source 6]:90~93.2, [Source 5]:139.6~187.4, [Source 1]:143.8~143.8, [Source 17]:112.4~208.6, [Source 8]:105.3~234.9, [Source 19]:97.6~136.6, [Source 13]:106.6~189.9	[Source 12]:93.3~96.5, [Source 5]:113.6~153.4, [Source 1]:73.9~73.9, [Source 8]:73.1~168.9, [Source 19]:85.7~122.8, [Source 13]:83.2~147.4
0	[Source 12]:7.7~7.83, [Source 22]:3.92~4.69, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 14]:3.5~4.2, [Source 1]:5.25, [Source 2]:5.37~10.2, [Source 8]:5, [Source 19]:2.71~5.33	[Source 12]:163.2~164.2, [Source 22]:264~268.7, [Source 6]:107.4~111.2, [Source 5]:166.6~223.7, [Source 14]:163~170.8, [Source 1]:171.6~171.6, [Source 8]:125.7~280.3, [Source 19]:116.5~163.0	[Source 12]:112.3~116.2, [Source 22]:213.6~219.9, [Source 5]:135.6~183.1, [Source 14]:155.4~158, [Source 1]:88.2~88.2, [Source 2]:141.5~203.8, [Source 8]:87.2~201.6, [Source 19]:102.3~146.6
5	[Source 12]:7.7~7.83, [Source 22]:3.92~4.69, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 15]:4.5, [Source 17]:5, [Source 8]:5, [Source 19]:2.71~5.33, [Source 13]:3.5	[Source 12]:220.1~221.4, [Source 22]:354.5~360.8, [Source 6]:144.1~149.3, [Source 5]:223.7~300.3, [Source 1]:230.4~230.4, [Source 15]:337.8, [Source 17]:180.0~334.2, [Source 8]:168.7~376.4, [Source 19]:156.4~218.8, [Source 13]:170.8~304.3	[Source 12]:152.3~157.4, [Source 22]:286.7~295.3, [Source 5]:182~245.8, [Source 1]:118.4~118.4, [Source 8]:117.1~270.6, [Source 19]:137.4~196.8, [Source 13]:133.4~236.2

UMa NLOS, Penetration margin 80 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 19]:2.71~5.33	[Source 19]:2.8~4.0	[Source 19]:2.5~3.6
0	[Source 19]:2.71~5.33	[Source 19]:3.4~4.8	[Source 19]:3.0~4.3
5	[Source 19]:2.71~5.33	[Source 19]:4.6~6.4	[Source 19]:4.0~5.7

Table 7.3.4.2.1-4: D2R - Device C, Reference data rate other value of 20kbps

UMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 8]:20	[Source 8]:193.0~375.9	[Source 8]:137.9~268.7
0	[Source 8]:20	[Source 8]:230.3~448.6	[Source 8]:164.6~320.6
5	[Source 8]:20,	[Source 8]:309.2~602.3	[Source 8]:221.0~430.5,
UMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 8]:20	[Source 8]:59.4~115.7	[Source 8]:42.4~82.7
0	[Source 8]:20	[Source 8]:70.9~138.1	[Source 8]:50.7~98.7
5	[Source 8]:20,	[Source 8]:95.2~185.4	[Source 8]:68.0~132.5,

7.3.4.2.2 RMa NLOS

Table 7.3.4.2.2-1: D2R - Device C, Reference data rate 0.1kbps

RMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 12]:0.15~0.391, [Source 5]:0.1~0.17,	[Source 12]:1335.1~1788.2, [Source 5]:1964.6~2743,	[Source 12]:927.8~1451.3, [Source 5]:1839.9~2478.7,
0	[Source 12]:0.15~0.391, [Source 22]:0.08~0.11, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104,	[Source 12]:1596.6~2138.4, [Source 22]:3247.8~3386.2, [Source 5]:2349.2~3280, [Source 4]:3505.2~3881.3,	[Source 12]:1109.7~1735.6, [Source 22]:2831.8~3078.2, [Source 5]:2200.1~2964,
5	[Source 12]:0.15~0.391, [Source 22]:0.08~0.11, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104,	[Source 12]:2151.1~2880.9, [Source 22]:4375.4~4561.8, [Source 5]:3164.8~4418.7, [Source 4]:4722.1~5228.8,	[Source 12]:1495.3~2338.4, [Source 22]:3814.9~4146.8, [Source 5]:2964~3992.9,
RMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**

-3	[Source 12]:0.15~0.391, [Source 5]:0.1~0.17,	[Source 12]:404.1~542, [Source 5]:596.5~832.8,	[Source 12]:279.9~439.5, [Source 5]:558.6~752.5,
0	[Source 12]:0.15~0.391, [Source 22]:0.08~0.11, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104,	[Source 12]:483.7~648.4, [Source 22]:986.1~1028.1, [Source 5]:713.2~995.8, [Source 4]:1064.2~1178.4,	[Source 12]:335.4~526, [Source 22]:859.7~934.6, [Source 5]:668~899.9,
5	[Source 12]:0.15~0.391, [Source 22]:0.08~0.11, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104,	[Source 12]:652.3~874.1, [Source 22]:1328.4~1385.0, [Source 5]:960.9~1341.6, [Source 4]:1433.7~1587.5,	[Source 12]:452.9~709.2, [Source 22]:1158.2~1259.0, [Source 5]:899.9~1212.3,

Table 7.3.4.2.2-2: D2R - Device C, Reference data rate 1kbps

RMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 19]:0.57~2.27, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 12]:1083.5~1367.3, [Source 21]:283.3~794.5, [Source 5]:1373.9~1764.7, [Source 1]:1212.3~1754.2, [Source 19]:653.3~1207.1, [Source 9]:867.8~1047.8, [Source 17]:1039.8~2002.9, [Source 18]:1090.5	[Source 12]:990.8, [Source 21]:123.7~532.9, [Source 5]:1149~1557.1, [Source 1]:899.9~1493.5, [Source 19]:590.3~1097.3, [Source 9]:665.7~771.3, [Source 18]:559.4
0	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 12]:1295.9~1635.2, [Source 21]:338.8~950.1, [Source 22]:2026.9~2088.2, [Source 5]:1642.9~2110.2, [Source 1]:1449.6~2097.7, [Source 4]:1967.4~1967.4, [Source 19]:781.2~1443.5, [Source 18]:1304.0	[Source 12]:1185, [Source 21]:148~637.3, [Source 22]:1685.0~1887.0, [Source 5]:1373.9~1862, [Source 1]:1076.1~1785.9, [Source 19]:705.9~1312.2, [Source 18]:668.9
5	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 12]:1746~2203.1, [Source 21]:456.4~1279.9, [Source 22]:2730.6~2813.2, [Source 5]:2213.3~2842.8, [Source 1]:1952.9~2825.9, [Source 4]:2650.4~2650.4, [Source 9]:1390.2~1678.6, [Source 17]:1675.0~3226.5, [Source 19]:1052.5~1944.6, [Source 18]:1756.8	[Source 12]:1596.6, [Source 21]:199.3~858.5, [Source 22]:2270.0~2524.1, [Source 5]:1850.9~2508.4, [Source 1]:1449.6~2405.9, [Source 9]:1066.4~1235.7, [Source 19]:951.0~1767.7, [Source 18]:901.2
RMa NLOS, Penetration margin 20 dB			
Device Tx	**Achievable cell edge data rate**	**Achievable maximum distance**	**Achievable maximum distance**

power (dBm)	(kbps)	at BLER 10% (meter)	at BLER 1% (meter)
-3	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 12]:327.4~413.9, [Source 21]:86~241.2, [Source 5]:417.1~535.8, [Source 1]:368.1~532.6, [Source 9]:267.1~322.5, [Source 17]:315.7~608.1, [Source 19]:198.3~366.4, [Source 18]:331.1	[Source 12]:299.1, [Source 21]:37.6~161.8, [Source 5]:348.8~472.8, [Source 1]:273.2~453.4, [Source 9]:204.9~237.4, [Source 19]:179.2~333.1, [Source 18]:169.8
0	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 12]:392.1~495.4, [Source 21]:102.9~288.4, [Source 22]:615.4~634.0, [Source 5]:498.8~640.7, [Source 1]:440.1~636.9, [Source 4]:597.3~597.3, [Source 19]:237.2~438.2, [Source 18]:395.9	[Source 12]:358.3, [Source 21]:44.9~193.5, [Source 22]:511.6~572.9, [Source 5]:417.1~565.3, [Source 1]:326.7~542.2, [Source 19]:214.3~398.3, [Source 18]:203.1
5	[Source 12]:0.782~0.962, [Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 16]:0.53~0.71, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 12]:529.1~668.1, [Source 21]:138.6~388.6, [Source 22]:829.0~854.1, [Source 16]:566.1~649.3, [Source 5]:672~863.1, [Source 1]:592.9~858, [Source 4]:804.7~804.7, [Source 9]:427.9~516.7, [Source 17]:508.5~979.6, [Source 19]:319.5~590.3, [Source 18]:533.4	[Source 12]:483.7, [Source 21]:60.5~260.6, [Source 22]:689.2~771.8, [Source 5]:562~761.6, [Source 1]:440.1~730.4, [Source 9]:328.3~380.4, [Source 19]:288.7~536.6, [Source 18]:273.6
RMa NLOS, Penetration margin 80 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 21]:0.53~1.14, [Source 19]:0.57~2.27	[Source 21]:2.4~6.8, [Source 19]:5.5~10.3	[Source 21]:1.1~4.5, [Source 19]:5.0~9.3
0	[Source 21]:0.53~1.14, [Source 19]:0.57~2.27	[Source 21]:2.9~8.1, [Source 19]:6.6~12.3	[Source 21]:1.3~5.4, [Source 19]:6.0~11.1
5	[Source 21]:0.53~1.14, [Source 19]:0.57~2.27	[Source 21]:3.9~10.9, [Source 19]:8.9~16.5	[Source 21]:1.7~7.3, [Source 19]:8.1~15.0

Table 7.3.4.2.2-3: D2R - Device C, Reference data rate 5~7kbps

RMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 12]:7.7~7.83, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 19]:2.71~5.33, [Source 17]:5,	[Source 12]:771.1~775.7, [Source 6]:500.1~518.3, [Source 5]:779.9~1050.7, [Source 1]:803.5~803.5, [Source	[Source 12]:533.9~551.8, [Source 5]:633.1~858, [Source 1]:409.7~409.7, [Source 19]:476.3~685.2, [Source

	[Source 8]:5	19]:543.1~762.8, [Source 17]:705.4~1318.9, [Source 8]:475.8~1061.4	8]:330.2~763.1
0	[Source 12]:7.7~7.83, [Source 22]:3.92~4.69, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 8]:5, [Source 19]:2.71~5.33	[Source 12]:922.3~927.8, [Source 22]:1485.8~1512.7, [Source 6]:598.1~619.8, [Source 5]:932.6~1256.4, [Source 1]:960.9~960.9, [Source 8]:567.8~1266.5, [Source 19]:649.4~912.2	[Source 12]:638.8~660.2, [Source 22]:1198.9~1235.2, [Source 5]:757~1026, [Source 1]:490~490, [Source 8]:394.1~910.6, [Source 19]:569.6~819.4
5	[Source 12]:7.7~7.83, [Source 22]:3.92~4.69, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 8]:5, [Source 19]:2.71~5.33	[Source 12]:1242.9~1250.3, [Source 22]:2001.7~2037.8, [Source 6]:805.7~835.1, [Source 5]:1256.4~1692.6, [Source 1]:1294.4~1294.4, [Source 17]:1136.3~2124.6, [Source 8]:762.3~1700.4, [Source 19]:874.9~1228.9	[Source 12]:861.1~889.9, [Source 22]:1615.1~1664.0, [Source 5]:1019.9~1382.1, [Source 1]:660.1~660.1, [Source 8]:529.0~1222.6, [Source 19]:767.4~1103.9
RMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 12]:7.7~7.83, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 8]:5, [Source 19]:2.71~5.33	[Source 12]:231.9~233.3, [Source 6]:151.8~157.4, [Source 5]:236.8~319, [Source 1]:244~244, [Source 17]:214.2~400.4, [Source 8]:146.5~326.7, [Source 19]:164.9~231.6	[Source 12]:158.9~164.5, [Source 5]:192.2~260.5, [Source 1]:124.4~124.4, [Source 8]:101.7~234.9, [Source 19]:144.6~208.0
0	[Source 12]:7.7~7.83, [Source 22]:3.92~4.69, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 14]:3.5~4.2, [Source 1]:5.25, [Source 8]:5, [Source 19]:2.71~5.33	[Source 12]:278.2~279.9, [Source 22]:451.1~459.3, [Source 6]:181.6~188.2, [Source 5]:283.2~381.5, [Source 14]:279.9~293.5, [Source 1]:291.7~291.7, [Source 8]:174.8~389.9, [Source 19]:197.1~276.9	[Source 12]:191.3~197.9, [Source 22]:364.0~375.0, [Source 5]:229.8~311.5, [Source 14]:266.7~271.3, [Source 1]:148.8~148.8, [Source 8]:121.3~280.3, [Source 19]:172.9~248.7
5	[Source 12]:7.7~7.83, [Source 22]:3.92~4.69, [Source 6]:3.86~5.49, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 8]:5, [Source 19]:2.71~5.33	[Source 12]:376~378.3, [Source 22]:607.7~618.7, [Source 6]:244.6~253.5, [Source 5]:381.5~513.9, [Source 1]:393~393, [Source 17]:345.0~645.0, [Source 8]:234.7~523.4, [Source 19]:265.6~373.1	[Source 12]:259.5~268.3, [Source 22]:490.4~505.2, [Source 5]:309.6~419.6, [Source 1]:200.4~200.4, [Source 8]:162.9~376.3, [Source 19]:233.0~335.1
RMa NLOS, Penetration margin 80 dB			
Device Tx	**Achievable cell edge data rate**	**Achievable maximum distance at**	**Achievable maximum distance at**

power (dBm)	(kbps)	BLER 10% (meter)	BLER 1% (meter)
-3	[Source 19]:2.71~5.33	[Source 19]:4.6~6.5	[Source 19]:4.0~5.8
0	[Source 19]:2.71~5.33	[Source 19]:5.5~7.7	[Source 19]:4.8~7.0
5	[Source 19]:2.71~5.33	[Source 19]:7.4~10.4	[Source 19]:6.5~9.4

Table 7.3.4.2.2-4: D2R - Device C, Reference data rate other value of 20kbps

RMa NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 8]:20	[Source 8]:268.4~522.8,	[Source 8]:191.8~373.7,
0	[Source 8]:20	[Source 8]:320.3~623.8,	[Source 8]:228.9~445.9,
5	[Source 8]:20	[Source 8]:430~837.5,	[Source 8]:307.3~598.6,
RMa NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 8]:20	[Source 8]:82.6~160.9,	[Source 8]:59.1~115.0,
0	[Source 8]:20	[Source 8]:98.6~192.0,	[Source 8]:70.5~137.3,
5	[Source 8]:20	[Source 8]:132.4~257.8,	[Source 8]:94.6~184.3,

7.3.4.2.3 UMi NLOS

Table 7.3.4.2.3-1: D2R - Device C, Reference data rate 0.1kbps

UMi NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 5]:0.1~0.17	[Source 5]:1200.6~1730.0	[Source 5]:1117.5~1548.4
0	[Source 22]:0.08~0.11, [Source 5]:0.1~0.17	[Source 22]:2081.3~2178.6, [Source 5]:1460.1~2103.9	[Source 22]:1791.4~1962.7, [Source 5]:1359.0~1883.1
5	[Source 22]:0.08~0.11, [Source 5]:0.1~0.17	[Source 22]:2883.9~3018.7, [Source 5]:2023.2~2915.2	[Source 22]:2482.2~2719.5, [Source 5]:1883.1~2609.3
UMi NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 5]:0.1~0.17	[Source 5]:325.7~469.3	[Source 5]:303.2~420.1
0	[Source 22]:0.08~0.11, [Source 5]:0.1~0.17	[Source 22]:564.6~591.0, [Source 5]:396.1~570.8	[Source 22]:486.0~532.4, [Source 5]:368.7~510.9
5	[Source 22]:0.08~0.11, [Source 5]:0.1~0.17	[Source 22]:782.4~818.9, [Source 5]:548.9~790.9	[Source 22]:673.4~737.8, [Source 5]:510.9~707.9

Table 7.3.4.2.3-2: D2R - Device C, Reference data rate 1kbps

UMi NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 21]:0.53~1.14, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 19]:0.57~2.27, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 21]:144.2~445.8, [Source 5]:811.8~1067.6, [Source 1]:707.9~1060.7, [Source 19]:359.8~704.4, [Source 9]:1022.1~1259.4, [Source 17]:525.2~1076.3, [Source 18]:1102.3	[Source 21]:58.2~287.9, [Source 5]:667.5~930.9, [Source 1]:510.9~889.4, [Source 19]:322.0~634.6, [Source 9]:762.1~897.1, [Source 18]:565.5
0	[Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96,[Source 19]:0.57~2.27, [Source 18]:1.17	[Source 21]:175.4~542.1, [Source 22]:1242.4~1283.6, [Source 5]:987.2~1298.4, [Source 1]:860.9~1289.9, [Source 19]:437.5~856.6, [Source 18]:1318.1	[Source 21]:70.8~350.2, [Source 22]:1014.9~1148.8, [Source 5]:811.8~1132.2, [Source 1]:621.3~1081.6, [Source 19]:391.6~771.7, [Source 18]:676.2
5	[Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 10]:1.1~1.2, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 21]:243~751.2, [Source 22]:1721.5~1778.5, [Source 10]:437~489, [Source 5]:1367.9~1799.0, [Source 1]:1192.8~1787.4, [Source 9]:1722.4~2122.2, [Source 17]:885.0~1813.7, [Source 19]:606.2~1186.9, [Source 18]:1775.7	[Source 21]:98.1~485.2, [Source 22]:1406.3~1591.9, [Source 5]:1124.8~1568.7, [Source 1]:860.9~1498.7, [Source 9]:1284.3~1511.7, [Source 19]:542.6~1069.3, [Source 18]:910.9
UMi NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 21]:0.53~1.14, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 21]:39.1~120.9, [Source 5]:220.2~289.6, [Source 1]:192~287.7, [Source 9]:277.3~341.7, [Source 17]:142.5~292.0, [Source 19]:97.6~191.1, [Source 18]:334.7	[Source 21]:15.8~78.1, [Source 5]:181.1~252.6, [Source 1]:138.6~241.3, [Source 9]:206.8~243.4, [Source 19]:87.4~172.1, [Source 18]:171.7
0	[Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 19]:0.57~2.27, [Source 18]:1.17	[Source 21]:47.6~147.1, [Source 22]:337~348.2, [Source 5]:267.8~352.2, [Source 1]:233.5~349.9, [Source 19]:118.7~232.4, [Source 18]:400.2	[Source 21]:19.2~95, [Source 22]:275.3~311.7, [Source 5]:220.2~307.1 [Source 1]:168.5~293.4, [Source 19]:130.0~209.4, [Source 18]:205.3
5	[Source 21]:0.53~1.14, [Source 22]:1.02~1.19, [Source 16]:0.53~0.71, [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 9]:0.76~1.12, [Source 17]:1.67~2.5,	[Source 21]:65.9~203.8, [Source 22]:467~482.5, [Source 16]:307.6~357.4, [Source 5]:371.1~488.1, [Source 1]:323.6~484.9, [Source	[Source 21]:26.6~131.6, [Source 22]:381.5~431.8, [Source 5]:305.1~425.6, [Source 1]:233.5~406.6, [Source 9]:348.4~410.1, [Source

	[Source 19]:0.57~2.27, [Source 18]:1.17	9]:467.3~575.7, [Source 17]:240.1~492.0, [Source 19]:164.5~322.0, [Source 18]:539.1	19]:147.2~290.1, [Source 18]:276.6
UMi NLOS, Penetration margin 80 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 21]:0.53~1.14, [Source 19]:0.57~2.27	[Source 21]:0.8~2.4, [Source 19]:1.9~3.8	[Source 21]:0.3~1.6, [Source 19]:1.7~3.4
0	[Source 21]:0.53~1.14, [Source 19]:0.57~2.27	[Source 21]:0.9~2.9, [Source 19]:2.4~4.6	[Source 21]:0.4~1.9, [Source 19]:2.4~4.2
5	[Source 21]:0.53~1.14, [Source 19]:0.57~2.27	[Source 21]:1.3~4.1, [Source 19]:3.3~6.4	[Source 21]:0.5~2.6, [Source 19]:2.9~5.8

Table 7.3.4.2.3-3: D2R - Device C, Reference data rate 5~7kbps

UMi NLOS, Penetration margin 0 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 5]:4.21~5.44, [Source 1]:5.25, [Source 19]:2.71~5.33, [Source 17]:5	[Source 5]:436.8~605.3, [Source 1]:451.3~451.3, [Source 19]:293.9~426.3, [Source 17]:343.5~681.3	[Source 5]:347.7~484.9, [Source 1]:216~216, [Source 19]:254.6~379.0
0	[Source 22]:3.92~4.69, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 19]:2.71~5.33	[Source 22]:884.4~901.9, [Source 5]:531.2~736.1, [Source 1]:548.9~548.9, [Source 19]:357.4~518.4	[Source 22]:699.3~722.5, [Source 5]:422.8~589.7, [Source 1]:262.6~262.6, [Source 19]:309.6~461.0
5	[Source 22]:3.92~4.69, [Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 19]:2.71~5.33	[Source 22]:1225.5~1249.7, [Source 5]:736.1~1020.0, [Source 1]:760.5~760.5, [Source 17]:578.8~1148.1, [Source 19]:495.2~718.3	[Source 22]:969~1001.1, [Source 5]:585.9~817.1, [Source 1]:363.9~363.9, [Source 19]:429.0~638.7
UMi NLOS, Penetration margin 20 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 19]:2.71~5.33	[Source 5]:118.5~164.2, [Source 1]:122.4~122.4, [Source 17]:93.2~184.8, [Source 19]:79.7~115.6	[Source 5]:94.3~131.5, [Source 1]:58.6~58.6, [Source 19]:69.1~102.8
0	[Source 22]:3.92~4.69, [Source 5]:4.21~5.44, [Source 14]:3.5~4.2, [Source 1]:5.25, [Source 19]:2.71~5.33	[Source 22]:239.9~244.7, [Source 5]:144.1~199.7, [Source 14]:139~146.4 [Source 1]:148.9~148.9, [Source 19]:97.0~140.6	[Source 22]:189.7~196, [Source 5]:114.7~160.0, [Source 14]:131.8~134.4 [Source 1]:71.2~71.2, [Source 19]:84.0~125.1
5	[Source 22]:3.92~4.69, [Source	[Source 22]:332.5~339, [Source	[Source 22]:262.9~271.6, [Source

	5]:4.21~5.44, [Source 1]:5.25, [Source 17]:5, [Source 19]:2.71~5.33	5]:199.7~276.7, [Source 1]:206.3~206.3, [Source 17]:157.0~311.5, [Source 19]:134.4~194.9	5]:158.9~221.7, [Source 1]:98.7~98.7, [Source 19]:116.4~173.3
UMi NLOS, Penetration margin 80 dB			
Device Tx power (dBm)	**Achievable cell edge data rate (kbps)**	**Achievable maximum distance at BLER 10% (meter)**	**Achievable maximum distance at BLER 1% (meter)**
-3	[Source 19]:2.71~5.33	[Source 19]:1.6~2.3	[Source 19]:1.4~2.1
0	[Source 19]:2.71~5.33	[Source 19]:1.9~2.8	[Source 19]:1.7~2.5
5	[Source 19]:2.71~5.33	[Source 19]:2.7~3.9	[Source 19]:2.3~3.5

7.3.5 LLS for R2D

7.3.5.1 Device 2b

Table 7.3.5.1-1: LLS R2D - Device 2b

Reference data rate (kbps)	Achievable cell edge data rate (kbps)	CNR at BLER 10% (dB)	CNR at BLER 1% (dB)
1	[Source 22]:1.04~1.12, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 4]:1.79~3.6, [Source 3]:1.84~2.46,	[Source 22]:-0.75~1.2, [Source 5]:21.41~N/A*, [Source 1]:2.39~7.09, [Source 4]:2.36~4.95, [Source 3]:0.9~1.12	[Source 22]:4.2~6.3, [Source 5]:30.21~N/A*, [Source 5]:5.39~12.89, [Source 3]:11.06~11.13
5~7	[Source 12]:4.67~14, [Source 21]:10.16~14.18, [Source 22]:3.7~4.14, [Source 11]:8.8~12, [Source 16]:9.33~13.33, [Source 10]:6~11.5, [Source 5]:4.12~6.6, [Source 14]:4.8~6.67, [Source 1]:5.25, [Source 17]:14, [Source 2]:5.37~10.2, [Source 9]:11.53~12.85, [Source 8]:14, [Source 15]:5.25, [Source 3]:3.69~7.37, [Source 18]:7	[Source 12]:-0.9~7.8, [Source 21]:14.9~18.2, [Source 22]:6~6.1, [Source 11]:0.5~2.15, [Source 16]:4.1~8.1, [Source 10]:6.65~7.4, [Source 5]:20.11~N/A, [Source 14]:15.97~18.81, [Source 1]:15.69, [Source 17]:11~12.8, [Source 9]:5.5~7.3, [Source 8]:4.7, [Source 15]:6.35, [Source 3]:2.15~5.73, [Source 18]:25.8	[Source 12]:9.2~21, [Source 21]:23.7~29.6, [Source 22]:11.5~14.3, [Source 11]:11~12.7, [Source 5]:31.01~N/A, [Source 14]:18~21.12, [Source 1]:24.89, [Source 2]:12.3~19.7, [Source 9]:15~17.5, [Source 8]:15.9, [Source 3]:12.01~16.03, [Source 18]:39.4
48~60	[Source 17]:42~84, [Source 8]: 42~84	[Source 17]:12~16, [Source 8]: 9.6~13.9	
Other value of 168~224	[Source 17]:168~224, [Source 8]:168	[Source 17]:19~24.5, [Source 8]:26.3,	
*Note: "N/A" means the BLER keeps nearly always 100% even for very high CNR/SNR.			

7.3.5.2 Device C

Table 7.3.5.2-1: LLS R2D - Device C

Reference data rate (kbps)	Achievable cell edge data rate (kbps)	CNR at BLER 10% (dB)	CNR at BLER 1% (dB)
1	[Source 22]:1.14~1.16, [Source 5]:0.35~1.05, [Source 1]:0.54~0.96, [Source 19]:1.21~3.59, [Source 4]:1.79~3.6	[Source 22]:-2.2~0.47, [Source 5]:-0.14~2.46, [Source 1]:-0.4~4.3, [Source 19]:4.8~8.8, [Source 4]:2.36~4.95	[Source 22]:1.1~7.5, [Source 5]:4.06~9.06, [Source 1]:2.6~10.1, [Source 19]:6.1~10.4
5~7	[Source 12]:4.67~14, [Source 21]:10.16~14.3, [Source 22]:4.3~4.5, [Source 6]:7, [Source 11]:8.8~12, [Source 16]:9.33~13.33, [Source 10]:6.2~12, [Source 5]:4.12~6.6, [Source 14]:4.8~6.67, [Source 1]:5.25, [Source 19]:3.59~13.46, [Source 2]:5.37~10.2, [Source 9]:11.53~14.42, [Source 8]:14, [Source 15]:5.25, [Source 13]:8.24~11.8, [Source 18]:7	[Source 12]:-1.1~7.5, [Source 21]:12.7~17.4, [Source 22]:5.3~6, [Source 6]:8.69~11.55, [Source 11]:0.5~2.4, [Source 16]:4.1~8.1, [Source 10]:6.5~7.2, [Source 5]:7.46~12.96, [Source 14]:14.06~16.93, [Source 1]:12.9, [Source 19]:8.4~14.1, [Source 9]:5.2~7.9, [Source 8]:4.5, [Source 15]:6.35, [Source 13]:-2.5~13.5, [Source 18]:16.8	[Source 12]:8.3~20, [Source 21]:22.8~27, [Source 22]:11~14.3, [Source 11]:11~11.2, [Source 5]:17.46~30.46, [Source 14]:15.45~20, [Source 1]:22.1, [Source 19]:9.5~15.3, [Source 2]:10.3~17.7, [Source 9]:14.6~18.4, [Source 8]:15, [Source 13]:9, [Source 18]:29.4
48~60	[Source 8]:42~84	[Source 8]:9.8~13.2	[Source 8]:30
Other value of 168~224	[Source 8]:168,	[Source 8]:20.3,	

7.3.6 LLS for D2R

7.3.6.1 Device 2b

Table 7.3.6.1-1: LLS D2R - Device 2b

Reference data rate (kbps)	Achievable cell edge data rate (kbps)	SNR at BLER 10% (dB)	SNR at BLER 1% (dB)
0.1	[Source 12]:0.083~0.391, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104, [Source 3]:0.28~0.38	[Source 12]:-8~-1.6, [Source 5]:0~0.8, [Source 4]:-13~11.2, [Source 3]:5.36~6.31	[Source 12]:-5~3.1, [Source 5]:4.1~5.3, [Source 3]:15.96~20.12
1	[Source 12]:0.301~0.962, [Source 21]:0.44~1.12, [Source 22]:0.9~1.03, [Source 16]:0.53~0.71, [Source	[Source 12]:-8~2, [Source 21]:6.8~24.5, [Source 22]:-5.6~-5.1, [Source 16]:-1.1~1.2, [Source 10]:4.6~6.4, [Source 5]:12.5~12.8,	[Source 12]:-5~error floor, [Source 21]:13.1~42, [Source 22]:-3.8~-1.9, [Source 5]:19.3~20.1, [Source 1]:0.3~8.8, [Source 9]:-3.9~-1.5,

	10]:0.98~1.12 [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 4]:0.72, [Source 9]:0.76~0.84, [Source 3]:0.57~1.14, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 1]:-2.4~3.8, [Source 4]:-1.5, [Source 9]:-9~-5.8, [Source 3]:8.06~18, [Source 17]:0~11, [Source 18]:1.3	[Source 3]:23.19~34.95, [Source 17]:9~18, [Source 18]:12.5
5~7	[Source 12]:6.51~7.83, [Source 22]:3.62~4.1, [Source 6]:3.86~5.49, [Source 11]:11~14, [Source 5]:4.21~5.44, [Source 14]:3.5~4.2, [Source 1]:5.25, [Source 2]:5.37~10.2, [Source 17]:5, [Source 8]:5	[Source 12]:-3.6~-2.6, [Source 22]:-3.2~-2.8, [Source 6]:10.7~17.3, [Source 11]:-10~-5.5, [Source 5]:13.8~16.7, [Source 14]:4.44~6.09, [Source 1]:10.7, [Source 17]:4~14.5, [Source 8]:-2.3~10.3	[Source 12]:2~3.9, [Source 22]:0.2~0.8, [Source 11]:0~7.4, [Source 5]:20.2~23.1, [Source 14]:9.65~9.8, [Source 1]:22, [Source 2]:5.7~12.9, [Source 17]:16~27, [Source 8]:3.2~17.6
Other value of 20	[Source 8]:20	[Source 8]:3.8~14.6	[Source 8]:9.3~21.2

7.3.6.2 Device C

Table 7.3.6.2-1: LLS D2R - Device C

Reference data rate (kbps)	Achievable cell edge data rate (kbps)	SNR at BLER 10% (dB)	SNR at BLER 1% (dB)
0.1	[Source 12]:0.077~0.391, [Source 22]:0.08~0.11, [Source 5]:0.1~0.17, [Source 4]:0.077~0.104	[Source 12]:-7~-2.1, [Source 22]:-10.8~-10.1, [Source 5]:-17.6~-15.8, [Source 4]:-16.1~-14.39	[Source 12]:-3.5~4, [Source 22]:-9.2~-7.8, [Source 5]:-15.9~-14.7
1	[Source 12]:0.301~0.962, [Source 21]:0.44~1.12, [Source 22]:1.02~1.19, [Source 16]:0.53~0.71, [Source 10]:0.98~1.12 [Source 5]:0.78~1.4, [Source 1]:0.54~0.96, [Source 19]:0.57~2.27, [Source 4]:0.72, [Source 9]:0.76~0.84, [Source 17]:1.67~2.5, [Source 18]:1.17	[Source 12]:-6.8~1.4, [Source 21]:6.2~23.5, [Source 22]:-5.7~-5.2, [Source 16]:-1.1~1.2, [Source 10]:4.5~6.2, [Source 5]:-10.2~-9.8, [Source 1]:-10.1~-3.9, [Source 19]:0.5~10.8, [Source 4]:-4.7, [Source 9]:-9.2~-6, [Source 17]:0~11, [Source 18]:1.3	[Source 12]:-3.7~error floor, [Source 21]:12.9~37.4, [Source 22]:-4~-2.1, [Source 5]:-8.1~-6.8, [Source 1]:-7.4~1.1, [Source 19]:-1, [Source 9]:-4~-1.5, [Source 17]:9~18, [Source 18]:12.5
5~7	[Source 12]:6.51~7.83, [Source 22]:3.92~4.69, [Source 6]:3.86~5.49, [Source 11]:11~14, [Source 5]:4.21~5.44, [Source 14]:3.5~4.2, [Source 1]:5.25, [Source 19]:4.67, [Source 2]:5.37~10.2, [Source 15]:4.5, [Source 17]:5, [Source 8]:5, [Source 13]:3.5	[Source 12]:-3.8~-2.65, [Source 22]:-3.3~-3, [Source 6]:6.3~6.9, [Source 11]:-10~-5.5, [Source 5]:-2.8~-0.3, [Source 14]:4.39~5.19, [Source 1]:3, [Source 19]:8.2~13.9, [Source 15]:-5, [Source 17]:4~14.5, [Source 8]:-2.7~8.7, [Source 13]:1.7~11.5	[Source 12]:1.85~3.8, [Source 22]:0.1~0.6, [Source 11]:0~5.3, [Source 5]:0.6~3.2, [Source 14]:5.71~6, [Source 1]:14.3, [Source 19]:10~16.1, [Source 2]:4.1~10.3, [Source 17]:16~27, [Source 8]:-2.7~8.7, [Source 13]:6~15.7
Other value of	[Source 8]:20,	[Source 8]:4.1~12.4	[Source 8]:9~18.1

20			

8 Conclusions and recommendations

8.1 Indoor scenarios

It is feasible to support a harmonized air interface design, with minimized differences (where necessary) for Ambient IoT, to enable the devices described in Clause 3 for addressing the objectives stated in [2]. The extent to which maximum distance targets, differentiated per device and deployment/topology scenario, as described in Clause 4.1, can be met under different sets of evaluation assumptions is reported in Clause 7.1. A latency target for inventory of a single device in TR 38.848, can be met under different evaluation assumptions according to evaluations reported in Clause 7.2.1. The addressed subset of design targets described in TR 38.848 can be met or taken into account in feasible designs, and the design targets left to WGs in Clause 5 of TR 38.848 have been refined. Evaluations have also reported the inventory completion time for multiple devices. Corresponding evaluation scenarios, assumptions, and methodologies have been developed.

The basic blocks of A-IoT device architectures based on RF-ED receiver for device 1, 2a, and 2b have been identified, and also for receiver architectures based on IF-ED or ZIF for device 2b. For device 2a, the characteristics of blocks representing reflection amplifier(s) and large frequency shifter have been studied. Aspects to consider further regarding whether the large frequency shifter block is feasible, and its necessity or effectiveness have also been identified.

Purposes for which clocks(s) may be used by the different A-IoT devices 1/2a/2b have been identified, together with descriptions of the corresponding power consumption, and clock accuracy characteristics.

Regarding the potential impact of device (un)availability due to energy harvesting, two 'Directions' were found, depending on whether the reader does not, or can, provide information to a device based on which the device may become available/unavailable. Company solutions are reported in Clause 6.2, grouped according to the Direction they apply to.

The physical channels studied are PRDCH in R2D, and PDRCH in D2R, for both of which the generation process has been described at high level. Feasible candidates for the methods involved in the processes have also been documented, together with evaluations and pros/cons reported by companies. Information related to R2D reception and D2R scheduling have been listed together with ways they could be conveyed.

The physical signals studied have potential purposes involving start indication, timing acquisition/tracking, calibration/frequency synchronization at the device and the reader, channel or interference estimation at the reader, and indicating the end of a R2D/D2R transmission. Some of those purposes could be provided by information in a PRDCH.

Pros and cons of multiplexing/multiple access options for R2D and D2R links have been studied by company reports. Options and relationships among the resource allocations for messages of the random access procedures have been identified. The needed timing relationships between corresponding transmissions have been captured, for example the minimum time between a R2D transmission and the corresponding D2R transmission following it; and others.

Carrier-wave waveforms and their characteristics have been identified, together with cases for how CW transmission could be deployed in terms of the CW node and the spectrum used. Two CW waveforms were studied: a single-tone unmodulated sinusoid; or two unmodulated single-tone sinusoids. Their characteristics were studied and compared in terms of D2R backscattering reception performance and CW interference suppression at the reader, spectrum utilization of the backscattered signal, and relative complexity of CW generation. The effects of frequency hopping and antenna hopping were included.

For Topology 2, RAN1 studied resource allocation and/or control of the intermediate UE via the Uu interface via higher-layer signaling, or L1 and higher-layer signaling.

The harmonized air interface design focuses on DT and DO-DTT use cases. The study also assessed whether the harmonized air interface design can address the DO-A use case, and concluded that it cannot. The part(s) which is/are not sufficient for the DO-A use case are summarized in Clause 6.10.

RAN1 recommends that in a normative phase, the design should start from the descriptions in this TR.

(1) The following techniques are recommended in the first normative phase:

- PRDCH and PDRCH, as respectively the single physical channels for R2D and D2R
- Modulations of OOK-1 and OOK-4 in R2D
- Multiplexing in time domain for R2D
- Convolutional FEC in D2R and no FEC in R2D
- R2D and D2R signal(s)

(2) At least the following aspects may be considered for inclusion or non-inclusion in a given first or subsequent Release:

- Device 1 and/or Device 2a (with or without large frequency shift) and/or Device 2b (with RF-ED and/or IF-ED and/or ZIF receiver architecture)
- D1T1 and/or D2T2
  - Scenario A1 and/or A2 and/or B and/or C
- Line code(s) Manchester/PIE for R2D, and Manchester / Miller / no line code for D2R
- Multiple access by TDMA and/or FDMA and/or CDMA for D2R; FDMA for device 2b (IF-ED/ZIF receiver) in R2D
- OOK and/or BPSK and/or MSK modulation for D2R
- Carrier-wave waveform 1 and/or 2, and transmission case(s)
- Device (un)availability direction 1 and/or 2
- Proximity determination solution 1 and/or 2

From RAN2 perspective, RAN2 has studied Topology 1 and Topology 2, and has concluded they are feasible. In detail:

- General AS procedures, A-IoT paging, A-IoT random access, A-IoT data transmission and Topology 2 aspects have been studied:
  - See clause 6.3.1 for the overall AS procedure and the information useful to be visible to the reader from CN, etc.
  - See clause 6.3.3 for the A-IoT paging function, including paging message content, paging monitoring behavior, multiple (subsequent) A-IoT paging messages for the same service, etc.
  - See clause 6.3.4) for the A-IoT random access, including the contention-free access and contention-based random access procedure, the message contents, the re-access solutions, etc.
  - See clause 6.3.2 and 6.3.5 for protocol stack for A-IoT radio interface between A-IoT device and reader, required functionalities, and A-IoT data transmission functionalities, etc.
  - See clause 6.3.6 for the Topology 2 specific aspects, including the architecture options and management of radio resources in A-IoT radio interface, etc.
- From RAN2 perspective, the descriptions under clause 6.3 and 6.10 are considered as the baseline conclusions.
- Following is further concluded:
  - At least following features are recommended for the normative phase in the A-IoT MAC layer:
    - A-IoT paging
    - A-IoT random access:
      - For contention-based random access, further down-selection between *Solution 1* (3step-only) and *Solution 3* (unified solution) in clause 6.3.4.
    - A-IoT data transmission

- For Topology 2, at least the following aspects are considered:
  - Forwarding of A-IoT upper layer data and A-IoT radio resource management.
- No further down-selection for Topology 2 architecture options (which are feasible) from RAN2 perspective (the final decision depends on TSGs/WGs outcome).

Logical system architectures have been studied, as documented in clause 6.4.

Impacts on CN-RAN interface, signaling and procedures have been studied to support A-IoT Inventory operation and Command operation for topology 1 and topology 2 respectively, as documented in clause 6.5.

Solutions for locating an Ambient IoT device with minimal specification impact are identified, as documented in clause 6.9.

In RAN3, no evaluation and down selection for T2 solutions have been performed which requires coordination among WGs.

RAN4 has performed adjacent channel co-existence studies between ambient IOT and NR for the scenario that NR macro BS outdoor and ambient IOT system indoor. RAN4 evaluated a total of 9 deployment scenarios as described in Table 6.6.2-1. For each deployment scenario, a total of 4 cases were performed as described in Table 6.6.2-1. The co-existence cases are described in Table 6.6.2-2. The performance metrics for NR system impacted by ambient IOT are throughput loss at the cell edge and cell average. The performance metric for ambient IOT system impacted by NR system is SINR degradation for inter-system interference impact analysis. Meanwhile, in addition to SINR degradation, ambient IOT intra-system performance evaluated by link level simulation is also one of the key aspects affecting ambient IOT system performance, which can be further analyzed in WI phase. The summary of co-existence evaluation can be found in clause 6.6.5.

RAN4 has studied the RF requirements impact for ambient IOT, including system parameters, ambient IoT BS, intermediate UE, devices (device 1, 2a and 2b) and CW node, which can be found in clause 6.7.1, 6.7.2, 6.7.3 and 6.7.4.

RAN4 has also studied the feasibility of CW interference cancellation for both ambient IoT BS and intermediate UE, and the feasible interference cancellation values can be found in clause 6.7.5.

Moreover, RAN4 also studied the testability issue for ambient IOT devices, which can be found in clause 6.7.4.4.

8.2 Outdoor scenarios

RAN1 defined necessary further evaluation assumptions of deployment scenarios for coverage and coexistence evaluations, as reported in Clause 4.2.2. The link budget calculation for coverage is reported in Clause 4.3.2. The achievable cell edge data rates, the corresponding maximum distances between reader and device, BS/device transmit power for device C/2b, etc., are reported in Clause 7.3. Values between -3 dBm and 5 dBm are feasible maximum transmit power for Device C, from device power consumption perspective. RAN1 observes 5 dBm as the maximum transmit power for Device C to achieve maximum coverage in outdoor scenarios. Values between -3dBm and 5dBm are feasible to achieve different coverage. This does not preclude RAN to select any value(s) between -3 dBm and 5 dBm as maximum transmit power for Device C. Assumptions on the energy harvesting/storage used to obtain the reported data rates, specifically on the energy source and the energy storage capacity, are reported in Clause 4.5.

For Device 2b and C, the maximum distance between reader and device, differentiated per pathloss model, BS transmit power, Device transmit power, data rate, penetration margin, BLER, etc. are reported in Clause 7.3.

Device C achieves better coverage than Device 2b for both R2D and D2R. Device 2b is applicable to outdoor scenarios for short distance e.g. 50-200m. Device C is applicable to outdoor scenarios for both short and large distance e.g. 50-500m. Necessity of Device 2b and C in terms of aspects other than coverage is up to RAN decision.

RAN1 studied and found necessary and feasible changes to the Rel-19 air interface to support A-IoT in outdoor scenarios under the conditions stated in [227].

Regarding the R2D modulation and waveform, shorter and longer chip durations as compared to the Rel-19 chip durations were studied, with their pros/cons reported by companies. R2D multiplexing using FDMA was also studied. Regarding the changes to the PRDCH in R2D, the necessity and feasibility of introducing FEC, interleaving, bit collection and repetition were presented, and feasible candidates for these enhancements have been compared and

documented, as reported by companies. There is no need to remove Manchester coding for Rel-20 A-IoT PRDCH. Necessity and feasibility of R2D scrambling were also studied.

The R-TAS was identified as requiring changes to support active device for outdoor scenarios, with a new SIP, where the feasible candidates for the SIP were studied, with the different design aspects identified. Candidate solutions for the device to determine the chip duration of the L1 R2D control information and the data payload of the PRDCH have been studied, with their respective pros/cons identified by company reports.

New R2D physical signals including periodic synchronization signal and CFO calibration signal, and periodic transmission of broadcast information have been identified as necessary for the functionalities of initial time/frequency synchronization, CFO estimation/LO calibration at the device and providing necessary information for subsequent transmission and reception, respectively. Candidate solutions for the functionality of SFO calibration were identified, without the need for a new SFO calibration signal.

L1 R2D control information with a separate CRC has been studied by company reports, including the aspects of potential content, including the TBS to indicate the end of the PRDCH transmission, time and frequency locations, and transmission parameters (CRC/FEC/repetition/payload size) of the L1 R2D control information.

Four D2R modulation schemes, i.e., BPSK/OOK/DBPSK/MSK, have been studied considering their performance, sensitivity to CFO, overhead, complexity, etc. Enhancements on D2R preamble and midamble have been identified, where the sequence type/length/number, insertion interval, etc. were studied. D2R interleaving and bit-collection were studied considering performance gain and required memory. The study assumes single-carrier waveform for D2R. Rel-19 SFS is not necessary in Rel-20 A-IoT D2R. 1SB/2SB without small frequency shift (SFS) were studied considering pros/cons, implementation complexity, etc. For D2R FEC scheme, in addition to 1/3 code rate, both 1/4 and 1/2 were studied considering their performance, complexity, etc. D2R scrambling is necessary for randomizing interference and preventing long runs of identical bits when Manchester code is not applied. For D2R repetition, the introduction of additional number of D2R repetitions larger than 2 was considered.

D2R CDM(A), i.e., D2R sequence based Msg1, has been studied, including sequence type/length/number, performance aspects, and potential impact to random access procedures. For CBRA for DT and DO-DTT, enhancements of D2R TDM(A), i.e., X1 > 2 for Msg1, X3 > 1 for Msg3 (X1/X3: number of time domain resources for Msg1/Msg3 transmission indicated by the corresponding R2D message), have been studied.

For R2D and D2R, the order of channel coding (including interleaver if supported) and block-level repetition (if supported) for complexity reduction, considering necessity and feasibility, has been studied.

Regarding the procedures required to support active devices in outdoor scenarios, both the initial frequency acquisition and reader identification/differentiation procedures have been studied, with the former using the periodic synchronization signal while for the latter, potential candidate solutions on how the device receives the reader ID-related information have been determined.

Support for the DO-A traffic type considering both periodic and event-triggered transmissions has been studied, taking into account the time and frequency synchronization required for the device to transmit DO-A traffic, the resource allocation/determination method used, how the necessary information is carried and provided to the device and the subsequent steps the device would need to perform, with candidate solutions for these aspects identified.

The necessity of D2R Tx power control for Device 2b and Device C for outdoor scenarios was studied and has majority support, and two methods of open loop power control are identified. Feasibility and necessity of closed-loop power control were also studied.

The changes to existing timing offsets, e.g., timing offset value between R2D and corresponding subsequent D2R, etc., have been studied.

According to the SID objective, outdoor device localization for Device 2b/Device C has been studied. D2R preamble / D2R midamble / PDRCH is reported as applicable D2R signal(s)/channel(s) for A-IoT device localization. D2R RSRP-like measurement has been studied. Evaluation results along with the device localization methods have also been reported, which achieve more accurate outdoor device localization than based on Reader-ID.

According to the SID objectives, RAN1 provides the following study outcome:

(1) At least the following changes to the Rel-19 A-IoT specifications are identified to be feasible and necessary to support outdoor scenarios:

R2D

- Periodic R2D sync signal (binary sequence based, with a default pre-defined transmission periodicity)
- Periodic transmission of broadcast information
- CFO calibration signal
- L1 R2D control information with a separate CRC from data (no new R2D physical channel for L1 R2D control information)
- Chip duration determination for L1 R2D control information
- Chip duration determination for R2D data payload
- Binary sequence-based SIP
- R2D FEC (including LTE bit collection with or without R2D interleaving) and/or block level repetitions for A-IoT device in R2D to achieve maximum coverage (e.g., 200 m with penetration loss for Device 2b, 500 m with penetration loss for Device C) as observed in Clause 6A.1.3.1.4, and it can be enabled under certain condition
- No R2D postamble

D2R

- Rel-19 SFS is not supported in Rel-20 A-IoT D2R, and support of FDMA without Rel-19 SFS
- Block-level repetitions for D2R for repetition numbers larger than 2
- PDRCH scrambling (applied after FEC and repetition, if any)
- D2R preamble/midamble(s) in terms of at least one of sequence length/number/type, midamble insertion interval, etc.
- D2R interleaving, and order of FEC and block level repetition

Initial timing and frequency acquisition (using periodic sync signal on predefined sync raster points (sync raster design is handled by RAN4))

Reader identification/differentiation

Support of DO-A, at least including:

- Configuration/signaling of necessary information and resource allocation/determination method for the first D2R transmission at least for DO-A traffic

Timing relationships

(2) At least the following changes to the Rel-19 A-IoT specifications are identified and may or may not be considered to support outdoor scenarios:

R2D

- Addition of $M_{\text{chip}}^{\text{symb}} = 1$
- R2D FDM

D2R

- Addition of DBPSK, removal of OOK
- Additional code rate(s) for D2R FEC (1/2 and/or 1/4)
- Enhancements on TDMA at least for CBRA for DT and DO-DTT (X1 > 2 for Msg1 and/or X3 > 1 for Msg3)
- CDMA (sequence-based Msg1 considering sequence type/length/number)

D2R Tx power control

At least the followings are identified to be feasible and necessary to support outdoor device localization more accurate than based on reader ID under the constraints listed in [227], at least including:

- D2R RSRP-like measurement based on one or more of Rel-20 D2R preamble/midamble/PDRCH

Annex A:
Device receiver sensitivity for *Budget-Alt1*

The receiver sensitivity for device 1, device 2a, and device 2b assuming *Budget-Alt1* is used in the coverage evaluation is summarized as follows.

Table A-1: Device 1 receiver sensitivity (*Budget-Alt1*)

Company	-40 dBm	-36 dBm	-30 dBm	Reference
[197]		×		
[199]	×	×		
[215]			×	
[200]		×		
[193]		×		
[207]		×		
[191]	×	×	×	[216], [217], [218]
[201]	×	×	×	
[203]		×		
[192]		×		
[208]			×	[219]
[212]		×	×	
[211]			×	
[204]			×	
[210]	×	×		
[190]			×	
[202]	×	×	×	
[209]		×	×	
[214]			×	
[185]			×	
[194]		×		
[195]		×		
[198]			×	
[213]		×		[216]
[205]		×		
[196]			×	

Table A-2: Device 2a receiver sensitivity (*Budget-Alt1*)

Company	-50 dBm	-45 dBm	-40 dBm	Reference
[197]		×		
[199]			×	
[215]			×	
[200]		×		
[193]		×		
[207]			×	
[191]	×	×	×	[216], [217], [218]
[201]			×	
[203]		×		
[192]		×		
[208]			×	[220], [221]
[212]			×	
[211]			×	
[204]			×	
[210]		×		
[190]		×		
[202]		×	×	
[209]		×	×	
[185]			×	
[194]		×	×	
[195]			×	
[198]		×		
[213]		×		[220], [222]
[205]			×	
[196]		×		
[214]			×	

Table A-3: Device 2b receiver sensitivity *(Budget-Alt 1)*

Company	-85 dBm	-60 dBm	-55 dBm	-50 dBm	-45 dBm	-40 dBm	Reference
[199]					×		
[215]					×		
[193]					×		
[191]				×			[216], [217], [218]
[201]						×	
[203]					×		
[192]					×		[220], [221]
[212]					×		
[211]					×		
[210]					×		
[190]					×		
[202]					×	×	
[209]					×	×	
[195]		×					

[198]					×		[223], [224], [225]
[213]			×				
[196]	×						
[214]						×	

Annex B (informative): Change history

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2024-02	RAN1#116	R1-2401795				TR skeleton	0.0.1
2024-08	RAN1#118	R1-2407489				Inclusion of agreements up to RAN1#117	0.1.0
2024-09	RAN#105	RP-241784				First version for information to RAN#105	1.0.0
2024-10	RAN#118b	R1-2409993				Addition of the decisions made in RAN1#118bis. Endorsed as v1.1.0 in post-meeting email discussion [Post-118bis-AIoT-01]	1.1.0
2024-11						MCC clean-up – mainly fixing formatting issues	1.1.1
2024-12	RAN#106	RP-242466				For approval to RAN#106	2.0.0
2024-12	RAN#106	RP-243319				Replacement of one FFS and one Editor's Note during RAN#106	2.0.1
2024-12	RAN#106					RAN#106 approval - TR under change control as part of Release 19	19.0.0
2026-03	RAN#111	RP-260581	0001	-	B	Inclusion of RAN4 outcomes of Study on enhancements for solutions for Ambient IoT in NR outdoor for active devices	20.0.0
2026-03	RAN#111	RP-260581	0002	-	B	Inclusion of RAN1 outcomes of Study on enhancements for solutions for Ambient IoT in NR outdoor for active devices	20.0.0